

Krishna's

Operations RESEARCH

*(For B.A. or B.Sc.-Statistics Part-III, Students of All Colleges Affiliated to
Universities in Uttar Pradesh.)*

As per U.P. UNIFIED Syllabus (w.e.f. 2012-2013)

By

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Jai Shri Radhey Shyam

Dedicated
to
Lord
Krishna

Author & Publishers

PREFACE

The book entitled "**Operations Research**" is written according to the latest U.P. UNIFIED syllabus prescribed for **B.A. & B.Sc.-Statistics, Part-III Students**. The book will serve all who are preparing for any examination in Operations Research.

The aim of the book is to explain the concepts systematically and develop understanding of the subject and problem solving methods. In this book all the topics and examples are discussed step by step in detail, without leaving aside anything, so that the reader may understand the subject themselves without the help of others. The book contains large number of Solved Examples for understanding and Unsolved Exercises for practice.

No claim to originality can be made but the treatment of the subject is in our own style.

The Author is grateful to his publishers **Mr. S.K. Rastogi (Managing Director)**, **Mr. Sugam Rastogi (Executive Director)**, **Mrs. Kanupriya Rastogi (Director)** & Entire Team of **Krishna Prakashan Media (P) Ltd.** Meerut for their full co-operation in bringing out the book in the present nice form.

Any suggestions for the improvement of the book will be welcomed, appreciated and acknowledged.

— **Dr. R.K. Gupta**

SYLLABUS

Operations Research

As per U.P. UNIFIED Syllabus (*w.e.f. 2012-2013*)

B.A. & B.Sc.-Statistics, Part-III

Unit-I

General linear programming problems and their formulations. Method for solving LPP : Graphical Method, Simplex method, Big – M method, Two phase Method Duality in LPP

Transportation problem: North-west corner rule, Least cost method, Vogel's approximation method. Optimum solution : Stepping stone method, Method of Multipliers. Assignment Problem : Hungarian Algorithm.

Unit-II

Replacement problems (individual and group). Queueing Models – M/M/1, M/M/C models waiting time distribution for M/M/1, Little's formulae. M/G/1 Queueing system, cost profit models in queueing theory.

Unit-III

Network Analysis : Minimal Spanning Tree, Model Shortest-route problems, Maximal Flow Model. Project Management : PERT/CPM determination of floats construction of time chart and resources labelling.

Unit-IV

Inventory Models : FOQ models, Non-zero, lead time, EOQ with shortages allowed. Dynamic Programming : Bellman's optimality principle. Applications. Job sequencing : n jobs – machines, n jobs – K machines, 2 jobs – n machines.

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UNIT

I



Chapters



1. General Linear Programming Problems
(Formulation and Graphical Solution)
2. Simplex Method
3. Duality in Linear Programming
4. Transportation Problem
5. Assignment Problem

UNIT-I

Chapter

1



General Linear Programming Problem

(Formulation and Graphical Solution)

1.1 Introduction

A common problem in business organization involve the allocation of resources such as men, machine, materials, capital etc. to various activities with the objective of maximizing profit or minimizing cost or both in the best interest of the organization. *Linear programming* is a mathematical tool (Technique) used in solving such problems since world war II. The first mathematical technique in this field, called the **simplex method** of linear programming, was developed in 1947 by an American mathematician George B. Dantzig, recognized as the father of linear programming. Since then, new techniques and applications have been developed through the efforts and co-operation of interested individuals in academic institutions and industry both.

1.2 Linear Programming and Linear Programming Problem (Definition)

Linear programming: *Linear programming is a mathematical technique for determining the optimum allocation of resources and obtaining a particular objective subject to satisfying certain number of given linear restrictions.*

The term '*linear*' means that all the mathematical relations used in the problem are linear relations and the term '*programming*' refers to the method of determining a particular programme or plan of action.

Some important definitions of linear programming are as follows:

1. *Linear programming is the analysis of problems in which linear function of a number of variables is to be maximized (or minimized) when those variables are subject to a number of restraints in the form of linear inequalities.*

[By Samuelson, Dorfman and Solow]

2. *Linear programming is a planning technique that permits some objective functions to be maximized or minimized within the framework of given situational restrictions.*

[By William M. Fox]

3. *Linear programming is an optimization technique for finding an optimal (maximum or minimum) value of a linear function of several independent variables which are subject to various restrictions or (constraints) expressed as equations or inequalities.*

[By David Smith]

Linear Programming Problem (L.P.P.): *The linear programming problem in general call for optimizing (maximizing or minimizing) a linear function of variables called the **objective function** subject to a set of linear equations and / or linear inequations called the **constraints** or restrictions.*

1.3 Basic Requirements of a Linear Programming Problem (Main Characteristics of Linear Programming Problem)

There are certain necessary basic requirements (characteristics) to be fulfilled before the linear programming techniques can be used for optimization problems. These are as follows:

1. **Well-defined objective function:** The objective function of a L.P.P. should be clear and well-defined, which may be either to maximize the profit by utilizing the available resources or it may be to minimize the cost by using a limited amounts of various resources required. The objective function should be expressed as a linear function of decision variables.
2. **Decision variables and their relationship:** The decision variables that may be the number of candidates, number of products, quantity of resources etc. should be clear in a problem and the inter-relationships between these variables should be linear.
3. **Presence of constraints and restrictions:** The available resources which are to be allocated among various competing activities should be limited. These constraints must be linear equalities or inequalities of decision variables.
4. **Alternative courses (lines) of action:** There must be different courses (lines) of actions available to the decision maker to perform the job.
5. **Linearity:** A primary requirement of a L.P.P. is that the objective function and all the constraints must be expressed in terms of linear equations and inequalities. Linearity means that the relationship among decision variables should be directly

proportional *i.e.*, if the decision variables are increased in a certain ratio then the objective function must also increase in the same ratio. For example if all the decision variables are doubled then the objection function is also doubled.

Thus if x and y are the decision variables in a L.P. problem then their products such as xy, x^2y, x^2, y^3 etc. and combination of the type $ax + be^y, ax + b \log y$ etc. are not allowed either in the objective function or in the constraints.

6. **Non-negative restrictions:** All decisions variables considered for decisions must assume non-negative values. *i.e.*, if x, y are the decision variables in a L.P.P., then $x \geq 0, y \geq 0$.

1.4 Basic Assumptions of Linear Programming

There are few basic assumptions, which are given below, are necessary for all linear programming problems.

1. **Proportionality:** In a L.P.P. we assume that proportionality exists in the objective function and constraints. For example, if we require the profit or loss to be doubled in a L.P.P. then all the required resources are also to be doubled.
2. **Additivity:** In a L.P.P. we assume that the total (sum) of the resources used by different activities must be equal to the total (sum) of resources used for each activity. In other words the resources are not consumed in changing the activity. For example, if it takes x hours on machine A to make product P and y hours to make product Q then the total time on machine A devoted to this production is $x + y$ *i.e.*, the time to change the machine frame product P to product Q is negligible.
3. **Multiplicativity:** Multiplicativity means that if the time required for the production of one unit is t hours then the time to produce x units will be xt hours.
4. **Divisibility:** Divisibility means that the fractional values of the decision variables must be permissible besides the integral values.
5. **Certainty:** In a L.P.P. we assume that the coefficients in the objective function and the constraints are completely known and do not change during the period of study.
6. **Finite number of constraints:** The number of activities in a L.P.P. are assumed to be finite *i.e.*, the number of constraints should be finite.

1.5 Mathematical Description of a General Linear Programming Problem

A general *linear programming problem* can be stated as follows:

Optimize (Maximize or minimize)

$$Z = c_1x_1 + c_2x_2 + \dots + c_nx_n \quad \dots(1)$$

$$\left. \begin{array}{l} a_{11} x_1 + a_{12} x_2 + + a_{1n} x_n (\leq, =, \geq) b_1 \\ a_{21} x_1 + a_{22} x_2 + + a_{2n} x_n (\leq, =, \geq) b_2 \\ \\ \\ a_{m1} x_1 + a_{m2} x_2 + + a_{mn} x_n (\leq, =, \geq) b_m \end{array} \right\} ... (2)$$
$$x_1, x_2, \dots, x_n \geq 0 \quad \dots(3)$$

1. $c_1, c_2, \dots, c_{xi}, a_{11}, a_{12}, \dots, a_{mn}$ and $b_1, b_2, \dots, b_m$ are constants.
2. $x_1, x_2, \dots, x_n$ are variables and are called **decision variables** whose values are to be determined.
3. The linear function Z given by (1) is called the **objective function**, which is to be optimized (*i.e.* maximized or minimized).
4. $c_1, c_2, \dots, c_n$ are called **profit or cost coefficients** *i.e.*, they represent the per unit contribution of profit or cost of decision variables $x_1, x_2, \dots, x_n$ to the value of objective function Z .
5. The coefficients $a_{ij} (i=1, 2, \dots, m; j=1, 2, \dots, n)$ represent the amount of resources consumed per unit of variable and are referred to as structural (or technological) coefficients.
6. The linear equations and inequalities given in (2) are called the **constraints**.
7. The conditions given in (3) are called the **non-negative restrictions** to decision variables $x_1, x_2, \dots, x_n$ *i.e.*, these variables cannot be negative.

If $c_1, c_2, \dots, c_n$ are the costs, per unit contribution of decision variables $x_1, x_2, \dots, x_n$ respectively, then the objective function is minimize $Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$.

3. **To formulate the constraints:** Identify all the limitations given in the problem and express them as linear equations or linear inequalities of the decision variables.
4. **To mention non-negative restrictions:** In a L.P.P. the values of decision variables cannot be negative, so give the non-negative restrictions of variables as $x_i \geq 0, \forall i = 1, 2, \dots, n$.

ILLUSTRATIVE EXAMPLES

Example 1: A factory produces two products A and B. Each of the product A requires 2 hrs of moulding, 3 hrs of grinding and 4 hrs for polishing and each of the product B requires 4 hrs for moulding, 2 hrs for grinding and 2 hrs for polishing. The moulding machine can work for 20 hrs, grinding machine for 24 hrs and polishing machine is available for 13 hrs. The profit is ₹ 5 per unit of A and ₹ 3 per unit of B. Assuming that the factory can sell all that it produces, formulate the problem as a L.P.P. to maximize the profit.

Solution: Let the factory produce x_1 units of product A and x_2 units of product B, for maximum profit.

All the given information in the problem can be put in a tabular form as follows:

Product	Moulding time in hrs.	Grinding time in hrs.	Polishing time in hrs.	Profit on one unit in ₹
$A(x_1)$	2	3	4	5
$B(x_2)$	4	2	2	3
Time available in hours	20	24	13	

Total moulding time required to produce x_1 units of A and x_2 unit of B is $2x_1 + 4x_2$ hrs and this cannot exceed 20 hrs which is the maximum moulding time.

$\therefore$ We have, $2x_1 + 4x_2 \leq 20$

Similarly, total time required for grinding $= 3x_1 + 2x_2 \leq 24$

and total time required for polishing $= 4x_1 + 2x_2 \leq 13$

To find objective function: If Z is profit in ₹ to produce x_1 units of product A and x_2 units of product B, then

$$\text{Profit } Z (\text{in ₹}) = 5x_1 + 3x_2.$$

Since the number of products to be produced cannot be negative, so the non-negative restrictions are $x_1 \geq 0$ and $x_2 \geq 0$.

Hence the L.P.P. for the given problem is as follows:

$$\text{Max. } Z = 5x_1 + 3x_2 \text{ (Objective function)}$$

Subject to the constraints

$$\left. \begin{array}{l} 2x_1 + 4x_2 \leq 20 \\ 3x_1 + 2x_2 \leq 24 \\ 4x_1 + 2x_2 \leq 13 \end{array} \right\} \text{ Constraints}$$

and $x_1 \geq 0, x_2 \geq 0$ (Non-negative restrictions)

Example 2: A firm manufactures two types of products A and B and sells them at a profit of ₹ 2 on type A and ₹ 3 on type B. Each product is processed on two machines E and F. Type A requires one minute of processing time on E and two minutes on F, type B requires one minute on E and one minute on F. The machine E is available for not more than 6 hours 40 minutes while machine F is available for 10 hours during any working day. Formulate the problem as a linear programming problem.

Solution: Let the firm manufacture x_1 units of product A and x_2 units of product B, for maximum profit.

All the informations given in the problem can be put in a tabular form as follows:

Machines	Processing time per unit (in minutes)		Available time (in minutes)
	Type A (x_1 units)	Type B (x_2 Units)	
E	1	1	400
F	2	1	600
Profit per unit (in ₹)	2	3	

Total time required on machine E for the manufacture of x_1 units of A and x_2 units of B is $1.x_1 + 1.x_2$ minutes and this time cannot exceed 6 hours 40 minutes i.e., 400 minutes which is the available time of machines A.

$\therefore$ We have, $x_1 + x_2 \leq 400$

Similarly total time required on machine F $= 2x_1 + x_2 \leq 600$.

To find objective function: If Z is the profit in ₹ by sell of x_1 units of A and x_2 units of B, then

$$Z = 2x_1 + 3x_2$$

Also, since the number of units of product A and B produced cannot be negative,

$$\therefore x_1 \geq 0, x_2 \geq 0.$$

Hence the L.P.P. of the given problem is as follows:

Maximize $Z = 2x_1 + 3x_2$ (objective function)

Subject to the constraints

$$\left. \begin{array}{l} x_1 + x_2 \leq 400 \\ 2x_1 + x_2 \leq 600 \end{array} \right\} \text{Constraints}$$

and $x_1 \geq 0, x_2 \geq 0$ (Non-negative restrictions)

Example 3: A toy company manufactures two types of doll; an ordinary-doll A and a deluxe doll B. Each type of doll B takes twice as long to produce as one of type A. It is given that company would have time to make a maximum of 2000 dolls per day if it produces only the ordinary version. The supply of plastic is sufficient to produce 1500 dolls per day (both A and B combined). The deluxe version requires a fancy dress of which there are only 600 pieces per day available. If the company makes profit of ₹ 3.00 and ₹ 5.00, per doll respectively on doll A and B. Formulate the problem as a linear programming problem to maximize the profit.

Solution: Let the company manufacture x_1 ordinary dolls A and x_2 deluxe dolls B per day.

Let an ordinary doll A be produced in time t , then the time required to produce a deluxe doll B will be $2t$.

$\therefore$ Total time required to produce x_1 ordinary dolls A and x_2 deluxe dolls B is $tx_1 + 2tx_2$.

Since the company can produce 2000 dolls per day of ordinary version, so the time available to the company per day is $2000t$.

$\therefore$ We have, $tx_1 + 2tx_2 \leq 2000t$ i.e., $x_1 + 2x_2 \leq 2000$.

Since the supply of plastic is sufficient to produce 1500 dolls per day both A and B combined

$$\therefore x_1 + x_2 \leq 1500$$

Since the fancy dress for deluxe version dolls B is available for 600 pieces per day only, so the company cannot produce more than 600 deluxe dolls B.

$$\therefore x_2 \leq 600$$

To find objective function: If Z is the total profit in ₹ to produce x_1 ordinary dolls A and x_2 deluxe dolls B, then $Z = 3x_1 + 5x_2$.

Also the number of dolls produced cannot be negative.

$$\therefore x_1 \geq 0, \quad x_2 \geq 0.$$

Hence the L.P.P. of the given problem is as follows:

$$\text{Max. } Z = 3x_1 + 5x_2 \quad \text{objective function}$$

Subject to the constraints,

$$\left. \begin{array}{l} x_1 + 2x_2 \leq 2000 \\ x_1 + x_2 \leq 1500 \\ x_2 \leq 600 \end{array} \right\} \text{Constraints}$$

$$\text{and } x_1, x_2 \geq 0 \quad (\text{Non-negative restrictions})$$

Example 4: A dietician decides a certain minimum intake of vitamins A, B and C for a family. The minimum daily needs of the vitamins A, B, C are 30, 20, 16 units respectively. For the supply of these, the dietician depends on two types of foods X and Y. The first one gives 7, 5, 2 units per gram of vitamins A, B, C, respectively. The second one gives 2, 4, 8 units per gram of these vitamins respectively. The first food costs ₹ 2 per gram and the second ₹ 1 per gram.

How many grams of each food stuff should the family buy every day to keep the food expense at a minimum? Formulate a L.P.P. of this problem.

Solution: Let the family buy x_1 grams of food X and x_2 grams of food Y every day to keep the food expense at a minimum.

The information given in the problem can be put in the tabular form as follows:

Foods	Vitamin A	Vitamin B	Vitamin C	Cost/gram. (in ₹)
X (x_1)	7	5	2	2
Y (x_2)	2	4	8	1
Min. Daily Needs	30	20	16	

In x_1 gram of food X and x_2 gram of food Y.

Available Vitamin A is $7x_1 + 2x_2$ Units

Vitamin B is $5x_1 + 4x_2$ Units

Vitamin C is $2x_1 + 8x_2$ Units

Since minimum daily requirements of vitamins A, B, and C are 30, 20 and 16 units respectively.

$$\therefore 7x_1 + 2x_2 \geq 30, \quad 5x_1 + 4x_2 \geq 20, \quad 2x_1 + 8x_2 \geq 16$$

If Z is the total expense in buying x_1 grams of food X and x_2 grams of food Y , then $Z = ₹ (2x_1 + 1x_2)$.

Since the quantity of foods cannot be negative

$$\therefore x_1 \geq 0, x_2 \geq 0.$$

Hence the L.P.P. of the given problem is

$$\text{Minimize } Z = 2x_1 + x_2$$

Subject to the constraints

$$7x_1 + 2x_2 \geq 30, 5x_1 + 4x_2 \geq 20, 2x_1 + 8x_2 \geq 16$$

$$\text{and } x_1 \geq 0, x_2 \geq 0$$

Example 5: A firm can produce three types of cloth, say A , B and C . Three kinds of wool are required for it, say red wool, green wool and blue wool. One unit length of type A cloth needs 2 yards of red wool and 3 yards of blue wool; one unit length of type B cloth needs 3 yards of red wool, 2 yards of green wool and 2 yards of blue wool and one unit length of type C cloth needs 5 yards of green wool and 4 yards of blue wool. The firm has only a stock of 8 yards of red wool, 10 yards of green wool and 15 yards of blue wool. It is assumed that the income obtained from one unit length of type A cloth is ₹ 3.00, of type B cloth is ₹ 5.00 and of type C cloth is ₹ 4.00. Formulate the problem as a linear programming model to maximize the income from the finished cloth.

Solution: Let the firm produce x_1, x_2, x_3 yards of cloths of types A, B and C respectively.

All the informations given in the problem can be put in a tabular form as follows:

Types of Cloth	Kinds of wool for unit length (In yards)			Income from one unit length of cloth in ₹
	Red	Green	Blue	
$A (x_1)$	2	0	3	3
$B (x_2)$	3	2	2	5
$C (x_3)$	0	5	4	4
Stock of wool available (In yards)	8	10	15	

$\therefore$ Red wool required (in yards) $= 2x_1 + 3x_2 + 0x_3$, which cannot be more than 8 yards available.

$$\therefore 2x_1 + 3x_2 + 0x_3 \leq 8.$$

Similarly, green wool required (in yards) $= 0x_1 + 2x_2 + 5x_3 \leq 10$

and blue wool required (in yards) $= 3x_1 + 2x_2 + 4x_3 \leq 15$.

To find objective function: If Z is the profit in ₹ to produce x_1, x_2, x_3 yards of these types of clothes respectively, then

$$Z = 3x_1 + 5x_2 + 4x_3$$

Hence the L.P.P. for the given problem is as follows.

Maximize, $Z = 3x_1 + 5x_2 + 4x_3$

Subject to the constraints

$$2x_1 + 3x_2 \leq 8, 2x_2 + 5x_3 \leq 10, 3x_1 + 2x_2 + 4x_3 \leq 15$$

and $x_1, x_2, x_3 \geq 0$.

Example 6: A city hospital has the following minimum daily requirements for nurses:

Period	Clock Time (24 hr. day)	Minimum number of nurses required
1	6 A.M. — 10 A.M.	2
2	10 A.M. — 2 P.M.	7
3	2 P.M. — 6 P.M.	15
4	6 P.M. — 10 P.M.	8
5	10 P.M. — 2 A.M.	20
6	2 A.M. — 6 A.M.	6

Nurses report to the hospital at the beginning of each period and work for 8 consecutive hours. The hospital wants to determine the minimum number of nurses available for each period. Formulate this as a L.P.P. by setting up appropriate constraints and objective function.

Solution: Let x_1 nurses report on duty at 6 A.M. x_2 nurses at 10 A.M. x_3 nurses at 2 P.M. x_4 nurses at 6 P.M. x_5 nurses at 10 P.M. and x_6 nurses at 2 A.M. x_1 nurses which report on duty at 6 A.M. will work for 8 hours upto 2 P.M., while x_2 more nurses will report at 10 A.M. and will work for 8 hours upto 6 P.M.

Thus number of nurses available between 10 A.M. to 2 P.M. is $x_1 + x_2$. Also x_2 nurses are already available between 2 P.M. to 6 P.M. while x_3 more nurses will report on duty at 2 P.M. and will work up to 10 P.M.

Thus number of nurses available between 2 P.M. to 6 P.M. is $x_2 + x_3$.

Similarly number of nurses available

Between 6 P.M. to 10 P.M. is $x_3 + x_4$

Between 10 P.M. to 2. A.M. is $x_4 + x_5$

Between 2 A.M. to 6 A.M. is $x_5 + x_6$

and Between 6 A.M. to 10 A.M. is $x_6 + x_1$

If Z is the total number of nurses reporting on duty in 24 hours, then

$$Z = x_1 + x_2 + x_3 + x_4 + x_5 + x_6$$

Since the number of nurses cannot be negative,

$$\therefore x_1, x_2, \dots, x_6 \geq 0.$$

Since the minimum number of nurses required in periods 6 A.M. to 10 A.M., 10 A.M. to 2 P.M. etc. are 2, 7, 15, 8, 20 and 6 respective, so the L.P.P. of the given problem is,

$$\text{Minimize } Z = x_1 + x_2 + x_3 + x_4 + x_5 + x_6$$

Subject to the constraints

$$x_1 + x_2 \geq 7, \quad x_2 + x_3 \geq 15, \quad x_3 + x_4 \geq 8,$$

$$x_4 + x_5 \geq 20, \quad x_5 + x_6 \geq 6, \quad x_6 + x_1 \geq 2$$

$$\text{and } x_1, x_2, \dots, x_6 \geq 0.$$

Exercise 1.1

1. A goldsmith manufactures necklaces and bracelets. The total number of necklaces and bracelets that he can handle per day is at most 24. It takes one hour to make a bracelet and half an hour to make a necklace. It is assumed that he can work for a maximum of 16 hours a day. Further the profit on a bracelet is ₹ 300 and the profit on a necklace is ₹ 100. Formulate this problem as a linear programming problem so as to maximize the profit.
2. A furniture dealer deals in two items, viz, tables and chairs. He has ₹ 10000 to invest and a space to store at most 60 pieces (including both tables and chairs). A table costs him ₹ 500 and a chair ₹ 100. He can sell all the items that he buys earning a profit of ₹ 50 for each table and ₹ 15 for each chair. Formulate this problem as a L.P.P. so that he maximizes the profit.
3. A furniture firm manufactures chairs and tables, each requiring the use of three machines A, B and C. Production of one chair requires 2 hours on machine A, 1 hour on machine B and 1 hour on machine C. Each table requires 1 hour each on

machines A and B and 3 hours on machine C. The profit realized by selling one chair is ₹ 30 while for a table the figure is ₹ 60. The total time available per week on machine A is 70 hours, on machine B is 40 hours, and on machine C is 90 hours. Formulate this problem as a linear programming problem to maximize the profit.

4. A company produces two types of hats. Each hat of the first type requires twice as much labour time as the second type. If all hats are of the second type only, the company can produce a total of 500 hats a day. The market limits daily sales of the first and second types to 150 and 250 hats. Assuming that the profits per hat are ₹ 8 for type I and ₹ 5 for type II, formulate the problem as a linear programming model in order to determine the number of hats to be produced of each type so as to maximize the profit. [B.B.A. III GNDU April 2002]
5. An animal feed manufacturer has to produce 200 kg. of a feed mixture consisting of two ingredients x_1 and x_2 . x_1 costs ₹ 6 per kg. and x_2 ₹ 16 per kg. Not more than 80 kg. of x_1 can be used and at least 60 kg. of x_2 must be used. Write this problem of linear programming in the form of mathematical formulation.
6. The maximum sum of two positive whole numbers is 15. The maximum difference of double of first number and the second number is 8. If their maximum product be P , then give mathematical formulation for this linear programming problem.
7. The HKD manufacturing company can make two products P_1 and P_2 . Each of the products requires time on a cutting machine and finishing machine.

Products	Time on Machine (In hrs)		Profit/Unit (In ₹)
	Cutting	Finishing	
P_1	2	1	6
P_2	3	3	4

If numbers of cutting hours available per week is 390 and number of finished hour available is 810, formulate L.P.P. for maximum profit.

8. A diet is to contain at least 4000 units of carbohydrates, 500 units of fat and 300 units of protein. Two foods A and B are available. Food A costs ₹ 2 per unit and food B costs ₹ 4 per unit. A unit of food A contains 10 units of carbohydrates, 20 units of fat and 15 units of protein. A unit of food B contains 25 units of carbohydrates, 10 units of fat and 20 units of protein. Formulate the problem as a L.P.P. so as to find the minimized cost for a diet that consists of a mixture of these two foods and also meets the minimum nutrition requirements.
9. A firm can produce two products A and B during a given period of time. Each of these products requires four different operations, viz., Grinding, Turning, Assembling

and Testing. The requirement in hours per unit of manufacturing of these products is as given:

Products	Grinding	Turning	Assembling	Testing
<i>A</i>	1	3	4	5
<i>B</i>	2	1	3	4

The available capacities of these operations in hours for the given time period are: 30 for grinding, 60 for turning, 200 for assembling and 200 for testing.

Profit on each unit of *A* is ₹ 3 and that for each unit of *B* is ₹ 2. Formulate the problem as a linear programming model to maximize the profit assuming that the firm can sell all the items that it produces at the prevailing market price.

10. A firm manufactures 3 products *A*, *B* and *C*. The profits are ₹ 3, ₹ 2 and ₹ 4 respectively. The firm has 2 machines and below is the required processing times in minutes for each machine on each product:

Machine	Product		
	<i>A</i>	<i>B</i>	<i>C</i>
<i>G</i>	4	3	5
<i>H</i>	2	2	4

Machines *G* and *H* have 2,000 and 2,500 machine minutes, respectively. The firm must manufacture 100 *A*'s, 200 *B*'s and 50 *C* 's, but not more than 150 *A*'s. Set up a linear programming problem to maximize profit.

11. A firm produces three products *A*, *B* and *C*, each of which passes through three departments: Fabrication, Finishing and Packing. Each unit of product *A* requires 3, 4 and 2 hours; a unit of *B* requires 5, 4 and 4 hours while each unit of product *C* requires 2, 4, 5 hours respectively in the three departments. Every day, 60 hours are available in the fabrication department, 72 hours in the finishing department and 100 hours in the packing department. If the unit contribution of product *A* is ₹ 5 of product *B* is ₹ 10 and of product *C* is ₹ 8 formulate the above problem as a L.P.P., the aim of the firm is to earn maximum profit.
12. According to the medical experts it is necessary for an adult to consume at least 75 g of proteins, 85 g of fats and 300 g of carbohydrates daily. The following table gives the analysis of the food items readily available in the market with their respective costs.

Food types	Food value (in g) per 100 g			Cost in ₹ per kg
	Proteins	Fats	Carbohydrates	
A	18.0	15.0	–	3.0
B	16.0	4.0	7.0	4.0
C	4.0	20.0	2.5	2.0
D	5.0	8.0	40.0	1.5
Minimum daily requirement	75.0	85.0	300.0	

Formulate a linear programming problem for an optimum diet.

Answers 1.1

1.	If x_1 necklaces and x_2 bracelets are manufactured per day, then L.P.P. is Max $Z = 100x_1 + 300x_2$. Subject to, $x_1 + 2x_2 \leq 32$, $x_1 + x_2 \leq 24$ and $x_1, x_2 \geq 0$.
2.	If x_1 tables and x_2 chairs are bought, then L.P.P. is Max. $Z = 50x_1 + 15x_2$. Subject to $x_1 + x_2 \leq 60$, $5x_1 + x_2 \leq 100$ and $x_1, x_2 \geq 0$.
3.	If the firm manufacture x_1 and x_2 numbers of chairs and tables per week, then L.P.P. is Maximize $Z = 30x_1 + 60x_2$ Subject to, $2x_1 + x_2 \leq 70$, $x_1 + x_2 \leq 40$, $x_1 + 3x_2 \leq 90$ and $x_1, x_2 \geq 0$.
4.	If x_1, x_2 are the numbers of hats of two types produced, then L.P.P. is Max $Z = 8x_1 + 5x_2$. Subject to $2x_1 + x_2 \leq 500$, $x_1 \leq 150$, $x_2 \leq 250$ and $x_1, x_2 \geq 0$.
5.	L.P.P. is Min. $Z = 6x_1 + 16x_2$, subject to $x_1 + x_2 \leq 200$, $x_1 \leq 80$, $x_2 \geq 60$ and $x_1, x_2 \geq 0$.
6.	If x_1, x_2 are two numbers, then L.P.P. is Max. $Z = x_1x_2$, Subject to $x_1 + x_2 \leq 15$, $2x_1 - x_2 \leq 8$ and $x_1, x_2 \geq 0$
7.	If x_1, x_2 are the numbers of products P_1 and P_2 produced, L.P.P. is Max. $Z = 6x_1 + 4x_2$ Subject to, $2x_1 + 3x_2 \leq 390$, $x_1 + 3x_2 \leq 810$ and $x_1, x_2 \geq 0$

8.	If diet consists of x_1 units of food A and x_2 units of food B , then L.P.P. is Minimize $Z = 2x_1 + 4x_2$, subject to $10x_1 + 25x_2 \geq 4000$, $20x_1 + 10x_2 \geq 500$, $15x_1 + 20x_2 \geq 300$ and $x_1, x_2 \geq 0$
9.	If the firm produce x_1 and x_2 units of two products A and B respectively, L.P.P. is Max. $Z = 3x_1 + 2x_2$, subject to $x_1 + 2x_2 \leq 30$, $3x_1 + x_2 \leq 60$, $4x_1 + 3x_2 \leq 200$, $5x_1 + 4x_2 \leq 200$, $x_1, x_2 \geq 0$
10.	If x_1, x_2 and x_3 are the numbers of products A, B, C produced, then L.P.P. is Max. $Z = 3x_1 + 2x_2 + 4x_3$ subject to $4x_1 + 3x_2 + 5x_3 \leq 2000$, $2x_1 + 2x_2 + 4x_3 \leq 2500$, $100 \leq x_1 \leq 150$, $0 \leq x_2 \leq 200$ and $0 \leq x_3 \leq 50$,
11.	If x_1, x_2, x_3 units of product A, B, C are produced, then L.P.P. is Max. $Z = 5x_1 + 10x_2 + 8x_3$, Subject to $3x_1 + 5x_2 + 2x_3 \leq 60$, $4x_1 + 4x_2 + 4x_3 \leq 72$, $2x_1 + 4x_2 + 5x_3 \leq 100$, $x_1, x_2, x_3 \geq 0$
12.	If daily diet consist of x_1 kg of food A , x_2 kg of food B , x_3 kg of food C and x_4 kg of food D , then L.P.P. is Minimize $Z = 3x_1 + 4x_2 + 2x_3 + 1.5x_4$. subject to $180x_1 + 160x_2 + 40x_3 + 50x_4 \geq 75$, $150x_1 + 40x_2 + 200x_3 + 80x_4 \geq 85$. $70x_2 + 25x_3 + 400x_4 \geq 300$ and $x_1, x_2, x_3, x_4 \geq 0$.

1.7 Some Important Definitions

1. **Solution of a L.P.P.** : A set of values of the variables satisfying the constraints of a L.P.P. is called a *solution* of the L.P.P.
2. **Feasible solution (F.S.) of a L.P.P.**: A set of values of the variables satisfying the constraints and the non-negative restrictions of a L.P.P. is called a *feasible solution* (F.S.) of a L.P.P.
3. **Optimal (or optimum) solution of a L.P.P.**: A feasible solution of a L.P.P. is said to be **optimal** (or **optimum**) if it also optimizes (*i.e.*, maximizes or minimizes as the case may be) the objective function of the problem.

or

A set of values of the variables satisfying the constraints and the non-negative restrictions of a L.P.P. is said to be its optimal (or optimum) solution if it optimizes (*i.e.*, maximizes or minimizes as the case may be) the objective function of the problem.

4. **Unbounded solution**: If the value of the objective function of a L.P.P. can be increased or decreased indefinitely, such solutions are called **unbounded solutions**. In this case we say that the problem has an unbounded solution.

1.8 Redundancy in Constraint Equations

By redundancy in constraint equations in a L.P.P. we mean that the problem has more than enough number of constraint equations. A constraint/constraints in a L.P.P. is / are said to be redundant if the feasible region of the problem is unchanged by deleting (leaving) that constraint/constraints.

1.9 Fundamental Extreme Point Theorem

Statement: An optimum solution of a L.P.P. if it exists, occurs at one of the extreme points (*i.e.*, corner points) of the convex polygon of the set of all feasible solutions.

1.10 Solution of a Linear Programming Problem

A linear programming problem can be solved by any of the following two methods.

1. Graphical Method and
2. Simplex Method

1.11 Graphical (or Geometrical) Method for the Solution of a Linear Programming Problem

A linear programming problem can be solved by graphical method when it involves only two decision variables. A problem of three variables can also be solved by this method, but it is complicated enough.

For the solution of a L.P.P. by graphical method, proceed as follows:

Step 1: First of all consider all the constraints as equalities *i.e.*, replace the signs $\geq$ or $\leq$ by $=$ in all constraints.

Step 2: Draw the lines in the plane corresponding to each equation obtained in step one.

Method to draw lines: Consider the equation $ax_1 + bx_2 = c$.

Putting $x_2 = 0$, find the value of x_1 and then putting $x_1 = 0$ find the value of x_2 . Thus we get two points $(-c/a, 0)$ and $(0, -c/b)$ which are the points of intersection of the line with the axes. Plot these two points on the axes and connect by a straight line. This represents the line $ax_1 + bx_2 = c$. Draw lines corresponding to each equation.

Thus we get as many straight lines, as there are the equations, each straight line representing one constraint.

Step 3: To find feasible region (permissible region): Then we find the permissible region (*i.e.*, feasible region) for the values of the variables, which is the set of points that simultaneously satisfy all the constraints and is the regions (polygon) in the first quadrant bounded by the co-ordinate axes and the lines drawn in step two. Method to find feasible region. (*i.e.*, *Permissible region*).

- (a) Consider the constraint $ax_1 + bx_2 \leq (\text{or } \geq) c$. The line $ax_1 + bx_2 = c$ (drawn in step 2) divides the x_1, x_2 plane in two regions, one containing the origin and the other not containing the origin. If the point $(0, 0)$ satisfy the constraint $ax_1 + bx_2 \leq (\text{or } \geq) c$, then the feasible region corresponding to this constraint is the region which contains the origin. Also if $(0,0)$ does not satisfy the constraint $ax_1 + bx_2 \leq (\text{or } \geq) c$, then the feasible region corresponding to this constraint is the region which does not contains the origin. For $\leq$ or $\geq$ signs in the constraint, the feasible region includes the points on the constraint line also.

- (b) Consider the constraint $x_2 - mx_1 \leq \text{or } \geq 0, m > 0$.

The line $y - mx = 0$ (or $x_2 - mx_1 = 0$) divide the xy plane (or $x_1 x_2$ plane) in two regions, one contains the positive x -axis (or x_1 axis). For the feasible constraint $y - mx < 0$ (or $x_2 - mx_1 < 0$), the feasible region is the region which contains the positive x -axis (or x_1 - axis) and for the constraint $y - mx > 0$ (or $x_2 - mx_1 > 0$), the feasible region is the region which contains the positive y -axis (or x_2 - axis) see the following figure.

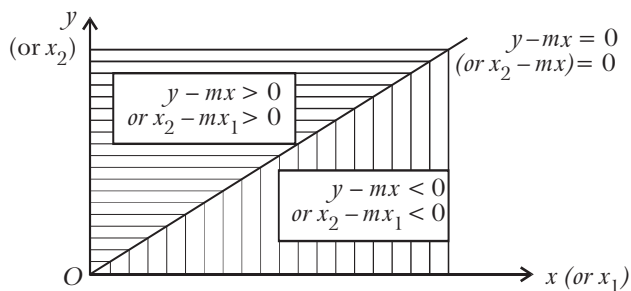


Fig. 1.1

Thus find feasible regions corresponding to all inequalities. Then the region which is common to all these regions is the convex polygon in the first quadrant and is the **permissible region (*i.e.*, feasible region)** for the values of the variables. This permissible region is shaded.

Step 4: Locate the extreme points (corner points) of the feasible region.

Step 5: Calculate the values of the objective function Z at all the extreme points.

Step 6: The optimum solution of the problem occurs at the extreme point giving the largest value of the objective function Z for maximization problem and at the extreme point giving the smallest value of the objective function Z for minimization problem.

The procedure of the method may be understood by the following examples.

ILLUSTRATIVE EXAMPLES

Example 1: Solve by graphical method the L.P.P.

Maximize, $Z = 100x_1 + 300x_2$

Subject to the constraints

$$x_1 + 2x_2 \leq 32$$

$$x_1 + x_2 \leq 24$$

and non-negative restrictions $x_1 \geq 0, x_2 \geq 0$

Solution: Step 1: Considering the constraints as equations, we get the following equations

$$x_1 + 2x_2 = 32 \quad \dots(1)$$

$$x_1 + x_2 = 24 \quad \dots(2)$$

Step 2: To draw the lines corresponding to equations (1) & (2), In equation (1), taking $x_2 = 0$, we get $x_1 = 32$, and taking $x_1 = 0$, we get $x_2 = 16$ i.e., line (1) intersect axes at the points (32,0) and (0, 16).

In equation (2), taking $x_2 = 0$ we get $x_1 = 24$ and taking $x_1 = 0$, we get $x_2 = 24$

i.e., line (2) intersect the axes at the points (24,0) and (0,24)

Plotting the points of intersection of lines to the axes, we join them to draw lines. (see fig. 1.2)

Step 3: Since (0,0) satisfy both given inequalities, so the regions of the given constraints are the regions containing the origin, as shown in the figure by arrows ($\rightarrow$).

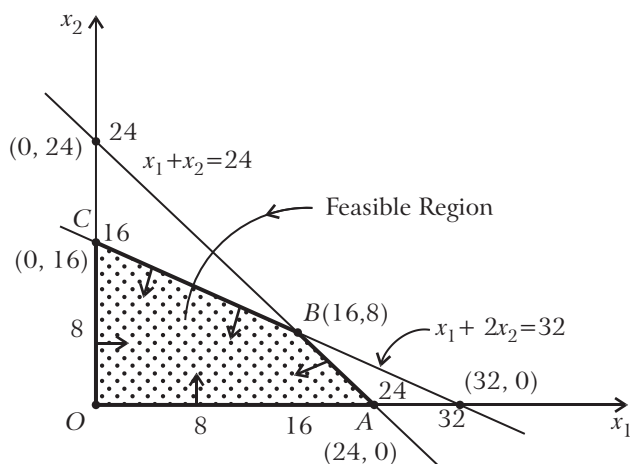


Fig. 1.2

Thus the feasible region for the variables x_1 and x_2 is the set of points enclosed in the polygon $OABCO$ (shaded) including the points on its edges.

Step 4: The extreme points (corner points) of the feasible region are $O(0, 0)$, $A(24, 0)$, $B(16, 8)$, $C(0, 16)$.

Step 5: The values of the objective function Z at the corner points are given in the following table.

Point (x_1, x_2)	Value of Obj. fun. $Z = 100x_1 + 300x_2$
$O(0, 0)$	$Z = 0 + 0 = 0$
$A(24, 0)$	$Z = 100 \times 24 + 300 \times 0 = 2400$
$B(16, 8)$	$Z = 100 \times 16 + 300 \times 8 = 4000$
$C(0, 16)$	$Z = 100 \times 0 + 300 \times 16 = 4800 \text{ (Max)}$

Obviously Z is maximum at the point $C(0, 16)$.

Hence the optimum solution is $x_1 = 0$, $x_2 = 16$ and optimum (maximum) value of $Z = 4800$.

Example 2: Solve the following L.P.P. by Graphical method.

Maximize $Z = 48x_1 + 40x_2$

Subject to the constraints

$$2x_1 + x_2 \leq 90$$

$$x_1 + 2x_2 \leq 80$$

$$x_1 + x_2 \leq 50$$

and $x_1 \geq 0, x_2 \geq 0$.

Solution: **Step 1:** Considering the constraints as equations, we get the following equations.

$$2x_1 + x_2 = 90 \quad \dots(1)$$

$$x_1 + 2x_2 = 80 \quad \dots(2)$$

$$x_1 + x_2 = 50 \quad \dots(3)$$

Step 2: In (1), $x_2 = 0$, gives $x_1 = 45$ and $x_1 = 0$ gives $x_2 = 90$

$\therefore$ line (1) intersect axes at the points $(45, 0)$ and $(0, 90)$.

In (2) $x_2 = 0$ gives $x_1 = 80$ and $x_1 = 0$ gives $x_2 = 40$

$\therefore$ lines (2) intersect axes at the points (80, 0) and (0, 40).

In (3), $x_2 = 0$, gives $x_1 = 50$ and $x_1 = 0$ gives $x_2 = 50$.

$\therefore$ line (3) intersect axes at the points (50,0) and (0,50).

Now we draw lines as shown in the figure.

Step 3: Since (0, 0) satisfy all the given constraints. So the feasible regions of all the three constraints contain the origin (0, 0).

Thus the feasible region for the variables x_1 and x_2 is the set of points on and within the polygon $OABCD$. (Shaded)

Step 4: The extreme points (corner points) of the feasible region are $O(0, 0)$, $A(45, 0)$, $B(40, 10)$, $C(20, 30)$, $D(0, 40)$.

Step 5: The values of the objective function Z at these corner points are given in the following table.

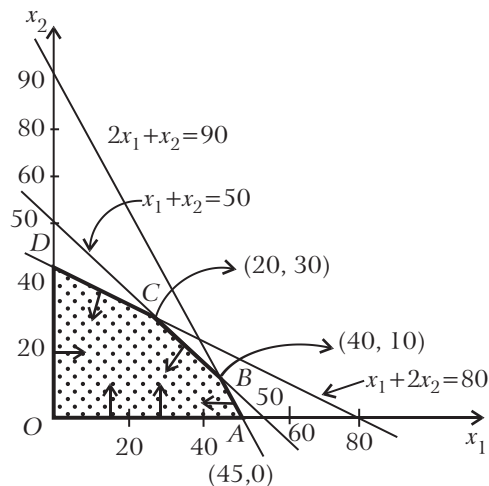


Fig. 1.3

Point (x_1, x_2)	Value of the Obj. fun. $Z = 48x_1 + 40x_2$
$O(0,0)$	$Z = 0 + 0 = 0$
$A(45,0)$	$Z = 48 \times 45 + 40 \times 0 = 2160$
$B(40,10)$	$Z = 48 \times 40 + 40 \times 10 = 2320$ (Max)
$C(20,30)$	$Z = 48 \times 20 + 40 \times 30 = 2160$
$D(0,40)$	$Z = 48 \times 0 + 40 \times 40 = 1600$

Obviously Z is maximum at the point $B(40,10)$

Hence the optimal solution is

$x_1 = 40$, $x_2 = 10$ and maximum $Z = 2320$

Example 3: Solve by graphical method, the L.P.P.

Minimize, $Z = 20x_1 + 10x_2$

Subject to the constraints

$$x_1 + 2x_2 \leq 40$$

$$3x_1 + x_2 \geq 30$$

$$4x_1 + 3x_2 \geq 60$$

and the non-negative restrictions $x_1, x_2 \geq 0$.

Solution: **Step 1:** Considering the constraints as equations, we get the following equations.

$$x_1 + 2x_2 = 40 \quad \dots(1)$$

$$3x_1 + x_2 = 30 \quad \dots(2)$$

$$4x_1 + 3x_2 = 60 \quad \dots(3)$$

Step 2: In (1), $x_2 = 0$, gives $x_1 = 40$ and $x_1 = 0$ gives $x_2 = 20$

$\therefore$ line (1) intersect axes at the points (40, 0) and (0, 20).

In (2), $x_2 = 0$ gives $x_1 = 10$ and $x_1 = 0$ gives $x_2 = 30$

$\therefore$ line (2) intersect axes at the points (10, 0) and (0, 30)

In (3), $x_2 = 0$ gives $x_1 = 15$ and $x_1 = 0$ gives $x_2 = 20$.

$\therefore$ line (3) intersect axes at the points (15, 0) and (0, 20) Now we draw lines as shown in the figure.

Step 3: Since (0,0) satisfy first constraint, so the feasible region of this constraint contain the origin.

Since (0, 0) does not satisfy 2nd and 3rd constraints, so their feasible regions do not contain the origin.

Thus the feasible region for the variables x_1 and x_2 is the set of points on and within the polygon $ABCD$. (Shaded region).

Step 4: The extreme points (corner points) of the feasible region are $A(15,0), B(40,0), C(4,18), D(6,12)$.

Step 5: The values of the objective function Z at these corner points are given in the following table.

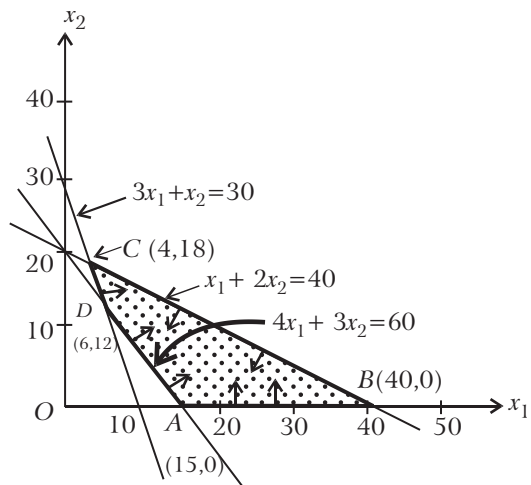


Fig. 1.4

Point (x_1, x_2)	Value of the Obj. fun. $Z = 20x_1 + 10x_2$
A (15,0)	$Z = 20 \times 15 + 10 \times 0 = 300$
B (40,0)	$Z = 20 \times 40 + 10 \times 0 = 800$
C (4,18)	$Z = 20 \times 4 + 10 \times 18 = 260$
D (6,12)	$Z = 20 \times 6 + 10 \times 12 = 240 \text{ (Min)}$

Obviously Z is minimum at the point $D(6,12)$. Hence the optimum solution is $x_1 = 6, x_2 = 12$ and minimum $Z = 240$.

Example 4: Old hens can be bought at ₹ 2 each and young ones at ₹ 5 each. The old hens lay 3 eggs per week and the young ones lay 5 eggs per week, each egg being worth 30 paise. A hen (young or old) costs ₹ 1 per week to feed. I have only ₹ 80 to spend for hens. How many of each kind should I buy to give me a maximum profit, assuming that I cannot house more than 20 hens?

Solution: Formulation of the problem as a L.P.P.

Let me buy x_1 old hens and x_2 young hens.

Since old hens lay 3 eggs per week and young ones lay 5 eggs per week, so total number of eggs. I have per week is $3x_1 + 5x_2$.

Since each egg work 30 paise,

$$\therefore \text{Total income per week} = ₹ 0.3 (3x_1 + 5x_2)$$

And total expenses for feeding $(x_1 + x_2)$ hens, at the rats of ₹ 1 per hen per week = ₹ $(x_1 + x_2)$.

$\therefore$ If Z is the profit per week in ₹ then

$$\begin{aligned} Z &= \text{total income} - \text{Total expenses} \\ &= 0.3 (3x_1 + 5x_2) - (x_1 + x_2) = -0.1x_1 + 0.5x_2 \end{aligned}$$

Now total cost for x_1 old hens and x_2 young hens is ₹ $(2x_1 + 5x_2)$

Which cannot be more than ₹ 80 which I have to spend

$$\therefore 2x_1 + 5x_2 \leq 80$$

Since I cannot house more than 20 hens, so total number of hens = $x_1 + x_2 \leq 20$.

Also the number of hens purchased (whether old or young) cannot be negative

$$\therefore x_1 \geq 0 \text{ and } x_2 \geq 0$$

Hence the problem formulated as L.P.P. is as follows:

$$\text{Maximize } Z = -0.1x_1 + 0.5x_2$$

Subject to the constraints

$$2x_1 + 5x_2 \leq 80, \quad x_1 + x_2 \leq 20$$

and non-negative restrictions $x_1 \geq 0, \quad x_2 \geq 0$.

Solution of the L.P.P.

Step 1: Considering the constraints as equations, we get the following equations

$$2x_1 + 5x_2 = 80 \quad \dots(1)$$

$$x_1 + x_2 = 20 \quad \dots(2)$$

Step 2: In (1), $x_2 = 0$ gives $x_1 = 40$ and $x_1 = 0$ gives $x_2 = 16$.

$\therefore$ line (1), intersect the axes at the points (40,0) and (0,16).

In (2), $x_2 = 0$ gives $x_1 = 20$ and $x_1 = 0$ gives $x_2 = 20$.

$\therefore$ line (2), intersect the axes at the points (20,0) and (0, 20)

Now we draw the lines as shown in the figure.

Step 3: The feasible region is the set of points on and within the polygonal $OABCO$. (Shaded region).

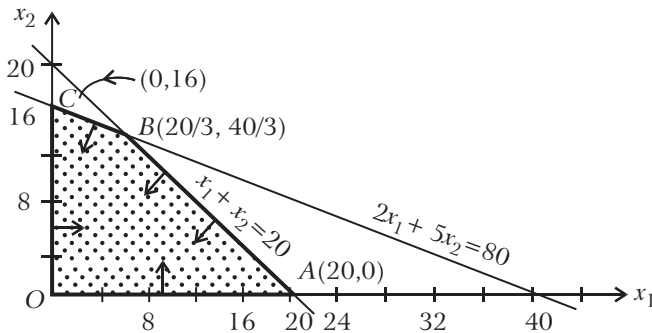


Fig. 1.5

Step 4: The extreme points (corner points) of the feasible region are $O(0,0)$, $A(20,0)$, $B(20/3, 40/3)$, $C(0,16)$.

Step 5: The values of the objective function Z at these corner points are given in the following table.

Point (x_1, x_2)	Value of the Obj. fun. $Z = -0.1x_1 + 0.5x_2$
$O(0,0)$	$Z = 0+0 = 0$
$A(20,0)$	$Z = -0.1 \times 20 + 0.5 \times 0 = -2$
$B(20/3, 40/3)$	$Z = -0.1 \times 20/3 + 0.5 \times 40/3 = 6$
$C(0,16)$	$Z = -0.1 \times 0 + 0.5 \times 16 = 8 \text{ (Max)}$

Obviously Z is maximum at the point $C(0,16)$.

Hence the optimal solution is $x_1 = 0$, $x_2 = 16$ and $\text{Max. } Z = ₹ 8$ i.e., I should buy only 16 young hens, which will give me the maximum profit of ₹ 8 per week.

Example 5: *The post master of a local post office wishes to hire extra helpers during the deepawali season, because of a large increase in the volume of mail handing and delivery. Keeping in view the limited office space and the budgetary conditions, the number of temporary helpers must not exceed 10. According to past experience, a man can handle 300 letters and 80 packages per day, and a woman can handle 400 letters and 50 packages per day. It is believed that the daily volume of extra mail and packages will be no less than 3400 and 680 respectively. A man receives ₹ 25 a day and a women receives ₹ 22 a day. How many men and women helpers should be hired to keep the payroll at a minimum?*

Solution: Formulation of the problem as a L.P.P.

Let x_1 men and x_2 women helpers be hired to keep the payroll at a minimum.

If Z is the daily payroll in ₹ for the extra helpers, then

$$Z = 25x_1 + 22x_2$$

Since the total number of temporary helpers cannot exceed 10,

$$\therefore x_1 + x_2 \leq 10$$

Given that a men can handle 300 letters and women can handle 400 letters daily, so extra temporary helpers can handle only $300x_1 + 400x_2$ letters daily. Since the number of extra letters daily is not less than 3400,

$$\therefore 300x_1 + 400x_2 \geq 3400$$

Similarly the total number of packages that can be handled daily is

$$80x_1 + 50x_2 \geq 680$$

As the number of men and women helpers hired cannot be negative,

$$\therefore x_1 \geq 0, x_2 \geq 0$$

Hence the problem formulated as L.P.P. is as follows:

$$\text{Minimize } Z = 25x_1 + 22x_2$$

Subject to the constraints

$$x_1 + x_2 \leq 10, 300x_1 + 400x_2 \geq 3400, 80x_1 + 50x_2 \geq 680$$

and non-negative restrictions $x_1 \geq 0, x_2 \geq 0$.

Solution of the L.P.P.

Step 1: Considering the constraints as equations, we get the following equations.

$$x_1 + x_2 = 10 \quad \dots(1)$$

$$300x_1 + 400x_2 = 3400 \quad \dots(2)$$

$$80x_1 + 50x_2 = 680 \quad \dots(3)$$

Step 2: In (1), $x_2 = 0$ gives $x_1 = 10$ and $x_1 = 0$ gives $x_2 = 10$.

$\therefore$ line (1) intersect the axes at the points (10,0) and (0,10).

In (2), $x_2 = 0$, gives $x_1 = 34/3$ and $x_1 = 0$, gives $x_2 = 17/2$.

$\therefore$ line (2) intersect the axes at the points (34/3,0) and (0,17/2).

In (3), $x_2 = 0$, gives $x_1 = 17/2$ and $x_1 = 0$, gives $x_2 = 68/5$.

∴ line (3) intersect the axes at the points $(17/2, 0)$ and $(0, 68/5)$.

Now we draw the lines as shown in the figure.

Step 3: Since $(0,0)$ satisfy first constraint so the feasible region of constraint (1) is the region from line (1) and containing the origin.

Since $(0,0)$ does not satisfy 2nd and 3rd constraints. So the feasible region of 2nd and 3rd constraints are the regions from the 2nd and 3rd line which does not contain origin.

Obviously the permissible region (region common to all above three regions) consists of a point $P(6,4)$ only.

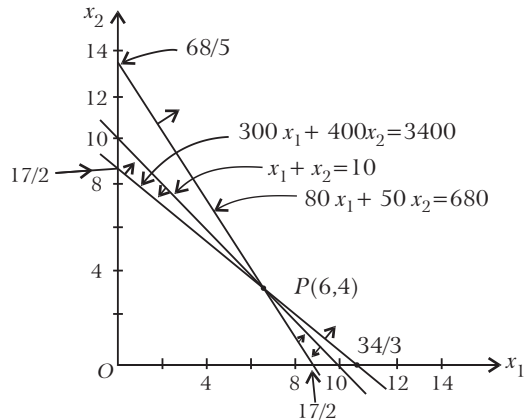


Fig. 1.6

Hence the optimal solution is $x_1 = 6$, $x_2 = 4$ and $\min. Z = ₹(25 \times 6 + 22 \times 4) = ₹ 238$.

i.e., the postmaster should hire 6 men and 4 women helpers to meet his seasonal requirements and to keep the payroll at a minimum of ₹ 238.

Example 6: Solve by graphical method the L.P.P.

Maximize $Z = 6x_1 - 2x_2$

Subject to the constraints

$$2x_1 - x_2 \leq 2$$

$$x_1 \leq 3$$

and the non-negative restriction $x_1 \geq 0$, $x_2 \geq 0$.

Solution: Step 1: Considering the constraints as equations, we get the following equations

$$2x_1 - x_2 = 2 \quad \dots(1)$$

$$x_1 = 3 \quad \dots(2)$$

Step 2: In (1), $x_2 = 0$ gives $x_1 = 1$ and $x_1 = 0$ gives $x_2 = -2$.

∴ Line (1) intersect the axes at the points $(1,0)$ and $(0, -2)$

Line (2) passes through the point $(3,0)$ and is parallel to Ox_2 axis.

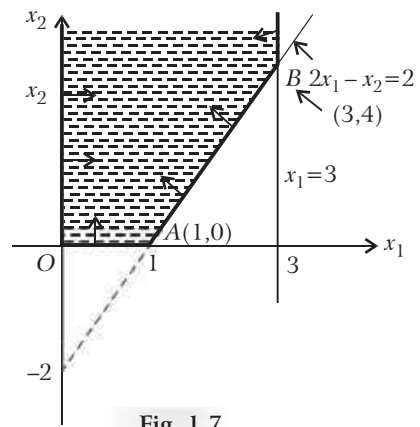


Fig. 1.7

Step 3: Since $(0,0)$ satisfy both constraints, so the feasible regions of both the constraints are the region from the corresponding line and containing the origin.

Thus the feasible region for the variables x_1 and x_2 is the set of points in the shaded region shown in the figure, which is unbounded on the upper side.

Step 4: The extreme points of the region are $A(1,0), B(3,4)$.

Step 5: At $A(1,0)$, $Z = 6x_1 - 2x_2 = 6 \times 1 - 0 = 6$

and at $B(3,4)$, $Z = 6x_1 - 2x_2 = 6 \times 3 - 2 \times 4 = 10$

On the line $Ox_2, x_1 = 0$ and $x_2 \rightarrow \infty \therefore Z \rightarrow -\infty$

and on the line $x_1 = 3, x_2 \rightarrow \infty \therefore Z \rightarrow -\infty$

Obviously Z is maximum at $B(3,4)$.

Hence the optimal solution is $x_1 = 3, x_2 = 4$ and $\max Z = 10$

Note: Here it is to be noted that a L.P.P. problem can have an optimal solution while the feasible region for the values of the variables may be unbounded.

Some Exceptional Cases

1. L.P.P. having Multiple Solutions

Example 7: Solve the following L.P.P. by Graphical method.

Maximize $Z = x_1 + x_2$

Subject to the constraints

$$x_1 + 2x_2 \leq 2000$$

$$x_1 + x_2 \leq 1500$$

$$x_2 \leq 600$$

and $x_1, x_2 \geq 0$.

Solution: **Step 1:** Considering the constraints as equations, we get the following equations.

$$x_1 + 2x_2 = 2000 \quad \dots(1)$$

$$x_1 + x_2 = 1500 \quad \dots(2)$$

$$x_2 = 600 \quad \dots(3)$$

Step 2: In (1), $x_2 = 0$ gives $x_1 = 2000$ and $x_1 = 0$ gives $x_2 = 1000$.

$\therefore$ lines (1) intersect the axes at the points (2000, 0) and (0, 1000).

In (2), $x_2 = 0$ gives $x_1 = 1500$ and $x_1 = 0$ gives $x_2 = 1500$

$\therefore$ line (2) intersect the axes at the points (1500,0) and (0,1500)

line (3) passes through the point (0,600) and is parallel to Ox_1 axis.

Now we draw the lines in the plane as shown in the figure.

Step 3: Since $(0, 0)$ satisfy all the constraints, so the feasible region of all constraints contain origin.

Thus the feasible region for the variables x_1 and x_2 is the set of points on and within the polygon $OABCD$ (shaded).

Step 4: The extreme points of the feasible region are $O(0,0)$, $A(1500,0)$, $B(1000,500)$, $C(800,600)$ and $D(0,600)$.

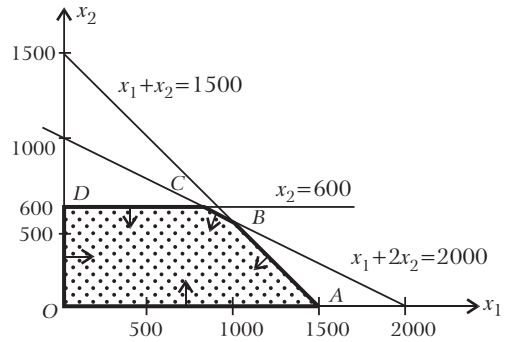


Fig. 1.8

Step 5: The values of the objective function Z at the extreme points are given in the following table.

Point (x_1, x_2)	Value of Obj. Fun. $Z = x_1 + x_2$
$O(0,0)$	$Z = 0 + 0 = 0$
$A(1500,0)$	$Z = 1500 + 0 = 1500$ (Max.)
$B(1000,500)$	$Z = 1000 + 500 = 1500$ (Max)
$C(800,600)$	$Z = 800 + 600 = 1400$
$D(0,600)$	$Z = 0 + 600 = 600$

Obviously Z is maximum at $A(1500,0)$ and $B(1000,500)$. Therefore each point of the line segment AB gives the optimal solution of the L.P.P.

Hence the problem have infinite number of solutions. Two of these solutions can be given as

$$x_1 = 1500, x_2 = 0 \text{ and } x_1 = 1000, x_2 = 500$$

and in both cases $\text{Max. } Z = 1500$.

2. L.P.P. having no Feasible Solution

Example 8: Solve by graphical method the L.P.P.

Maximize $Z = 3x_1 + x_2$

S.t. $x_1 - x_2 \leq -1$

$-x_1 + x_2 \leq 0$

and $x_1, x_2 \geq 0$.

Solution: Step 1: Considering the constraints as equation, we get the following equations

$$x_1 - x_2 = -1 \quad \dots(1)$$

$$-x_1 + x_2 = 0 \quad \dots(2)$$

Step 2: In (1), $x_2 = 0$ gives $x_1 = -1$ and $x_1 = 0$ gives $x_2 = 1$

$\therefore$ line (1) intersect the axes at the points $(-1, 0)$ and $(0, 1)$.

line (2) $x_2 - x_1 = 0$ passes through the origin and bisect the angle between the axes in first quadrant.

Now we draw the lines in the plane.

Step 3: The feasible regions of the two constraints are shown in the figure by shaded regions. Obviously there is no point common in the two regions, i.e., there is no point (x_1, x_2) which simultaneously satisfy both the constraints.

Hence the problem has no feasible solution.

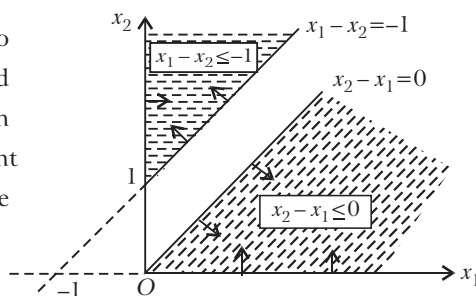


Fig. 1.9

3. L.P.P. having Unbounded Solution

Example 9: Solve the following L.P.P. by Graphical method.

Maximize $Z = 0.75x_1 + x_2$

s.t. $x_1 - x_2 \geq 0$

$$-0.5x_1 + x_2 \leq 1$$

and $x_1, x_2 \geq 0.$

Solution: **Step 1:** Considering the constraints as equations, we get the following equations.

$$x_1 - x_2 = 0 \quad \dots(1)$$

$$-0.5x_1 + x_2 = 1 \quad \dots(2)$$

Step 2: Equation (1) represents a line through the origin and bisection the angle between the co-ordinate axes in first quadrant.

In line (2), $x_2 = 0$ gives $x_1 = -2$ and $x_1 = 0$ gives $x_2 = 1$.

$\therefore$ line (2) intersect the axes at the points $(-2, 0)$ and $(0, 1)$

Step 3: Since $(0, 0)$ satisfy (2), so the feasible region of 2nd constraint contain origin.

The first constrain can be written as $x_2 - x_1 \leq 0$, so its feasible region is the region from the line and containing positive x_1 axis.

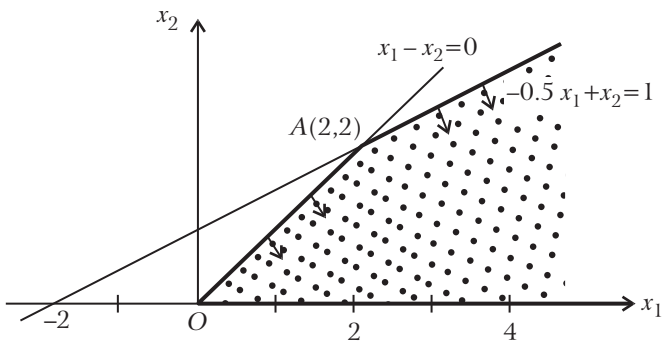


Fig. 1.10

Thus the feasible region for values of x_1 and x_2 is the shaded region shown in the figure which is unbounded on the right.

Since both x_1 and $x_2 \rightarrow \infty$ in this feasible region, so the value of Z can be made arbitrarily large by taking x_1 and x_2 sufficiently large.

Hence the solution of the problem is unbounded.

Exercise 1.2

Solve the Following L.P.P. by Graphical Method

1.	Maximize, $Z = 2x_1 + 10x_2$ subject to the constraints $2x_1 + 5x_2 \leq 16$ $6x_1 \leq 30$ and $x_1, x_2 \geq 0$	2.	Maximize, $Z = 6x_1 + 11x_2$ subject to the constraints $2x_1 + x_2 \leq 104$ $x_1 + 2x_2 \leq 76$ and $x_1, x_2 \geq 0$
3.	Maximize, $Z = 2x_1 + 3x_2$ subject to the constraints $x_1 + x_2 \leq 1$ $3x_1 + x_2 \leq 4$ and $x_1, x_2 \geq 0$	4.	Maximize, $Z = 5x_1 + 4x_2$ subject to the constraints $4x_1 + x_2 \leq 40$ $2x_1 + 3x_2 \leq 90$ and $x_1, x_2 \geq 0$

5.	Minimize, $Z = -x_1 + 2x_2$ subject to the constraints $-x_1 + 3x_2 \leq 10$ $x_1 + x_2 \leq 6$ $x_1 - x_2 \leq 2$ and $x_1, x_2 \geq 0$	6.	Minimize, $Z = 20x_1 + 40x_2$ subject to the constraints $36x_1 + 6x_2 \geq 108$ $3x_1 + 12x_2 \geq 36$ $20x_1 + 10x_2 \geq 100$ and $x_1, x_2 \geq 0$
7.	Minimize, $Z = 2x_1 + x_2$ subject to the constraints $5x_1 + 10x_2 \leq 50$ $x_1 + x_2 \geq 1$ $x_1 \leq 4$ and $x_1, x_2 \geq 0$	8.	Minimize, $Z = x_1 + x_2$ subject to the constraints $5x_1 + 10x_2 \leq 50$ $x_1 + x_2 \geq 1$ $x_2 \leq 4$ and $x_1, x_2 \geq 0$
9.	Maximize, $Z = x_1 + x_2$ subject to the constraints $x_1 - x_2 \geq 0$ $-3x_1 + x_2 \geq 3$ and $x_1, x_2 \geq 0$	10.	Minimize, $Z = 1.5x_1 + 2.5x_2$ subject to the constraints $x_1 + 3x_2 \geq 3$ $x_1 + x_2 \geq 2$ and $x_1, x_2 \geq 0$
11.	Maximize, $Z = 2x_1 + x_2$ subject to the constraints $x_1 - x_2 \leq 10$ $x_1 \leq 20$ and $x_1, x_2 \geq 0$	12.	Maximize, $Z = -3x_1 + 2x_2$ subject to the constraints $x_1 \leq 3$ $x_1 - x_2 \leq 0$ and $x_1, x_2 \geq 0$

13. An animal feed company must produced 200kg of mixture consisting of ingredients x_1 and x_2 , x_1 costs ₹ 3 per kg and x_2 costs ₹ 8 per kg. Not more than 80 kg of x_1 can be used and at least 60 kg of x_2 must be used. Formulate a L.P.P. and solve it graphically.
14. A dietician mixes two types of food in such a way that the vitamin contents of the mixture contain at least 8 units of vitamin A and 10 units of vitamin C. Food X contains 2 units/kg of vitamin A and 1 unit/kg of vitamin C while food Y contains 1 unit/kg of vitamin A and 2 units/kg of vitamin C. One kg of food X costs. ₹ 5 whereas one kg of food Y costs ₹ 7. Determine the minimum cost of such a mixture.
15. A company sells two different products A and B. The company makes a profit of ₹ 40 and ₹ 30 per unit on products A and B respectively. The products are produced in a common production process and are sold in two different markets. The

production process has a capability of 30,000 man-hours. It takes 3 hours to produce one unit of A and one hour to produce one unit of B. The market has been surveyed, and company officials feel that the maximum number of units of A that can be sold is 8,000 and the maximum of B is 12,000 units. Subject to these limitations, the products can be sold in any convex combination. Formulate the above problem as a L.P.P. and solve it by graphical method.

16. A publisher sells a hard cover edition of a text book for ₹ 72 and a paperback edition of the same text for ₹ 40. Costs to the publisher are ₹ 56 and ₹ 28 per book respectively in addition to weekly costs of ₹ 9600. Both types of books require 5 minutes of printing time, although hard cover requires 10 minutes binding time and the paperback requires only 2 minutes. Both the binding and printing operations have 4800 minutes available each week. How many of each type of books should be produced in order to maximize profit?
 17. An electric company manufactures two radio models, each on a separate production machine. The daily capacity of first machines is 60 radios and that of second is 75 radios. Each unit of first model uses 10 pieces of certain electronic component whereas each unit of second model uses 8 pieces of it. Maximum ready availability of the component is 800 pieces. Profit per unit of models is ₹ 30 and ₹ 20 respectively. Give a mathematical formulation of linear program plane to get maximum profit and solve by the graphical method.
 18. **Diet Problem:** Consider two different types of foodstuffs, say F_1 and F_2 . Assume that these foodstuffs contain vitamins V_1 , V_2 and V_3 respectively. Minimum daily requirement of three vitamins are 1 mg. of V_1 , 50 mg. of V_2 and 10 mg. of V_3 . Suppose that the foodstuff F_1 contains 1 mg. of V_1 , 100 mg of V_2 and 10 mg. of V_3 . Whereas the foodstuff F_2 contains 1 mg. of V_1 , 10 mg. of V_2 , 100 mg. of V_3 . Cost of one unit of foodstuff F_1 is ₹ 1 and that of F_2 is ₹ 1.5. Find the minimum cost diet that would supply the body at least the minimum requirements of each vitamin by graphical method.
 19. A company produces two type of dolls A and B. Doll A is a superior quality and doll B is a lower quality. Profit on doll A and B are ₹ 5 and ₹ 3 per piece respectively, raw material required for each doll A is twice as that for doll B. The supply of raw material is sufficient only for 1000 dolls of type B per day. Doll A requires a special clip and only 400 such clips are available per day. For doll B only 700 clips are available per day. Find graphically the product mix so that the company can make maximum profit.
 20. An automobile manufacturer makes automobiles and trucks in a factory that is divided into two shops. Shop A, which performs the basic assembly operation, must work 5 man-days on each truck but only 2 man-days on each automobile. Shop B, which performs finishing operation, must work 3 man-days for each automobile or truck that it produces. Because of men and machine limitations, shop A has 180 man-days per week available while shop B has 135 man-days per week. If the manufacturer makes a profit. of ₹ 300 on each truck and ₹ 200 on each automobile, how many of each should he produce to maximize his profit?
-

Answers 1.2

1.	$x_1 = 0, x_2 = 3.2$, Max. $Z = 32$	2.	$x_1 = 44, x_2 = 16$, Max. $Z = 440$
3.	$x_1 = 0, x_2 = 1$, Max. $Z = 3$	4.	$x_1 = 3, x_2 = 28$, Max. $Z = 127$
5.	$x_1 = 2, x_2 = 0$, Min. $Z = -2$	6.	$x_1 = 4, x_2 = 2$, Min. $Z = 160$
7.	$x_1 = 0, x_2 = 1$, Min. $Z = 1$	8.	Infinite, $x_1 = 0, x_2 = 1; x_1 = 1, x_2 = 0$ All points on line segment joining points (0,1) and (1,0), Min. $Z = 1$.
9.	No feasible solution	10.	$x_1 = 3/2, x_2 = 1/2$, Min. $Z = 7/2$
11.	Unbounded	12.	Unbounded
13.	L.P.P. Min. $Z = 3x_1 + 8x_2$, s.t. $x_1 + x_2 = 200, x_1 \leq 80, x_2 \geq 60$ and $x_1, x_2 \geq 0$. Solution: $x_1 = 80, x_2 = 120$, Min. $Z = ₹ 1200$.		
14.	L.P.P. Min. $Z = 5x_1 + 7x_2$, s.t. $2x_1 + x_2 \geq 8, x_1 + 2x_2 \geq 10$ and $x_1, x_2 \geq 0$. where x_1 kg of food X and x_2 kg of food Y are mixed. Solution: $x_1 = 2$ kg, $x_2 = 4$ kg, min $Z = ₹ 38$.		
15.	L.P.P. Max. $Z = 40x_1 + 30x_2$ s.t. $3x_1 + x_2 \leq 30,000, x_1 \leq 8000, x_2 \leq 12000$ and $x_1, x_2 \geq 0$ where company sells x_1 and x_2 units of products A and B resp. Solution: $x_1 = 6,000, x_2 = 12,000$, max. $Z = 600,000$.		
16.	Hard cover edition = 360, paper back edition = 600, max. profit = ₹ 3360.		
17.	L.P.P. Max. $Z = 30x_1 + 20x_2$, s.t. $x_1 \leq 60, x_2 \leq 75, 10x_1 + 8x_2 \leq 800$ and $x_1, x_2 \geq 0$ Where x_1 and x_2 are the number of two radio models manufactured. Solution: $x_1 = 60, x_2 = 25$, max $Z = ₹ 2850$.		
18.	$F_1 = 1$ unit, $F_2 = 0$ unit, min cost = ₹ 1.		
19.	Doll A = 150, Doll B = 700, Max. Profit = ₹ 2850.		
20.	Trucks = 30, Automobiles = 15 per week, max. profit = ₹ 12000.		

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1. Which of the following is related to L.P.P.?
(a) Objective function (b) Constraints
(c) Requirement vector (d) All of these.
 2. Each constraint in a L.P.P. is expressed as an:
(a) Equation with $=$ sign (b) Inequality with $\geq$ sign
(c) Inequality with $\leq$ sign (d) None of these.
 3. A constraint that does not affect the feasible region is a:
(a) Non-negative constraint (b) Redundant constraint
(c) Standard constraint (d) Slack constraint.
 4. Infeasibility means that the number of solutions of the L.P. model that satisfies all the constraints is:
(a) At least 1 (b) Zero
(c) Infinite (d) At least 2.
 5. If in the graphical solution of a L.P.P., two constraints do not intersect in the positive quadrant, then:
(a) The problem has no solution
(b) One of the constraints is redundant
(c) Problem has unbounded solution
(d) Both constraints are redundant.
 6. In the graphical method of a L.P.P. the following is/are used:
(a) Linear equations (b) Objective function equation
(c) Constraint equations (d) None of these.
 7. In the degenerate basic solution of a system of equations:
(a) No basic variable is zero
(b) At least one basic variable is zero
(c) At least one basic variable is negative
(d) No basic variable is negative.
 8. The variables associated to B.F.S. are called:
(a) Basic variables (b) Non-basic variables
(c) Slack variables (d) Surplus variables.
-

9. If there is no feasible solution to a L.P.P. then the problem is said to have:
- (a) Unbounded solution
 - (b) Infinite number of solutions
 - (c) No solution
 - (d) An unique solution.
10. Which of the following statements is not true for a L.P.P. ?
- (a) A feasible solution satisfies all constraints.
 - (b) An optimal solution satisfies all constraints.
 - (c) An infeasible solution violates all constraints.
 - (d) A feasible solution point does not have to lie on the boundary of the feasible region.
11. All L.P.P. have all of the following properties, except:
- (a) A linear objective function that is to be maximized or minimized.
 - (b) A set of linear equations.
 - (c) Alternative optimal solution.
 - (d) Variables that are all restricted to non-negative values.

Fill in the Blank

1. In a L.P.P. the linear function which is to be optimized is called the
 2. The conditions in a L.P.P. expressed as simultaneous linear equations or inequalities are known as the
 3. In a L.P.P. if $Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$ is the function to be optimized, then $c = (c_1, c_2, \dots, c_n)$ is known as the
 4. The positive variables which are added to the L.H.S. of the constraints to convert them into equalities are called the variables.
 5. The positive variables which are subtracted from the L.H.S. of the constraints to convert them into equalities are called the variables.
 6. The set of values of the variables which satisfies the set of constraints and the non-negative restrictions in a L.P.P. is called its
 7. A feasible solution to a L.P.P. which also optimizes the objective function of the problem is called its solution.
-

8. For the system $Ax = b$ of m equations in n unknowns ($n \geq m$), the maximum number of basic solutions is
9. A B.F.S. of a L.P.P. is said to be of none of the basic variables is zero.
10. In the standard form of a L.P.P. the right hand side of each constraint is

True/False

1. If the value of the objective function can be increased or decreased indefinitely, the solution is called an unbounded solution.
 2. In a basic feasible solution of a L.P.P. the values of all non-basic variables is zero.
 3. A solution of a system $Ax = b$ of m equations in n unknowns ($n > m$), obtained by setting $n - m$ variables to zero is called a basic solution, provided the determinant of the coefficients of the remaining m variables is zero.
 4. A B.F.S. of a L.P.P. is said to be degenerate B.F.S. if at least one of the basic variables is zero.
 5. A feasible solution to a L.P.P. is said to be basic feasible solution if the vectors associated to the feasible solution is L.I.
 6. A L.P.P. of two variables can be solved graphically.
 7. The optimal solutions of all L.P.P. are integral valued.
 8. In a L.P.P. the objective function and all constraints are linear.
-

Answers

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1.	(d)	2.	(d)	3.	(b)
4.	(b)	5.	(b)	6.	(c)
7.	(b)	8.	(a)	9.	(c)
10.	(c)	11.	(c)		

Fill in the Blank

1.	objective function	2.	constraints	3.	price vector
4.	slack	5.	surplus	6.	feasible solution
7.	optimal	8.	nC_m	9.	non-degenerate
10.	non-negative				

True/False

1.	T	2.	T	3.	F
4.	T	5.	T	6.	T
7.	F	8.	T		

UNIT-I

Chapter

2



Simplex Method

2.1 Introduction

In the previous chapter, we have discussed the formulation of a linear programming problems and the graphical method of solving L.P.P. involving two variables only. The graphical method suffers from the great limitation that it can handle problems involving only two decision variables whereas in real world situations, we frequently encounter cases where more than two variables are involved. To overcome this difficulty a mathematical iterative procedure known as “**Simplex Method**” was developed in 1947 by **George B. Dantzig** which can be used to solve a L.P.P. with any number of decision variables (unknowns) and any number of constraints.

The simplex method is an iterative procedure, for finding, in a systematic manner, the optimal solution to a linear programming problem. This method is based on *fundamental theorem of linear programming* which states that “*whenever there exists an optimal solution of a L.P.P. then at least one basic feasible solution must be optimal*”.

The simplex method consists of the following steps:

1. To find a trial basic feasible solution of the given problem.
2. To test whether this B.F.S. is optimal or not.
3. To improve the first trial B.F.S. (if it is not optimal) by a set of rules.
4. Repeat the above procedure till an optimal solution is attained.

In this chapter we shall concentrate on maximization problems only because the minimization problem can easily be converted to maximization problem.

2.2 Slack and Surplus Variables

1. **Slack Variables:** If a constraint has a sign $\leq$; then in order to make it an equality, we have to add something positive on its left hand side.

The non-negative variables which are added to the L.H. sides of the constraints with ' $\leq$ ' sign, to convert them into equalities are called “Slack variables”.

For example, consider the constraints

$$x_1 + 2x_2 + x_3 \leq 4 \quad \text{and} \quad 2x_1 + 5x_2 - x_3 \leq 5$$

Then let s_1, s_2 be two non-negative variables to be added on the L.H. sides of these constraints, to convert them into equalities *i.e.*, we have

$$x_1 + 2x_2 + x_3 + s_1 = 4 \quad \text{and} \quad 2x_1 + 5x_2 - x_3 + s_2 = 5$$

These non-negative variables s_1 and s_2 are called **Slack variables**.

2. **Surplus Variables:** If a constraint has a sign $\geq$, then in order to make it an equality, we have to subtract something positive from its left hand side.

The non-negative variables which are subtracted from the L.H. sides of the constraints with ' $\geq$ ' sign, to convert them into equalities, are called “Surplus variables”.

For example, consider the constraints

$$x_1 - 2x_2 + 3x_3 \geq 6 \quad \text{and} \quad 2x_1 + 3x_2 - 7x_3 \geq 9$$

Then let s_1, s_2 be two non-negative variables to be subtracted from the L.H. sides of the constraints to convert them into equalities *i.e.*, we have

$$x_1 - 2x_2 + 3x_3 - s_1 = 6 \quad \text{and} \quad 2x_1 + 3x_2 - 7x_3 - s_2 = 9$$

These non-negative variables s_1 and s_2 are called **Surplus variables**.

2.3 Some Important Definitions

1. **Basic Solution (B.S.):** Consider a system of m equations in n unknown ($n > m$) such that, rank of matrix $A = m$, where A is the matrix of the coefficients of the unknowns. A solution of this system by setting any $(n - m)$ variables equal to zero is called a **Basic solution**, provided the determinant of the coefficients of the remaining ' m ' variables is not zero. Such ' m ' variables (any of them may have zero value) are called the **Basic variables** are remaining $(n - m)$ zero valued variables are called **Non-basic variables**.

Thus for a solution for the above system of equations to be basic, at least $(n - m)$ variables must be zero.

Note: Number of basic solutions of the above system of equations can be at the most nC_m .

2. **Degenerate Basic Solution:** A basic solution is said to be *degenerate basic solution* if at least one of the basic variables is zero.
3. **Non-degenerate Basic Solution:** A basic solution is said to be *non-degenerate basic solution* if none of the basic variables is zero.
4. **A Basic Feasible Solution (B.F.S.):** In a L.P.P. a feasible solution which is also basic is called a *basic feasible solution*.

In other words a basic solution to a L.P.P. is called basic feasible solution if it also satisfy the non-negative restrictions of the variables.

Hence the B.F.S. of a L.P.P. with m constraints cannot have more than m non-zero (positive) variables.

5. **Degenerate B.F.S.:** A B.F.S. of a L.P.P. is said to be degenerate B.F.S. if at least one of the basic variable is zero.
6. **Non-degenerate B.F.S.:** A B.F.S. of a L.P.P. is said to be non-degenerate B.F.S. if none of the basic variables is zero.

2.4 Computational Procedure of Simplex Method (For Maximization Problem)

The various steps involved to solve the maximization linear programming problem by simplex method are as follows:

Step 1: *If the given problem is of minimization, convert it into the maximization problem.*

In a L.P.P. if the objective function is $\text{Min. } Z = x_1 - 3x_2$ then take another function $Z' = -Z = -x_1 + 3x_2$.

Obviously, Maximum $Z' = -\text{Minimum } Z$.

Step 2: *To make all the bi's non-negative.*

The R.H. sides of each constraint should be non-negative. If R.H.S. of some constraint in a L.P.P. is negative, then multiply both sides of such constraint (or constraints) by -1 , to convert the R.H.S. to positive value. Note that by doing so, the sign of inequality will be reversed.

For example, if $2x_1 - 5x_2 \geq -7$ is a constraint in a L.P.P. then multiplying both sides by -1 , to reduce the R.H.S. to positive, this constraint will change to $-2x_1 + 5x_2 \leq 7$.

Step 3: *To convert the inequalities into equations.*

Convert all the inequalities into equations by the introduction of slack or/and surplus variables in the constraints as follows:

Constraints	Method	Equation
$a_1x + b_1y \leq b_1$	Add a slack variable s_1 on L.H.S.	$a_1x + b_1y + s_1 = b_1$
$a_2x + b_2y \geq b_2$	Subtract a surplus variable s_2 from L.H.S.	$a_2x + b_2y - s_2 = b_2$

Step 4: To find Initial basic feasible solution (B.F.S.).

Initial B.F.S. can be obtained as follows:

- (i) If only slack variables say $s_1, s_2, \dots, s_m$ are introduced in the m constraints in n variables $x_1, x_2, \dots, x_n$, then putting $x_1 = 0 = x_2 = \dots = x_n$, we get $s_1 = b_1, s_2 = b_2, \dots, s_m = b_m$. ($b_1, b_2, \dots, b_m$ are R.H.S. of equations obtained in step 3) which constitute the initial B.F.S. of the problem.
- (ii) If surplus and artificial variables are also used in some constraints, then to obtain initial B.F.S. put the given decision variables and the surplus variables equal to zero.

Step 5: To construct the initial (starting) simplex table.

The initial simplex table is constructed as follows:

Table 2.1

B	$\mathbf{c}_B$	$c_j \rightarrow$	c_1	c_2	$\dots$	c_k	$\dots$	c_n	0	$\dots$	0	Min. Ratio $\mathbf{x}_B / Y_k,$ $y_{ik} > 0$	
		$\mathbf{x}_B$	Y_1	Y_2	$\dots$	Y_k	$\dots$	Y_n	S_1	S_2	$\dots$		S_m
S_1	$c_{B_1} = 0$	$x_{B_1} = b_1$	a_{11}	a_{12}	$\dots$	a_{1k}	$\dots$	a_{1n}	1	0	$\dots$	0	$\mathbf{x}_{B_r} / a_{rk}$ (Min.) $\longrightarrow$
S_2	$c_{B_2} = 0$	$x_{B_2} = b_2$	a_{21}	a_{22}	$\dots$	a_{2k}	$\dots$	a_{2n}	0	1	$\dots$	0	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$		
S_r	$c_{B_r} = 0$	$x_{B_r} = b_r$	a_{r1}	a_{r2}	$\dots$	a_{rk}	$\dots$	a_{rn}	0	0	$\dots$	0	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$		
S_m	$c_{B_m} = 0$	$x_{B_m} = b_m$	a_{m1}	a_{m2}	$\dots$	a_{mk}	$\dots$	a_{mn}	0	0	$\dots$	1	
$Z = \mathbf{c}_B \mathbf{x}_B$	Δ_j	Δ_1	Δ_2	$\dots$	Δ_k $\uparrow$	$\dots$	Δ_n	Δ_{n+1}	Δ_{n+2}	$\dots$	Δ_{n+m}		

All entries in the above table are given as follows.

- (i) The first row called *price row* (or *objective row*) represented by c_j is the row of prices i.e., row of the coefficients of decision variables (unknown variables) $x_1, x_2, \dots, x_n$ and slack variables in the objective function Z .

Note that the coefficients of all slack variables are taken 0 (zero) in the objective function.

- (ii) (a) The first column represented by B is the column of columns in the basis matrix.
- (b) The second column represented by c_B is the column of coefficients of current basic variables in the objective function Z .
- (c) The third column represented by x_B is the column of the values of the basic variables. Note that the value of each non-basic variable is zero.

- (d) The columns $Y_1, \dots, Y_n$ are the columns of the coefficients of $x_1, x_2, \dots, x_n$ and columns $S_1, S_2 \dots S_m$ are the columns of the coefficients of the slack variables. Note that the columns $S_1, S_2 \dots S_m$ form a unit matrix of order $m \times m$.
- (e) Last column is the column of Minimum ratio x_B / Y_k , $Y_{ik} > 0$, where Y_k is the incoming vector.
- (f) Last row is the row of Δ_j 's such that $\Delta_j = c_j - c_B Y_j$, $j = 1, 2, \dots, n + m$.

Step 6: To test the initial B.F.S. for optimality.

Compute the net evaluation Δ_j for each variable x_j ($j = 1, 2, \dots, n$) and slack variable $s_1, \dots, s_m$ by using the formula $\Delta_j = c_j - c_B Y_j$;

and make entries in the last row of Δ_j 's in the simplex table.

Note that in the starting table Δ_j for x_j is equal to b_j , $j = 1, 2, \dots, n$ and Δ_j for each basic variable i.e. slack variable is zero.

Optimality Test

- (i) If $\Delta_j \leq 0$ for all j , the solution under consideration is **optimal**.
 - (a) **Alternative optimal** solution will exist if any Δ_j (for non-basic variable) is also zero.
 - (b) If $\Delta_j < 0$ for all j , corresponding to non-basic variables, then the solution is **unique optimal solution**.
- (ii) If $\Delta_j > 0$ for any j i.e., if at least one Δ_j is positive the solution under test is **not optimal** and can be improved. In this case, we must proceed to *improve the solution* (step 7).
- (iii) If corresponding to maximum positive Δ_j , all elements of the column Y_j are negative or zero, the solution under test will be **unbounded**.
- (iv) If optimality condition is satisfied and at least one artificial variable is present in the basis whose value is non-zero, the problem will have no **feasible solution**.

Step 7: To select the entering (incoming) vector and departing (outgoing) vector.

To improve the initial B.F.S. we select the vector entering the basis matrix and the vector to be removed from the basis matrix by the following rules:

To find Incoming Vector: The incoming vector Y_k is always selected corresponding to the largest positive value of Δ_j . This column with largest value of Δ_j is called the **key-column** and is indicated by arrow $\uparrow$ in the simplex table. This indicates the variable to be introduced into the basis.

If largest Δ_k is not unique and occurs in more than one column i.e., there is a tie in the column vector to be introduced in the basis matrix then in order to minimize the number of iterations required to reach the optimal solution the following rules are followed for the selection of one of these to enter the basis matrix.

- (i) If maximum value of Δ_k occurs for more than one decision variable columns, then select any one of these.

- (ii) If maximum value of Δ_k occurs for decision and slack (surplus) variable columns then select the decision variable column.
- (iii) If maximum value of Δ_k occurs for more than one slack (surplus) variable columns only then select any one of these.

To find Outgoing Vector: The departing vector β_r is selected corresponding to that value of r for which

$$\frac{x_{Br}}{y_{rk}} = \min_i \left\{ \frac{x_{Bi}}{y_{ik}}, y_{ik} > 0 \right\},$$

if $\alpha_k(Y_k)$ is the incoming vector.

This row which correspond to minimum ratio is called the **key row** and is indicated by arrow $\rightarrow$ in the simplex table.

If above mentioned minimum value is not unique then there is a choice between basic variables (more than one) that will leave the basis. In this case to reach the optimal solution quickly in minimum number of iterations the following rules are followed for the selection of a basic variable to leave the basis.

- (i) If minimum ratio occurs for more than one decision variable, then select any one of these.
- (ii) If minimum ratio occurs for a decision variable and a slack (or surplus) variables, then select the slack (or surplus) variable.
- (iii) If minimum ratio occurs for two or more slack (or surplus) variables, then select any one of these.
- (iv) If minimum ratio occurs for decision (or slack or surplus) and artificial variable, then select the artificial variable.

Step 8: *To find key element.*

If $Y_k(\alpha_k)$ is the column vector entering the basis matrix (i.e., x_k is the non-basic variable to become the basic variable) and $\beta_r(Y_r)$ is the vector leaving the basis matrix (i.e., x_r is basic variable leaving the basis) then the element $y_{rk}(a_{rk})$ which lies at the intersection of outgoing vector arrow $\rightarrow$ and incoming vector arrow $\uparrow$ is called the **pivot** (or **key**) **element** and this element is put in $\square$. Note that key element is always positive.

Step 9: *Construction of the next simplex table.*

After identifying the key element, we construct the next simplex table with the following changes in the above simplex table.

- (i) The column $Y_k(\alpha_k)$ is reduced to a column of unit matrix with 1 at the place of key element and 0 in all other rows of this column, for which we proceed as follows.

Divide each element of the key row (here r -th row) by the key element $y_{rk}(a_{rk})$ so that it reduces to 1. Then all other elements in the key column (here k -th column) are reduced to 0 by subtracting suitable multipliers of elements of this new changed key row.

Thus, if y_{ik} (a_{ik}) is the element in i -th row of k -th key column, then we shall subtract y_{ik} (a_{ik}) times the elements of new key row from all the corresponding elements of i -th row, to get 0 in i -th row of key column Y_k (α_k). Similarly all other elements of key column are reduced to zero.

- (ii) In the first column B of simplex table replace Y_r by Y_k .
- (iii) In the second column c_B of simplex table replace c_r by c_k .

Step 10: Test the new B.F.S. obtained in this new table for optimality by computing $\Delta_j = c_j - c_B y_j$ for all columns as in step 6.

If this solution is not optimal then repeat steps 7 to 9, until we obtain an optimal solution.

Note: In place of computing Δ_j for all columns Y_j ($j = 1, 2, \dots, m + n$) we can find new Δ_j row as follows.

If $\Delta_k = t$ (i.e., the value of Δ_j in entering column Y_k is t), then subtract t times the elements of the new key row from all the corresponding elements of Δ_j row.

ILLUSTRATIVE EXAMPLES

Example 1: Use Simplex method to solve the following L.P.P.

$$\begin{aligned}
 &\text{Maximize} && Z = 6x_1 + 8x_2 \\
 &\text{Subject to,} && 30x_1 + 20x_2 \leq 300 \\
 &&& 5x_1 + 10x_2 \leq 110 \\
 &\text{and} && x_1, x_2 \geq 0.
 \end{aligned}$$

Solution: We shall solve this problem stepwise in detail, so that the readers may understand the method.

Step 1: Given problem is already a maximization problem.

Step 2: R.H.S. of all constraints are already non-negative.

Step 3: Introducing the slack variables s_1 and s_2 , in the constraints the problem reduces to

$$\begin{aligned}
 \text{Max.} &&& Z = 6x_1 + 8x_2 + 0.s_1 + 0.s_2 \\
 \text{s.t.} &&& 30x_1 + 20x_2 + s_1 + 0.s_2 = 300 \\
 &&& 5x_1 + 10x_2 + 0.s_1 + s_2 = 110 \\
 \text{and} &&& x_1, x_2, s_1, s_2 \geq 0
 \end{aligned}$$

Step 4: To find initial B.F.S.

Taking $x_1 = 0 = x_2$, we get $s_1 = 300$, $s_2 = 110$ which is the initial B.F.S. of the given problem. Here s_1 and s_2 are basic variables while x_1 and x_2 are non-basic variables.

Step 5: The starting *simplex table* is as follows:

Table 2.2

B	c_j		6	8	0	0	Min. Ratio $x_B / Y_2, y_{i2} > 0$
	x_B		Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	
$Y_3(S_1) \quad c_{B_1} = 0$	$s_1 = 300$		30	20	1	0	$300/20 = 15$
$Y_4(S_2) \quad c_{B_2} = 0$	$s_2 = 110$		5	10	0	1	$110/10 = 11 \rightarrow$ (min.)
$Z = c_B x_B = 0$	Δ_j		6	8 $\uparrow$	0	0	

Step 6: To test solution for optimality

Calculations of Δ_j . We have $\Delta_j = c_j - c_B Y_j$.

$$\therefore \Delta_1 = c_1 - c_B Y_1 = 6 - (0, 0) \cdot [30, 5] = 6 - (0 \times 30 + 0 \times 5) = 6$$

$$\Delta_2 = c_2 - c_B Y_2 = 8 - (0, 0) \cdot [20, 10] = 8 - (0 \times 20 + 0 \times 10) = 8$$

$$\Delta_3 = c_3 - c_B Y_3 = 0 - (0, 0) \cdot [1, 0] = 0 - (0 \times 1 + 0 \times 0) = 0$$

$$\Delta_4 = c_4 - c_B Y_4 = 0 - (0, 0) \cdot [0, 1] = 0 - (0 \times 0 + 0 \times 1) = 0$$

Note that in the starting simplex table as the basic variables are only slack variables whose coefficients in the objective function are zero,

$$\therefore \Delta_1 = c_1 = 6, \Delta_2 = c_2 = 8, \Delta_3 = c_3 = 0, \Delta_4 = c_4 = 0$$

Make these entries in the last row of Δ_j 's in the above simplex table (table 2.2).

The value of objective function Z .

$$\text{We have } Z = c_B x_B = (0, 0) \cdot [300, 110] = 0 \times 300 + 0 \times 110 = 0$$

Since in the above table all Δ_j are not ≤ 0 , i.e., few $\Delta_j > 0$, so this solution is not optimal and can be improved further.

Step 7: Since the largest positive value of Δ_j is $\Delta_2 = 8$, so the vector Y_2 (column Y_2) is the *key column* denoted by $\uparrow$ in the table and will enter the basis matrix B i.e., the variable x_2 enters the basis.

By minimum ratio rule (shown in the last column of table 2.2) smallest non-negative ratio is 11 in 2nd row, so the second row $Y_4(S_2)$ is the *key row* denoted by $\rightarrow$ in the table and it will leave the basis matrix B i.e., the variable s_2 will leave the basis.

Thus variable x_2 will replace the variable s_2 in the basis i.e., x_2 becomes the basic and s_2 becomes the non-basic variable in the next table.

Step 8: The key column and the key row intersect at the element $y_{22} = 10$, which is the *key element* and we enclose it in $\square$.

Step 9: Now we construct the new (second) simplex table as follows:

Since the key column Y_2 in the second simplex table enters the basis so it should be $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Note that there should be 1 at the place of key element y_{22} and 0 at all other places in this column.

Calculations for the entries in the rows of *second simplex table*.

Since the key element is $y_{22} = 10$ in the second row so first, we divide all elements of 2nd row by 10 to obtain unity at this place.

$$\begin{array}{l} \therefore \text{new 2nd row entries are} \\ \text{i.e.,} \end{array} \quad \begin{array}{cccccc} 110/10 & : & 5/10 & 10/10 & 0/10 & 1/10 \\ \hline 11 & : & 1/2 & 1 & 0 & 1/10 \\ \hline \end{array}$$

Now, new 1st row is obtained as follows:

$$\begin{array}{l} \text{Old 1st row} \\ -20 \text{ times of new 2nd row} \\ \text{Adding, new 1st row,} \end{array} \quad \begin{array}{cccccc} 300 & : & 30 & 20 & 1 & 0 \\ -20 \times 11 & : & -20 \times 1/2 & -20 \times 1 & -20 \times 0 & -20 \times 1/10 \\ \hline 80 & : & 20 & 0 & 1 & -2 \\ \hline \end{array}$$

Also in the 1st column B replace $Y_4(S_2)$ by Y_2

in the 2nd column c_B , replace $c_{B_2} = c_2 = 8$

and in the 3rd column x_B , replace $x_{B_2} = S_2$ by x_2 .

Making all the above changes in the first simplex table (table 2.2), i.e., replacing column $Y_4(S_1)$ by Y_2 in the basis B , we get the following second simplex table (table 2.3)

Second Simplex Table

Table 2.3

B	c_B	c_j	6	8	0	0	Min. Ratio $x_B/Y_1, J_{i1} > 0$
			Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	
$Y_3(S_1)$	$c_{B_1} = 0$	$s_1 = 80$	20	0	1	-2	$80/20 = 4 \rightarrow$ (min.)
Y_2	$c_{B_2} = 8$	$x_2 = 11$	1/2	1	0	1/10	$11/(1/2) = 22$
$Z = c_B \cdot x_B = 88$		Δ_j	2 ↑	0	0	-4/5	

To test the optimality of the solution in second simplex table (table 2.3)

Now $Z = \mathbf{c}_B \mathbf{x}_B = (0, 8) \cdot [80, 11] = 0 \times 80 + 8 \times 11 = 88$ (greater than Z in table 2.2)

To compute the values of Δ_j for $j = 1, 2, 3, 4$

$$\Delta_1 = c_1 - \mathbf{c}_B Y_1 = 6 - (0, 8) \cdot [20, 1/2] = 6 - (0 + 4) = 2,$$

$$\Delta_2 = c_2 - \mathbf{c}_B Y_2 = 8 - (0, 8) \cdot [0, 1] = 8 - (0 + 8) = 0$$

$$\Delta_3 = c_3 - \mathbf{c}_B Y_3 = 0 - (0, 8) \cdot [1, 0] = 0 - (0 + 0) = 0,$$

$$\Delta_4 = c_4 - \mathbf{c}_B Y_4 = 0 - (0, 8) \cdot [-2, 1/10] = 0 - (0 + 8/10) = -4/5$$

The values of Δ_j 's can also be computed as follows (since Δ_2 in key column two should be zero).

Old Δ_j row (in table 2.2)	6	8	0	0
-8 times of new key row	$-8 \times 1/2$	-8×1	-8×0	$-8 \times 1/10$
Adding, new Δ_j row, (in table 2.3)	2	0	0	$-4/5$

We make these entries in the last row of Δ_j 's in table 2.3.

Now we see that the value of the objective function Z has been improved from 0 to 88 for the solution in the above table 2.3. However $\Delta_1 = 2 > 0$. So the above solution is also not optimal and can be improved further.

Step 10: Since largest positive value of Δ_j (in table 2.3) is $\Delta_1 = 2$, so the vector Y_1 (column Y_1) is the key column denoted by $\uparrow$ will enter the basis matrix B i.e., the variable x_1 will enter the basis.

By minimum ratio rule (shown in the last column of table 2.3) smallest non-negative ratio is 4 in first row, so the 1st row $Y_3(S_1)$ is the key row denoted by $\rightarrow$ in the table and it will leave the basis B so the column vector $Y_3(S_1)$ will be replaced by column Y_1 , in the basis matrix B i.e., the variable x_1 enters the basis for variable s_1 .

The key column and the key rows in second simplex table (table 2.3) intersect at the element $y_{11} = 20$, which is the key element and we enclose it in $\square$.

Since the key column Y_1 enters the basis and key element is y_{11} , so Y_1 should be $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ in the new table.

Now, we construct the third simplex table (table 2.4) with the following changes in table 2.3.

To find new 1st row

Since key element in the 1st row is $y_{11} = 20$, so the new 1st row is obtained by dividing all elements $80 : 20, 0, 1, -2$ of 1st row by 20 to obtain 1 at the place of key element.

$$\begin{array}{lcl}
 \text{i.e., new 1st row is} & 80/20 & : \quad 20/20 \quad 0/20 \quad 1/20 \quad -2/20 \\
 & \text{i.e.,} & 4 \quad : \quad 1 \quad 0 \quad 1/20 \quad -1/10
 \end{array}$$

The **new 2nd row** is obtained as follows.

$$\begin{array}{lcl}
 \text{Old 2nd row (in table 2.3)} & 11 & : \quad 1/2 \quad 1 \quad 0 \quad 1/10 \\
 (-1/2) \text{ times of new 1st row} & -1/2 \times 4 & : \quad -1/2 \times 1 \quad -1/2 \times 0 \quad -1/2 \times 1/20 \quad (-1/2)(-1/10) \\
 \text{Adding, new 2nd row} & 9 & : \quad 0 \quad 1 \quad -1/40 \quad 3/20
 \end{array}$$

To find new Δ_j row. Since key element is y_{11} is in 1st column, so Δ_1 will be zero in the next table which is obtained as follows.

$$\begin{array}{lcl}
 \text{Old } \Delta_j \text{ row (in table 2.3)} & 2 & 0 \quad 0 \quad -4/5 \\
 -2 \text{ times of new 1st key row,} & -2 \times 1 & -2 \times 0 \quad -2 \times 1/20 \quad -2 \times (-1/10) \\
 \text{Adding new } \Delta_j \text{ row} & 0 & 0 \quad -1/10 \quad -3/5
 \end{array}$$

$\therefore$ Entering Y_1 in place of $Y_3(S_1)$ in the basis matrix B_1 the **third simplex table** is as follows:

Table 2.4

B	c_B	x_B	c_j				
			6	8	0	0	
			Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	
Y_1	$c_{B_1} = 6$	$x_1 = 4$	1	0	1/20	-1/10	
Y_2	$c_{B_2} = 8$	$x_2 = 9$	0	1	-1/40	3/20	
$Z = c_B \cdot x_B = 96$		Δ_j	0	0	-1/10	-3/5	

Now $Z = c_B \cdot x_B = (6, 8) \cdot [4, 9] = 24 + 72 = 96$ (greater than Z in table 2.4)

Since in the above table all $\Delta_j \leq 0$, so the solution given in the above table is optimal and can not be improved further.

Hence the optimal solution to the given problem is $x_1 = 4$, $x_2 = 9$ and Max. $Z = 96$.

All tables 2.2, 2.3 and 2.4 can be computed in a single table as in the table 2.5.

Table 2.5

B	c_B	x_B	c_j				Min. Ratio $x_B/Y_2, y_{i2} > 0$
			6	8	0	0	
			Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	
$Y_3(S_1)$	0	$s_1 = 300$	30	20	1	0	$300/20 = 15$
$Y_4(S_2)$	0	$s_2 = 110$	5	10	0	1	$110/10 = 11 \rightarrow$ (min.)
$Z = c_B x_B = 0$		Δ_j	6	8 $\uparrow$	0	0	x_B/Y_1 $y_{i1} > 0$
$Y_3(S_1)$	0	$s_1 = 80$	20	0	1	-2	$80/20 = 4 \rightarrow$ (min.)
Y_2	8	$x_2 = 11$	1/2	1	0	1/10	$11/(1/2) = 22$
$Z = c_B x_B = 88$		Δ_j	2 $\uparrow$	0	0	-4/5	
Y_1	6	$x_1 = 4$	1	0	1/20	-1/10	
Y_2	8	$x_2 = 9$	0	1	-1/40	3/20	
$Z = c_B x_B = 96$		Δ_j	0	0	-1/10	-3/5	

Example 2: Use simplex method to solve the following L.P.P.

Maximize, $Z = 20x_1 + 30x_2$

subject to, $3x_1 + 3x_2 \leq 36, \quad 5x_1 + 2x_2 \leq 50$

$2x_1 + 6x_2 \leq 60$ and $x_1, x_2 \geq 0$.

Solution: The given problem is a maximization problem and all $b_i \geq 0$.

Introducing the slack variables s_1, s_2, s_3 , in the successive constraints, the given problem reduces to

Max. $Z = 20x_1 + 30x_2 + 0 \cdot s_1 + 0 \cdot s_2 + 0 \cdot s_3$

s.t. $3x_1 + 3x_2 + s_1 + 0 \cdot s_2 + 0 \cdot s_3 = 36$

$5x_1 + 2x_2 + 0 \cdot s_1 + s_2 + 0 \cdot s_3 = 50$

$2x_1 + 6x_2 + 0 \cdot s_1 + 0 \cdot s_2 + s_3 = 60$

and $x_1, x_2, s_1, s_2, s_3 \geq 0$

Taking $x_1 = 0 = x_2$, we get $s_1 = 36, s_2 = 50, s_3 = 60$, which is the initial B.F.S. of the given problem. Here s_1, s_2, s_3 are basic variables.

The solution of the problem by simplex method is shown in the following table:

Table 2.6

	B	c_B	x_B	c_j					Min. Ratio $x_B/Y_2, j_{i2} > 0$
				Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	$Y_5(S_3)$	
I	$Y_3(S_1)$	0	$s_1 = 36$	3	3	1	0	0	$36/3 = 12$
	$Y_4(S_2)$	0	$s_2 = 50$	5	2	0	1	0	$50/2 = 25$
	$Y_5(S_3)$	0	$s_3 = 60$	2	6	0	0	1	$60/6 = 10 \rightarrow$ (min.)
	$Z = c_B x_B = 0$		Δ_j	20	30 $\uparrow$	0	0	0	x_B/Y_1 $j_{i1} > 0$
II	$Y_3(S_1)$	0	$s_1 = 6$	2	0	1	0	$-1/2$	$6/2 = 3 \rightarrow$ (min.)
	$Y_4(S_2)$	0	$s_2 = 30$	$13/3$	0	0	1	$-1/3$	$30/(13/3) = 90/13$
	Y_2	30	$x_2 = 10$	$1/3$	1	0	0	$1/6$	$10/(1/3) = 30$
	$Z = c_B x_B = 300$		Δ_j	10 $\uparrow$	0	0	0	-5	
III	Y_1	20	$x_1 = 3$	1	0	$1/2$	0	$-1/4$	
	$Y_4(S_2)$	0	$s_2 = 17$	0	0	$-13/6$	1	$3/4$	
	Y_2	30	$x_2 = 9$	0	1	$-1/6$	0	$1/4$	
	$Z = c_B x_B = 330$		Δ_j	0	0	-5	0	$-5/2$	

Calculation Work

For table I. $\Delta_1 = c_1 = 20$, $\Delta_2 = c_2 = 30$, $\Delta_3 = 0 = \Delta_4 = \Delta_5$ ($\because$ All basic variables are slack variables)

$\therefore$ Max. positive ($\Delta_j = 30 = \Delta_2$), $\therefore Y_2$ is the key column to enter the basis matrix.

By minimum ratio rule (shown in last column); $Y_5(S_3)$ is the key row, to leave the basis matrix.

$\therefore$ key element = $j_{32} = 6$

For table II. Since Y_2 is the column entering the basis matrix and key element is $j_{32} = 6$. So in table II, there should be 1 at the place of key element j_{32} and all other entries zero in this key column II including Δ_2 .

New 3rd row of table II is obtained by dividing all elements of this key row (in table I) by key element 6.

$\therefore$ 3rd row in table II is

$$60/6 : 2/6, 6/6, 0/6, 0/6, 1/6 \text{ i.e., } 10 : 1/3, 1, 0, 0, 1/6.$$

To find 1st row

1st row in table I is	36	:	3	3	1	0	0
-3 times of new 3rd row	-3×10	:	$-3 \times 1/3$	-3×1	-3×0	-3×0	$-3 \times 1/6$
Adding 1st row of table II	6	:	2	0	1	0	-1/2

To find 2nd row

2nd row in table I is	50	:	5	2	0	1	0
-2 times of new 3rd row	-2×10	:	$-2 \times 1/3$	-2×1	-2×0	-2×0	$-2 \times 1/6$
Adding 1st row of table II	30	:	13/3	0	0	1	-1/3

To find Δ_j row

Δ_j row in table I is	20		30	0	0	0	0
-30 times of new 3rd row,	$-30 \times 1/3$		-30×1	-30×0	-30×0	-30×0	$-30 \times 1/6$
Adding Δ_j row of table II	10		0	0	0	0	-5

Since max. positive Δ_j is $\Delta_1 = 10$, so Y_1 is the key column, to enter the basis matrix.

By minimum ratio rule shown in last column of 2nd table first row is the key row to leave the basis matrix.

$\therefore$ key element = $y_{11} = 2$.

For table III. Since Y_1 is the column entering the basis matrix and key element is $y_{11} = 2$. So in table III, there should be 1 at the place of key element y_{11} and all other entries zero in this key column I including Δ_j .

$\therefore$ New 1st row of table III is obtained by dividing all elements of this key row (in table II) by the key element 2,

$\therefore$ 1st row in table III's	6/2	:	2/2	0/2	1/2	0/2	$(-1/2)/2$
i.e.,	3	:	1	0	1/2	0	-1/4

To find 2nd row.

2nd row in table II	30	:	13/3	0	0	1	-1/3
-13/3 times the new 1st row	$(-13/3)$	:	$-13/3$	$-13/3$	$(-13/3)$	$(-13/3)$	$(-13/3)$
	$\times 3$		$\times 1$	$\times 0$	$\times 1/2$	$\times 0$	$\times (-1/4)$
Adding 2nd row of table III	17	:	0	0	-13/6	1	3/4

To find 3rd row

3rd row in table II	10	:	1/3	1	0	0	1/6
$-(1/3)$ times the new 1st row	$(-1/3) \cdot 3$	:	$-(1/3)$ $\times 1$	$-(1/3)$ $\times 0$	$(-1/3)$ $\times (1/2)$	$(-1/3)$ $\times 0$	$(-1/3)$ $\times (-1/4)$
Adding 3rd row of table III	9	:	0	1	$-1/6$	0	$1/4$

To find Δ_j row

Δ_j row in table II,	10	0	0	0	-5
-10 times the new 1st row	-10×1	-10×0	$-10 \times 1/2$	-10×0	$-10 \times (-1/4)$
Adding Δ_j row of table III	0	0	-5	0	$-5/2$

Since all $\Delta_j \leq 0$ in last table III, so the optimal solution from the table is

$$x_1 = 3, x_2 = 9, s_1 = 0, s_2 = 17, \text{Max. } Z = 330.$$

Hence the optimal solution of the problem is

$$x_1 = 3, x_2 = 9 \text{ and Max. } Z = 330.$$

Example 3: Solve by simplex method the following L.P.P.

$$\begin{aligned} \text{Maximize} \quad & Z = 7x_1 + 5x_2 \\ \text{subject to} \quad & -x_1 - 2x_2 \geq -6 \\ & 4x_1 + 3x_2 \leq 12 \\ & x_1, x_2 \geq 0. \end{aligned}$$

Solution: The problem is of maximization.

In the first constraint b_1 is negative, so we make it positive by multiplying the inequality by -1 , so the first constraint changes to $x_1 + 2x_2 \leq 6$.

Now converting the inequalities of the constraints into equations by introducing the slack variables s_3, s_4 , the given problem becomes:

$$\begin{aligned} \text{Max.} \quad & Z = 7x_1 + 5x_2 + 0 \cdot s_3 + 0 \cdot s_4 \\ \text{s.t.} \quad & x_1 + 2x_2 + s_3 + 0 \cdot s_4 = 6 \\ & 4x_1 + 3x_2 + 0 \cdot s_3 + s_4 = 12 \\ & x_1, x_2, s_3, s_4 \geq 0 \end{aligned}$$

Taking $x_1 = 0, x_2 = 0$, we get $s_3 = 6, s_4 = 12$ which is the initial B.F.S.

The solution to the problem using simplex method is given in the following table:

Table 2.7

B	c_B	x_B	c_j				Min. Ratio $x_B/Y_1, Y_{i1} > 0$
			7	5	0	0	
			Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	
$Y_3(S_1)$	0	$s_1 = 6$	1	2	1	0	$6/1 = 6$
$Y_4(S_2)$	0	$s_2 = 12$	4	3	0	1	$12/4 = 3 \rightarrow$ (min.)
$Z = c_B x_B = 0$		Δ_j	7 $\uparrow$	5	0	0 $\downarrow$	
$Y_3(S_1)$	0	$s_1 = 3$	0	$5/4$	1	$-1/4$	
Y_1	7	$x_1 = 3$	1	$3/4$	0	$1/4$	
$Z = c_B x_B = 21$		Δ_j	0	$-1/4$	0	$-7/4$	

In the last table all $\Delta_j \leq 0$ therefore, it gives the optimal solution.

Hence the optimal solution is

$$x_1 = 3, x_2 = 0 \text{ and Max. } Z = 21.$$

Example 4: Solve the following L.P.P. by Simplex method.

$$\begin{aligned}
 &\text{Minimize} && Z = x_1 - 3x_2 + 2x_3 \\
 &\text{subject to} && 3x_1 - x_2 + 2x_3 \leq 7 \\
 &&& -2x_1 + 4x_2 \leq 12 \\
 &&& -4x_1 + 3x_2 + 8x_3 \leq 10 \\
 &\text{and} && x_1, x_2, x_3 \geq 0.
 \end{aligned}$$

Solution: The given problem is of minimization. Convert it to, maximization problem by taking the objective function as $Z' = -Z$.

$\therefore$ The objective function becomes, $\text{Max } Z' = -Z = -x_1 + 3x_2 - 2x_3$.

Converting the inequalities of the constraints into equations by the introduction of slack variables s_1, s_2, s_3 , the given problem becomes.

$$\begin{aligned}
 &\text{Max.} && Z' = -x_1 + 3x_2 - 2x_3 + 0.s_1 + 0.s_2 + 0.s_3 \\
 &\text{s.t.} && 3x_1 - x_2 + 2x_3 + s_1 + 0.s_2 + 0.s_3 = 7 \\
 &&& -2x_1 + 4x_2 + 0.x_3 + 0.s_1 + s_2 + 0.s_3 = 12 \\
 &&& -4x_1 + 3x_2 + 8x_3 + 0.s_1 + 0.s_2 + s_3 = 10 \\
 &\text{and} && x_1, x_2, x_3, s_1, s_2, s_3 \geq 0.
 \end{aligned}$$

Taking $x_1 = 0 = x_2 = x_3$, we get $s_1 = 7, s_2 = 12, s_3 = 10$, which is the initial B.F.S.

The solution to the problem by simplex method is given as follows.

Table 2.8

B	c_B	x_B	c_j						Min. Ratio $x_B / Y_{i2}, Y_{i2} > 0$
			-1	3	-2	0	0	0	
$Y_4(S_1)$	0	$s_1 = 7$	Y_1	Y_2	Y_3	$Y_4(S_1)$	$Y_5(S_2)$	$Y_6(S_3)$	
$Y_4(S_1)$	0	$s_1 = 7$	3	-1	2	1	0	0	—
$Y_5(S_2)$	0	$s_2 = 12$	-2	4	0	0	1	0	$12/4 = 3 \rightarrow$ (min.)
$Y_6(S_3)$	0	$s_3 = 10$	-4	3	8	0	0	1	$10/3$
$Z' = c_B x_B = 0$		Δ_j	-1	3 $\uparrow$	-2	0	0	0	x_B / Y_{i1} $Y_{i1} > 0$
$Y_4(S_1)$	0	$s_1 = 10$	5/2	0	2	1	$1/4$	0	$10/(5/2) = 4 \rightarrow$ (min.)
Y_2	3	$x_2 = 3$	$-1/2$	1	0	0	$1/4$	0	—
$Y_6(S_3)$	0	$s_3 = 1$	$-5/2$	0	8	0	$-3/4$	1	—
$Z' = c_B x_B = 9$		Δ_j	$1/2$ $\uparrow$	0	-2	0	$-3/4$	0	
Y_1	-1	$x_1 = 4$	1	0	$4/5$	$2/5$	$1/10$	0	
Y_2	3	$x_2 = 5$	0	1	$2/5$	$1/5$	$3/10$	0	
$Y_6(S_3)$	0	$s_3 = 11$	0	0	10	1	$-1/2$	1	
$Z' = c_B x_B = 11$		Δ_j	0	0	$-12/5$	$-1/5$	$-4/5$	0	

Since all $\Delta_j \leq 0$, so the solution given in the above table is optimal.

Hence the optimal solution is $x_1 = 4, x_2 = 5, x_3 = 0$ and Min. $Z = -$ Max. $Z' = -11$.

Example 5: Solve by simplex method the following L.P.P.:

$$\begin{aligned}
 &\text{Maximize} && Z = 3x_1 + 2x_2 + 5x_3 \\
 &\text{subject to} && x_1 + 2x_2 + x_3 \leq 430 \\
 &&& 3x_1 + 2x_3 \leq 460 \\
 &&& x_1 + 4x_2 \leq 420, \\
 &&& x_1, x_2, x_3 \geq 0.
 \end{aligned}$$

Solution: The given problem is of maximization and all the b_i 's are positive.

Converting the inequalities of the constraints into equations by introducing slack variables s_1, s_2 and s_3 the given problem becomes

$$\text{Max. } Z = 3x_1 + 2x_2 + 5x_3 + 0s_1 + 0s_2 + 0s_3$$

$$\text{s.t.} \quad x_1 + 2x_2 + x_3 + s_1 + 0.s_2 + 0.s_3 = 430$$

$$3x_1 + 0x_2 + 2x_3 + 0.s_1 + s_2 + 0.s_3 = 460$$

$$x_1 + 4x_2 + 0x_3 + 0.s_1 + 0.s_2 + s_3 = 420$$

$$\text{and} \quad x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

Taking $x_1 = 0, x_2 = 0, x_3 = 0$, we get $s_1 = 430, s_2 = 460, s_3 = 420$ which is the initial B.F.S.

The solution to the problem using simplex algorithm is given in the following table :

Table 2.9

B	c_B	x_B	c_j						Min. Ratio $x_B/Y_3, Y_{i3} > 0$
			Y_1	Y_2	Y_3	$Y_4(S_1)$	$Y_5(S_2)$	$Y_6(S_3)$	
$Y_4(S_1)$	0	$s_1 = 430$	1	2	1	1	0	0	430
$Y_5(S_2)$	0	$s_2 = 460$	3	0	2	0	1	0	230 (min.) →
$Y_6(S_3)$	0	$s_3 = 420$	1	4	0	0	0	1	—
$Z = c_B x_B = 0$		Δ_j	3	2	5 ↑	0	0 ↓	0	x_B/Y_2 $y_{i2} > 0$
$Y_4(S_1)$	0	$s_1 = 200$	-1/2	2	0	1	-1/2	0	100 (min.) →
Y_3	5	$x_3 = 230$	3/2	0	1	0	1/2	0	—
$Y_6(S_3)$	0	$s_3 = 420$	1	4	0	0	0	1	420/4
$Z = 1150$		Δ_j	-9/2	2 ↑	0	0 ↓	-5/2	0	
Y_2	2	$x_2 = 100$	-1/4	1	0	1/2	-1/4	0	
Y_3	5	$x_3 = 230$	3/2	0	1	0	1/2	0	
$Y_6(S_3)$	0	$s_3 = 20$	2	0	0	-2	1	1	
$Z = 1350$		Δ_j	-4	0	0	-1	-2	0	

In the last table all $\Delta_j \leq 0$, therefore the solution is optimal.

Hence, the optimal solution is

$$x_1 = 0, x_2 = 100, x_3 = 230 \text{ and Max. } Z = 1350.$$

Exercise 2.1

Solve the following L.P.P. by simplex method.

1. Maximize $Z = 3x_1 + 2x_2$ subject to $x_1 + x_2 \leq 4$ $x_1 - x_2 \leq 2$ and $x_1, x_2 \geq 0$	2. Maximize $Z = 5x_1 + 3x_2$ Subject to $3x_1 + 5x_2 \leq 15$ $5x_1 + 2x_2 \leq 10$ and $x_1, x_2 \geq 0$
3. Maximize $Z = 21x_1 + 15x_2$ subject to $-x_1 - 2x_2 \geq -6$ $x_1 + 3x_2 \leq 12$ and $x_1, x_2 \geq 0$	4. Maximize $Z = 2x_1 + 3x_2$ subject to $x_1 + x_2 \leq 400$ $2x_1 + x_2 \leq 600$ and $x_1, x_2 \geq 0$
5. Maximize $Z = 2x_1 + 4x_2$ subject to $2x_1 + 3x_2 \leq 48$ $x_1 + 3x_2 \leq 42$ $x_1 + x_2 \leq 21$ and $x_1, x_2 \geq 0$	6. Maximize $Z = x_1 + 3x_2$ subject to $x_2 \leq 10$ $x_1 + x_2 \leq 16$ $8x_1 + 3x_2 \leq 96$
7. Maximize $Z = 5x_1 + 3x_2$ subject to $x_1 + x_2 \leq 2$ $5x_1 + 2x_2 \leq 10$ $3x_1 + 8x_2 \leq 12$ and $x_1, x_2 \geq 0$	8. Maximize $Z = x_1 + 2x_2$ subject to $2x_1 + x_2 \leq 70$ $x_1 + x_2 \leq 40$ $x_1 + 3x_2 \leq 90$ and $x_1, x_2 \geq 0$
9. Maximize $Z = x_1 + x_2 + 3x_3$ subject to $3x_1 + 2x_2 + x_3 \leq 3$ $2x_1 + x_2 + 2x_3 \leq 2$ and $x_1, x_2, x_3 \geq 0$	10. Maximize $Z = 3x_1 + 5x_2 + 4x_3$ subject to $2x_1 + 3x_2 \leq 18$ $2x_2 + 5x_3 \leq 18$ $3x_1 + 2x_2 + 4x_3 \leq 25$ and $x_1, x_2, x_3 \geq 0$
11. Maximize $Z = 2x_1 + 5x_2 + 7x_3$ subject to $3x_1 + 2x_2 + 4x_3 \leq 100$ $x_1 + 4x_2 + 2x_3 \leq 100$ $x_1 + x_2 + 3x_3 \leq 100$ and $x_1, x_2, x_3 \geq 0$	12. Maximize $Z = 2x_1 + 4x_2 + 3x_3$ subject to $3x_1 + 4x_2 + 2x_3 \leq 60$ $2x_1 + x_2 + 2x_3 \leq 40$ $x_1 + 3x_2 + 2x_3 \leq 80$ and $x_1, x_2, x_3 \geq 0$

13. A company manufactures two items A and B and they are sold at a profit of ₹ 30 and ₹ 20 per unit respectively. A requires 2 kg of material, 3 man hours and 1 machine hour per unit. B requires 1 kg of material, 2 man hours and 3 machine

hours per unit. During each production run there are 280 kg material available, 500 labour hours and 420 hours of machine can be used. How much of the two items should be produced to maximize the profit ? Formulate the problem as L.P.P. and solve by simplex method.

14. A company produces two types of products say type A and B. Product B is of superior quality and product A is of a lower quality. Profits on the two types of products are ₹ 30 and 40 respectively. The data on resources required and capacity available is given below:

	Requirements		Capacity available per month
	Product A	Product B	
Raw materials (kg)	60	120	12000
Machining (hours per piece)	8	5	600
Assembly (man hour)	3	4	500

How should the company manufacture the two types of products in order to have a maximum overall profit ? Use simplex method.

Answers 2.1

1. $x_1 = 3, x_2 = 1$, Max. $Z = 11$	2. $x_1 = 20/19, x_2 = 45/19$, Max. $Z = 235/19$
3. $x_1 = 6, x_2 = 0$, Max. $Z = 126$	4. $x_1 = 0, x_2 = 400$, Max. $Z = 1200$
5. $x_1 = 6, x_2 = 12$, Max. $Z = 60$	6. $x_1 = 6, x_2 = 10$, Max. $Z = 36$
7. $x_1 = 2, x_2 = 0$, Max. $Z = 10$	8. $x_1 = 15, x_2 = 25$, Max. $Z = 65$
9. $x_1 = 0 = x_2, x_3 = 1$, Max. $Z = 3$	10. $x_1 = 3, x_2 = 4, x_3 = 2$, Max. $Z = 37$
11. $x_1 = 0, x_2 = 50/3, x_3 = 50/3$, Max. $Z = 200$	12. $x_1 = 0, x_2 = 20/3, x_3 = 50/3$, Max. $Z = 230/3$
13. If x_1, x_2 units of items A and B are produced, then L.P.P.'s Max. $Z = 30x_1 + 20x_2$. subject to $2x_1 + x_2 \leq 280, 3x_1 + 2x_2 \leq 500, x_1 + 3x_2 \leq 420, x_1, x_2, x_3 \geq 0$ Sol. $x_1 = 84, x_2 = 112$, Max. $Z = ₹ 4760$.	
14. If x_1 and x_2 and the numbers of two products A and B produced then L.P.P. is Max. $Z = 30x_1 + 40x_2$ subject to, $60x_1 + 120x_2 \leq 12000, 8x_1 + 5x_2 \leq 600, 3x_1 + 4x_2 \leq 500, x_1, x_2 \geq 0$. Sol. $x_1 = 200/11, x_2 = 1000/11$, Max. $Z = ₹ 46000/11$.	

25 Artificial Variable Technique

Sometimes in linear programming problems constraints may also have $\geq$ and $=$ signs after ensuring that all $b_i \geq 0$. In such problems we first introduce *surplus variables* to convert inequalities into equations. In such cases basis matrix is not obtained as an identity matrix in the starting simplex table. To overcome this difficulty we introduce new variables called, the *artificial variables* to each of such constraints. These variables are fictitious and cannot have any physical meaning.

The artificial variables are introduced for the limited purpose of obtaining an *initial solution*. It is not relevant whether the objective function is of the maximization or minimization type. Since artificial variables do not represent any quantity relating to the decision problem they must be driven out of the system and must not be present in the final solution (if at all they do, it represents a situation of infeasibility which is discussed later in this chapter).

The L.P.P. involving artificial variables can be solved by two methods:

1. Big M-Method (Carners M-Method)
2. Two Phase Method

2.5.1 Method 1. Big-M Method (Carners M-method)

For the solution of the L.P.P. involving artificial variables a method known as **Big-M Method or Carners M-Method** was developed by A. Carners. In this method, a very large negative price say $-M$ is assigned to each artificial variable in the objective function.

$\therefore$ the objective function is written as

$$Z = \mathbf{c} \mathbf{x} + 0 \cdot \mathbf{x}_{\text{slack}, V} + 0 \cdot \mathbf{x}_{\text{surplus}, V} - M \cdot \mathbf{x}_{\text{art}, V}$$

Due to this large negative price $-M$ in the objective function, it cannot be improved in the presence of the artificial variable. So first we shall have to remove these artificial vector from the basis matrix. Once artificial column vectors corresponding to the artificial variables leave the basis, we forget about it forever and never consider it as a vector to enter into the basis at any iteration.

For clear understanding of the procedure see following examples.

Note: A solution to the problem which does not contain an artificial variable in the basis, represents a feasible solution to the problem.

ILLUSTRATIVE EXAMPLES

Example 1: Applying Big-M method to solve the following L.P.P.

$$\begin{array}{ll} \text{Maximize} & Z = 4x_1 + 2x_2 \\ \text{subject to} & 3x_1 + x_2 \leq 27 \\ & x_1 + x_2 \geq 21 \\ & x_1, x_2 \geq 0 \end{array}$$

Solution: The problem is of maximization and all b_i 's are positive.

Introducing the necessary slack variable s_1 , in first constraint, surplus variable s_2 in second constraint and artificial variable x_a in second constraints to obtain a unit matrix I_2 , the problem reduces to the following form:

$$\begin{aligned} \text{Max.} \quad & Z = 4x_1 + 2x_2 + 0 \cdot s_1 + 0 \cdot s_2 - M \cdot x_a \\ \text{s.t.} \quad & 3x_1 + x_2 + s_1 + 0 \cdot s_2 + 0 \cdot x_a = 27 \\ & x_1 + x_2 + 0 \cdot s_1 - s_2 + x_a = 21 \\ & x_1, x_2, s_1, s_2, x_a \geq 0. \end{aligned}$$

Note that the columns of the coefficients of s_1 and x_a in the equations form a unit matrix I_2 , so these variables are taken as basic variables.

Taking $x_1 = 0$, $x_2 = 0$, $s_2 = 0$, we get $s_1 = 27$, $x_a = 21$ which is the initial B.F.S.

The solution to the problem using simplex method is given in the following table.

Table 2.10

	c_j		c_B	x_B	4	2	0	0	$-M$	Min. Ratio $\frac{x_B/Y_1}{y_{i1}} > 0$
					Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	A	
I	$Y_3(S_1)$	0	$s_1 = 27$	3		1	1	0	0	$27/3 \rightarrow$ (min.)
	A	$-M$	$x_a = 21$	1	1	0	0	-1	1	21/1
	$Z = c_B x_B$ $= -21M$	Δ_j		$4 + M$ $\uparrow$	$2 + M$	0	$-M$	0		x_B/Y_2 $y_{i2} > 0$
II	Y_1	4	$x_1 = 9$	1	1/3	1/3	0	0		27
	A	$-M$	$x_a = 12$	0	2/3	-1/3	-1	1		$18 \rightarrow$ (min.)
	$Z = c_B x_B$ $= -36 - 12M$	Δ_j		0	$\frac{2}{3}(M+1)$ $\uparrow$	$-\frac{1}{3}(M+4)$	$-M$	0		x_B/Y_4 $y_{i4} > 0$
III	Y_1	4	$x_1 = 3$	1	0	1/2	1/2			$6 \text{ (min.)} \rightarrow$
	Y_2	2	$x_2 = 18$	0	1	-1/2	-3/2			—
	$Z = c_B x_B = 48$	Δ_j		0	0	-1	1 $\uparrow$			
IV	Y_4	0	$s_2 = 6$	2	0	1	1			
	Y_2	2	$x_2 = 27$	3	1	1	0			
	$Z = c_B x_B = 54$	Δ_j		-2	0	-2	0			

Computation of Δ_j . For the first table I

$$\Delta_1 = c_1 - c_B Y_1 = 4 - (0, -M) [3, 1] = 4 + M$$

$$\Delta_2 = c_2 - c_B Y_2 = 2 - (0, -M) [1, 1] = 2 + M$$

$$\Delta_4 = c_4 - c_B Y_4 = 0 - (0, -M) [0, -1] = -M$$

$$\Delta_3 = 0 = \Delta_5$$

For the table II

$$\Delta_1 = 0 = \Delta_5, \Delta_2 = c_2 - c_B Y_2 = 2 - (4 - M) [1/3, 2/3] = \frac{2}{3} (M + 1)$$

$$\Delta_3 = c_3 - c_B Y_3 = 0 - (4 - M) [1/3, -1/3] = -(4 + M)/3,$$

$$\Delta_4 = c_4 - c_B Y_4 = 0 - (4, -M) [0, -1] = -M$$

For the table III

$$\Delta_1 = 0 = \Delta_2, \Delta_3 = c_3 - c_B Y_3 = 0 - (4, 2) [1/2, -1/2] = -1,$$

$$\Delta_4 = c_4 - c_B Y_4 = 0 - (4, 2) [1/2, -3/2] = 1$$

For the table IV

$$\Delta_1 = c_1 - c_B Y_1 = 4 - (0, 2) [2, 3] = -2, \Delta_2 = 0$$

$$\Delta_3 = c_3 - c_B Y_3 = 0 - (0, 2) [1, 1] = -2, \Delta_4 = 0$$

In the last table all $\Delta_j \leq 0$ and no artificial variable appears in the basis, so the optimal solution of the given problem is

$$x_1 = 0, x_2 = 27, \text{ and Max. } Z = 54.$$

Example 2: Apply Big-M Method to solve the following L.P.P.

$$\text{Maximize} \quad Z = -2x_1 - x_2$$

$$\text{subject to} \quad 3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4, x_1, x_2 \geq 0.$$

Solution: The problem is of maximization and all b_i 's are positive. Introducing the surplus variable s_1 , slack variable s_2 and artificial variables x_{a_1}, x_{a_2} , to obtain unit matrix I_3 the problem reduces to the form

$$\begin{aligned}
 \text{Max.} \quad & Z = -2x_1 - x_2 + 0.s_1 + 0.s_2 - M.x_{a_1} - M.x_{a_2} \\
 \text{s.t.} \quad & 3x_1 + x_2 + x_{a_1} = 3 \\
 & 4x_1 + 3x_2 - s_1 + x_{a_2} = 6 \\
 & x_1 + 2x_2 + s_2 = 4 \\
 \text{and} \quad & x_1, x_2, s_1, s_2, x_{a_1}, x_{a_2} \geq 0
 \end{aligned}$$

Taking $x_1 = 0, x_2 = 0$, we get $x_{a_1} = 3, x_{a_2} = 6, s_2 = 4$, which is the starting B.F.S.

The solution of the problem by simplex method is given in the following table:

Table 2.11

	B	$\mathbf{c}_B$	c_j	$\mathbf{x}_B$	-2	-1	0	0	$-M$	$-M$	Min. Ratio $\mathbf{x}_B/Y_1$ $y_{i1} > 0$
					Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	A_1	A_2	
I	A_1	$-M$	$x_{a_1} = 3$	3	1	0	0	1	0		$3/3 = 1 \rightarrow$ (min.)
	A_2	$-M$	$x_{a_2} = 6$	4	3	-1	0	0	1		$6/4 = 3/2$
	$Y_4(S_2)$	0	$s_2 = 4$	1	2	0	1	0	0		$4/1 = 4$
	$Z = \mathbf{c}_B \mathbf{x}_B$ $= -9\ M$	Δ_j	$-2 + 7M$ $\uparrow$	$-1 + 4M$	$-M$	0	0 $\downarrow$	0		$\mathbf{x}_B/Y_2$ $y_{i2} > 0$	
II	Y_1	-2	$x_1 = 1$	1	$1/3$	0	0	0			3
	A_2	$-M$	$x_{a_2} = 2$	0	5/3	-1	0	1			$6/5$ (min.) $\rightarrow$
	$Y_4(S_2)$	0	$s_2 = 3$	0	$5/3$	0	1	0			$9/5$
	$Z = \mathbf{c}_B \mathbf{x}_B$ $= -2 - 2\ M$	Δ_j	0	$\frac{1}{3}(-1 + 5M)$ $\uparrow$	$-M$	0	0 $\downarrow$				
III	Y_1	-2	$x_1 = 3/5$	1	0	$1/5$	0				
	Y_2	-1	$x_2 = 6/5$	0	1	$-3/5$	0				
	$Y_4(S_2)$	0	$s_2 = 1$	0	0	1	1				
	$Z = \mathbf{c}_B \mathbf{x}_B$ $= -12/5$	Δ_j	0	0	$-1/5$	0					

Computation of Δ_j . For the table I

$$\Delta_1 = c_1 - c_B Y_1 = -2 - (-M, -M, 0) [3, 4, 1] = -2 + 7M$$

$$\Delta_2 = c_2 - c_B Y_2 = -1 - (-M, -M, 0) [1, 3, 2] = -1 + 4M$$

$$\Delta_3 = c_3 - c_B Y_3 = 0 - (-M, -M, 0) [0, -1, 0] = -M$$

$$\Delta_4 = 0 = \Delta_5 = \Delta_6$$

For the table II

$$\Delta_1 = 0 = \Delta_4 = \Delta_6$$

$$\Delta_2 = c_2 - c_B Y_2 = -1 - (-2, -M, 0) [1/3, 5/3, 5/3] = \frac{1}{3}(-1 + 5M)$$

$$\Delta_3 - c_3 - c_B Y_3 = 0 - (-2, -M, 0) [0, -1, 0] = -M$$

For the table III

$$\Delta_1 = 0 = \Delta_2 = \Delta_4$$

$$\Delta_3 = c_3 - c_B Y_3 = 0 - (-2, -1, 0) [1/5, -3/5, 1] = -1/5$$

In the last table all $\Delta_j \leq 0$, and no artificial variable appears in the basis, therefore this solution is optimal.

Hence the optimal solution is

$$x_1 = 3/5, x_2 = 6/5 \text{ and Max. } Z = -12/5.$$

Example 3: Apply Big-M Method to solve the following L.P.P.

$$\text{Maximize} \quad Z = x_1 + 2x_2 + 3x_3 - x_4$$

$$\text{subject to} \quad x_1 + 2x_2 + 3x_3 = 15$$

$$2x_1 + x_2 + 5x_3 = 20$$

$$x_1 + 2x_2 + x_3 + x_4 = 10,$$

$$x_1, x_2, x_3, x_4 \geq 0.$$

Solution: The given problem is of maximization and all b_i 's are positive. Also the constraints are equations. Examining the constraints we observe that in order to obtain a unit matrix of order 3 we need two more unit vectors as one unit vector is formed by the coefficients of x_4 . Therefore, introducing two artificial variables x_{a_1} and x_{a_2} in the first two constraints and assigning large negative cost $-M$ to the artificial variables, the given problem becomes

$$\text{Max.} \quad Z = x_1 + 2x_2 + 3x_3 - x_4 - M.x_{a_1} - M.x_{a_2}$$

$$\text{subject to} \quad x_1 + 2x_2 + 3x_3 + 0x_4 + x_{a_1} + 0.x_{a_2} = 15$$

$$2x_1 + x_2 + 5x_3 + 0x_4 + 0x_{a_1} + x_{a_2} = 20$$

$$x_1 + 2x_2 + x_3 + x_4 + 0.x_{a_1} + 0.x_{a_2} = 10$$

Taking $x_1 = 0, x_2 = 0, x_3 = 0$, we get $x_{a_1} = 15, x_{a_2} = 20, x_4 = 10$, which is the initial B.F.S.

The solution to the problem using simplex method is given in the following table:

Table 2.12

	B	c_B	c_j	x_B	1	2	3	-1	-M	-M	Min. ratio
					Y_1	Y_2	Y_3	Y_4	A_1	A_2	$x_B / Y_3, j_{i3} > 0$
I	A_1	-M	15		1	2	3	0	1	0	5
	A_2	-M	20		2	1	5	0	0	1	$4 \rightarrow$ (min)
	Y_4	-1	10		1	2	1	1	0	0	10
	$Z = c_B x_B$ $= -35M - 10$		Δ_j		$3M + 2$	$3M + 4$	$8M + 4$ $\uparrow$	0	0	0	$x_B / Y_2, j_{i2} > 0$
II	A_1	-M	3		-1/5	7/5	0	0	1		$15/7 \rightarrow$ (min)
	Y_3	3	4		2/5	1/5	1	0	0		20
	Y_4	-1	6		3/5	9/5	0	1	0		10/3
	$Z = c_B x_B$ $= -3M + 6$		Δ_j		$\frac{(-M+2)}{5}$	$\frac{7M+16}{5}$ $\uparrow$	0	0	0		$x_B / Y_i, j_{i1} > 0$
III	Y_2	2	15/7		-1/7	1	0	0			—
	Y_3	3	25/7		3/7	0	1	0			25/3
	Y_4	-1	15/7		6/7	0	0	1			$5/2$ (min.) $\rightarrow$
	$Z = c_B x_B$ $= 90/7$		Δ_j		$6/7$ $\uparrow$	0	0	0 $\downarrow$			
IV	Y_2	2	5/2		0	1	0	1/6			
	Y_3	3	5/2		0	0	1	-1/2			
	Y_1	1	5/2		1	0	0	7/6			
	$Z = c_B x_B$ $= 15$		Δ_j		0	0	0	-1			

Computation of Δ_j . By $\Delta_j = c_j - c_B Y_j$

For the table I.

$$\Delta_1 = c_1 - c_B Y_1 = 1 - (-M, -M, -1)(1, 2, 1) = 3M + 2,$$

$$\Delta_2 = c_2 - c_B Y_2 = 2 - (-M, -M, -1)(2, 1, 2) = 3M + 4$$

$$\Delta_3 = c_3 - c_B Y_3 = 3 - (-M, -M, -1)(3, 5, 1) = 8M + 4.$$

$$\Delta_4 = 0 = \Delta_5 = \Delta_6$$

For the table II.

$$\Delta_1 = 1 - (-M, 3, -1)(-1/5, 2/5, 3/5) = (-M + 2)/5.$$

$$\Delta_2 = 2 - (-M, 3, -1)(7/5, 1/5, 9/5) = (7M + 16)/5.$$

$$\Delta_3 = 0 = \Delta_4 = \Delta_5$$

For the table III.

$$\Delta_1 = 1 - (2, 3, -1)(-1/7, 3/7, 6/7) = 6/7, \Delta_2 = 0 = \Delta_3 = \Delta_4$$

For the table IV.

$$\Delta_1 = 0 = \Delta_2 = \Delta_3$$

$$\Delta_4 = c_4 - c_B Y_4 = -1 - (2, 3, 1)[1/6, -1/2, 7/6] = -1$$

In the last table all $\Delta_j \leq 0$ and no artificial variable appears in the basis, therefore this solution is optimal.

Hence, the optimal solution is

$$x_1 = 5/2 = x_2 = x_3, x_4 = 0 \text{ and Max. } Z = 15.$$

2.5.2 Method 2. Two Phase Method

As an alternative to the Big- M method, there is another method for dealing with linear programming problems involving artificial variables. This is called the **two phase method** and as its name suggests it separates the solution procedure into two phases.

In phase I, all the artificial variables are eliminated from the basis.

In phase II, we use the solution from phase I as the initial basic feasible solution and use the simplex method to determine the optimal solution.

Computational Procedure of Two Phase Method

Phase I: Step 1: After making all b_i 's positive, convert each of the constraints into equations, by introducing slack, surplus and artificial variables.

Step 2: Take new objective function Z' by, assigning zero coefficient to each of the primary, slack and surplus variables and the coefficient (-1) to each of the artificial variables. As a result the new objective function is

$$Z' = -(\text{sum of the artificial variables})$$

Step 3: Solve the problem formed in step 2 by applying the simplex method. If the original problem has a feasible solution then this problem shall have an optimal solution with optimal value of the objective function Z' equal to zero as each of the artificial variables will be equal to zero.

If $\max Z' < 0$ and at least one artificial variable appears in the optimum basis at a positive level then the given problem *does not possess any feasible solution*.

If $\max Z' = 0$ and at least one artificial variable appears in the optimum basis at zero level, we proceed to phase II.

If $\max Z' = 0$ and no artificial variable appears in the optimum basis then also we proceed to phase II.

Phase II: In phase II start with the (optimal) solution contained in the final simplex table of the phase I. Assign the actual costs to the variables in the objective function and a zero cost to every slack and surplus variable. Eliminate the artificial variables which are non-basic at the end of the phase-I. Remove c_j row values of the optimum table and replace them by c_j values of the original problem. Now apply simplex algorithm to the problem contained in the new table to obtain the optimal solution.

Disadvantages of Big-M Method Over Two Phase Method

1. We can always use Big-M method to check the existence of a feasible solution but its computational procedure may be inconvenient because of the manipulation of the constant M . Two phase method eliminates the artificial variables in the beginning.
2. When we solve a problem on a digital computer, we have to assign some numerical value to M which must be larger than the values $c_1, c_2, \dots$ present in the objective function. But a computer has only a fixed number of digits.

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Example 1: Solve the following problem by using the two phase method:

$$\begin{array}{ll} \text{Minimize} & Z = 40x_1 + 24x_2 \\ \text{subject to} & 20x_1 + 50x_2 \geq 4800 \\ & 80x_1 + 50x_2 \geq 7200, \quad x_1, x_2 \geq 0. \end{array}$$

Solution: The given problem is of minimization. Converting it to maximization by taking the objective function as

$$Z' = -Z = -40x_1 - 24x_2$$

Introducing the surplus variables s_1, s_2 and artificial variables x_{a1}, x_{a2} , the constraint inequalities, reduce to the following equations:

$$\begin{aligned} 20x_1 + 50x_2 - s_1 + 0s_2 + x_{a1} + 0x_{a2} &= 4800 \\ 80x_1 + 50x_2 + 0s_1 - s_2 + 0x_{a1} + x_{a2} &= 7200, \\ x_1, x_2, s_1, s_2, x_{a1}, x_{a2} &\geq 0. \end{aligned}$$

Phase I: Assigning cost -1 to artificial variables, cost 0 to all other variables, the new objective function of auxiliary problem becomes

$$\text{Max. } Z'' = 0x_1 + 0x_2 + 0x_3 + 0x_4 - x_{a_1} - x_{a_2}$$

subject to the constraints given above.

Now applying the simplex method in the usual manner, we have the following table:

Table 2.13

B	c_B	x_B	c_j						Min. Ratio $x_B/Y_1, Y_{i1} > 0$
			0	0	0	0	-1	-1	
A_1	-1	4800	20	50	-1	0	1	0	4800/20 = 240
A_2	-1	7200	80	50	0	-1	0	1	7200/80 = 90 → (min.)
$Z'' = c_B x_B = -12000$	Δ_j		100 ↑	100	-1	-1	0	0 ↓	x_B/Y_2 $y_{i2} > 0$
A_1	-1	3000	0	75/2	-1	1/4	1		80 → (min.)
Y_1	0	90	1	5/8	0	-1/80	0		144
$Z'' = -2910$	Δ_j		0	75/2 ↑	0	1/4	0 ↓		
Y_2	0	80	0	1	-2/75	1/50			
Y_1	0	40	1	0	1/60	-1/60			
$Z'' = c_B x_B = 0$	Δ_j		0	0	0	0			

Since all $\Delta_j \leq 0$, $\text{max. } Z'' = 0$ and no artificial variable appears in the basis, therefore an optimum solution to the auxiliary problem has been attained.

Phase II: Now assign the actual costs to the original variables and cost zero to the surplus variables, the objective function becomes

$$\text{Max. } Z' = -40x_1 - 24x_2 + 0s_1 + 0s_2$$

Replace the c_j row values in the final simplex table of phase I by the c_j values of the objective function Z' .

Also deleting the artificial variables from the final simplex table of phase I, we write the first simplex table of phase II. Now solution of the problem applying simplex method in the usual manner is given in the following table:

Table 2.14

B	c_B	c_j	-40	-24	0	0	Min. Ratio $x_B/Y_3, y_{i3} > 0$
		x_B	Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	
Y_2	-24	80	0	1	-2/75	1/50	2400 → (min.)
Y_1	-40	40	1	0	1/60	-1/60	
$Z' = c_B x_B = -3520$		Δ_j	0 ↓	0	2/75 ↑	-38/75	
Y_2	-24	144	8/5	1	0	-1/50	
$Y_3(S_1)$	0	2400	60	0	1	-1	
$Z' = c_B x_B = -3456$		Δ_j	-8/5	0	0	-12/25	

Since all $\Delta_j \leq 0$, therefore the solution obtained is optimal.

Hence, the optimal solution is $x_1 = 0$, $x_2 = 144$ and Min. $Z = -\text{Max. } Z' = 3456$.

Example 2: Solve the following L.P.P. by using the two phase method.

$$\begin{aligned}
 &\text{Maximize} && Z = 5x_1 + 8x_2 \\
 &\text{subject to} && 3x_1 + 2x_2 \geq 3 \\
 &&& x_1 + 4x_2 \geq 4 \\
 &&& x_1 + x_2 \leq 5 \\
 &&& x_1, x_2 \geq 0.
 \end{aligned}$$

Solution: Introducing the surplus variables s_1, s_2 in first two constraints, slack variable s_3 in the third constraint and artificial variables x_{a_1}, x_{a_2} the constraint inequalities reduces to the following equations

$$\begin{aligned}
 3x_1 + 2x_2 - s_1 + 0.s_2 + 0.s_3 + x_{a_1} + 0.x_{a_2} &= 3 \\
 x_1 + 4x_2 + 0s_1 - s_2 + 0s_3 + 0.x_{a_1} + x_{a_2} &= 4 \\
 x_1 + x_2 + 0.s_1 + 0.s_2 + s_3 + 0.x_{a_1} + 0.x_{a_2} &= 5 \\
 x_1, x_2, s_1, s_2, s_3, x_{a_1}, x_{a_2} &\geq 0.
 \end{aligned}$$

Phase I. Assigning costs -1 to artificial variables and cost 0 to all other variables, the new objective function of the auxiliary problem, becomes

$$\text{Max. } Z' = 0.x_1 + 0.x_2 + 0.s_1 + 0.s_2 + 0.s_3 - x_{a_1} - x_{a_2},$$

subject to the constraint equations given above.

Taking $x_1=0, x_2=0, s_1=0, s_2=0$, we get $x_{a_1}=3, x_{a_2}=4, s_3=5$, which is the starting B.F.S.

Now applying the simplex method to this auxiliary problem in the usual manner, we have the following table:

Table 2.15

	B	c_B	c_j	0	0	0	0	0	-1	-1	Min. Ratio
			x_B	Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	$Y_5(S_3)$	A_1	A_2	$x_B / Y_2, \quad J_{i2} > 0$
I	A_1	-1	3	3	2	-1	0	0	1	0	3/2
	A_2	-1	4	1	4	0	-1	0	0	1	4/4 → (min.)
	$Y_5(S_3)$	0	5	1	1	0	0	1	0	0	5/1
	$Z' = c_B x_B = -7$		Δ_j	4	6 ↑	-1	-1	0	0	0	$x_B / Y_1 \quad J_{i1} > 0$
II	A_1	-1	1	5/2	0	-1	1/2	0	1		2/5 → (min.)
	Y_2	0	1	1/4	1	0	-1/4	0	0		4
	$Y_5(S_3)$	0	4	3/4	0	0	1/4	1	0		16/3
	$Z' = c_B x_B = -1$		Δ_j	5/2 ↑	0	-1	1/2	0	0		
III	Y_1	0	2/5	1	0	-2/5	1/5	0			
	Y_2	0	9/10	0	1	1/10	-3/10	0			
	$Y_5(S_3)$	0	37/10	0	0	3/10	1/10	1			
	$Z' = c_B x_B = 0$		Δ_j	0	0	0	0	0			

Computation of Δ_j . By $\Delta_j = c_j - c_B Y_j$

For the table I.

$$\Delta_1 = c_1 - c_B Y_1 = 0 - (-1, -1, 0)(3, 1, 1) = 4$$

$$\Delta_2 = 0 - (-1, -1, 0)(2, 4, 1) = 6, \quad \Delta_3 = 0 - (-1, -1, 0)(-1, 0, 0) = -1$$

$$\Delta_4 = 0 - (-1, -1, 0)(0, -1, 0) = -1, \quad \Delta_5 = 0 = \Delta_6 = \Delta_7$$

In the last table all $\Delta_j \leq 0$, therefore this solution is optimal solution to the auxiliary problem.

Phase II. Now the first simplex table of phase II is written as follows.

Assigning the original costs to the original variables, cost zero to slack and surplus variables, the objective function becomes

$$\text{Max. } Z = 5x_1 + 8x_2 + 0 \cdot s_1 + 0 \cdot s_2 + s_3 \cdot x_5.$$

Then deleting the artificial variable column from the last simplex table of phase I, we write the first simplex table of phase II.

The solution of the problem applying simplex method is given in the following table:

Table 2.16

B	c_B	c_j	5	8	0	0	0	Min. Ratio $x_B/Y_4, y_{i4} > 0$
		x_B	Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	$Y_5(S_3)$	
Y_1	5	2/5	1	0	-2/5	1/5	0	2 (min.) $\rightarrow$
Y_2	8	9/10	0	1	1/10	-3/10	0	—
$Y_5(S_3)$	0	37/10	0	0	3/10	1/10	1	37
$Z = c_B x_B = 46/5$		Δ_j	0	0	6/5	7/5 $\uparrow$	0	x_B/Y_3
Y_4	0	2	5	0	-2	1	0	—
Y_2	8	3/2	3/2	1	-1/2	0	0	—
$Y_5(S_3)$	0	7/2	-1/2	0	1/2	0	1	7 (min.) $\rightarrow$
$Z = c_B x_B = 12$		Δ_j	-7	0	4 $\uparrow$	0	0	
Y_4	0	16	3	0	0	1	4	
Y_2	8	5	1	1	0	0	1	
Y_3	0	7	-1	0	1	0	2	
$Z = c_B x_B = 40$		Δ_j	-3	0	0	0	-8	

Computation of Δ_j . By $\Delta_j = c_j - c_B Y_j, \Delta_1 = 0 = \Delta_2 = \Delta_5$

$$\Delta_3 = c_3 - c_B Y_3 = 0 - (5, 8, 0) (-2/5, 1/10, 3/10) = 6/5,$$

$$\Delta_4 = 0 - (5, 8, 0) (1/5, -3/10, 1/10) = 7/5$$

In the last table all $\Delta_j \leq 0$, therefore this solution is optimal.

Hence, the optimal solution is

$$x_1 = 0, x_2 = 5 \text{ and } \text{Max. } Z = 40.$$

Exercise 2.2

Solve the following L.P.P. by Big M-Method

1. Minimize $Z = 5x_1 + 6x_2$ subject to $3x_1 + x_2 \geq 1200$ $2x_1 + 5x_2 \geq 1500$ and $x_1, x_2 \geq 0$	2. Minimize $Z = 4x_1 + 6x_2$ subject to $x_1 + 2x_2 \geq 80$ $3x_1 + x_2 \geq 75$ and $x_1, x_2 \geq 0$
3. Minimize $Z = 60x_1 + 80x_2$ subject to $x_1 + x_2 = 500$ $x_1 \leq 400$ $x_2 \geq 200$ and $x_1, x_2 \geq 0$	4. Maximize $Z = x_1 + x_2 + 3x_3$ subject to $3x_1 + 2x_2 + x_3 \leq 3$ $2x_1 + x_2 + 2x_3 \geq 3$ and $x_1, x_2, x_3 \geq 0$
5. Maximize $Z = x_1 + 3x_2 - 2x_3$ subject to $-x_1 - 2x_2 - 2x_3 = -6$ $-x_1 - x_2 + x_3 \leq -2$ and $x_1, x_2, x_3 \geq 0$	6. Minimize $Z = x_1 - x_2 - 3x_3$ subject to $x_1 + 2x_2 + x_3 = 4$ $x_2 + x_3 = 1$ $x_1 + x_2 \leq 6$ and $x_1, x_2, x_3 \geq 0$

Solve the following L.P.P. by two phase method.

7. Minimize $Z = x_1 + x_2$ subject to $2x_1 + x_2 \geq 4$ $x_1 + 7x_2 \geq 7$ and $x_1, x_2 \geq 0$	8. Maximize $Z = 3x_1 - x_2$ subject to $2x_1 + x_2 \geq 2$ $x_1 + 3x_2 \leq 2$ $x_2 \leq 4$ and $x_1, x_2 \geq 0$
9. Minimize $Z = 2x_1 + 3x_2$ subject to $x_1 + x_2 \geq 5$ $x_1 + 2x_2 \geq 6$ and $x_1, x_2 \geq 0$	10. Maximize $Z = 30x_1 + 10x_2$ subject to $2x_1 + 8x_2 \geq 80$ $5x_1 + 2x_2 \geq 50$ and $x_1, x_2 \geq 0$

11. A finished product must weight exactly 150 gms. The two raw materials used in manufacturing the product are : A with cost of ₹ 2 per unit and B with a cost of ₹ 8 per unit. At least 14 units of B and not more than 20 units of A must be used. Each unit of A and B weights 5 and 10 grams respectively.

How much of each type of raw material should be used for each unit of the final product in order to minimize the cost?

Answers 2.2

1.	$x_1 = 4500/13, x_2 = 2100/13,$ Min. $Z = 2700$	2.	$x_1 = 14, x_2 = 33,$ Min. $Z = 254$
3.	$x_1 = 300, x_2 = 200,$ Min. $Z = 3400$	4.	$x_1 = 3/4, x_2 = 0, x_3 = 3/4,$ Max. $Z = 3$
5.	$x_1 = 0, x_2 = 3, x_3 = 0,$ Max. $Z = 9$	6.	$x_1 = 3, x_2 = 0, x_3 = 1,$ Min. $Z = 0$
7.	$x_1 = 21/13, x_2 = 10/13,$ Min. $Z = 31/13$	8.	$x_1 = 2, x_2 = 0,$ Max. $Z = 6$
9.	$x_1 = 4, x_2 = 1,$ Min. $Z = 11$	10.	$x_1 = 0, x_2 = 25,$ Min. $Z = 250$
11.	A, 2 units; B, 14 units. Min. cost = ₹ 116.		

2.6 Linear Programming Problem (Some Special Cases)

2.6.1 L.P.P. having no Feasible Solution (Infeasibility)

A L.P.P. is said to have a feasible solution if it have a solution which satisfies all the constraints and the non-negative restrictions. Some times the constraints may be inconsistent so that there is no feasible solution to the problem. Such a situation is called *infeasibility*. *In the case of infeasibility one or more artificial variables appear in the basis at positive level in the final simplex table.*

ILLUSTRATIVE EXAMPLES

Example 1: Solve the following L.P.P. by using the two phase method

$$\begin{aligned}
 &\text{Minimize} && Z = x_1 - 2x_2 - 3x_3 \\
 &\text{subject to} && -2x_1 + x_2 + 3x_3 = 2 \\
 &&& 2x_1 + 3x_2 + 4x_3 = 1, \\
 &&& x_1, x_2, x_3 \geq 0.
 \end{aligned}$$

Solution: Converting the objective function to maximization form by substituting $Z' = -Z$, we get

$$\text{Max. } Z' = -x_1 + 2x_2 + 3x_3$$

Introducing the artificial variables x_{a_1} and x_{a_2} , the constraint equations become

$$-2x_1 + x_2 + 3x_3 + x_{a_1} = 2$$

$$2x_1 + 3x_2 + 4x_3 + x_{a_2} = 1$$

where

$$x_1, x_2, x_3, x_{a_1}, x_{a_2} \geq 0.$$

Phase I. Assigning costs – 1 to artificial variables and costs 0 to all other variables, the new objective function of auxiliary problem is

$$\text{Max. } Z' = 0x_1 + 0x_2 + 0x_3 - x_{a_1} - x_{a_2}$$

subject to the constraints mentioned above.

Taking $x_1 = 0, x_2 = 0, x_3 = 0$, we get $x_{a_1} = 2, x_{a_2} = 1$, which is the starting B.F.S.

Now apply simplex method in the usual manner to remove artificial variables.

Table 2.17

	c_j		0	0	0	– 1	– 1	Min. Ratio
B	c_B	x_B	Y_1	Y_2	Y_3	A_1	A_2	$x_B / Y_3, Y_3 > 0$
A_1	–1	2	–2	1	3	1	0	2/3
A_2	–1	1	2	3	4	0	1	1/4 → (min.)
$Z' = -3$	Δ_j		0	4	7 ↑	0	0	
A_1	–1	5/4	–7/2	–5/4	0	1	–3/4	
Y_3	0	1/4	1/2	3/4	1	0	1/4	
$Z' = -5/4$	Δ_j		–7/4	–5/4	0	0	–3/4	

Since all Δ_j 's are negative or zero, an optimum basic feasible solution to the auxiliary problem has been attained. But the artificial variable x_{a_1} (column vector A_1) appears in the basic solution at a positive level. Hence the original L.P.P. does not possess any feasible solution.

Note: Students are advised to solve the problem by Big-M method also. (For practice)

Example 2: Apply Big-M method to solve the L.P.P.

$$\text{Maximize.} \quad Z = -x_1 - x_2$$

subject to

$$3x_1 + 2x_2 \geq 30$$

$$-2x_1 + 3x_2 \leq -30$$

$$x_1 + x_2 \leq 5,$$

$$x_1, x_2 \geq 0.$$

Solution: The given problem is of maximization.

Since b_i in second constraint is negative, therefore multiplying both sides by -1 , we get

$$2x_1 - 3x_2 \geq 30.$$

To convert inequalities of constraints into equations, introducing slack, surplus and artificial variables wherever necessary, the given problem becomes

$$\text{Max. } Z = -x_1 - x_2 + 0s_1 + 0s_2 + 0s_3 - Mx_{a_1} - Mx_{a_2}$$

subject to

$$3x_1 + 2x_2 - s_1 + 0s_2 + 0s_3 + x_{a_1} + 0x_{a_2} = 30$$

$$2x_1 - 3x_2 + 0s_1 - s_2 + 0s_3 + 0x_{a_1} + x_{a_2} = 30$$

$$x_1 + x_2 + 0s_1 + 0s_2 + s_3 + 0x_{a_1} + 0x_{a_2} = 5$$

where

$$x_1, x_2, s_1, s_2, s_3, x_{a_1}, x_{a_2} \geq 0.$$

Taking, $x_1 = 0, x_2 = 0, x_3 = 0, x_4 = 0$, we get $x_5 = 5, x_{a_1} = 30, x_{a_2} = 30$ which is the initial B.F.S.

The solution to the problem using simplex method is given below:

Table 2.18

B	c_B	c_j	x_B	-1	-1	0	0	0	$-M$	$-M$	Min. Ratio x_B / y_1 , $y_{i1} > 0$
				Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	$Y_5(S_3)$	A_1	A_2	
A_1	$-M$		$x_{a_1} = 30$	3	2	-1	0	0	1	0	10
A_2	$-M$		$x_{a_2} = 30$	2	-3	0	-1	0	0	1	15
$Y_5(S_3)$	0		$s_3 = 5$	1	1	0	0	1	0	0	5 (min.) →
$Z = c_B x_B$ $= -60M$		Δ_j		$5M - 1$ ↑	$-M$	$-M$	$-M$	0 ↓	0	0	
A_1	$-M$		$x_{a_1} = 15$	0	-1	-1	0	-3	1	0	
A_2	$-M$		$x_{a_2} = 20$	0	-5	0	-1	-2	0	1	
Y_1	-1		$x_1 = 5$	1	1	0	0	1	0	0	
$Z =$ $-35M - 5$		Δ_j		0	$-6M$	$-M$	$-M$	$-5M$	0	0	

Since all $\Delta_j \leq 0$, so the solution given in the final simplex table is optimal. But the artificial variables x_{a_1} and x_{a_2} appear in the basis having positive values which implies that the given L.P.P. has no *feasible solution*.

Note: Students are advised to solve the problem by Two-Phase Method also. (For practice)

2.6.2 L.P.P. having Unbounded Solution

A L.P.P. has unbounded solution if the feasible region is unbounded such that the value of the objective function can be increased or decreased (as required) indefinitely. In other words the solution of a L.P.P. is unbounded if any one or more of the decision variables can be increased or decreased (as the case may be) indefinitely without violating any of the constraints.

In the solution of a L.P.P. by simplex method, we select the variable to go out of the basis in the key row which correspond to min. ratio x_B/Y_k , $y_{ik} > 0$, where Y_k is the key column (corresponding to $\Delta_k = \max. \Delta_j > 0$) to enter the basis matrix. If all $y_{ik} \leq 0$, then we cannot select the key row *i.e.*, we cannot select the variable to leave the basis. In such situation we say that the given L.P.P. have **unbounded solution**. In other words, we say that a L.P.P. have an **unbounded solution** if all the elements in the key column are either zero or negative.

For clear understanding of the method see the following example.

Example 3: Solve the following L.P.P.

$$\begin{aligned} \text{Maximize} \quad & Z = 3x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & -3x_1 + 2x_2 + 2x_3 = 8 \\ & -3x_1 + 4x_2 + x_3 = 7 \\ \text{and} \quad & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Solution: Introducing the artificial variables x_{a_1}, x_{a_2} the given L.P.P. reduces to the following form

$$\begin{aligned} \text{Max.} \quad & Z = 3x_1 + 2x_2 + x_3 - Mx_{a_1} - Mx_{a_2} \\ & -3x_1 + 2x_2 + 2x_3 + x_{a_1} = 8 \\ & -3x_1 + 4x_2 + x_3 + x_{a_2} = 7 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \end{aligned}$$

Taking $x_1 = 0, x_2 = 0, x_3 = 0$, we get $x_{a_1} = 8, x_{a_2} = 7$ which is the starting B.F.S.

All computation work is done in the following table.

Table 2.19

B	c_B	x_B	c_j	3	2	1	$-M$	$-M$	Min. Ratio
			x_B	Y_1	Y_2	Y_3	A_1	A_2	x_B/Y_2 $y_{i2} > 0$
A_1	$-M$	$x_{a_1} = 8$		-3	2	2	1	0	4
A_2	$-M$	$x_{a_2} = 7$		-3	4	1	0	1	7/4 (Min.) →
$Z = c_B x_B$ $= -15M$		Δ_j		$-6M + 3$	$6M + 2$ ↑	$3M + 1$	0	0	X_B/Y_3 $y_{i3} \geq 0$
A_1	$-M$	9/2		-3/2	0	3/2	1		3 (Min.) →
Y_2	2	7/4		-3/4	1	1/4	0		7
$Z = c_B x_B$ $= 7/2 - 9M/2$		Δ_j		$(-3M + 9)/2$	0	$(3M + 1)/2$ ↑	0		X_B/Y_1
Y_3	1	3		-1	0	1			
Y_2	2	1		-1/2	1	0			
$Z = c_B x_B = 5$		Δ_j		5 ↑	0	0			

Here Y_1 is the incoming vector. All the elements of this column are negative. Thus, we cannot select the outgoing vector.

Hence, in this case the solution is unbounded.

2.6.3 L.P.P. with Unrestricted Variables

If in a L.P.P. a variable say x_i is unrestricted in sign *i.e.*, it can take positive, negative or zero value, then substitute $x_i = x_i' - x_i''$ where $x_i', x_i'' \geq 0$ and solve the problem as usual. In case more than one variable is unrestricted in sign, then we replace all these variable by the difference of two non-negative variables and solve as usual L.P.P. method.

For clear understanding of method see the following example.

Example 4: Maximize $Z = 2x_1 + 3x_2$

subject to

$$-x_1 + 2x_2 \leq 4$$

$$x_1 + x_2 \leq 6$$

$$x_1 + 3x_2 \leq 9, \quad x_1, x_2 \text{ are unrestricted.}$$

Solution: Taking $x_1 = x_1' - x_1''$, $x_2 = x_2' - x_2''$ s.t. $x_1', x_1'', x_2', x_2'' \geq 0$ and introducing the slack variables s_1, s_2 and s_3 the given problem reduces to

$$\begin{aligned}
 \text{Max.} \quad & Z = 2x_1' - 2x_1'' + 3x_2' - 3x_2'' + 0x_3 + 0x_4 + 0x_5 \\
 & -x_1' + x_1'' + 2x_2' - 2x_2'' + s_1 = 4 \\
 & x_1' - x_1'' + x_2' - x_2'' + s_2 = 6 \\
 & x_1' - x_1'' + 3x_2' - 3x_2'' + s_3 = 9
 \end{aligned}$$

Taking $x_1' = 0 = x_1'' = x_2' = x_2''$, we get $x_3 = 4, x_4 = 6, x_5 = 9$ which is the starting B.F.S. All computation work is done in the following table.

Table 2.20

B	c_B	c_j	2	-2	3	-3	0	0	0	Min. Ratio $x_B / Y_{2'}, y'_{i2} > 0$
		x_B	Y_1'	Y_1''	Y_2'	Y_2''	$Y_3(S_1)$	$Y_4(S_2)$	$Y_5(S_3)$	
Y_3	0	4	-1	1	2	-2	1	0	0	2 (min.) →
Y_4	0	6	1	-1	1	-1	0	1	0	6
Y_5	0	9	1	-1	3	-3	0	0	1	3
$Z = c_B x_B = 0$		Δ_j	2	-2	3 ↑	-3	0 ↓	0	0	$x_B / Y_1' y'_{i1} > 0$
Y_2'	3	2	-1/2	1/2	1	-1	1/2	0	0	—
Y_4	0	4	3/2	-3/2	0	0	-1/2	1	0	8/3
Y_5	0	3	5/2	-5/2	0	0	-3/2	0	1	6/5 (min.) →
$Z = c_B x_B = 6$		Δ_j	7/2 ↑	-7/2	0	0	-3/2	0	0 ↓	$x_B / Y_3 y'_{i3} > 0$
Y_2'	3	13/5	0	0	1	-1	1/5	0	1/5	13
Y_4	0	11/5	0	0	0	0	2/5	1	-3/5	11/2 (min.) →
Y_1'	2	6/5	1	-1	0	0	-3/5	0	2/5	
$Z = c_B x_B = 51/5$		Δ_j	0	0	0	0	3/5 ↑	0 ↓	-7/5	
Y_2'	3	3/2	0	0	1	-1	0	-1/2	1/2	
Y_3	0	11/2	0	0	0	0	1	5/2	-3/2	
Y_1'	2	9/2	1	-1	0	0	0	3/2	-1/2	
$Z = c_B x_B = 27/2$		Δ_j	0	0	0	0	0	-3/2	-1/2	

Since all $\Delta_j \leq 0$, so this solution is optimal. The optimal solution to the given L.P.P. is $x_1' = 9/2, x_1'' = 0, x_2' = 3/2, x_2'' = 0$ and Max. $Z = 27/2$

i.e., $x_1 = x_1' - x_1'' = 9/2, x_2 = x_2' - x_2'' = 3/2$, Max. $Z = 27/2$.

2.6.4 L.P.P. with Some Constant term in the Objective Function

In a L.P.P. when the objective function contains a constant term, then the problem is solved for a new objective function obtained by leaving this constant term. In the end the optimal value of the given objective function is obtained by adding this constant value in the optimal value of the new objective function.

For clear understanding of the method see the following example.

Example 5: Solve the following L.P.P. by simplex method

$$\text{Maximize} \quad Z = x_1 + x_2 + 3x_3 + 10$$

$$\text{s.t.} \quad x_1 + x_2 \leq 2$$

$$-x_1 + x_2 + 3x_3 \leq 3$$

$$\text{and} \quad x_1, x_2, x_3 \geq 0$$

Solution: Taking the objective function $Z' = Z - 10 = x_1 + x_2 + 3x_3$, and introducing the slack variables s_1 and s_2 , the given problem reduces to the problem

$$\text{Max} \quad Z' = x_1 + x_2 + 3x_3 + 0.s_1 + 0.s_2$$

$$\text{s.t.} \quad x_1 + x_2 + 0.x_3 + s_1 + 0.s_2 = 2$$

$$-x_1 + x_2 + 3x_3 + 0.s_1 + s_2 = 3$$

$$\text{and} \quad x_1, x_2, x_3, s_1, s_2 \geq 0.$$

Taking $x_1 = 0 = x_2 = x_3$, we get $s_1 = 2$ and $s_2 = 3$, which is the initial B.F.S.

The solution of the problem using simplex method is given in the following table.

Table 2.21

B	c_B	x_B	c_j					Min. Ratio $x_B / Y_3, y_{i3} > 0$
			Y_1	Y_2	Y_3	$Y_4(S_1)$	$Y_5(S_2)$	
Y_4	0	$s_1 = 2$	1	1	0	1	0	—
Y_5	0	$s_2 = 3$	-1	1	3	0	1	$3/3 = 1 \xrightarrow{\rightarrow} (\text{min.})$
$Z' = c_B x_B = 0$			Δ_j	1	1	3	0	$x_B / Y_1, y_{i1} > 0$
Y_4	0	$s_1 = 2$	1	1	0	1	0	$2/1 \xrightarrow{\rightarrow} (\text{min.})$
Y_3	3	$x_3 = 1$	-1/3	1/3	1	0	1/3	—
$Z' = c_B x_B = 3$			Δ_j	2	0	0	-1	
Y_1	1	$x_1 = 2$	1	1	0	1	0	
Y_3	3	$x_3 = 5/3$	0	2/3	1	1/3	1/3	
$Z' = c_B x_B = 7$			Δ_j	0	-2	0	-2	-1

In the last table all $\Delta_j \leq 0$, so the solution is optimal.

Hence the optimal solution is

$$x_1 = 2, x_2 = 0, x_3 = 5/3 \text{ and Max. } Z = \text{Max. } Z' + 10 = 17.$$

2.6.5 L.P.P. with Lower Bounds of Some or All Variables, Other than Zero

In some L.P.P., the lower bounds may be specified. For examples, in a L.P.P. it may be stipulated that $x_1 \geq c_1$, $x_2 \geq c_2$ etc. In such problems, we substitute $x_1 = c_1 + y_1$, $x_2 = c_2 + y_2$ etc., and then solve the problem in terms of y_1 and y_2 , where $y_1 \geq 0$, $y_2 \geq 0$ etc.

For clear understanding of the method see the following example:

Example 6: Solve the L.P.P.

$$\begin{aligned} \text{Maximize} \quad & Z = 3x_1 + 5x_2 + 4x_3 \\ \text{s.t.} \quad & 2x_1 - 3x_2 \leq 8 \\ & 2x_2 + 5x_3 \leq 10 \\ & 3x_1 + 2x_2 + 4x_3 \leq 15 \\ & x_1 \geq 2, x_2 \geq 4, x_3 \geq 0. \end{aligned}$$

Solution: Taking $x_1 = y_1 + 2$, $x_2 = y_2 + 4$, $x_3 = y_3$; the given problem reduces to

$$\begin{aligned} \text{Max.} \quad & Z = 3y_1 + 5y_2 + 4y_3 + 26 \\ \text{s.t.} \quad & 2y_1 - 3y_2 \leq 16, 2y_2 + 5y_3 \leq 2, \\ & 3y_1 + 2y_2 + 4y_3 \leq 1 \text{ and } y_1, y_2, y_3 \geq 0. \end{aligned}$$

Introducing the slack variables s_1, s_2, s_3 and leaving constant term 26 from Z in the beginning, the above L.P.P. reduces to

$$\begin{aligned} \text{Max. } Z' &= 3y_1 + 5y_2 + 4y_3 \text{ where } Z = Z' + 26 \\ \text{s.t.} \quad & 2y_1 - 3y_2 + s_1 = 16 \\ & 2y_2 + 5y_3 + s_2 = 2 \\ & 3y_1 + 2y_2 + 4y_3 + s_3 = 1 \\ \text{and} \quad & y_1, y_2, y_3, s_1, s_2, s_3 \geq 0 \end{aligned}$$

Taking $y_1 = 0 = y_2 = y_3$, we get $s_1 = 16, s_2 = 2, s_3 = 1$, which is the starting B.F.S.

The solution by simplex method is shown in the following table:

Table 2.22

B	c_B	c_j	3	5	4	0	0	0	Min. Ratio x_B/Y_2 , $y_{i2} > 0$
		x_B	Y_1	Y_2	Y_3	$Y_4(S_1)$	$Y_5(S_2)$	$Y_6(S_3)$	
Y_4	0	$s_1 = 16$	2	-3	0	1	0	0	—
Y_5	0	$s_2 = 2$	0	2	5	0	1	0	2/2
Y_6	0	$s_3 = 1$	3	2	4	0	0	1	1/2 (min.) $\rightarrow$
$Z' = c_B x_B$ $= 0$		Δ_j	3	5 $\uparrow$	4	0	0	0	
Y_4	0	$s_1 = 35/2$	13/2	0	6	1	0	3/2	
Y_5	0	$s_2 = 1$	-3	0	1	0	1	-1	
Y_2	5	$y_2 = 1/2$	3/2	1	2	0	0	1/2	
$Z' = c_B x_B$ $= 5/2$		Δ_j	-9/2	0	-6	0	0	-5/2	

Since all $\Delta_j \leq 0$, so this solution is optimal.

$\therefore$ Optimal solution is

$$y_1 = 0, y_2 = 1/2, y_3 = 0, \text{Max. } Z' = 5/2$$

$$\therefore x_1 = y_1 + 2 = 2, x_2 = y_2 + 4 = 9/2, x_3 = y_3 = 0, \text{Max. } Z = Z' + 26 = 57/2$$

$$\text{i.e., } x_1 = 2, x_2 = 9/2, x_3 = 0, \text{Max. } Z = 57/2$$

2.6.6 L.P.P. having More than One Optimum Solution

Some L.P.P. have more than one optimum solution. If in the final simplex table when an optimal solution is obtained we notice that, for a non-basic variable say x_j the value of $\Delta_j = c_j - c_B Y_j = 0$, then an alternate optimal solution to the L.P.P. exists. The alternate optimal solution is obtained by taking the corresponding column vector Y_j as the entering vector (key column) and finding outgoing vector (key row) by the usual minimum ratio rule.

For clear understanding of the procedure see the following example.

Example 7: Solve the following L.P.P. by simplex method:

$$\text{Maximize } Z = 4x_1 + 10x_2$$

$$\text{subject to } 2x_1 + x_2 \leq 50$$

$$2x_1 + 5x_2 \leq 100$$

$$2x_1 + 3x_2 \leq 90$$

$$\text{and } x_1, x_2 \geq 0.$$

Also find the alternative optimum solution if exists.

Solution: Introducing the slack variables s_1, s_2, s_3 the given problem becomes

$$\begin{aligned} \text{Max. } Z &= 4x_1 + 10x_2 + 0.s_1 + 0.s_2 + 0.s_3 \\ \text{s.t. } 2x_1 + x_2 + s_1 &= 50 \\ 2x_1 + 5x_2 + s_2 &= 100 \\ 2x_1 + 3x_2 + s_3 &= 90 \\ x_1, x_2, s_1, s_2, s_3 &\geq 0. \end{aligned}$$

Taking, $x_1=0, x_2=0$, we get $s_1=50, s_2=100, s_3=90$, which is the initial B.F.S.

The solution to the problem using simplex method is given in the following table:

Table 2.23

B	c_B	c_j	4	10	0	0	0	Min. Ratio
		x_B	Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	$Y_5(S_3)$	$x_B / Y_2, y_{i2} > 0$
Y_3	0	$s_1 = 50$	2	1	1	0	0	50/1
Y_4	0	$s_2 = 100$	2	5	0	1	0	100/5 (min.) $\rightarrow$
Y_5	0	$s_3 = 90$	2	3	0	0	1	90/3
$Z = c_B x_B = 0$		Δ_j	4	10 $\uparrow$	0	0	0	$x_B / Y_1, y_{i1} > 0$
Y_3	0	$s_1 = 30$	8/5	0	1	-1/5	0	75/4 (min.) $\rightarrow$
Y_2	10	$x_2 = 20$	2/5	1	0	1/5	0	50
Y_5	0	$s_3 = 30$	4/5	0	0	-3/5	1	75/2
$Z = c_B x_B = 200$		Δ_j	0 $\uparrow$	0	0	-2	0	

In the last table all $\Delta_j \leq 0$, therefore this solution is optimal.

The optimal solution is $x_1=0, x_2=20$ and Max. $Z=200$.

Here, corresponding to non-basic variable x_1 in the last table $\Delta_1=0$ and Y_1 is not in the basis B , therefore an alternative solution also exists. Thus the problem does not have unique solution.

Now taking Y_1 as incoming vector (key column) and using minimum ratio rule, we find that Y_3 is the outgoing vector (key row).

Taking column Y_1 in the basis matrix in place of Y_3 we obtain the next simplex table as follows:

Table 2.24

B	c_B	c_j	4	10	0	0	0	Min. Ratio
		x_B	Y_1	Y_2	$Y_3(S_1)$	$Y_4(S_2)$	$Y_5(S_3)$	
Y_1	4	$x_1 = 75/4$	1	0	$5/8$	$-1/8$	0	
Y_2	10	$x_2 = 25/2$	0	1	$-1/4$	$1/4$	0	
Y_5	0	$s_3 = 15$	0	0	$-1/2$	$1/10$	1	
$Z = c_B x_B = 200$		Δ_j	0	0	0	-2	0	

Here all $\Delta_j \leq 0$, so this solution is also optimal having the same maximum value of Z .

$\therefore$ Second optimal solution is $x_1 = 75/4, x_2 = 25/2$ and $\text{Max. } Z = 200$.

Hence the two optimal solutions of the problem are

- (i) $x_1 = 0, x_2 = 20, \text{Max. } Z = 200$ and
- (ii) $x_1 = 75/4, x_2 = 25/2, \text{Max. } Z = 200$.

Actually, all the points on the line segment joining the points $(0, 20)$ and $(75/4, 25/2)$ is the solution of the problem which are infinite in number having the same max. value of $Z = 200$.

For any arbitrary value of λ such that $0 \leq \lambda \leq 1$ the following table gives different optimal solutions which are infinite in number.

Variables	I Sol.	II Sol.	General Sol.
x_1	0	$75/4$	$0 \cdot \lambda + (75/4)(1 - \lambda)$
x_2	20	$25/2$	$20\lambda + (25/2)(1 - \lambda)$

Taking a particular value of λ , s.t. $0 \leq \lambda \leq 1$, say $\lambda = 1/2$, the third optimal solution is

$$x_1 = 75/8, x_2 = 65/4 \text{ and } \text{Max. } Z = 200.$$

Exercise 2.3

Solve the following L.P.P. by Simplex Method

1.	Maximize $Z = 3x_1 + 2x_2$ subject to $-x_1 + x_2 \geq 3$ $x_1 + 2x_2 \leq 2$ and $x_1, x_2 \geq 0$	2.	Maximize $Z = 3x_1 + 2x_2$ subject to $2x_1 + x_2 \leq 2$ $3x_1 + 4x_2 \geq 12$ and $x_1, x_2 \geq 0$
3.	Maximize $Z = 4x_1 + 3x_2$ subject to $x_1 + x_2 \leq 50$ $x_1 + 2x_2 \geq 80$ $3x_1 + 2x_2 \geq 140$ and $x_1, x_2 \geq 0$	4.	Maximize $Z = 10x_1 + 20x_2$ subject to $2x_1 + 4x_2 \geq 16$ $x_1 + 5x_2 \geq 15$ and $x_1, x_2 \geq 0$
5.	Maximize $Z = 5x_1 + 4x_2$ subject to $x_1 - x_2 \leq 8$ $x_1 \leq 7$ and $x_1, x_2 \geq 0$	6.	Maximize $Z = 2x_1 + x_2$ subject to $x_1 - x_2 \leq 10$ $2x_1 - x_2 \leq 40$ and $x_1, x_2 \geq 0$
7.	Maximize $Z = 3x_1 + 4x_2$ subject to $x_1 - x_2 \leq 1$ $-x_1 + x_2 \leq 2$ and $x_1, x_2 \geq 0$	8.	Maximize $Z = 5x_1 + 6x_2 + x_3$ subject to $9x_1 + 3x_2 - 2x_3 \leq 5$ $4x_1 + 2x_2 - x_3 \leq 2$ $x_1 - 4x_2 + x_3 \leq 3$ and $x_1, x_2, x_3 \geq 0$
9.	Maximize $Z = 8x_2$ subject to $x_1 - x_2 \geq 0$ $2x_1 + 3x_2 \leq -6$ and x_1, x_2 are unrestricted in sign.	10.	Maximize $Z = 8x_1 - 4x_2$ subject to $4x_1 + 5x_2 \leq 20$ $x_1 - 3x_2 \geq 23$ and $x_1 \geq 0, x_2$ is unrestricted in sign.
11.	Minimize $Z = x_1 + x_2 + x_3$ subject to $x_1 - 3x_2 + 4x_3 = 5$ $x_1 - 2x_2 \leq 3$ $2x_1 - x_3 \geq 4$ $x_1, x_2 \geq 0$ and x_3 is unrestricted in sign.	12.	Maximize $Z = 7x_1 + x_2 + 2x_3 + 7$ subject to $x_1 + x_2 - 2x_3 \leq 10$ $4x_1 + x_2 + x_3 \leq 20$ and $x_1, x_2, x_3 \geq 0$

13.	Maximize $Z = 6x_1 + 4x_2$ subject to $2x_1 + 3x_2 \leq 30$ $3x_1 + 2x_2 \leq 24$ $x_1 + x_2 \geq 3$ and $x_1, x_2 \geq 0$	14.	Maximize $Z = 9x_1 + 6x_2$ subject to $x_1 + 2x_2 \leq 8$ $3x_1 + 2x_2 \leq 12$ and $x_1, x_2 \geq 0$
15.	Maximize $Z = 5x_1 + 10x_2 + 8x_3$ subject to $3x_1 + 5x_2 + 2x_3 \leq 60$, $4x_1 + 4x_2 + 4x_3 \leq 72$, $2x_1 + 4x_2 + 5x_3 \leq 100$, $x_1 \geq 0$, $x_2 \geq 2$, $x_3 \geq 5$		

Answers 2.3

1.	Infeasible solution	2.	Infeasible solution
3.	Infeasible solution	4.	Unbounded
5.	Unbounded	6.	Unbounded
7.	Unbounded	8.	Unbounded
9.	$x_1 = -6/5$, $x_2 = -6/5$, Max. $Z = -48/5$	10.	$x_1 = 175/17$, $x_2 = -72/17$, Max. $Z = 1688/17$
11.	$x_1 = 0$, $x_2 = 21/5$, $x_3 = 22/5$, Min. $Z = 43/5$	12.	$x_1 = 0$, $x_2 = 0$, $x_3 = 20$, Max. $Z = 47$
13.	Infinite, $x_1 = 8\lambda + (12/5)(1 - \lambda)$ $x_2 = 0.\lambda + (42/5)(1 - \lambda)$ $0 \leq \lambda \leq 1$, Max. $Z = 48$	14.	Infinite, $x_1 = 4\lambda + 2(1 - \lambda)$ $x_2 = 0.\lambda + 3(1 - \lambda)$ $0 \leq \lambda \leq 1$, Max. $Z = 36$
15.	$x_1 = 0$, $x_2 = 8$, $x_3 = 10$, Max. $Z = 160$		

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

- Simplex Method (algorithm) was developed by:
 - George Dantzig
 - Hungarian
 - John Von Neumann
 - Morgenstern.
- To solve a L.P.P. by simplex method all x_i are:
 - Positive
 - Negative
 - Non-negative
 - None of these.

3. If in the final simplex table of a L.P.P. optimality condition is satisfied and at least one artificial variable appears in the basis with non-zero value, then the problem has:
 - (a) A feasible solution
 - (b) No feasible solution
 - (c) Has unbounded solution
 - (d) None of these.
4. To improve the solution in a simplex table of a maximization L.P.P., the vector α_k is selected as the incoming vector if $\Delta_k = c_k - Z_k$ is:
 - (a) Minimum
 - (b) Maximum and > 0
 - (c) Negative
 - (d) None of these.
5. For a maximization L.P.P., the simplex table is terminated i.e. final optimal solution is obtained when all:
 - (a) $c_j - Z_j = 0$
 - (b) $c_j - Z_j < 0$
 - (c) $c_j - Z_j \leq 0$
 - (d) $c_j - Z_j \geq 0$.
6. To solve a L.P.P. by simplex method all the variables x_j for all j are:
 - (a) Negative
 - (b) Non-negative
 - (c) Positive
 - (d) None of these.
7. In a maximization L.P.P., the coefficients of artificial variables in the objective function are taken as:
 - (a) Zero
 - (b) M
 - (c) $-M$
 - (d) None of these.
8. The value of all the variables that does not appear in the basis variables column B of the last simplex table is:
 - (a) Zero
 - (b) Negative
 - (c) Positive
 - (d) Non-equal to zero.
9. To solve a L.P.P. by simplex method artificial variables are added to some constraints to:
 - (a) Obtain initial B.F.S.
 - (b) Obtain F.S.
 - (c) Obtain non-negative solution
 - (d) None of these.
10. In the solution of a L.P.P. by simplex method the variable in the i -th row of basis variable column is selected for which x_{Bi}/y_{ik} , where x_k is the entering variable is:
 - (a) Maximum
 - (b) Minimum
 - (c) Minimum positive
 - (d) None of these.

Fill in the Blank

1. For the solution of a L.P.P. in which artificial variables are used to find the initial basic feasible solution, any of the following two methods, Big-M Method and can be applied.
2. In a L.P.P. in which the objective function is to be maximized, for any basic feasible solution there is some α_j in A but not in B for which $c_j - Z_j > 0$, and $y_{ij} \leq 0, \forall i=1,2,\dots,m$ then the problem has an solution.

3. If in the final simplex table $c_j - Z_j < 0$ for all j the optimal solution is
4. If in the final simplex table with all $c_j - Z_j \leq 0$ for all j and $c_j - Z_j = 0$ for at least one non-basic variable, then optimal solution of the problem will exist.
5. To convert a minimization L.P.P. to maximization problem multiply both sides by sign.

True/False

1. If a L.P. problem, has an optimal solution, then at least one basic feasible solution must be optimal.
2. For the solution of a linear programming problem with mixed constraints (constraints with $\geq, =$ and $\leq$ signs), Big-M Method can be applied.
3. For the solution of a linear programming problem with mixed constraints (constraints with $\geq, =$ and $\leq$ signs), only Big-M Method can be applied.
4. A L.P.P. with three variables and two constraints can be solved by graphical method.
5. In the solution process of a L.P.P. by simplex method using Big-M Method the column corresponding to the artificial variable that leaves the basis variable column can be deleted.

Answers

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1.	(a)	2.	(c)	3.	(b)
4.	(b)	5.	(d)	6.	(b)
7.	(b)	8.	(a)	9.	(b)
10.	(c)				

Fill in the Blank

1.	two phase method	2.	unbounded	3.	unique
4.	alternative	5.	negative		

True/False

1.	T	2.	T	3.	F
4.	F	5.	T		

UNIT-I

Chapter

3



Duality in Linear Programming

3.1 Duality in L.P. Problem

Every linear programming problem is associated with another linear programming problem called the **dual** of the problem. The original problem is called '**primal**' while the other is called its '**dual**'. The optimal solution of either problem reveals information concerning the optimal solution of the other. If the optimal solution of either problem (primal or its dual) is known then the optimal solution of the other is also available. This fact is important because sometimes it is easier to solve the dual than the primal.

3.2 Standard Form of the Primal

- (i) All the constraints involve the sign $\leq$ if it is a problem of maximization.
- or
- (ii) All the constraints involve the sign $\geq$ if it is a problem of minimization.

3.3 Symmetric Dual Problem

A symmetric relation between a primal and its dual problem exists.

Consider a L.P. problem.

Find $x_1, x_2, \dots, x_n$, which

...(1)

The dual problem of the above L.P. problem is obtained by:

- (i) Transposing the coefficient matrix.
- (ii) Interchanging the role of constant terms and the coefficients of the objective function.
- (iii) Reverting the inequalities and
- (iv) Minimizing the objective function instead of maximizing it.

Find $w_1, w_2, \dots, w_m$, which

...(2)

Primal Problem: Find a column vector $\mathbf{x}$, which

where, $\mathbf{x} = [x_1, x_2, \dots, x_n]$, $\underline{c} = [c_1, c_2, \dots, c_n]$, $\underline{b} = [b_1, b_2, \dots, b_n]$

Min. $Z_D = \mathbf{b}' \mathbf{w}$

and $w \geq 0$

and A' , $\mathbf{b}'$, $\mathbf{c}'$ are the transposes of A , $\mathbf{b}$ and $\mathbf{c}$ respectively.

3.4 Unsymmetric Dual Problem

Primal Problem : Find a column vector $\mathbf{x}$ which

$$\begin{array}{ll} \text{Max.} & Z_p = \mathbf{c} \mathbf{x} \\ \text{s.t.} & A \mathbf{x} = \mathbf{b} \\ \text{and} & \mathbf{x} \geq 0 \end{array}$$

Dual Problem : Find a column vector $\mathbf{w}$, which

$$\begin{array}{ll} \text{Min.} & Z_D = \mathbf{b}' \mathbf{w} \\ \text{s.t.} & A' \mathbf{w} \geq \mathbf{c}' \end{array}$$

It is important to note that the dual variables are unrestricted in sign.

3.5 The Dual of a Mixed System

If a system consists of a mixture of equations, inequalities (in either direction), non-negative variables and unrestricted variables then the dual of the problem can be obtained by reducing the problem to the standard form, article 3.2. The following procedure is adopted:

- (i) In maximization problem, if a constraint has a sign $\geq$, then multiply both sides by -1 and make the sign $\leq$.
- (ii) In minimization problem, if a constraint has a sign $\leq$, then multiply both sides by -1 and make the sign $\geq$.
- (iii) If a constraint has a sign $=$, i.e., an equation, it is replaced by two constraints involving the inequalities going in opposite directions.

For Ex. In maximization L.P.P an equation $\sum_{j=1}^n a_{ij}x_j = b_i$ is replaced by two constraints (inequalities)

$$\sum_{j=1}^n a_{ij}x_j \leq b_i \quad \text{and} \quad \sum_{j=1}^n a_{ij}x_j \geq b_i$$

and the second inequality is written as

$$-\sum_{j=1}^n a_{ij}x_j \leq -b_i$$

while, in minimization problem $\sum_{j=1}^n a_{ij}x_j = b_i$ is replaced by two inequalities.

$$\sum_{j=1}^n a_{ij}x_j \geq b_i \quad \text{and} \quad \sum_{j=1}^n a_{ij}x_j \leq b_i$$

and the second inequality is written as

$$-\sum_{j=1}^n a_{ij}x_j \geq -b_i$$

The above order should be noted carefully.

(iv) Replace every unrestricted variable by the difference of two non-negative variables.

Note: The dual variables which correspond to primal equality constraints, must be unrestricted in sign and those associated with the primal inequalities must be non-negative.

ILLUSTRATIVE EXAMPLES

Example 1 : Write the dual of the problem.

$$\begin{array}{ll}
 \text{Min.} & Z = 3x_1 + x_2 \\
 \text{s.t.} & 2x_1 + 3x_2 \geq 2 \\
 & x_1 + x_2 \geq 1 \\
 \text{and} & x_1, x_2 \geq 0
 \end{array}$$

Solution : The given L.P.P. is in the standard primal form. In matrix form, the given problem can be written as

$$\text{Min.} \quad Z = (3, 1)[x_1, x_2] = \mathbf{c} \mathbf{x}$$

$$\text{s.t.} \quad \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\text{i.e.,} \quad \mathbf{Ax} \geq \mathbf{b}$$

$\therefore$ The dual of the given problem is

$$\text{Max.} \quad Z_D = \mathbf{b}' \mathbf{w} = (2, 1)[w_1, w_2] = 2w_1 + w_2$$

$$\text{s.t.} \quad \mathbf{A}' \mathbf{w} \leq \mathbf{c}'$$

$$\text{or} \quad \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \leq \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$\text{or} \quad 2w_1 + w_2 \leq 3$$

$$3w_1 + w_2 \leq 1$$

$$\text{and} \quad w_1, w_2 \geq 0$$

Example 2 : Give the dual of the following L.P.P.

$$\text{Max.} \quad Z = 2x_1 + 3x_2 + x_3$$

$$\text{s.t.} \quad 4x_1 + 3x_2 + x_3 = 6$$

$$x_1 + 2x_2 + 5x_3 = 4$$

$$\text{and} \quad x_1, x_2, x_3 \geq 0$$

[C.S.J.M., Kanpur 1995]

Solution : First we shall write the given problem in the standard primal form as follows:

(i) Since it is a maximization problem, all the constraints must involve the sign $\leq$.

(ii) The first equation (constraint) is equivalent to

$$4x_1 + 3x_2 + x_3 \leq 6$$

and $4x_1 + 3x_2 + x_3 \geq 6$

the second can be written as

$$-4x_1 - 3x_2 - x_3 \leq -6$$

(iii) The second equation (constraint) is equivalent to

$$x_1 + 2x_2 + 5x_3 \leq 4$$

and $x_1 + 2x_2 + 5x_3 \geq 4$

the second can be written as

$$-x_1 - 2x_2 - 5x_3 \leq -4$$

Thus, the given L.P.P. in the standard primal form is

Max. $Z = 2x_1 + 3x_2 + x_3 = (2, 3, 1) [x_1, x_2, x_3] = \mathbf{c} \mathbf{x}$

s.t. $4x_1 + 3x_2 + x_3 \leq 6$

$$-4x_1 - 3x_2 - x_3 \leq -6$$

$$x_1 + 2x_2 + 5x_3 \leq 4$$

$$-x_1 - 2x_2 - 5x_3 \leq -4$$

or
$$\begin{bmatrix} 4 & 3 & 1 \\ -4 & -3 & -1 \\ 1 & 2 & 5 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \leq \begin{bmatrix} 6 \\ -6 \\ 4 \\ -4 \end{bmatrix}$$

or $A\mathbf{x} \leq \mathbf{b}$ and $x_1, x_2, x_3 \geq 0$

∴ The dual of the given problem is

Min. $Z_D = \mathbf{b}' \mathbf{w} = (6, -6, 4, -4) [w_1', w_1'', w_2', w_2'']$

$$= 6w_1' - 6w_1'' + 4w_2' - 4w_2''$$

s.t. $A' \mathbf{w} \geq \mathbf{c}'$

or
$$\begin{bmatrix} 4 & -4 & 1 & -1 \\ 3 & -3 & 2 & -2 \\ 1 & -1 & 5 & -5 \end{bmatrix} \begin{bmatrix} w_1' \\ w_1'' \\ w_2' \\ w_2'' \end{bmatrix} \geq \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

or $4w_1' - 4w_1'' + w_2' - w_2'' \geq 2$

$$3w_1' - 3w_1'' + 2w_2' - 2w_2'' \geq 3$$

$$w_1' - w_1'' + 5w_2' - 5w_2'' \geq 1$$

and $w_1', w_1'', w_2', w_2'' \geq 0$

writing $w_1' - w_1'' = w_1$, and $w_2' - w_2'' = w_2$, the dual of the given problem is

Min. $Z_D = 6w_1 + 4w_2$

$$\begin{aligned}
 \text{s.t.} \quad & 4w_1 + w_2 \geq 2 \\
 & 3w_1 + 2w_2 \geq 3 \\
 & w_1 + 5w_2 \geq 1 \text{ and } w_1, w_2 \text{ are both unrestricted in sign.}
 \end{aligned}$$

Example 3 : Find the dual of the following L.P.P.

$$\text{Min.} \quad Z = x_1 + x_2 + x_3$$

$$\text{s.t.} \quad x_1 - 3x_2 + 4x_3 = 5$$

$$x_1 - 2x_2 \leq 3$$

$$2x_2 - x_3 \geq 4$$

$$x_1, x_2 \geq 0, x_3 \text{ is unrestricted in sign.}$$

Solution : First, we shall write the given problem in the standard primal form as follows:

- (i) It is a minimization problem, all the constraints must contain the sign $\geq$.
(ii) The variable x_3 is unrestricted in sign.

$$\therefore \text{ We write } x_3 = x_3' - x_3''$$

$$\text{where } x_3', x_3'' \geq 0$$

- (iii) The first constraint (equation) is equivalent to

$$x_1 - 3x_2 + 4(x_3' - x_3'') \geq 5$$

$$\text{and } x_1 - 3x_2 + 4(x_3' - x_3'') \leq 5$$

The second can be written as

$$-x_1 + 3x_2 - 4x_3' + 4x_3'' \geq -5$$

- (iv) Multiplying second constraint by -1 , we have

$$x_1 - 2x_2 + 0 \cdot (x_3' - x_3'') \geq -3$$

Thus, the given problem in the standard primal form is

$$\text{Min.} \quad Z = x_1 + x_2 + x_3' - x_3'' = (1, 1, 1, -1)[x_1, x_2, x_3', x_3'']$$

$$= \mathbf{c} \mathbf{x}$$

$$\text{s.t.} \quad x_1 - 3x_2 + 4x_3' - 4x_3'' \geq 5$$

$$-x_1 + 3x_2 - 4x_3' + 4x_3'' \geq -5$$

$$-x_1 + 2x_2 + 0 \cdot x_3' - 0 \cdot x_3'' \geq -3$$

$$0 \cdot x_1 + 2x_2 - x_3' + x_3'' \geq 4$$

$$\text{or} \quad \begin{bmatrix} 1 & -3 & 4 & -4 \\ -1 & 3 & -4 & 4 \\ -1 & 2 & 0 & 0 \\ 0 & 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3' \\ x_3'' \end{bmatrix} \geq \begin{bmatrix} 5 \\ -5 \\ -3 \\ 4 \end{bmatrix}$$

$$\text{or} \quad A \mathbf{x} \geq \mathbf{b} \text{ and } x_1, x_2, x_3', x_3'' \geq 0.$$

∴ The dual of the given problem is

$$\begin{aligned} \text{Max. } Z_D = \mathbf{b}' \mathbf{w} &= (5, -5, -3, 4) [w_1', w_1'', w_2, w_3] \\ &= 5w_1' - 5w_1'' - 3w_2 + 4w_3 \end{aligned}$$

$$\text{s.t. } \mathbf{A}' \mathbf{w} \leq \mathbf{c}'$$

$$\text{or } \begin{bmatrix} 1 & -1 & -1 & 0 \\ -3 & 3 & 2 & 2 \\ 4 & -4 & 0 & -1 \\ -4 & 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} w_1' \\ w_1'' \\ w_2 \\ w_3 \end{bmatrix} \leq \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\begin{aligned} \text{or } \quad & w_1' - w_1'' - w_2 + 0w_3 \leq 1 \\ & -3w_1' + 3w_1'' + 2w_2 + 2w_3 \leq 1 \\ & 4w_1' - 4w_1'' + 0w_2 - w_3 \leq 1 \\ & -4w_1' + 4w_1'' + 0w_2 + w_3 \leq -1 \end{aligned}$$

$$\text{and } w_1', w_1'', w_2, w_3 \geq 0.$$

Writing $w_1' - w_1'' = w_1$, the dual problem is

$$\text{Max. } Z_D = 5w_1 - 3w_2 + 4w_3$$

$$\text{s.t. } w_1 - w_2 \leq 1$$

$$-3w_1 + 2w_2 + 2w_3 \leq 1$$

$$4w_1 - w_3 = 1$$

$$w_2, w_3 \geq 0, w_1 \text{ unrestricted in sign.}$$

3.6 Duality Theorems

Here, we shall give a number of theorems to describe the relation between the primal and its dual. The relationship between the primal and the dual is extremely useful. The proofs of the theorems are out of scope of this book. For the proof see authors book of Linear Programming.

Theorem 1 : Dual of the dual of a given primal, is the primal itself.

Proof : Consider the L.P. problem

$$\begin{aligned} \text{Primal : Max. } & Z_p = c_1x_1 + c_2x_2 + \dots + c_nx_n \\ \text{s.t. } & \left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &\leq b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &\leq b_2 \\ \dots &\dots \dots \dots \dots \dots \dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &\leq b_m \end{aligned} \right\} \dots(1) \\ \text{and } & x_1, x_2, \dots, x_n \geq 0. \end{aligned}$$

Dual : The dual of the above primal (1) is given by

$$\begin{array}{ll}
 \text{Min.} & Z_D = b_1 w_1 + b_2 w_2 + \dots + b_m w_m \\
 \text{s. t.} & \left. \begin{array}{l}
 a_{11} w_1 + a_{21} w_2 + \dots + a_{m1} w_m \geq c_1 \\
 a_{12} w_1 + a_{22} w_2 + \dots + a_{m2} w_m \geq c_2 \\
 \dots \dots \dots \dots \dots \dots \dots \dots \\
 \dots \dots \dots \dots \dots \dots + \dots \\
 a_{1n} w_1 + a_{2n} w_2 + \dots + a_{mn} w_m \geq c_n
 \end{array} \right\} \dots(2) \\
 \text{and} & w_1, w_2, \dots, w_m \geq 0.
 \end{array}$$

Now to write the dual of the dual (2), we first write (2) in the standard primal form (1). In standard primal form, (1) the dual (2) can be written as

$$\begin{array}{ll}
 \text{Max.} & (-Z_D) = -b_1 w_1 - b_2 w_2 - \dots - b_m w_m \\
 \text{s. t.} & \left. \begin{array}{l}
 -a_{11} w_1 - a_{21} w_2 - \dots - a_{m1} w_m \leq -c_1 \\
 -a_{12} w_1 - a_{22} w_2 - \dots - a_{m2} w_m \leq -c_2 \\
 \dots \dots \dots \dots \dots \dots \dots \dots \\
 \dots \dots \dots \dots \dots \dots \dots \dots \\
 -a_{1n} w_1 - a_{2n} w_2 - \dots - a_{mn} w_m \leq -c_n
 \end{array} \right\} \dots(3) \\
 \text{and} & w_1, w_2, \dots, w_m \geq 0.
 \end{array}$$

Dual of the dual : Now (3) is of the form (1)

$\therefore$ Considering (3) as primal its dual is given by

$$\begin{array}{ll}
 \text{Min.} & Z_{D1} = -c_1 v_1 - c_2 v_2 - \dots - c_n v_n \\
 \text{s. t.} & \left. \begin{array}{l}
 -a_{11} v_1 - a_{12} v_2 - \dots - a_{1n} v_n \geq -b_1 \\
 -a_{21} v_1 - a_{22} v_2 - \dots - a_{2n} v_n \geq -b_2 \\
 \dots \dots \dots \dots \dots \dots \dots \dots \\
 \dots \dots \dots \dots \dots \dots \dots \dots \\
 -a_{m1} v_1 - a_{m2} v_2 - \dots - a_{mn} v_n \geq -b_m
 \end{array} \right\} \dots(4) \\
 \text{and} & v_1, v_2, \dots, v_n \geq 0.
 \end{array}$$

or the dual (4) of the dual (2) can be written as

$$\begin{array}{ll}
 \text{Max.} & Z_{D2} = c_1 v_1 + c_2 v_2 + \dots + c_n v_n \\
 \text{s. t.} & \left. \begin{array}{l}
 a_{11} v_1 + a_{12} v_2 + \dots + a_{1n} v_n \leq b_1 \\
 a_{21} v_1 + a_{22} v_2 + \dots + a_{2n} v_n \leq b_2 \\
 \dots \dots \dots \dots \dots \dots \dots \dots \\
 \dots \dots \dots \dots \dots \dots \dots \dots \\
 a_{m1} v_1 + a_{m2} v_2 + \dots + a_{mn} v_n \leq b_m
 \end{array} \right\} \dots(5) \\
 \text{and} & v_1, v_2, \dots, v_n \geq 0.
 \end{array}$$

which is identical to (1).

Hence, the dual of the dual is the primal.

Note: Instead of writing the primal [dual (2)] in the standard primal form (1), the dual of (2) can directly be written as (5).

Thus, we conclude that in the sets (1) and (2) if either problem is considered the primal, the other is its dual.

Aliter : Consider the following L.P.P. in matrix form:

$$\begin{array}{lll} \text{Primal :} & \text{Max.} & Z_p = \mathbf{c}\mathbf{x} \\ & \text{s. t.} & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ & \text{and} & \mathbf{x} \geq \mathbf{0} \end{array} \quad \dots(6)$$

Dual : The dual of the primal (6) is

$$\begin{array}{lll} & \text{Min.} & Z_D = \mathbf{b}'\mathbf{w} \\ & \text{s. t.} & \mathbf{A}'\mathbf{w} \geq \mathbf{c}' \\ & \text{and} & \mathbf{w} \geq \mathbf{0} \end{array} \quad \dots(7)$$

The dual (7) can be written as

$$\begin{array}{lll} & \text{Max.} & (-Z_D) = -\mathbf{b}'\mathbf{w} \\ & \text{s. t.} & -\mathbf{A}'\mathbf{w} \leq -\mathbf{c}' \\ & \text{and} & \mathbf{w} \geq \mathbf{0} \end{array} \quad \dots(8)$$

Dual of the dual : Now (8) is of the form (6).

$\therefore$ Considering (8) as a primal its dual is given by

$$\begin{array}{lll} & \text{Min.} & Z_{D1} = (-\mathbf{c}')'\mathbf{v} = -\mathbf{c}\mathbf{v} \\ & \text{s. t.} & (-\mathbf{A}')'\mathbf{v} \geq (-\mathbf{b}')' \\ & \text{and} & \mathbf{v} \geq \mathbf{0} \end{array} \quad \dots(9)$$

(9) can be written as

$$\begin{array}{lll} & \text{Max.} & Z_{D2} = \mathbf{c}\mathbf{v} \\ & \text{s. t.} & \mathbf{A}\mathbf{v} \leq \mathbf{b} \\ & \text{and} & \mathbf{v} \geq \mathbf{0} \end{array} \quad \dots(10)$$

which is identical to (6).

Hence, the dual of the dual is the primal.

Theorem 2 : If $\mathbf{x}$ is any feasible solution of the primal Max. $Z_p = \mathbf{c}\mathbf{x}$ s.t. $\mathbf{A}\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}$ and $\mathbf{w}$ is any feasible solution to its dual problem Min. $Z_D = \mathbf{b}'\mathbf{w}$ s.t. $\mathbf{A}'\mathbf{w} \geq \mathbf{c}', \mathbf{w} \geq \mathbf{0}$, then $\mathbf{c}\mathbf{x} \leq \mathbf{b}'\mathbf{w}$ i.e., $Z_p \leq Z_D$.

Theorem 3 : If $\hat{\mathbf{x}}$ is a feasible solution to the primal Max. $Z_p = \mathbf{c}\mathbf{x}$ s.t. $\mathbf{A}\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}$ and $\hat{\mathbf{w}}$ is a feasible solution to its dual problem Min. $Z_D = \mathbf{b}'\mathbf{w}$ s.t. $\mathbf{A}'\mathbf{w} \geq \mathbf{c}', \mathbf{w} \geq \mathbf{0}$, such that $\mathbf{c}\hat{\mathbf{x}} = \mathbf{b}'\hat{\mathbf{w}}$ then $\hat{\mathbf{x}}$ and $\hat{\mathbf{w}}$ are the optimal solutions of the primal and the dual problems respectively.

Or

Alternative Statement : The necessary and sufficient condition for any L.P.P. and its dual to have optimal solution is that both have feasible solutions.

Theorem 4 : Basic Duality Theorem : If $\mathbf{x}_0$ is an optimum solution to the primal, then there exists a feasible solution $\mathbf{w}_0$ to the dual such that $\mathbf{c}\mathbf{x}_0 = \mathbf{b}'\mathbf{w}_0$, where $\mathbf{b}'$ is the transpose of $\mathbf{b}$.

Theorem 5 : Fundamental Duality Theorem : *If either the primal or the dual problem has a finite optimal solution, then the other problem also has a finite optimal solution and the values of the two objective functions are equal.*

Theorem 6 : *If the primal problem has an unbounded solution, then the dual has either no solution or an unbounded solution.*

Theorem 7 : *If any of the constraints in the primal is a perfect equality, the corresponding dual variable is unrestricted in sign.*

Theorem 8 : *If any variable of the primal is unrestricted in sign, the corresponding constraint in the dual will be strict equality.*

Theorem 9 : Complementary Slackness Theorem : *For the optimal feasible solution of the primal and dual system.*

- (a) *Whatever inequality occurs in the i -th relation of either system, if the corresponding slack or surplus variable is positive, then the i -th variable of its dual is zero.*
- (b) *If the j -th variable is positive in either system, the j -th relation of its dual holds as a strict equality i.e., the corresponding slack or surplus variable w_{m+j} vanishes.*

3.7 Correspondence between Primal and Dual

The following correspondence rules between the primal and the dual problem hold:

	Primal	Dual
1.	Objective function Max. Z_p .	Objective function Min. Z_D .
2.	Requirement vector.	Price vector.
3.	Coefficient matrix.	Transpose of the coefficient matrix.
4.	Constants with $\leq$ sign.	Constraints with $\geq$ sign.
5.	Relation.	Variable
6.	i -th inequality.	i -th variable $w_i \geq 0$.
7.	i -th constraint an equality.	i -th variable w_i unrestricted in sign.
8.	Variable.	Relation
9.	i -th variable $x_i > 0$.	i -th relation a strict inequality.
10.	i -th variable x unrestricted in sign.	i -th constraint a strict equality.
11.	i -th slack variable positive.	i -th variable zero.
12.	i -th variable zero.	i -th surplus variable positive.
13.	Finite optimal solution.	Finite optimal solution with equal optimal value of objective function.
14.	Unbounded solution.	No solution or an unbounded solution.

Exercise 3.1

Obtain the dual problem of the following LPP

1. Max. $Z = 3x_1 + x_2 + 4x_3 + x_4 + 9x_5$ s.t. $4x_1 - 5x_2 - 9x_3 + x_4 - 2x_5 \leq 6$ $2x_1 + 3x_2 + 4x_3 - 5x_4 + x_5 \leq 9$ $x_1 + x_2 - 5x_3 - 7x_4 - 11x_5 \leq 10$ and $x_1, x_2, x_3, x_4, x_5 \geq 0$	5. Min. $Z = x_1 - 3x_2 - 2x_3$ s.t. $3x_1 - x_2 + 2x_3 \leq 7$ $2x_1 - 4x_2 \geq 12$ $-4x_1 + 3x_2 + 8x_3 = 10$ $x_1, x_2 \geq 0$ and x_3 is unrestricted in sign.
2. Min. $Z = 3x_1 - 2x_2 + 4x_3$ s.t. $3x_1 + 5x_2 + 4x_3 \geq 7$ $6x_1 + x_2 + 3x_3 \geq 4$ $7x_1 - 2x_2 - x_3 \leq 10$ $x_1 - 2x_2 + 5x_3 \geq 3$ $4x_1 + 7x_2 - 2x_3 \geq 2$ and $x_1, x_2, x_3 \geq 0$ [Agra 2000]	6. Min. $Z = x_1 + x_2 + x_3$ s.t. $x_1 - 3x_2 + 4x_3 = 5$ $x_1 - 2x_2 \leq 3$ $2x_2 - x_3 \geq 4$ $x_1, x_3 \geq 0$ x_2 is unrestricted.
3. Min. $Z = 2x_2 + 5x_3$ s.t. $x_1 + x_2 \geq 2$ $2x_1 + x_2 + 6x_3 \leq 6$ $x_1 - x_2 + 3x_3 = 4$ and $x_1, x_2, x_3 \geq 0$	7. Max. $Z = 3x_1 + x_2 + 2x_3 - x_4$ s.t. $2x_1 - x_2 + 3x_3 + x_4 = 1$ $x_1 + x_2 - x_3 + x_4 = 3$ $x_1, x_2, x_3 \geq 0$, x_4 is unrestricted.
4. Min. $Z = 40x + 24y$ s.t. $20x + 50y \geq 4800$ $80x + 50y \geq 7200$ and $x, y \geq 0$	8. Max. $Z = -3x_1 + 4x_2$ s.t. $2x_1 + 3x_2 \leq 16$ $5x_1 + 2x_2 \geq 20$ and $x_1, x_2 \geq 0$

Answers 3.1

1. Min. $Z_D = 6w_1 + 9w_2 + 10w_3$ s.t. $4w_1 + 2w_2 + w_3 \geq 1$ $-5w_1 + 3w_2 + w_3 \geq 1$ $-9w_1 + 4w_2 - 5w_3 \geq 4$ $w_1 - 5w_2 - 7w_3 \geq 1$ $-2w_1 + w_2 + 11w_3 \geq 9$ and $w_1, w_2, w_3 \geq 0$.	2. Max. $Z_D = 7w_1 + 4w_2 - 10w_3 + 3w_4 + 2w_5$ s.t. $3w_1 + 6w_2 - 7w_3 + w_4 + w_5 \leq 3$ $5w_1 + w_2 + 2w_3 - 2w_4 + 7w_5 \leq -2$ $4w_1 + 3w_2 + w_3 + 5w_4 - 2w_5 \leq 4$ and $w_1, w_2, \dots, w_5 \geq 0$.
---	--

3.	$\text{Max. } Z_D = 2w_1 - 6w_2 + 4w_3$ $\text{s.t. } w_1 - 2w_2 + w_3 \leq 0$ $w_1 - w_2 - w_3 \leq 2$ $-6w_2 + 3w_3 \leq 5$ $\text{and } w_1, w_2 \geq 0,$ $w_3 \text{ is unrestricted.}$	4.	$\text{Max. } Z_D = 4800u + 7200v$ $\text{s.t. } 20u + 80v \geq 40$ $80u + 50v \geq 24$ $\text{and } u, v \geq 0.$
5.	$\text{Max. } Z_D = -7w_1 + 12w_2 + 10w_3$ $\text{s.t. } -3w_1 + 2w_2 - 4w_3 \leq 1$ $-w_1 + 4w_2 - 3w_3 \geq 3$ $2w_1 - 8w_3 \geq 2$ $w_1, w_2 \geq 0,$ $w_3 \text{ is unrestricted.}$	6.	$\text{Max. } Z_D = 5w_1 - 3w_2 + 4w_3$ $\text{s.t. } w_1 - w_2 \leq 1,$ $-3w_1 + 2w_2 + 2w_3 = 1$ $4w_1 - w_3 \leq 1,$ $w_2, w_3 \geq 0, w_1 \text{ is unrestricted.}$
7.	$\text{Min } Z_D = w_1 + 3w_2$ $\text{s.t. } 2w_1 + w_2 \geq 3, -w_1 + w_2 \geq 1,$ $3w_1 - w_2 \geq 2, w_1 + w_2 = -1$ $w_1, w_2 \text{ unrestricted in sign.}$	8.	$\text{Min. } Z_D = 16w_1 - 20w_2$ $\text{s.t. } 2w_1 - 5w_2 \geq -3$ $3w_1 - 2w_2 \geq 4$ $\text{and } w_1, w_2 \geq 0$

3.8 To Read the Solution to the Dual from the Final Simplex Table of the Primal and Vice-Versa

It is important to note that the final simplex table of the primal problem also contains, the optimal solution of the dual and vice-versa. For this observe the following rules.

1. The optimal value of primal objective function is equal to the optimal value of the dual optimal function.
i.e. $\text{Max. } Z_p = \text{Min. } Z_D$
2. The value of the optimal dual variables are the values of Δ_j with sign changed, under the columns of the corresponding slack variables (if present in the constraints), or under the columns of the corresponding surplus variables (if slack variable not present in the constraint).

If no slack variable is present in the constraint then the value of Δ_j , leaving M , under the column of the corresponding artificial variable also give the value of the corresponding optimal dual variable.

For example : The value of the first optimal dual variable w_1 is the value of Δ_j with sign changed under the column corresponding to the slack variable (if present) or surplus variable (if slack variable not present) in the first constraint of the primal. Also the value of Δ_j leaving M under the column corresponding to the artificial variable (if slack variable not present) in the first constraint of the primal, is the value of the first optimal dual variable w_1 .

3. If either problem has unbounded solution, then the other will have no feasible solution or an unbounded solution.

Note: It is always advantageous to apply the simplex method to the problem having lesser number of constraints. Hence, it saves a lot of time, if the problem (primal or dual) with lesser number of constraints is solved by the simplex method and then the solution of the other problem is read from the final simplex table of the first.

ILLUSTRATIVE EXAMPLES

Example 1 : Apply the simplex method to solve the following L.P.P.

$$\begin{array}{ll} \text{Max. } Z = 30x_1 + 23x_2 + 29x_3 & \\ \text{s.t.} & 6x_1 + 5x_2 + 3x_3 \leq 26 \\ & 4x_1 + 2x_2 + 5x_3 \leq 7 \\ \text{and} & x_1, x_2, x_3 \geq 0 \end{array} \quad \dots(1)$$

Also write the dual of the problem and solve. Read the solution of each problem from the final simplex table of the other.

Solution : Introducing the slack variables x_4, x_5 the given problem reduces to,

$$\begin{array}{ll} \text{Max. } Z = 30x_1 + 23x_2 + 29x_3 + 0 \cdot x_4 + 0 \cdot x_5 & \\ \text{s.t.} & 6x_1 + 5x_2 + 3x_3 + x_4 = 26 \\ & 4x_1 + 2x_2 + 5x_3 + x_5 = 7 \end{array}$$

Taking $x_1 = 0, x_2 = 0, x_3 = 0$, we have $x_4 = 26, x_5 = 7$ which is the starting B.F.S.

All computation work is done in the following table :

Table 3.1

B	C_B	c_j	30	23	29	0	0	Min. Ratio X_B/Y_1 $y_{1l} > 0$
		X_B	Y_1	Y_2	Y_3	Y_4	Y_5	
Y_4	0	26	6	5	3	1	0	13/3
Y_5	0	7	4	2	5	0	1	7/4 (Min.)→
$Z = C_B X_B = 0$		Δ_j	30 ↑	23	29	0	0 ↓	X_B/Y_2 $y_{2l} > 0$
Y_4	0	31/2	0	2	-9/2	1	-3/2	31/4
Y_1	30	7/4	1	1/2	5/4	0	1/4	7/2 (Min.)→
$Z = C_B X_B = 105/2$		Δ_j	0	8 ↑	-17/2	0	-15/2	
Y_4	0	17/2	-4	0	-19/2	1	-5/2	
Y_2	23	7/2	2	1	5/2	0	1/2	
$Z = C_B X_B = 161/2$		Δ_j	-16	0	-57/2	0	-23/2	

Since No. $\Delta_j > 0$, so this solution is optimal.

$\therefore$ Optimal solution is $x_1 = 0, x_2 = 7/2, x_3 = 0$,

and $\text{Max. } Z = 161/2$.

To write the dual of the problem.

The given problem can be written as

$$\text{Max. } Z = (30, 23, 29) [x_1, x_2, x_3]$$

$$\text{s.t. } \begin{bmatrix} 6 & 5 & 3 \\ 4 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \leq \begin{bmatrix} 26 \\ 7 \end{bmatrix}$$

and $x_1, x_2, x_3 \geq 0$

$\therefore$ the dual of the given problem is given by

$$\text{Min. } Z_D = (26, 7) [w_1, w_2]$$

$$\text{s.t. } \begin{bmatrix} 6 & 4 \\ 5 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \geq \begin{bmatrix} 30 \\ 23 \\ 29 \end{bmatrix}$$

and $w_1, w_2 \geq 0$

$$\begin{aligned} \text{Or } \text{Min. } Z_D &= 26w_1 + 7w_2 \\ \text{s.t. } \quad 6w_1 + 4w_2 &\geq 30 \\ \quad 5w_1 + 2w_2 &\geq 23 \\ \quad 3w_1 + 5w_2 &\geq 29 \end{aligned} \quad \dots(2)$$

and $w_1, w_2 \geq 0$

Solution of the dual (2) of the given problem (1);

The dual problem (2), can be written as

$$\text{Max. } Z'_D = -Z_D = -26w_1 - 7w_2 + 0.w_3 + 0.w_4 + 0.w_5 - Mw_{a_1} - Mw_{a_2} - Mw_{a_3}$$

$$\text{s.t. } 6w_1 + 4w_2 - w_3 + w_{a_1} = 30$$

$$5w_1 + 2w_2 - w_4 + w_{a_2} = 23$$

$$3w_1 + 5w_2 - w_5 + w_{a_3} = 29$$

and $w_1, w_2, w_3, w_4, w_5, w_{a_1}, w_{a_2}, w_{a_3} \geq 0$

where, w_3, w_4, w_5 are surplus variables and $w_{a_1}, w_{a_2}, w_{a_3}$ are the artificial variables.

Now proceeding as usual by simplex method (by Big. M-Method) we get the following simplex table:

Table 3.2

	C_j		-26	-7	0	0	0	-M	-M	-M	Min. Ratio
B	C_B	W_B	W_1	W_2	W_3	W_4	W_5	W_{a_1}	W_{a_2}	W_{a_3}	$\frac{W_B}{W_1}$ $W_{i_1} > 0$
W_{a_1}	-M	30	6	4	-1	0	0	1	0	0	30/6
W_{a_2}	-M	23	5	2	0	-1	0	0	1	0	23/5(Min.)→
W_{a_3}	-M	29	3	5	0	0	-1	0	0	1	29/3
$Z'_D = -82M$	Δ_j		$-26+14M$ ↑	$-7+11M$	-M	-M	-M	0	0 ↓	0	$\frac{W_B}{W_2}$ $W_{i_2} > 0$
W_{a_1}	-M	12/5	0	8/5	-1	6/5	0	1	-6/5	0	12/8(Min.)→
W_1	-26	23/5	1	2/5	0	-1/5	0	0	1/5	0	23/2
W_{a_3}	-M	76/5	0	19/5	0	3/5	-1	0	-3/5	1	76/19
$Z'_D = -(88 + 598)/5$	Δ_j		0	$\frac{1}{5}(27M+17)$ ↑	-M	$\frac{1}{5}(9M-26)$	-M	0 ↓	$\frac{1}{5}(-14M+26)$	0	$\frac{W_B}{W_3}$ $W_{i_3} > 0$
W_2	-7	3/2	0	1	-5/8	3/4	0	5/8	-3/4	0	—
W_1	-26	4	1	0	1/4	-1/2	0	-1/4	1/2	0	16
W_{a_3}	-M	19/2	0	0	19/8	-9/4	-1	-19/8	9/4	1	4 (Min.)→
$Z'_D = \frac{-19M+229}{2}$	Δ_j		0	0	$\frac{(17+19M)}{8}$ ↑	$-\frac{(9M+31)}{4}$	-M	$-\frac{(27M+17)}{8}$	$\frac{(5M+31)}{4}$	0 ↓	$\frac{W_B}{W_5}$ $W_{i_5} > 0$
W_2	-7	4	0	1	0	3/19	-5/19	0	-3/19	5/19	—
W_1	-26	3	1	0	0	-5/19	2/19	0	5/19	-2/19	57/2(Min.)→
W_3	0	4	0	0	1	-18/19	-8/19	-1	18/19	8/19	—
$Z'_D = -106$	Δ_j		0 ↓	0	0	$-\frac{109}{19}$	$\frac{17}{19}$ ↑	-M	$-M + \frac{109}{19}$	$-M - \frac{17}{19}$	
W_2	-7	23/2	5/2	1	0	-1/2	0	0	1/2	0	
W_5	0	57/2	19/2	0	0	-5/2	1	0	5/2	-1	
W_3	0	16	4	0	1	-2	0	-1	2	0	
$Z'_D = -\frac{116}{2}$	Δ_j		$-\frac{17}{2}$	0	0	$-\frac{7}{2}$	0	0	$-M + \frac{7}{2}$	-M	

From the above table the solution of the dual problem is $w_1 = 0, w_2 = 23/2$ and $\text{Min. } Z_D = -Z'_D = 161/2$.

To read the solution of dual problem from the final simplex table of the primal and vice-versa.

To read solution of dual problem from the final simplex table of the primal on page 99.

In this table, since x_4 and x_5 are slack variable and Δ_j below the corresponding column Y_4 and Y_5 are Δ_4 and Δ_5 ,

$$\therefore w_1 = -\Delta_4 = 0 \quad \text{and} \quad w_2 = -\Delta_5 = 23/2$$

$$\text{and Min.} \quad Z_D = \text{Max. } Z = 161/2$$

To read the solution of the primal problem from the final simplex table of the dual on page 101.

In this table since w_3, w_4, w_5 are the surplus variables (slack variables are not present in the constraints) and Δ_j below the corresponding columns W_3, W_4, W_5 and $\Delta_3, \Delta_4, \Delta_5$.

$$\therefore x_1 = -\Delta_3 = 0, x_2 = -\Delta_4 = 7/2, x_3 = -\Delta_5 = 0$$

$$\text{and Max.} \quad Z = \text{Min. } Z_D = 161/2.$$

Also note that leaving M in Δ_j 's below the columns W_{a1}, W_{a2}, W_{a3} corresponding to the artificial variables w_{a1}, w_{a2}, w_{a3} respectively also give the same values of x_1, x_2 and x_3 .

Example 2 : Using the dual, solve the following L.P.P.

$$\text{Max.} \quad Z_p = 3x_1 - 2x_2$$

$$\text{s.t.} \quad x_1 \leq 4$$

$$x_2 \leq 6$$

$$x_1 + x_2 \leq 5$$

$$-x_2 \leq -1$$

$$\text{and} \quad x_1, x_2 \geq 0$$

Solution : The dual of the given problem is given by

$$\text{Min.} \quad Z_D = 4w_1 + 6w_2 + 5w_3 - w_4$$

$$\text{s.t.} \quad w_1 + 0.w_2 + 1.w_3 + 0.w_4 \geq 3$$

$$0.w_1 + 1.w_2 + 1.w_3 - 1.w_4 \geq -2$$

$$\text{and} \quad w_1, w_2, w_3, w_4 \geq 0$$

Changing the minimization objective function to maximization function and the requirements to positive requirements, the dual problem can be written as

$$\text{Max.} \quad Z'_D = -Z_D = -4w_1 - 6w_2 - 5w_3 + w_4$$

$$\text{s.t.} \quad w_1 + 0.w_2 + 1.w_3 + 0.w_4 \geq 3$$

$$0.w_1 - 1.w_2 - 1.w_3 + 1.w_4 \leq 2$$

$$\text{and} \quad w_1, w_2, w_3, w_4 \geq 0$$

Introducing the surplus variable w_5 in the first constraint and the slack variable w_6 in the second constraint, the dual, problem reduces to the following form:

$$\text{Max. } Z'_D = -4w_1 - 6w_2 - 5w_3 + w_4 + 0.w_5 + 0.w_6$$

$$\text{s.t. } w_1 + 0.w_2 + 1.w_3 + 0.w_4 - 1.w_5 = 3$$

$$0.w_1 - 1.w_2 - 1.w_3 + 1.w_4 + w_6 = 2$$

$$\text{and } w_1, w_2, \dots, w_6 \geq 0$$

Since first column of identity matrix I_2 is present in the coefficient matrix column of (coefficients of w_1 so there is no need to introduce artificial variable in the first constraint.

Taking $w_2 = 0 = w_3 = w_4 = w_5$ (non-basic variables), we have $w_1 = 3, w_6 = 2$ which is the starting basic feasible solution of the dual.

All computation work is done in the following table:

Table 3.3

B	C_B	c_j	-4	-6	-5	1	0	0	Min. Ratio W_B / W_4 $w_{i4} > 0$
		W_B	W_1	W_2	W_3	W_4	W_5	W_6	
W_1	-4	3	1	0	1	0	-1	0	-
W_6	0	2	0	-1	-1	1	0	1	2/1 (Min.)→
$Z'_D = C_B X_B$ =-12	Δ_j		0	-6	-1	1 ↑	-4	0 ↓	
W_1	-4	3	1	0	1	0	-1	0	
W_4	1	2	0	-1	-1	1	0	1	
$Z'_D = C_B W_B$ =-10	Δ_j		0	-5	0	0	-4	-1	

In the start of the table 3.3

$$\Delta_2 = c_2 - C_B W_2 = -6 - (-4, 0) \cdot (0, 1) = -6$$

$$\Delta_3 = c_3 - C_B W_3 = -5 - (-4, 0) \cdot (1, -1) = -1$$

$$\Delta_4 = c_4 - C_B W_4 = 1 - (-4, 0) \cdot (0, 1) = 1$$

$$\Delta_5 = c_5 - C_B W_5 = 0 - (-4, 0) \cdot (-1, 0) = -4$$

$$\text{Since Max. } \Delta_j = 1 = \Delta_4$$

∴ W_4 is entering vector and by minimum ratio rule outgoing vector is $W_6 (\beta_2)$.

∴ Key element $y_{24} = 1$

Proceeding as above, finally since no $\Delta_j > 0$, so this solution of the dual problem is optimal.

i.e., Optimal solution of the dual problem is

$$w_1 = 3, w_2 = 0, w_3 = 0, w_4 = 2$$

$$\text{Min. } Z_D = -\text{Max. } Z'_D = 10$$

Solution of the original problem (Primal).

From the final simplex table 6.3, for the dual the solution of the primal is read as follows :

Here, w_5 is slack and w_6 the surplus (slack variable not present in 2nd constraint) variables in the two constraints and $\Delta_5 = -4$ and $\Delta_6 = -1$ are the values of Δ_j below the columns w_5, w_6 corresponding to these variables.

$$\therefore x_1 = -\Delta_5 = 4, x_2 = -\Delta_6 = 1$$

$$\text{and Max. } Z = \text{Min. } Z_D = 10$$

Exercise 3.2

1. Use the simplex method to solve the following problem. Also verify your answers using the dual of the problem:

$$\text{Max. } Z = 40x_1 + 50x_2$$

$$\text{s.t. } 2x_1 + 3x_2 \leq 3, 8x_1 + 4x_2 \leq 5 \text{ and } x_1, x_2 \geq 0$$

2. Solve the following L.P.P. by simplex method:

$$\text{Min. } Z = (15/2)x_1 - 2x_2$$

$$\text{s.t. } 3x_1 - x_2 - x_3 \geq 3, x_1 - x_2 + x_3 + 2 \geq 0, x_1 - x_2 + x_3 \geq 2 \text{ and } x_1, x_2, x_3 \geq 0$$

Write the dual of the above problem and solve. Read the solution of each problem from the final simplex table of the other.

3. Write the dual of the following and solve it:

$$\text{Max. } Z = 4x_1 + 2x_2$$

$$\text{s.t. } -x_1 - x_2 \leq -3, -x_1 + x_2 \leq -2 \text{ and } x_1, x_2 \geq 0.$$

Hence, or otherwise write down the solution of the primal.

4. Use principle of duality to solve the following L.P.P.

$$\text{Max. } Z = 3x_1 + 2x_2$$

$$\text{subject to } x_1 + x_2 \geq 1, x_1 + x_2 \leq 7, x_1 + 2x_2 \leq 10$$

$$x_2 \leq 3 \text{ and } x_1, x_2 \geq 0.$$

5. Write the dual of the following problem:

$$\text{Max. } Z = 5x_1 - 2x_2 + 3x_3$$

$$\text{subject to, } 2x_1 + 2x_2 - x_3 \geq 2, 3x_1 - 4x_2 \leq 3, x_2 + 2x_3 \leq 5$$

$$\text{and } x_1, x_2, x_3 \geq 0$$

Hence, or otherwise find the solution of the given problem.

6. A company makes three products X, Y, Z out of three raw materials A, B, C . The number of units of raw-materials required to produce one unit of the product is as given in the following table:

	X	Y	Z
A	1	2	1
B	2	1	4
C	2	5	1

The unit profit contribution of the products X, Y and Z are ₹ 40, 25 and 50 respectively. The number of units of raw-materials available are 36, 60 and 45 respectively.

- Determine the products mix that will maximize the total profit.
- Write the dual of the problem and through the final simplex table write the solution of the dual problem.

Answers 3.2

1.	$x_1 = 3/16, x_2 = 7/8, \text{ Max. } Z = 205/4$
2.	$x_1 = 5/4, x_2 = 0, x_3 = 3/4, \text{ Min. } Z = 75/8$ Dual : Max. $Z_D = 3w_1 + 2w_3$ s.t. $2w_1 + w_2 + w_3 \leq 15/2, w_1 + w_2 + w_3 \geq 2, -w_1 + w_2 + w_3 \leq 0$ and $w_1, w_2, w_3 \geq 0$.
3.	Dual : Min. $Z_D = -3w_1 - 2w_2$ s.t. $w_1 + w_2 \leq -4, w_1 - w_2 \leq -2$ and $w_1, w_2 \geq 0$ Neither primal nor Dual problem possess optimal B.F.S.

4.	$x_1 = 7, x_2 = 0$, Max. $Z = 21$.
5.	Dual : Min. $Z_D = -2w_1 + 3w_2 + 5w_3$ s.t. $-2w_1 + 3w_2 \geq 5$, $-2w_1 - 4w_2 + w_3 \geq -2$, $w_1 + 2w_3 \geq 3$ and $w_1, w_2, w_3 \geq 0$. Solution of the given problem, $x_1 = 23/3, x_2 = 5, x_3 = 0$, Max. $Z = 85/3$.
6.	(i) 20 units of X , no unit of Y , 5 units of Z , Max. profit = ₹ 1050. Solution of the dual, $w_1 = 0, w_2 = 10, w_3 = 10$. Min. $Z_D = ₹ 1050$.

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

- If the standard primal problem is of minimization, then all the constraints involve the sign:
 - $\leq$
 - $\geq$
 - $=$
 - None of these.
- If the standard primal problem is of maximization then all the constraints involve the sign:
 - $\leq$
 - $\geq$
 - $=$
 - None of these.
- If the i -th, slack variable of the primal is positive, then the i -th variable of the dual is:
 - Positive
 - Negative
 - Zero
 - None of these.
- If Z_P and Z_D are the optimal values of the primal and its dual, then:
 - $Z_P > Z_D$
 - $Z_P < Z_D$
 - $Z_P = -Z_D$
 - $Z_P = Z_D$.
- The right hand side constants of the constraints in a primal in standard form appears in its dual problem as:
 - Right hand constants
 - The coefficients in the objective function
 - The coefficients of an extra constraint
 - None of these.

Fill in the Blank

1. The dual of a dual is the itself.
2. If the primal problem is a minimization problem then its dual problem is problem.
3. The coefficient matrix of the dual is obtained by transposing the coefficient matrix of the
4. Requirement vector of the dual problem is price vector of the
5. If in the primal problem i -th constraint is an inequality then i -th variable w_i of the dual is
6. If, there does not exist any finite optimal solution to the primal problem then these exist any feasible solution to the dual problem.
7. Every unrestricted variable in the primal is replaced by the of two non-negative variables before writing its dual.

True/False

1. If the primal problem is a maximization problem then its dual problem is a minimization problem.
 2. If in the primal problem i -th constraint is an equality then i -th variable w_i of dual problem is unrestricted in sign.
 3. If the primal problem has unbounded solution then its dual problem has either no solution or an unbounded solution.
 4. If in the primal problem i -th variable n_i is unrestricted in sign, then the i -th constraint of its dual problem is strict inequality.
 5. If the coefficient matrix in the primal is of order $m \times n$ then the coefficient matrix of its dual is $n \times m$.
-

Answers

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1.	(b)	2.	(a)	3.	(c)
4.	(d)	5.	(b)		

Fill in the Blank

1.	primal	2.	maximization	3.	primal
4.	primal	5.	non-negative	6.	does not
7.	difference				

True/False

1.	T	2.	T	3.	T
4.	F	5.	T		

○○○

UNIT-I

Chapter

4



Transportation Problem

4.1 Introduction

The transportation problem is a particular class of linear programming problem in which the objective is to transport a commodity or to provide services from several supply origins to different demand destinations at a minimum total cost. The transportation problem can easily be expressed by linear relations as a linear programming problem and can be solved by simplex method. But it takes a long time due to the presence of a large number of variables and constraints. To overcome this difficulty the transportation algorithm have been developed for the solution of a transportation problem.

4.2 Transportation Problem/Model

In transportation problem/model, there are more than one centres, called "*origins*" or '*sources*' from when the various amounts of single homogeneous commodity need to be transported to more than one places called '*destinations*'. Here each origin (or source) has a certain capacity (*i.e.*, availability) and each destination has a certain requirement such that the sum of capacities (*i.e.*, availables) at all origins (sources) is equal to the sum of requirements of all the destinations. The costs of transporting from each of the origins to each of the destinations are different and known. The object is to transport the total amount available at all sources to all the destinations to fulfill their requirements in such a way that the total cost of transportation is minimum.

Thus transportation problem (or model) deals in minimizing the cost of transportation by satisfying the requirements of all destinations by the total amount available at all the sources.

4.3 Notations used in Transportation Problem

We shall use the following notations in transportation problem.

m = the number of sources

n = the number of destinations

a_i = the availability (or supply) at the i -th source

b_j = the requirement (or demand) at the j -th destination

c_{ij} = the cost of transportation of one unit from i -th source to j -th destination

x_{ij} = the number of units to be transported from the i -th source to j -th destination.

4.4 Tabular Representation of Transportation Problem

The transportation problem as described above can be represented in a tabular form known as *transportation table*, as follows:

Destinations → Sources ↓	W_1	W_2	...	W_j	...	W_n	Capacities of the sources
F_1	c_{11}	c_{12}	...	c_{1j}	...	c_{1n}	a_1
F_2	c_{21}	c_{22}	...	c_{2j}	...	c_{2n}	a_2
⋮	...	...	...	...	...	...	⋮
F_i	c_{i1}	c_{i2}	...	c_{ij}	...	c_{in}	a_i
⋮	...	...	...	...	...	...	⋮
F_m	c_{m1}	c_{m2}	...	c_{mj}	...	c_{mn}	a_m
Requirements →	b_1	b_2	...	b_j	...	b_n	$\sum_{i=1}^m a_i = \sum_{j=1}^n b_j$

The calculations are made directly on the 'transportation array' given below which gives the current trial solution.

Destinations → Sources ↓	W_1	W_2	...	W_j	...	W_n	Capacities of the sources
F_1	x_{11}	x_{12}	...	x_{1j}	...	x_{1n}	a_1
F_2	x_{21}	x_{22}	...	x_{2j}	...	x_{2n}	a_2
⋮	...	...	...	...	...	...	⋮
⋮	...	...	...	...	...	...	⋮
F_i	x_{i1}	x_{i2}	...	x_{ij}	...	x_{in}	a_i
⋮	...	...	...	...	...	...	⋮
⋮	...	...	...	...	...	...	⋮
F_m	x_{m1}	x_{m2}	...	x_{mj}	...	x_{mn}	a_m
Requirements →	b_1	b_2	...	b_j	...	b_n	$\sum_{i=1}^m a_i = \sum_{j=1}^n b_j$

The above two tables can be combined together by writing the costs c_{ij} within the bracket (), as follows:

Destinations → Sources ↓	W_1	W_2		W_j		W_n	Capacities of the sources
F_1	$x_{11}(c_{11})$	$x_{12}(c_{12})$	...	$x_{1j}(c_{1j})$	...	$x_{1n}(c_{1n})$	a_1
F_2	$x_{21}(c_{21})$	$x_{22}(c_{22})$	...	$x_{2j}(c_{2j})$	...	$x_{2n}(c_{2n})$	a_2
⋮	...	...	...	...	...	...	⋮
⋮	...	...	...	...	...	...	⋮
F_i	$x_{i1}(c_{i1})$	$x_{i2}(c_{i2})$	...	$x_{ij}(c_{ij})$	...	$x_{in}(c_{in})$	a_i
⋮	...	...	...	...	...	...	⋮
⋮	...	...	...	...	...	...	⋮
F_m	$x_{m1}(c_{m1})$	$x_{m2}(c_{m2})$	...	$x_{mj}(c_{mj})$	...	$x_{mn}(c_{mn})$	a_m
Requirements	b_1	b_2	...	b_j	...	b_n	$\sum_{i=1}^m a_i = \sum_{j=1}^n b_j$

Mathematically, a transportation problem can be stated as a linear programming problem as follows:

$$\text{Min. } Z = \sum_{i=1}^m \sum_{j=1}^n c_{ij} \cdot x_{ij} \quad \dots(1)$$

Supply constraints

[illegible]

$$\left. \begin{array}{l} x_{11} + x_{21} + \dots + x_{m1} = b_1 \\ x_{12} + x_{22} + \dots + x_{m2} = b_2 \\ \dots\dots\dots \\ x_{1n} + x_{2n} + \dots + x_{mn} = b_n \end{array} \right\} \begin{array}{l} i.e. \sum_{i=1}^m x_{ij} = b_j \\ \text{for } j = 1, 2, \dots, n. \end{array} \dots(3)$$

$$a_1 + a_2 + \dots + a_m = b_1 + b_2 + \dots + b_n \quad \text{i.e.,} \quad \sum_{i=1}^m a_i = \sum_{j=1}^n b_j \quad \dots(4)$$

and $x_{ij} \geq 0$ for all $i = 1, 2, \dots, m, j = 1, 2, \dots, n$.

Thus the transportation problem is a L.P.P. of special type, where we are required to find the values of $m.n$ variable that minimize the objective function Z given by (1), satisfying $(m+n)$ constraints given in (2) and (3), constraint (4) and the non-negative restriction of variables.

Note: The restriction (4) indicates that one of the $(m + n)$ constraints given in (2) and (3) is redundant. Thus in $m \times n$ transportation problem only $m + n - 1$ equations form a linearly independent set.

4.6 Terminology (Important Terms/Definitions) Used in Transportation Problem

Here we shall define few terms that are used in transportation problems.

1. A Feasible Solution (F.S.)

A *feasible solution* to a transportation problem is a set of non-negative individual allocations ($x_{ij} \geq 0$) which satisfies the row and column sum restrictions *i.e.*, the constraints of the problem/model. For feasible solution to exist, it is necessary that $\sum_{i=1}^m a_i = \sum_{j=1}^n b_j$.

2. Basic Feasible Solution (B.F.S.)

A feasible solution of m by n transportation problem is said to be a *basic feasible solution* if the total number of positive allocations x_{ij} is less or equal to $m + n - 1$ (one less than the sum of the number of rows and columns). *i.e.*, a B.F.S. of a T.P. does not contain more than $m + n - 1$, positive allocations.

3. Optimal Solution

A feasible solution (not necessarily basic) is said to be *optimal* if it minimizes the total transportation cost.

4. Non-degenerate Basic Feasible Solution

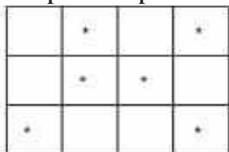
A feasible solution of m by n transportation problem is said to be *non-degenerate basic feasible solution* if

- (i) Total number of positive allocations is exactly equal to $(m + n - 1)$ and
- (ii) These allocations are in independent positions.

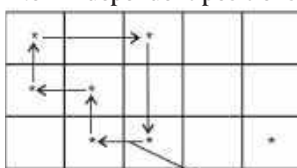
In other words, if a F.S. involving exactly $(m + n - 1)$ individual positive allocations, is said to be a *non-degenerate B.F.S.* if these allocations use in independent positions, otherwise, it is said to be *degenerate B.F.S.*

Here by independent positions of the allocation we mean that it is always impossible to form any closed circuit (loop) by joining some or all of these allocations by horizontal and vertical lines only *i.e.*, for the allocations in independent positions it is impossible to travel from any allocation, back to itself by a series of horizontal and vertical jumps from one occupied all to another occupied cell, without a direct reversal of the route. See the following tables in which the positions of allocations are indicated by

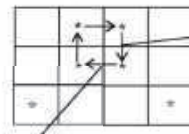
Independent positions



Non-Independent positions



Non-Independent positions



Closed circular loop

5. Balanced Transportation Problem

A transportation problem is said to be *balanced* if the total availables (supply) at all the sources (origins) is equal to the total requirements at all the destinations.

i.e.,

$$\text{if } \sum_{i=1}^m a_i = \sum_{j=1}^n b_j.$$

6. Unbalanced Transportation Problem

A transportation problem is said to be *unbalanced* if the total availables (supply) at all the sources (origins) is not equal to the total requirements at all the destinations.

$$\text{i.e.,} \quad \text{if} \quad \sum_{i=1}^m a_i \neq \sum_{j=1}^n b_j.$$

7. Rim Requirements

In a transportation problem, the quantity required for distribution i.e., $\sum_{j=1}^n b_j$ is known as Rim requirement.

4.7 Some Important Theorems

Theorem 1: Existence of feasible solution

A necessary and sufficient condition for the existence of feasible solution of a $m \times n$ transportation problem is

$$\sum_{i=1}^m a_i = \sum_{j=1}^n b_j \quad (\text{Rim condition})$$

Theorem 2: Out of $(m + n)$ equations (constraints) in a $m \times n$ transportation problem, one (any) is redundant and remaining $(m + n - 1)$ equations form a linearly independent set.

Theorem 3: Existence of an optimal solution

There always exists an optimal solution to balanced transportation problem.

4.8 Solution of a Transportation Problem

A transportation problem is solved, in general, by the following step by step procedure.

Step 1: To make a transportation table

Set up the problem in the form of a transportation table (see table in article 4.4 on page 110) if not given so.

Step 2: To check the balance in supply and requirements (demands)

Check whether the given T.P. is balanced or not.

$$\text{For a balanced } m \times n \text{ T.P.} \quad \sum_{i=1}^m a_i = \sum_{j=1}^n b_j$$

i.e., the total supply (availability) at the sources must be equal to the total requirements (demands) at all destinations whenever it is not so, a dummy origin (sources) or destination (as the case may be) is created to balance the supply and demands. (see article 4.15 on page 148).

Step 3: To find an initial basic feasible solution (B.F.S.)

First we find an initial B.F.S. of the given T.P. by any of the methods given in article 4.9. It is in general better to find initial B.F.S. by VAM which will save their valuable time to reach the optimal solution of the problem. Then

- (i) Check whether the B.F.S. has allocations in exactly $(m + n - 1)$ cells or not. If the number of allocations is less than $(m + n - 1)$, then it is case of degeneracy *i.e.*, the B.F.S. is degenerate. In this case proceed as in article 4.14 on page 147 to resolve degeneracy.
- (ii) Check whether the B.F.S. has $(m + n - 1)$ allocations in independent positions or not. If not then either shift source allocation from an occupied cell to an occupied (empty) cell or find B.F.S. by other method to get exactly $(m + n - 1)$ allocations in independent positions.

Step 4: To check the solution for optimality

Make optimality test (see article 4.13 on page 129) to check the above non-degenerate solution obtained in step 3 for optimality. If this solution is not optimal then proceed to the next step.

Step 5: To find the modified (revised) solution

If the solution in step 3 is not optimal, then modify the solution by shifting an allocation from an occupied all (cell having allocation) to an unoccupied cell (cell having no allocation) so that total transportation cost is not increased and the allocations remain in independent positions.

Step 6: Repeat steps 4 and 5 until an optimal solution is obtained.

4.9 Methods to find an Initial Feasible Solution

There are several methods for finding the initial basic feasible solution of the given transportation problem. Here we describe the following three simple methods.

1. **Method 1:** North-West Corner Method (NWCN)
2. **Method 2:** Lowest Cost Entry Method (LCEM) or Matrix Minima Method (MMM)
3. **Method 3:** Vogel's Approximation Method (VAM) or Unit Cost Penalty Method (UCPM)

4.9.1 Method 1: North-West Corner Method (NWCN)

In this method, we have the following steps:

Step 1: Start with the $(1, 1)$ at the north-west corner *i.e.*, the upper left corner of the transportation table and allocate it the maximum possible amount x_{11} which is equal to the

minimum of available supply a_1 in row 1 and the demand required b_1 in the column 1 *i.e.*, $x_{11} = \min. (a_1, b_1)$.

i.e., if $a_1 < b_1$, then allocate $x_{11} = a_1$ and move to cell (2, 1) vertically below cell (1, 1) as the supply in row 1 is exhausted.

if $a_1 > b_1$, then allocate $x_{11} = b_1$ and move to cell (1, 2) horizontally right to cell (1, 1) as the requirement in column one is exhausted.

if $a_1 = b_1$, then allocate $x_{11} = a_1 = b_1$ and move to cell (2, 2), diagonally to cell (1, 1) as the supply in row 1 and requirement in column 1 are exhausted simultaneously.

Step 2: Adjust the supply and demand units in the respective rows and columns *i.e.*, in row 1 and column 1 through the cell (1, 1).

Step 3: Repeat the steps 1 and 2 with the new cell moved.

Step 4: Continue in this manner, step by step until the total available supply is fully allocated to the cells destinations as required.

Note that the procedure will end at the cell in the south-east corner (*i.e.*, lowest right corner).

Advantages of the Method

1. The method is simple and easy to compute.
2. The allocations obtained by this method are always at independent positions.

Disadvantages of the Method

In this method the costs of transportation are not taken in to consideration, so the method does not give the solution close to the optimal solution.

The method is well explained by taking the following numerical example:

	W_1	W_2	W_3	Supply
F_1	(2)	(7)	(4)	5
F_2	(3)	(3)	(1)	8
F_3	(5)	(4)	(7)	7
F_4	(1)	(6)	(2)	14
Demand	7	9	18	34

Since $\sum a_i = 34 = \sum b_j$, so there exists a feasible solution to the above transportation problem.

	W_1	W_2	W_3	
F_1	5(2)			$a_1 = 5 - 5 = 0$
F_2	2(3)	6(3)		$a_2 = 8 - 2 = 6 - 6 = 0$
F_3		3(4)	4(7)	$a_3 = 7 - 3 = 4 - 4 = 0$
F_4			14(2)	$a_4 = 14 - 14 = 0$

$$\begin{aligned}
 b_1 &= 7 - 5, & b_2 &= 9 - 6, & b_3 &= 18 - 4 \\
 &= 2 - 2 & &= 3 - 3 & &= 14 - 14 = 0 \\
 &= 0 & &= 0 & &
 \end{aligned}$$

- (i) We start with the north-west corner *i.e.*, (1, 1) cell and allocate it maximum possible amount 5 *i.e.*, $x_{11} = 5$, which is equal to the minimum of $a_1 = 5$ and $b_1 = 7$ and adjust the supply and demand in column 1 and row 1.
- (ii) Adjusting the supply and demand in row 1 and column 1 through the cell (1, 1), we find that no supply is left in row 1 while the demand $7 - 5 = 2$ is still required in column 1. So we move to the cell (2, 1) below the cell (1, 1) in column 1.
- (iii) Since the minimum of the supply $a_2 = 8$ in row 2 and the demand left $7 - 5 = 2$ in column 1 through the cell (2, 1) is 2, so the maximum amount 2 can be allocated to the cell (2, 1) *i.e.* $x_{21} = 2$. Allocating the maximum amount 2 to the cell (2, 1), we adjust the supply and demand in row 2 and column 1 through cell (2, 1).
- (iv) Adjusting the supply and demand in row and column through the cell (2, 1), we find that no demand is left in column 1 while the supply $8 - 2 = 6$ is left in row 2. So we move to the cell (2, 2) to the right of the cell (2, 1).
- (v) Since the minimum of the remaining supply 6 in row 2 and demand 9 in column 2, through the cell (2, 2) is 6, so we can allocate maximum amount 6 to the cell (2, 2). Allocating the maximum amount 6 to the cell (2, 2) *i.e.* $x_{22} = 6$, we adjust the supply and demand in row and column through this cell.
- (vi) Adjusting the supply and demand in row and column through the cell (2, 2) we find that no supply is left in row 2, while the demand of $9 - 6 = 3$ is left in column 2. So we move to the cell (3, 2) below the cell (2, 2).
- (vii) Since the minimum of the supply 7 in row 3 and remaining demand 3 in column 2, through the cell (3, 2), is 3, so we can allocate maximum amount 3 to the cell (3, 2). Allocating the maximum amount 3 to the cell (3, 2) *i.e.* $x_{32} = 3$ we adjust the supply and demand in row 3 and column 2 through this cell.
- (viii) Adjusting the supply and demand in row and column through the cell (3, 2), we find that no demand is left in column 2 while supply $7 - 3 = 4$ is left in row 3. So we move to the cell (3, 3) to the right of the cell (3, 2).

- (ix) Since the minimum of supply 4 in row 3 and demand 18 in column 3 through the cell (3, 3) is 4, so we can allocate a maximum amount 4 to the cell (3, 3). Allocating the maximum amount 4 to the cell (3, 3) i.e. $x_{33} = 4$, we adjust the supply and demand in row 3 and column 3 through this cell.
- (x) Adjusting the supply and demand in row and column through the cell (3, 3) we find that no supply is left in row 3, while a demand $18 - 4 = 14$ is left in column 3. So we move to the cell (4, 3) below the cell (3, 3).
- (xi) Since supply and demand in row 4 and column 3 through the cell (4, 3) are both equal to 14, so we allocate amount 14 to this cell (4, 3) i.e. $x_{43} = 14$ which completes allocations.

All allocations are shown in the above table.

Multiplying each individual allocation by its corresponding unit cost in (), and adding, the total cost corresponding to this basic feasible solution.

$$= ₹ (5 \cdot 2 + 2 \cdot 3 + 6 \cdot 3 + 3 \cdot 4 + 4 \cdot 7 + 14 \cdot 2) = ₹ 102.$$

Note: In this method, we always move to the right or down, so no loop can be formulated by drawing horizontal and vertical lines to the allocations. Also at each step (allocation) at least one row or column is discarded from further consideration, while the last allocation discards both a row and a column simultaneously, so we cannot get more than $(m + n - 1)$ individual positive allocation. Thus we always get a non-degenerate basic feasible solution by the north-west corner rule.

4.9.2 Method 2: Lowest Cost Entry Method (LCEM) or Matrix Minima Method (MMM)

In this method, we have the following steps:

Step 1: Write all the costs within the brackets () in the transportation table.

Step 2: Examine the cost matrix carefully and find the lowest cost. Let it be c_{ij} . Then allocate x_{ij} as much as possible in the cell (i, j) . i.e. $x_{ij} = \min(a_i, b_j)$.

- (i) If $a_i < b_j$, $x_{ij} = a_i$, then the capacity of the i -th origin is completely exhausted. In this case cross out the i -th row of the transportation table and decrease the requirement b_j by a_i . Now go to step 3.
- (ii) If $a_i > b_j$, $x_{ij} = b_j$, then the requirement of j -th destination is completely satisfied. In this case cross out the j -th column of the transportation table and decrease a_i by b_j . Now go to step 3.
- (iii) If $x_{ij} = a_i = b_j$ then either cross-out the i -th row or j -th column but not both. Now go to step 3.

If such cell of lowest cost is not unique, select the cell where we can allocate more amount.

Step 3: Adjust the supply and demand in the row and column through the cell in which allocation is made.

Step 4: Repeat the steps 2 and 3 leaving the cost of the cells in the row or column already crossed, untill all the supply is exhausted or all the requirements are satisfied.

Advantages of the Method

1. In this method, the costs are considered for allocations, so usually here we get a better result than in NWCM.
2. In this method, the lowest costs are considered for allocations, so here the basic feasible solution is more near to the optimal solution.

The method is well explained by taking the same numerical example as in method 1.

		To			Supply
		W_1	W_2	W_3	
From	F_1	2	7	4	5
	F_2	3	3	1	8
	F_3	5	4	7	7
	F_4	1	6	2	14
Demand		7	9	18	34

Since $\sum a_i = 34 = \sum b_j$, so there exists a feasible solution to the above transportation problem.

		To			Supply
		W_1	W_2	W_3	
From	F_1	(2)	(7)	(4)	$a_1 = 5$
	F_2	(3)	(3)	(1)	$a_2 = 8$
	F_3	(5)	(4)	(7)	$a_3 = 7$
	F_4	(1)	(6)	(2)	$a_4 = 14$
Demand		$b_1 = 7$	$b_2 = 9$	$b_3 = 18$	

Table-1

- (i) Writing all the costs within the brackets (), given transportation problem is as shown in table 1.
- (ii) Examining the above cost matrix we find that there is lowest cost 1 in cell (2, 3) and in cell (4, 1). Here we choose cell (2, 3) in which we can allocate more amount 8 which is the minimum of $a_2 = 8$ and $b_3 = 18$, demand and supply in row and column

through the cell (2, 3). *i.e.* $x_{23} = 8$. Since supply in row 2 is exhausted, so we cross out (shown by shade in table 2) the row 2 and adjust the demand in column 3.

- (iii) Neglecting the costs in crossed row 2, the minimum of the remaining costs is 1 in cell (4, 1). Here we allocate the maximum amount 7 which is minimum of supply and demand in row and column through this cell *i.e.* $x_{41} = 7$. Since the demand in column 1 is exhausted, so we cross out (shown by shade in table 2) the column 1 and adjust the supply in row 4.

	W_1	W_2	W_3	Supply left
F_1	(2)	(7)	(4)	5
F_2	(3)	(3)	8(1)	0
F_3	(5)	(4)	(7)	7
F_4	7(1)	(6)	(2)	7
Demand left	0	9	10	

Table 2

	W_1	W_2	W_3	Supply left
F_1	(2)	(7)	(4)	5
F_2	(3)	(3)	8(1)	0
F_3	(5)	(4)	(7)	7
F_4	7(1)	(6)	7(2)	0
Demand left	0	9	3	

Table 3

- (iv) Neglecting the costs in crossed row 2 and column 1 (see table 2) the minimum of the remaining costs is 2 in cell (4, 3) in which we allocate the maximum amount 7, which is minimum of remaining supply and demand in row and column through this cell. *i.e.* $x_{43} = 7$. Since the remaining supply is exhausted in row 4, so we cross out (shown by shade in table 3), this row 4 and adjust the supply in column 3.

	W_1	W_2	W_3	Supply left
F_1	(2)	(7)	(4)	5
F_2	(3)	(3)	8(1)	0
F_3	(5)	7(4)	(7)	0
F_4	7(1)	(6)	7(2)	0
Demand left	0	2	3	

Table 4

	W_1	W_2	W_3	Supply left
F_1	(2)	2(7)	3(4)	5
F_2	(3)	(3)	8(1)	8
F_3	(5)	7(4)	(7)	7
F_4	7(1)	(6)	7(2)	14
Demand left	7	9	18	

Table 5

- (v) Neglecting the costs in crossed row 2, row 4 and column 1 (see table 3, the minimum of the remaining costs is 4 in cell (1, 3) and in cell (3, 2). Here we choose cell (3, 2) in

which we can allocated more amount 7 which is minimum of supply and demand in row and column through this cell. i.e. $x_{32} = 7$. Since the supply in row 3 is exhausted, so we cross this row 3 (shown by shade in table 4) and adjust the remaining demand in column 2.

- (vi) Finally, we allocate the demand 2 in cell (1, 2) and demand 3 in cell (1, 3), i.e. $x_{12} = 2, x_{13} = 3$, so that the supply 5 in row 1 is exhausted and get the final table 5.

All allocations are shown in table 5.

Multiplying each individual allocation by the corresponding cost in (), and adding, the total cost corresponding to this basic feasible solution

$$= ₹ (2 \cdot 7 + 3 \cdot 4 + 8 \cdot 1 + 7 \cdot 4 + 7 \cdot 1 + 7 \cdot 2) = ₹ 83.$$

Note 1: The total transportation cost of allocations obtained by this method is less than that obtained by NWCM.

Note 2: After some practice all the above calculations may be done in the same table as shown below.

			Supply
(2)	2(7)	3(4)	5 - 3 = 2 - 2 = 0
(3)	(3)	8(1)	8 - 8 = 0
(5)	7(4)	(7)	7 - 7 = 0
7(1)	(6)	7(2)	14 - 7 = 7 - 7 = 0
Demand	7 - 7 = 0	9 - 7 = 2 - 2 = 0	18 - 8 = 10 - 7 = 3

4.9.3 Method 3: Vogel's Approximation Method (VAM)

In this method, each allocation is made on the basis of the penalty (or opportunity) cost that would have incurred if allocation in certain cells with minimum cost were missed. In this method, we have the following steps:

Step 1: Write the differences of the smallest and the second smallest costs (i.e. penalties) in each row to the right of the corresponding row and write the similar differences (penalties) of each column below the corresponding column.

Step 2: Select the row or column for which the penalty is the largest and allocate the maximum possible amount to the cell with lowest cost in that particular row or column.

If the largest penalty among rows and columns is not unique, select that row or column in which we can allocate more amount in the lowest cost cell of that row or column.

Step 3: Adjust the supply and demand units in the respective row and column through the cell in which allocation is made and cross (or leave) out the row (or column) in which the supply (or demand) is exhausted.

Step 4: Repeat the steps 1, 2 and 3 with the costs in the remaining rows and columns left after crossing the exhausted row or column in previous step till all the supply and demands are exhausted.

Advantage of the Method

In this method, we proceed to avoid as many penalties as possible, so the B.F.S. obtained by this method is very close to the optimal solution.

Disadvantage of the Method

In this method, more time is consumed to compute the initial B.F.S. as compared to NWCM or LCEM.

The method is well explained by taking the same numerical example as in method 1.

(i) First we write the cost matrix as follows:

	W_1	W_2	W_3	Supply	Penalties
F_1	(2)	(7)	(4)	5	(2)
F_2	(3)	(3)	8(1)	8-8=0	(2) ←
F_3	(5)	(4)	(7)	7	(1)
F_4	(1)	(6)	(2)	14	(1)
Demand	7	9	18-8 =10	34	
Penalties	(1)	(1)	(1)		

Table-1

And write the penalties (difference between the lowest and the second lowest cost) in each row and column. We find that the largest penalty is 2, which is in row 1 and 2 both. Here we select the row 2, since here we can allocate maximum amount 8 to the cell (2, 3) with lowest cost 1, which is minimum of supply 8 and demand 18 in row and column respectively through the cell (2, 3).

(ii) Adjusting the supply and demand we find that the supply in row 2 is exhausted, so we cross (leave) this row 2 and get the table 2. Then we write the penalties in each row and column as shown in table 2. Here the largest penalty is 2 in row 2, column 2 and 3. Here we select the column 3, since here we can allocate maximum amount 10 to the cell (3, 3) with lowest cost.

Note that we will not select row 1, with the same largest penalty 2 cell (1, 1), as we can allocate only amount 5 to the same lowest cost 2. Also we will not select column 2 with the same largest penalty 2, as this column the smallest cost is 4 which is more than the lowest cost 2 in other two cases.

	W_1	W_2	W_3	Supply	Penalties
F_1	(2)	(7)	(4)	5	(2)
F_3	(5)	(4)	(7)	7	(1)
F_4	(1)	(6)	10(2)	$14 - 10 = 4$	(1)
Demand	7	9	$10 - 10 = 0$		
Penalties	(1)	(2)	(2)		

↑

Table-2

- (iii) Adjusting the supply and demand in the row and column through the cell (3, 3) in table 2 in which allocation is made, we find that the column 3 (W_3) is exhausted, so we cross (or leave) column 3 (W_3) in table 2 and get the table 3.

	W_1	W_2	Supply	Penalties
F_1	(2)	(7)	5	(5)
F_3	(5)	(4)	7	(1)
F_4	4(1)	(6)	$4 - 4 = 0$	(5)←
Demand	$7 - 4 = 3$	9		
Penalties	(1)	(2)		

Table-3

- (iv) The largest penalty in table 3 is 5 in row 1 and 3. Here we shall select row 3 and allocate maximum possible amount 4 to cell (3, 1) with lowest cost 1. We shall not select the row 1 as here the lowest cost is 2 in cell (1, 1) which is more than the cost in selected cell (3, 1) in row 3.
- (v) Adjusting the supply and demands in the row 3 and column 1 through the cell (3, 1) in table 3 in which allocation is made, the row 3 (row F_3) is exhausted, so we cross (or leave) row 3 (row F_4) in table 3 and get the table 4.
- (vi) The largest penalty in table 4 is 5 in row 1 (row F_1), so we allocate the maximum possible amount 3 to lowest cost cell (1, 1) in this row. The remaining amount 2 is allocated to cell (1, 2) in row 1 (F_1) and then the remaining amount 7 is allocated to cell (2,2) in column 2 (W_2).

	W_1	W_2	Supply	Penalties
F_1	3(2)	2(7)	5	(5) ←
F_3	(5)	7(4)	7	(1)
Demand	3	9		
Penalties	(3)	(3)		

Table-4

Thus, we get the required basic feasible solution as shown in the following table:

	W_1	W_2	W_3	Supply
F_1	3(2)	2(7)	(4)	5
F_2	(3)	(3)	8(1)	8
F_3	(5)	7(4)	(7)	7
F_4	4(1)	(6)	10(2)	14
Demand	7	9	18	

Table-5

The total transportation cost to this B.F.S.

$$= ₹ (3 \times 2 + 2 \times 7 + 8 \times 1 + 7 \times 4 + 4 \times 1 + 10 \times 2) = ₹ 80.$$

which is less than the costs associated to the B.F.S. obtained by the previous methods N.W.C.M. and L.C.E.M.

All the calculations for B.F.S. done in the above four tables can be shown in the form of one table as follows:

	W_1	W_2	W_3	Supply	Penalties			
					P_1	P_2	P_3	P_4
F_1	3(2)	2(7)	(4)	5	(2)	(2)	(5)	(5) ←
F_2	(3)	(3)	8(1)	8-8=0	(2) ←	×	×	×
F_3	(5)	7(4)	(7)	7-7=0	(1)	(1)	(1)	(1)
F_4	4(1)	(6)	10(2)	14-10=4-4=0	(1)	(1)	(5) ←	×
Demand	7-4=3-3=0	9-2=7-7=0	18-8=10-10=0					
P_1	(1)	(1)	(1)					
P_2	(1)	(2)	(2)					
P_3	(1)	(2)	×					
P_4	(3)	(3)	×					

Exercise 4.1

- ❖ Determine an initial basic feasible solution to the following transportation problems using north-west corner method.

1.						2.						
To						Destination						
	I	II	III	IV	Supply		D_1	D_2	D_3	D_4	Supply	
A	13	11	15	20	2		O_1	6	4	1	5	14
From B	17	14	12	13	6		Origin O_2	8	9	2	7	16
C	18	18	15	12	7		O_3	4	3	6	2	5
Demand	3	3	4	5			Demand	6	10	15	4	

3.						4.					
To						To					
	I	II	III	IV	Capacity		I	II	III	Capacity	
A	14	25	45	5	6		A	19	16	12	14
From B	65	25	35	55	8		From B	22	13	19	16
C	35	3	65	15	16		C	14	28	8	12
Demand	7	7	6	13			Demand	10	15	17	

- ❖ Determine an initial basic feasible solution to the following transportation problems using lowest cost entry method.

5.

		Destinations				
		A	B	C	D	Supply
Origin	I	1	5	3	3	34
	II	3	3	1	2	15
	III	0	2	2	3	12
	IV	2	7	2	4	19
Demand		21	25	17	17	

6.

		Warehouse				
		W ₁	W ₂	W ₃	W ₄	Capacity
Factory	F ₁	19	39	50	10	7
	F ₂	70	30	40	60	9
	F ₃	40	8	70	20	18
Requirement		5	8	7	14	

7.	Warehouse					
	W_1	W_2	W_3	W_4	Capacity	
	F_1	3	1	7	4	300
	Factory F_2	2	6	5	9	400
	F_3	8	3	3	2	500
Requirement		250	350	400	200	

8.	To					
	W_1	W_2	W_3	W_4	Capacity	
	F_1	90	30	50	10	7
	From F_2	70	30	40	60	9
	F_3	40	8	70	20	18
Demand		5	8	7	14	

❖ Determine an initial basic feasible solution to the following transportation problems using vogel's approximation method (VAM).

9.	Destination					
	D_1	D_2	D_3	D_4	Supply	
	O_1	5	8	3	6	30
	Sources O_2	4	5	7	4	50
	O_3	6	2	4	6	20
Demand		30	40	20	10	

10.	Destination					
	D_1	D_2	D_3	D_4	Supply	
	S_1	3	7	6	4	5
	Sources S_2	2	4	3	2	2
	S_3	4	3	8	5	3
Demand		3	3	2	2	

11.	Destinations					
	D_1	D_2	D_3	D_4	Supply	
	O_1	21	16	25	13	11
	Origin O_2	17	18	14	23	13
	O_3	32	27	18	41	19
Demand		6	10	12	15	

12.	Destinations						
	D_1	D_2	D_3	D_4	Available		
	O_1	1	2	1	4	20	
	Origin O_2	3	3	2	1	40	
	O_3	4	2	5	9	20	
		O_4	5	3	6	10	20
Requirements		20	40	30	10		

11.	Destinations					Supply
	D_1	D_2	D_3	D_4		
	O_1	21	16	25	13	
	Origin O_2	17	18	14	23	
	O_3	32	27	18	41	
Demand		6	10	12	15	

12.	Destinations					Available
	D_1	D_2	D_3	D_4		
	O_1	1	2	1	4	
	Origin O_2	3	3	2	1	
	O_3	4	2	5	9	
	O_4	5	3	6	10	
Requirements		20	40	30	10	

Answers 4.1

1.	$x_{11} = 2, x_{21} = 1, x_{22} = 3, x_{23} = 2, x_{33} = 2, x_{34} = 5$. Total T.C. = ₹ 199.
2.	$x_{11} = 6, x_{12} = 8, x_{22} = 2, x_{23} = 14, x_{33} = 1, x_{34} = 4$. Total T.C. = ₹ 128.
3.	$x_{11} = 4, x_{12} = 2, x_{22} = 5, x_{23} = 3, x_{33} = 3, x_{34} = 13$, Total T.C. = ₹ 726.
4.	$x_{11} = 10, x_{12} = 4, x_{22} = 11, x_{23} = 5, x_{33} = 12$. Total T.C. = ₹ 588.
5.	$x_{11} = 9, x_{12} = 8, x_{14} = 17, x_{23} = 15, x_{31} = 12, x_{42} = 17, x_{43} = 2$. Total T.C. = ₹ 238.
6.	$x_{14} = 7, x_{21} = 2, x_{23} = 7, x_{31} = 3, x_{32} = 8, x_{34} = 7$. Total T.C. = ₹ 814.
7.	$x_{12} = 300, x_{21} = 250, x_{23} = 150, x_{32} = 50, x_{33} = 250, x_{34} = 200$. Total T.C. = ₹ 2850.
8.	$x_{14} = 7, x_{21} = 2, x_{23} = 7, x_{31} = 3, x_{32} = 8, x_{34} = 7$. Total T.C. = ₹ 814.
9.	$x_{11} = 10, x_{13} = 20, x_{21} = 20, x_{22} = 20, x_{24} = 10, x_{32} = 20$. Total T.C. = ₹ 370.
10.	$x_{11} = 3, x_{14} = 2, x_{23} = 2, x_{32} = 3$. Total T.C. = ₹ 32.
11.	$x_{14} = 11, x_{21} = 6, x_{22} = 3, x_{24} = 4, x_{32} = 7, x_{33} = 12$. Total T.C. = ₹ 796.
12.	$x_{11} = 20, x_{23} = 30, x_{24} = 10, x_{32} = 20, x_{42} = 20$. Total T.C. = ₹ 190.

4.10 Optimality Test

After getting the initial F.S., of a transportation problem, we test this solution for optimality *i.e.*, we check whether the feasible solution obtained, minimizes the total transportation cost or not. There we start the optimality test to a F.S. consisting of $(m + n - 1)$ allocations in independent positions *i.e.*, to a non-degenerated B.F.S.

In general, there are following two methods used for the test of optimality of the solution:

- (i) The stepping-stone method.
- (ii) The modified distribution (MODI) method or $u-v$ method.

4.11 The Stepping-Stone Method

Consider the matrix giving the basic feasible solution (feasible solution with exactly $m + n - 1$ allocations) of the transportation problem where m is the number of rows and n is the number of columns.

To test the B.F.S. for optimality by **stepping-stone method**, proceed stepwise as follows:

Step 1: Select one of the empty cells (*i.e.*, a cell in which there is no allocation). Beginning with this cell, trace a closed path using the route through at least three occupied cells, used in the solution, by moving horizontally and vertically and then back to this cell. Here in this closed path only the cells at the turning points can be considered, skipping aside the other occupied and unoccupied cells that come in the route. These cells at the turning points are called **stepping-stones** on the path.

Step 2: Allocating $+1$ unit to this selected unoccupied cell, allocate -1 and $+1$ units alternately on each corner cells of the above closed path so that the sum of the allocations in the rows and columns remain unchanged.

Step 3: Compute the net change in the cost along this closed path by adding the unit costs in the cells with $+1$ allocations and subtracting the unit costs in the cells with -1 allocations. This net change in the cost is called the **cell evaluation** of the unoccupied cell selected in step 1 and it may be positive or negative. The positive **cell evaluation** implies that the above adjustment would increase the total cost while the negative **cell evaluation** implies that this adjustment would decrease the total cost.

Step 4: Repeat steps 1 to 4 for all unoccupied cells in the solution.

Step 5: Check the signs of all cell evaluations. If all are greater or equal to zero, then the solution under consideration is optimal and cannot be improved further. If at least one of these cell evaluations is negative, then the current solution is not optimal and can be improved further. In this, case proceed to the next step 6.

Step 6: Select the unoccupied cell having the greatest negative cell evaluation and allocate maximum possible number of units to this cell and make adjustments in units in other occupied cells so that the sum of the allocations in the rows and columns remain unchanged.

Step 7: Repeat the steps 1 to 6 for all unoccupied cells in this new solution and repeat the procedure untill we get an optimal solution.

Since there are $mn - (m + n - 1) = (m - 1)(n - 1)$ unoccupied cells, so for each solution we require to calculate $(m - 1)(n - 1)$ such cell evaluations, which is very complicated.

To avoid this complication, we give the following theorem (without proof) for the calculation of cell evaluations for all the unoccupied cell simultaneously.

4.12 Theorem

If we have a B.F.S. consisting of $(m + n - 1)$ independent positive allocations, and a set of arbitrary numbers u_i and v_j , $i = 1, 2, \dots, m$; $j = 1, 2, \dots, n$; such that

$$c_{rs} = u_r + v_s$$

for all occupied cells (r, s) then the cell evaluations d_{ij} corresponding to each empty cell (i, j) is given by

$$d_{ij} = c_{ij} - (u_i + v_j).$$

4.13 MODI(Modified Distribution) Method or u-v Method

The iterative procedure of determining an optimal solution of a minimization transportation problem by **MODI Method** is as follows:

Step 1: Construct a transportation table entering the capacities $a_1, a_2, \dots, a_m$ of the sources and the requirements $b_1, b_2, \dots, b_n$. Enter the various costs c_{ij} at the upper left corners of all the cells. Find an initial B.F. solution (allocation in independent positions) of the problem by any of the methods given in article 4.9. Enter the allocations at the centres of the cells.

Step 2: Determine a set of $(m + n)$ numbers

$$u_i, \quad i = 1, 2, \dots, m$$

$$v_j, \quad j = 1, 2, \dots, n$$

such that, for each occupied cell (r, s)

$$c_{rs} = u_r + v_s$$

For this, we assign an arbitrary value to one of the u_i 's or v_j 's then the rest $(m + n - 1)$ of them can easily be solved algebraically from the relation $c_{rs} = u_r + v_s$, for occupied cells. Generally, we choose that u_i or $v_j = 0$ for which the corresponding row or column have the maximum number of individual allocations.

Step 3: Find the *cell evaluation* $u_i + v_j$ for each unoccupied cell (i, j) and enter at the upper right corner of the corresponding cell (i, j) .

Step 4: Find the *cell evaluations* $d_{ij} = c_{ij} - (u_i + v_j)$ for each unoccupied cell (i, j) and enter at the lower right corners of the corresponding cells.

Step 5: Examine the cell evaluations d_{ij} for unoccupied cells and conclude that

- (i) if all $d_{ij} > 0$, then the solution under test is **optimal** and **unique**.
- (ii) if all $d_{ij} \geq 0$, with at least one $d_{ij} = 0$, then the solution under test is **optimal** and *alternative optimal solution exists*.
- (iii) if at least one $d_{ij} < 0$, then the solution is **not optimal**. In the last case proceed to the next step 6.

Step 6: Form a new B.F.S. by giving maximum allocation to the cell for which d_{ij} is greatest negative, by making an occupied cell empty and adjusting the units in the other occupied cells so that the sum of the allocations in the rows and columns remain unchanged.

Step 7: Then repeat the step (2) to (5) to test the optimality to this new B.F. solution. Continue improving the B.F.S. interactively using the step 2 to (6) till an optimal solution is attained.

Thus, in the table after making all the entries the occupied (cells having allocations) and the unoccupied cells (Cells having no allocation) will be as follows:

(Cells having no allocation)

(c_{ij})
Allocation

(c_{ij})	$(u_i + v_j)$
$(d_{ij} = c_{ij} - (u_i + v_j))$	

Step 2: Here we determine a set of u_i and v_j s.t. for each occupied cell (r, s) , $c_{rs} = u_r + v_s$. See table below.

For this we choose $u_2 = 0$ (since row 2 contains maximum number of allocations).

Since $c_{21} = 15 = u_2 + v_1$, $c_{22} = 16 = u_2 + v_2$, $c_{24} = 21 = u_2 + v_4$

$\therefore v_1 = 15, v_2 = 16, v_4 = 21$

Also $c_{14} = 11 = u_1 + v_4$, $c_{32} = 25 = u_3 + v_2$, $c_{33} = 16 = u_3 + v_3$

$\therefore u_1 = 11 - v_4 = -10$, $u_3 = 25 - v_2 = 9$, $v_3 = 16 - u_3 = 7$

Step 3: Now we find the cell evaluations $u_i + v_j$ for each unoccupied cell (i, j) and enter at the upper right corner of the corresponding unoccupied cell.

Step 4: Then we find the cell evaluations $d_{ij} = c_{ij} - (u_i + v_j)$ (i.e., the difference of the upper right corner entry from the upper left corner entry) for each unoccupied cell (i, j) and enter at the lower right corner of the corresponding unoccupied cell.

Thus, we get the following table:

				u_i				
(19)	(5)	(14)	(6)	(23)	(- 3)	(11)	$- 10 \ (u_1)$	
		(14)			(8)	(26)		11
(15)		(16)		(12)		(7)		(21)
6		3					4	$0 \ (u_2)$
						(5)		
(30)	(24)	(25)		(16)		(39)	(30)	$9 \ (u_3)$
		7		12				
		(6)				(9)		
				v_j				
15		16		7		21		
(v_1)		(v_2)		(v_3)		(v_4)		

Step 5: Since all $d_{ij} > 0$ in the above table. Hence the B.F.S. shown by the table is an optimal solution and the optimal transportation cost = ₹ 710.

Thus the solution of the given transportation problem is

from plant P_1 transport 6 units to market M_2

from plant P_2 transport 3 and 7 units to markets M_2 and M_3 respectively.

from plant P_3 transport 12 units to market M_3

and from plant P_4 transport 11 and 4 units to markets M_1 and M_2 respectively.

Example 2: Solve the following transportation problem

		To			Supply
		1	2	3	
From	1	2	7	4	5
	2	3	3	1	8
	3	5	4	7	7
	4	1	6	2	14
Demand		7	9	18	34

Solution:

Step 1:

The initial B.F.S. of the above problem (by VAM) is given in the following table.

Total transportation cost

$$= ₹ (5 \times 2 + 2 \times 1 + 7 \times 4 + 2 \times 6 + 8 \times 1 + 10 \times 2) = ₹ 80$$

			a_i
(2)	(7)	(4)	
5			5
(3)	(3)	(1)	
		8	8
(5)	(4)	(7)	
	7		7
(1)	(6)	(2)	
2	2	10	14
b_j	7	9	18

Step 2: Now we determine a set of u_i and v_j s.t. for each occupied cell (r, s) , $c_{rs} = u_r + v_s$.

For this we choose $u_4 = 0$ (since row 4 contains maximum number of allocations).

Since $c_{41} = 1 = u_4 + v_1, c_{42} = 6 = u_4 + v_2, c_{43} = 2 = u_4 + v_3$

$\therefore v_1 = c_{41} - u_4 = 1, v_2 = 6 - u_4 = 6, v_3 = 2 - u_4 = 2$

Also $c_{11} = 2 = u_1 + v_1, c_{23} = 1 = u_2 + v_3, c_{32} = 4 = u_3 + v_2$

$\therefore u_1 = 2 - v_1 = 1, u_2 = 1 - v_3 = -1, u_3 = 4 - v_2 = -2$

Step 3: Then we find the cell evaluations $u_i + v_j$ for each unoccupied cell (i, j) and enter at the upper right corner of the corresponding unoccupied cell.

Step 4: Then we find the cell evaluations $d_{ij} = c_{ij} - (u_i + v_j)$ (i.e., the difference of the upper right corner entry from the upper left corner entry) for each unoccupied cell (i, j) and enter at the lower right corner of the corresponding unoccupied cell.

Thus, we get the following table:

			u_i		
(2)	(7)	(7)	(4)	(3)	
5					
		(0)		(1)	
(3)	(0)	(3)	(5)	(1)	
	(3)		(- 2*)	8	
(5)	(-1)	(4)	(7)	(0)	
	(6)	7		(7)	
(1)	(6)	(2)		0 (u_4)	
2	2	10			
v_j			1	6	2
			(v_1)	(v_2)	(v_3)

Step 5: Since cell evaluation $d_{22} = -2 < 0$, so the solution under test is not optimal.

Step 6: Since minimum d_{ij} is $d_{22} = -2$ (negative), so we give maximum allocation θ to this cell from an occupied cell and make the necessary changes in other allocations as shown in the following table.

5		
	$+\theta$	$8 - \theta$
	7	
2	$2 - \theta$	$10 + \theta$

Since minimum allocation containing $-\theta$ is $2 - \theta$

$\therefore$ Taking $2 - \theta = 0$, we get $\theta = 2$

Step 7: The new B.F.S. (allocations in independent positions) thus obtained is shown in the following table. For this B.F.S. total transportation cost

$$= ₹ (5 \times 2 + 2 \times 1 + 2 \times 3 + 7 \times 4 + 6 \times 1 + 12 \times 2) = ₹ 76$$

which is less than that for the initial B.F.S.

	a_i			
(2)	(7)	(4)		
5			5	
(5)	(3)	(1)		
	(2)	(6)	8	
(5)	(4)	(7)		
	7		7	
(1)	(6)	(2)		
(2)		(12)	14	
	b_j			
	7	9	18	

Step 8: Proceeding as in step 2, 3 and 4 (to test the optimality of the above B.F.S.) we get the following table:

	u_i			
(2)	(7)	(5)	(4)	(3)
5				
		(2)		(1)
(3)	(0)	(3)	(1)	
	(3)	2	6	
(5)	(1)	(4)	(7)	(2)
	(4)	7		(5)
(1)	(6)	(4)	(2)	
2			12	
		(2)		
	v_j			
	1	4	2	
	(v_1)	(v_2)	(v_3)	

Since all $d_{ij} > 0$. Hence, the B.F.S. shown by table in step 8 is an optimal solution which is also unique.

Thus, the solution of the given transportation problem is

from source 1 transport 5 units to destination 1.

from source 2 transport 2 and 6 units to destinations 2 and 3 respectively.

from source 3 transport 7 units to destination 2.

And from source 4 transport 2 and 12 units to destinations 1 and 3 respectively

which can also be written as

$$x_{11} = 5, x_{22} = 2, x_{23} = 6, x_{32} = 7, x_{41} = 2, x_{43} = 12$$

where x_{ij} is the number of units to be transported from i -th source to j -th destination.

And the total transportation cost (optimal) = ₹ 76.

Example 3: A whole sale company has three warehouses from which supplies are drawn to four retail customers. The company deals in a single product, the supplies of which at each house are:

Warehouse No.	Supply Units	Customer No.	Demand (Units)
1	20	1	15
2	28	2	19
3	17	3	13
		4	18

Conveniently total supply at the warehouses is equal to total demand from the customers. The following table gives the transportation costs per unit shipment from each warehouse to each customer:

Warehouse	Customer			
	1	2	3	4
1	3	6	8	5
2	6	1	2	5
3	7	8	3	9

Determine, what supplies to dispatch from each of the warehouses to each customer so as to minimize overall transportation cost? Test the optimality also.

Solution: The given transportation problem, with all supply and demand, can be expressed in the form of cost matrix as follows:

Warehouse ↓	Customer				Supply
	1	2	3	4	
1	3	6	8	5	20
2	6	1	2	5	28
3	7	8	3	9	17
Demand	15	19	13	18	65

Step 1: By VAM an *initial B.F.S.* of the problem is given in the following table:

						Supply Penalties			
		1	2	3	4		P_1	P_2	P_3
	1	2(3)	(6)	(8)	18(5)	20-18=2	5-3=2	5-3=2	5-3=2
	2	9(6)	19(1)	(2)	(5)	28-19=9	2-1=1	5-2=3	6-5=1
	3	4(7)	(8)	13(3)	(9)	17-13=4	7-3=4	7-3=4 ←	9-7=2
Demand		15	19-19=0	13-13=0	18-18=0				
Penalties	P_1	6-3=3	6-1=5 ↑	3-2=1	9-5=4				
	P_2	6-3=3	×	3-2=1	9-5=4				
	P_3	6-3=3	×	×	9-5=4 ↑				

The total transportation cost = ₹ $(2 \times 3 + 18 \times 5 + 9 \times 6 + 19 \times 1 + 4 \times 7 + 13 \times 3) = ₹ 236$

Step 2: To test the solution for optimality

Here total number of allocations is $m + n - 1 = 3 + 4 - 1 = 6$ at independent positions.

Now we determine a set of u_i and v_j ($i = 1, 2, 3$) ($j = 1, 2, 3, 4$) such that for each occupied cell (r, s) , $c_{rs} = u_r + v_s$.

Since first column contains maximum number of allocations, so we choose $v_1 = 0$ (see the table).

Now, Since $c_{11} = 3 = u_1 + v_1, c_{21} = 6 = u_2 + v_1, c_{31} = 7 = u_3 + v_1$

$\therefore u_1 = 3, u_2 = 6, u_3 = 7$

Also $c_{22} = 1 = u_2 + v_2, c_{33} = 3 = u_3 + v_3, c_{14} = 5 = u_1 + v_4$

$\therefore v_2 = -5, v_3 = -4, v_4 = 2.$

Step 3: Now we find cell evaluations $u_i + v_j$ for each unoccupied cell (i, j) and enter at the upper right corner of the corresponding unoccupied cell, (See the table).

Step 4: Then we find the cell evaluations $d_{ij} = c_{ij} - (u_i + v_j)$ (i.e., the difference of the upper right corner entry from the upper left corner entry) for each unoccupied cells (i, j) and enter at the lower right corner of the corresponding unoccupied cell.

The calculations of step 2, 3 and 4 are shown in the following table:

	u_i				
(3)	(6)	(-2)	(8)	(-1)	(5)
2				18	3 (= u_1)
		(8)		(9)	
(6)	(1)		(2)	(2)	(5)
9	19			(8)	6 (= u_2)
				(0)	(-3)
(7)	(8)	(2)	(3)	(9)	(9)
4			13		7 (= u_3)
		(6)		(0)	
v_j	0	-5	-4	2	
	(= v_1)	(= v_2)	(= v_3)	(= v_4)	

Step 5: Since cell evaluation $d_{24} = -3 < 0$, so the solution under test is not optimum.

Step 6: Since minimum d_{ij} is $d_{24} = -3$ (negative), so we give maximum allocation θ to this cell from an occupied cell and make the necessary changes in other allocations as shown in the following table:

2 + θ			18 - θ
9 - θ	19		+ θ
4		13	

Since minimum allocation containing $-\theta$ is $9 - \theta$.

$\therefore$ Taking $9 - \theta = 0$, we get $\theta = 9$.

Step 7: The new B.F.S. ($m + n - 1$ allocations at independent positions) is shown in the following table:

For this B.F.S. the total transportation cost.

$$= ₹ (11 \times 3 + 9 \times 5 + 19 \times 1 + 9 \times 5 + 4 \times 7 + 13 \times 3) = ₹ 209$$

which is less than that for the initial B.F.S.

Step 8: Proceeding as in step 2, 3 and 4, (to test the optimality of the above B.F.S.), we get the following table:

				$u_i \downarrow$	
(3)	(6)	(1)	(8)	(-1)	(5)
11					9
		(5)		(9)	
(6)	(3)	(1)	(2)	(-1)	(5)
		19			9
	(3)			(3)	
(7)	(8)	(5)	(3)		(9)
4			13		(9)
		(3)			(0)
v_j	3	1	-1	5	
	(= v_1)	(= v_2)	(= v_3)	(= v_4)	

Since all $d_{ij} > 0$, hence the solution shown in the last table is optimal. Also $d_{34} = 0$, implies that alternative optimal solution also exists.

Thus the optimal solution of the problem is

$$x_{11} = 11, x_{14} = 9, x_{22} = 19, x_{24} = 9, x_{31} = 7, x_{33} = 13$$

where x_{ij} = Amount to be transported from i -th warehouse to j -th customer.

And optimal transportation cost = ₹ 209.

Example 4: Solve the following transportation problem.

	S_1	S_2	S_3	S_4	a_i
D_1	1	2	1	4	30
D_2	3	3	2	1	50
D_3	4	2	5	9	20
b_j	20	40	30	10	100

Solution: By 'Lowest Cost Entry' method, we get the following B.F.S. of the problem.

	S_1	S_2	S_3	S_4	a_i
D_1	(1) 20	(2)	(1) 10	(4)	30
D_2	(3)	(3) 20	(2) 20	(1) 10	50
D_3	(4)	(2) 20	(5)	(9)	20
b_j	20	40	30	10	

and total transportation cost

$$= ₹ (20 \times 1 + 10 \times 1 + 20 \times 3 + 20 \times 2 + 10 \times 1 + 20 \times 2) = ₹ 180$$

To test the solution for optimality

Finding the set of $u_i (i = 1, 2, 3), v_j (j = 1, 2, 3, 4)$ such that $c_{rs} = u_r + v_s$ for occupied cells and entering $u_i + v_j$ and d_{ij} in the unoccupied cells, the table giving the optimal solution is as follows:

	u_i				
(1) 20	(2)	(2) (0)	(1) 10	(4) (0) (4)	- 1 (u_1)
(3) (1)	(2) 20	(3) 20	(2) 20	(1) 10	0 (u_2)
(4) (3)	(1) 20	(2) (4)	(5) (1)	(9) (0) (9)	- 1 (u_3)
v_j	2 (v_1)	3 (v_2)	2 (v_3)	1 (v_4)	

Since all $d_{ij} \geq 0$, the solution given by the above table is an optimal solution. Since $d_{12} = 0$ it implies that solution is not unique but an alternative optimal solution will exist. Optimal transportation cost is ₹ 180.

Thus, the solution of the problem is

from D_1 transport 20 and 10 units to destinations S_1 and S_3 respectively.

from D_2 transport 20, 20 and 10 units to destinations S_2, S_3 and S_4 respectively.

from D_3 transport 20 units to destination S_2 .

And the total transportation cost = ₹ 180.

Example 5: Solve the following transportation problem.

		To				
		1	2	3	4	Supply
1		21	16	25	13	11
2		17	18	14	23	13
3		32	27	18	41	19
	Demand	6	10	12	15	43

Solution: The final table giving optimal solution is as follows:

		To				
		1	2	3	4	u_i
From	1	(21) (7) (14)	(16) (8) (8)	(25) (-1) (26)	(13) 11	-10 (u_1)
	2	(17) 6 6	(18) 3 3	(14) (9) (5)	(23) 4 4	0 (u_2)
	3	(32) (26) (6)	(27) 7 7	(18) 12 12	(41) (32) (9)	9 (u_3)
		v_j 17 (v_1)	1 (v_2)	9 (v_3)	23 (v_4)	

Hence, the solution of the given problem is

from source 1 transport 11 units to destination 4

from source 2 transport 6, 3 and 4 units to destinations 1, 2 and 4 respectively.

from source 3 transport 7 and 12 units to destinations 2 and 3 respectively.

Optimal transportation cost

$$= ₹ (6 \times 17 + 3 \times 18 + 7 \times 27 + 12 \times 18 + 11 \times 13 + 4 \times 23) = ₹ 796.$$

Exercise 4.2

1. Describe the methods to obtain an initial feasible solution for a transportation problem.
2. Explain the term 'Degeneracy' in the context of transportation problem.
3. What is mean by unbalanced transportation problem? Explain the method for solving such a problem.
4. Is $x_{13} = 50, x_{14} = 20, x_{21} = 55, x_{31} = 30, x_{32} = 35, x_{34} = 25$ an optimal solution of the following transportation problem?

				Available Units
6	1	9	3	70
11	5	2	8	55
10	12	4	7	90
Required Units	85	35	50	45

If not, modify it to obtain an optimal basic feasible solution.

- ❖ Solve the following transportation problems for minimum cost.

5.

		To				
		I	II	III	IV	
From	A	15	10	17	18	2
	B	16	13	12	13	6
	C	12	17	20	11	7
		3	3	4	5	

6.

Sources	Destination			Supply
	X	Y	Z	
<i>A</i>	2	7	4	50
<i>B</i>	3	3	7	70
<i>C</i>	5	4	1	80
<i>D</i>	1	6	2	140
Demand	70	90	180	340

7.

Warehouse → Factory ↓	W_1	W_2	W_3	W_4	Factory capacity
F_1	19	39	50	10	7
F_2	70	30	40	60	9
F_3	40	8	70	20	18
Warehouse Requirement	5	8	7	14	34

8.

Warehouse → Factory ↓		To					Available
		W_1	W_2	W_3	W_4	W_5	
From	F_1	3	4	6	8	9	20
	F_2	2	10	1	5	8	30
	F_3	7	11	20	40	3	15
	F_4	2	1	9	14	16	13
Required		40	6	8	18	6	78

9.

	D_1	D_2	D_3	D_4	a_i
O_1	5	3	6	2	19
O_2	4	7	9	1	37
O_3	3	4	7	5	34
b_j	16	18	31	25	90

10.

Plant	Customer					Available
	C_1	C_2	C_3	C_4	C_5	
P_1	4	1	3	4	4	60
P_2	2	3	2	2	3	35
P_3	3	5	2	4	4	40
Required	22	45	20	18	30	135

11.

	D_1	D_2	D_3	D_4	D_5	D_6	a_i
O_1	9	12	9	6	9	10	5
O_2	7	3	7	7	5	5	6
O_3	6	5	9	11	3	11	2
O_4	6	8	11	2	2	10	9
b_j	4	4	6	2	4	2	22

12.

Plant	Market				Available
	A	B	C	D	
X	10	22	10	20	8
Y	15	20	12	8	13
Z	20	12	10	15	11
Required	5	11	8	8	32

13.

Plant	Market				Available
	A	B	C	D	
X	14	9	18	6	11
Y	10	11	7	16	13
Z	25	20	11	34	19
Required	6	10	12	15	43

14.

Plant	Warehouse				Available at plant
	A	B	C	D	
X	30	20	50	20	75
Y	20	10	20	40	120
Z	40	30	40	30	105
Required at warehouse	65	60	80	95	300

15.

Plant	Market			Available at plant
	A	B	C	
X	11	21	16	14
Y	7	17	13	26
Z	11	23	21	36
Required at market	18	28	25	

16. A textile firm has three factories F_1, F_2, F_3 and four warehouses W_1, W_2, W_3, W_4 . The transportation costs, factory capacities and warehouse requirements are given in the following table:

	W_1	W_2	W_3	W_4	Capacity
F_1	15	24	11	12	5000
F_2	25	20	14	16	1000
F_3	12	16	22	13	7000
Requirements:	3000	2500	3500	4000	13000

Find the least transportation schedule.

17. Goods to be transported from factories A, B, C and D to warehouse X, Y and Z. The transportation cost per unit, capacities of the factories and requirements of the warehouses are given in the following table. Find the distribution with minimum cost.

Factories Warehouses					Requirement
	A	B	C	D	
X	15	24	11	12	2000
Y	25	20	14	16	4000
Z	12	16	22	13	7000
Capacity	3000	2500	3500	4000	13000

18. Solve the following transportation problems:

		To			Supply
		D_1	D_2	D_3	
From	O_1	2	4	1	40
	O_2	6	3	2	50
	O_3	4	5	6	20
	O_4	3	2	1	30
	O_5	5	2	5	10
Demand		50	60	40	150

19. Determine the optimum basic feasible solution to the following transportation problem:

	D_1	D_2	D_3	D_4	D_5	Capacity
O_1	5	5	6	4	2	9
O_2	6	9	7	8	5	13
O_3	5	6	4	6	3	9
Demand	4	4	6	2	4	

where O_i and D_j denote i -th origin and j -th destination respectively.

Answers 4.2

4.	No; $x_{12} = 30, x_{14} = 40, x_{22} = 5, x_{23} = 50, x_{31} = 85, x_{34} = 5$. Min. cost = ₹ 1160
5.	$x_{12} = 2, x_{22} = 1, x_{23} = 4, x_{24} = 1, x_{31} = 3, x_{34} = 4$. Min. cost = ₹ 174
6.	$x_{11} = 50, x_{21} = 70, x_{32} = 20, x_{33} = 60, x_{41} = 20, x_{43} = 120$, OPTC = ₹ 710
7.	$x_{11} = 5, x_{14} = 2, x_{22} = 2, x_{23} = 7, x_{32} = 6, x_{34} = 12$. OP.T.C. = ₹ 743
8.	$x_{11} = 20, x_{21} = 4, x_{23} = 8, x_{24} = 18, x_{31} = 9, x_{35} = 6, x_{41} = 7, x_{42} = 6$. OP.T.C. = 267
9.	$x_{12} = 18, x_{13} = 1, x_{21} = 12, x_{24} = 25, x_{31} = 4, x_{33} = 30$. OP.T.C. = ₹ 355

10.	$x_{12} = 45, x_{15} = 15, x_{21} = 17, x_{24} = 18, x_{31} = 5, x_{33} = 20, x_{35} = 15$. OP.T.C. = ₹ 290
11.	$x_{13} = 5, x_{22} = 4, x_{26} = 2, x_{31} = 1, x_{33} = 1, x_{41} = 3, x_{44} = 2, x_{45} = 4$. OP.T.C. = ₹ 112
12.	$x_{11} = 5, x_{13} = 3, x_{23} = 5, x_{24} = 8, x_{32} = 11$. OP.T.C. = ₹ 336
13.	$x_{14} = 11, x_{21} = 6, x_{22} = 3, x_{24} = 4, x_{32} = 7, x_{23} = 12$. OP.T.C. = ₹ 495
14.	$x_{14} = 75, x_{22} = 40, x_{23} = 80, x_{31} = 65, x_{32} = 20, x_{34} = 20$. or $x_{11} = 65, x_{12} = 10, x_{22} = 40, x_{23} = 80, x_{32} = 10, x_{34} = 95$. or $x_{11} = 55, x_{12} = 20, x_{22} = 40, x_{23} = 80, x_{31} = 10, x_{34} = 95$. OP.T.C. = ₹ 7300
15.	$x_{13} = 14, x_{22} = 15, x_{23} = 11, x_{31} = 18, x_{32} = 13$, 5 units are left at plant Z. OP.T.C. = ₹ 1119
16.	$x_{13} = 2500, x_{14} = 2500, x_{23} = 1000, x_{31} = 3000, x_{32} = 2500, x_{34} = 1500$, OP.T.C. = ₹ 16700.
17.	$x_{13} = 2000, x_{23} = 1500, x_{24} = 2500, x_{31} = 3000, x_{32} = 2500, x_{34} = 1500$. OP.T.C. = ₹ 176500.
18.	$x_{11} = 30, x_{13} = 10, x_{22} = 50, x_{31} = 20, x_{43} = 30, x_{52} = 10$, OP.T.C. ₹ 350.
19.	$x_{12} = 4, x_{14} = 5, x_{21} = 3, x_{22} = 2, x_{25} = 8, x_{32} = 1, x_{33} = 8$. OP.T.C. = ₹ 154.

4.14 Degeneracy in Transportation Problems

In a transportation problem, degeneracy occurs whenever the number of independent individual allocations is less than $(m + n - 1)$. Degeneracy may occur either at the initial stage or at an intermediate stage at some subsequent iteration.

In such cases, to resolve degeneracy, we allocate an extremely small amount (close to zero) to one or more empty cells of the matrix (generally lowest cost cells if possible), so that the total number of occupied (allocated) cells becomes $(m + n - 1)$ at independent positions. We denote this small amount by Δ (delta) or ϵ (epsilon) satisfying the following conditions.

1. $0 < \Delta x_{ij}$, for all $x_{ij} > 0$
2. $\Delta + 0 = \Delta = 0 + \Delta$
3. $x_{ij} \pm \Delta = x_{ij}$, for all $x_{ij} > 0$

4. If there are more than one Δ 's introduced in the solution, then
- (i) if Δ, Δ' are in the same row, $\Delta < \Delta'$ when Δ is to the left of Δ'
 - and (ii) if Δ, Δ' are in the same column, $\Delta < \Delta'$ when Δ is above Δ' .

It is clear that after introducing Δ satisfying the above conditions, the original solution of the problem is not changed. It is merely a technique to apply the optimality test and is omitted ultimately.

For the clear understanding of the method, see examples 1 and 2.

4.15 Unbalanced Transportation Problem

A transportation problem is said to be an **unbalanced transportation problem**, if the sum of all available amount (quantities) is not equal to the sum of all requirements,

$$\text{i.e., if } \sum_{i=1}^m a_i \neq \sum_{j=1}^n b_j$$

An unbalanced transportation problem is converted into a balanced transportation problem, by introducing a fictitious source or destination which will provide the surplus supply or demand. The cost of transporting an unit from the fictitious source (or to the fictitious destination) is taken to be zero. After converting the unbalanced transportation problem to balanced transportation problem, by the introduction of fictitious source or destination, it is solved by the earlier methods.

For the clear understanding of the method, see examples 3 and 4.

ILLUSTRATIVE EXAMPLES

❖ Problems of Degeneracy

Example 1: Solve the following transportation problem (cell entries represent unit cost).

						a_i	
	5	3	7	3	8	5	3
	5	6	12	5	7	11	4
	2	1	3	4	8	2	2
	9	6	10	5	10	9	8
b_j	3	3	6	2	1	2	17

Solution: By V.A.M. the initial B.F.S. of the given problem is given by the following table:

						a_i
(5)	(3) 1	(7)	(3)	(8)	(5) 2	3
(5) 3	(6)	(12)	(5) Δ	(7) 1	(11)	4
(2)	(1)	(3) 2	(4)	(8)	(2)	2
(9)	(6) 2	(10) 4	(5) 2	(10)	(9)	8
b_j	3	3	6	2	1	2

Since the total number of allocations is 8 which is one less than $m + n - 1 = 9$.

Hence this solution is degenerate solution.

Now to resolve this degeneracy we allocate a very small amount Δ to the cell (2, 4) getting 9 allocations at independent positions.

To test solution for optimality

Now to test the solution for optimality we construct the following table:

												u_i		
(5)	(2)	(3)	1		(7)	(7)	(3)	(2)	(8)	(4)	(5)	2		- 3
(3)				(0)		(1)		(4)						
(5)	(6)		(6)	(12)	(10)	(5)	Δ	(7)	(11)	(8)	0			
3		(0)		(2)				1		(3)				
(2)	(-2)	(1)	(-1)	(3)	2		(4)	(-2)	(8)	(0)	(2)	(1)	- 7	
(4)		(2)				(6)		(8)		(1)				
(9)	(5)	(6)	2		(10)	4		(5)	2		(10)	(7)	(9)	(8)
(4)										(3)		(1)		0
5		6		10		5		7		8				

Since all $d_{ij} \geq 0$, so the solution under test is optimal.

Thus the optimal solution is $x_{12} = 1, x_{16} = 2, x_{21} = 3, x_{25} = 1, x_{33} = 2, x_{42} = 2, x_{43} = 4, x_{44} = 2$ and the optimal cost

$$= ₹ (1 \times 3 + 2 \times 5 + 3 \times 5 + 1 \times 7 + 2 \times 3 + 2 \times 6 + 4 \times 10 + 2 \times 5) = ₹ 103$$

Example 2: Given below the unit costs array with supplies a_i ; $i = 1, 2, 3$ and demands b_j ; $j = 1, 2, 3, 4$.

		Sink				
Source		1	2	3	4	a_i
	1	8	10	7	6	50
	2	12	9	4	7	40
	3	9	11	10	8	30
	b_j	25	32	40	23	120

Find the optimal solution to the above Hitchcock problem.

Solution: By "Lowest Cost Entry Method" the initial B.F.S. of the given problem is given by the following table:

Since the total number of allocation is 5 which is one less than $m + n - 1 = 6$.

Hence, this solution is a degenerate solution.

	1	2	3	4	a_i
1	(8) 25	(10) 2	(7)	(6) 23	50
2	(12)	(9)	(4) 40	(7) Δ	40
3	(9)	(11) 30	(10)	(8)	30
b_j	25	32	40	23	

Now to resolve this degeneracy we allocate a very small amount Δ to the cell (2, 4) getting 6 allocations at independent positions.

To test the solution for optimality.

Now to test the solution for optimality we construct the following table:

	1	2	3	4	u_i
1	(8) 25	(10) 2	(7) (3)	(6) 23	0 (u_1)
2	(12) (9)	(9) (11)	(4) 40	(7) Δ	1 (u_2)
3	(9) (9)	(11) 30	(10) (4)	(8) (7)	1 (u_3)
v_j	8 (v_1)	10 (v_2)	3 (v_3)	6 (v_4)	

Since $d_{22} = -2 < 0$, so the solution under test is not optimal. Now we shall allocate to this cell (2, 2) as much as possible. Thus, we take Δ from cell (2, 4) to cell (2, 2) and form the new table to check the optimality of the solution.

	1	2	3	4	u_i
1	(8) 25	(10) 2	(7) (5)	(6) 23	0
2	(12) (7)	(9) Δ	(4) 40	(7) (5)	- 1
3	(9) (9)	(11) 30	(10) (6)	(8) (7)	1
v_j	8	10	3	6	

Since all $d_{ij} \geq 0$, so the solution under test is optimal.

Also $d_{31} = 0$, implies that an alternative optimal solution exists.

Thus, the optimal solution is $x_{11} = 25$, $x_{12} = 2$, $x_{14} = 23$, $x_{23} = 40$, $x_{32} = 30$

and the optimal cost

$$= ₹ (25 \times 8 + 2 \times 10 + 30 \times 11 + 40 \times 4 + 23 \times 6) = ₹ 848.$$

❖ Problems on Unbalanced Transportation Problem

Example 3: Determine the optimal transportation plan from the following table given the plant to market shipping costs and quantities at each market and available at each plant:

Plant	W_1	W_2	W_3	W_4	Availability
F_1	11	20	7	8	50
F_2	21	16	10	12	40
F_3	8	12	18	9	70
Requirement	30	25	35	40	

Solution: Here total requirement of the market = $30+25+35+40 = 130$

and total availability at the plants = $50+40+70 = 160$.

Since the total availability at three plants is 30 more than the total requirements in four markets W_1, W_2, W_3, W_4 . Therefore, this transportation problem is unbalanced and so we convert this problem to a balanced one by introducing a fictitious, market W_5 with requirement 30 such that the cost of transportation from plants to this market W_5 are zero.

Thus, the balanced transportation problem is given by the following table:

Plant	Market					Availability
	W_1	W_2	W_3	W_4	W_5	
F_1	11	20	7	8	0	50
F_2	21	16	10	12	0	40
F_3	8	12	18	9	0	70
Requirement	30	25	35	40	30	(Total) 160

By 'VAM', we get the following B.F. solution of the problem:

Plant	Market						u_i
	W_1	W_2	W_3	W_4	W_5	a_i	
F_1	(11) (7) (4)	(20) (11) (9)	(7) 25	(8) 25	(0) (-3) (3)	50	-1
F_2	(21) (10) (11)	(16) (14) (2)	(10) 10	(12) (11) (1)	(0) 30	40	2
F_3	(8) 30	(12) 25	(18) (8) (10)	(9) 15	(0) (-2) (2)	70	0
b_j	30	25	35	40	30	160	
v_j	8	12	8	9	-2		

The solution given in the above table is an optimal solution as all $d_{ij} \geq 0$.

Thus the optimal solution is

transport from plant F_1 to market W_3 , 25 units.

transport from plant F_1 to market W_4 , 25 units.

transport from plant F_2 to market W_3 , 10 units.

transport from plant F_3 to market W_1 , 30 units.

transport from plant F_3 to market W_2 , 25 units.

transport from plant F_3 to market W_4 , 15 units.

i.e. $x_{13} = 25, x_{14} = 25, x_{23} = 10, x_{31} = 30, x_{32} = 25, x_{34} = 15$

Total transportation cost

$$= ₹ (25 \times 7 + 25 \times 8 + 10 \times 10 + 30 \times 8 + 25 \times 12 + 15 \times 9) = ₹ 1150$$

It is important to note that 30 units are dispatched from plant F_2 to market (Fictitious) W_5 . In other words, we can say that 30 units are left undispached at the plant F_2 .

Example 4: A steel company has three open hearth furnaces and five rolling mills. Transportation cost (rupees per quintal) for shipping steel from furnaces to rolling mills are shown in the following table:

	M_1	M_2	M_3	M_4	M_5	Capacity (in quintal)
F_1	4	2	3	2	6	8
<i>Furnaces</i> F_2	5	4	5	2	1	12
F_3	6	5	4	7	3	14
<i>Requirement</i> (in quintal)	4	4	6	8	8	

What is an optimal shipping schedule?

Solution: Here total requirement of mills = 30 quintals

and total capacity of all furnaces = 34 quintals

Since total requirement and total capacity are not equal, so the problem is an unbalanced transportation problem.

Since the total capacity is 4 quintals more than the total requirement, so we convert this problem to a balanced one by introducing a fictitious mill M_6 of requirement 4 quintals with all transportation costs to this mill as zero.

The balanced transportation problem is given by the following table:

	M_1	M_2	M_3	M_4	M_5	M_6	Capacity
F_1	4	2	3	2	6	0	8
F_2	5	4	5	2	1	0	12
F_3	6	5	4	7	3	0	14
Requirement	4	4	6	8	8	4	34

By VAM, we get the following initial B.F.S. of the problem shown in this table:

	M_1	M_2	M_3	M_4	M_5	M_6	
F_1	(4)	4 (2)	(3)	4 (2)	(6)	(0)	8
F_2	(5)	(4)	(5)	4 (2)	8 (1)	(0)	12
F_3	4 (6)	(5)	6 (4)	(7)	(3)	4 (0)	14
	4	4	6	8	8	4	

To test the solution for optimality, we construct the following table with all the entries u_i and v_j such that for each occupied cell $c_{rs} = u_r + v_s$ and make the entries $u_i + v_j$ and $d_{ij} = c_{ij} - (u_i + v_j)$ at appropriate corners in empty cells.

	M_1	M_2	M_3	M_4	M_5	M_6	u_i
F_1	(4) (4) (2)	(3) (2) (2)	(6) (1) (0)	(-2)			-2
F_2	(5) (4) (4) (2) (5) (2) (2)	(1) (2) (3)	(8) (1) (0) (-2)	(2)			-2
F_3	(6) (4) (5) (4) (4) (7) (4) (3)	(0) (4)					0
$v_j \rightarrow$	6	4	4	4	3	0	

Since all $d_{ij} \geq 0$, so this solution is optima *i.e.* optimal solution is $x_{12} = 4, x_{14} = 4, x_{24} = 4, x_{25} = 8, x_{31} = 4, x_{33} = 6, x_{36} = 4$ and the minimum T.C.

$$= ₹(2 \times 4 + 2 \times 4 + 2 \times 4 + 1 \times 8 + 6 \times 4 + 4 \times 6 + 0 \times 4) = ₹ 80$$

Exercise 4.3

Solve the following cost minimizing transportation problems:

1.						2.					
	D_1	D_2	D_3	D_4	Availability		D_1	D_2	D_3	D_4	Availability
O_1	5	3	6	5	15	O_1	1	2	3	4	6
O_2	10	7	12	4	11	O_2	4	3	2	0	8
O_3	7	5	8	4	13	O_3	0	2	2	1	10
Demand	8	12	13	6		Demand	4	6	8	6	

3.						4.						
	Destinations						Destinations					
	D_1	D_2	D_3	$a_i \downarrow$			D_1	D_2	D_3	D_4	$a_i \downarrow$	
Sources	O_1	7	4	0	5	F_1	4	3	1	2	80	
	O_2	6	8	0	15		F_2	5	2	3	4	60
	O_3	3	9	0	9		F_3	3	5	6	3	40
	$b_j \rightarrow$	15	6	8			$b_j \rightarrow$	50	60	20	50	

5. A manufacturer wants to ship 8 loads of his product as shown in the table. The matrix gives the mileage from origin O to destination D. Shipping costs are ₹ 10 per load per mile. What shipping schedule be used?

	D_1	D_2	D_3	Available
O_1	50	30	220	1
O_2	90	45	170	3
O_3	250	200	50	4
Required	4	2	2	

6. Obtain an optimum B.F.S. to the following degenerate T.P.

	To			Available
From	7	3	4	2
	2	1	3	3
	3	4	6	5
Demand	4	1	5	10

7. A company has three plants A, B, C and three were houses X, Y and Z . Number of units, available at the plants are 60, 70 and 80 respectively. Demands at X, Y and Z are 50, 80 and 80 respectively. Units costs of transportation are as follows:

	X	Y	Z
A	8	7	3
B	3	8	9
C	11	3	5

Find the allocation so that total transportation cost is minimum.

8. Three factories A, B and C produce a certain product and supply to four warehouses D, E, F and G . The manufacturing capacities of the factories are 325, 375 and 500 units respectively. The requirement at the warehouses are 225, 300, 350 and 325 units respectively. The data regarding the unit transportation costs in ₹ from factories to warehouses are as follows:

	To			
From	D	E	F	G
A	19	18	18	14
B	11	27	20	26
C	22	15	12	19

- ❖ Solve the following cost minimizing transportation problems:

- 9.

	D_1	D_2	D_3	Available
O_1	30	20	10	800
O_2	5	15	25	500
Demand	300	300	400	

10.

	D_1	D_2	D_3	$a_i \downarrow$
O_1	4	3	2	10
O_2	2	5	0	13
O_3	3	8	6	12
$b_j \rightarrow$	8	5	4	

11.

	Origin				
Destination	A	B	C	D	Requirements
1	7	4	3	4	15
2	3	2	7	5	25
3	4	4	3	7	20
4	9	7	5	3	40
Available	12	8	35	25	

12.

		To				
		D_1	D_2	D_3	D_4	$a_i \downarrow$
	O_1	5	3	6	2	19
From	O_2	4	7	9	1	37
	O_3	3	4	7	5	34
	$b_j \rightarrow$	16	18	32	25	

13. A company has factories A, B and C which supply warehouses at D, E, F and G . Monthly factory capacities are 160, 150 and 190 units respectively. Monthly warehouse requirements are 80, 90, 110 and 160 units respectively. Unit shipping costs (in rupees) are follows:

		To			
		D	E	F	G
	A	42	48	38	37
From	B	40	49	52	51
	C	39	38	40	43

Determine an optimal transportation plan.

14. At the beginning to next week there will be a surplus of 6, 9, 7 and 5 trailers in cities 1, 2, 3 and 4 respectively. Cities A , B and C will have a deficit of 8, 7 and 9 trailers respectively. The cost for moving from each surplus city to each deficit city is given in rupee in the table.

Find the minimum cost transportation schedule.

	Deficit		
	A	B	C
1	26	32	28
2	19	27	16
Surplus 3	39	21	32
4	18	24	23

Answers 4.3

1.	$x_{11} = 8, x_{12} = 7, x_{22} = 5, x_{24} = 6, x_{33} = 13$. Min. T. cost = ₹ = 224
2.	$x_{12} = 6, x_{23} = 2, x_{24} = 6, x_{31} = 4, x_{33} = 6$. Min. T. cost = ₹ 28
3.	$x_{12} = 5, x_{21} = 6, x_{22} = 1, x_{23} = 8, x_{31} = 9$. Min. T. cost = ₹ 91
4.	$x_{11} = 10, x_{13} = 20, x_{14} = 50, x_{22} = 60, x_{31} = 40$. Min. T cost = ₹ 400.
5.	$x_{11} = 1, x_{21} = 3, x_{32} = 2, x_{33} = 2$. Mini. Mileage = 820 mile, Min. T. cost = ₹ 8200
6.	$x_{13} = 2, x_{22} = 1, x_{23} = 2, x_{31} = 4, x_{33} = 1$. Mini. T. cost = ₹ 33
7.	$x_{13} = 60, x_{21} = 50, x_{23} = 20, x_{32} = 80$. Min. T. cost = ₹ 750
8.	$x_{14} = 325, x_{21} = 225, x_{23} = 150, x_{32} = 300, x_{33} = 200$.
9.	$x_{12} = 100, x_{13} = 400, x_{21} = 300, x_{22} = 200$. 300 units are left at O_1 Min. T. cost = ₹ 10500.

10.	$x_{12} = 5, x_{21} = 8, x_{23} = 4$. 5, 1 and 12 units are left at sources O_1, O_2, O_3 respectively. Min. T. cost = ₹ 31
11.	$x_{A2} = 12, x_{B2} = 8, x_{C1} = 15, x_{C3} = 20, x_{D4} = 25$. Requirement of 5 and 15 units unfulfilled at destinations 2 and 4 respectively.
12.	$x_{12} = 18, x_{13} = 1, x_{21} = 12, x_{24} = 25, x_{31} = 4, x_{33} = 30$. D_3 gets one unit less than requires. Min. T. cost = ₹ 355.
13.	$x_{14} = 160, x_{21} = 80, x_{22} = 10, x_{32} = 80, x_{33} = 110$, 60 units one left with factory B. Min. T. cost = ₹ 17050.
14.	$x_{11} = 3, x_{23} = 9, x_{32} = 7, x_{41} = 5$. 3 units remain surplus at city 1. Min. T.C. = ₹ 459.

4.16 Profit Maximization Problems

Till now we have discussed cost minimization transportation problems. But in some cases in the transportation matrix, the profits are given in place of costs and we are required to find the allocations to maximize the profit. Such problems can be termed as **profit maximization problems**.

Such problems can be solved in any of the following two ways.

1. Change the profit maximization problem to cost minimization problem by multiplying each element of the profit matrix by -1 and then proceed to solve as usual, see example 1.
2. Change the profit maximization problem to cost minimization problem by multiplying each element of the profit matrix by -1 and then to avoid work on negative values add the highest negative value in the matrix, with positive sign, to each element of the matrix. Thus, all entries in the corresponding cost matrix will be positive. Now proceed as usual. This cost matrix may also be obtained by subtracting each element of the profit matrix from the greatest (highest) value, element in the table, see example 2.

ILLUSTRATIVE EXAMPLES

❖ Profit Maximization Problems

Example 1: Solve the following transportation problem to maximize profit and give criteria for optimality.

Origin	Profit (₹)/Unit. Destination				Supply
	1	2	3	4	
A	40	25	22	33	100
B	44	35	30	30	30
C	38	38	28	30	70
Demand	40	20	60	30	

Solution: In the given problem, the objective is to maximize the profit. First we convert the given profit matrix to loss matrix by multiplying all the entries in the given matrix by minus sign. Thus, the loss matrix is as follows:

Origin	Destination				Supply
	1	2	3	4	
A	– 40	– 25	– 22	– 33	100
B	– 44	– 35	– 30	– 30	30
C	– 38	– 38	– 28	– 30	70
Demand	40	20	60	30	

Since here the supply $200 > 150$, demand, so the above T.P. is not balanced one, therefore we introduce a fictitious (dummy) destination 5 with demand (requirement) 50 such that the cost of transportation from any plant to destination 5 is zero.

∴ The balanced T.P. is given in the following table:

Origin	Destination					Supply
	1	2	3	4	5	
A	– 40	– 25	– 22	– 33	0	100
B	– 44	– 35	– 30	– 30	0	30
C	– 38	– 38	– 28	– 30	0	70
Demand	40	20	60	30	50	200 (Total)

By VAM, we get the following initial B.F. solution of the problem:

Origin	1	2	3	4	5	Supply
A	(- 40) 30	(- 25)	(- 22) 20	(- 33) 30	(0) 50	100
B	(- 44) 30	(- 35)	(- 30)	(- 30)	(0)	30
C	(- 38) 10	(- 38) 20	(- 28) 40	(- 30)	(0)	70
Demand	40	20	60	30	50	

For this B.F.S. total T.C. = ₹ (- 440 - 990 + 0 - 1320 - 380 - 760 - 1120)
= - (₹ 5010).

To test the solution for optimality

Now to test the solution for optimality we construct the following table:

	1	2	3	4	5	
A	(- 40) (-32) (- 8)	(- 25) (-32) (7)	(- 22) 20	(- 33) 30	(0) 50	6
B	(- 44) 30	(- 35) (- 44) (9)	(- 30) (-34) (4)	(- 30) (-45) (15)	(0) (-12) (12)	- 6
C	(- 38) 10	(- 38) 20	(- 28) 40	(-30) (-39) (9)	(0) (- 6) (6)	0
	- 38	- 38	- 28	- 39	- 6	

Since $d_{11} = -8 < 0$, so the solution under test is not optimal.

Since minimum $d_{11} = -8$ (negative), so we give maximum allocation θ to this cell from a occupied cell and make the necessary changes in other allocations as shown in the table:

	1	2	3	4	5
A	θ ↑		$20 - \theta$ →	30	50
B		30			
C	$10 - \theta$ ←	20	$40 + \theta$ ↓		

$\therefore$ Minimum of the allocations containing $-\theta$ is $10 - \theta$

$\therefore$ taking $10 - \theta = 0$, we get $\theta = 10$

The new B.F.S. is shown in the following table:

	1	2	3	4	5
A	(- 40) 10	(- 25)	(- 22) 10	(- 33) 30	(0) 50
B	(- 44) 30	(- 35)	(- 30)	(- 30)	(0)
C	(- 38)	(- 38) 20	(- 28) 50	(- 30)	(0)

For this B.F.S. total T.C. = ₹ $(- 400 - 220 - 990 - 1320 - 760 - 1400)$

= - ₹ 5090 < - ₹ 5010 (T.C. for initial B.F.S.)

To test this solution for optimality we construct the following table:

	1	2	3	4	5	u_i
A	(- 40) 10	(- 25) (- 32) (7)	(- 22) 10	(- 33) 30	(0) 50	0
B	(- 44) 30	(- 35) (- 36) (1)	(- 30) (- 26) (- 4)	(- 30) (- 37) (7)	(0) (- 4) (4)	- 4
C	(- 38) (- 46) (8)	(- 38) 20	(- 28) 50	(- 30) (- 39) (9)	(0) (- 6) (6)	- 6
v_j	- 40	- 32	- 22	- 33	0	

Since $d_{23} = -4 < 0$, so this solution also is not optimal.

As minimum $d_{23} = -4 < 0$, so giving the maximum allocation θ to this cell from occupied cell we make the necessary changes in other allocations as follows.

	1	2	3	4	5
A	10 + θ		10 - θ	30	50
B	30 - θ		θ		
C		20	50		

$\therefore$ minimum of the allocations containing $-\theta$ is $10 - \theta$

$\therefore$ taking $10 - \theta = 0$, we get $\theta = 10$

Thus, the new B.F.S. is shown in the following table:

	1	2	3	4	5
A	(- 40) 20	(- 25)	(- 22)	(- 33) 30	(0) 50
B	(- 44) 20	(- 35)	(- 30) 10	(- 30)	(0)
C	(- 38)	(- 38) 20	(- 28) 50	(- 30)	(0)

For this B.F.S. total T.C. = ₹ (- 800 - 990 - 880 - 300 - 760 - 1400)

$$= -\text{₹ } 5130 < -\text{₹ } 5090 \text{ (T.C. for the last B.F.S.)}$$

To test this solution for optimality we construct the following table:

	1	2	3	4	5	u_i
A	(- 40) 20	(- 25) (- 36) (11)	(- 22) (- 26) (4)	(- 33) 30	(0) 50	0
B	(- 44) 20	(- 35) (- 40) (5)	(- 30) 10	(- 30) (- 37) (7)	(0) (- 4) (4)	- 4
C	(- 38) (- 42) (4)	(- 38) 20	(- 28) 50	(- 30) (- 35) (5)	(0) (- 2) (2)	- 2
v_j	- 40	- 36	- 26	- 33	0	

Since all $d_{ij} > 0$. Hence, the solution shown in this table is optimal.

After converting the loss matrix into profit matrix the final allocation are given by $x_{11} = 20$, $x_{14} = 30$, $x_{21} = 20$, $x_{23} = 10$, $x_{32} = 20$, $x_{33} = 50$ and 50 units are left with origin A.

$$\text{Maximum Profit} = \text{₹ } (800 + 990 + 880 + 300 + 760 + 1400) = \text{₹ } 5130$$

Example 2: A department store wishes to purchase the following quantities of dresses.

Dress Type	A	B	C	D	E
Quantity	150	100	75	250	200

Tenders are submitted by four different manufacturers who undertake to supply not more than the quantities as:

Manufacture	X	Y	Z	W
Quantity	250	150	200	300

The estimated profits (in ₹) per dress is given in the profit matrix:

	A	B	C	D	E
X	6.00	6.25	7.50	4.75	4.00
Y	5.50	6.50	7.75	5.00	4.25
Z	6.25	5.75	7.00	5.50	4.75
W	5.75	6.50	7.25	5.25	4.50

How should the orders be placed to maximize the profit?

Solution: According to the given data the transportation matrix (profit matrix) is as follows:

	A	B	C	D	E	Supply
X	6.00	6.25	7.50	4.75	4.00	250
Y	5.50	6.50	7.75	5.00	4.25	150
Z	6.25	5.75	7.00	5.50	4.75	200
W	5.75	6.50	7.25	5.25	4.50	300
Demand	150	100	75	250	200	

The given problem is a maximization problem. First we convert the given profit matrix to loss matrix by multiplying each element of the matrix by -1 . Thus, the corresponding loss matrix is as follows and here we require the minimization of loss.

	A	B	C	D	E	
X	- 6.00	- 6.25	- 7.50	- 4.75	- 4.00	250
Y	- 5.50	- 6.50	- 7.75	- 5.00	- 4.25	150
Z	- 6.25	- 5.75	- 7.00	- 5.50	- 4.75	200
W	- 5.75	- 6.50	- 7.25	- 5.25	- 4.50	300
	150	100	75	250	200	

To avoid the work on negative and fractional values, first we multiply each cell value by 100 to get the integral values and then add 775 (the highest –ve value) to each changed cell value. This we get the following loss matrix with positive integral values.

	A	B	C	D	E	
X	175	150	25	300	375	250
Y	225	125	0	275	350	150
Z	150	200	75	225	300	200
W	200	125	50	250	325	300
	150	100	75	250	200	

This may also be obtained by subtracting each element of the profit matrix from the highest element 7.75 of the table and then multiplying each element by 100 to avoid fractions.

Here, the total supply = 250+150+200+300 = 900 is more than the total demand = 150 + 100 + 75 + 250 + 200 = 775 by 900 – 775 = 125 units.

So we add a dummy department store *F* with demand of 125 units with costs 0 and get a balanced transportation problem. Initial B.F.S. by VAM is given in this table.

	A	B	C	D	E	F	
X	150 (175)	(150)	(25)	(300)	100 (375)	(0)	250
Y	(225)	75 (125)	75 (0)	(275)	(350)	(0)	150
Z	(150)	(200)	(75)	75 (225)	(300)	125 (0)	200
W	(200)	25 (125)	(50)	175 (250)	100 (325)	(0)	300
	150	100	75	250	200	125	

For this B.F.S. total cost (taking 100 as a unit for transportation)

$$= ₹ (175 \times 150 + 375 \times 100 + 125 \times 75 + 0 \times 75 + 225 \times 75 + 0 \times 125 + 125 \times 25 + 250 \times 175 + 325 \times 100) = ₹ 169375$$

Also the total number of allocations is 9 ($= m + n - 1$) at independent positions.

To test the solution for optimality

To test this solution for optimality we construct the following table in which for occupied cell $c_{rs} = u_r + v_s$ and in unoccupied cells we enter the evaluations $u_i + v_j$ and $d_{ij} = c_{ij} - (u_i + v_j)$.

										$u_i \downarrow$
175	150	125	25	50	300	300	375	0	75	
150							100	$-\theta$	$+\theta$	50
		25		-25		0			-75	
225	125	125	0		275	250	350	325	0	25
		75		75						0
100						25		25		-25
150	100	200	100	75	-25	225	300	300	0	
						75				-25
	50		100		100	$+\theta$		0		125
										$-\theta$
200	125	125	50	0	250		325		0	25
		25				175				
	75			50		$-\theta$			100	
								$+\theta$		-25
$v_j \rightarrow$	125	125	0		250		325		25	

In the above table, all d_{ij} are not non-negative. So the solution in this table is not optimal as the most negative d_{ij} is $d_{16} = -75 < 0$, so we allocate maximum amount say θ to this cell and make necessary shifting of allocations in the table. Since minimum of the allocations containing $-\theta$ is $100 - \theta$ $\therefore$ taking $100 - \theta = 0$, we get $\theta = 100$. Thus, the table with the new allocations is as follows:

							$u_i \downarrow$				
175	150	100	25	-25	300	225	375	300	0	100 + θ	-25
150 - θ											

Here, the number of allocations is $9 (= m + n - 1)$ in independent positions. For these allocations T.C. = ₹ 161875, which is less than the previous cost.

To test this solution for optimality all the entries for $u_i, v_j, u_i + v_j$ and d_{ij} are shown in the last table on page 160. In this table all d_{ij} are not non-negative. So this solution is not optimal. $\therefore$ the most negative d_{ij} is $d_{31} = -25 = d_{26} = d_{46}$, so we allocate maximum amount say θ to one of these cells say in the cell (3, 1) and make necessary shifting of the allocations in the table. Since minimum of the allocations containing $-\theta$ is $25 - \theta \therefore$ taking $25 - \theta = 0$, we get $\theta = 25$. Thus, the table with new allocations is as follows.

	A	B	C	D	E	F	$u_i \downarrow$
X	175 125	150 125 25	25 0 25	300 250 50	375 325 50	0 125	0
Y	225 125 100	125 75	0 75	275 250 25	350 325 25	0 0 0	0
Z	150 25	200 100 100	75 -25 100	225 175	300 300 0	0 -25 25	- 25
W	200 175 25	125 25	50 0 50	250 75	325 200	0 0 0	0
$v_j \rightarrow$	175	125	0	250	325	0	

Here, the number of allocations is $9 (= m + n - 1)$ in independent positions. For these allocations T.C. = ₹ 161250, which is less than the previous cost.

To test this solution for optimality, all the entries for $u_i, v_j, u_i + v_j$ and d_{ij} are shown in the above table.

In this table all $d_{ij} \geq 0$, so this solution is optimal.

$\therefore$ The optimal solution for maximum profit (minimum cost) is

supply 125 units from manufacturer X to store A, 75 units from Y to B, 75 units

from Y to C, 25 units from Z to A, 175 units from Z to D, 25 units from W to B, 75 units from W to D and 200 units from W to E. Note that 125 units will remain with X, unsupplied to any store. i.e. $x_{11} = 125, x_{16} = 125, x_{22} = 75, x_{23} = 75, x_{31} = 25, x_{34} = 175, x_{42} = 25, x_{44} = 25, x_{44} = 75, x_{45} = 200$.

$$\begin{aligned}
 \text{Total Profit} &= ₹ (125 \times 6.00 + 75 \times 6.50 + 75 \times 7.75 + 25 \times 6.25 \\
 &\quad + 175 \times 5.50 + 25 \times 6.50 + 75 \times 5.25 + 200 \times 4.50) \\
 &= ₹ 4393.75
 \end{aligned}$$

4.17 Prohibited Transportation Route

Some times it is not possible to transport goods from certain sources to certain destinations due to road blockade or for any reasons, such type of problems can be handled by assigning a very large cost say M (or ∞) to that route or cell.

Example 1: Given the following data:

	Destinations			Capacity
	1	2	3	
Sources	1	2	3	10
	2	4	1	15
	3	1	3	40
Demand		20	15	30

The cost of shipment from third source to the third destination is not known. How many units should be transported from sources to the destinations so that the total cost of transporting all the units to their destinations is a minimum?

Solution: Since cost C_{33} in cell (3, 3) not known, so we assign a very large cost say M to this cell. By VAM an initial B.F.S. is shown in the following table.

Also all the entries of the optimality test are shown in this table:

	1	2	3	$u_i \downarrow$
1	(2) (4 - M) (M - 2)	(2) (6 - M) (M - 4)	(3) 10	3 - M
2	(4) (3 - M) (M + 1)	(1) (5 - M) (M - 4)	(2) 15	2 - M
3	(1) 20	(3) 15	(M) 5	0
$v_j \rightarrow$	1	3	M	

Here all $d_{ij} \geq 0$, since M is very large. So this solution is optimal.

Hence the optimal solution is

$$x_{13} = 10, x_{23} = 15, x_{31} = 20, x_{32} = 15, x_{33} = 5$$

Note: The cell (3, 3) also appears in the solution for which the cost of shipment is not known. This is known as pseudo optimum basic feasible solution.

Exercise 4.4

1. Solve the following transportation problem for maximum profit.

Per unit profit (₹)					
Market					
Warehouse	A	B	C	D	Available
X	12	18	6	25	200
Y	8	7	10	18	500
Z	14	3	11	20	300
Demand	180	320	100	400	

2. A company manufacturing coolers has two plants located at Mumbai and Kolkata with a capacity of 200 units and 100 units per week respectively. The company supplies the coolers to its four showrooms situated at Agra, Delhi, Lucknow and Kanpur which have a maximum demand of 75, 100, 100 and 30 units respectively.

Due to the differences in raw material cost and transportation cost, the profit per unit in ₹ differs which are shown in the following table:

	Agra	Delhi	Lucknow	Kanpur
Mumbai	90	90	100	110
Kolkata	50	70	130	85

Plan the production plan so as to maximize the profit. The company may have its production capacity at both plants partly or wholly unused.

3. A company has factories at four different places which supply warehouses A, B, C, D and E. Monthly factory capacities are 200, 175, 150 and 325 respectively.

Monthly warehouse requirements are 110, 90, 120, 230 and 160 respectively. Unit shipping costs are given in the following table. The costs are in rupees.

		To				
		A	B	C	D	E
From	1	13	–	31	8	20
	2	14	9	17	6	10
	3	25	11	12	17	15
	4	10	21	13	–	17

Shipments from 1 to B and from 4 to D is not possible. Determine the optimum distribution to minimize shipping costs.

4. Solve the following transportation problem with the restriction that nothing can be transported from factory F_2 to warehouse W_3 .

Factory	Warehouse				Available
	W_1	W_2	W_3	W_4	
F_1	40	60	60	10	15
F_2	70	30	–	20	4
F_3	30	90	20	40	11
Requirements	10	5	10	5	

Answers 4.4

1.	$x_{12} = 200, x_{22} = 120, x_{23} = 100, x_{24} = 280, x_{31} = 180, x_{34} = 120$. Max. T. cost = ₹ 15,400.
2.	$x_{BA} = 70, x_{BD} = 100, x_{BK} = 30, x_{CL} = 100$. Demand of 5 remain unfulfilled at Agra. Max. Profit = ₹ 31600
3.	$x_{ID} = 200, x_{2D} = 30, x_{2E} = 145, x_{3B} = 10, x_{3C} = 45, x_{3E} = 15, x_{4A} = 110, x_{4C} = 75$. 140 units are left with factory 4. Max. Profit = ₹ 7060.
4.	$x_{11} = 9, x_{12} = 1, x_{14} = 5, x_{22} = 4, x_{31} = 1, x_{33} = 10$. Total T. cost = ₹ 820

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1. A necessary and sufficient condition for the existence of feasible solution of a m by n transportation problem for $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$ is that:
(a) $\sum a_i = \sum b_j$ (b) $\sum a_i < \sum b_j$
(c) $\sum a_i > \sum b_j$ (d) None of these.
2. The feasible solution of a transportation problem can be obtained by the method:
(a) North-West corner method (b) Lowest cost entry method
(c) Vogel's approximation method (d) All of these.
3. Which of the following is used to calculate the feasible solution in transportation problem?
(a) Vogel's approximation method (b) Stepping stone method
(c) North-West corner method (d) Both (a) and (c).
4. To find feasible solution of a transportation problem by North-West corner method, we start with:
(a) (1,1) cell (b) Any cell in first row
(c) Any cell in first column (d) None of these.
5. In a B.F.S. of a m by n transportation problem the total number of positive allocations is at most:
(a) $m - n$ (b) $m + n$
(c) $m + n - 1$ (d) None of these.
6. In a m by n transportation problem, degeneracy is said to occur when the number of allocations is:
(a) Greater than $m + n - 1$ (b) Less than $m + n - 1$
(c) Equal to $m + n - 1$ (d) Less than $m - n - 1$.
7. A transportation problem is said to be unbalanced if:
(a) $\sum a_i = \sum b_j$ (b) $\sum a_i < \sum b_j$
(c) $\sum a_i > \sum b_j$ (d) (b) or (c).
8. In a 5 by 6 transportation problem, for a feasible solution to be non-degenerate, the total number of allocations is:
(a) 30 (b) 11
(c) 10 (d) 12.

Fill in the Blank

1. A feasible solution of a m by n transportation problem is said to be non-degenerate basic feasible solution if it has total number of $m + n - 1$ positive allocations in positions.

2. A feasible solution of a transportation problem is said to be optimal if it the total transportation cost.
3. In a transportation problem the solution under test is optimal and unique if for all empty cells $d_{ij} - (u_i + v_j) > \dots\dots\dots$.
4. In computational procedure of optimality test we choose that u_i or $v_j = 0$ for which the corresponding row or column has the number of individual allocations.
5. In Vogel's approximation method the difference of the smallest and second smallest costs in each row and column are called

True/False

1. The problems in which the amounts of commodity from various origins are transported to various destinations in such a way that the total cost of transportations is minimum are called transportation problems.
2. A transportation problem can not be expressed as a L.P. problem.
3. A balanced transportation problem always has a F.S.
4. A balanced transportation problem always has a B.F.S.
5. In computational procedure of optimality test, for each occupied cell $c_{rs} = u_r + v_s$.

Answers

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1.	(a)	2.	(d)	3.	(b)
4.	(a)	5.	(c)	6.	(b)
7.	(d)	8.	(b)		

Fill in the Blank

1.	independent	2.	minimizes	3.	0
4.	maximum	5.	penalties		

True/False

1.	T	2.	F	3.	T
4.	T	5.	T		

UNIT-I

Chapter

5



Assignment Problem

5.1 Introduction

It is a special type of linear programming problem in which the objective is to find the optimum allocation of a number of tasks (jobs) to an equal number of facilities (persons). Here we make the assumption that each person can perform each job but with varying degree of efficiency. For example, a departmental head may have four persons available for assignment and four jobs to fill. Then his interest is to find the best assignment which will be in the best interest of the department.

Assignment problem finds many applications in allocation. For example, in assigning men to works; trucks or cars or drivers to different roots; planes or pilots to different commercial flights etc.

Although simplex method is powerful enough to solve all the L.P. problems, but the above type of the problems may be solved by special procedures which are described in the following sections.

5.2 Assignment Problem

The assignment problem can be stated in the form of $n \times n$, matrix $[c_{ij}]$ called the **cost or effectiveness matrix**, where c_{ij} is the cost of assigning i -th facility (person) to the j -th job

Effectiveness matrix

		Jobs.						
		1	2	3	...	j	...	n
Persons	1	c_{11}	c_{12}	c_{13}	...	c_{1j}	...	c_{1n}
	2	c_{21}	c_{22}	c_{23}	...	c_{2j}	...	c_{2n}
	3	...	...	...	...	...	...	...
	...	...	...	...	...	...	...	...
	...	...	...	...	...	...	...	...
	i	c_{i1}	c_{i2}	c_{i3}	...	c_{ij}	...	c_{in}
	...	...	...	...	...	...	...	...
	n	c_{n1}	c_{n2}	c_{n3}	...	c_{nj}	...	c_{nn}

n persons can be assigned to n jobs in $n!$ possible ways. One method may be to find all possible $n!$ assignments and evaluate total costs in all cases. Then the assignment with minimum cost (as required) will give the optimal assignment. But this method is extremely laborious. For example if $n = 8$, then the number of such possible assignments is $8! = 40320$. The evaluation of costs for all these allocations will take a large time. Thus, there is a need to develop an easy computational technique for the solution of assignment problems.

Mathematical Formulation of Assignment Problem

[Rohd.Univ. 2000]

Mathematically an assignment problem can be stated as follows :

Minimize the total cost

$$Z = \sum_{i=1}^n \sum_{j=1}^n c_{ij} \cdot x_{ij}$$

where $x_{ij} = \begin{cases} 1, & \text{if } i\text{-th person is assigned to the } j\text{-th job} \\ 0, & \text{if } i\text{-th person is not assigned to the } j\text{-th job} \end{cases}$

Subject to the conditions

$$(i) \quad \sum_{i=1}^n x_{ij} = 1, j = 1, 2, \dots, n$$

which means that only one job is done by the i -th person, $i = 1, 2, \dots, n$

$$(ii) \quad \sum_{j=1}^n x_{ij} = 1, i = 1, 2, \dots, n$$

which means that only one-person should be assigned to the j -th job, $j = 1, 2, \dots, n$.

5.3 Difference between a Transportation and an Assignment Problem

An assignment problem is a special case of transportation problem in which $m = n$ and all a_i 's and b_j 's ($i = 1, 2, \dots, m; j = 1, 2, \dots, n$) are unity and each x_{ij} is limited to one of the two values 0 and 1.

In these circumstances, exactly n of x_{ij} can be non-zero, one in each row of the array and one in each column showing that only one source (person) can be assigned to each destination (job).

5.4 Important Theorem

Now we shall prove the following two important theorems on which the procedure of solution of an assignment problem is based.

Theorem 1: (Reduction theorem) : If, in an assignment problem, a constant is added or subtracted to every element of a row (or column) of the cost matrix $[c_{ij}]$, then an assignment which minimizes the total cost for one matrix, also minimizes the total cost for the other matrix.

or

Mathematically the theorem may be stated as follows :

If $x_{ij} = X_{ij}$, minimizes $Z = \sum_{i=1}^n \sum_{j=1}^n c_{ij} x_{ij}$ over all x_{ij} s.t.

$$\sum_{i=1}^n x_{ij} = 1 = \sum_{j=1}^n x_{ij} \text{ and } x_{ij} \geq 0$$

then $x_{ij} = X_{ij}$ also minimizes $Z' = \sum_{i=1}^n \sum_{j=1}^n c'_{ij} \cdot x_{ij}$

where $c'_{ij} = c_{ij} \pm a_i \pm b_j$.

a_i, b_j are constants, $i = 1, 2, \dots, n; j = 1, 2, \dots, n$

Proof : We have

$$\begin{aligned} Z' &= \sum_{i=1}^n \sum_{j=1}^n c'_{ij} x_{ij} \\ &= \sum_{i=1}^n \sum_{j=1}^n (c_{ij} \pm a_i \pm b_j) \cdot x_{ij} \\ &= \sum_{i=1}^n \sum_{j=1}^n c_{ij} x_{ij} \pm \sum_{i=1}^n \sum_{j=1}^n a_i x_{ij} \pm \sum_{i=1}^n \sum_{j=1}^n b_j x_{ij} \end{aligned}$$

$$\begin{aligned}
&= Z \pm \sum_{i=1}^n a_i \left(\sum_{j=1}^n x_{ij} \right) \pm \sum_{j=1}^n b_j \left(\sum_{i=1}^n x_{ij} \right) \\
&= Z \pm \sum_{i=1}^n a_i \cdot 1 \pm \sum_{j=1}^n b_j \cdot 1 \\
&= Z \pm \sum_{i=1}^n a_i \pm \sum_{j=1}^n b_j
\end{aligned}$$

Since, $\sum_{i=1}^n a_i$, $\sum_{j=1}^n b_j$ are independent of x_{ij} , it follows that Z' is minimized when Z is minimized.

Hence, $x_{ij} = X_{ij}$ which minimizes Z also minimizes Z' .

Theorem 2 : If all $c_{ij} \geq 0$ and there exists a solution

$$x_{ij} = X_{ij} \quad \text{s.t.} \quad \sum_{i=1}^n \sum_{j=1}^n c_{ij} x_{ij} = 0$$

then this solution is an optimal solution (i.e. this solution minimizes Z).

Proof : Since all $c_{ij} \geq 0$

$$\therefore Z = \sum_{i=1}^n \sum_{j=1}^n c_{ij} x_{ij} \text{ cannot be negative.}$$

Thus, its minimum value is 0, when $x_{ij} = X_{ij}$.

Hence, the solution $x_{ij} = X_{ij}$ for which $\sum_{i=1}^n \sum_{j=1}^n c_{ij} x_{ij} = 0$ is an optimal solution.

5.5 Hungarian Method (Reduced Matrix Method)

For solving a minimal assignment problem (Assignment algorithm).

From the above two theorems we get a powerful method termed as "**assignment algorithm**" (Hungarian Method) for the solution of minimal assignment problem which was developed and presented by Hungarian Mathematician D.Konig. The procedure for the solution is as follows :

Step 1 : Subtract the minimum element of each row in the cost matrix $[c_{ij}]$ from every element of the corresponding row.

Step 2 : Subtract the minimum element of each column in the reduced matrix obtained in the step 1 from every element of the corresponding column.

Step 3 : (a) Starting with row 1 of the matrix obtained in step 2, examine rows successively until a row with exactly one zero element is found. Mark ($\square$), at this zero, as an assignment will be made there. Mark ($\times$) at all other zeros in the column (in which we mark $\square$) to show

that they cannot be used to make other assignments. Proceed in this way until the last row is examined.

(b) After examining all the rows completely, proceed similarly examining the columns. Examine columns starting with column 1 until a column containing exactly one unmarked zero is found. Mark ($\square$) at this zero and cross (mark $\times$) at all zeros of the row in which $\square$ is marked. Proceed in this way until the last column is examined.

(c) Continue these operations (a) and (b) successively until we reach to any of the two situations.

(i) All the zeros are marked $\square$ or crossed.

or

(ii) The remaining unmarked zeros lies at least two in each row and column.

In case (i), we have a maximal assignment (assignment as much as we can) and in case (ii) still we have some zeros to be treated for which we use the trial and error method to avoid the use of highly complicated algorithm.

Now there are two possibilities :

- (i) If it has an assignment in every row and every column (*i.e.*, total number of marked $\square$ zeros is exactly n), then the complete optimal assignment is obtained. (See example 1)
- (ii) If it does not contain assignment in every row and every column (*i.e.*, the total number of marked $\square$ zeros is less than n), then one has to modify the cost (effectiveness) matrix by adding or subtracting to create some more zeros in it. For this proceed to the next step 4.

Step 4 : When the matrix obtained in step 3 does not contain assignment in every row and every column then we draw the minimum number of horizontal and vertical lines necessary to cover all zeros at least once. For this the following procedure is adopted.

- (i) Mark ($\checkmark$) all rows for which assignment have not been made.
- (ii) Mark ($\checkmark$) column which have zeros in marked rows.
- (iii) Mark ($\checkmark$) rows (not already marked) which have assignment in marked columns.
- (iv) Repeat step (ii) and (iii) until the chain of marking ends.
- (v) Draw minimum number of lines through unmarked rows and through marked columns to cover all the zeros.

This procedure will yield the minimum number of lines (equal to the number of assignments in the maximal assignment obtained in step 3) that will pass through all zeros.

Step 5 : Select the smallest of the elements that do not have a line through them, subtract it from all these elements that do not have a line through them, add it to every element that lies at the intersection of two lines and leave the remaining elements of the matrix unchanged.

Step 6 : At the end of step 5, number of zeros are increased (never decreased) in the matrix than that in step 3.

Now re-apply the step 3 to the modified matrix obtained in step 5, to obtain the desired solution. For the clear understanding of the procedure see Example 4.

ILLUSTRATIVE EXAMPLES

Example 1: Solve the following minimal assignment problem :

Man →	1	2	3	4
Job ↓				
I	12	30	21	15
II	18	33	9	31
III	44	25	24	21
IV	23	30	28	14

Solution: For the clear understanding, this example is solved step by step systematically.

Step 1 : Subtracting the smallest element of each row from every element of the corresponding row, we get the following matrix :

	1	2	3	4
I	0	18	9	3
II	9	24	0	22
III	23	4	3	0
IV	9	16	14	0

Step 2 : Subtracting smallest element of each column from every element of the corresponding column, we get the following matrix :

	1	2	3	4
I	0	14	9	3
II	9	20	0	22
III	23	0	3	0
IV	9	12	14	0

Step 3 : Now we test whether it is possible to make an assignment using the zeros by the method described in step 3 in § 5.5

Starting with row I, we mark $\square$ (i.e., make assignment) in the row containing only one zero and cross ($\times$) the zeros in the corresponding column in which $\square$ lies. Thus, we get the following table.

	1	2	3	4
I	$\square$	14	9	3
II	9	20	$\square$	22
III	23	0	3	$\times$
IV	9	12	14	$\square$

Again starting with column 1, we mark $\square$ (i.e., make assignment) in the column containing only one unmarked or uncrossed zero in the above table and cross out the zeros in the corresponding row in which this assignment $\square$ is marked. Thus, we get the following table.

	1	2	3	4
I	$\square$	14	9	3
II	9	20	$\square$	22
III	23	$\square$	3	8
IV	9	12	14	$\square$

Since in the last table, every row and every column have one assignment, so we have the complete optimal zero assignment.

Job :	I	II	III	IV
Man :	1	3	2	4

i.e., $I \rightarrow 1, II \rightarrow 3, III \rightarrow 2, IV \rightarrow 4$.

Which is the optimal assignment.

Example 2 : A department head has four subordinates, and four tasks to be performed. The subordinates differ in efficiency and the tasks differ in their intrinsic difficulty. His estimate of the times each man would take to perform each task is given in the effectiveness matrix below. How should the task be allocated, one to a man, so as to minimize the total man hour?

Subordinates

	I	II	III	IV
A	8	26	17	11
B	13	28	4	26
C	38	19	18	15
D	19	26	24	10

Solution : Step 1 : Subtracting the smallest element in each row from every element of the corresponding row, we get the following matrix.

	I	II	III	IV
A	0	18	9	3
B	9	24	0	22
C	23	4	3	0
D	9	16	14	0

Step 2: Subtracting the smallest element in each column of the above matrix from every element of the corresponding column, we get the following matrix.

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>
<i>A</i>	0	14	9	3
<i>B</i>	9	20	0	22
<i>C</i>	23	0	3	0
<i>D</i>	9	12	14	0

The above matrix is the same as obtained in step 3 in example 1, therefore for minimum man hours the allotment should be as follows :

Tasks	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
Subordinates	<i>I</i>	<i>III</i>	<i>II</i>	<i>IV</i>
Man hours	8	4	19	10

i.e., $A \rightarrow I, B \rightarrow III, C \rightarrow II, D \rightarrow IV$.

and the total Man hours are $8 + 4 + 19 + 10 = 41$.

Example 3: Solve minimal assignment problem whose effectiveness matrix is

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>
<i>A</i>	2	3	4	5
<i>B</i>	4	5	6	7
<i>C</i>	7	8	9	8
<i>D</i>	3	5	8	4

Solution: **Step 1 :** Subtracting the smallest element of each row from every element of the corresponding row, we get the following matrix.

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>
<i>A</i>	0	1	2	2
<i>B</i>	0	1	2	3
<i>C</i>	0	1	2	1
<i>D</i>	0	2	5	1

Step 2: Subtracting the smallest element of each column of the above matrix from the corresponding column, we get the following matrix.

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>
<i>A</i>	0	0	0	2
<i>B</i>	0	0	0	2
<i>C</i>	0	0	0	0
<i>D</i>	0	1	3	0

Step 3 : Now we test whether it is possible to make an assignment using the zeros of the above table by the method described in step 3 in § 5.5

Since none of the rows or columns contain exactly one zero, therefore the trial and error method is followed. Now we start searching two zeros. Starting with row 1, we find row 4 which contain two zeros. We make the assignment at the first zero as shown in table 1 (on page 176) and cross out the other zero in this row and all the other zeros of the first column in which we have made the assignment $\square$. Now starting with column 1 we find column 4 which contain one zero and make assignment $\square$ at this zero of row 3 and cross out all other zeros of this row. Now again starting with row 1 we search row containing only one unmarked zero but we find no such row. The same is true when we check the columns for one unmarked zero. Again we start with row 1 searching two zeros and find the row (row 1) containing two unmarked zeros. We can make an assignment at any one of these zeros and cross out the other zero in this row and zeros of the corresponding column in which assignment is made (see table 1 and 2). If we made an assignment in first row, second column and crossed other zero and zeros of second column then the second row contain only one zero in third column where we can make an assignment. See table 1. Similarly if we made an assignment in first row, third column and crossed the other zero and zeros of third column then the second row contain only one zero in second column where we can make an assignment. See table 2.

Table 1

	I	II	III	IV
A	0	$\square$	0	2
B	0	0	$\square$	2
C	0	0	0	$\square$
D	$\square$	1	3	0

Table 2

	I	II	III	IV
A	0	0	$\square$	2
B	0	$\square$	0	2
C	0	0	0	$\square$
D	$\square$	1	3	0

Thus, we get the following two assignments.

$$A \rightarrow II, B \rightarrow III, C \rightarrow IV, D \rightarrow I$$

$$\text{and } A \rightarrow III, B \rightarrow II, C \rightarrow IV, D \rightarrow I$$

In both cases minimum cost = ₹ 20

Note: Other assignments also exist. Students may try to find them.

Example 4 : Solve the assignment problem represented by the following matrix.

	I	II	III	IV	V	VI
A	9	22	58	11	19	27
B	43	78	72	50	63	48
C	41	28	91	37	45	33
D	74	42	27	49	39	32
E	36	11	57	22	25	18
F	3	56	53	31	17	28

Solution : Step 1 : Subtracting the smallest element of each row from every element of the corresponding row, we get the following reduced matrix.

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>	<i>VI</i>
<i>A</i>	0	13	49	2	10	18
<i>B</i>	0	35	29	7	20	5
<i>C</i>	13	0	63	9	17	5
<i>D</i>	47	15	0	22	12	5
<i>E</i>	25	0	46	11	14	7
<i>F</i>	0	53	50	28	14	25

Step 2 : Subtracting the smallest element of each column from every element of the corresponding column, we get the following reduced matrix.

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>	<i>VI</i>
<i>A</i>	0	13	49	0	0	13
<i>B</i>	0	35	29	5	10	0
<i>C</i>	13	0	63	7	7	0
<i>D</i>	47	15	0	20	2	0
<i>E</i>	25	0	46	9	4	2
<i>F</i>	0	53	50	26	4	20

Step 3 : Now we give the zero assignments in our usual manner and get the following matrix.

0	13	49	0	0	13
0	35	29	5	10	0
13	0	63	7	7	0
47	15	0	20	2	0
25	0	46	9	4	2
0	53	50	26	4	20

Since row 3 and column 5 have no assignments so we proceed to the next step.

Step 4 : In this step we draw minimum number of lines to cover all zeros at least once. For this we proceed as follows.

	I_1	I_2		I_3	
L_4	18	13	49	<u>0</u>	13
	18	35	29	5	10
	13	18	63	7	7
L_5	47	15	<u>0</u>	20	2
	25	<u>0</u>	46	9	4
	<u>0</u>	53	50	26	4
	✓ (6)	✓ (2)			✓ (3)
					✓ (5)
					✓ (1)
					✓ (4)
					✓ (7)

- (i) We mark (✓) row 3 in which there is no assignment.
- (ii) Then we mark (✓) columns 2 and 6 which have zeros in marked row 3.
- (iii) Then we mark (✓) rows 5 and 2 which have assignments in the marked columns 2 and 6.
- (iv) Then we mark column 1 (not already marked) which has zero in the marked row 2.
- (v) Then we mark row 6 which has assignment in the marked column 1.
- (vi) Now we draw lines through all marked columns 1, 2, 6. Then we draw lines through unmarked row 1 and 4 having zeros through which there is no line. Thus, we get five lines (minimum number) to cover all the zeros.

Step 5 : Now the smallest of the elements that do not have a line through them is 4. Subtracting this element 4 from all the elements that do not have a line through them and adding to every element that lies at the intersection of two lines and leaving the remaining elements unchanged, we get the following matrix.

	I	II	III	IV	V	VI
A	4	17	49	<u>0</u>	18	17
B	<u>0</u>	35	25	1	6	18
C	13	18	59	3	3	<u>0</u>
D	51	19	<u>0</u>	20	2	4
E	25	<u>0</u>	42	5	18	2
F	18	53	46	22	<u>0</u>	20

Step 6 : Again repeating the step 3 we make the zero assignments and get the following matrix (see above table).

Thus, the optimal assignment is

$$A \rightarrow IV, B \rightarrow I, C \rightarrow VI, D \rightarrow III, E \rightarrow II, F \rightarrow V$$

and minimum cost, $Z = 11 + 43 + 33 + 27 + 11 + 17 = 142$

(from the given cost matrix)

Another optimal solution of the assignment problem is as follows :

$A \rightarrow IV, B \rightarrow VI, C \rightarrow II, D \rightarrow III, E \rightarrow V, F \rightarrow I$

	I	II	III	IV	V	VI
A	4	17	49	0	11	17
B	35	35	25	1	6	0
C	13	0	59	3	3	17
D	51	19	0	20	2	4
E	25	43	42	5	0	2
F	0	53	46	22	27	20

In this case minimum cost

$$Z = 11 + 48 + 28 + 27 + 25 + 3$$

= 142 (from the given cost matrix) which is equal to the minimum cost for the first solution.

Example 5: Solve the following assignment problem.

		Man				
		I	II	III	IV	V
Task	A	1	3	2	3	6
	B	2	4	3	1	5
	C	5	6	3	4	6
	D	3	1	4	2	2
	E	1	5	6	5	4

Solution: Step 1 and 2 : Applying step 1 and 2 we obtain the following matrix.

	I	II	III	IV	V
A	0	2	1	2	4
B	1	3	2	0	3
C	2	3	0	1	2
D	2	0	3	1	0
E	0	4	5	4	2

Step 3 : Giving the zero assignment in our usual manner we get the following matrix.

	I_{L_1}	II	III	IV	V	
A	0	2	1	2	4	✓ (3)
B	1	3	2	0	3	L_2
C	2	3	0	1	2	I_3
D	2	0	3	1	2	I_4
E	2	4	5	4	2	✓ (1)
	✓ (2)					

Since row 5 and column 5 have no assignments, so we proceed to the next step.

Step 4 : The minimum number of lines drawn in the usual manner are 4 (see above table).

Step 5 : Now the smallest of the elements that do not contains line through them is 1. Subtracting this element 1 from the elements that do not have a line through them, adding to every element that lies at the intersection of two lines and leaving the remaining elements unchanged, we get the following matrix.

	I_{L_1}	II	I_{L_2}	III	IV	V	
A	2	1	2	1	3		✓ (1)
B	2	3	2	0	3	L_3	
C	3	3	0	1	2		✓ (4)
D	2	0	3	1	2	L_4	
E	0	3	4	3	1		✓ (5)
	✓ (3)		✓ (2)				

Step 6 : Again repeating the step 3 we make the zero assignments in matrix (above table) and see that even now the row 1 and column 5 do not contain any assignments. Therefore, we again repeat step 4 of drawing lines.

Step 7 : According to our usual manner the minimum number of lines drawn are 4 (see table in step 5).

Step 8 : Again repeating step 5, we get the following matrix.

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>
<i>A</i>	0	3	4	5	2
<i>B</i>	3	3	2	0	3
<i>C</i>	3	2	0	4	1
<i>D</i>	4	0	4	1	5
<i>E</i>	5	2	4	2	0

Step 9 : Repeating the step 3 we make the zero assignments and get the following optimal assignments, (see above table)

$$A \rightarrow I, B \rightarrow IV, C \rightarrow III, D \rightarrow II, E \rightarrow V$$

The minimum cost, $Z = 1 + 1 + 3 + 1 + 4 = 10$

Exercise 5.1

1. Define assignment problem.
2. Give an algorithm to solve an assignment problem.
3. Give a mathematical formulation of the assignment problem.
4. Show that if in an assignment problem, we add a constant to every element of row (or column) in the effectiveness matrix, then an assignment that minimizes the total effectiveness in one matrix also minimizes the total effectiveness in the other matrix.
5. Explain the difference between a transportation problem and an assignment problem.
6. Explain Hungarian method for solving an assignment problem.
7. Solve the following minimal assignment problem :

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>
<i>A</i>	1	4	6	3
<i>B</i>	9	7	10	9
<i>C</i>	4	5	11	7
<i>D</i>	8	7	8	5

8. Solve the following assignment problem.

Works	Time (In minutes)		
	Job 1	Job 2	Job 3
A	4	2	7
B	8	5	3
C	4	5	6

9. A computer centre has got three expert programmers. The centre needs three application programmes to be developed. The head of the computer centre, after studying carefully the programmes to be developed, estimates the computer time in minutes required by the experts to the application programmes as follows :

		Programme		
		A	B	C
Programmer	1	120	100	80
	2	70	90	110
	3	110	140	120

assign the programmers to the programmes in such a way that the total computer time is least.

10. There are five jobs to be assigned, one each to five machines and the associated cost matrix is as follows :

Jobs ↓	Machine				
	1	2	3	4	5
A	11	17	8	16	20
B	9	7	12	6	15
C	13	16	15	12	16
D	21	24	17	28	26
E	14	10	12	11	15

Solve the following minimal assignment problem.

11. An automobile dealer wishes to put four repairmen to four different jobs. The repairman have somewhat different kinds of skills and they exhibit different levels of efficiency from one job to the other. The dealer has estimated the number of man-hours that would be required for each job-man combination. This is given in matrix form in the following table :

		Job			
		A	B	C	D
Man	1	5	3	2	8
	2	7	9	2	6
	3	6	4	5	7
	4	5	7	7	8

Find the optimal assignment that will result in minimum man-hours needed.

12. Find the optimal assignment for the problem with the following matrix :

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>
<i>A</i>	5	3	1	8
<i>B</i>	7	9	2	6
<i>C</i>	6	4	5	7
<i>D</i>	5	7	7	6

13. Find the optimum solution to the assignment problem having the following cost matrix :

Sales Territories
(Cost in rupees thousands)

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>
<i>A</i>	30	25	26	28
<i>B</i>	26	32	24	20
<i>C</i>	20	22	18	27
<i>D</i>	23	20	21	19

14. Solve the following assignment problem represented by the matrix :

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>
<i>A</i>	6	5	8	11	16
<i>B</i>	1	13	16	1	10
<i>C</i>	16	11	8	8	8
<i>D</i>	9	14	12	10	16
<i>E</i>	10	13	11	8	16

15. One car is available at each of the stations 1, 2, 3, 4, 5, 6, and one car is required at each of the stations 7, 8, 9, 10, 11, 12. The distance between the various stations are given in the matrix below. How should the cars be despatched so as to minimize the total mileage travelled ?

	7	8	9	10	11	12
1	42	72	39	52	25	51
2	22	29	49	65	81	50
3	27	39	60	51	32	32
4	45	50	48	52	37	43
5	29	40	39	26	30	33
6	82	40	40	60	51	30

16. Solve the following assignment problem.

Typist	Job (Time)			
	P	Q	R	S
A	85	50	30	40
B	90	40	70	45
C	70	60	60	50
D	75	45	35	55

17. Consider the problem of assigning five jobs to five persons. The assignment costs are given as follows.

		Job				
		1	2	3	4	5
Person	A	8	4	2	6	1
	B	0	9	5	5	4
	C	3	8	9	2	6
	D	4	3	1	0	3
	E	9	5	8	9	5

Determine the optimum assignment schedule.

18. Solve the following minimal assignment problem.

Man →	1	2	3	4	5
Job ↓					
I	12	8	7	15	14
II	7	9	17	14	10
III	9	6	12	6	7
IV	7	6	14	6	10
V	9	6	12	10	6

19. Solve the following minimal assignment problem.

	1	2	3	4
A	10	12	19	11
B	5	10	7	8
C	12	14	13	11
D	8	15	11	9

20. Find the minimum cost solution for the 5×5 assignment problem whose coefficients are as given below :

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>
1	- 2	- 4	- 8	- 6	- 1
2	0	- 9	- 5	- 5	- 4
3	- 3	- 8	- 9	- 2	- 6
4	- 4	- 3	- 1	0	- 3
5	- 9	- 5	- 8	- 9	- 5

21. Five men are available to do five different jobs. From past records, the time (in hours) that each man takes to do each job is known and is given in the following table.

		Job				
		<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>
Man	1	2	9	2	7	1
	2	6	8	7	6	1
	3	4	6	5	3	1
	4	4	2	7	3	1
	5	5	3	9	5	1

Find the assignments of men to jobs that will minimize the total time taken.

Answers 5.1

7.	A $\rightarrow$ I, B $\rightarrow$ III, C $\rightarrow$ II, D $\rightarrow$ IV
8.	Work $\rightarrow$ Job: A $\rightarrow$ 2, B $\rightarrow$ 3, C $\rightarrow$ I. Min. Time = 9m.
9.	Programmer $\rightarrow$ Programme : 1 $\rightarrow$ C, 2 $\rightarrow$ B, 3 $\rightarrow$ A. Min. Time = 280m
10.	Job $\rightarrow$ Machine : A $\rightarrow$ 1, B $\rightarrow$ 4, C $\rightarrow$ 5, D $\rightarrow$ 3, E $\rightarrow$ 2, Min. Cost = ₹ 60
11.	Man $\rightarrow$ Job : 1 $\rightarrow$ B, 2 $\rightarrow$ C, 3 $\rightarrow$ D, 4 $\rightarrow$ A or 1 $\rightarrow$ C, 2 $\rightarrow$ D, 3 $\rightarrow$ B, 4 $\rightarrow$ A Min. Time = 17 h
12.	A $\rightarrow$ III, B $\rightarrow$ IV, C $\rightarrow$ II, D $\rightarrow$ I Min. Cost = ₹ 16
13.	Sales man $\rightarrow$ Territory : A $\rightarrow$ II, B $\rightarrow$ IV, C $\rightarrow$ I, D $\rightarrow$ III or A $\rightarrow$ III, B $\rightarrow$ IV, C $\rightarrow$ I, D $\rightarrow$ II Min. Cost = ₹ 86

14.	$A \rightarrow II, B \rightarrow I, C \rightarrow V, D \rightarrow III, E \rightarrow IV$ or $A \rightarrow II, B \rightarrow IV, C \rightarrow V, D \rightarrow I, E \rightarrow III$ Min. Cost = ₹ 34
15.	Station $\rightarrow$ Station : $1 \rightarrow 11, 2 \rightarrow 8, 3 \rightarrow 7, 4 \rightarrow 9, 5 \rightarrow 10, 6 \rightarrow 12$ Min. Mil = 185
16.	$A \rightarrow S, B \rightarrow Q, C \rightarrow P, D \rightarrow R$ Min. $Z^* = 185$
17.	Person $\rightarrow$ Job : $A \rightarrow 5, B \rightarrow 1, C \rightarrow 4, D \rightarrow 3, E \rightarrow 2$ Min. Cost = ₹ 9
18.	Job $\rightarrow$ Man. : $I \rightarrow 3, II \rightarrow 1, III \rightarrow 2, IV \rightarrow 4, V \rightarrow 5$ or $I \rightarrow 3, II \rightarrow 1, III \rightarrow 4, IV \rightarrow 2, V \rightarrow 5$ Min. Cost = ₹ 32
19.	$A \rightarrow 2, B \rightarrow 3, C \rightarrow 4, D \rightarrow 1$, Min. $Z^* = 38$
20.	$1 \rightarrow III, 2 \rightarrow II, 3 \rightarrow V, 4 \rightarrow I, 5 \rightarrow IV$ or $1 \rightarrow IV, 2 \rightarrow II, 3 \rightarrow III, 4 \rightarrow V, 5 \rightarrow I$ Min. Cost = ₹ 36
21.	Man $\rightarrow$ Job : $1 \rightarrow III, 2 \rightarrow V, 3 \rightarrow I, 4 \rightarrow IV, 5 \rightarrow II$ or $1 \rightarrow III, 2 \rightarrow V, 3 \rightarrow IV, 4 \rightarrow II, 5 \rightarrow I$ or $1 \rightarrow III, 2 \rightarrow V, 3 \rightarrow IV, 4 \rightarrow I, 5 \rightarrow II$ Min. Time = 13 hours

5.6 Maximization Assignment Problem

Some times the assignment problem deals with the maximization of the objective function *i.e.*, the problem may be to assign persons to the jobs in such a way that the expected profit is maximized. Such maximization problem may be solved by converting it to minimization problem. This is done by converting the profit matrix to the cost (*i.e.*, loss) matrix by either of the following two methods.

Method 1: Subtract each element of the given matrix (Profit matrix) from the greatest element of the matrix to get the equivalent cost (*i.e.*, loss) matrix.

Method 2: Place minus sign before each element of the profit matrix to get the equivalent cost matrix.

ILLUSTRATIVE EXAMPLES

Example 1: Alpha corporation has four plants each of which can manufacture any of the four products. Production costs differ from plant to plant as do sales revenue. From the following data, obtain which product each plant should produce to maximize profit ?

		<i>Sales revenue</i> (₹ 1000)			
		<i>Product</i>			
Plant ↓		1	2	3	4
A		50	68	49	62
B		60	70	51	74
C		55	67	53	70
D		58	65	54	69

		<i>Production cost</i> (₹ 1000)			
		<i>Product</i>			
Plant ↓		1	2	3	4
A		49	60	45	61
B		55	63	45	69
C		52	62	49	68
D		55	64	48	66

Solution: Since, Profit = Sales revenue – Product cost, so the profit matrix is as follows.

	1	2	3	4
A	1	8	4	1
B	5	7	6	5
C	3	5	4	2
D	3	1	6	3

Which is a maximization problem. We shall solve this problem by converting it to minimization problem by both methods discussed in article 5.6.

1. By Method 1: Subtracting each element of the above matrix from the greatest element 8 of the matrix, the equivalent loss matrix is

	1	2	3	4
A	7	0	4	7
B	3	1	2	3
C	5	3	4	6
D	5	7	2	5

Step 1 and 2: Subtracting the minimum element of each row from all the elements of the corresponding row and then subtracting minimum element of each column from all the elements of the corresponding column, we get the following matrix.

	1	2	3	4
A	5	0	4	5
B	0	0	1	0
C	0	0	1	1
D	1	5	0	1

Step 3: Giving zero assignments in the usual manner, we get the following matrix.

	1	2	3	4
A	5	0	4	5
B	∞	∞	1	0
C	0	∞	1	1
D	1	5	0	1

In the above table there is an assignment in each row and each column. Hence the optimal assignment for maximum profit is

$$A \rightarrow 2, B \rightarrow 4, C \rightarrow 1, D \rightarrow 3$$

and Max. Profit = ₹ $(8 + 5 + 3 + 6) \times 1000 = ₹ 22000$

2. By Method 2: Placing negative sign before each element of the profit matrix, the equivalent loss matrix is

	1	2	3	4
A	- 1	- 8	- 4	- 1
B	- 5	- 7	- 6	- 5
C	- 3	- 5	- 4	- 2
D	- 3	- 1	- 6	- 3

Now subtracting the minimum element of each row from every elements of the corresponding row and then subtracting the minimum element of each column from every element of the corresponding column, we get the following matrix.

	1	2	3	4
A	5	0	4	5
B	0	0	1	0
C	0	0	1	1
D	1	5	0	1

Which is the same matrix as obtained in step 1 and 2 in methods 1. Hence giving zero assignments we get the same optimal solution as in method 1.

Example 2: Suggest optimum solution to the following assignment problem and also the maximum sales :

Salesmen	Markets (Sales in Lakhs ₹)			
	I	II	III	IV
A	44	80	52	60
B	60	56	40	72
C	36	60	48	48
D	52	76	36	40

Solution: This is a maximization assignment problem. Placing negative sign before each element of the given matrix, the corresponding loss (or minimization) assignment problem is given by

	I	II	III	IV
A	- 44	- 80	- 52	- 60
B	- 60	- 56	- 40	-72
C	- 36	- 60	- 48	- 48
D	- 52	- 76	- 36	- 40

Step 1 and 2 : Subtracting the minimum element of each row from all the elements of the corresponding row and then subtracting the minimum element of each column from all the elements of the corresponding column, we get the following matrix.

	I	II	III	IV
A	24	0	16	20
B	0	16	20	0
C	12	0	0	12
D	12	0	28	36

Step 3 : Giving zero assignments in the usual manner, we get the following matrix.

	I	II	III	IV	
A	24	0	16	20	✓ ③
B	0	16	20	0	
C	12	0	0	12	
D	12	0	28	36	✓ ①

②
 ↓
 L₁
 L₂
 L₃

In the above table, we observe that the row IV and column IV has no zero assignment. So we proceed to the next step.

Step 4 : Here we draw minimum number of lines (horizontal and vertical) to cover all the zeros at least once. The number of such lines is three. See table in step 3.

Step 5 : Since the smallest element among all uncovered element in the above table is 12, so subtracting 12 from all uncovered elements adding it to every element that lies at the

intersection of two lines and leaving remaining elements unchanged, the above table reduces to the following form :

	I	II	III	IV
A	12	0	4	8
B	∞	28	20	0
C	12	12	0	12
D	0	∞	16	24

Step 6 : Giving zero assignment in the usual manner, in the table obtained in step 5, we observe that each row and each column has assignment. (See table in step 5)

Hence the required optimum solution is

$$A \rightarrow II, B \rightarrow IV, C \rightarrow III, D \rightarrow I$$

The maximum sales for the above assignments is

$$₹ (80+72+48+52) = ₹ 252$$

5.7 Unbalanced Assignment Problem

In case the number of tasks (jobs) is not equal to the number of facilities (persons), the assignment problem is called an *Unbalanced assignment problem*. Thus the cost matrix of an unbalanced assignment problem is not a square matrix. For the solution of such problem we add the dummy (fictitious) rows or columns to the given matrix with zero costs to form it a square matrix. Then the usual assignment algorithm can be applied to this resulting balanced assignment problem.

ILLUSTRATIVE EXAMPLES

Example 1: A department head has four tasks to be performed and three subordinates. The subordinates differ in efficiency. The estimates of the time, each subordinate would take to perform, are given below in the matrix. How should he allocate the tasks, one to each man, so as to minimize the total man hours?

		Subordinates		
		1	2	3
Tasks	I	9	26	15
	II	13	27	6
	III	35	20	15
	IV	18	30	20

Solution: Since the matrix is not square, it is an unbalanced assignment problem. We introduce one fictitious subordinate (4th column with zero costs) to get a square matrix. Thus the resulting matrix is shown in the following table. Now the problem can be solved by usual method.

	1	2	3	4
I	9	26	15	0
II	13	27	6	0
III	35	20	15	0
IV	18	30	20	0

Step1: Subtracting the minimum element of each row from every element of the corresponding row and then subtracting the minimum element of each column from every element of the corresponding column, the matrix reduces to

0	6	9	0
4	7	0	0
26	0	9	0
9	10	14	0

Step2: Giving zero assignments in the usual manner, we observe that, each row and each column have zero assignments.

	1	2	3	4
I	0	6	9	0
II	4	7	0	0
III	26	0	9	0
IV	9	10	14	0

Hence the optimal assignment is as follows.

Tasks → subordinates, I → 1, II → 3, III → 2.

Task IV remains unassigned.

From the original matrix, the total time (man hours) = 9 + 6 + 20 = 35 hours.

Exercise 5.2

1. Solve the following assignment problem:

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>	<i>VI</i>
<i>A</i>	12	10	15	22	18	8
<i>B</i>	10	18	25	15	16	12
<i>C</i>	11	10	3	8	5	9
<i>D</i>	6	14	10	13	13	12
<i>E</i>	8	12	11	7	13	10

2. Solve the following unbalanced assignment problem of minimizing total time for doing all the jobs.

		Jobs				
		1	2	3	4	5
Operators	1	6	2	5	2	6
	2	2	5	8	7	7
	3	7	8	6	9	8
	4	6	2	3	4	5
	5	9	3	8	9	0
	6	4	7	5	6	8

3. Use the Hungarian Method to find which of the two jobs should be left undone when each of the four persons will do only one job in the following cost minimizing assignment problem.

		Jobs					
		<i>J</i> ₁	<i>J</i> ₂	<i>J</i> ₃	<i>J</i> ₄	<i>J</i> ₅	<i>J</i> ₆
Persons	<i>P</i> ₁	10	9	11	12	8	5
	<i>P</i> ₂	12	10	9	11	9	4
	<i>P</i> ₃	8	11	10	7	12	6
	<i>P</i> ₄	10	7	8	10	10	5

4. A company has 4 machines to do 3 jobs. Each job can be assigned to one and only one machine. The cost of each job on each machine is given in the following table :

		Machines			
		W	X	Y	Z
Jobs	A	18	24	28	32
	B	8	13	17	19
	C	10	15	19	22

What are the job assignments which will minimize the cost ?

5. There are 3 persons P_1, P_2 and P_3 and 5 jobs $J_1, J_2, \dots, J_5$. Each persons can do only one job and a job is to be done by one person only. Using Hungarian method, find which 2 jobs should be left undone in the following cost minimizing problem.

	J_1	J_2	J_3	J_4	J_5
P_1	7	8	6	5	9
P_2	9	6	7	6	10
P_3	8	7	9	5	6

6. In a machine shop, a supervisor wishes to assign five jobs among six machines. Any one of the jobs can be processed completely by any one of the machines as given below :

		Machines					
		A	B	C	D	E	F
Jobs	1	13	13	16	23	19	9
	2	11	19	26	16	17	18
	3	12	11	4	9	6	10
	4	7	15	9	14	14	13
	5	9	13	12	8	14	11

The assignment of jobs to machines be on a one to one basis. Assign the jobs to machines so that the total is minimum. Find the minimum total cost.

7. A marketing manager has 5 salesmen and 5 sales-districts. Considering the capabilities of the salesmen and the nature of districts, the marketing manager estimates that sales per month (in hundred rupees) for each salesman in each district would be as follows :

		Districts				
		A	B	C	D	E
Salesmen	1	32	38	40	28	40
	2	40	24	28	21	36
	3	41	27	33	30	37
	4	22	38	41	36	36
	5	29	33	40	35	39

Find the assignment of salesman to districts that will result in maximum sales.

[U.P.T.U. MBA 2005-06]

8. The owner of a small machine shop has four persons available to assign to jobs for the day. Five jobs are offered with the expected profit in rupees for each person on each job being as follows :

		Jobs				
		A	B	C	D	E
Persons	1	6.20	7.80	5.00	10.10	8.20
	2	7.10	8.40	6.10	7.30	5.90
	3	8.70	9.20	11.10	7.10	8.10
	4	4.80	6.40	8.70	7.70	8.00

Find the assignments of persons to jobs that will result in a maximum profit. Which job should be declined ?

[C.C.S. Meerut 2006]

[Hint : Introduce 5th fictitious row with zero costs to make the unbalanced problem to balanced one. Multiply each element by 10 to change the numbers to complete number and also multiply each element by - 1 to change it to minimization problem. Now proceed by Hungarian Method].

9. A company is faced with the problem of assigning 4 machines to 6 different jobs (one machine to one job only). The profits are estimated as follows :

		Machines			
Jobs		A	B	C	D
↓					
1		3	6	2	6
2		7	1	4	4
3		3	8	5	8
4		6	4	3	7
5		5	2	4	3
6		5	7	6	4

Answers 5.2

1.	$A \rightarrow \text{II}, B \rightarrow \text{VI}, C \rightarrow \text{III}, D \rightarrow \text{I}, E \rightarrow \text{IV}, V$ is not assigned.
2.	Operator $\rightarrow$ Job: $1 \rightarrow 4, 2 \rightarrow 1, 4 \rightarrow 2, 5 \rightarrow 5, 6 \rightarrow 3$. Operator 3 do no job. Min. Total Time = 11 Units.
3.	Person $\rightarrow$ Job: $P_1 \rightarrow J_5, P_2 \rightarrow J_6, P_3 \rightarrow J_4, P_4 \rightarrow J_2$ Jobs J_1 and J_3 are left undone.
4.	Job $\rightarrow$ Machine : $A \rightarrow W, B \rightarrow X, C \rightarrow Y$ or $A \rightarrow W, B \rightarrow Y, C \rightarrow X$ In both cases no job is assigned to machine Z.
5.	Person $\rightarrow$ Job : $P_1 \rightarrow J_3, P_2 \rightarrow J_2, P_3 \rightarrow J_4$; Jobs J_1 and J_5 left undone or $P_1 \rightarrow J_4, P_2 \rightarrow J_2, P_3 \rightarrow J_5$; Jobs J_1 and J_3 left undone. In both cases minimum cost is 17 units.
6.	Job $\rightarrow$ Machine : $1 \rightarrow F, 2 \rightarrow A, 3 \rightarrow E, 4 \rightarrow C, 5 \rightarrow D$, Min. Cost = 43 units.
7.	Salesman $\rightarrow$ District : $1 \rightarrow B, 2 \rightarrow A, 3 \rightarrow E, 4 \rightarrow C, 5 \rightarrow D$, Max. Sales = ₹19100.
8.	Person $\rightarrow$ Job : $1 \rightarrow D, 2 \rightarrow B, 3 \rightarrow C, 4 \rightarrow E$, Job A should be declined.
9.	Job $\rightarrow$ Machine : $2 \rightarrow A, 3 \rightarrow B, 4 \rightarrow D, 6 \rightarrow C$, Jobs 1 and 5 are left undone: Max Profit = 28 units.

5.8 Restrictions on Assignment

Sometimes due to some restrictions (technical, legal or others) the assignment of a particular facility to a particular job is not permitted. To overcome such difficulty we assign a very high cost (say, infinite cost) to the corresponding cell, so that the activity will be automatically excluded from the optimal solution.

ILLUSTRATIVE EXAMPLE

Example 1: Four engineers are available to design four projects. Engineer 2 is not competent to design the project B. Given the following time estimates needed to each engineer to design a given project, find how should the engineers be assigned to projects so as to minimize the total design time of four projects.

		<i>Projects</i>			
		<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>Engineers</i>	1	12	10	10	8
	2	14	<i>Not suitable</i>	15	11
	3	6	10	16	4
	4	8	10	9	7

Solution: To avoid the assignment $2 \rightarrow B$, we take its time to be very large (say ∞). Then the cost matrix of the resulting assignment problem is shown in the following

12	10	10	8
14	∞	15	11
6	10	16	4
8	10	9	7

Now we apply the assignment technique in the usual manner.

Step1: Subtracting the minimum element of each row from every element of the corresponding row and then subtracting minimum element of each column from every element of the corresponding column, the reduced matrix is

3	0	0	0
2	∞	2	0
1	4	10	0
0	1	0	0

Step 2: Giving zero assignments in the usual manner, we observe that row 3 and column 3 have no zero assignments. So we draw minimum number of lines to cover all zeros at least once. Number of such zeros is 3.

				L_1	
L_2	3	0	$\times$	$\times$	
	2	∞	2	0	✓ (3)
	1	4	10	$\times$	✓ (1)
L_3	0	1	$\times$	$\times$	
					✓ (2)

Step 3: In the above table, the smallest of the uncovered elements is 1. Subtracting this element 1 from all uncovered elements, adding to each element that lies at the intersection of two lines and leaving remaining elements unchanged we get the following matrix.

1	3	0	0	1
2	1	∞	1	0
3	0	3	9	0
4	0	1	0	1

Step 4: Giving zero assignments in the usual manner, we observe that each row and each column have a zero assignment.

	A	B	C	D
1	3	0	$\times$	1
2	1	∞	1	0
3	0	3	9	$\times$
4	$\times$	1	0	1

Hence the optimal assignment is

Engineer $\rightarrow$ Project: $1 \rightarrow B, 2 \rightarrow D, 3 \rightarrow A, 4 \rightarrow C$

From the given matrix total minimum time = $10 + 11 + 6 + 9 = 36$.

Exercise 5.3

- Four engineers are available to design four projects. Engineer 2 is not competent to design the project B. Given the following time estimates needed to each engineer to design a given project, find how should the engineers be assigned to projects so as to minimise the total design time of four projects.

		Projects			
		A	B	C	D
Engineers	1	16	14	14	12
	2	16	—	17	13
	3	11	15	21	9
	4	8	10	9	7

- Five operators have to be assigned to five machines. The assignment cost are given in the table below :

		Machines				
		<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>
Operators	<i>A</i>	5	5	–	2	6
	<i>B</i>	7	4	2	3	4
	<i>C</i>	9	3	5	–	3
	<i>D</i>	7	2	6	7	2
	<i>E</i>	6	5	7	9	1

Operator *A* cannot operate machine *III* and operator *C* cannot operate machine *IV*. Find the optimal assignment schedule.

3. The Secretary of a school is taking bids on the city's four school bus routes. Four companies have made the bids as detailed in the following table.

		Bids			
Company ↓		Route 1	Route 2	Route 3	Route 4
1		₹ 4000	₹ 5000	–	–
2		–	₹ 4000	–	₹ 4000
3		₹ 3000	–	₹ 2000	–
4		–	–	₹ 4000	₹ 5000

Suppose each bidder can be assigned only one route. Use the assignment model to minimize the school's cost of running the four bus routes.

4. Solve the following minimal assignment problem.

	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>
<i>A</i>	9	11	15	10	11
<i>B</i>	12	9	–	10	9
<i>C</i>	–	11	14	11	7
<i>D</i>	14	8	12	7	8

Answers 5.3

1.	Engineer $\rightarrow$ Project: $1 \rightarrow B, 2 \rightarrow D, 3 \rightarrow A, 4 \rightarrow C$. Min. Total Time = 47 hours
2.	Operator $\rightarrow$ Machine (i) $A \rightarrow IV, B \rightarrow III, C \rightarrow II, D \rightarrow I, E \rightarrow V$ or (ii) $A \rightarrow IV, B \rightarrow III, C \rightarrow II, D \rightarrow V, E \rightarrow I$ or (iii) $A \rightarrow IV, B \rightarrow III, C \rightarrow V, D \rightarrow II, E \rightarrow I$
3.	Company $\rightarrow$ Route: $1 \rightarrow 1, 2 \rightarrow 2, 3 \rightarrow 3, 4 \rightarrow 4$. Min. Cost = ₹15000
4.	$A \rightarrow I, B \rightarrow II, C \rightarrow V, D \rightarrow IV$. Job III remains undone. Min. Cost = 32 units.

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

- Which of the following method is not used for solving an assignment problem?
 - Hungarian method
 - Transportation method
 - Simplex method
 - None of the above.
- In an assignment problem if x_{ij} denotes the i -th person to be assigned to j th job, then if $\sum_{i=1}^n x_{ij} = a$ and $\sum_{j=1}^n x_{ij} = b$, then (a, b) :
 - $(0, 1)$
 - $(1, 0)$
 - $(0, 0)$
 - $(1, 1)$.
- The total number of marked in the final reduced cost matrix of order n when an optimal assignment is obtained is:
 - $n - 1$
 - n
 - $n + 1$
 - None of these.
- In the process of drawing minimum number of lines to cover all the zeros of the reduced matrix the lines are draw through:
 - Marked rows
 - Marked columns
 - Unmarked columns
 - All of these.
- Profit maximization assignment problem can be solved by Hungarian method by multiplying each element of pay off matrix by:
 - -1
 - 2
 - -2
 - None of these.
- An assignment problem is called an unbalanced assignment problem when in its cost matrix:
 - Number of rows = number of columns

- (b) Number of rows < number of columns
 - (c) Number of rows > number of columns
 - (d) (b) or (c).
7. All costs in the dummy (fictitious) row or column added to convert an unbalanced assignment problem to balanced assignment problem is taken equal to:
- (a) 1
 - (b) - 1
 - (c) 0
 - (d) None of these.
8. A maximization assignment problem is transformed to minimization assignment problem by:
- (a) Subtracting each element in rows from the maximum element of the corresponding rows
 - (b) Subtracting each element in columns from the maximum element of the corresponding columns
 - (c) Subtracting each entry in the table from the maximum entry of the table
 - (d) None of these.

Fill in the Blank

1. In an assignment problem with all costs $C_{ij} \geq 0$, a feasible solution x_{ij} that satisfies $\sum_{i=1}^n \sum_{j=1}^n C_{ij} x_{ij} = 0$, is solution for the problem.
2. The assignment problem is solved by method.
3. The assignment of persons to jobs in an assignment problem remains when a constant value is subtracted from all the elements the cost matrix.
4. Hungarian method is applied to a minimizing assignment problem.
5. If the number of rows is not equal to the number of columns in the cost matrix of an assignment problem then it is called assignment problem.
6. Dummy row or column is added in assignment problem.

True/False

1. Assignment problems are the problems where the objective is to assign a number of origins to the equal number of destinations at a minimum cost.
2. An assignment problem is considered as a particular case of a transportation problem.
3. In an assignment problem, the assignment of persons to jobs is changed when a constant is added in all the elements of the cost matrix.
4. The assignment problem requires that only one activity be assigned to each resource.
-

Answers

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1.	(b)	2.	(d)	3.	(b)
4.	(b)	5.	(a)	6.	(d)
7.	(c)	8.	(c)		

Fill in the Blank

1.	an optimal	2.	hungarian	3.	unchanged
4.	cost	5.	unbalanced	6.	unbalanced

True/False

1.	T	2.	T	3.	F
4.	T				

○○○

UNIT

II



Chapters

6. Replacement Problems

7. Queueing Theory (Queueing Models)

UNIT-II

Chapter

6



Replacement Problems

6.1 Introduction (Replacement and Replacement Situations)

The efficiency of all industrial and military equipments deteriorates with time. Sometimes the equipment fails completely and effects the whole system. The maintenance costs (running costs) of an equipment also go on increasing with time. Thus, it becomes more economical to replace an old equipment with the new one. Hence, there is a need to formulate a most economical replacement policy which is in the best interest of the system. In this chapter, we shall study the following types of the replacement problems :

1. Replacement of major or capital item (equipment) that deteriorates (become worse) with time; *e. g.*, machines, trucks, buses etc.
2. Replacement of item (equipment) in anticipation of complete failure, the probability of which increases with time (or age).

6.2 Replacement of Major or Capital Item (Equipment) that Deteriorates with Time

For a machine, the maintenance cost always increase with time and a stage comes when the maintenance cost becomes so large that it is economical to replace the machine with a new one. Thus, the problem of replacement in this case, is to find the best time (age of machine) at which the old machine should be replaced by the new one. The policy adopted to find this replacement age is as proved in the next article.

6.3 To Find the Best Replacement Age (Time) of a Machine

When:

- (i) Its maintenance cost is given by a Function Increasing with time.
- (ii) Its scrap Value is Constant and,
- (iii) The money Value is Not Considered.

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Let C = Cost of the machine

S = Scrap value of the machine which is constant

and $C_m(t)$ = Maintenance cost of the machine at time t , [$C_m(t)$ increase with t]

Case I : When time 't' is a continuous variable

Total maintenance cost of the machine in n years

$$= \int_0^n C_m(t) dt$$

$\therefore$ The total cost of the machine in n years = Cost of the machine – scrap value of machine
+ total maintenance cost in n years

$$= C - S + \int_0^n C_m(t) dt$$

The average cost of the machine per year during n years is given by

$$A(n) = \frac{C - S + \int_0^n C_m(t) dt}{n}$$

$$\text{or } A(n) = \frac{C - S}{n} + \frac{1}{n} \int_0^n C_m(t) dt \quad \dots(1)$$

Now for $A(n)$ to be minimum,

$$\frac{d}{dn} (A(n)) = -\frac{C - S}{n^2} - \frac{1}{n^2} \int_0^n C_m(t) dt + \frac{1}{n} C_m(t) = 0$$

$$\therefore C_m(t) = \frac{C - S + \int_0^n C_m(t) dt}{n} = A(n) \quad [\text{from (1)}]$$

i.e., Maintenance cost of the machine at time t = Average cost of the machine per year.

$$\text{Also } \frac{d^2 A(n)}{dn^2} = \frac{2(C - S)}{n^3} + \frac{2}{n^3} \int_0^n C_m(t) dt > 0$$

Hence, the average cost of the machine per year is minimized by replacing it when the average cost to date becomes equal to the current maintenance cost.

Case II : When time 't' is a discrete variable

When time t takes the value 1, 2, 3,etc., then the total maintenance cost of the machine in n years

$$= \sum_{m=1}^n C_m(t)$$

∴ In this case, the average cost of the machine per year during n years is given by

$$A(n) = \frac{C - S + \sum_{m=1}^n C_m(t)}{n} = \frac{C - S}{n} + \frac{1}{n} \sum_{m=1}^n C_m(t) \quad \dots(2)$$

But $A(n)$ is minimum for that value of n for which

$$\Delta A(n-1) < 0 < \Delta A(n) \quad \dots(3)$$

Now $\Delta A(n) = A(n+1) - A(n)$

$$\begin{aligned} &= \left[\frac{C - S}{n+1} + \frac{1}{n+1} \sum_{m=1}^{n+1} C_m(t) \right] - \left[\frac{C - S}{n} + \frac{1}{n} \sum_{m=1}^n C_m(t) \right] \\ &= (C - S) \left(\frac{1}{n+1} - \frac{1}{n} \right) + \left(\frac{1}{n+1} \right) \left[\sum_{m=1}^n C_m(t) + C_{n+1}(t) \right] - \frac{1}{n} \sum_{m=1}^n C_m(t) \\ &= -\frac{(C - S)}{n(n+1)} + \left(\frac{1}{n+1} - \frac{1}{n} \right) \sum_{m=1}^n C_m(t) + \frac{1}{n+1} C_{n+1}(t) \\ &= -\left[\frac{(C - S) + \sum_{m=1}^n C_m(t)}{n(n+1)} \right] + \frac{1}{n+1} C_{n+1}(t) \\ &= -\frac{A(n)}{n+1} + \frac{1}{n+1} C_{n+1}(t) \quad \text{[from (2)]} \end{aligned}$$

$$\text{Similarly } \Delta A(n-1) = -\frac{A(n-1)}{n} + \frac{1}{n} C_n(t)$$

∴ from (2), $A(n)$ is minimum for value of n for which

$$\Delta A(n) > 0 \quad \text{and} \quad \Delta A(n-1) < 0$$

$$\text{i.e.,} \quad -\frac{A(n)}{n+1} + \frac{1}{n+1} C_{n+1}(t) > 0$$

$$\text{and} \quad -\frac{A(n-1)}{n} + \frac{1}{n} C_n(t) < 0$$

$$\left. \begin{array}{l} \text{or } C_{n+1}(t) > A(n) \\ \text{and } C_n(t) < A(n-1) \end{array} \right\} \quad \dots(4)$$

From (4), we conclude that

1. *Do not replace if the next years maintenance cost is less than the previous years average total cost.*
2. *Replace if the next years maintenance cost is greater than the previous years average total cost.*

While solving the numerical problems we adopt the following policy :

A machine is continued upto the time the average cost per year (include cost of new machine, maintenance cost etc.) of the machine decrease and is replaced at the time when this cost begins to increase.

Proof : From (4), the replacement age of the machine is n years, if

$$C_{n+1}(t) > A(n)$$

$$C_{n+1}(t) > \frac{1}{n} [C - S + C_1 + C_2 + \dots + C_n]$$

$$\text{or } \frac{C_{n+1}}{n+1} > \frac{1}{n(n+1)} [C - S + C_1 + C_2 + \dots + C_n] \quad [\text{writing } C_{n+1}(t) \text{ as } C_{n+1}]$$

Adding, $(C - S + C_1 + C_2 + \dots + C_n)/(n+1)$, on both sides, we have

$$\begin{aligned} \frac{1}{n+1} [C - S + C_1 + C_2 + \dots + C_{n+1}] &> \frac{1}{n(n+1)} [C - S + C_1 + \dots + C_n] \\ &\quad + \frac{1}{n+1} [C - S + C_1 + \dots + C_n] \end{aligned}$$

$$\text{or } \frac{1}{n+1} [C - S + C_1 + C_2 + \dots + C_{n+1}] > [C - S + C_1 + \dots + C_n] \left(\frac{1}{n+1} \right) \left(\frac{1}{n} + 1 \right)$$

$$\text{or } \frac{1}{n+1} [C - S + C_1 + C_2 + \dots + C_{n+1}] > \frac{1}{n} [C - S + C_1 + \dots + C_n]$$

$$\text{or } A(n+1) > A(n)$$

i.e., average cost in $(n+1)$ years $>$ average cost in n years.

From which it follows that the best replacement age of the machine is n years if the average cost of $(n+1)$ years is greater than the average cost of n years *i.e., the machine should be replaced when the average cost begins to increase.*

The same type of reasoning is used to decide which of the several machines should be purchased.

"The average cost per year for each machine, assuming replacement of the optimal age (it may not be same for all machines under consideration) is computed and the machine with the lowest average cost is purchased."

The procedure is illustrated by the following examples :

ILLUSTRATIVE EXAMPLES

Example 1 : The cost of a truck is ₹ 3000. The salvage value (resale value) and the running cost are given as under. Find the most economical replacement age of the truck :

Years	1	2	3	4	5	6	7
Running cost	600	700	800	900	1000	1200	1500
Resale value	2000	1333	1000	750	500	300	300

Solution : To find the average cost per year of the machine, we prepare the following table :

Age of Replacement in years	Total Running cost ₹	Depreciation = Cost of truck – Resale price ₹	Total cost ₹	Average cost per year ₹
1	600	1000	1600	1600.00
2	1300	1667	2967	1483.50
3	2100	2000	4100	1366.67
4	3000	2250	5250	1312.50
5*	4000	2500	6500	1300.00*
6	5200	2700	7900	1316.67

It is clear from the table, that the average cost per year decrease upto the end of 5th year and then this cost begin to increase. Hence, the lowest average cost per year is achieved by replacing the truck at the end of every 5 years.

Note: Depreciation = Cost of new truck – Resale price of the truck.

Example 2 : The cost of a machine is ₹ 6100.00 and its scrap value (resale value) is only ₹ 100.00. The maintenance costs are found from experience to be as under:

Years	1	2	3	4	5	6	7	8
Maintenance cost in ₹	100	250	400	600	900	1250	1600	2000

When should the machine be replaced ?

Solution : Average cost per year of the machine are computed in the following table :

Age of Replacement	Total Running cost ₹	Depreciation ₹	Total cost ₹	Average cost per year ₹
1	100	6000	6100	6100.00
2	350	6000	6350	3175.00
3	750	6000	6750	2250.00
4	1350	6000	7350	1837.50
5	2250	6000	8250	1650.00
6*	3500	6000	9500	1583.33*
7	5100	6000	11100	1585.71

It is clear from the table, that lowest average cost per year is achieved by replacing the machine at the end of 6th year.

Example 3 :

- (a) Machine A costs ₹ 9000. Annual operating costs are ₹ 200 for the first year, and then increase by ₹ 2000 every year. Determine the best age at which to replace the machine. If the optimum replacement policy is followed, what will be the average yearly cost of owning and operating the machine ? (Assume that the machine has no resale value when replaced and that future costs are not discounted).
- (b) Machine B costs ₹ 10000. Annual operating costs are ₹ 400 for the first year and then increase by ₹ 800 every year. You have now a machine of type A, which is one year old. Should you replace it with B and if so when ?
- (c) Suppose you are, just ready to replace machine A with another machine of the same type, when you hear that machine B will become available in a year. What should you do ?

Solution : (a) The maintenance costs of machine A in successive years are as follows :

Years	1	2	3	4	5
Maintenance costs in ₹	200	2200	4200	6200	8200

To find the average cost per year of machine A, we prepare the following table :

Replacement at the end of year	Total Running cost ₹	Depreciation ₹	Total cost ₹	Average cost per year ₹
1	200	9000	9200	9200
2	2400	9000	11400	5700
3*	6600	9000	15600	5200*
4	12,800	9000	21800	5450

It is clear from the table that machine A should be replaced at the end of third year. The average yearly cost in this situation, will be ₹ 5200.

(b) The maintenance costs of machine B are as follows :

Years	1	2	3	4	5	6	7
Maintenance cost ₹	400	1200	2000	2800	3600	4400	5200

To find the average yearly cost of machine B, we prepare the following table :

Replacement at the end of year	Total running cost ₹	Depreciation ₹	Total cost ₹	Average cost per year ₹
1	400	10000	10400	10400.00
2	1600	10000	11600	5800.00
3	3600	10000	13600	4533.33
4	6400	10000	16400	4100.00
5*	10000	10000	20000	4000.00*
6	14400	10000	24400	4066.66

It is clear from the above table that the machine *B* should be replaced after 5 years and then its average cost per year will be ₹ 4000.

Since the lowest average cost ₹ 4000 for machine *B* is less than the lowest average cost ₹ 5200 for machine *A*, the machine *A* should be replaced by machine *B*.

To find the time of replacement of machine *A* by machine *B* :

The machine *A* is replaced by machine *B* at the time (age) when its running cost of the next year exceeds the lowest average yearly cost ₹ 4000 of machine *B*.

Total cost of the machine *A* in the successive years are as follows :

Years	1	2	3	4	5
Total cost in the year	₹ 9200	₹ 11400 – 9200 = ₹ 2200	₹ 15600 – 11400 = ₹ 4200	₹ 21800 – 15600 = ₹ 6200	₹ 30000 – 21800 = ₹ 8200

The running cost of third year of machine *A* is ₹ 4200 which is more than the lowest average yearly cost ₹ 4000 of machine *B*. Therefore, the machine *A* should be replaced by machine *B* when its age is 2 years. Since the machine *A* is one year old now, therefore, it should be replaced after one year from now.

(c) As shown in (b), the machine *A* should be replaced after one year from now and the machine *B* is also available at that time. Therefore, the machine *A* should be replaced by machine *B* after one year from now.

Example 4 : (a) A machine owner finds from his past records that the cost per year of maintaining a machine *A* whose purchase price is ₹ 6000.00 are as given below :

Years	1	2	3	4	5	6	7	8
Maintenance cost ₹	1000	1200	1400	1800	2300	2800	3400	4000
Resale price ₹	3000	1500	750	375	200	200	200	200

At what age is the replacement due ?

- (b) The cost of a machine B whose capacity is 50% more than machine A is ₹ 8000.00. The estimates of maintaining cost and resale price for this machine are given below :

Years	1	2	3	4	5	6	7	8
Maintenance cost ₹	1200	1500	1800	2400	3100	4000	5000	6100
Resale price ₹	4000	2000	1000	500	300	300	300	300

What is the replacement age of machine B ?

- (c) A machine owner has three machines of type A. Two of these machines are two years old and the third one year old. He is considering to purchase a new machine of type B. Assuming that the loss of flexibility due to fewer machines is of no importance; and he continues to have sufficient work for three of the old machines, what should his policy be ?

Solution : (a) To find average yearly cost of machine A, we prepare the following table :

Age of Replacement Years	Total Running cost ₹	Depreciation = Cost of Machine – resale price ₹	Total Cost ₹	Average cost per year ₹
1	1000	3000	4000	4000
2	2200	4500	6700	3350
3	3600	5250	8850	2950
4	5400	5625	11025	2756
5*	7700	5800	13500	2700*
6	10500	5800	16300	2717

It is clear from the table that replacement age of machine A is 5 years and minimum average yearly cost is ₹ 2700.

- (b) To find average yearly cost of machine B, we prepare the following table :

Age of Replacement years	Total Running cost ₹	Depreciation = Cost of Machine – resale price ₹	Total cost ₹	Average cost per year ₹
1	1200	4000	5200	5200.00
2	2700	6000	8700	4350.00
3	4500	7000	11500	3833.33
4	6900	7500	14400	3600.00
5*	10000	7700	17700	3540.00*
6	14000	7700	21700	3616.66

It is clear from the table that replacement age of machine B is 5 years and minimum average yearly cost is ₹ 3540.

(c) To decide if the replacement of machine A by machine B is justified or not :

Since the capacity of machine B is 50% more than that of machine A . Therefore, 2 larger machines B are equivalent to 3 smaller machine A . The average yearly cost of machine B is ₹ 3540 which is equivalent to ₹ $\frac{2 \times 3540}{3}$ i.e., ₹ 2360.00 of one smaller machine A . Since this is less than ₹ 2700.00, the lowest average yearly cost of one smaller machine A , therefore, the smaller machines A should be replaced by the larger machines B .

To find the time of replacement :

The three smaller machines A should be replaced by the two larger machines B at the time when the maintenance cost for the next year of the three smaller machines A exceeds the average yearly cost of two larger machines B .

Total average cost of two machines $B = ₹ 2 \times 3540 = ₹ 7080.00$

Total cost of a machine A in the successive years are as follows :

Years	1	2	3	4	5
Total cost in the year ₹	4000	6700 – 4000 = 2700	8850 – 6700 = 2150	11025 – 8850 = 2175	13500 – 11025 = 2475

It is given that the two machines are two years old and one, one year old.

∴ Total cost next year from now for three machines A

= Total cost of two machines A in the third year

+ Total cost of one machine A in the second year

= ₹ $(2 \times 2150 + 2700) = ₹ 7000$

and Total cost during the second year from now for three machines A

= Total cost of two machines A in the 4th year

+ Total cost of one machine A in the 3rd year

= ₹ $(2 \times 2175 + 2150) = ₹ 6500$

Similarly, Total cost of three machines in the third year from now

= ₹ $(2 \times 2475 + 2175) = 7125$ etc.

Since the total cost of three machines in the third year from now, is ₹ 7125 which is more than ₹ 7080.00 the total average cost of two machines B .

Hence, three machines A should be replaced by two machines B after two years from now without waiting for their normal replacement age.

Example 5 : For a machine, the following data are available :

Years	0	1	2	3	4	5	6
Cost of spares (₹)	–	200	400	700	1000	1400	1600
Salary Maintenance staff (₹)	–	1200	1200	1400	1600	2000	2600
Losses due to break down (₹)	–	600	800	700	1000	1200	1600
Resale value (₹)	12000	6000	3000	1500	800	400	400

Determine the optimum period for replacement of the above machine.

Solution : Average cost per year of the machine are computed in the following table :

Replacement at the end of year	Cost of Spares (₹)	Salary of Main. Staff (₹)	Losses due to break-down (₹)	Depreciation (₹)	Total cost (₹)	Average cost per year (₹)
1	200	1200	600	6000	8000	8000
2	600	2400	1400	9000	13400	6700
3	1300	3800	2100	10500	17700	5900
4	2300	5400	3100	11200	22000	5500
5	3700	7400	4300	11600	27000	5400
6	5300	10000	5900	11600	32800	5466.66

Yearly average cost is minimum at the end of 5th year. Hence, the optimum period of replacement of the machine is at the end of every 5 years.

Exercise 6.1

1. The cost of a machine is ₹ 5000. The running cost and the salvage value of the machine are given as under. Find the optimal replacement policy :

Years	1	2	3	4	5	6	7	8
Running cost in ₹	1500	1600	1800	2100	2500	2900	3400	4000
Salvage value in ₹	3500	2500	1700	1200	800	500	500	500

Salvage value means the cost at which the machine could be sold at the end of the year.

2. A firm is considering when to replace its machine whose price is ₹ 12200. The scrap value of the machine is ₹ 200 only. From past experience, the maintenance costs of the machine are as under :

Years	1	2	3	4	5	6	7	8
Maintenance cost in ₹	200	500	800	1200	1800	2500	3200	4000

Find when the new machine should be purchased.

3. A truck owner finds from his past records that the maintenance costs per year, of a truck whose purchase price is ₹ 8000, are as given below :

Years	1	2	3	4	5	6	7	8
Maintenance costs	1000	1300	1700	2200	2900	3800	4800	6000
Resale price	4000	2000	1200	600	500	400	400	400

Determine at which time it is profitable to replace the truck.

4. The running cost per year of a truck whose purchase price is ₹ 30000 are as follows. Determine when is the replacement due :

Years	1	2	3	4	5	6	7
Running cost (₹)	5000	6000	7000	9000	11500	14000	17000
Resale value (₹)	15000	7500	3750	1875	1000	1000	1000

5. The following table gives the running costs per year and resale values of a certain equipment whose purchase price is ₹ 6500. At what year is the replacement due optimally,

Years	1	2	3	4	5	6	7	8
Running cost (₹)	1400	1500	1700	2000	2400	2800	3300	3900
Resale value (₹)	4000	3000	2200	1700	1300	1000	1000	1000

6. The cost of a truck is ₹ 3500. The salvage value and the running costs are given below. Find the most economical age for replacement :

Years	1	2	3	4	5	6	7
Running cost	600	700	800	900	1000	1200	1500
Resale value	2500	1833	1500	1250	1000	800	800

7. The following table gives the running costs per year and resale prices of a certain equipment whose purchase price is ₹ 10000 :

Years	1	2	3	4	5	6	7
Running cost (₹)	3000	3200	3600	4200	5000	5800	6800
Resale value (₹)	7000	5000	3400	2400	1600	1000	1000

8. The maintenance cost and resale value per year of machine whose purchase price is ₹ 7000 are given below :

Years	1	2	3	4
Maintenance cost (In ₹)	900	1200	1600	2100
Resale value (In ₹)	4000	2000	1200	600

When should the machine be replaced ?

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9. A manufacturing firm has come to know from his past records that a machine costing ₹ 56000/- is not working satisfactorily in spite of its regular maintenance. With a view to replacing this machine the following facts were obtained :

Years	Annual running cost (₹)	Resale value (₹)
1	7,000	28,000
2	9,100	14,000
3	11,900	8,400
4	15,400	4,200
5	20,300	3,500
6	26,600	3,000
7	33,600	3,000
8	42,000	3,000

When should the machine be replaced ?

10. The data of the operating costs per year and resale prices of equipment A whose purchase price is ₹ 10000 are given below :

Years	1	2	3	4	5	6	7
Operating cost (₹)	1500	1900	2300	2900	3600	4500	5500
Resale value (₹)	5000	2500	1250	600	400	400	400

- What is the optimum period for replacement ?
- When Equipment A is 2 years old, Equipment B, which is a new model for the same usage is available. The optimum period for replacement is 4 years with an average cost of ₹ 3600. Should we change equipment A with that of B ? If so, when ?

11. XYZ manufacturing company is using a machine whose purchase price is ₹ 65,000. The installation charges amount to ₹ 18,000 and the machine has a scrap value of only ₹ 8,000 because of firm has a monopoly of this type of work. The maintenance cost in various years is given in the following table :

Years	Maintenance Cost (₹)
1	1250
2	3750
3	5000
4	7500
5	10500
6	14500
7	20000
8	24000
9	30000

Determine after how many years should the machine be replaced, assuming that the machine replacement can be done only at the year end.

Answers 6.1

1.	After 4 years;	2.	After 6 years;	3.	After 5 years;	4.	After 5 years;
5.	After 5 years;	6.	After 5 years;	7.	After 4 years;	8.	4 years;
9.	5 years;	10.	(i) 5 years (ii) Change when equipment A is 4 years old				
11.	6 years						

6.4 Few Important Terms

Now we shall explain few important terms which are used in replacement problems :

1. **Money value** : *The value of money changes with time.* This statement is explained by the following example :

If we borrow ₹ 100 at the interest of 10% per year then after one year we have to return ₹ 110. Thus, ₹ 110 after one year from now are equivalent to ₹ 100 today.

Therefore, ₹ 1.00 after one year from now is equivalent to ₹ $\frac{100}{110}$ i.e., $(1.1)^{-1}$ today at

the rate of 10%. So, ₹ 1.00, n years after from now is equivalent to ₹ $(1.1)^{-n}$ today at the rate of 10%. Thus, we can say that the value of money changes with time.

2. **Present value (or present worth):** We have shown that ₹ 1.00, after n years from now is equivalent to ₹ $(1.1)^{-n}$ today at the rate of 10%. In other words, we can say that $(1.1)^{-n}$ is the **present value** (or present worth) of one rupee spent after n years from now.

In general, if the interest on ₹ 1 is Rs. i per year, then the present value of ₹ 1 to be spent after n years from now is ₹ $\frac{1}{(1+i)^n}$.

3. **Discount rate (or Depreciation ratio) :** The value of the money decreases with a constant ratio which is known as its **discount rate** or **depreciation ratio**.

If r is the rate of interest per year on ₹ 1.00. Then the present value of ₹ 1.00 to be spent after one year from now is ₹ $\frac{1}{1+r}$.

This ratio $v = \frac{1}{1+r}$ is the **discount rate** or **depreciation ratio** which is less than one.

6.5 To Determine the Best Replacement Age of Items whose Maintenance Costs Increase with Time and the Value of Money also Changes with Time

Let Rs. A be the cost of a machine and Rs. $C_1, C_2, C_3, \dots, C_n$ be the maintenance costs in year 1, 2, 3, ..., n , respectively. If v is the depreciation ratio during a year then to find the best replacement age of the machine which minimizes the total of all future discounted costs.

Present value of maintenance costs $C_1, C_2, \dots, C_n$ in n years 1, 2, 3, ..., n , are $C_1, C_2v, C_3v^2, \dots, C_nv^{n-1}$ respectively.

Therefore, the present value of the total expenditure on the machine in n years is given by

$$P(n) = A + C_1 + C_2v + C_3v^2 + \dots + C_nv^{n-1} = A + \sum_{r=1}^n C_r v^{r-1}$$

If the machine is replaced after n years from now with a similar new machine then the expenditures on the new machine in years $n+1, n+2, \dots, 2n$, from now are $A + C_1, C_2, C_3, \dots, C_n$. (Note that years $n+1, n+2, \dots$, are years 1, 2, ..., for the new machine).

The present value of all these expenditures on the machine are given by

$$(A + C_1)v^n + C_2v^{n+1} + C_3v^{n+2} + \dots + C_nv^{2n-1}$$

Similarly, we can find the present value of the total expenditures on the machines in the successive cycles of n years.

Therefore, the present value of the total expenditures on the machine with a replacement policy of n years is given by

$$\begin{aligned}
 C(n) &= [A + C_1 + C_2 v + C_3 v^2 + \dots + C_n v^{n-1}] \\
 &\quad + [(A + C_1)v^n + C_2 v^{n+1} + C_3 v^{n+2} + \dots + C_n v^{2n-1}] \\
 &\quad + [(A + C_1)v^{2n} + C_2 v^{2n+1} + \dots + C_n v^{3n-1}] + \dots
 \end{aligned}$$

$$= \left[A + \sum_{r=1}^n C_r v^{r-1} \right] + v^n \left[A + \sum_{r=1}^n C_r v^{r-1} \right] + v^{2n} \left[A + \sum_{r=1}^n C_r v^{r-1} \right] + \dots$$

$$= \left[A + \sum_{r=1}^n C_r v^{r-1} \right] [1 + v^n + v^{2n} + \dots]$$

$$\text{or } C(n) = \frac{A + \sum_{r=1}^n C_r v^{r-1}}{1 - v^n} = \frac{P(n)}{1 - v^n} \quad \dots(1)$$

Thus, (1), gives the total amount of the money required at present to meet all the future purchasing and maintaining expenses on the machines with a replacement policy of every n years.

Now, we are interested to find that value of n for which $C(n)$ is minimum.

$C(n)$ is minimum for that value of n for which

$$\Delta C(n-1) < 0 < \Delta C(n) \quad \dots(2)$$

$$\text{Now, } \Delta C(n) = C(n+1) - C(n)$$

$$\begin{aligned}
 &= \frac{A + \sum_{r=1}^{n+1} C_r v^{r-1}}{1 - v^{n+1}} - \frac{A + \sum_{r=1}^n C_r v^{r-1}}{1 - v^n} \\
 &= \frac{\left(A + \sum_{r=1}^{n+1} C_r v^{r-1} \right) (1 - v^n) - \left(A + \sum_{r=1}^n C_r v^{r-1} \right) (1 - v^{n+1})}{(1 - v^{n+1})(1 - v^n)} \\
 &= \frac{A(v^{n+1} - v^n) + \left(\sum_{r=1}^{n+1} C_r v^{r-1} \right) (1 - v^n) - \left(\sum_{r=1}^n C_r v^{r-1} \right) (1 - v^{n+1})}{(1 - v^{n+1})(1 - v^n)} \\
 &= \frac{-A(1 - v)v^n + \sum_{r=1}^n C_r v^{r-1} \{ (1 - v^n) - (1 - v^{n+1}) \} + C_{n+1} v^n (1 - v^n)}{(1 - v^{n+1})(1 - v^n)} \\
 &= \frac{v^n}{(1 - v^{n+1})(1 - v^n)} \left[(1 - v^n) C_{n+1} - (1 - v) \left(A + \sum_{r=1}^n C_r v^{r-1} \right) \right]
 \end{aligned}$$

$$\text{or} \quad \Delta C(n) = \frac{v^n(1-v)}{(1-v^{n+1})(1-v^n)} \cdot \left[\frac{(1-v^n)}{(1-v)} C_{n+1} - \left(A + \sum_{r=1}^n C_r v^{r-1} \right) \right]$$

Similarly,

$$\Delta C(n-1) = \frac{v^{n-1}(1-v)}{(1-v^n)(1-v^{n-1})} \cdot \left[\frac{(1-v^{n-1})}{(1-v)} C_n - \left(A + \sum_{r=1}^{n-1} C_r v^{r-1} \right) \right]$$

$\therefore$ from (2), $C(n)$ is minimum for that value of n for which

$$\begin{aligned} \frac{v^{n-1}(1-v)}{(1-v^n)(1-v^{n-1})} \left[\frac{1-v^{n-1}}{1-v} \cdot C_n - \left(A + \sum_{r=1}^{n-1} C_r v^{r-1} \right) \right] &< 0 \\ &< \frac{v^n(1-v)}{(1-v^{n+1})(1-v^n)} \cdot \left[\frac{1-v^n}{1-v} C_{n+1} - \left(A + \sum_{r=1}^n C_r v^{r-1} \right) \right] \end{aligned}$$

$$\begin{aligned} \text{or} \quad \frac{1-v^{n-1}}{1-v} C_n - \left(A + \sum_{r=1}^{n-1} C_r v^{r-1} \right) &< 0 < \frac{1-v^n}{1-v} C_{n+1} - \left(A + \sum_{r=1}^n C_r v^{r-1} \right) \\ \left[\text{Since } \frac{v^{n-1}(1-v)}{(1-v^n)(1-v^{n-1})} \text{ and } \frac{v^n(1-v)}{(1-v^{n+1})(1-v^n)} \text{ are both positive as } v < 1 \right] \end{aligned}$$

$$\text{or} \quad \frac{1-v^{n-1}}{1-v} \cdot C_n - P(n-1) < 0 < \frac{1-v^n}{1-v} \cdot C_{n+1} - P(n) \quad \dots(3)$$

Thus, the best replacement age of the machine is n years for which inequality (3) holds.

On further simplification from (3), we have

$$\frac{1-v^n}{1-v} \cdot C_{n+1} - \left(A + \sum_{r=1}^n C_r \cdot v^{r-1} \right) > 0$$

$$\text{and} \quad \frac{1-v^{n-1}}{1-v} C_n - \left(A + \sum_{r=1}^{n-1} C_r \cdot v^{r-1} \right) < 0$$

$$\text{or} \quad C_{n+1} > \frac{A + \sum_{r=1}^n C_r \cdot v^{r-1}}{\frac{1-v^n}{1-v}} \quad \text{and} \quad C_n < \frac{A + \sum_{r=1}^{n-1} C_r \cdot v^{r-1}}{\frac{1-v^{n-1}}{1-v}}$$

$$\text{or} \quad C_{n+1} > \frac{A + \sum_{r=1}^n C_r \cdot v^{r-1}}{1+v+v^2+\dots+v^{n-1}} \quad \text{and} \quad C_n < \frac{A + \sum_{r=1}^{n-1} C_r \cdot v^{r-1}}{1+v+v^2+\dots+v^{n-1}}$$

$$\text{Since } \frac{1-v^n}{1-v} = 1+v+v^2+\dots+v^{n-1}$$

$$\text{or} \quad C_{n+1} > R(n) \quad \text{and} \quad C_n < R(n-1) \quad \dots(4)$$

$$A + \sum_{r=1}^n C_r \cdot v^{r-1}$$

Where $R(n) = \frac{A + \sum_{r=1}^n C_r \cdot v^{r-1}}{1 + v + v^2 + v^3 + \dots + v^{n-1}} = \text{weighted average of costs in } n \text{ years.}$

From (4) we conclude that

1. Do not replace if the operating cost of the next period (year) is less than the weighted average of previous costs.
2. Replace if the operating cost of the next period (year) is greater than the weighted average of the previous costs.

Note : When money value is not considered i.e., when $v = 1$

$$\begin{aligned} \text{and } R(n) &= \frac{A + \sum_{r=1}^n C_r}{1 + 1 + 1 + \dots + n \text{ terms}} \\ &= \frac{A + C_1 + C_2 + \dots + C_n}{n} = \text{Average yearly cost with no resale value of the machine.} \end{aligned}$$

$\therefore C_{n+1} > R(n)$ gives $C_{n+1} > \text{Average yearly cost with no resale value of the machine.}$

i.e., replace the machine when the maintenance cost of the next year is greater than the average yearly cost upto this year with no resale value of the machine. This replacement policy is same as given in article 6.3.

6.6 A Discounted Cost $P(n)$ is Invested by taking Loan at the Interest Rate r ; and the Loan is Repaid by Fixed Annual Payments Say x , Throughout the Life of the Machine. To Find the Minimum Value of x for Optimum Period n at which to Replace the Machine

[C.C.S., Meerut 1994(P), 98]

The present worth of fixed annual payments x for n years must be equal to $P(n)$ the sum borrowed.

$$\therefore P(n) = x + vx + v^2x + \dots + v^{n-1}x \quad \dots(1)$$

where $v = \frac{1}{1+r}$. (Discount rate)

$$= (1 + v + v^2 + \dots + v^{n-1})x$$

$$= \frac{1 - v^n}{1 - v} \cdot x$$

$$\text{or} \quad x = \frac{(1 - v)}{1 - v^n} \cdot P(n) \quad \dots(2)$$

Since $(1 - v)$ is constant, independent of n .

$\therefore x$ is minimum when

$$C(n) = \frac{P(n)}{1 - v^n} \quad \dots(3)$$

is minimum.

$C(n)$ given by (3) is minimum, when

$$\Delta C(n-1) < 0 < \Delta C(n) \quad \dots(4)$$

From (3), $\Delta C(n) = C(n+1) - C(n)$

$$\begin{aligned} &= \frac{P(n+1)}{1 - v^{n+1}} - \frac{P(n)}{1 - v^n} = \frac{(1 - v^n) P(n+1) - (1 - v^{n+1}) P(n)}{(1 - v^{n+1})(1 - v^n)} \\ &= \frac{P(n+1) - P(n) - v^n [P(n+1) - v \cdot P(n)]}{(1 - v^{n+1})(1 - v^n)} \quad \dots(5) \end{aligned}$$

But if A is the cost of the machine and $C_1, C_2, \dots, C_n$ are the running costs in years 1, 2, ..., n then the present value of the total expenditure in n years must be equal to $P(n)$.

$$\therefore P(n) = A + C_1 + C_2 v + C_3 v^2 + \dots + C_n v^{n-1}$$

$$\begin{aligned} \therefore P(n+1) &= A + C_1 + C_2 v + C_3 v^2 + \dots + C_n v^{n-1} + C_{n+1} v^n \\ &= P(n) + C_{n+1} v^n \end{aligned}$$

$\therefore$ From (5), we have

$$\begin{aligned} \Delta C(n) &= \frac{C_{n+1} v^n - v^n [P(n) + C_{n+1} v^n - v \cdot P(n)]}{(1 - v^{n+1})(1 - v^n)} \\ &= \frac{(1 - v^n) C_{n+1} v^n - (1 - v) P(n) \cdot v^n}{(1 - v^{n+1})(1 - v^n)} \\ &= \frac{(1 - v) v^n}{(1 - v^{n+1})(1 - v^n)} \cdot \left[\frac{1 - v^n}{1 - v} C_{n+1} - P(n) \right] \end{aligned}$$

$\therefore \Delta C(n) > 0$, gives

$$\frac{(1 - v) v^n}{(1 - v^{n+1})(1 - v^n)} \cdot \left[\frac{1 - v^n}{1 - v} C_{n+1} - P(n) \right] > 0$$

or $\frac{1 - v^n}{1 - v} C_{n+1} - P(n) > 0$ since $\frac{(1 - v) v^n}{(1 - v^{n+1})(1 - v^n)}$ is positive as $0 < v < 1$

$$\text{or } C_{n+1} > \frac{P(n)}{\left(\frac{1 - v^n}{1 - v} \right)} \text{ or } C_{n+1} > \frac{A + C_1 + C_2 v + C_3 v^2 + \dots + C_n v^{n-1}}{1 + v + v^2 + \dots + v^{n-1}} \quad \dots(6)$$

Similarly, $\Delta C(n-1) < 0$, on simplification gives

$$\frac{1 - v^{n-1}}{1 - v} C_n - P(n-1) < 0$$

or

$$C_n < \frac{A + C_1 + C_2 v + C_3 v^2 + \dots + C_{n-1} v^{n-2}}{1 + v + v^2 + \dots + v^{n-2}} \quad \dots(7)$$

From (6) and (7) we conclude the same replacement policy as in article 6.5.

$\therefore$ (4) reduces to

$$\frac{1 - v^{n-1}}{1 - v} C_n - P(n-1) < 0 < \frac{1 - v^n}{1 - v} C_{n+1} - P(n) \quad \dots(8)$$

which gives the best replacement age n of machine for which $C(n)$ or $P(n)$ is minimum.

Then minimum value of x is obtained from (2).

ILLUSTRATIVE EXAMPLES

Example 1 : The cost pattern for two machines A and B when money value is not considered, is given as follows :

Years	Cost at the beginning of year in ₹	
	Machine A	Machine B
1	900	1400
2	600	100
3	700	700
Total	2200	2200

Find the cost pattern for each machine, when money is worth 10% per year, and hence find which machine is less costly.

Solution : The total expenditure for each machine in three years is equal to ₹ 2200 when the money value is not taken into consideration. Thus, the two machines are equally good if the money has no value over time.

When money value is 10% per year, the discount rate

$$v = \frac{100}{100 + 10} = 0.9091$$

The present value of the maintenance costs in the year 1, 2, 3 for two machines are shown in the following table :

Years	Machine A (₹)	Machine B (₹)
1	900	1400
2	$600 \times (0.9091) = 545.45$	$100 \times (0.9091) = 90.91$
3	$700 \times (0.9091)^2 = 578.52$	$700 \times (0.9091)^2 = 578.25$
Total	₹ 2023.97	₹ 2069.43

The present value of the total expenditure for machine *A* in three years is less than that for machine *B*. Hence, machine *A* is less costly.

Example 2 : Let the value of money be assumed to be 10% per year and suppose that machine *A* is replaced after every 3 years whereas machine *B* is replaced after every six years. The yearly costs of both the machines are given as under :

Years :	1	2	3	4	5	6
Machine <i>A</i> :	1000	200	400	1000	200	400
Machine <i>B</i> :	1700	100	200	300	400	500

Determine which machine should be purchased.

Solution : Since the value of money is 10% per year, the discount rate, $v = \frac{100}{100 + 10} = \frac{10}{11}$

∴ The total discount cost (present worth) of costs of machine *A* for 3 years is

$$= 1000 + 200 \times (10/11) + 400 \times (10/11)^2 = ₹ 1512 \text{ approx.}$$

and the total discount cost (present worth) of costs of machine *B* for 6 years

$$= 1700 + 100 \times (10/11) + 200 \times (10/11)^2 + 300 \times (10/11)^3 + 400 \times (10/11)^4 + 500 \times (10/11)^5 = ₹ 2765 \text{ approx.}$$

∴ Average yearly cost of machine *A* = $1512 / 3 = ₹ 504$

and average yearly cost of machine *B* = $2765 / 6 = ₹ 460.83$

From this, machine *B* looks to be less costly than machine *A*. But it is not true since the periods of two machines considered are not the same.

Now, the total discount cost of machine *A* for 6 years

$$= 1000 + 200 \times (10/11) + 400 \times (10/11)^2 + 1000 \times (10/11)^3 + 200 \times (10/11)^4 + 400 \times (10/11)^5 = ₹ 2647 \text{ approx.}$$

The present value of the total expenditure in 6 years on machine *A* is less than that for machine *B*. Thus, the machine *A* is less costly and hence machine *A* should be purchased.

Example 3 : If you wish to have a return of 10% per annum on your investment, which of the following planes would you prefer ?

	Plan <i>A</i> (₹)	Plan <i>B</i> (₹)
<i>Ist cost</i>	200000	250000
<i>Scrap value after 15 years</i>	150000	180000
<i>Excess of annual revenue over annual disbursement</i>	25000	30000

Solution : Discount rate at 10% per annum, $v = \frac{100}{100 + 10} = 0.9091$

Single payment present worth factor (pwf) at 10% for 15 years

$$= v^{15} = (0.9091)^{15} = 0.2394$$

and annual series present worth factor (pwf) at 10% for 15 years

$$= v + v^2 + \dots + v^{15} = v(1 - v^{14}) / (1 - v)$$

$$= (0.9091) \{1 - (0.9091)^{15}\} / (1 - 0.9091) = 7.6066$$

The present value of the total cost on the two plans for a period of 15 years are tabulated in the following table :

Present value of	Plan A in (₹)	Plan B in (₹)
Ist cost (1)	200000	250000
Scrap value after 15 year (2)	$150000 \times 0.2394 = 35910$	$180000 \times 0.2394 = 43092$
Excess of annual revenue over annual disbursement (3)	$25000 \times 7.6066 = 190165$	$30000 \times 7.6066 = 228198$
Present value of total return in 15 years in ₹ (2) + (3) - (1)	26075	21290

Since the present value of the total revenue in 15 years on plan A is more than that on plan B. Hence, the plan A should be preferred.

Example 4 : The cost of new car is ₹ 10000. Compare the optimum moment of replacement assuming the following cost informations :

Age of car, n	Repair cost in n-th year	Salvage value at the end of the n-th year
1	5000	8000
2	10000	6400
3	10000	5120

Assume that repairs are made at the end of each year only if the car is to be retained and are not necessary if the car to be sold for its salvage value. Also assume that the rate of discount is 10%.

Solution : Here, we shall compare the present value of the total investments on the car in 6 years with replacement policy after each year, two years and three years. Note that the period of 6 years is taken as the L.C.M. of 1, 2, 3 is 6.

Since the rate of discount is 10%,

$$\therefore \text{discount rate, } v = \frac{100}{100 + 10} = 0.9091$$

Determination of discounted cost i.e., the present value of total investments.

Year n	Cost of the new car at the end of n -th year in ₹	Repair cost at the end of n -th year in ₹	Salvage value at the end of n -th year in ₹	Total cost in ₹ (v)=(ii) + (iii) - (iv)	Discount factor v^n	$(v) \times v^n$
(i)	(ii)	(iii)	(iv)			
<i>Replacement after each year.</i>						
0	10000	—	—	10000	1.0000	10000.00
1	10000	—	8000	2000	0.9091	1818.20
2	10000	—	8000	2000	0.8265	1653.00
3	10000	—	8000	2000	0.7513	1502.60
4	10000	—	8000	2000	0.6830	1366.08
5	10000	—	8000	2000	0.6209	1241.80
6		—	8000	— 8000	0.5645	— 4516.00
Total = 13065.68						
<i>Replacement after 2 years.</i>						
0	10000	—	—	10000	1.0000	10000.00
1	—	5000	—	5000	0.9091	4545.50
2	10000	—	6400	3600	0.8265	2975.40
3	—	5000	—	5000	0.7513	3756.50
4	10000	—	6400	3600	0.6830	2458.80
5	—	5000	—	5000	0.6209	3104.50
6	—	—	6400	— 6400	0.5645	—3612.80
Total = 23227.90						
<i>Replacement after 3 years.</i>						
0	10000	—	—	10000	1.0000	10000.00
1	—	5000	—	5000	0.9091	4545.00
2	—	10000	—	10000	0.8265	8265.00
3	10000	—	5120	4880	0.7513	3666.34
4	—	5000	—	5000	0.6830	3415.00
5	—	10000	—	10000	0.6209	6209.00
6	—	—	5120	—5120	0.5645	—2890.24
Total = 33210.60						

Since the total discounted cost (*i.e.*, the present value of costs) in 6 years with three replacement policies, that with replacement annually (*i.e.*, every year) is minimum, hence yearly replacement policy is the optimum policy.

Example 5 : The cost of a new machine is ₹ 5000. The maintenance cost of n -th year is given by $C_n = 500 (n - 1)$; $n = 1, 2, \dots$. Suppose that the discount rate per year is 0.5. After how many years it will be economical to replace the machine by new one ?

Solution : Here $v = 0.5$, $A = ₹ 5000$ and $C_n = 500 (n - 1)$, $n = 1, 2, \dots$, $1 - v = 0.5$. If n years is the optimum replacement age, then from (3), article 6.5, we have

$$\frac{1 - v^{n-1}}{1 - v} C_n - P(n-1) < 0 < \frac{1 - v^n}{1 - v} C_{n+1} - P(n) \quad \dots(1)$$

where $P(n) = A + C_1 + C_2 v + C_3 v^2 + \dots + C_n v^{n-1}$

To find the value of n , satisfying (1), we compute the following table :

Year n	C_n ₹	v^{n-1} $= (0.5)^{n-1}$	$v^{n-1} C_n$	$(1 - v^{n-1})$ $\cdot C_n$	$\frac{(1 - v^n)}{1 - v}$ $\cdot C_{n+1}$	$P(n)$	$\frac{1 - v^n}{1 - v}$ $\cdot C_{n+1}$ $- P(n)$
(1)	(2)	(3)	(4) = (2) $\times$ (3)	(5) = (2) - (4)	(6)	(7)	(8) = (6) - (7)
1	0	1.0000	00.00	00.00	500.00	5000.00	< 0
2	500	0.5000	250.00	250.00	1500.00	5250.00	< 0
3	1000	0.2500	250.00	750.00	2625.00	5500.00	< 0
4	1500	0.1250	187.50	1312.50	3750.00	5687.50	< 0
5	2000	0.0625	125.00	1875.00	4844.00	5812.50	< 0
*6	2500	0.0312	78.00	2422.00	5906.40	5890.50	> 0
7	3000	0.0156	46.80	2953.20	—	—	

It is clear from the table that for $n = 6$.

$$\begin{aligned}
 \frac{1 - v^5}{1 - v} C_6 - P(5) &= ₹ 4844.00 - ₹ 5812.50 \\
 &= - ₹ 968.50 < 0 < \frac{1 - v^6}{1 - v} C_7 - P(6) \\
 &= ₹ 5906.40 - ₹ 5890.50 \\
 &= ₹ 15.90
 \end{aligned}$$

Hence, it will be economical to replace the machine by a new one after every six years.

Exercise 6.2

1. Assuming that the present value of one rupee to be spent in a year's time is ₹ 0.9 and $c = ₹ 3,000$, capital cost of equipment and the running cost are given in the table. When should the machine be replaced ?

Years	1	2	3	4	5	6	7
Running cost (₹)	500	600	800	1000	1300	1600	2000

2. A truck is priced at ₹ 60,000 and running costs are estimated at ₹ 6,000 for each of the first four years, increasing by ₹ 2,000 per year in the fifth and subsequent years. If money is worth 10% per year, when should the truck be replaced ? Assume that the truck will eventually be sold for scrap at a negligible price.
3. The cost of a new machine is ₹ 500. The maintenance cost of n -th year is given by $R_n = 50(n - 1)$, for $n = 1, 2, 3, \dots$. Suppose that the discount rate per year is 0.05 (or 5%). After how many years, it will be economical to replace the machine by new one?

Answers 6.2

1.	7 years.	2.	9 years.	3.	7 years.
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6.7 Replacement of Items in Anticipation of Complete Failure the Probability of which Increases with Time

In general, the probability of failure of an item increases with time. The complete failure of an item (machine) may result in complete break down of the system which puts the organisation to a heavy loss. Thus, there is a need of some replacement policy of such items to avoid the possibility of complete break down. Such items may be replaced even when they are in working order.

The following two policies are followed:

- Individual replacement policy:** According to this policy the failed item is immediately replaced by the new one. *e.g.*, in houses, fused bulb is replaced by the new one.
- Group replacement policy:** In general, a system contains a large number of identical low-cost items such that the probability of their failure increases with time or age. In such cases, there is a set up cost for replacement which is independent of

the number replaced. Here, in such systems, it may be advantageous to replace all the items at some fixed interval. Such policy is called the **group replacement policy**. This policy is followed in the systems where the cost of an individual item is so small that the cost of keeping records of individual's age cannot be justified.

6.8 To Determine the Interval of Optimum Replacement

Assume that the items of the system that fail during a period of fixed time t_1 (which may be a year, month or day etc.) are replaced individually at the end of that period and all the items of the system (including those which may be almost new) are replaced at intervals $t (= nt_1)$ consisting of n periods, i.e., all the items of the system are replaced after every n periods. Thus, we have to find the value of n for which average cost per period is minimum.

Let N = Total number of items in the system.

$N(X)$ = Number of items failed during X th period.

$$X = 1, 2, 3, \dots, (n-1)$$

C_g = The cost per item when all the items are replaced as a group.

C_i = The cost of replacing an individual item on its failure.

$C(n)$ = Total cost in the interval t (including the costs of replacing individuals at the end of each period) consisting of n periods.

$$= N \cdot C_g + C_i \cdot [\text{Total number of failures in the periods } 1, 2, 3, \dots, (n-1)]$$

$$= N \cdot C_g + C_i [N(1) + N(2) + N(3) + \dots + N(n-1)]$$

$$\text{or} \quad C(n) = N \cdot C_g + C_i \sum_{X=1}^{n-1} N(X) \quad \dots(1)$$

$$\therefore \text{Average cost per period is given by, } A(n) = \frac{C(n)}{n} \quad \dots(2)$$

$A(n)$ is minimum for that value of n , for which

$$\Delta A(n-1) < 0 < \Delta A(n) \quad \dots(3)$$

Now, $\Delta A(n) = A(n+1) - A(n)$

$$= \frac{C(n+1)}{n+1} - \frac{C(n)}{n} = \frac{C(n) + C_i N(n)}{n+1} - \frac{C(n)}{n}$$

$$= \frac{n C_i N(n) - C(n)}{n(n+1)} = \frac{C_i \cdot N(n) - \frac{C(n)}{n}}{n+1}$$

$$\text{Similarly } \Delta A(n-1) = \frac{C_i \cdot N(n-1) - \frac{C(n-1)}{n-1}}{n}$$

$\therefore$ From (3), we have

$$\frac{C_i \cdot N(n-1) - \frac{C(n-1)}{n-1}}{n} < 0 < \frac{C_i N(n) - \frac{C(n)}{n}}{n+1}$$

$$\text{or} \quad C_i \cdot N(n-1) - \frac{C(n-1)}{n-1} < 0 < C_i N(n) - \frac{C(n)}{n} \quad \dots(4)$$

From (4), we have

$$\left. \begin{array}{l} C_i N(n) > \frac{C(n)}{n} \\ \text{and } C_i N(n-1) < \frac{C(n-1)}{n-1} \end{array} \right\} \quad \dots(5)$$

From (5), we conclude the following group replacement policy :

1. Group replacement should be made at the end of n -th period if the cost of individual replacement for the n -th period is greater than the average cost per period by the end of n periods.
2. Group replacement should not be made at the end of n -th period if the cost of individual replacement at the end of $(n-1)$ th period is not less than the average cost per period by the end of $(n-1)$ periods.

ILLUSTRATIVE EXAMPLES

Example 1 : The following mortality rates have been observed for a certain type of light bulb :

Weeks	1	2	3	4	5
Percent failing by end of week	10	25	50	80	100

There are 1000 bulbs in use and it costs ₹ 1.00 to replace an individual bulb which has burnt out. If all bulbs were replaced simultaneously it would cost 25 paise per bulb. It is proposed to replace all bulbs at fixed intervals, whether or not they have burnt out and to continue replacing burnt out bulbs as they fail. At what intervals should all the bulbs be replaced ?

Solution : If P_i is the probability of failure of a new bulb in the i -th week, then

$$P_1 = \frac{10}{100} = 0.10, P_2 = \frac{25-10}{100} = 0.15, P_3 = \frac{50-25}{100} = 0.25$$

$$P_4 = \frac{80-50}{100} = 0.30, P_5 = \frac{100-80}{100} = 0.20$$

Here, $P_1 + P_2 + P_3 + P_4 + P_5 = 1.00$

Since the sum of all probabilities is equal to 1.

$\therefore P_6, P_7, \dots$ are all zero.

Therefore, a bulb cannot survive for more than five weeks *i.e.*, a bulb which has survived for four weeks is sure to fail in the fifth week.

Assuming that the burnt out bulbs in any week are replaced just at the end of that week, if N_i be the number of replacements at the end of i -th week, while all 1000 bulbs were new initially, then we have

$$N_0 = \text{Number of bulbs in the beginning} = 1000$$

$$\begin{aligned} N_1 &= \text{Number of burnt out bulbs replaced at the end of first week} \\ &= N_0 P_1 = 1000 \times 0.10 = 100 \end{aligned}$$

$$\begin{aligned} N_2 &= \text{Number of burnt out bulbs replaced at the end of second week} \\ &= N_0 P_2 + N_1 P_1 = 1000 \times 0.15 + 100 \times 0.10 = 160 \end{aligned}$$

$$\begin{aligned} N_3 &= \text{Number of burnt out bulbs replaced at the end of the third week} \\ &= N_0 P_3 + N_1 P_2 + N_2 P_1 = 1000 \times 0.25 + 100 \times 0.15 + 160 \times 0.10 = 281 \end{aligned}$$

$$\begin{aligned} N_4 &= \text{Number of burnt out bulbs replaced at the end of fourth week} \\ &= N_0 P_4 + N_1 P_3 + N_2 P_2 + N_3 P_1 \\ &= 1000 \times 0.30 + 100 \times 0.25 + 160 \times 0.15 + 281 \times 0.10 = 377 \end{aligned}$$

$$\begin{aligned} N_5 &= \text{Number of burnt out bulbs replaced at the end of fifth week} \\ &= N_0 P_5 + N_1 P_4 + N_2 P_3 + N_3 P_2 + N_4 P_1 \\ &= 1000 \times 0.20 + 100 \times 0.30 + 160 \times 0.25 + 281 \times 0.15 + 377 \times 0.10 = 350 \end{aligned}$$

$$\begin{aligned} N_6 &= \text{Number of burnt out bulbs replaced at the end of sixth week} \\ &= N_0 P_6 + N_1 P_5 + N_2 P_4 + N_3 P_3 + N_4 P_2 + N_5 P_1 \\ &= 0 + 100 \times 0.20 + 160 \times 0.15 + 281 \times 0.25 + 377 \times 0.15 + 350 \times 0.10 = 230 \end{aligned}$$

$$\begin{aligned} N_7 &= \text{Number of burnt out bulbs replaced at the end of seventh week} \\ &= N_0 P_7 + N_1 P_6 + N_2 P_5 + N_3 P_4 + N_4 P_3 + N_5 P_2 + N_6 P_1 \\ &= 0 + 0 + 160 \times 0.20 + 281 \times 0.30 + 377 \times 0.25 + 350 \times 0.15 \\ &\quad + 230 \times 0.10 = 286 \end{aligned}$$

$$\begin{aligned} N_8 &= \text{Number of burnt out bulbs replaced at the end of eighth week} \\ &= N_0 P_8 + N_1 P_7 + N_2 P_6 + N_3 P_5 + N_4 P_4 + N_5 P_3 + N_6 P_2 + N_7 P_1 \\ &= 0 + 0 + 0 + 281 \times 0.20 + 377 \times 0.30 + 350 \times 0.25 + 230 \times 0.15 \\ &\quad + 289 \times 0.10 = 320 \end{aligned}$$

and so on.

From the above calculations, we see that N_i the expected number of bulbs failing each week increases upto fourth week and then decreases upto 6th week and then increases again. In this way, N_i oscillates till the system settles down to a steady state.

The average life of a bulb

$$= \sum X_i P_i \text{ where } X_i \text{ is the week and } P_i \text{ the probability of failure in the } i\text{-th week.}$$

$$= 1P_1 + 2P_2 + 3P_3 + 4P_4 + 5P_5 + 6P_6 + \dots$$

$$= 0.10 + 2 \times 0.15 + 3 \times 0.25 + 4 \times 0.30 + 5 \times 0.20 = 3.35$$

$$\therefore \text{Average number of replacement every week} = \frac{N}{(\text{Mean age})} = \frac{1000}{3.35} = 299$$

$$\therefore \text{Average cost of weekly individual replacement policy} = ₹ 299$$

(Since cost of individual replacement of a bulb costs ₹ 1)

Now we consider the case of group replacement

End of week	Total cost of group Replacement in ₹	Average cost per week in ₹
1	$1000 \times 0.25 + 100 \times 1 = 350$	350
*2	$1000 \times 0.25 + (100 + 160) \times 1 = 510$	255
3	$1000 \times 0.25 + (100 + 160 + 281) \times 1 = 791$	263.66

Thus, the minimum cost per week is Rs. 255.00 if the bulbs are replaced as a group after every two weeks and this cost is also less than the average cost of weekly individual replacement policy.

Hence, for minimum monthly cost all the bulbs should be replaced after every two weeks.

Example 2 : Let $p(t)$ be the probability that a machine in a group of 30 machines would breakdown in period t . The cost of repairing broken machine is ₹ 200.00. Preventive maintenance is performed by servicing all the 300 machines at the end of T units of time. Preventive maintenance cost is ₹ 15 per machine. Find optimal T which will minimise the expected cost per period of servicing, given that

$$p(t) = \begin{cases} 0.03 & , \text{ for } t = 1 \\ p(t-1) + 0.01 & , \text{ for } t = 2, 3, \dots, 10 \\ 0.13 & , \text{ for } t = 11 \end{cases}$$

Solution: For $t=1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8 \quad 9 \quad 10 \quad 11 \quad 12 \quad 13 \dots$

$$p_i = 0.03 \quad 0.04 \quad 0.05 \quad 0.06 \quad 0.07 \quad 0.08 \quad 0.09 \quad 0.10 \quad 0.11 \quad 0.12 \quad 0.13 \quad 0 \quad 0 \dots$$

$$\therefore \sum_{i=1}^{11} p_i = 1, \Rightarrow p_{12} = 0 = p_{13} = \dots$$

Here, $N_0 = 30, N_1 = N_0 p_1 = 1, N_2 = N_0 p_2 + N_1 p_1 = 1$, Similarly, $N_3 = 2, N_4 = 2, N_5 = 2, N_6 = 3$, etc. Now average costs are, For $T=1, 30 \times 15 + 200 \times 1 = 650$, For $T=2, (1/2) [30 \times 15 + 200 \times (1+1)] = 425$, $T=3, (1/3) [30 \times 15 + 200 \times (1+1+2)] = 416$, For $T=4, 412$, For $T=5, 410$, For $T=6, 442$

Since minimum is 410, For $T=5$. Hence, optimal policy is to perform preventive maintenance of all the machines at the end of 5th period. Thus, average cost per period = ₹ 410.00.

Exercise 6.3

1. What is replacement? Describe some important replacement situations and policies.
2. Write short note on Group replacement policy.
3. The mortality rates as obtained for an electric compound are given as under :

Years	1	2	3	4	5	6
% failure at the end of the year	8	22	45	70	85	100

There are 1500 items in operation. It costs ₹ 20.00 to replace an individual item and ₹ 0.50 per items, if all are replaced simultaneously. If the policy is either to replace the items whenever they fail or a policy of group replacement along with the individuals. Suggest which should be better economically.

[Hint : $P_1 = 0.08, P_2 = 0.14, P_3 = 0.23, P_4 = 0.25, P_5 = 0.15, P_6 = 0.15$

$$P_1 + P_2 + P_3 + P_4 + P_5 + P_6 = 1, P_7 = 0 = P_8 = \dots$$

$$N_0 = 1500, N_1 = 120, N_2 = 306, N_3 = 386, N_4 = 476, N_5 = 410 \text{ etc.}$$

$$\text{Average life of the component} = \sum X_i P_i = 3.70$$

$$\text{Average number of replacement every year} = 1500/3.70 = 405]$$

4. The following failure rates have been observed for certain items :

End of month	1	2	3	4	5
Prob. of failure of date	0.10	0.30	0.55	0.85	1.00

The cost of replacement of an individual item is ₹ 1.25. The decision is made to replace all items simultaneously at fixed intervals and also to replace individual items as they fail. If the cost of group replacements is 50 paise, what is the best interval of group replacement ? At what group replacements price per item, would a policy of strictly individual replacement become preferable to be adopted policy ?

5. The following failure rates have been observed for a certain type of light bulb :

End of week	1	2	3	4	5	6	7	8
Prob. of failure to date	0.05	0.13	0.25	0.43	0.68	0.88	0.96	1.00

The cost of replacing an individual failed bulb is ₹ 1.25. The decision is made to replace all bulbs simultaneously at fixed intervals and also to replace individual bulbs as they fail in service. If the cost of group replacement is 30 paise per bulb and the total number of bulbs is 1000, what is the best intervals between group replacement ?

At what group replacement price per bulb would a policy of strictly individual replacement become preferable to be adopted policy ?

6. A system has a large number of light bulbs, all of which, we must keep in working order. If a bulb fails in service, it cost ₹ 1 to replace, but if we replace all the bulbs in the same operation we can do for only ₹ 0.35 a bulb. Find the optimum period for group replacement policy. The life distribution of the bulbs is as follows :

Weeks	1	2	3	4	5	6
Prob. of the bulb failing in the week	0.09	0.16	0.24	0.36	0.12	0.03

7. The prob. p_n of failure just before age n are shown below. If individual replacement costs ₹ 1.25 and group replacement costs ₹ 0.50 per item, find the optimal group replacement policy (assuming that there are 1000 bulbs in use).

n	1	2	3	4	5	6	7	8	9	10	11
p_n	0.01	0.03	0.05	0.07	0.10	0.15	0.20	0.15	0.11	0.08	0.05

Answers 6.3

3.	Group replacement yearly	4.	Group replacement, 3 weeks
5.	Group replacement 3 weeks	6.	3 weeks
7.	5 weeks.		

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

- The problem of replacement is most concerned about the :
 - Item that fail suddenly
 - Maintenance of an item work out profitability
 - Item that deteriorate graphically
 - Determination of optimum replacement interval.
- Which cost of the following is irrelevant to replacement analysis ?
 - Purchase cost of machine
 - Operating cost of machine
 - Maintenance cost of machine
 - None of these.

3. Replacement models discuss the problem where we wish to :
 - (a) Maximize operational (maintenance) cost
 - (b) Maximize step up cost
 - (c) Minimize operational cost
 - (d) Minimize set up cost.
4. When the equipment starts deteriorating with respect to time, its maintenance cost gradually starts :
 - (a) Increasing
 - (b) Decreasing
 - (c) Constant
 - (d) None of these.
5. A certain items set up cost and scrap value is constant with respect to time. If $F(n)$ is the maintenance cost function and $A(n)$ is the average total annual cost, then it is better to replace that item, when :
 - (a) $A(n) = \text{const}$
 - (b) $F(n) = \text{const}$
 - (c) $F(n) < A(n)$
 - (d) $F(n) = A(n)$.
6. Assuming that the item has no resale price at the time of replacement, if $P(n)$ is the present worth of all future discounted costs, then replace the item at the end of every n years of :
 - (a) $P(n+1) - P(n) > 0$
 - (b) $P(n-1) - P(n) < 0$
 - (c) If both (a) and (b) hold
 - (d) None of these.
7. Replacement model is a :
 - (a) Static model
 - (b) Dynamic model
 - (c) Both (a) and (b)
 - (d) None of these.
8. What is concerned with the prediction replacement costs and determination of the most-economic replacement policy ?
 - (a) Search theory
 - (b) Theory of replacement
 - (c) Probabilistic Programming
 - (d) None of these.
9. The replacement policy in which the failed item is immediately replaced by the new one is called :
 - (a) Replacement policy
 - (b) Individual replacement policy
 - (c) Group replacement policy
 - (d) None of these.

Fill in the Blank

1. Replacement policy for items whose maintenance cost increases with time and money value is constant is that to replace the machine when the average cost becomes than the current average cost.
 2. If r is the interest rate, n is the number of years then present worth factor is given by..... .
-

3. The group replacement should be made at the end of n -th period if the cost of individual replacement for the n -th period is than the average cost per period by the end of n periods.

True/False

1. In some systems in which all the items are replaced at some fixed interval is called the group replacement policy.
2. The group replacement should be made at the end of (n) th period if the cost of individual replacement at the end of $(n-1)$ th period, is not less than the average cost per period by the end of $(n-1)$ periods.

Answers

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1.	(b)	2.	(d)	3.	(c)
4.	(a)	5.	(d)	6.	(c)
7.	(b)	8.	(b)	9.	(b)

Fill in the Blank

1.	greater	2.	$(1+r)^{-n}$	3.	greater
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True/False

1.	T	2.	F
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UNIT-II

Chapter

7



Queueing Theory (Queueing Models)

7.1 Introduction

A group of items waiting to receive service, including those receiving the service is known as a **waiting line** or a **queue**. Queueing theory involves the mathematical study of “queues” or waiting lines. The formation of waiting lines (or queues) is a common phenomenon which occurs whenever the current demand for a service exceeds the current capacity to provide that service. The queues of peoples can be seen at a cinema ticket window, bus stop, reservation office, counters of super market etc. The person waiting in a queue or receiving the service is called the **customer** and the person by whom he is serviced is called a **server**.

For example: The customers arriving for service may form one queue and be serviced through only one station, as in a doctor’s clinic; they may form one queue and be serviced through several stations, as in barber shop; or they may form several queues and be served through as many stations, as at the check out counters of a super market.

7.2 Basic Queueing Process(system) and its Characteristics

The basic queueing process can be described as a process in which the customers arrive for service at a service counter (or station), wait for their turn in the queue if the server is busy in the service of the other customer and are served when the server gets free. Finally the customer leave the system as soon as he is served.

Characteristics of Queueing System

1. The input (or arrival pattern).
2. Queue (or waiting line).
3. The service discipline (or queue discipline).
4. The service mechanism (or service pattern).



Fig 7.1: The basic queueing process

1. The input (or arrival pattern)

The input describes the pattern in which the customers arrive for service. Since the units for service (customers) arrive in a random fashion therefore, their arrival pattern can be described in terms of probabilities. Here, in this chapter, we assume that they arrive according to Poisson process *i.e.*, the number of units arriving until any specific time has a Poisson distribution. This is the case where arrivals to the queueing system occur at random, but at a certain average rate.

2. Queue (or waiting line)

The units requiring service enter the queueing system on their arrival and join a “queue” which is characterized by the maximum permissible number of units that it can contain called the **capacity of the system**. A queue is called **finite** if the number of units in it is **finite** otherwise it is called **infinite**.

3. The service discipline (or queue discipline)

The service discipline refers to the manner in which the members in the queue are chosen for service. The following service disciplines are seen in common practice:

- (i) **First come, first served (FCFS):** According to this discipline the customers are served in the order of their arrival. This service discipline may be seen at a cinema ticket window, at a railway ticket window etc. Here, in this chapter, we shall deal with the queueing models in which the service discipline in *FCFS*.
- (ii) **Last come, first served (LCFS):** This discipline may be seen in big godowns where the units (items) which come last are taken out (served) first.
- (iii) **Service in random order.**
- (iv) **Service on some priority-procedure:** Some customers are served before the others without considering their order of arrival *i.e.*, some customers are served on priority basis.

4. The service mechanism (or service pattern)

The service mechanism refers to :

- (i) **The pattern according to which the customers are served:** Here, in this chapter, we shall deal with the queueing models in which the service time follows the ‘Exponential’ and ‘Erlang’ probability distributions.

- (ii) **Facilities given to the customers:** The facilities given to the customers can be classified in two categories :
- (a) **Single channel :** Here, the customers are served by one counter only *i.e.*, only one customer can be served at a time *e.g.*, at cinema ticket window.
 - (b) **Multi-channel :** Here, the customers are served by several (more than one) counters *i.e.*, more than one customer can be served at a time *e.g.*, in the barber shop.

7.3 Customers Behaviour in a Queue

1. **Balking :** A customer may not like to wait in a queue due to lack of time or space or otherwise.
2. **Reneging :** A customer may leave the queue due to impatience.
3. **Collusion :** Some customers may collaborate and only one of them may join the queue. As at the cinema ticket window one person may join the queue and purchase tickets for his friends.
4. **Jockeying :** If there are more than one queues then one customer may leave one queue and join the other. This occurs generally in the super market.

7.4 Important definitions in Queueing Problem

Here, we give the definitions of various terms used in this chapter:

1. **Queue Length :** Queue length is defined by the number of persons (customers) waiting in the line at any time.
 2. **Average Length of Line :** Average length of line (or Queue) is defined by the number of customers in the queue per unit time.
 3. **Waiting Time :** It is time upto which a unit has to wait in the queue before it is taken into service.
 4. **Servicing Time :** The time taken for servicing of a unit is called its servicing time.
 5. **Busy Period :** Busy-period of a server is the time during which he remains busy in servicing. Thus, it is the time between the start of service of the first unit to the end of service of the last unit in the queue.
 6. **Idle Period :** When all the units in the queue are served. The idle period of the server begins and it continues upto the time of arrival of the unit (customer). The idle period of a server is the time during which he remains free because there is no customer present in the system.
 7. **Mean Arrival Rate :** The mean arrival rate in a waiting-line situation is defined as the expected number of arrivals occurring in a time interval of length unity.
-

8. **Mean Servicing Rate :** The mean servicing rate for a particular servicing station is defined as the expected number of services completed in a time interval of length unity, given that the servicing is going on throughout the entire time unit.
9. **Traffic Intensity :** In case of a simple queue the traffic intensity is the ratio of mean arrival rate and the mean servicing rate.

$$\text{i.e., Traffic intensity} = \frac{\text{Mean arrival rate}}{\text{Mean servicing rate}}$$

7.5 The State of the System

A basic concept in the analysis of a queueing theory is that of a state of the system. It involves the study of a system's behaviour over time. The state of a system may be classified as follows :

1. **Transient state :** *A system is said to be in transient state when its operating characteristics are dependent on time.* Thus, a queueing system is in transient state when the probability distributions of arrivals, waiting time and servicing time of the customers are dependent on time. This state occurs at the beginning of the operation of the system.
2. **Steady state :** *A system is said to be in steady state when its operating characteristics becomes independent of time.* Thus, a queueing system acquires steady state when the probability distributions of arrivals, waiting time and servicing time of the customers are independent of the time. This state occurs in the long run of the system.

Let $P_n(t)$ denote the probability that there are n units in the system at time t , then the system acquires steady state as $t \rightarrow \infty$, i.e.

$$\lim_{t \rightarrow \infty} P_n(t) = P_n \text{ (independent of time } t\text{)}.$$

In most of the queueing problems, steady-state solutions exist independent of the initial state of the queue.

3. **Explosive state :** If the arrival rate of the system is more than its servicing rate, the length of the queue will go on increasing with time and will tend to infinity as $t \rightarrow \infty$. This state of the system is said to be *explosive state*.

7.6 Poisson Process

In Poisson process, the probability of n arrivals during time intervals of length t is given by

$$P_n(t) = \frac{(\lambda t)^n \cdot e^{-\lambda t}}{n!} \quad \dots(1)$$

where λt is the parameter.

Case I : When $n = 0$

From (1), the probability of no arrival in time Δt is given by

$$\begin{aligned}
 P_0(\Delta t) &= \frac{(\lambda \Delta t)^0 \cdot e^{-\lambda \Delta t}}{0!} = e^{-\lambda \Delta t} \\
 &= 1 - \lambda \Delta t + \frac{\lambda^2}{2!} (\Delta t)^2 \dots = 1 - \lambda \Delta t + 0 (\Delta t)
 \end{aligned}$$

where $0 (\Delta t)$ denotes a quantity which is of smaller order of magnitude than Δt .

If Δt is very small then $0 (\Delta t) = 0$

$$\therefore P_0(\Delta t) = 1 - \lambda \Delta t$$

i.e., the probability of no arrival in time $\Delta t = 1 - \lambda \Delta t$.

Case II : When $n = 1$

From (1), the probability of one arrival in time Δt is given by

$$\begin{aligned}
 P_1(\Delta t) &= \frac{(\lambda \Delta t)^1 \cdot e^{-\lambda \Delta t}}{1!} \\
 &= \lambda \Delta t \cdot \left\{ 1 - \lambda \Delta t + \frac{\lambda^2}{2!} (\Delta t)^2 \dots \right\} = \lambda \Delta t + 0 (\Delta t)
 \end{aligned}$$

$$\therefore P_1(\Delta t) = \lambda \Delta t. \quad (\text{If } \Delta t \text{ is very small})$$

i.e., the probability of one arrival in time $\Delta t = \lambda \Delta t$.

Case III : When $n = m > 1$

From (1), the probability of m (more than one) arrivals in time Δt is given by

$$\begin{aligned}
 P_m(\Delta t) &= \frac{(\lambda \Delta t)^m \cdot e^{-\lambda \Delta t}}{m!} \\
 &= \frac{\lambda^m (\Delta t)^m}{m!} \left\{ 1 - \lambda \Delta t + \frac{\lambda^2 (\Delta t)^2}{2!} \dots \right\} \\
 &= \frac{\lambda^m}{m!} \{ (\Delta t)^m - \lambda (\Delta t)^{m+1} \dots \}
 \end{aligned}$$

$$\therefore P_m(\Delta t) = 0. \quad (\text{If } \Delta t \text{ is very small})$$

$(m > 1)$

Thus, the probability of more than one arrival in time $\Delta t = 0$.

(If the terms of second and higher powers of Δt are negligibly small).

From the above discussion, we arrive to the following result which is sometimes known as *postulates for Poisson Process*:

Postulates for the Poisson Process

Postulate 1

The number of arrivals in non-overlapping intervals are statistically independent, *i.e.,* the process has independent increments.

Postulate 2

The probability that an arrival occurs between time t and time $t + \Delta t$ is equal to $\lambda \Delta t + 0 (\Delta t)$.

$$*i.e.,* \quad P_1(\Delta t) = \lambda \Delta t + 0 (\Delta t).$$

where λ is a constant independent of $P_n(t)$, Δt is small increment in t and $O(\Delta t)$ denotes a quantity which is of smaller order of magnitude than Δt s.t., $\lim_{\Delta t \rightarrow 0} \frac{O(\Delta t)}{\Delta t} = 0$.

Postulate 3

The probability of more than one arrival between time t and time $t + \Delta t$ is $O(\Delta t)$, i.e., the probability of two or more arrivals during the small time interval Δt is negligible.

Thus, $P_0(\Delta t) + P_1(\Delta t) + O(\Delta t) = 1$.

7.7 Poisson Arrivals

In general, arrivals in a queueing system do not occur at regular intervals, but tend to be scattered in some fashion. The "Poisson assumption" specifies the behaviour of arrivals, by postulating the existence of constant λ , which is independent of time, queue length, or any other random property of the queue, such that

$$P(\text{an arrival occurs between time } t \text{ and } t + \Delta t) = \lambda \Delta t \quad \dots(1)$$

(Probability)

if the interval Δt is sufficient small.

A waiting line for which arrivals occur in accordance with (1) is called a queue with *Poisson arrivals*.

7.8 Theorem

Let n be a random variable (discrete) representing the number of arrivals in some time interval of length t , then if equation (1) is satisfied, n obeys a Poisson distribution.

$$P_n(t) = \frac{(\lambda t)^n \cdot e^{-\lambda t}}{n!}$$

(Probability of n arrivals in time t)

where λt is a parameter.

Proof : Here the arrivals are Poisson and follow the postulate for the Poisson Process (see 7.6).

Let $(0, t)$ and $(t, t + \Delta t)$ be two consecutive time intervals where Δt is sufficiently small.

Case I : When $n \geq 1$

There are n arrivals in the interval $(0, t + \Delta t)$ in the following ways:

1. There are n arrivals in the interval $(0, t)$ and no arrival in the consecutive time interval $(t, t + \Delta t)$.

The probability in this case will be $P_n(t) \cdot P_0(\Delta t)$

(Since the events are mutually exclusive)

2. There are $(n-1)$ arrivals in the interval $(0, t)$ and one arrival in the consecutive time interval $(t, t + \Delta t)$. The probability in this case will be $P_{n-1}(t) \cdot P_1(\Delta t)$.
3. There are $(n-k)$ arrival in the interval $(0, t)$ and $k > 1$ arrivals in the consecutive time interval $(t, t + \Delta t)$. The probability in this case will be $P_{n-k}(t) \cdot P_k(\Delta t)$.

Thus, adding above independent probabilities the probability of n arrivals in the time interval $(0, t + \Delta t)$ is given by

$$P_n(t + \Delta t) = P_n(t) \cdot P_0(\Delta t) + P_{n-1}(t) \cdot P_1(\Delta t) + P_{n-k}(t) \cdot P_k(\Delta t) \quad \dots(1)$$

Now $P_0(\Delta t)$ = Probability that there is no arrival in time interval $(t, t + \Delta t)$
 $= 1 - \text{probability that there is an arrival in } (t, t + \Delta t)$
 $= 1 - \lambda \Delta t + 0(\Delta t)$

Also from the postulate of Poisson process

$$P_1(\Delta t) = \lambda \Delta t + 0(\Delta t)$$

and $P_k(\Delta t) = 0(\Delta t) \quad k > 1$

$\therefore$ From (1), we have

$$P_n(t + \Delta t) = P_n(t) \cdot [1 - \lambda \Delta t + 0(\Delta t)] + P_{n-1}(t) \cdot [\lambda \Delta t + 0(\Delta t)] + P_{n-k}(t) \cdot [0(\Delta t)]$$

$$= P_n(t) - \lambda P_n(t) \Delta t + \lambda P_{n-1}(t) \Delta t + 0(\Delta t)$$

or $\frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} = -\lambda P_n(t) + \lambda P_{n-1}(t) + \frac{0(\Delta t)}{\Delta t}$

Now taking limit as $\Delta t \rightarrow 0$, we have

$$P'_n(t) = -\lambda P_n(t) + \lambda P_{n-1}(t). \quad \dots(2)$$

Case II : When $n = 0$

$$P_0(t + \Delta t) = \text{Probability of no arrival in time interval } (0, t + \Delta t)$$

$$= \text{Probability of no arrival in time interval } (0, t)$$

$$\times \text{Probability of no arrival in consecutive time interval } (t, t + \Delta t)$$

$$= P_0(t) \cdot P_0(\Delta t)$$

$$= P_0(t) \cdot [1 - \lambda \Delta t + 0(\Delta t)]$$

$\therefore \frac{P_0(t + \Delta t) - P_0(t)}{\Delta t} = -\lambda P_0(t) + \frac{0(\Delta t)}{\Delta t}$

Taking limit as $\Delta t \rightarrow 0$, we have

$$P'_0(t) = -\lambda P_0(t) \quad \dots(3)$$

or $\frac{P'_0(t)}{P_0(t)} = -\lambda$

Integrating, $\log P_0(t) = -\lambda t + A$.

When $t = 0, P_0(0) = 1$, so that $A = 0$

$$\therefore \log P_0(t) = -\lambda t$$

$$\text{or } P_0(t) = e^{-\lambda t} \quad \dots(4)$$

Now putting $n = 1$ in (2). We have

$$P'_1(t) = -\lambda P_1(t) + \lambda P_0(t)$$

$$\text{or } P'_1(t) + \lambda P_1(t) = \lambda e^{-\lambda t} \quad [\text{Substituting } P_0(t) = e^{-\lambda t} \text{ from (4)}]$$

which is linear differential of 1st degree.

$$\therefore \text{I.F.} = e^{\int \lambda dt} = e^{\lambda t}$$

Multiplying both sides by I.F. $e^{\lambda t}$ and then integrating, we have

$$e^{\lambda t} \cdot P_1(t) = C + \int \lambda e^{-\lambda t} \cdot e^{\lambda t} dt = C + \lambda t$$

when $t = 0, P_1(0) = 0$, so that $C = 0$

$$\therefore P_1(t) = \lambda t \cdot e^{-\lambda t} \quad \dots(5)$$

Again putting $n = 2$, in the equation (2), we have

$$P'_2(t) = -\lambda P_2(t) + \lambda P_1(t)$$

$$= -\lambda P_2(t) + \lambda \cdot \lambda t e^{-\lambda t} \quad [\text{Substituting from (5)}]$$

$$\text{or } P'_2(t) + \lambda P_2(t) = \lambda^2 t e^{-\lambda t}$$

which is L.D.E. proceeding as before, we have

$$P_2(t) = \frac{\lambda^2 t^2}{2!} \cdot e^{-\lambda t} = \frac{(\lambda t)^2 \cdot e^{-\lambda t}}{2!}$$

Proceeding in the similar way for $n = 3, 4, \dots$, we arrive to the result

$$P_n(t) = \frac{(\lambda t)^n e^{-\lambda t}}{n!} \quad \dots(6)$$

i.e., n arrivals in the system follows Poisson Process.

Note: If the arrivals follow Poisson Process, from (6)

Mean (expected number of arrivals in a time interval of length t) $= \lambda t$.

Taking $t = 1$ it follows that the mean arrival rate for Poisson arrivals is just λ .

7.9 Some Distributions

Here, we give some distributions which are used in queueing theory:

1. **Exponential distribution** : If T is a random variable representing the inter-arrival time (the time between consecutive arrivals), then T obeys an exponential distribution *i.e.*, if $f(T)$ denotes the probability density function of T , then

$$f(T) = \lambda e^{-\lambda T}$$

where λ is parameter.

2. **Regular distribution** : Here, the inter-arrival time is assumed to be constant. This distribution is defined by

$$\begin{aligned} f(T) &= 0 \quad \text{for } T < a \\ &= 1 \quad \text{for } T \geq a. \end{aligned}$$

a is constant.

7.10 An Important Theorem

If the arrival pattern in a queueing problem follows a Poisson Process, then the random variable T representing the inter arrival time follows the exponential distribution and vice-versa.

Proof : If n is the number of arrivals during time interval of length t , then

$$P_n(t) = \frac{(\lambda t)^n \cdot e^{-\lambda t}}{n!} \quad \dots(1)$$

Let t_0 = instant of an arrival initially

The random variable T represents the inter arrival time, therefore, there is no arrival in the interval $(t_0, t_0 + T)$ and $t_0 + T + \Delta T$ will be the instant of consecutive arrival.

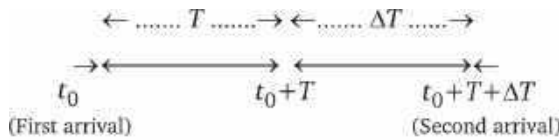


Fig 7.2

From (1),

$$\begin{aligned} P_0(T) &= \text{Probability of no arrival in time } T \\ &= \frac{(\lambda T)^0 \cdot e^{-\lambda T}}{0!} = e^{-\lambda T} \end{aligned} \quad \dots(2)$$

and $P_0(T + \Delta T)$ = Probability of no arrival in time $T + \Delta T$

$$\begin{aligned} &= \frac{\{\lambda (T + \Delta T)\}^0 \cdot e^{-\lambda (T + \Delta T)}}{0!} \\ &= e^{-\lambda (T + \Delta T)} = e^{-\lambda T} \cdot e^{-\lambda \Delta T} \\ &= P_0(T) \cdot [1 - \lambda \Delta T + 0 (\Delta T)^2] \end{aligned}$$

where $0 (\Delta T)^2$ contain the terms of second or higher power of ΔT .

$$\therefore P(T + \Delta T) - P_0(T) = -\lambda P_0(T) \Delta T + 0 (\Delta T)^2$$

or
$$\frac{P_0(T + \Delta T) - P_0(T)}{\Delta T} = -\lambda P_0(T) + \frac{0 (\Delta T)^2}{\Delta T}$$

Taking limit as $\Delta T \rightarrow 0$, we have

$$\frac{d}{dT} P_0(T) = -\lambda P_0(T) \quad \therefore \frac{d}{dT} P_0(T) = f(T) = \lambda e^{-\lambda T} \quad \text{from (2)}$$

(Probability density function)

(Leaving -ve sign since probability density function is always non-negative).

Hence, T follows the exponential distribution law.

Converse : The converse of the theorem may be proved in the similar way.

Remark : In the similar way, we can show that, if the time t to complete service on a unit follows the Exponential distribution

i.e., $f(t) = \text{the density function of } t = \mu e^{-\mu t}$

where μ is the mean servicing rate for a particular servicing station then n the number of units the servicing station could serve in time t , if the servicing is going on throughout t will follow the Poisson distribution.

i.e., $\phi_t(n) = \text{Probability } [n \text{ servicing in time } t, \text{ if there were no enforced idle time}]$

$$= \frac{(\mu t)^n e^{-\mu t}}{n!}$$

7.11 The list of Formulae

If λ is the mean arrival rate of customers and μ the mean service rate then for probabilistic queueing models in which the arrivals and service of customers follow Poisson distribution, then

1. $P_n(t) = \text{Prob. of } n \text{ arrivals in time } t = \frac{(\lambda t)^n}{n!} e^{-\lambda t}$
2. $P_0(\Delta t) = \text{Prob. of one arrival in time } \Delta t \text{ (small)} = 1 - \lambda \Delta t$
3. $P_1(\Delta t) = \text{Prob. of one arrival in time } \Delta t = \lambda \Delta t$
4. $P_k(\Delta t) = \text{Prob. of } k \text{ arrivals in time } \Delta t = 0, k > 1$
5. $P(\text{a customer will be served in time less than } T) = 1 - e^{-\mu T}$
6. $P(\text{a customer will be served in time more than } T) = e^{-\mu T}$
7. $\Phi_t(n) = \text{Prof. of } n \text{ services in time } t = \frac{(\mu t)^n}{n!} e^{-\mu t}$
8. $\Phi_{\Delta t}(0) = \text{Prob. of no service in time } \Delta t = 1 - \mu \Delta t$
9. $\Phi_{\Delta t}(1) = \text{Prob. of one service in time } \Delta t = \mu \Delta t$
10. $\Phi_{\Delta t}(k) = \text{Prob. of } k > 1 \text{ services in time } \Delta t = 0$

7.12 Notations

The following notations will be used throughout the chapter:

T	The inter arrival time between two successive customers.
n	Number of customers in the queueing system.
$P_n(t)$	Transient state-probability that there are exactly n units in the system at any time t .
P_n	Steady state probability of having n units in the system $\left(\lim_{t \rightarrow \infty} P_n(t) = P_n \right)$
λ_n	Mean arrival rate of customers when there are n units present in the system.
μ_n	Mean service rate when there are n units in the system.
λ	Mean arrival rate of customer (independent of n).
μ	Mean service rate (independent of n).
s	Number of parallel service stations.
$\rho = \frac{\lambda}{\mu}$	Traffic intensity (or utilization factor).
$O(\Delta t)$	A quantity which is of smaller order of magnitude than Δt .
$FCFS$	First come first served (service discipline).
$\psi(\omega)$	Probability density function of waiting time in the system.
$E(L_s)$	Expected number of customers in the system (waiting and in service) <i>i.e.</i> , expected or mean line length.
$E(L_q)$	Expected number of customers in the queue <i>i.e.</i> , expected or mean queue length (excluding the number of units in service).
$E(W_s)$	Expected waiting time per customer in the system (including service time).
$E(W_q)$	Expected waiting time per customer in the queue (excluding service time).
$E(L/L > 0)$	Expected length of the non-empty queue.
$E(W/W > 0)$	Expected waiting time of a customer who has to wait.

$$\binom{m}{n} = {}^m C_n$$

Note: The value of $\rho = \lambda/\mu$ must always be less than 1, so that the steady state condition may be obtained. If $\rho > 1$ *i.e.*, $\lambda > \mu$, then extremely large queues will be obtained.

7.13 Classification of Queueing Models

A queueing model is symbolically represented as $(A/B/C) : (D/E)$, where,

- A Arrival pattern of the units which is given by the probability distribution of inter arrival time of units.
- B The probability distribution of servicing time of the unit (which is actually serviced).
- C The number of servicing channels (stations) in the system.
- D Capacity of the system *i.e.*, maximum number allowed in the system (in service + waiting).
- E The manner in which units are selected for service in the system (service discipline).

The various types of the Queueing model discussed in this chapter are as follows:

Model I : (Erlang model); $(M/M/1) : (\infty/FCFS)$

Here, first M stands for Poisson arrival

second M stands for Poisson service

1 stands for single server.

∞ stands for infinite capacity of the system.

$FCFS$ notation representing first come first served (service discipline).

Model II : $(M/M/1) : (\infty/FCFS)$ (General single station queueing model)

This model is same as Model I except that here the rate of arrival and service depend on the length of the line *i.e.*, here some person interested in joining the queue may not join due to long queue and the servicing rate of the server (servicing station) is also effected by the length of the queue.

Model III : $(M/M/1) : (N/FCFS)$

This is the model with Poisson arrival, Poisson service, single channel, capacity of the system is N (finite) and first come first served service discipline. Here, the number of arrival will not exceed N (capacity of the system).

Model IV : $(M/M/S) : (\infty/FCFS)$

This is the model with Poisson arrival, Poisson service, number of channels is S (finite) and service discipline is first come first served.

Model V : $M/G/1$ Queueing System

This is the queueing system with Poisson arrival, general service time distribution, single server.

Model VI: Cost Profit Model

7.14 Solution of Queue Models

The solution of a queueing model consists of the following steps :

- Step one :** First of all we shall obtain the system of steady state equations governing the queue.
- Step two :** In step two, we solve the equations obtained in step one and find the probability distribution of queue length.
- Step three :** Then we obtain various probability density functions and derive formulae for

$$E(L_s), E(L_q), E(W_s), E(W_q), E(L/L > 0) \text{ and } E(W/W > 0).$$

7.15 Model I (M/M/1): (∞ /FCFS) (Birth and Death Model)

This is the queueing model, with Poisson arrival, Poisson service, single servicing channel, with infinite capacity. The service discipline is first come first serve. Here λ = mean arrival rate of units and μ = mean service rate.

Step one : To find the system of steady state equations.

There are $n > 0$ units in the system at time $(t + \Delta t)$ in the following ways :

- (i) $(n - 1)$ units in the system at time t with probability $P_{n-1}(t)$

one arrival in time Δt with probability $P_1(\Delta t) = \lambda \Delta t$

no service in time Δt with probability $\phi_{\Delta t}(0) = 1 - \mu \Delta t$

The probability in this case is $P_{n-1}(t) \cdot \lambda \Delta t (1 - \mu \Delta t)$.

- (ii) n units in the system at time t , with probability $P_n(t)$

no arrival in time Δt , with probability $P_0(\Delta t) = 1 - \lambda \Delta t$

no service in time Δt , with probability $\phi_{\Delta t}(0) = 1 - \mu \Delta t$

The probability in this case is $P_n(t) \cdot (1 - \lambda \Delta t) \cdot (1 - \mu \Delta t)$

- and (iii) $(n + 1)$ units in the system at time t , with probability $P_{n+1}(t)$

no arrival in time Δt , with probability $P_0(\Delta t) = 1 - \lambda \Delta t$

one service in time Δt , with probability $\phi_{\Delta t}(1) = \mu \Delta t$.

The probability in this case is $P_{n+1}(t) \cdot (1 - \lambda \Delta t) \cdot \mu \Delta t$.

Since all the above three cases are mutually exclusive, therefore, $P_n(t + \Delta t)$, the probability of n units in the system at time $(t + \Delta t)$ is obtained by adding the probabilities in the above three cases.

$$\begin{aligned} \text{i.e., } P_n(t + \Delta t) = & P_{n-1}(t) \lambda \Delta t \cdot (1 - \mu \Delta t) + P_n(t) \cdot (1 - \lambda \Delta t) (1 - \mu \Delta t) \\ & + P_{n+1}(t) \cdot (1 - \lambda \Delta t) \cdot \mu \Delta t \end{aligned}$$

or $P_n(t + \Delta t) = [\lambda P_{n-1}(t) - (\lambda + \mu) P_n(t) + \mu P_{n+1}(t)] \Delta t + P_n(t) + 0(\Delta t)$

$\therefore \frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} = \lambda P_{n-1}(t) - (\lambda + \mu) P_n(t) + \mu P_{n+1}(t) + \frac{0(\Delta t)}{\Delta t} \quad \dots (1)$

Similarly there is no unit in the system at time $(t + \Delta t)$ in the following two ways :

- (i) no unit in the system at time t , with probability $P_0(t)$
 no arrival in time Δt , with probability $P_0(\Delta t) = 1 - \lambda \Delta t$
 The probability in this case is $P_0(t) \cdot (1 - \lambda \Delta t)$

- and (ii) one unit in the system at time t , with probability $P_1(t)$
 no arrival in time Δt , with probability $P_0(\Delta t) = 1 - \lambda \Delta t$
 one service in time Δt , with probability $\phi_{\Delta t}(1) = \mu \Delta t$
 The probability in this case is $P_1(t) \cdot (1 - \lambda \Delta t) \cdot \mu \Delta t$

$\therefore$ Adding the probabilities in the above two cases, we get $P_0(t + \Delta t)$, the probability of no unit in the system at time $(t + \Delta t)$.

i.e., $P_0(t + \Delta t) = P_0(t) \cdot (1 - \lambda \Delta t) + P_1(t) \cdot (1 - \lambda \Delta t) \cdot \mu \Delta t$

or $P_0(t + \Delta t) = P_0(t) + [-\lambda P_0(t) + \mu P_1(t)] \Delta t + 0(\Delta t)$

$\therefore \frac{P_0(t + \Delta t) - P_0(t)}{\Delta t} = -\lambda P_0(t) + \mu P_1(t) + \frac{0(\Delta t)}{\Delta t} \quad \dots (2)$

Taking limit as $\Delta t \rightarrow 0$, from (1) and (2), we have

$$\frac{d}{dt} P_n(t) = \lambda P_{n-1}(t) - (\lambda + \mu) P_n(t) + \mu P_{n+1}(t) \quad \dots (3)$$

and $\frac{d}{dt} P_0(t) = -\lambda P_0(t) + \mu P_1(t). \quad \dots (4)$

Now when the system reaches steady state,

$$\lim_{t \rightarrow \infty} P_n(t) = P_n \text{ and } \lim_{t \rightarrow \infty} P'_n(t) = 0$$

$\therefore$ Under steady state of the system, the equations (3) and (4) reduce to

$$\lambda P_{n-1} - (\lambda + \mu) P_n + \mu P_{n+1} = 0 \quad \dots (5)$$

$(n > 0)$

and $-\lambda P_0 + \mu P_1 = 0 \quad \dots (6)$

These equations (5) and (6) are the *steady state equations of the system*.

Note: The system will settle to steady state only if $\lambda < \mu$ i.e., $\rho = \frac{\lambda}{\mu} < 1$.

Step two : To solve the steady state equations (5) and (6) obtained in step one.

From (6),

$$P_1 = \frac{\lambda}{\mu} P_0 = \rho P_0, \quad \frac{\lambda}{\mu} = \rho \text{ (Traffic intensity)} < 1$$

Putting $n = 1$ in (5),

$$P_2 = -\frac{\lambda}{\mu} P_0 + \left(\frac{\lambda}{\mu} + 1\right) P_1 = \left(\frac{\lambda}{\mu}\right) P_1 = \rho P_1 = \rho^2 P_0$$

Putting $n = 2$ in (5),

$$P_3 = -\frac{\lambda}{\mu} P_1 + \left(\frac{\lambda}{\mu} + 1\right) P_2 = -\rho^2 P_0 + (\rho + 1) \cdot \rho^2 P_0 = \rho^3 P_0$$

Proceeding in this way, we have

$$P_n = \rho^n P_0 \quad \text{for } n \geq 0$$

But
$$\sum_{n=0}^{\infty} P_n = 1$$

or
$$P_0 + P_1 + P_2 + P_3 + \dots = 1$$

or
$$(1 + \rho + \rho^2 + \rho^3 + \dots) P_0 = 1$$

or
$$\frac{1}{1 - \rho} \cdot P_0 = 1$$

or
$$P_0 = 1 - \rho \quad \dots(7)$$

$\therefore P_n = \rho^n (1 - \rho) \quad \dots(8)$

which is the probability that there are n units in the system at any time.

Step three : (a) To find Probability (queue size $\geq N$)

Probability (queue size $\geq N$)

$$\begin{aligned} &= \sum_{n=N}^{\infty} P_n = \sum_{n=0}^{\infty} P_n - \sum_{n=0}^{N-1} P_n \\ &= 1 - [P_0 + P_1 + P_2 + \dots + P_{N-1}] \\ &= 1 - [1 + \rho + \rho^2 + \dots + \rho^{N-1}] P_0 \\ &= 1 - \frac{1 \cdot (1 - \rho^N)}{(1 - \rho)} \cdot (1 - \rho) = \rho^N \end{aligned}$$

or **Probability (queue size $\geq N$)** $= \rho^N = \left(\frac{\lambda}{\mu}\right)^N \quad \dots(9)$

(b) To find $E(L_s)$, expected line length (average number of customers) in the system.

We have,

$$\begin{aligned} E(L_s) &= \sum_{n=0}^{\infty} n P_n = \sum_{n=0}^{\infty} n \cdot (1 - \rho) \rho^n \\ &= (1 - \rho) \cdot \rho \sum_{n=1}^{\infty} n \cdot \rho^{n-1} \end{aligned}$$

$$\begin{aligned}
 &= \rho (1 - \rho) \cdot [1 + 2\rho + 3\rho^2 + \dots] \\
 &= \rho (1 - \rho) \cdot [1 - \rho]^{-2}
 \end{aligned}$$

$$\therefore E(L_s) = \frac{\rho}{1 - \rho} = \frac{\lambda}{\mu - \lambda} \quad \because \rho = \frac{\lambda}{\mu} \quad \dots(10)$$

To find the variance of queue length.

By definition, we have

$$\begin{aligned}
 \text{Var}(n) &= E(n^2) - [E(n)]^2 = \sum_{n=1}^{\infty} n^2 P_n - \left[\sum_{n=1}^{\infty} n P_n \right]^2 \\
 &= \sum_{n=1}^{\infty} n^2 (1 - \rho) \rho^n - \left(\frac{\rho}{1 - \rho} \right)^2 \quad \text{Since } \sum_{n=1}^{\infty} n P_n = E(L_s) = \frac{\rho}{1 - \rho} \\
 &= \rho (1 - \rho) \cdot [1 + 2^2 \rho + 3^2 \rho^2 + \dots] - \frac{\rho^2}{(1 - \rho)^2}
 \end{aligned}$$

$$\text{Now let } S = 1 + 2^2 \rho + 3^2 \rho^2 + \dots$$

Integrating both sides w.r.t. ρ from 0 to ρ , we have

$$\begin{aligned}
 \int_{\rho=0}^{\rho} S d\rho &= [\rho + 2 \rho^2 + 3 \rho^3 + \dots]_0^{\rho} \\
 &= \rho + 2 \rho^2 + 3 \rho^3 + \dots \\
 &= \rho (1 + 2 \rho + 3 \rho^2 + \dots) \\
 &= \rho (1 - \rho)^{-2}
 \end{aligned}$$

Now differentiating both sides w.r.t. ρ , we have

$$S = \frac{1}{(1 - \rho)^2} + \frac{2\rho}{(1 - \rho)^3} = \frac{1 + \rho}{(1 - \rho)^3}$$

$$\therefore \text{Var}(n) = \rho (1 - \rho) \cdot \frac{(1 + \rho)}{(1 - \rho)^3} - \frac{\rho^2}{(1 - \rho)^2}$$

$$\text{or } \text{Var}(n) = \frac{\rho}{(1 - \rho)^2} \quad \dots(11)$$

(c) To find $E(L_q)$, expected queue length (average number of customers in the queue or average length of waiting line).

We have,

$$E(L_q) = \sum_{n=1}^{\infty} (n - 1) P_n$$

(Since there are $(n - 1)$ units in the queue excluding one in service)

$$\begin{aligned}
&= \sum_{n=1}^{\infty} n P_n - \sum_{n=1}^{\infty} P_n \\
&= E(L_s) - \left[\sum_{n=0}^{\infty} P_n - P_0 \right] = E(L_s) - 1 + P_0 \\
&= \frac{\rho}{1-\rho} - 1 + (1-\rho)
\end{aligned}$$

or $E(L_q) = \frac{\rho^2}{1-\rho} = \frac{\lambda^2}{\mu(\mu-\lambda)}$... (12)

(d) To find $E(L/L > 0)$ expected length of non empty queue, i.e., average length of non-empty queue.

We have,

$$\begin{aligned}
E(L/L > 0) &= \frac{E(L_q)}{\text{prob. [an arrival has to wait, } L > 0]} \\
&= \frac{E(L_q)}{P[(n-1) > 0]} = \frac{E(L_q)}{P(n > 1)} \\
&= \frac{E(L_q)}{\sum_{n=2}^{\infty} P_n} = \frac{E(L_q)}{P_2 + P_3 + P_4 + \dots} = \frac{E(L_q)}{(\rho^2 + \rho^3 + \rho^4 + \dots) P_0} \\
&= \frac{\frac{\rho^2}{1-\rho}}{\frac{\rho^2}{1-\rho} \cdot (1-\rho)} = \frac{1}{1-\rho}
\end{aligned}$$

or $E(L/L > 0) = \frac{\rho}{1-\rho} = \frac{\mu}{\mu-\lambda}$... (13)

(e) To find probability density function of waiting time (excluding service) distribution.

At any time, let there be n units in the system, $(n-1)$ waiting and one in service.

If one unit arrive at this time, then it has to wait for time between ω and $\omega + d\omega$ only when $(n-1)$ waiting units are served in time ω and one unit in service is served in time $d\omega$.

Since servicing distribution is also Poisson.

$\therefore$ Probability $[(n-1)$ units waiting are served in time $\omega]$

$$= \frac{(\mu \omega)^{n-1} \cdot e^{-\mu \omega}}{(n-1)!}$$

and Probability [one unit in service is served in time $d\omega]$

$$= \mu d\omega$$

$$\begin{aligned}
 \therefore \quad \psi_n(\omega) d\omega &= \text{Probability that a new arrival is taken in to service after a time} \\
 &\quad \text{lying between } \omega \text{ and } \omega + d\omega . \\
 &= \text{Probability } [(n-1) \text{ units waiting are served in time } \omega] \\
 &\quad \times \text{Probability [one unit in service is served in time } d\omega] \\
 &= \frac{(\mu \omega)^{n-1} \cdot e^{-\mu \omega}}{(n-1)!} \cdot \mu d\omega
 \end{aligned}$$

Let W be the waiting time of a unit who has to wait s.t. $\omega \leq W \leq \omega + d\omega$.

Since the queue length can vary between 1 and ∞ , therefore, the probability density function of the waiting time is given by

$$\begin{aligned}
 \psi(\omega) d\omega &= \sum_{n=1}^{\infty} [\text{Probability that there are } n \text{ units in the system}] \times \psi_n(\omega) d\omega \\
 &= \sum_{n=1}^{\infty} P_n \cdot \psi_n(\omega) d\omega = \sum_{n=1}^{\infty} (1-\rho) \rho^n \cdot \frac{(\mu \omega)^{n-1} \cdot e^{-\mu \omega}}{(n-1)!} \cdot \mu d\omega \\
 &= \mu \rho (1-\rho) \cdot e^{-\mu \omega} \sum_{n=1}^{\infty} \frac{(p \mu \omega)^{n-1}}{(n-1)!} d\omega \\
 &= \mu \rho (1-\rho) e^{-\mu \omega} \cdot e^{\rho \mu \omega} d\omega \\
 &= \mu \rho (1-\rho) \cdot e^{-(1-\rho)\mu \omega} \cdot d\omega \\
 &= \mu \rho \left(1 - \frac{\lambda}{\mu}\right) \cdot e^{-(\mu - \lambda) \omega} \cdot d\omega
 \end{aligned}$$

$$\text{or } \psi(\omega) d\omega = \rho (\mu - \lambda) e^{-(\mu - \lambda) \omega} d\omega \quad \omega > 0 \quad \dots(14)$$

$$\text{Hence } \int_0^{\infty} \psi(\omega) d\omega = \rho = \frac{\lambda}{\mu} \neq 1.$$

This is because the case for which $\omega = 0$ is excluded.

$$\text{Probability } [W > 0] = \int_0^{\infty} \psi(\omega) d\omega = \rho$$

$$\begin{aligned}
 \text{Probability } [W = 0] &= \text{Probability [no unit in the system]} \\
 &= P_0 = 1 - \rho.
 \end{aligned}$$

Since the sum of these probabilities for waiting time is 1.

$\therefore$ Complete distribution of the waiting time is given by

(i) part continuous for $\omega \leq W \leq \omega + d\omega$

with probability density function $\psi(\omega)$ given by

$$\psi(\omega) d\omega = \mu \rho (1-\rho) \cdot e^{-(1-\rho)\mu \omega} \cdot d\omega$$

$$= \lambda \left(1 - \frac{\lambda}{\mu} \right) \cdot e^{-(\mu - \lambda) \omega} d\omega$$

and (ii) part discrete for $W = 0$

$$\text{with probability } (W = 0) = 1 - \rho = 1 - \frac{\lambda}{\mu}$$

Also Probability $(W > t_0)$ = the probability that an arrival has to wait at least for t_0 units of time

$$= \int_{t_0}^{\infty} \frac{\lambda}{\mu} (\mu - \lambda) e^{-(\mu - \lambda) \omega} d\omega = \frac{\lambda}{\mu} e^{-(\mu - \lambda) t_0}$$

(f) To find busy period distribution *i.e.*, $\psi(\omega/\omega > 0)$ (conditional density function for waiting time, given that a person has to wait).

$\psi(\omega/\omega > 0)$ = Probability density function for waiting time such that a person has to wait

$$= \frac{\psi(\omega)}{\text{Prob. } (W > 0)} = \frac{\psi(\omega)}{\int_0^{\infty} \psi(\omega) d\omega}$$

$$= \frac{\mu \rho (1 - \rho) \cdot e^{-(1 - \rho) \mu \omega}}{\rho}$$

$$\therefore \psi(\omega/\omega > 0) = \mu (1 - \rho) e^{-(1 - \rho) \mu \omega} = (\mu - \lambda) e^{-(\mu - \lambda) \omega} \quad \dots(15)$$

which gives the conditional probability density function for waiting time, given that a person has to wait (p.d.f. for busy period).

(g) To find $E(W_q)$ expected waiting time in the queue (excluded service time), *i.e.*, average waiting time of an arrival in the queue.

$$\text{We have } E(W_q) = \int_0^{\infty} \omega \psi(\omega) d\omega$$

$$= \int_0^{\infty} \omega \mu \rho (1 - \rho) e^{-(1 - \rho) \mu \omega} d\omega$$

$$= \mu \rho (1 - \rho) \cdot \int_0^{\infty} \omega \cdot e^{-(1 - \rho) \mu \omega} d\omega$$

Integrating by parts taking ω as first function

$$= \mu \rho (1 - \rho) \left[\frac{\omega \cdot e^{-(1 - \rho) \mu \omega}}{-(1 - \rho) \mu} - \frac{e^{-(1 - \rho) \mu \omega}}{(1 - \rho)^2 \mu^2} \right]_0^{\infty}$$

$$= \mu \rho (1 - \rho) \left[- \lim_{\omega \rightarrow \infty} \frac{\omega}{\mu (1 - \rho) e^{(1 - \rho) \mu \omega}} + \frac{1}{\mu^2 (1 - \rho^2)} \right]$$

$$= \mu \rho (1 - \rho) \left[0 + \frac{1}{\mu^2 (1 - \rho)^2} \right] = \frac{\rho}{\mu (1 - \rho)}$$

$$\therefore E(W_q) = \frac{\rho}{\mu(1-\rho)} = \frac{\lambda}{\mu(\mu-\lambda)}. \quad \dots(16)$$

(h) To find probability density function of the time that an arriving unit has to spend in the system (waiting time + service time).

Let $\theta_n(v)$ be the probability that an arriving unit spends time v in the system (waiting + service), when there are already n units in the system at the time of its arrival.

$\therefore$ as in (e), we have

$$\begin{aligned} \theta_n(v) &= \text{Probability [that } n \text{ units are served in the time } v] \\ &\quad \times \text{Probability [one unit is served in time } dv] \\ &= \frac{(\mu v)^n \cdot e^{-\mu v}}{n!} \cdot \mu dv, \quad n=0, 1, 2, 3, \dots, \infty. \end{aligned}$$

$$\begin{aligned} \therefore \theta(v) dv &= \sum_{n=0}^{\infty} [\text{Probability that there are } n \text{ units in the system}] \\ &\quad \times [\text{Probability that new arrival spends time } v \text{ in the system}] \\ &= \sum_{n=0}^{\infty} P_n \cdot \theta_n(v) dv = \sum_{n=0}^{\infty} (1-\rho) \rho^n \cdot \frac{(\mu v)^n e^{-\mu v}}{n!} \cdot \mu dv \\ &= \mu(1-\rho) e^{-\mu v} \sum_{n=0}^{\infty} \frac{(\rho \mu v)^n}{n!} dv \\ &= \mu(1-\rho) e^{-\mu v} \cdot e^{\rho \mu v} dv \end{aligned}$$

$$\text{or} \quad \theta(v) dv = \mu(1-\rho) e^{-(1-\rho)\mu v} \cdot dv = (\mu - \lambda) e^{-(\mu-\lambda)v} dv \quad \dots(17)$$

(i) To find $E(W_s)$ expected waiting time in the system (including service time), i.e., average time that an arrival spends in the system.

$$\begin{aligned} \text{We have, } E(W_s) &= \int_0^{\infty} v \cdot \theta(v) dv \\ &= \int_0^{\infty} v \cdot \mu(1-\rho) e^{-(1-\rho)\mu v} \cdot dv \\ &= \mu(1-\rho) \int_0^{\infty} v \cdot e^{-(1-\rho)\mu v} dv = \frac{1}{\mu(1-\rho)}. \end{aligned}$$

[Integrating by parts taking v as first function as in (g)]

$$\therefore E(W_s) = \frac{1}{\mu(1-\rho)} = \frac{1}{\mu - \lambda} \quad \dots(18)$$

- (j) To find $E(W/W > 0)$ expected waiting time of a customer who has to wait, *i.e.*, average waiting time of an arrival who waits.

We have,

$$\begin{aligned} E(W/W > 0) &= \int_0^{\infty} \omega \cdot \psi(\omega/\omega > 0) d\omega \\ &= \int_0^{\infty} \omega \cdot \mu (1 - \rho) e^{-(1 - \rho)\mu \omega} d\omega \\ &= \mu (1 - \rho) \int_0^{\infty} \omega e^{-(1 - \rho)\mu \omega} d\omega = \frac{1}{\mu (1 - \rho)} \end{aligned}$$

$$\therefore E(W/W > 0) = \frac{1}{\mu (1 - \rho)} = \frac{1}{\mu - \lambda} \quad \dots(19)$$

- (k) To find the probability of arrivals during the service time of any given customer.

We know that arrivals and service times follows Poisson and exponential distribution laws.

$\therefore$ The probability of m arrivals during the service time of any customer is

$$\begin{aligned} &= \int_0^{\infty} P_m(t) \cdot f(t) dt = \int_0^{\infty} \frac{e^{-\lambda t} \cdot (\lambda t)^m}{m!} \mu e^{-\mu t} dt \\ &= \frac{\lambda^m \mu}{m!} \int_0^{\infty} e^{-(\lambda + \mu)t} t^m dt \\ &= \frac{\lambda^m \mu}{m!} \cdot \frac{\Gamma(m+1)}{(\lambda + \mu)^{m+1}} \quad \left[\text{Since } \int_0^{\infty} e^{-at} \cdot t^n dt = \frac{\Gamma(n+1)}{a^{n+1}} \right] \\ &= \left(\frac{\lambda}{\lambda + \mu} \right)^m \cdot \frac{\mu}{(\lambda + \mu)} \quad \dots(20) \end{aligned}$$

7.16 Little's Formulae Relationship between $E(L_s)$, $E(L_q)$, $E(W_s)$ and $E(W_q)$.

In a steady state queuing process, it can be proved that, under essentially general conditions

$$E(L_s) = \lambda \cdot E(W_s) \quad \dots(1)$$

and $E(L_q) = \lambda \cdot E(W_q) \quad \dots(2)$

Also it is obvious that

$$E(W_s) = E(W_q) + \frac{1}{\mu} \quad \dots(3)$$

These relations (1) to (3) are very important since they enable us to determine all the four fundamental quantities

$E(L_s)$, $E(L_q)$, $E(W_s)$ and $E(W_q)$ as soon as one of these is determined.

7.17 Model II (General Erlang Queueing Model)

This model is same as model I with the difference that here mean arrival rate and mean service rate are not constant but depend on n , the number of units present in the system. *i.e.*, Here

λ_n = mean arrival rate of customers when there are n units present in the system

and μ_n = mean service rate when there are n units in the system.

Step one : To find the system of steady state equations.

Proceeding similarly as in Model I, the probability of n units in the system at time $(t + \Delta t)$ is given by

$$\begin{aligned} P_n(t + \Delta t) &= P_{n-1}(t) \cdot \lambda_{n-1} \Delta t \cdot (1 - \mu_{n-1} \Delta t) \\ &\quad + P_n(t) \cdot (1 - \lambda_n \Delta t) \cdot (1 - \mu_n \Delta t) \\ &\quad + P_{n+1}(t) \cdot (1 - \lambda_{n+1} \Delta t) \mu_{n+1} \Delta t \end{aligned}$$

$$\begin{aligned} \text{or } P_n(t + \Delta t) &= [\lambda_{n-1} P_{n-1}(t) - (\lambda_n + \mu_n) P_n(t) \\ &\quad + \mu_{n+1} P_{n+1}(t)] \Delta t + P_n(t) + O(\Delta t) \end{aligned}$$

$$\begin{aligned} \therefore \frac{P_n(t + \Delta t) - P_n(t)}{\Delta t} &= \lambda_{n-1} P_{n-1}(t) - (\lambda_n + \mu_n) P_n(t) \\ &\quad + \mu_{n+1} P_{n+1}(t) + \frac{O(\Delta t)}{\Delta t} \end{aligned} \quad \dots(1)$$

for $n > 0$

and probability of no unit in the system at time $(t + \Delta t)$ is given by

$$P_0(t + \Delta t) = P_0(t) \cdot (1 - \lambda_0 \Delta t) + P_1(t) \cdot (1 - \lambda_1 \Delta t) \cdot \mu_1 \Delta t$$

$$\text{or } P_0(t + \Delta t) = [-\lambda_0 P_0(t) + \mu_1 P_1(t)] \Delta t + P_0(t) + O(\Delta t)$$

$$\begin{aligned} \therefore \frac{P_0(t + \Delta t) - P_0(t)}{\Delta t} &= -\lambda_0 P_0(t) + \mu_1 P_1(t) + \frac{O(\Delta t)}{\Delta t} \end{aligned} \quad \dots(2)$$

for $n = 0$

Taking limit as $\Delta t \rightarrow 0$, from (1) and (2), we have

$$\begin{aligned} \frac{d}{dt} P_n(t) &= \lambda_{n-1} P_{n-1}(t) - (\lambda_n + \mu_n) P_n(t) + \mu_{n+1} P_{n+1}(t) \end{aligned} \quad \dots(3)$$

for $n > 0$

and
$$\frac{d}{dt} P_0(t) = -\lambda_0 P_0(t) + \mu_1 P_1(t) \quad \dots(4)$$

When the system reaches steady state

$$\lim_{t \rightarrow \infty} P_n(t) = P_n \text{ and } \lim_{t \rightarrow \infty} P'_n(t) = 0$$

$\therefore$ Under steady state of the system, the equations (3) and (4) reduce to

$$\lambda_{n-1} P_{n-1} - (\lambda_n + \mu_n) P_n + \mu_{n+1} P_{n+1} = 0 \quad \dots(5)$$

for $n > 0$

and
$$-\lambda_0 P_0 + \mu_1 P_1 = 0 \quad \dots(6)$$

for $n = 0$

These equations (5) and (6) are the steady state equations of the system.

Step two : To solve the steady state equations (5) and (6) obtained in step one.

From (6),

$$P_1 = \frac{\lambda_0}{\mu_1} P_0$$

Putting $n=1$, in (5)

$$\begin{aligned} P_2 &= \frac{\lambda_0}{\mu_2} P_0 + \frac{(\lambda_1 + \mu_1)}{\mu_2} P_1 = -\frac{\mu_1}{\mu_2} P_1 + \left(\frac{\lambda_1}{\mu_2} + \frac{\mu_1}{\mu_2} \right) P_1 \\ &= \frac{\lambda_1}{\mu_2} P_1 = \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2} P_0 \end{aligned}$$

Putting $n=2$, in (5),

$$\begin{aligned} P_3 &= -\frac{\lambda_1}{\mu_3} P_1 + \left(\frac{\lambda_2 + \mu_2}{\mu_3} \right) P_2 = -\frac{\mu_2}{\mu_3} \cdot P_2 + \left(\frac{\lambda_2}{\mu_3} + \frac{\mu_2}{\mu_3} \right) P_2 \\ &= \frac{\lambda_2}{\mu_3} P_2 = \frac{\lambda_0 \lambda_1 \lambda_2}{\mu_1 \mu_2 \mu_3} P_0 \end{aligned}$$

Proceeding in this way, we have

$$P_n = \frac{\lambda_0 \lambda_1 \lambda_2 \dots \lambda_{n-1}}{\mu_1 \mu_2 \mu_3 \dots \mu_n} \cdot P_0 \quad \dots(7)$$

But
$$\sum_{n=0}^{\infty} P_n = 1$$

or
$$P_0 + P_1 + P_2 + P_3 + \dots = 1$$

or
$$\left[1 + \frac{\lambda_0}{\mu_1} + \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2} + \dots \right] P_0 = 1$$

$$\text{or } P_0 = \frac{1}{S} \quad \dots (8)$$

$$\text{where } S = 1 + \frac{\lambda_0}{\mu_1} + \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2} + \dots \quad \dots (9)$$

If S is a divergent series then $P_0 = 0$, which is meaningless.

$\therefore$ we assume that S is convergent series.

Particular Cases : Now we shall discuss the following particular cases :

Case I: If $\lambda_n = \lambda$ and $\mu_n = \mu$ } (Both independent of n)

$$\begin{aligned} \text{then } S &= 1 + \frac{\lambda}{\mu} + \left(\frac{\lambda}{\mu}\right)^2 + \left(\frac{\lambda}{\mu}\right)^3 + \dots \\ &= \frac{1}{1 - \frac{\lambda}{\mu}} = \frac{1}{1 - \rho} \quad \text{where } \frac{\lambda}{\mu} = \rho < 1. \end{aligned}$$

$$\therefore P_0 = (1 - \rho) \text{ and so, } P_n = \left(\frac{\lambda}{\mu}\right)^n P_0 = \rho^n (1 - \rho).$$

These correspond to the results of Model I.

Note: In this case model II is the same as model I.

Case II : If $\lambda_n = \frac{\lambda}{n+1}$ and $\mu_n = \mu$ (constant).

In this case λ_n decreases as n increases, that is arrival rate decreases with the increase in queue length, and the rate of service $\mu_n = \mu$ is independent of n . This is called the case of "Queue with discouragement."

Substituting the values of λ_n ($n=0, 1, 2, \dots$) and μ_n in (9), we have

$$\begin{aligned} S &= 1 + \frac{\lambda}{\mu} + \frac{\lambda^2}{1 \cdot 2 \mu^2} + \frac{\lambda^3}{2 \cdot 3 \mu^3} + \dots \\ &= 1 + \rho + \frac{\rho^2}{2!} + \frac{\rho^3}{3!} + \dots \\ &= e^\rho, \text{ which is always convergent for } \rho \text{ to be finite.} \end{aligned} \quad \dots (10)$$

$\therefore$ From (8),

$$\begin{aligned} P_0 &= (1/S) = e^{-\rho} \text{ and } P_1 = \frac{\lambda_0}{\mu_1} P_0 = \rho e^{-\rho} \\ P_2 &= \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2} P_0 = \frac{\rho^2}{2!} e^{-\rho} \end{aligned}$$

.....

.....

$$P_n = \frac{\rho^n}{n!} e^{-\rho} \quad \dots (11)$$

Which implies that P_n follows the Poisson distribution.

Case III : If $\lambda_n = \lambda$ (constant) and $\mu_n = n\mu$.

In this case arrival rate $\lambda_n = \lambda$ is independent of n while the service rate $\mu_n = n\mu$ increases with the increase of n . This is known as a problem of “*queue with infinite number of channels*” i.e., if n customers arrive then n service stations will be available for all $n = 1, 2, \dots, \infty$. Thus, each arrival will enter the service facility without waiting and so no queue will be formed.

Substituting the values of $\lambda_n = \lambda$ and $\mu_n = n\mu$ ($n = 1, 2, \dots$) in (9), we have

$$S = 1 + \frac{\lambda}{\mu} + \frac{\lambda^2}{1 \cdot 2 \mu^2} + \frac{\lambda^3}{1 \cdot 2 \cdot 3 \mu^3} + \dots = 1 + \rho + \frac{\rho^2}{2!} + \frac{\rho^3}{3!} + \dots = e^\rho$$

$$\therefore P_0 = 1/S = e^{-\rho} \quad \dots(12)$$

$$\text{and } P_1 = \rho e^{-\rho}$$

$$P_2 = \frac{\rho^2}{2!} e^{-\rho}$$

.....

.....

$$P_n = \frac{\rho^n}{n!} \cdot e^{-\rho} \quad \dots(13)$$

Which implies that P_n follows the Poisson distribution in this case also.

Step three : In case II or III

$$P_n = \frac{\rho^n \cdot e^{-\rho}}{n!}.$$

(a) To find $E(L_s)$ expected line length (average number of customers) in the system.

We have,

$$\begin{aligned} E(L_s) &= \sum_{n=0}^{\infty} n P_n = \sum_{n=0}^{\infty} n \cdot \frac{\rho^n e^{-\rho}}{n!} \\ &= \rho \cdot e^{-\rho} \sum_{n=0}^{\infty} \frac{\rho^{n-1}}{(n-1)!} = \rho e^{-\rho} \cdot e^\rho = \rho = \frac{\lambda}{\mu} \quad \because \sum_{n=1}^{\infty} \frac{\rho^{n-1}}{(n-1)!} = e^\rho \end{aligned}$$

(b) Now with the help of relations (1), (2) and (3) of article 7.15, $E(L_q)$, $E(W_s)$ and $E(W_q)$ may be obtained.

ILLUSTRATIVE EXAMPLES

Examples on Model I

Example 1: Customers arrive at a sales counter manned by a single person according to a Poisson process with a mean rate of 20 per hour. The time required to serve a customer has an exponential distribution with a mean of 100 seconds. Find the average waiting time of a customer and queue length.

Solution: It is $(M/M/1) : (\infty/FCFS)$ problem.

Here, mean arrival rate $\lambda = 20$ per hour

and mean service rate $\mu = \frac{60 \times 60}{100} = 36$ per hour

$\therefore$ The average waiting time to a customer in the queue

$$= E(W_q) = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{20}{36(36 - 20)} \text{ hours} = 125 \text{ seconds}$$

The average waiting time of a customer in the system

$$= E(W_s) = \frac{1}{\mu - \lambda} = \frac{1}{36 - 20} \text{ hours} = 225 \text{ seconds}$$

And average queue length $= E(L_q) = \frac{\rho^2}{1 - \rho} = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{25}{36} = 0.7$ (appr.) customers.

Example 2: Arrivals at a telephone booth are considered to be Poisson, with an average time of 10 minutes between one arrival and the next. The length of a phone call assumed to be distributed exponentially, with mean 3 minutes. Find the followings :

- What is the probability that a person arriving at the booth will have to wait ?
- What is the average length of the queues that form from time to time ?
- The telephone company will install a second booth when convinced that an arrival would expect to have to wait at least three minutes for the phone. By how much must the flow of arrivals be increased in order to justify a second booth ?
- Find the average number of units in the system.
- What is the probability that it will take, an arrival, more than 10 minutes altogether to wait for the phone and complete his call ?
- What is the probability that an arrival will have to wait more than 10 minutes before the phone is free ?
- Estimate the fraction of a day that the phone will be in use.

Solution : It is $(M/M/1) : (\infty/FCFS)$ problem.

Here, mean arrival rate $\lambda = 1/10$,

and mean service rate $\mu = 1/3$.

- (a) Probability (arrival will have to wait)

$$= 1 - P_0 = \lambda/\mu = 3/10 = 0.3$$

- (b) Average length of the queue $= E(L/L > 0)$

$$= \frac{\mu}{\mu - \lambda} = \frac{1/3}{(1/3) - (1/10)} = \frac{10}{7} = 1.43 \text{ customers}$$

- (c) Average waiting time of an arrival in the queue

$$= E(W_q) = \frac{\lambda}{\mu(\mu - \lambda)}$$

Here $E(W_q) = 3, \mu = 1/3$, to find mean arrival rate $\lambda = \lambda'$ (say)

$$\therefore 3 = \frac{\lambda'}{(1/3) [(1/3) - \lambda']}$$

$$\text{or } 1/3 - \lambda' = \lambda' \quad \text{or } \lambda' = 1/6$$

Increase in mean arrival rate $= \lambda' - \lambda = \frac{1}{6} - \frac{1}{10} = \frac{1}{15} = 0.067$ arrivals per minute.

$$\therefore \text{Increase in the flow of arrivals} = \frac{1}{15} \times 60 = 4 \text{ per hour.}$$

$\therefore$ The second booth is justified if the increase in arrival rate is 0.067 customers per minute (or 4 customers per hour).

(d) Average number of units in the system

$$E(L_s) = \frac{\lambda}{\mu - \lambda} = \frac{(1/10)}{(1/3) - (1/10)} = \frac{3}{7} = 0.43 \text{ customers.}$$

(e) Probability [time that a unit spend in the system ≥ 10]

$$\begin{aligned} &= \int_{10}^{\infty} \theta(v) dv \\ &= \int_{10}^{\infty} (\mu - \lambda) e^{-(\mu - \lambda)v} dv \text{ from equation (19) of article 7.14} \\ &= \left[-e^{-(\mu - \lambda)v} \right]_{10}^{\infty} = e^{-10(\mu - \lambda)} \\ &= e^{-10(1/3 - 1/10)} = e^{-(7/3)} = e^{-2.33} = 0.10 \end{aligned}$$

(f) Probability [Waiting time of an arrival in queue ≥ 10]

$$\begin{aligned} &= \int_{10}^{\infty} \psi(\omega) d\omega \\ &= \int_{10}^{\infty} \frac{\lambda}{\mu} (\mu - \lambda) e^{-(\mu - \lambda)\omega} d\omega \\ &= \frac{\lambda}{\mu} e^{-10(\mu - \lambda)} = \frac{3}{10} \cdot e^{-10 \cdot [(1/3) - (1/10)]} \\ &= \frac{3}{10} \cdot e^{-2.33} = \frac{3}{10} \cdot (0.10) = 0.03 \end{aligned}$$

(g) Expected fraction of a day that the phone will be in use

= Expected fraction of any other time interval that the phone will be in use.

= Expected fraction of a small time interval Δt that the phone will be in use.

(in particular)

= Probability that the booth is occupied at random instant

$$= P[n > 0] = 1 - P_0 = 1 - \left(1 - \frac{\lambda}{\mu}\right) = \frac{\lambda}{\mu} = \frac{3}{10} = 0.3$$

Example 3: In a railway marshalling yard, goods trains arrive at a rate of 30 trains per day. Assuming that the inner-arrival time follows an exponential distribution and the service time distribution is also exponential with an average 36 minutes. Calculate the following :

(a) The mean queue size (line length)

and (b) The probability that the queue size exceeds 10.

Solution : It is $(M/M/1) : (\infty/FCFS)$ problem.

Here, mean arrival rate;

$$\lambda = \frac{30}{60 \times 24} = \frac{1}{48} \text{ trains/minute}$$

and mean service rate, $\mu = \frac{1}{36}$ trains/minute.

$$\rho = \frac{\lambda}{\mu} = \frac{3}{4}$$

(a) The mean queue size $= E(L_s) = \frac{\rho}{1 - \rho} = \frac{(3/4)}{1 - (3/4)} = 3$ trains

and (b) Probability [queue size ≥ 10] $= \rho^{10}$ [From article 7.15 equation (9)]

$$= \left(\frac{3}{4}\right)^{10} = 0.06$$

Example 4 : A TV repairman finds that the time spent on his jobs has an exponential distribution with mean 30 minutes. If he repairs sets in the order in which they came in, and if the arrival of sets is approximately Poisson with an average rate of 10 per 8 hour day, what is repairman's expected idle time each day ? How many jobs are ahead of the average set just brought?

Solution : It is $(M/M/1) : (FCFS)$ problem.

Here, $\lambda = \frac{10}{8 \times 60} = \frac{1}{48}$ sets/minute

and $\mu = 1/30$ sets/minute

Probability that there is no unit in the system

$$= P_0 = 1 - \lambda/\mu = 1 - 5/8 = 3/8$$

$\therefore$ Repairman's expected idle time in 8 hour day

$$= (3/8) \times 8 = 3 \text{ hours}$$

and expected (average) number of jobs

$$E(L_s) = \frac{\lambda}{\mu - \lambda} = \frac{2}{3} \text{ Jobs}$$

Example 5 : On average 96 patient per 24 hour day require the service of an emergency clinic. Also on average, a patient requires 10 minutes of active attention. Assume that the facility can handle only one emergency at a time. Suppose that it costs the clinic ₹100 per patient treated to obtain an average serving time of 10 patients, and that each minute of decrease in the average time would cost by ₹10 per patient treated. How much would have to be budgeted by the clinic to decrease the average size of the queue from one and one-third patients to half a patient ?

Solution : It is a $(M/M/1) : (\infty/FCFS)$ problem.

Here,
$$\lambda = \frac{96}{24 \times 60} = \frac{1}{15} \text{ patients/minute}$$

$$\mu = 1/10 \text{ patients/minute}$$

$\therefore \rho = \lambda / \mu = 2/3$

Expected (average) number of patients in the waiting line

$$= \frac{\rho^2}{1-\rho} = \frac{(2/3)^2}{1-(2/3)} = \frac{4}{3} = 1\frac{1}{3}.$$

Fraction of the time during which there are no patients $= 1 - \rho = 1/3$

Average size of the queue

$$= E(L_q) = \frac{\rho^2}{1-\rho} = \frac{\lambda^2}{\mu(\mu-\lambda)}.$$

Here, $E(L_q) = 1/2, \lambda = 1/15$

$\therefore \frac{1}{2} = \frac{(1/15)^2}{\mu [\mu - (1/15)]}$

$\Rightarrow \mu = 2/15 \text{ patients/minute}$

$\therefore$ average rate of treatment required

$$= \frac{1}{\mu} = \frac{15}{2} = 7\frac{1}{2} \text{ minutes.}$$

So the decrease in the average rate of treatment

$$= 10 - 7\frac{1}{2} = 2\frac{1}{2} \text{ minutes.}$$

$\therefore$ Budget per patient $= ₹ 100 + (5/2) \times 10 = ₹ 125$

$\therefore$ In order to get the required size of the queue the budget should be increased from ₹ 100 per patient to ₹ 125 per patient.

Example 6 : A fertilizer company distributes its products by trucks loaded at its only loading station. Both, company trucks and contractor's trucks are used for this purpose. It was found out that on an average every 5 minutes one truck arrived and the average loading time was 3 minutes. 40% of the trucks belong to the contractors. Making suitable assumptions determine :

- (i) The probability that a truck has to wait.
- (ii) The waiting time of a truck that waits.
- (iii) The expected waiting time of contractor's trucks per day.

Solution : The assumptions for solving the given problem are made as follows :

1. The arrival rate is randomly distributed according to Poisson distribution and mean arrival rate is λ .
2. The service time follows Exponential distribution and mean service rate is μ .
3. $\mu > \lambda$ (i.e., service rate is greater than arrival rate).
4. There is single service channel i.e., the trucks are loaded at one place only.
5. The number of trucks being loaded is infinite i.e., capacity of the system is infinite, and
6. The service discipline is *FCFS*.

Here, $\lambda = \frac{60}{5} = 12$ trucks/hour

$$\mu = \frac{60}{3} = 20 \text{ trucks/hour}$$

- (i) The probability that a truck has to wait

$$= 1 - P_0 = \rho = \frac{\lambda}{\mu} = \frac{12}{20} = 0.6$$

- (ii) The waiting time of a truck that waits

$$= \frac{1}{\mu - \lambda} = \frac{1}{20 - 12} = \frac{1}{8} \text{ hours} = 7.5 \text{ minutes}$$

- (iii) Expected number of contractor's trucks waiting per day

$$= 40\% \text{ of the number of trucks per day}$$

$$= \frac{40}{100} \times 12 \times 24 = 115.2$$

- $\therefore$ Expected waiting time of contractor's trucks per day

$$= \text{Expected number of contractors truck per day} \times \text{Expected waiting time of a truck}$$

$$= 115.2 \times \frac{\lambda}{\mu(\mu - \lambda)}$$

$$= 115.2 \times \frac{12}{20(20 - 12)} = 8.64 \text{ hours per day}$$

Example 7: Customers arrive at a one window drive in bank according to a Poisson distribution with mean 10 per hour. Service time per customer is exponential with mean 5 minutes. The car space in front of the window, including that for the service can accommodate maximum of 3 cars. Other cars can wait outside this space.

- (i) What is the probability that an arriving customer car drive directly to the space in front of the window ?
- (ii) What is the probability that an arrival will have to wait outside the indicated space ?
- (iii) How long is an arriving customer expected to wait before starting service ?

Solution : Here, $\lambda = 10$ customers/hour

$$\mu = 60/5 = 12 \text{ customers/hour}$$

$$\therefore \rho = \lambda/\mu = 10/12 = 5/6$$

$$\text{and } P_0 = 1 - \rho = 1/6$$

$$(i) \text{ Required probability} = P_0 + P_1 + P_2$$

$$= (1 + \rho + \rho^2) P_0 = (1 + 5/6 + 25/36) (1/6) = 0.42$$

$$(ii) \text{ Probability that an arrival will have to wait outside the indicated space}$$

$$= 1 - (P_0 + P_1 + P_2 + P_3)$$

$$= 1 - (P_0 + P_1 + P_2) - \rho^3 P_0$$

$$= 1 - 0.42 - (5/6)^3 \cdot (1/6) = 0.48$$

$$\text{Aliter. Required Prob.} = \text{Prob. (Queue size} \geq 4) = \sigma^4 = (5/6)^4 = 0.48$$

$$(iii) E(W_q) = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{10}{12(12 - 10)} = 0.417$$

Example 8: A warehouse has only one loading dock manned by a three persons crew. Trucks arrive at the loading dock at an average rate of 4 trucks per hour and the arrival rate is Poisson distributed. The loading of a truck takes 10 minutes on an average and can be assumed to be exponentially distributed. The operating cost of a truck is ₹ 20 per hour and the members of the loading crew are paid ₹ 6 each per hour. Would you advise the truck owner to add another crew of three persons ?

Solution: Here, mean arrival rate,

$$\lambda = 4 \text{ trucks per hour}$$

For present crew of 3 persons,

Mean service rate, $\mu = 60/10 = 6$ trucks per hour.

$$\therefore E(W_s) = \text{Average waiting time per truck, per hour}$$

$$= \frac{1}{\mu - \lambda} = \frac{1}{6 - 4} = \frac{1}{2} \text{ hour}$$

$$\therefore \text{Total cost per hour}$$

$$= \text{Loading crew cost} + \text{waiting time cost}$$

$$= ₹ 6 \times 3 + E(W_s) \times \text{number of trucks arrive in one hour}$$

$$\times \text{waiting time cost per hour}$$

$$= ₹ 18 + ₹ \{(1/2) \times 4 \times 20\} = ₹ 58.$$

After the addition of another crew of 3 persons.

Mean service rate $\mu = 2 \times 6 = 12$ trucks per hour.

$$\therefore E(W_s) = \frac{1}{\mu - \lambda} = \frac{1}{12 - 4} = \frac{1}{8} \text{ hour}$$

$$\begin{aligned} \therefore \text{Total cost per hour, (after the addition of another crew)} \\ &= \text{Loading crew cost} + \text{waiting time cost} \\ &= ₹ 6 \times 6 + ₹ \left\{ (1/8) \times 4 \times 20 \right\} = ₹ 46 . \end{aligned}$$

Since total cost per hour is the second case is less than that in the previous case, hence we advice the truck owner to add another crew of three persons.

Examples on Model II

Example 9 : A shipping company has single unloading berth with ships arriving in a Poisson fashion at an average rate of three per day. The unloading time distribution for a ship with n unloading crews is found to be exponential with average unloading time $\frac{1}{2n}$ days. The company has a large labour supply without regular working hours and to avoid long waiting lines the company has a policy of using as many unloading crews as there are ships waiting in line or being unloading. Under these conditions, find

- (i) The average number of unloading crews working at any time,
and (ii) Probability that more than four crews will be needed.

Solution : Here, the mean service rate is not constant as in model I.

Here $\lambda = 3$ ships/day
and mean service rate with one unloading crew $= \mu = 2$ ships/day

$$P_n = \frac{e^{-\lambda/\mu}}{n!} \left(\frac{\lambda}{\mu} \right)^n \quad (\text{From model II. Case III})$$

- (i) The expected number of crews working at any specified instant

$$= E(L_s) \quad (\text{According to the question})$$

$$\begin{aligned} &= \sum_{n=0}^{\infty} n P_n = \sum_{n=0}^{\infty} n \cdot \frac{e^{-\lambda/\mu}}{n!} \left(\frac{\lambda}{\mu} \right)^n \\ &= e^{-\lambda/\mu} \left(\frac{\lambda}{\mu} \right) \cdot \sum_{n=1}^{\infty} \frac{(\lambda/\mu)^{n-1}}{(n-1)!} \\ &= e^{-\lambda/\mu} \left(\frac{\lambda}{\mu} \right) e^{\lambda/\mu} = \frac{\lambda}{\mu} = \frac{3}{2} = 1.5 \text{ crews} \end{aligned}$$

- (ii) Probability that a ship entering service will require more than four crews.
= Probability that there are at least five ships in the system at any instant

$$= \sum_{n=5}^{\infty} P_n = \sum_{n=0}^{\infty} P_n - \sum_{n=0}^4 P_n$$

$$\begin{aligned}
&= 1 - (P_0 + P_1 + P_2 + P_3 + P_4) \\
&= 1 - \left[1 + \frac{\lambda}{\mu} + \frac{(\lambda/\mu)^2}{2!} + \frac{(\lambda/\mu)^3}{3!} + \frac{(\lambda/\mu)^4}{4!} \right] \cdot e^{-\lambda/\mu} \\
&= 1 - \left[1 + \frac{3}{2} + \frac{(3/2)^2}{2!} + \frac{(3/2)^3}{3!} + \frac{(3/2)^4}{4!} \right] \cdot e^{-3/2} \\
&= 0.019
\end{aligned}$$

Example 10 : Problems arrive at a computing centre in a Poisson fashion at an average rate of five per day. The rules of the computing centre are that man waiting to get his problem solved must aid the man whose problem is being solved. If the time to solve a problem with one man has an exponential distribution with mean time of 1/3 day and if the average solving time is inversely proportional to the number of peoples working on the problem, approximate the expected time in the centre of a person entering the line.

Solution : It is (M/M/1) : (∞ /FCFS) problem (general) queueing model.

Here, $\lambda = 5$ problems/day

and μ (mean service rate with one unsolved problem)

$= 3$ problems/day

and $P_n = \frac{e^{-\lambda/\mu}}{n!} \cdot \left(\frac{\lambda}{\mu} \right)^n$. (Case II of model II)

$\therefore$ Expected number of persons working on the problem at any instant

$$\begin{aligned}
&= E(L_s) = \sum_{n=0}^{\infty} n \cdot P_n \\
&= \sum_{n=0}^{\infty} n \cdot \frac{e^{-\lambda/\mu}}{n!} \cdot \left(\frac{\lambda}{\mu} \right)^n \\
&= e^{-\lambda/\mu} \cdot \left(\frac{\lambda}{\mu} \right) \sum_{n=1}^{\infty} \left\{ \frac{(\lambda/\mu)^{n-1}}{(n-1)!} \right\} \\
&= e^{-\lambda/\mu} \cdot \left(\frac{\lambda}{\mu} \right) \cdot e^{\lambda/\mu} = \frac{\lambda}{\mu} = \frac{5}{3} \text{ persons}
\end{aligned}$$

Average solving time (which is inversely proportional to the number of peoples working on the problem)

$= 1/5$ day/problems

$\therefore$ Expected time in the centre for a person entering the line.

$$= \frac{1}{5} \cdot E(L_s) = \frac{1}{5} \cdot \frac{5}{3} = \frac{1}{3} \text{ days} = 8 \text{ hours}$$

Exercise 7.1

1. In a public telephone booth, the arrivals are on the average 15 per hour. A call on the average takes 3 minutes. If there is just one phone:
 - (i) What is the expected number of callers in the booth at any time?
 - (ii) For what proportion of time is the booth expected to be idle?
2. At a one-man barber shop, customers arrive cutting time was exponentially distributed with an average hair cut taking 10 minutes. It is assumed that because of his excellent reputation, customers were always willing to wait. Calculate the following:
 - (i) Average number of customers in the shop and average number of customers waiting for a hair cut.
 - (ii) The percentage of time an arrival can walk right in without having to wait.
 - (iii) The percentage of customers who have to wait prior to getting into the barber's chair.

[Hint: $\lambda = 1/12$ cust./min, $\mu = 1/10$ cust./min, $\rho = 5/6$

$$(a) E(L_s) = \frac{\lambda}{\mu - \lambda} = 5, E(L_q) = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{26}{6}$$

$$(b) P_0 = 1 - \rho = 1/6. \quad \therefore \text{Req. \%} = (1/6) \times 100 = 16.7\%$$

$$(c) P(n > 0) = 1 - P_0 = 0.833, \quad \therefore \text{Req. \%} = 83.3\%$$

3. Consider a box office ticket window being manned by a single individual. Customers arrive to purchase tickets according to a Poisson input process with a mean rate of 30 per hour. The time required to serve a customer has an exponential distribution with a mean of 90 seconds. Find the following:
 - (i) Expected line length.
 - (ii) Expected queue length.
 - (iii) Expected waiting time in the system.
 - (iv) Expected waiting time in the queue.
4. In a bank, cheques are cashed at a single 'teller' counter. Customers arrive at the counter in a Position manner at an average rate of 30 customers per hour. The teller takes, on an average a minute and a half to cash cheque. The service time has been shown to be exponentially distributed.
 - (i) Calculate the percentage of time the teller is busy.
 - (ii) Calculate the average time a customer is expected to wait.

[Hint: $\lambda = 30$ customer/h, $\mu = 40$ customer/h, find

$$(i) 1 - P_0 = \rho \quad (ii) E(W_s) = 1/(\mu - \lambda)]$$

5. In a car manufacturing plant, a loading crane takes exactly 10 minutes to load a car into a wagon and again came back to position to load another car. If the arrival of cars a Poisson stream at an average of one every 20 minutes, calculate the average waiting time of a car.
6. Problems arrived at a computing centre in Poisson fashion with a mean arrival rate of 25 per hour. The average computing job requires 2 minutes of terminal time. Calculate the followings:
 - (i) The average number of problems for computer use.
 - (ii) The percent of times an arrival can walk right in without having to wait.
7. A child care slop dealing with children's requirements has one cashier who handles all customer's payments. The cashier takes on an average 4 minutes per customer. Customers come to cashier's area in random manner at an average rate of 10 per hour. The management received a large number of customer's complaints and decided to investigate the following questions:
 - (i) What is the average length of waiting line?
 - (ii) What proportion of time the cashier is idle?
 - (iii) What is the expected waiting time of a customer?
8. A repair shop attended by a single mechanic has an average of four customers per hour who bring small appliances for repair. The mechanic inspects them for defects and quite often can fix them right away or otherwise render a diagnosis. This takes him six minutes on the average. Arrivals are Poisson and service time has the exponential distribution. Find:
 - (i) The proportion of time during which the shop is empty.
 - (ii) Probability of finding at least one customer in the shop.
 - (iii) The average number of customers in the system.
 - (iv) The average time spent including service.
9. Customers arrive at a first class ticket counter of a theatre in a Poisson distributed arrival rate of 25 per hour. Service time is constant at 2 minutes. Calculate:
 - (i) The mean number in the waiting line.
 - (ii) The mean waiting time.

[Hint: $\lambda = 25/60 = 5/12$ customers/min, $\mu = 1/2$ customer/min, $\rho = 5/6$]

 - (a) $E(L_s) = 5$ customers, $E(L_q) = 25/6 = 4$ customers (appr.).
 - (b) $E(W_q) = \frac{\lambda}{\mu(\mu - \lambda)} = 10$ min, $E(W_s) = \frac{1}{\mu - \lambda} = 12$ min]
10. In a Bank, every 15 minutes one customer arrives for cashing the cheque. The staff in the payment counter takes 10 minutes for serving a customer on an average. find:
 - (i) The average queue length.
 - (ii) The average waiting time.

and (iii) The increase in the arrival rate in order to justify a second counter (when the waiting time of a customer is at least 25 minutes the management will increase one more counter).

11. At a one man barber shop, customers arrive according to the Poisson distribution with a mean arrival rate of 15 per hour and his hair cutting takes exactly 5/2 minutes for one hair cut, how long on the average a customer must wait for the service?
12. A barber with one man shop takes exactly 40 minutes to complete one haircut. If the inter arrival time of the customer follows an exponential distribution with average one in every 50 minutes, how long a customer must wait for service?
13. In a Bhawan cafeteria, it was observed that there is only one bearer who takes exactly 4 minutes to serve a cup of coffee once the order has been placed with him. If the students arrive in the cafeteria at an average rate of 10 per hour, how much time one is expected to spend waiting for his turn to place the order?

Answers 7.1

1.	(i) 3, (ii) 1/4	2.	(i) 5, 4.17%, (ii) 61.7%, (iii) 83.3%	3.	(i) 3, (ii) 2.25, (iii) 6 min- utes/customer, (iv) 4.5 min- utes/customer.
4.	1.33	5.	1/6 hrs	6.	(i) 5, (ii) 16.67%
7.	(i) 4/3, (ii) 33%, (iii) 2/15 hrs.	8.	(i) 0.6, (ii) 0.4, (iii) 2/3, (iv) 1/6 hrs.	9.	(i) $E(L_s) = 5, E(L_q) = 4$ (appr) (ii) $E(W_s) = 12$ minutes, $E(W_q) = 15$ minutes
10.	(i) 1.33 units (ii) 20 minutes (iii) 2/7 customers per hours	11.	25/6 minutes	12.	2 hour and 40 minutes
13.	8 minutes				

7.18 Model III (M/M/1):(N/FCFS)

This is the queueing model with Poisson arrival. Poisson service, single channel, capacity of the system equal to N (fixed) and the service discipline is first come first served.

Step one : To find the system of steady state equations.

In this model, $\lambda_n = \begin{cases} \lambda & \text{when } n < N \text{ i.e., } (n = 0, 1, 2, \dots, N-1) \\ 0 & \text{when } n \geq N. \end{cases}$

and $\mu_n = \mu, n = 1, 2, \dots$

Proceeding similarly as in Model I, the probabilities of 0 and n units in the system at time $(t + \Delta t)$ are given by

$$P_0(t + \Delta t) = P_0(t) \cdot (1 - \lambda \Delta t) + P_1(t) \cdot (1 - \lambda \Delta t) \cdot \mu \Delta t$$

For $n = 0$

$$\text{or } P_0(t + \Delta t) - P_0(t) = [-\lambda P_0(t) + \mu P_1(t)] \Delta t + O(\Delta t) \quad \dots(1)$$

For $n = 0$

$$\begin{aligned} \text{and } P_n(t + \Delta t) &= P_{n-1}(t) \cdot \lambda_{n-1} \Delta t (1 - \mu \Delta t) + P_n(t) \cdot (1 - \lambda_n \Delta t) \\ &\quad \times (1 - \mu \Delta t) + P_{n+1}(t) \cdot (1 - \lambda_{n+1} \Delta t) \mu \Delta t \end{aligned}$$

For $n < N$

$$\begin{aligned} \text{or } P_n(t + \Delta t) - P_n(t) &= [\lambda_{n-1} P_{n-1}(t) - (\lambda_n + \mu) P_n(t) \\ &\quad + \mu P_{n+1}(t)] \Delta t + O(\Delta t) \end{aligned} \quad \dots(2)$$

When $n < N, \lambda_{n-1} = \lambda = \lambda_n$

$\therefore$ From (2), we have

$$\begin{aligned} P_n(t + \Delta t) - P_n(t) &= [\lambda P_{n-1}(t) - (\lambda + \mu) P_n(t) + \mu P_{n+1}(t)] \Delta t + O(\Delta t) \\ \text{For } n < N \end{aligned} \quad \dots(3)$$

When $n = N$,

$$P_{N+1}(t) = 0, \lambda_{N-1} = \lambda, \lambda_N = 0$$

$\therefore$ From (2), we have

$$\begin{aligned} P_N(t + \Delta t) - P_N(t) &= [\lambda P_{N-1}(t) - \mu P_N(t)] \Delta t + O(\Delta t) \\ \text{For } n = N. \end{aligned} \quad \dots(4)$$

Dividing equation (1), (3) and (4) by Δt and taking limit as $\Delta t \rightarrow 0$, we have

$$\frac{d}{dt} P_0(t) = -\lambda P_0(t) + \mu P_1(t) \quad \text{for } n = 0$$

$$\frac{d}{dt} P_n(t) = \lambda P_{n-1}(t) - (\lambda + \mu) P_n(t) + \mu P_{n+1}(t) \quad \text{for } 1 \leq n < N$$

$$\frac{d}{dt} P_N(t) = \lambda P_{N-1}(t) - \mu P_N(t) \quad \text{for } n = N$$

Under steady state of the system these equations reduce to

$$-\lambda P_0 + \mu P_1 = 0, \quad (\text{for } n = 0) \quad \dots(5)$$

$$\begin{aligned} \lambda P_{n-1} - (\lambda + \mu) P_n + \mu P_{n+1} &= 0 \\ \text{for } 1 \leq n < N \end{aligned} \quad \dots(6)$$

$$\text{and } \lambda P_{N-1} - \mu P_N = 0 \quad (\text{for } n = N) \quad \dots(7)$$

which are the steady state equations of the system.

Step two : To solve the steady state equation (5), (6) and (7) obtained in step one.

From (5), we have

$$P_1 = \frac{\lambda}{\mu} P_0 = \rho P_0$$

Putting $n = 1$ in (6),

$$P_2 = -\frac{\lambda}{\mu} P_0 + \left(1 + \frac{\lambda}{\mu}\right) P_1 = \frac{\lambda}{\mu} P_1 = \left(\frac{\lambda}{\mu}\right)^2 P_0 = \rho^2 P_0$$

Putting $n = 2$ in (6),

$$P_3 = -\frac{\lambda}{\mu} P_1 + \left(1 + \frac{\lambda}{\mu}\right) P_2 = \left(\frac{\lambda}{\mu}\right) P_2 = \left(\frac{\lambda}{\mu}\right)^3 P_0 = \rho^3 P_0$$

.....
.....

Proceeding in this way, we have

$$P_n = \left(\frac{\lambda}{\mu}\right)^n P_0 = \rho^n \cdot P_0 \quad \dots(8)$$

$1 \leq n < N$

From (7), $P_N = \frac{\lambda}{\mu} P_{N-1} = \left(\frac{\lambda}{\mu}\right) \cdot \left(\frac{\lambda}{\mu}\right)^{N-1} P_0$ [from (8)]

or $P_N = \left(\frac{\lambda}{\mu}\right)^N P_0 = \rho^N P_0 \quad \dots(9)$

But since the capacity of the system is N

$$\therefore \sum_{n=0}^N P_n = 1$$

or $P_0 + P_1 + P_2 + \dots + P_N = 1$

or $(1 + \rho + \rho^2 + \dots + \rho^N) P_0 = 1$

or $\frac{1 - \rho^{N+1}}{1 - \rho} \cdot P_0 = 1$

$$\therefore P_0 = \frac{1 - \rho}{1 - \rho^{N+1}} \quad \dots(10)$$

$\therefore$ From (8) and (9), we have,

$$P_n = \left(\frac{1 - \rho}{1 - \rho^{N+1}} \right) \rho^n \quad \dots(11)$$

$$0 \leq n \leq N$$

Note: Here we do not require the condition that $\rho < 1$ i.e., the result holds even if $\rho > 1$.

Step three :

(a) To find $E(L_s)$, expected line length (average number of customers in the system).

We have,

$$\begin{aligned} E(L_s) &= \sum_{n=0}^N n P_n = \sum_{n=0}^N n \left(\frac{1-\rho}{1-\rho^{N+1}} \right) \rho^n \\ &= \frac{1-\rho}{1-\rho^{N+1}} \cdot \sum_{n=0}^N n \rho^n \\ &= \frac{1-\rho}{1-\rho^{N+1}} \cdot [\rho + 2\rho^2 + 3\rho^3 + \dots + N\rho^N] \end{aligned} \quad \dots (12)$$

Now let $S = \rho + 2\rho^2 + 3\rho^3 + \dots + N\rho^N$

$$\therefore \rho S = \rho^2 + 2\rho^3 + \dots + (N-1)\rho^N + N\rho^{N+1}$$

Substituting we have

$$\begin{aligned} (1-\rho) S &= \rho + \rho^2 + \rho^3 + \dots + \rho^N - N\rho^{N+1} \\ &= \frac{\rho(1-\rho^N)}{1-\rho} - N\rho^{N+1} \\ \therefore S &= \frac{\rho(1-\rho^N)}{(1-\rho)^2} - \frac{N\rho^{N+1}}{(1-\rho)} = \frac{\rho(1-\rho^N - N\rho^N + N\rho^{N+1})}{(1-\rho)^2} \end{aligned}$$

Substituting in (12), we have

$$E(L_s) = \frac{\rho [1 - (1+N)\rho^N + N\rho^{N+1}]}{(1-\rho)(1-\rho^{N+1})} \quad \dots (13)$$

(b) Now with the help of relations (1), (2) and (3) of article 7.16, $E(L_q)$, $E(W_s)$ and $E(W_q)$ may be obtained.

7.19 Model IV (M/M/S):(∞ /FCFS)

This is the queueing model with Poisson arrival, Poisson service. There are $S > 1$ channels, service rate at each channel is the same (μ). The service discipline is first come first serve. In this model the length of the waiting line will depend on the number of occupied channels.

If $n < S$, then there will be no customer waiting in queue as all of them will be served simultaneously and in this case $S - n$ service channels will remain idle. Here the rate of service $\mu_n = n\mu$.

If $n = S$, then all the service channels will be busy and the rate of service $\mu_n = S\mu$.

If $n > S$, then all the service channels will be busy, while $n - S$ customers will be waiting in queue and the rate of service $\mu_n = S\mu$.

Step one : To find the system of steady state equations :

In this model the arrival and service rates are as follows :

$$\lambda_n = \lambda \quad \text{for } n = 0, 1, 2, \dots$$

$$\text{and} \quad \mu_n = \begin{cases} n\mu & 0 \leq n < S \\ S\mu & n \geq S \end{cases}$$

Proceeding similarly as in model I, the probabilities of 0 and n units (customers) in the system at time $(t + \Delta t)$ are given by

$$P_0(t + \Delta t) = P_0(t) (1 - \lambda \Delta t) + P_1(t) \cdot (1 - \lambda \Delta t) \mu_1 \Delta t$$

$$\text{or} \quad P_0(t + \Delta t) - P_0(t) = [-\lambda P_0(t) + \mu_1 P_1(t)] \Delta t + O(\Delta t) \quad \dots(1)$$

$$\text{and} \quad P_n(t + \Delta t) = P_{n-1}(t) \cdot \lambda \Delta t \{1 - \mu_{n-1} \Delta t\} + P_n(t) (1 - \lambda \Delta t) \cdot \{1 - \mu_n \Delta t\} + P_{n+1}(t) \cdot (1 - \lambda \Delta t) \mu_{n+1} \Delta t$$

$$\text{or} \quad P_n(t + \Delta t) - P_n(t) = [\lambda P_{n-1}(t) - (\lambda + \mu_n) P_n(t) + \mu_{n+1} P_{n+1}(t)] \Delta t + O(\Delta t) \quad \dots(2)$$

When $1 \leq n < S$, $\mu_n = n\mu$, $\mu_{n+1} = (n+1)\mu$

$\therefore$ from (2), we have

$$P_n(t + \Delta t) - P_n(t) = [\lambda P_{n-1}(t) - (\lambda + n\mu) P_n(t) + (n+1)\mu P_{n+1}(t)] \Delta t + O(\Delta t) \quad \dots(3)$$

for $1 \leq n < S$

When $n \geq S$, $\mu_n = S\mu = \mu_{n+1}$

$\therefore$ from (2), we have

$$\text{or} \quad P_n(t + \Delta t) - P_n(t) = [\lambda P_{n-1}(t) - (\lambda + S\mu) P_n(t) + S\mu P_{n+1}(t)] \Delta t + O(\Delta t) \quad \dots(4)$$

for $n \geq S$

Dividing equations (1), (3) and (4) by Δt and taking limit as $\Delta t \rightarrow 0$, we have

$$\frac{d}{dt} P_0(t) = -\lambda P_0(t) + \mu P_1(t) \quad \text{for } n = 0$$

$$\frac{d}{dt} P_n(t) = \lambda P_{n-1}(t) - (\lambda + n\mu) P_n(t) + (n+1)\mu P_{n+1}(t),$$

$$\text{for } 1 \leq n < S$$

and
$$\frac{d}{dt} P_n(t) = \lambda P_{n-1}(t) - (\lambda + S\mu) P_n(t) + S\mu P_{n+1}(t) \quad \text{for } n \geq S$$

Under steady state of the system these equations reduce to

$$-\lambda P_0 + \mu P_1 = 0 \quad \text{for } n=0 \quad \dots(5)$$

$$\lambda P_{n-1} - (\lambda + n\mu) P_n + (n+1)\mu \cdot P_{n+1} = 0 \quad \dots(6)$$

for $1 \leq n < S$

and
$$\lambda P_{n-1} - (\lambda + S\mu) P_n + S\mu \cdot P_{n+1} = 0 \quad \dots(7)$$

for $n \geq S$

which are the steady state equations of the system.

Step two : To solve the steady state equations obtained in step one.

From (5), $P_1 = \frac{\lambda}{\mu} P_0$

Putting $n=1$ in (6),

$$P_2 = -\frac{\lambda}{2\mu} P_0 + \frac{(\lambda + \mu)}{2\mu} P_1 = \frac{\lambda}{2\mu} P_1 = \frac{1}{2!} \left(\frac{\lambda}{\mu} \right)^2 P_0$$

Putting $n=2$ in (6),

$$P_3 = \frac{\lambda}{3\mu} P_1 + \left(\frac{\lambda}{3\mu} + \frac{2}{3} \right) P_2 = \frac{\lambda}{3\mu} P_2 = \frac{1}{3!} \left(\frac{\lambda}{\mu} \right)^3 \cdot P_0$$

... ..
... ..

Proceeding in this way, we have

$$P_n = \frac{\lambda}{n\mu} P_{n-1} = \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n \cdot P_0 \quad \text{for } 0 \leq n < S$$

Putting $n=S-1$ in (6)

$$P_S = \frac{\lambda}{S\mu} \cdot P_{S-1} = \frac{1}{S!} \left(\frac{\lambda}{\mu} \right)^S \cdot P_0$$

Putting $n=S$ in (7),

$$P_{S+1} = -\frac{\lambda}{S\mu} P_{S-1} + \left(\frac{\lambda}{S\mu} + 1 \right) P_S = \frac{\lambda}{S\mu} \cdot P_S = \frac{1}{S \cdot S!} \left(\frac{\lambda}{\mu} \right)^{S+1} \cdot P_0$$

Putting $n=S+1$ in (7),

$$P_{S+2} = \frac{\lambda}{S\mu} P_{S+1} = \frac{1}{S^2 \cdot S!} \left(\frac{\lambda}{\mu} \right)^{S+2} \cdot P_0$$

... ..
... ..

In general

$$P_n = P_{S + (n - S)} = \frac{1}{S^n - S} \cdot \left(\frac{\lambda}{\mu}\right)^n \cdot P_0, \text{ for } n \geq S$$

But we have,

$$\sum_{n=0}^{\infty} P_n = 1$$

$$\text{or } \sum_{n=0}^{S-1} P_n + \sum_{n=S}^{\infty} P_n = 1$$

$$\text{or } \sum_{n=0}^{S-1} \left[\frac{1}{n!} \left(\frac{\lambda}{\mu}\right)^n P_0 \right] + \sum_{n=S}^{\infty} \left[\frac{1}{S^n - S} \left(\frac{\lambda}{\mu}\right)^n P_0 \right] = 1$$

$$\text{or } P_0 \left[\sum_{n=0}^{S-1} \frac{(S\rho)^n}{n!} + \sum_{n=S}^{\infty} \frac{S^S \cdot \rho^n}{S!} \right] = 1 \quad \text{Since } \rho = \frac{\lambda}{\mu S}$$

$$\text{or } P_0 \left[\sum_{n=0}^{S-1} \frac{(S\rho)^n}{n!} + \frac{S^S}{S!} \cdot (\rho^S + \rho^{S+1} + \rho^{S+2} + \dots) \right] = 1$$

$$\text{or } P_0 \left[\sum_{n=0}^{S-1} \frac{(S\rho)^n}{n!} + \frac{S^S}{S!} \cdot \frac{\rho^S}{1 - \rho} \right] = 1 \quad \rho < 1$$

$$\therefore P_0 = \frac{1}{\left[\sum_{n=0}^{S-1} \frac{(S\rho)^n}{n!} + \frac{(S\rho)^S}{S! (1 - \rho)} \right]} \quad \dots (8)$$

$$\text{Hence, } P_n = \begin{cases} \frac{(n\rho)^n}{n!} P_0, & 1 \leq n \leq S \\ \frac{S^S \cdot \rho^n}{S!} \cdot P_0, & n \geq S \end{cases} \quad \dots (9)$$

where P_0 is given by (8).

Step three :

(a) To find $E(L_q)$, expected queue length (average number of customers in the queue).

If $n > S$, then $n - S$ customers will be waiting in queue.

$$\therefore E(L_q) = \sum_{n=S}^{\infty} (n - S) P_n = \sum_{n=S}^{\infty} (n - S) \frac{S^S \rho^n}{S!} P_0$$

$$= \frac{(S\rho)^S}{S!} P_0 \sum_{n=S}^{\infty} (n-S) \rho^{n-S}.$$

$$= P_S [0 + \rho + 2\rho^2 + 3\rho^3 + \dots]$$

$$\text{Since } P_S = \frac{(S\rho)^S \cdot P_0}{S!}$$

$$= \rho P_S (1-\rho)^{-2}$$

[from (9)]

$$\therefore E(L_q) = \frac{\rho P_S}{(1-\rho)^2} \quad \dots(10)$$

(b) To find $E(L_s)$

From article 7.15, we have

$$E(W_s) = E(W_q) + \frac{1}{\mu}$$

$$\therefore \lambda E(W_s) = \lambda E(W_q) + \frac{\lambda}{\mu}$$

$$\text{or } E(L_s) = E(L_q) + \frac{\lambda}{\mu}$$

$$\text{or } E(L_s) = \frac{\rho P_S}{(1-\rho)^2} + S\rho \quad \dots(11)$$

(c) To find $E(W_s)$ From article 7.15, we have $E(W_s) = \frac{1}{\lambda} E(L_s)$

$$\therefore E(W_s) = \frac{\rho P_S}{\lambda (1-\rho)^2} + \frac{S\rho}{\lambda} \quad \dots(12)$$

(d) To find $E(W_q)$ From article 7.15, we have $E(W_q) = \frac{1}{\lambda} E(L_q)$

$$\therefore E(W_q) = \frac{\rho P_S}{\lambda (1-\rho)^2} \quad \dots(13)$$

(e) To find $E(L/L > 0)$

$$\text{We have, } E(L/L > 0) = \frac{\sum_{n=S+1}^{\infty} (n-S) P_n}{\sum_{n=S+1}^{\infty} P_n}$$

$$\text{or } E(L/L > 0) = \frac{1}{1-\rho} \quad (\text{On simplification}) \quad \dots(14)$$

(f) Probability that some customer has to wait.

$$= P[n > S] = \frac{\rho P_S}{1-\rho} \quad \dots(15)$$

(g) To find Probability of Busy Period.

All channels will be busy if $n \geq S$.

∴ **Probability of Busy Period** : (Probability that all the channels are occupied)

= Probability ($n \geq S$)

$$= \sum_{n=S}^{\infty} P_n = \sum_{n=S}^{\infty} \frac{1}{S^n - S} \frac{1}{S!} \left(\frac{\lambda}{\mu}\right)^n P_0$$

$$= \frac{(\rho S)^S P_0}{S!} \cdot \frac{1}{1-\rho} \quad [\text{As in step two on page 281}]$$

$$= \frac{P_S}{1-\rho} \quad \dots(16)$$

$$(h) E(W/W > 0) = \frac{1}{S\mu - \lambda} \quad \dots(17)$$

(i) Average number of items served.

$$= \sum_{n=1}^{S-1} n P_n + \sum_{n=S}^{\infty} P_n.$$

ILLUSTRATIVE EXAMPLES

Example on Model III

Example 1: If for a period of 2 hours in the day (8–10 A.M.) trains arrive at the yard every 20 minutes but the service time continues to remain 36 minutes, then calculate for this period:

(i) the probability that yard is empty. (ii) average queue length, on the assumption that the line capacity of the yard is limited to 4 trains only.

Solution : (a) It is $(M/M/1) : (N/FCFS)$ problem.

Here, $\lambda = 1/20$ trains/minute,

$\mu = 1/36$ trains/minute, $N = 4$

$\rho = \lambda/\mu = 36/20 = 1.8 > 1$

∴ Probability that the yard is empty = P_0

$$= \frac{-1+\rho}{-1+\rho^{N+1}}$$

[From eqns. (10) of article 7.18]

$$= \frac{-1+1.8}{-1+(1.8)^5} = 0.04$$

and (b) Average queue length (units in the system)

$$\begin{aligned}
 &= E(L_s) = \sum_{n=0}^4 n P_n \\
 &= P_0 \sum_{n=0}^4 n \rho^n \quad \because P_n = \rho^n P_0
 \end{aligned}$$

[From equation (8) of article 7.18]

$$\begin{aligned}
 &= P_0 [\rho + 2 \rho^2 + 3 \rho^3 + 4 \rho^4] \\
 &= (0.04) [(1.8) + 2 (1.8)^2 + 3 (1.8)^3 + 4 (1.8)^4] \\
 &= (0.072) [1 + 3.6 + 9.72 + 23.328] \\
 &= (0.072) \times (37.648) = 2.7 \text{ trains}
 \end{aligned}$$

Examples on Model IV

Example 2: A telephone exchange has two long distance operators. The telephone company finds that, during the peak load, long distance calls arrive in Poisson fashion at an average rate of 15 per hour. The length of service on these calls is approximately exponentially distributed with mean length 5 minutes.

- (i) What is the probability that a subscriber will have to wait for his long distance call during the peak hours of the day?
- (ii) If the subscribers will wait and serviced in turn, what is the expected waiting time?

Solution : It is $(M/M/S) : (\infty/FCFS)$ Problem.

Here, $\lambda = 15/60 = 1/4$ calls /minute.

$\mu = 1/5$ calls/minute.

$$\therefore \rho = \frac{\lambda}{\mu S} = \frac{1/4}{(1/5) \times 2} = \frac{5}{8} \quad \because S = 2$$

From equation (8) article 7.19, we have

$$\begin{aligned}
 P_0 &= \frac{1}{\left[\sum_{n=0}^{S-1} \frac{(S\rho)^n}{n!} + \frac{(S\rho)^S}{S! \cdot (1-\rho)} \right]} \\
 &= \frac{1}{\left[\sum_{n=0}^1 \frac{(5/4)^n}{n!} + \frac{(5/4)^2}{2! \cdot [1 - (5/8)]} \right]} = \frac{1}{1 + \frac{(5/4)}{1!} + \frac{25 \cdot 8}{32 \cdot 3}} = \frac{3}{13}
 \end{aligned}$$

- (i) Probability that a subscriber will have to wait for his long distance call

$$\begin{aligned}
 &= \text{Probability } (n \geq 2) = \sum_{n=2}^{\infty} P_n \\
 &= \sum_{n=0}^{\infty} P_n - \sum_{n=0}^1 P_n = 1 - P_0 - P_1 \\
 &= 1 - \frac{3}{13} - \frac{2 \rho S}{1!} \cdot P_0 \quad \because P_n = \frac{(S \cdot \rho)^n}{n!} P_0 \quad \text{for } n \leq S (=2) \\
 &= 1 - \frac{3}{13} - \frac{5}{4} \times \frac{3}{13} = \frac{25}{52} = 0.48
 \end{aligned}$$

(ii) Expected waiting time

$$\begin{aligned}
 &= E(W_q) = \frac{\rho \cdot (S\rho)^S}{\lambda \cdot S!(1-\rho)^2} \cdot P_0 \quad [\text{From eqn. (13) of article 7.19}] \\
 &= \frac{\frac{5}{8} \left(\frac{5}{4}\right)^2}{\frac{1}{4} \cdot 2! \left(1 - \frac{5}{8}\right)^2} \cdot \frac{3}{13} = \frac{125}{39} = 3.2 \text{ minutes}
 \end{aligned}$$

Example 3: Four counters are being run on the frontier of a country to check the passports and necessary papers of the tourists. The tourists choose counter at random. If the arrivals at the frontier is Poisson at the rate λ and the service time is exponential with parameter $\lambda/2$, what is the steady-stage average queue at each counter?

Solution : It is $(M/M/S) : (\infty/FCFS)$ problem.

Here $\lambda = \lambda, \mu = \lambda/2, S = 4 \quad \therefore \rho = \lambda/\mu S = 1/2$

From eqn. (8), article 7.19, we have

$$\begin{aligned}
 P_0 &= \frac{1}{\left[\sum_{n=0}^{S-1} \frac{(S\rho)^n}{n!} + \frac{(S\rho)^S}{S!(1-\rho)} \right]} \\
 &= \frac{1}{\left[\sum_{n=0}^3 \frac{2^n}{n!} + \frac{2^4}{4! \cdot [1 - (1/2)]} \right]} \\
 &= \frac{1}{1 + \frac{2}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} + \frac{4}{3}} = \frac{3}{23}
 \end{aligned}$$

$$\text{Average queue length } E(L_q) = \frac{\rho}{(1-\rho)^2} \cdot \frac{(S\rho)^S}{S!} \cdot P_0$$

[From equation (10) of article 7.19]

$$= \frac{(1/2)}{[1 - (1/2)]^2} \cdot \frac{2^4}{4!} \cdot \frac{3}{23} = \frac{4}{23}$$

Example 4: A super market has two girls ringing sales at the counters. If the service time for each customer is exponential with mean 4 minutes and if people arrive in a Poisson fashion at the counter at the rate of 10 an hour :

- (i) What is the probability of having to wait for service ?
- (ii) What is the expected percentage of idle time for each girl ?
- (iii) If a customer has to wait, what is the expected length of his waiting time ?

Solution : It is $(M/M/S) : (\infty / FCFS)$ problem

Here, $\lambda = 10 / 60 = 1/6$ people/minute

$\mu = 1/4$ people / minute. $S = 2$

$\therefore \rho = \lambda / \mu S = 1/3$

From eqn. (8) article 7.19, we have

$$\begin{aligned}
 P_0 &= \frac{1}{\left[\sum_{n=0}^{S-1} \frac{(S\rho)^n}{n!} + \frac{(S\rho)^S}{S!(1-\rho)} \right]} \\
 &= \frac{1}{\left[\sum_{n=0}^1 \frac{(2/3)^n}{n!} + \frac{(2/3)^2}{2! \cdot [1 - (1/3)]} \right]} \\
 &= \frac{1}{\left[1 + \frac{(2/3)}{1!} + \frac{1}{3} \right]} = \frac{1}{2}
 \end{aligned}$$

- (i) Probability of having to wait for service

$$= P(n \geq 2) = \sum_{n=2}^{\infty} P_n = 1 - P_0 - P_1$$

$$= 1 - \frac{1}{2} - \frac{(2/3)}{1!} P_0$$

$$= \frac{1}{2} - \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{6} = 0.167$$

$$\therefore P_n = \frac{(S\rho)^n}{n!} \cdot P_0, \text{ for } n \leq S$$

- (ii) Fraction of the time the service remains busy

$$= \rho \text{ (traffic intensity)} = 1/3$$

$\therefore$ The fraction of the time the service remains idle

$$= 1 - 1/3 = 2/3$$

i.e., Expected percentage of idle time for each girl

$$= \frac{2}{3} \times 100\% = 67\%$$

(iii) Expected length of the customers waiting time

$$= E(W/W > 0) = \frac{1}{S\mu - \lambda} \quad (\text{Equation (17) of article 7.19})$$

$$= \frac{1}{2 \cdot (1/4) - (1/6)} = 3 \text{ minutes.}$$

Exercise 7.2

- Trains arrive at the yard every 12 minutes and the service time is 30 minutes. If the line capacity of the yard is limited to 4 trains. Find:
 - The probability that the yard is empty
 - The average number of trains in the system.
- In a bank with a single server, there are two chairs for waiting customers. On an average one customer arrives every 10 minutes and each customer takes 5 minutes for getting served. Find :
 - The probability that an arrival will get a chair to sit down.
 - The probability that an arrival will have to stand.
- At a railway station, only one train is handled at a time. The railway yard is sufficient only for two trains to wait while other is given signal to leave the station. Trains arrive at the station at an average rate of 6 per hour and the railway station can handle them on an average of 12 per hour. Assuming Poisson arrivals and exponential service distribution, find the steady-state probabilities for the various number of trains in the system. Also find the average waiting time of a new train coming into the yard.

[Hint: $\rho = \lambda / \mu = 0.5$, N = Maximum number of trains in the system = 3]

$$P_0 = \frac{1-\rho}{1-\rho^{3+1}} = 0.53$$

$$\text{Use } P_n = P_0 \rho^n.$$

$$E(L_s) = 1P_1 + 2P_2 + 3P_3 = 0.73$$

$$[E(W_s / \lambda) = (1/\lambda) E(L_s) = 7.3 \text{ minutes.}]$$

- Consider a single server queueing system with Poisson input, exponential service times. Suppose the mean arrival rate is 3 calling units per hour, the expected service time is 0.25 hour and the maximum permissible calling units in the system is two. Derive the steady state probability distribution of the number of calling units in the system, and calculate the expected number in the system.
- A supermarket has two girls ringing of sales. The people arrive at a rate of 15 an hour. The service time for each customer is exponential with mean 3 minutes. If arrival of units obey Poissons process, find :

- (i) The probability to wait for service.
 - (ii) The expected waiting time of a customer who waits for his term.
 - (iii) The expected percentage of idle time for each salesgirl.
- 6.. A petrol pump station has two pumps. The service time follow the exponential distribution with a mean of 4 minutes and cars arrive for service in a Poission process at the rate ten cars per hour. Find the probability that a customer has to wait for service. What proportion of time the pumps remain idle?
7. Customers arrive at a barber's shop according to Poisson distribution. There are three barbers, all of same efficiency. The service time of the customers is exponentially distributed with a mean equal to 10 minutes per customer. find :
- (i) The expected number of customers in the shop.
 - (ii) The expected time a customer spends in the shop.
 - (iii) The average time a customer has to wait in the queue.
 - (iv) The expected number of idle barbers.

Answers 7.2

1.	(i) 0.0155 (ii) 3.38	2.	(i) 0.875 (ii) 0.125
3.	$P_1 = 0.27, P_2 = 0.13, P_3 = 0.07$ etc, 7.3 minutes	4.	$P_n = (0.43) \cdot (0.75)^n, 0.81$ units
5.	(i) 0.204 (ii) 0.491 minutes (iii) 62.5%	6.	0.167, 67% each pump.
7.	(i) 0.3747 (ii) 0.20414 hours. (ii) 0.03747 hour (iv) 1.333		

7.20 Model V: M/G/I Queueing System

This is the queueing system, with Poisson arrival, general service time distribution involving a single server. Here the probability that a customer leaves the system in time interval $(t, t + \Delta t)$ is a quantity dependent on when the service of that particular customer begins, *i.e.*, they would involve not only, but also another time variable, say t' measured from the beginning of the current service. Let $p(t')$ denote the probability of a departure from the system in time interval $(t, t + \Delta t)$.

The system is considered only at the points when the service terminates.

Let q_0 denote the queue length when the service of customer n terminates and q_1 the length when the service of customer $(n+1)$ terminates let k ($k = 0, 1, 2, \dots$) denote the number of customers who arrive during the service of customer $(n+1)$. Thus, we have

$$q_1 = \begin{cases} k & , \text{ if } q_0 = 0 \text{ i.e., queue length was zero after the service of customer } n. \\ (q_0 - 1) + k & , \text{ if } q_0 > 0, \text{ i.e., if there } q_0 \text{ customers in the queue after the service of customer } n \end{cases}$$

$$\text{or } q_1 = q_0 - \delta + k, \quad \dots(1)$$

$$\text{where } \delta = \begin{cases} 0, & \text{ If } q_0 = 0 \\ 1, & \text{ If } q_0 > 0 \end{cases}$$

Now, expected value of relation (1) is given by

$$E(q_1) = E(q_0) - E(\delta) + E(k) \quad \dots(2)$$

In steady state, we have $E(q_1) = E(q_0)$

$$\therefore (2) \Rightarrow E(\delta) = E(k) \quad \dots(3)$$

Now, from (1), $q_1^2 = (q_0 - \delta + k)^2$

$$= q_0^2 + \delta^2 + k^2 + 2kq_0 - 2\delta q_0 - 2k\delta$$

$$= q_0^2 + \delta + k^2 + 2kq_0 - 2q_0 - 2k\delta$$

$$(\because \delta^2 = \delta \text{ and } \delta q_0 = q_0)$$

$$\Rightarrow E(q_1^2) = E(q_0^2) + E(\delta) + E(k^2) + 2E(kq_0) - 2E(q_0) - 2E(k\delta)$$

$$\Rightarrow 2E(q_0) - 2E(kq_0) = E(k^2) + E(\delta) - 2E(k\delta)$$

$$\Rightarrow 2[1 - E(k)]E(q_0) = E(k^2) + E(k) - 2E^2(k)$$

$$\therefore E(kq_0) = E(k)E(q_0), E(\delta) = E(k), E(k\delta) = E(k) = E(\delta) = E^2(k)$$

as q_0, k and δ are independent.

$$\Rightarrow E(q_0) = \frac{E(k^2) + E(k) - 2E^2(k)}{2[1 - E(k)]} \quad \dots(4)$$

If λ is the mean arrival rate, the number of arrivals in a service time of length t has a Poisson distribution with mean λt .

$\therefore$ The probability of k arrivals in time t

$$= \frac{(\lambda t)^k \cdot e^{-\lambda t}}{k!}, \quad k \geq 1$$

$\Rightarrow$ the probability of k arrivals during the service time of a customer

$$= \int_0^{\infty} \frac{(\lambda t)^k \cdot e^{-\lambda t}}{k!} p(t) dt$$

where $p(t)$ is the probability density function of service time.

∴ The **mean number of arrivals** in the service time of a customer, is given by

$$\begin{aligned} E(k) &= \sum_{k=1}^{\infty} k \cdot \int_0^{\infty} \frac{(\lambda t)^k e^{-\lambda t}}{k!} p(t) dt \\ &= \int_0^{\infty} (\lambda t) \cdot \sum_{k=1}^{\infty} \frac{(\lambda t)^{k-1}}{(k-1)!} \cdot e^{-\lambda t} p(t) dt \\ &= \int_0^{\infty} (\lambda t) \cdot \sum_{k=0}^{\infty} \frac{(\lambda t)^k}{k!} \cdot e^{-\lambda t} p(t) dt \end{aligned}$$

$$= \int_0^{\infty} (\lambda t) \cdot e^{\lambda t} \cdot e^{-\lambda t} p(t) dt \quad \because \sum_{x=0}^{\infty} \frac{x^m}{m!} = e^x$$

$$\Rightarrow E(k) = \int_0^{\infty} (\lambda t) p(t) dt = \lambda \int_0^{\infty} t p(t) dt = \lambda E(t) \quad \dots(5)$$

$$\Rightarrow E(k) = \lambda \cdot (1/\mu) = \rho, \text{ if } \mu \text{ is the mean service rate.} \quad \dots(6)$$

And the **variance of arrivals** in the service time of a customer is given by

$$\begin{aligned} E(k^2) &= \sum_{k=1}^{\infty} k^2 \cdot \int_0^{\infty} \frac{(\lambda t)^k e^{-\lambda t}}{k!} \cdot p(t) dt \\ &= \int_0^{\infty} \sum_{k=1}^{\infty} \frac{k}{(k-1)!} (\lambda t)^k \cdot e^{-\lambda t} p(t) dt \\ &= \int_0^{\infty} \sum_{k=1}^{\infty} \frac{1+(k-1)}{(k-1)!} (\lambda t)^k \cdot e^{-\lambda t} p(t) dt \\ &= \int_0^{\infty} \left[(\lambda t) \sum_{k=1}^{\infty} \frac{(\lambda t)^{k-1}}{(k-1)!} \cdot e^{-\lambda t} + (\lambda t)^2 \sum_{k=2}^{\infty} \frac{(\lambda t)^{k-2}}{(k-2)!} \cdot e^{-\lambda t} \right] \cdot p(t) dt \\ &= \int_0^{\infty} \left[(\lambda t) \cdot \sum_{k=0}^{\infty} \frac{(\lambda t)^k}{k!} + (\lambda t)^2 \cdot \sum_{k=0}^{\infty} \frac{(\lambda t)^k}{k!} \right] \cdot e^{-\lambda t} p(t) dt \end{aligned}$$

$$\begin{aligned}
&= \int_0^{\infty} [(\lambda t) \cdot e^{\lambda t} + (\lambda t)^2 \cdot e^{\lambda t}] \cdot e^{-\lambda t} p(t) dt = \int_0^{\infty} (\lambda t + \lambda^2 t^2) p(t) dt \\
&= \lambda \int_0^{\infty} t p(t) dt + \lambda^2 \int_0^{\infty} t^2 p(t) dt \\
&= \lambda E(t) + \lambda^2 E(t^2) \\
&= \lambda \cdot (\text{mean service time}) + \lambda^2 \cdot [\text{variance of service time} + \text{square of mean service time}] \\
&= \lambda \cdot (1/\mu) + \lambda^2 \cdot [\sigma^2 + 1/\mu^2] \\
\Rightarrow E(k^2) &= \rho + \lambda^2 \sigma^2 + \rho^2. \quad \dots(7)
\end{aligned}$$

Using (6) and (7), from (4), we get

$$E(q_0) = \frac{(\rho + \lambda^2 \sigma^2 + \rho^2) + \rho - 2\rho^2}{2(1-\rho)} = \frac{2\rho(1-\rho) + (\lambda^2 \sigma^2 + \rho^2)}{2(1-\rho)}$$

$$i.e. \quad E(q_0) = \rho + \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)} \quad \dots(8)$$

If the customer $(n+1)$ has to wait for time w before being taken for service, then during the time w , the length of the queue is q_1

$$\begin{aligned}
\text{and } E(q_1) &= \lambda E(w+t) = \lambda[E(w) + E(t)] \\
&= \lambda E(w) + \lambda \cdot 1/\mu = \lambda E(w) + \rho
\end{aligned}$$

$$\Rightarrow E(w) = \frac{1}{\lambda} [E(q_1) - \rho]$$

In steady state, $E(q_1) = E(q_0)$

$$\therefore E(w) = \frac{1}{\lambda} [E(q_0) - \rho] = \frac{1}{\lambda} [\rho + \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)} - \rho]$$

$$\Rightarrow E(w) = \frac{\lambda^2 \sigma^2 - \rho^2}{2\lambda(1-\rho)}$$

Hence, for the model $(M/G/1) : (\infty/GD)$,

we obtain the following results:

1. *Average number of customers in the system*

$$E(L_s) = \rho + \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)}$$

2. *Average queue length*

$$E(L_q) = \left[\rho + \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)} \right] - \rho = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)}$$

3. Average waiting time of a customer in the queue

$$E(W_q) = \frac{\lambda^2 \sigma^2 + \rho^2}{2\lambda(1-\rho)}$$

4. Average waiting time that a customer spends in the system

$$E(W_S) = E(W_q) + E(t) = \frac{\lambda^2 \sigma^2 + \rho^2}{2\lambda(1-\rho)} = \frac{1}{\mu}$$

Note : The formulae given in (1) and (2) are known as *Pollaczek-Khintchine Cor (P.K) formulae*.

7.21 Mode VI: Cost Profit Model in Queueing

Here, we consider the *cost or profit function* for the steady-state queueing system. Here the arrival rate, the service rate, the number of servers and the accommodating capacity of the system are taken as the decision variables and it is assumed that these variables are independent of time and the state of the system.

1. To find optimal Service rate

Consider M/M/1 queueing system with constant arrival rate λ .

Here the cost function is given by

$$T = K\mu + CE(n). \quad \dots(1)$$

where T = total cost, K = the cost per unit service rate

C = holding cost of a customer in the system per unit time

and $E(n)$ = Average number of customers in the system.

Since the service rate μ is continuous, so its optimum value can be obtained by differentiating the cost function T , given by (1), with respect to μ

Case I. Model (M/M/1): (∞ / FIFO):

$$\text{In this model } E(n) = \frac{\lambda}{\mu - \lambda}$$

$$\therefore \text{ from (1), } T = K\mu + \frac{C\lambda}{\mu - \lambda}$$

$$\Rightarrow \frac{dT}{d\mu} = K - \frac{C\lambda}{(\mu - \lambda)^2} \text{ and } \frac{d^2T}{d\mu^2} = \frac{2C\lambda}{(\mu - \lambda)^3}.$$

$$\text{For maximum or minimum of } T, \frac{dT}{d\mu} = 0$$

$$\Rightarrow K - \frac{C\lambda}{(\mu - \lambda)^2} = 0 \Rightarrow \mu = \lambda + \sqrt{(C\lambda/K)} = \mu^* \text{ /say}$$

$$\text{for this value of } \mu, \frac{d^2T}{d\mu^2} > 0 \text{ for } \mu > \lambda.$$

i.e. that cost T is minimum for

$$\mu = \lambda + \sqrt{(C\lambda/k)} = \mu^*$$

From which it is obvious that optical value of μ i.e. μ^* is dependent on C, λ and K .

Case II. Model (M/M/1): (N/FIFO):

In this model the larger the value of N (accommodating capacity of the system). The smaller will be the value of lost customers who do not get the opportunity to join the system because of fixed accommodating capacity of the system.

Here the cost function is given by

$$T = K\mu + CE(n) + D_1N + D_2p_N. \quad \dots(2)$$

Where D_1 = the cost per unit time per additional accommodating unit

D_2 = the cost per lost customer

λp_N = number of lost customers per unit time.

Here we consider N as the decision variable and can be determined, together with μ , by minimizing T given by (2).

2. To find optimal Number of Services:

Consider M/M/S queueing system with constant arrival rate λ and constant service rate μ .

Here the cost function is given by

$$T(S) = KS + C \cdot L(S) \quad \dots(3)$$

Where K = the cost of per server per unit time

C = holding cost of a server in the system per unit time

and $L(S)$ = Average number of customers in the system when there are S servers.

Here S can be measured in discrete unit only, so the cost function $T(S)$ will have a local minima at $S = S_0$, if

$$\Delta T(S_0 - 1) < 0 < \Delta T(S_0) \quad \dots(4)$$

Now $\Delta T(S_0 - 1) < 0 \Rightarrow T(S_0) - T(S_0 - 1) < 0$

$$\Rightarrow [KS_0 + CL(S_0)] - [K(S_0 - 1) + CL(S_0 - 1)] < 0$$

$$\Rightarrow L(S_0) - L(S_0 - 1) + K/C < 0$$

$$\Rightarrow L(S_0 - 1) - L(S_0) > K/C \quad \dots(5)$$

$$\text{Similarly } \Delta T(S_0) > 0 \Rightarrow L(S_0) - L(S_0 + 1) < K/C \quad \dots(6)$$

Using (5) and (6) from (4), we get

$$L(S_0) - L(S_0 + 1) < K/C < L(S_0 - 1) - L(S_0). \quad \dots(7)$$

Thus the optimal number of services i.e., S_0 is the discrete value of S_0 , which satisfy (7).

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter (a), (b) (c) or (d)

1. A person waiting in a queue or receiving the service is called a:
(a) Customer (b) Waiter
(c) Server (d) None of these.
 2. A person by which a person waiting in a queue is serviced is called a:
(a) Headmaster (b) Server
(c) Heedful (d) None of these.
 3. Which of the followings are the service discipline?
(a) First come, first served (FCFS) (b) Last come, first served
(c) Both (a) and (b) (d) None of these.
 4. The behaviour of a customer in a queue where he leave the queue due to impatience in multi-channel system is:
(a) Jockeying (b) Renaging
(c) Balking (d) None of these.
 5. The calling population is assumed to be infinite when:
(a) All customers arrive at once
(b) Capacity of the system is infinite
(c) Arrivals are independent of each other
(d) Arrivals are dependent upon each other.
 6. The customers arrive in a queue according to:
(a) Poisson process (b) Regular distribution
(c) Exponential distribution (d) None of these.
 7. In poisson process, the probability of no (zero) arrival in time Δt , $P_0(\Delta t) =$
(a) $\lambda \Delta t$ (b) $-\lambda \Delta t$
(c) $1 - \lambda \Delta t$ (d) $1 + \lambda \Delta t$.
 8. If λ is the mean arrival rate and μ the mean service rate, then the traffic intensity $\rho =$
(a) $\lambda - \mu$ (b) λ/μ
(c) $\lambda + \mu$ (d) $\lambda \mu$.
-

9. In the queuing model $(M/M/S):(\infty/FCFS)$, S stands for:
- (a) No a of service channels (b) Arrival rate
(c) Service rate (d) None of these.
10. In queuing model $(M/M/1) : (\infty/FCFS)$, the probability of n units (customers) in the system at time t , $P_n =$
- (a) ρ^n (b) $1 - \rho$
(c) $\rho^n (1 - \rho)$ (d) $\rho^n (1 + \rho)$.
11. In queueing model $(M/M/1) : (\infty/FCFS)$, the expected line length in the system i.e. $E(L_s) =$
- (a) $\frac{\lambda}{\mu - \lambda}$ (b) $\frac{\lambda}{\mu + \lambda}$
(c) $\frac{\mu}{\mu - \lambda}$ (d) $\frac{\mu}{\mu + \lambda}$.
12. Which of these is true?
- (a) $E(L_s) = \mu \cdot E(W_s)$ (b) $E(L_s) = \lambda E(W_s)$
(c) $E(L_s) = \mu E(W_q)$ (d) $E(L_s) = \lambda E(W_q)$.
13. Which of these is true?
- (a) $E(W_s) = E(W_q) + 1/\mu$ (b) $E(W_s) = E(W_q) - 1/\mu$
(c) $E(W_s) = E(W_q) + \mu$ (d) $E(W_s) = E(W_q) + \mu$.
14. Which of the following relationship is not true?
- (a) $E(W_s) = E(W_q) + 1/\mu$ (b) $E(L_s) = \lambda E(W_s)$
(c) $E(L_s) = E(L_q) + 1/\lambda$ (d) $E(L_s) = \lambda E(W_q)$.

Fill in the Blank(s)

Fill in the blank's ".....", so that the following statements are complete and correct

- The customers behaviour in which he moves a from a queue to another in a multi-channel situation is
- Service discipline in which some customers are served before the others without considering their order of arrival is called service on procedure.
- The service facilities given to the customers can be classified in two categories and
- The state of the system in which the arrival rate is more than its service rate, is said to be

5. In Poisson's process probability of one arrival in time Δt is given by $P_1(\Delta t) = \dots\dots\dots$.
6. If the arrival pattern in a queueing problem follows a poisson process, then the random variable T representing the inter arrival time follows the distribution.
7. In the queueing model $(M/M/1) : (\infty/\text{FCFS})$, if P_0 is the probability of no unit in the system at time t and ρ the traffic intensity, then $P_0 = \dots\dots\dots$.
8. In queueing model $(M/M/1) : 1 (\infty/\text{FCFS})$, the average number of customers in the queue is given by $E(L_q) = \dots\dots\dots$.

True/False

Write 'T' for true and 'F' for false statement

1. In poisson process the probability of n arrivals during time interval of length t is given by $P_n(t) = \frac{(\lambda t)^n \cdot e^{-\lambda t}}{n!}$.
2. A system is said to be in steady state when the probability distribution of arrivals, waiting time and servicing time of the customers are not independent of the time.
3. The probability of $m (> 1)$ arrivals in Poisson's process in time Δt is zero.
4. In queueing model $(M/M/1) : (\infty/\text{FCFS})$ first M stands for Poisson arrival and second M for Poisson departure.

Answers

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1.	(a)	2.	(b)	3.	(c)
4.	(b)	5.	(c)	6.	(a)
7.	(c)	8.	(b)	9.	(a)
10.	(c)	11.	(a)	12.	(b)
13.	(a)	14.	(c)		

Fill in the Blank(s)

1.	jockeying	2.	some priority	3.	singlechannel, multichannel
4.	explosive state	5.	$\lambda \Delta t$	6.	exponential
7.	$1 - \rho$	8.	$\frac{\lambda^2 t}{\mu(\mu - \lambda)}$		

True/False

1.	T	2.	F	3.	T
4.	T				

○○○

UNIT

III



Chapters



8. Network Analysis

9. Project Management (PERT & CPM)

UNIT-III

Chapter

8



Network Analysis

8.1 Introduction

Network analysis is the general name given to a particular technique which can be used for planning, management and control of projects. Network analysis is the mathematical analysis of complex working procedures in terms of network of related activities. It is the mathematical method of analysing complex problems by representing the problem as a network of lines and nodes.

8.2 Network

A *network* is defined as a set of points (or *nodes*) which are connected by *lines* (or *links*). A way of going from one node (the origin) to another (the destination) is called a *route* (or *path*). The links in the network may be one way in either direction or two way in both directions. The numbers on the links in the network represent the time, cost or distance involved in traversing them.

The *network diagram* represents clearly and concisely the entire project enabling us to get a pictorial overview of what to happen, the various activities and the relationship of one to another.

8.2.1 Loops (or Cycles) in a Minimum Network

Consider the following graph with nodes *A*, *B* and *C* and the branches connecting them. The lengths of the branches are shown on the respective branches. Obviously, the minimal

network (*i.e.*, the network for which the sum of lengths of the chosen branches is minimum) occurs when the node C is connected to both nodes A and B . In this case, the total minimum length of branches is equal to $3 + 8 = 11$.

If the node A and B are also connected, then a loop (or cycle) results between the nodes A, B and C and the resulting network is not minimized. Thus in a network minimization problem, it is not optimum to include loops (or cycles) in the solution.

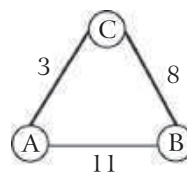


Fig. 8.1

8.3 Network Minimization

The network minimization deals with the determination of the branches that can join all the nodes of a network (*i.e.* every pair of nodes is connected directly or indirectly by branch/branches) such that the sum of the lengths of the chosen branches is **minimized**.

8.4 Tree and Spanning Tree

8.4.1 A Tree

A tree is a connected undirected graph with no cycles. A single graph can have many different trees.

For example, consider the following graph. One tree of this graph can be taken as $1 - 3, 3 - 4, 3 - 5, 4 - 7$ (shown by thick black branch lines).

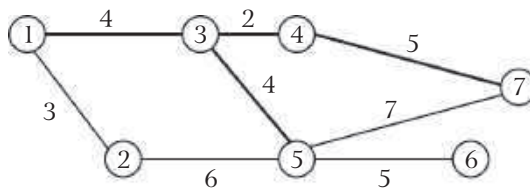


Fig. 8.2

Other trees also exist, such as

$1 - 3, 3 - 5, 5 - 6, 1 - 2, 2 - 5, 5 - 7$ etc.

8.4.2 Spanning Tree

A **spanning tree** of a connected undirected graph G is a tree that include all the vertices (nodes) and some or all of the edges (branches) of G . In other words, a tree of a graph G is called its **spanning tree** if it spans G (*i.e.* it includes every vertex of G) and is a subset of G (every edge in the tree belongs to G).

A *spanning tree* of a connected graph G can also be defined as a maximal set of edges of G that contains no cycle or as a minimal set of edges of G that contains all vertices.

A connected graph can have many different *spanning trees*.

According to **Cayley's formula**, the total number of *spanning trees* on a complete graph with n vertices (nodes) is n^{n-2} .

1. For $n=2$, the total number of spanning trees $= 2^{2-2} = 1$, as shown in fig. 8.3.



Fig. 8.3

2. For $n=3$, the total number of spanning trees $= 3^{3-2} = 3$, as shown in fig. 8.4.

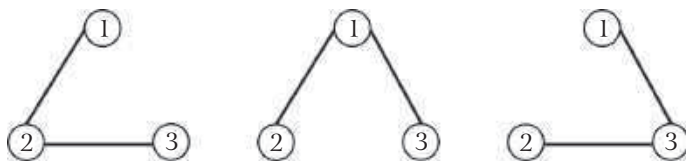


Fig. 8.4

3. For $n=4$, the total number of spanning trees $= 4^{4-2} = 16$, as shown in fig. 8.5.

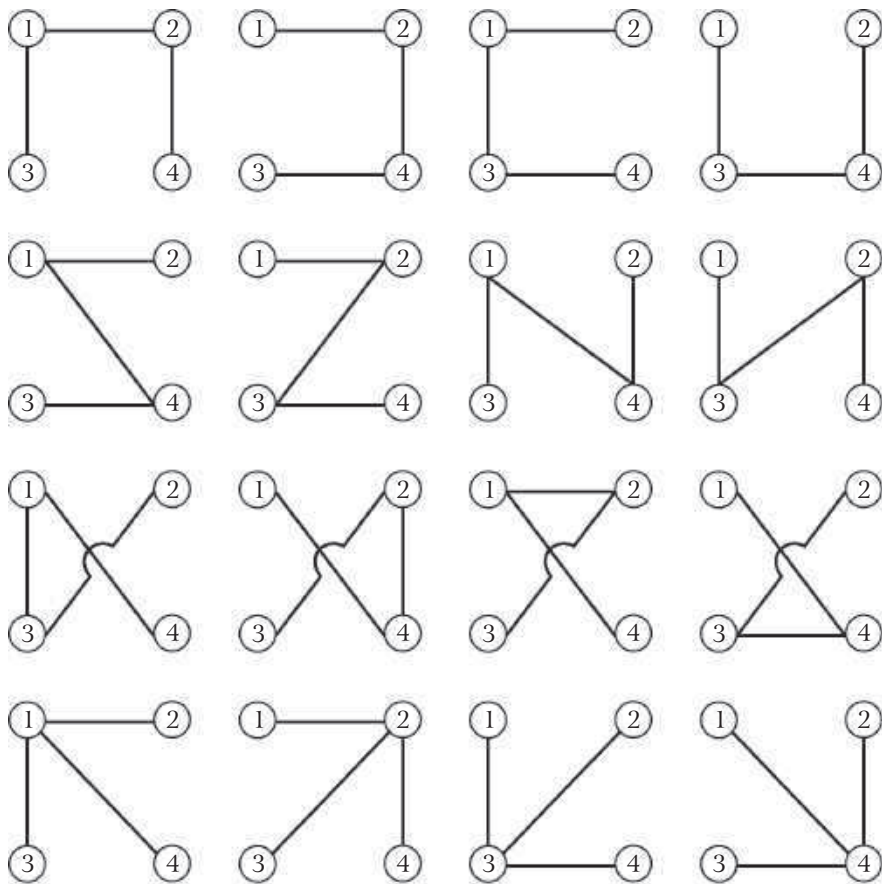


Fig. 8.5

8.5 Minimal Spanning Tree (M.S.T.)

Minimal spanning tree of a connected graph G is the connected undirected graph with no cycles that include all its vertices (nodes) such that the sum of lengths of the chosen branches is minimized.

In other words, the **minimal spanning tree** of a connected graph G is its spanning tree such that the sum of lengths of its branches is minimized.

Method to Find M.S.T.: The minimal spanning tree (M.S.T.) of a connected graph (network) is determined stepwise as follows:

Step 1: Start with any node (vertex) and join it to its *closed* (nearest) node in the network. The two nodes will form a connected set C and the remaining nodes will form the unconnected set $\bar{C}$.

Step 2: Choose a node from the unconnected set $\bar{C}$, that is closest (nearest) directly (or indirectly) to any node in the connected set C . Include this node in the connected set C and exclude from unconnected set C' .

Step 3: Repeat step 2 for the newly defined connected and unconnected sets.

Step 4: Repeat the steps 2 and 3 until the connected set includes all the nodes of the network.

If there is a tie at any stage (step), then it indicates the existence of alternative minimal spanning trees.

ILLUSTRATIVE EXAMPLES

Example 1: Find the minimal spanning tree of the following network:

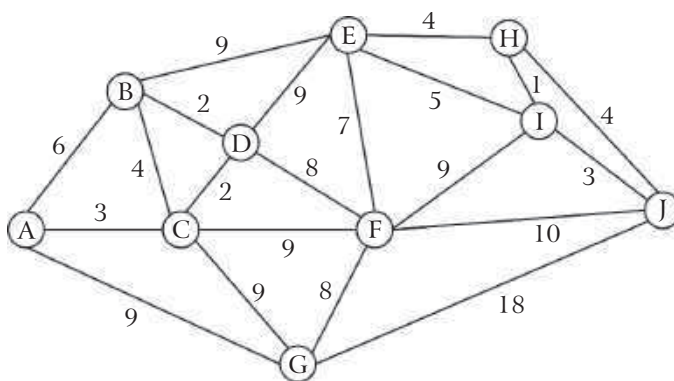


Fig.8.6

Solution: We shall find M.S.T. for the above problem stepwise so that the students may understand the procedure well.

Here, the nodes in the connected set C and unconnected set $\bar{C}$ will be represented by shaded and unshaded nodes respectively and the dark black line will represent the chosen branch of M.S.T.

Step 1: Starting with node (A) , we join it to its nearest node (C) . Thus, the connected set $C = \{(A), (C)\}$ and so unconnected set $\bar{C} = \{(B), (D), (E), (F), (G), (H), (I), (J)\}$. Thus, the branch joining nodes (A) and (C) is the chosen branch shown as dark black line in fig. 8.7.

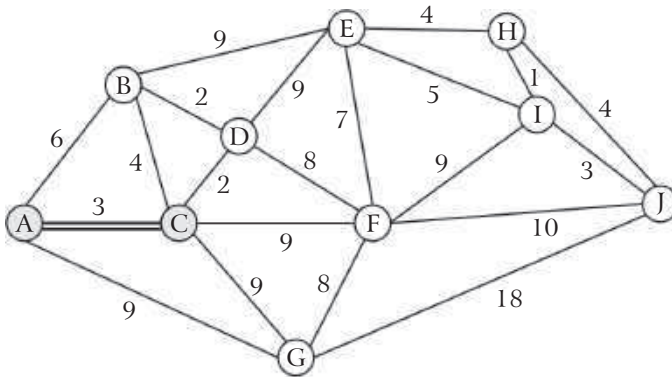


Fig. 8.7

Step 2: Now we choose a node from $\bar{C}$ which is nearest (directly or indirectly) to any node (A) or (C) in set C . Obviously, such a node is (D) and the branch joining nodes (A) and (D) is other chosen branch for M.S.T.. We transfer this node (D) from set $\bar{C}$ to set C and get

$$C = \{(A), (C), (D)\}, \quad \bar{C} = \{(B), (E), (F), (G), (H), (I), (J)\}.$$

These nodes and chosen branches are shown in fig. 8.8.

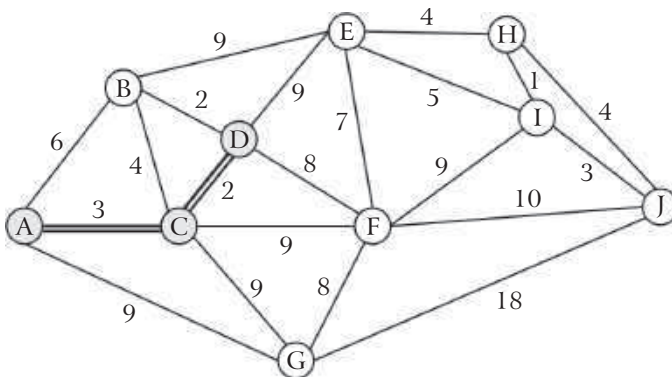


Fig. 8.8

Step 3: The node $\textcircled{B}$ of set $\bar{C}$ is nearest to the node $\textcircled{D}$ of set C . So we choose this node $\textcircled{B}$ and transfer it from the set $\bar{C}$ to the set C and choose the branch joining these two nodes $\textcircled{B}$ and $\textcircled{D}$ for M.S.T. These are shown in the fig. 8.9.

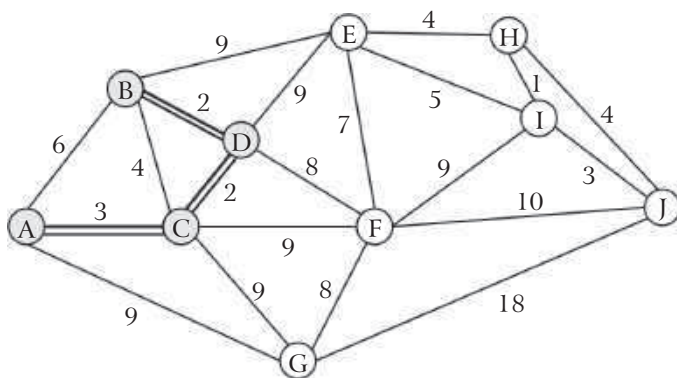


Fig. 8.9

Now $C = \{ \textcircled{A}, \textcircled{B}, \textcircled{C}, \textcircled{D} \}$, $\bar{C} = \{ \textcircled{E}, \textcircled{F}, \textcircled{G}, \textcircled{H}, \textcircled{I}, \textcircled{J} \}$

Step 4: The node $\textcircled{F}$ of set $\bar{C}$ is nearest to the node $\textcircled{D}$ of set C . So we choose this node $\textcircled{F}$ and transfer it from the set $\bar{C}$ to the set C and choose the branch joining these two nodes $\textcircled{F}$ and $\textcircled{D}$ for M.S.T.. These are shown in fig. 8.10.

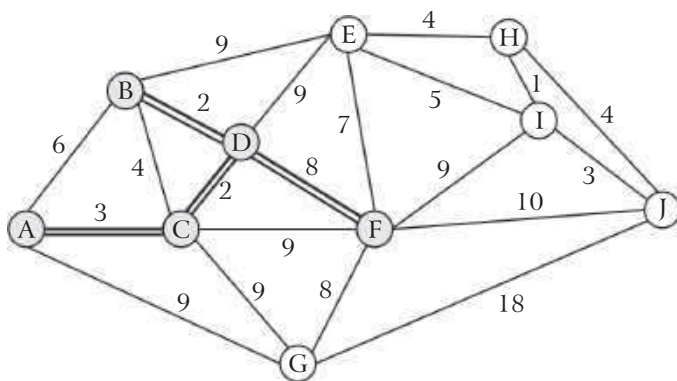


Fig. 8.10

Now, $C = \{ \textcircled{A}, \textcircled{B}, \textcircled{C}, \textcircled{D}, \textcircled{F} \}$, $\bar{C} = \{ \textcircled{E}, \textcircled{G}, \textcircled{H}, \textcircled{I}, \textcircled{J} \}$

Step 5: The node $\textcircled{E}$ of set $\bar{C}$ is nearest to the node $\textcircled{F}$ of set C . So we choose this node $\textcircled{E}$ and transfer it from the set $\bar{C}$ to the set C and choose the branch joining these two nodes $\textcircled{E}$ and $\textcircled{F}$ for M.S.T. . These are shown in fig. 8.11.

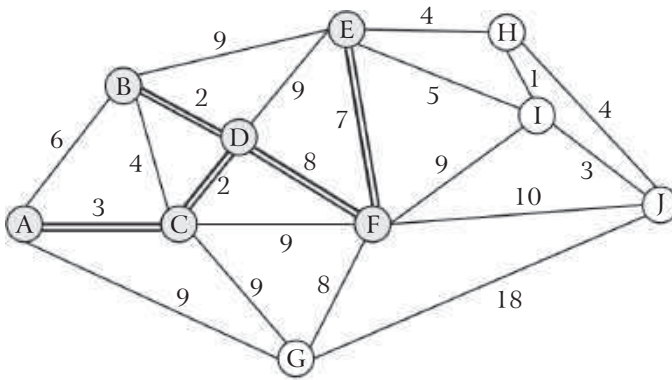


Fig. 8.11

Now $C = \{ \textcircled{A}, \textcircled{B}, \textcircled{C}, \textcircled{D}, \textcircled{E}, \textcircled{F} \}$, $\bar{C} = \{ \textcircled{G}, \textcircled{H}, \textcircled{I}, \textcircled{J} \}$

Step 6: The node $\textcircled{H}$ of the set $\bar{C}$ is nearest to the node $\textcircled{E}$ of set C . So we choose this node $\textcircled{H}$ and transfer it from the set $\bar{C}$ to the set C and choose the branch joining these two nodes $\textcircled{E}$ and $\textcircled{H}$ for M.S.T.. These are shown in fig. 8.12.

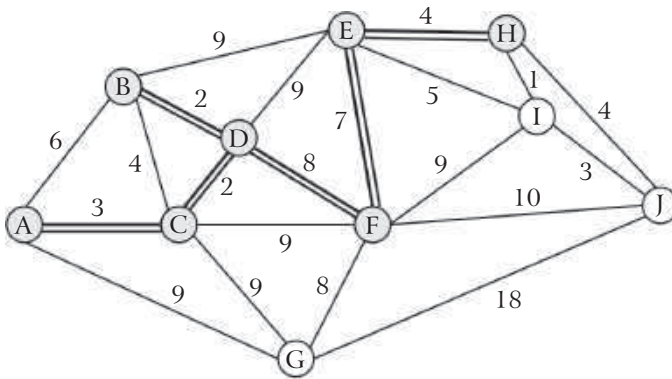


Fig. 8.12

Now $C = \{ \textcircled{A}, \textcircled{B}, \textcircled{C}, \textcircled{D}, \textcircled{E}, \textcircled{F}, \textcircled{H} \}$, $\bar{C} = \{ \textcircled{G}, \textcircled{I}, \textcircled{J} \}$

Step 7: The node $\textcircled{I}$ of the set $\bar{C}$ is nearest to the node $\textcircled{H}$ of the set C . So we choose this node $\textcircled{I}$ and transfer it from the set $\bar{C}$ to the set C and choose the branch joining these two nodes $\textcircled{H}$ and $\textcircled{I}$ for M.S.T.. These are shown in fig. 8.13.

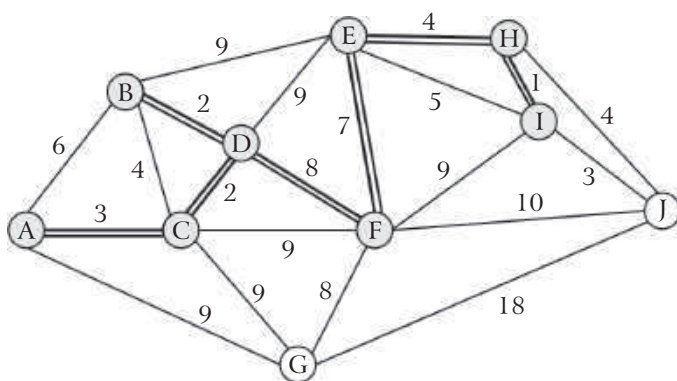


Fig. 8.13

Now $C = \{ \textcircled{A}, \textcircled{B}, \textcircled{C}, \textcircled{D}, \textcircled{E}, \textcircled{F}, \textcircled{H}, \textcircled{I} \}$, $\bar{C} = \{ \textcircled{G}, \textcircled{J} \}$

Step 8: The node $\textcircled{J}$ of the set $\bar{C}$ is nearest to the node $\textcircled{I}$ of the set C . So we choose this node $\textcircled{J}$ and transfer it from the set $\bar{C}$ to the set C and choose the branch joining these two nodes $\textcircled{I}$ and $\textcircled{J}$ for M.S.T.. These are shown in fig. 8.14.

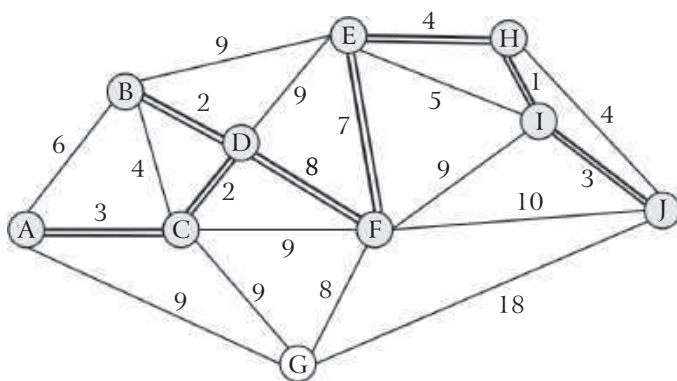


Fig. 8.14

Now $C = \{ \textcircled{A}, \textcircled{B}, \textcircled{C}, \textcircled{D}, \textcircled{E}, \textcircled{F}, \textcircled{H}, \textcircled{I}, \textcircled{J} \}$, $\bar{C} = \{ \textcircled{G} \}$.

Step 9: There is only one node $\textcircled{G}$ remaining in set $\bar{C}$, which is nearest to the node $\textcircled{F}$ of set C . So we choose this node $\textcircled{G}$ and transfer it from the set $\bar{C}$ to the set C and choose the branch joining these two nodes $\textcircled{G}$ and $\textcircled{F}$ for M.S.T. These are shown in fig. 8.15.

$C = \text{set of all the nodes}$ and $\bar{C} = \phi$

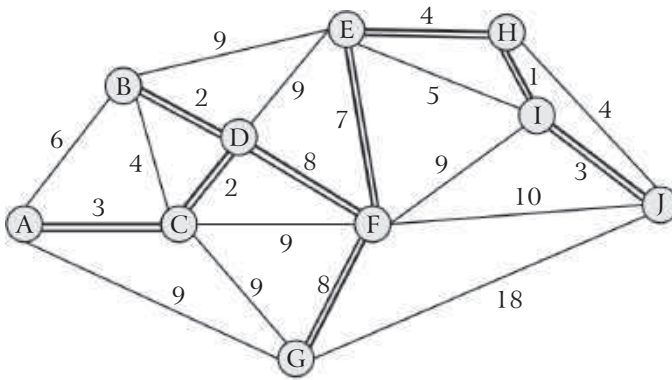


Fig. 8.15

Since all the nodes are connected, so the procedure is completed.

Hence, minimal spanning tree consists of all nodes and branches

$$\textcircled{A} - \textcircled{C}, \textcircled{C} - \textcircled{D}, \textcircled{D} - \textcircled{B}, \textcircled{D} - \textcircled{F}, \textcircled{F} - \textcircled{G}, \textcircled{F} - \textcircled{E}, \textcircled{E} - \textcircled{H}, \textcircled{H} - \textcircled{I}$$

and $\textcircled{I} - \textcircled{J}$ as shown in fig. 8.15.

and minimum length = $3 + 2 + 2 + 8 + 8 + 7 + 4 + 1 + 3 = 38$

8.6 Shortest-Route Problem

Shortest-route problem is the problem of finding a path between a source (vertex or node) and a destination (vertex or nodes) in a graph such that the sum of the weights (distances or costs) of its constituent edges is minimized.

A shortest-route problem (or minimal path problem or shortest A cyclic route model) arises when we wish to go from one place to another or to several other places and we are to select the shortest route (involving least distance or time or cost) out of many alternatives, to reach the desired place.

Such acyclic* route network problems can be solved by two methods.

1. By graphic method 2. By recursive computation method.

For clear understanding of the two methods, we shall solve the following examples by both methods:

***Acyclic route network:** A network is *acyclic network* in which there exists no chains connected a node to itself.

ILLUSTRATIVE EXAMPLES

Example 1: The network in the following figure 8.16 represents the distance in kilometers between various cities i , $i = 1, 2, \dots, 8$. Find the shortest route/routes between city 1 and city 8.

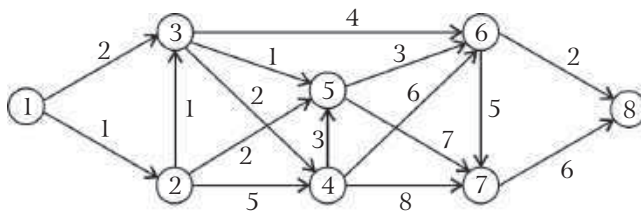


Fig. 8.16

The numbers on the links represent the distance in kilometers in going from one city to another.

Solution: (a) **By Graphical Method:** The shortest distance *i.e.* shortest route from city 1 to city i will be written in bracket [] above that city i in the network. The graphic method consists of the following steps:

Step 1: Starting with the origin 1 draw all links by which one can go from city 1 to other adjacent cities (here city 2 and 3) and represent the distance from city 1 to each of these cities on the links is shown in fig. 8.17.

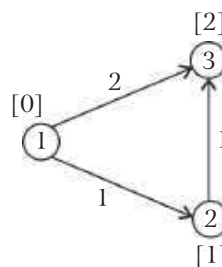


Fig. 8.17

Here distance of the direct route from city 1 to 3 is same as the distance of indirect route ($1 \rightarrow 2, 2 \rightarrow 3$) from city 1 to 3 (*i.e.* sum of distances from city 1 to 2 then from city 2 to 3), so both routes are drawn by solid lines and distance of city 3 from city 1 is also 2, shown on this city.

Distance of city 2 from city 1 is 1, which is shown on node 2. All these entries are shown in fig. 8.17.

Step 2: Add cities 4, 5 and 6 to which one can go from the cities 2 and 3 represented in fig. 8.17.

The distance of city 4 from city 1 through city 3 is 4 and that through city 2 is $1 + 5 = 6$. As the smaller of these *i.e.* 4, so shortest distance of city 4 from city 1 is taken as 4 and shown on city 4 and the larger distance link from city 2 to 4 is shown by dotted line. Also the distance of city 5 from 1, through city 3 as well as through city 2 is same equal to 3. Also distance of city 5 from city 1 through cities 2 and 4 is $1 + 5 + 3 = 9$, which is larger. So we leave it and the shorter distance [3] is written on city 5. The larger distance line from city 2 to 4 and then from city 4 to 5 are shown by dotted lines. All these entries are shown in fig. 8.18.

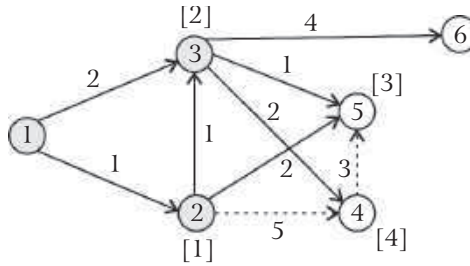


Fig. 8.18

Step 3: Now we add cities 7 and 8 to which one can go from the cities 1, 2, 3, 4, 5, 6 represented in fig. 8.18.

Distance of city 6 from 1 through city 3 is 6, that through city 5 is 6 and that through city 4 is 10. Smaller of these is 6 through city 3 as well as through city 5. So this shorter distance [6] is written on city 6. The larger distance link from node 4 to 6 is shown by dotted line.

Distance of city 7 from 1 through city 6 is 11, that through city 5 is 10 and that through city 4 is 12. Since smallest of these is 10 through city 5, so distance [10] is written on city 7 and larger distance links from city 4 to 7 and from city 6 to 7 are shown by dotted lines.

Distance of city 8 from 1 through city 6 is 8 and that through city 7 is 16. So the smaller distance 8 is written on city 8 and this link is shown by solid line while the larger distance link from city 7 to 8 is shown by dotted line.

All these entries are shown in fig. 8.19. In this fig. 8.19, shorter distance routes are shown by solid lines and larger distance routes by dotted lines.

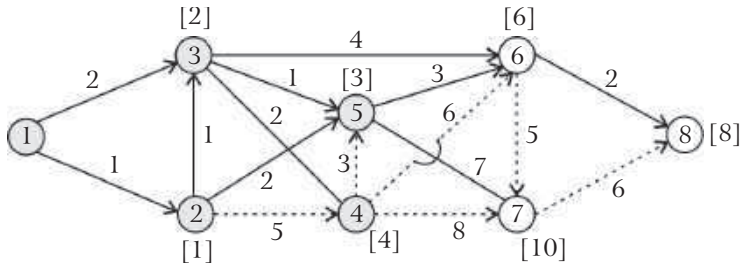


Fig. 8.19

Now we select the shortest distance route as follows:

Take the last city 8 and shorter route cities preceding this city, which is city 6. So we write this as

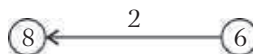


Fig. 8.20

Now consider cities 3 and 5 which are shorter route cities preceding to city 6. So we have the shorter paths as follows:

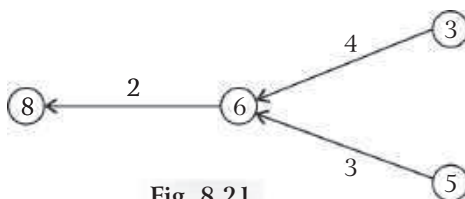


Fig. 8.21

Now consider shorter route cities preceding to cities 3 and 5 and continue similarly. All routes are shown in fig. 8.22. by double black lines.

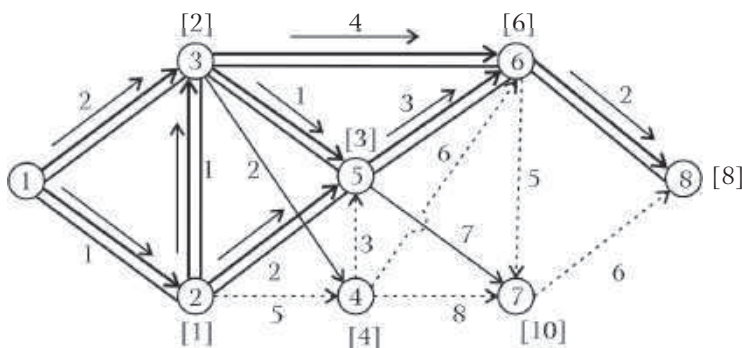


Fig. 8.22

Thus, we get five shortest distance routes between city 1 and 8 as

$$\textcircled{1} \rightarrow \textcircled{3} \rightarrow \textcircled{6} \rightarrow \textcircled{8}, \quad \textcircled{1} \rightarrow \textcircled{3} \rightarrow \textcircled{5} \rightarrow \textcircled{6} \rightarrow \textcircled{8},$$

$$\textcircled{1} \rightarrow \textcircled{2} \rightarrow \textcircled{5} \rightarrow \textcircled{6} \rightarrow \textcircled{8}, \quad \textcircled{1} \rightarrow \textcircled{2} \rightarrow \textcircled{3} \rightarrow \textcircled{6} \rightarrow \textcircled{8}$$

and $\textcircled{1} \rightarrow \textcircled{2} \rightarrow \textcircled{3} \rightarrow \textcircled{5} \rightarrow \textcircled{6} \rightarrow \textcircled{8}.$

In all these routes, the total shortest distance from city 1 to city 8 is 8-kilometers.

(b) By Recursive Computation Method: Here, we define the following two notations:

u_j = shortest distance from city 1 to city j , $j = 1, 2, \dots, 8$. $u_1 = 0$.

d_{ij} = network distance between adjacent cities i and j .

Thus
$$u_j = \min_i (u_i + d_{ij})$$

where city i , immediately precedes city j . The procedure will be completed when u_8 is computed.

Now starting at city 1, we see that only city 2 and 3 immediately follow city 1. So we proceed as follows:

Step 1: $u_1 = 0$.

Step 2: $u_2 = u_1 + d_{12} = 0 + 1 = 1$, (from city 1 to 2)

$$u_3 = \min. \{u_1 + d_{13}, u_2 + d_{23}\} = \min. \{0 + 2, 1 + 1\} = 2 \text{ (from city 1 and 2 both)}$$

Step 3: $u_4 = \min. \{u_2 + d_{24}, u_3 + d_{34}\} = \min. \{1 + 5, 2 + 2\} = 4$ (from city 3)

$$u_5 = \min. \{u_2 + d_{25}, u_3 + d_{35}\} = \min. \{1 + 2, 2 + 1\} = 3 \text{ (from city 2 and 3 both)}$$

Step 4: $u_6 = \min. \{u_3 + d_{36}, u_4 + d_{46}, u_5 + d_{56}\} = \min. \{2 + 4, 4 + 6, 3 + 3\}$

$$= 6 \text{ (from city 3 and 5 both).}$$

Step 5: $u_7 = \min. \{u_4 + d_{47}, u_5 + d_{57}, u_6 + d_{67}\} = \min. \{4 + 8, 3 + 7, 6 + 5\}$

$$= 10 \text{ (from city 5)}$$

Step 6: $u_8 = \min. \{u_6 + d_{68}, u_7 + d_{78}\} = \min. \{6 + 2, 10 + 6\} = 8$ (from city 6)

Conclusion of steps from 1 to 6. The distances (links) of city 2 from city 1, city 3 from city 1 and 2 both (step 2); city 4 from city 3, city 5 from city 2 and 3 both (step 3); city 6 from city 3 and 5 both (step 4); city 7 from city 5 (step 5) and city 8 from city 6 (step 6) are joined by solid line and other links are joined by dotted lines. [See fig. 8.22].

Obviously, the shortest routes from city 1 to city 8 (shown by thick black lines) are

- (i) $1 \rightarrow 3 \rightarrow 6 \rightarrow 8$; (ii) $1 \rightarrow 3 \rightarrow 5 \rightarrow 6 \rightarrow 8$; (iii) $1 \rightarrow 2 \rightarrow 5 \rightarrow 6 \rightarrow 8$;
 (iv) $1 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 8$ and (v) $1 \rightarrow 2 \rightarrow 3 \rightarrow 5 \rightarrow 6 \rightarrow 8$.

In all cases, mini distance = 8 kilometers.

Example 2: A person wishes to reach the destination 'k' while starting from the station 'a' in the network shown in figure 8.23. The links work in both directions. The numbers on the links represent cost (in ₹) of going from one station to another. Then

(a) find the route that involves the least cost.

(b) find the route that involves the least cost by visiting minimum number of stations before reaching station 'k'.

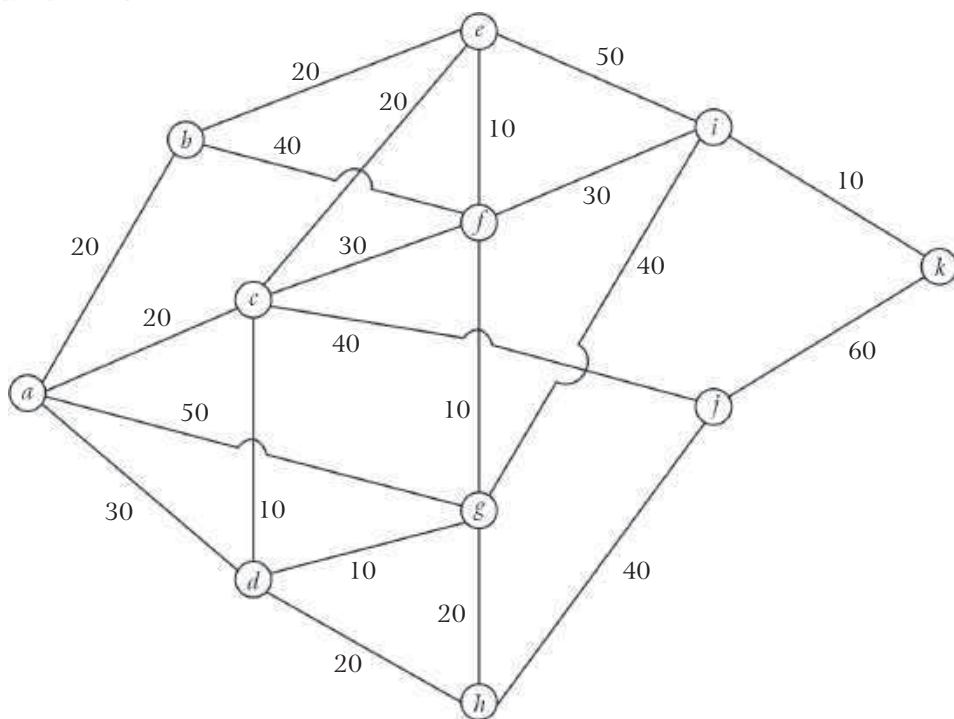


Fig. 8.23

Solution: The least cost from station 'a' to any station will be written on that station in bracket [] in the network. The graphical method consists of the following steps:

Step 1: Starting from the station *a* draw all links by which one can go from station *a* to other adjacent stations (here stations *b*, *c*, *d* and *g*) and represent the costs from station *a* to each of these station on the links and the corresponding station. This is shown in fig. 8.24.

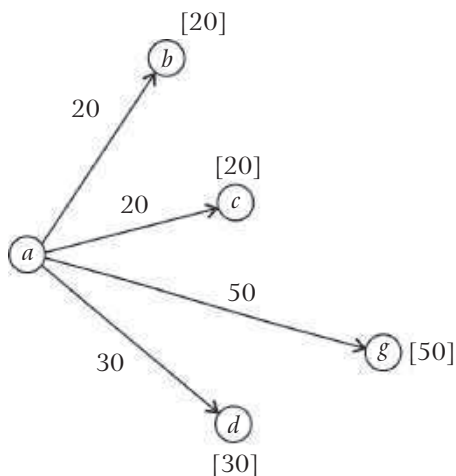


Fig. 8.24

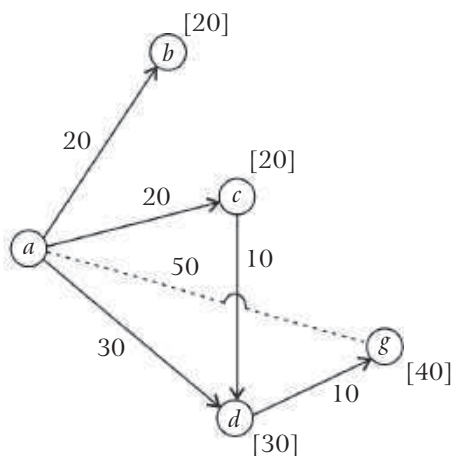


Fig. 8.25

Step 2: In case, there are links between any of the stations obtained in step 1, determine each of these links. If for the indirect route from station a to any of these stations, cost is less than the direct route, draw the shorter cost route as a solid line and the longer cost route as the dotted line and write the lesser cost on these stations in brackets [].

Here, the cost from station a to station d by indirect route ($a \rightarrow c$ then $c \rightarrow d$), is $20 + 10 = 30$ which is equal to the direct cost 30 between these two stations. So we write cost [30] on station d and draw solid lines of the direct as well as link between a and d . Also the cost from station a to station g by indirect route ($a \rightarrow d$ then $d \rightarrow g$) is $30 + 10 = 40$ which is less than the cost 50 in the direct route between these two stations. So we write lesser cost 40 in [g] on station g and draw dotted line of the direct link of larger cost between station a and g [See fig. 8.25].

Step 3: Now add stations e, f, j and h to which one can go from the stations b, c, d, g taken in step 2 and draw the links by solid and dotted lines by the method explained in step 2. Also write the least cost from station a to that station on the corresponding station. All these entries are done in fig. 8.26.

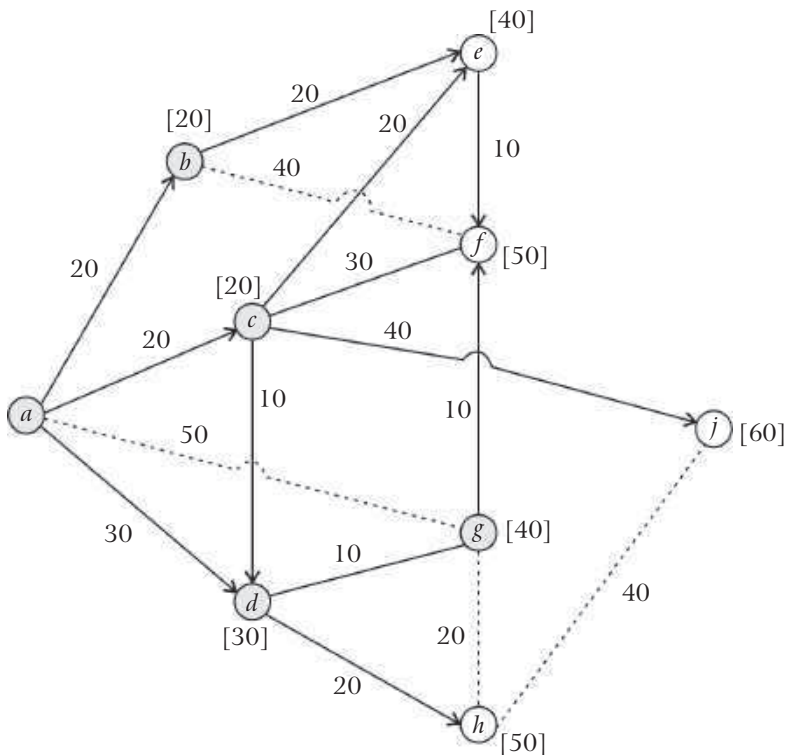


Fig. 8.26

Here, cost in indirect route from station b to f ($b \rightarrow e$ then $e \rightarrow f$) is $20 + 10 = 30$ which is less than that cost 40 in direct route between these two stations. So we draw dotted line in direct route between b and f . Similarly, cost in indirect route from a to j ($a \rightarrow c$ then $c \rightarrow j$)

$20 + 40 = 60$ is less than that from a to j ($a \rightarrow d, d \rightarrow h$, then $h \rightarrow j$) $30 + 20 + 40 = 90$ so the cost at j is put [60] and greater cost route h to j is drawn with dotted line.

Step 4: Now add stations i and k to which one can go from the stations taken in step 3 and draw the links by solid and dotted lines by the method explain in step 2. Also write the least cost from station a to that station on the corresponding station. All these entries are done in fig. 8.27.

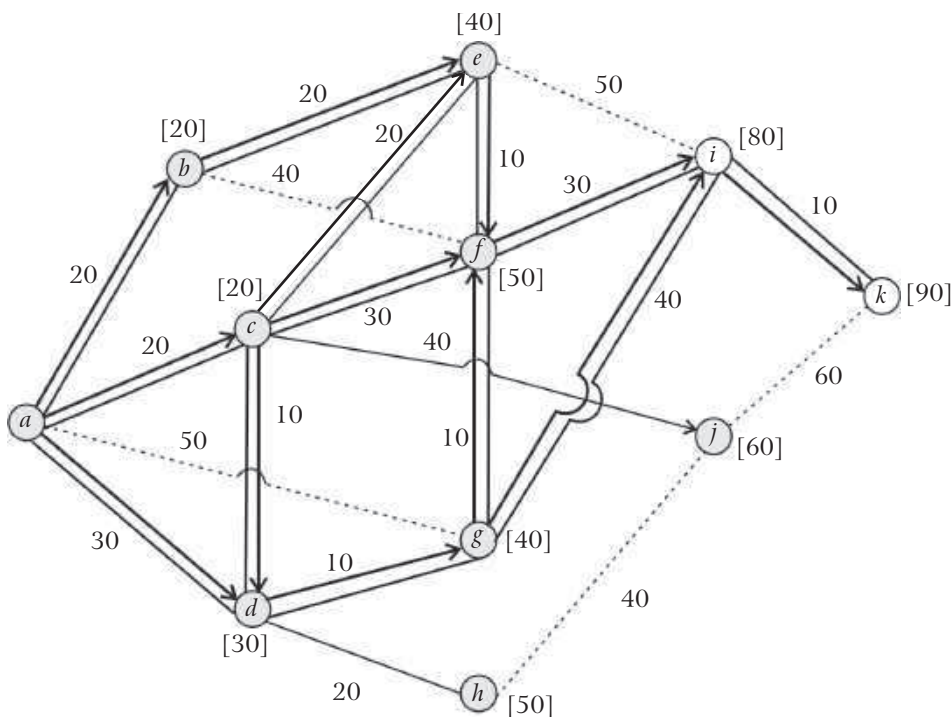


Fig. 8.27

Now starting from the destination station k , we proceed back to starting station a through all possible routes.

$\therefore$ The least cost path from station a to station k are as follows, which are shown by dark black lines.

- (i) $a \rightarrow b \rightarrow e \rightarrow f \rightarrow i \rightarrow k$ (Involving 6 stations)
- (ii) $a \rightarrow c \rightarrow f \rightarrow i \rightarrow k$ (Involving 5 stations)
- (iii) $a \rightarrow c \rightarrow d \rightarrow g \rightarrow f \rightarrow i \rightarrow k$ (Involving 7 stations)
- (iv) $a \rightarrow c \rightarrow d \rightarrow g \rightarrow i \rightarrow k$ (Involving 6 stations)
- (v) $a \rightarrow c \rightarrow e \rightarrow f \rightarrow i \rightarrow k$ (Involving 6 stations)
- (vi) $a \rightarrow d \rightarrow g \rightarrow f \rightarrow i \rightarrow k$ (Involving 6 stations)
- (vii) $a \rightarrow d \rightarrow g \rightarrow i \rightarrow k$ (Involving 5 stations)

All the seven routes have the same cost of ₹ 90 (minimum) in going from station a to station k .

In the above six routes, the person have to visit at least 5 stations. Hence, the routes involving minimum number of stations in going from station a to k are $a \rightarrow c \rightarrow f \rightarrow i \rightarrow k$ and $a \rightarrow d \rightarrow g \rightarrow i \rightarrow k$. Minimum travelling cost = ₹ 90.

8.7 Maximal Flow Model

The **maximal flow model** was first formulated in 1954 by T.E. Harris and T.S. Ross as a simplified model of Soviet railway traffic flow. In 1956, Lester R. Fort Jr. and Delbert R. Fulkerson created the first algorithm, the *Fort Ford Fulkerson algorithm* which computes the maximum flow in a flow network.

8.8 Fort Ford Fulkerson Algorithm

The idea behind the algorithm is as follows:

As long as there is a path from the origin (source) to the destination (sink) with available capacity on all edges in the path, we send flow along one of these paths. Then we find another path and so on. A path with available capacity is called an *augmenting path*.

The main idea is careful book keeping of the flows assigned to different routes from the origin node (source) to the destination node (sink). The steps in the method are as follows:

Step 1: Find any path from the origin node to the destination node that has a strictly positive flow capacity remaining. If there are no more such path, exit.

Step 2: Determine f , the maximum flow along this path, which will be equal to the smallest flow capacity on any arc in the path (the bottleneck arc).

Step 3: Subtract f from the remaining flow capacity in the forward direction for each arc in the path. Add f to the remaining flow capacity in the backwards direction for each arc in the path.

Step 4: Repeat steps 1 to 3 for all the paths.

On termination, the sum of the flows along the paths found in step 1 gives the maximum total flow between the origin and destination nodes.

8.9 Maximal-Flow Minimal-Cut Theorem (or Max-Flow Min-Cut Theorem)

The theorem states that *in a flow network, the maximum amount of flow passing from the source (origin) to the sink (destination) is equal to the minimum capacity over all cuts that, when removed in a specified way from the network causes the situation that no flow can pass from the source to the sink.*

The minimum cut is actually simple to find after the Ford-Fulkerson algorithm has been completed. Simply mark the arcs that are carrying a flow equal to their maximum flow capacity and look for a cut that consists of marked arcs and no other arcs. It often happens that all of the marked arcs are not used in the cut and it may also happen that there are multiple cuts-that can be made in this way. Not surprisingly, the cut is the same set of arcs whose flow capacities were forced to obey the Ford-Fulkerson algorithm, hence forcing the termination.

The heavy arcs on the figure are called the mini-cuts. These arcs are the bottlenecks that are restricting the maximum flow.

8.10 Residue Graph

The *residue graph* of a flow network is a graph which indicates additional possible flow. If there is a path from source to the sink in residual path, then it is possible to add flow. Every edge of a residue graph has a value called *residual capacity* which is equal to the original capacity of the edge minus the current flow. *Residual capacity* is basically the current capacity of the edge.

ILLUSTRATIVE EXAMPLES

Example 1: Find the maximum flow from node 1 to node 5 in the following unidirectional network. The label on the arcs indicates the flow capacity of the corresponding arc.

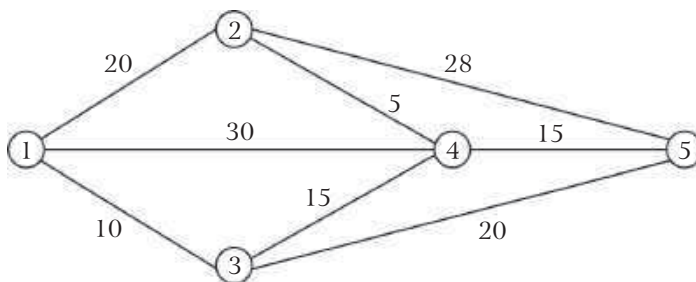


Fig. 8.28

Solution: We solve the problem by Ford-Fulkerson algorithm stepwise as follows:

Step 1: Consider the route 1 – 2 – 5.

In this route, the flow capacity in arc 1 – 2 is 20 and that in arc 2 – 5 is 28. Minimum of these is 20, so the maximum flow from node 1 to node 5 along this arc can be $f_1 = 20$ units. Here arc 1 – 2 having smallest flow capacity is the bottleneck in this route.

Step 2: Subtracting f_1 (i.e. 20) from the flow capacities of each arc in route 1 – 2 – 5, we get the following network (residue graph) in fig. 8.29.

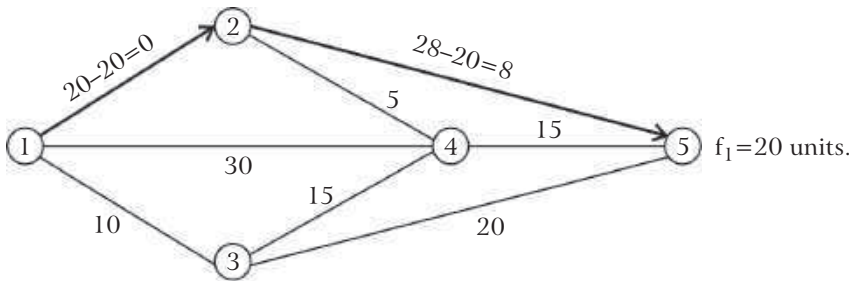


Fig. 8.29

Step 3: Consider the route 1 – 4 – 5. Then proceeding as in step 1 and 2, we get the following network (residue graph) fig. 8.30 and the maximum flow in this route is $f_2 = 15$ units. Here arc 4 – 5 is the bottleneck in this route.

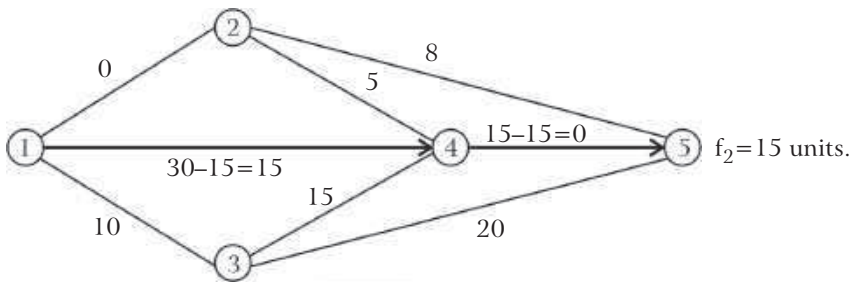


Fig. 8.30

Step 4: Now consider the route 1 – 3 – 5.

Then proceeding as in step 1 and 2, we get the following network (residue graph) fig. 8.31 and the maximum flow in this route is $f_3 = 10$ units.

Here arc 1 – 3, having smallest flow capacity is the bottleneck in this route.

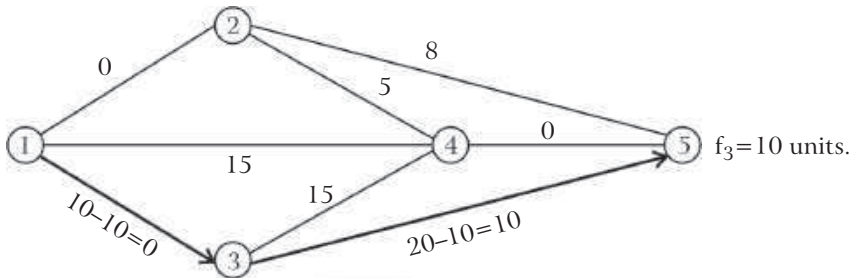


Fig. 8.31

Step 5: Now from the fig. 8.31, we note that we cannot consider any other route for flow from node 1 to node 5, as the capacity of flow of at least one arc in every route is exhausted (*i.e.* current capacity reaches to zero).

Thus the algorithm terminates summing up all the flows from step 1 to step 4 the pattern of flows and their directions which gives the maximum simultaneous flow is shown in fig. 8.32.

The arc labels in fig. 8.32 show the amount of flow in each arc. Note that the principle of flow conservation at every node is respected. For example, the flow entering the node 2 is 20 which is equal to the flow leaving node 2.

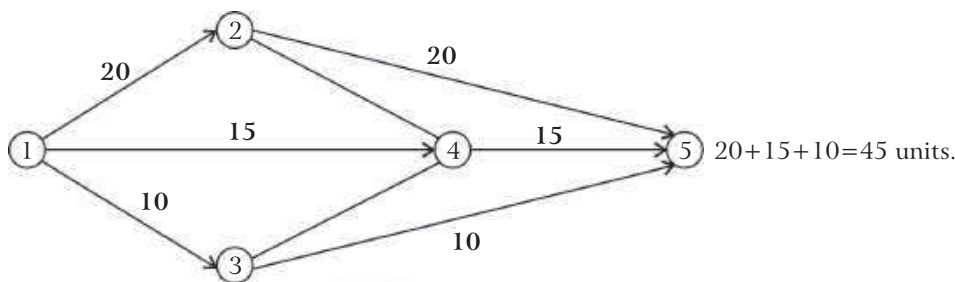


Fig. 8.32

The maximum total simultaneous flow from node 1 to node 5 is given by summing the flows on the 3 routes we selected and is equal to $20 + 15 + 10 = 45$ units.

At this stage, we cannot conclude that we have reached the end of calculations *i.e.* we cannot claim that this solution is optimal. It will be optimal only if this max. flow value is equal to the minimal capacity over all cuts that, when removed in the specified way from the network, causes the situation that no flow can pass from the source (node 1) to the sink (node 5). Minimal cut generally involve the bottlenecks in the network.

Some cuts can be identified as follows:

S.No.	Cut Set	Capacity
1.	1 - 2, 1 - 3, 1 - 4	$20 + 10 + 30 = 60$
2.	2 - 5, 3 - 5, 4 - 5	$28 + 20 + 15 = 63$
3.	1 - 2, 1 - 3, 4 - 5	$20 + 10 + 15 = 45$ (minimum)

Minimum capacity over these cuts is 45 (on cut 3) and is equal to the max flow from node 1 to node 5. Hence, by *max-flow min-cut theorem* the maximum flow from node 1 to node 5 is 45 units.

Example 2: Find the maximum flow from A to G in the following network. In the network, the labels on the arc indicate the flow capacities in both directions. For example, on the arc AB the label near node A indicates the flow capacity in A to B direction and that near the node B indicates the flow capacity in B to A direction.

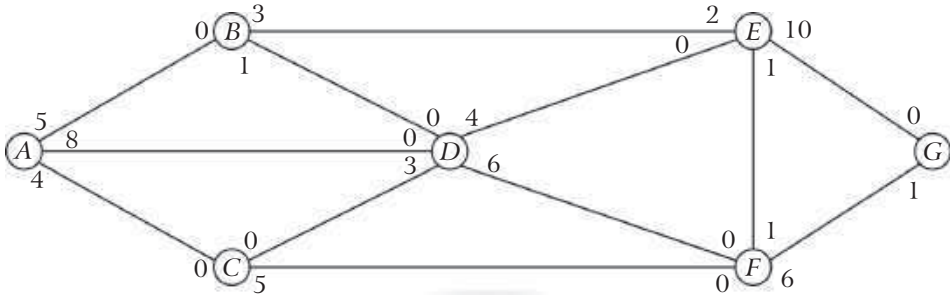


Fig. 8.33

Solution: For the solution of the problem by Ford-Fulkerson algorithm, we proceed stepwise as follows:

Step 1: Consider the route $A - D - E - G$, from origin A to destination G.

Step 2: In this route, the flow capacity in arc AD is 8, in arc DE is 4 and in arc EG is 10. Minimum of these capacities is 4. So the maximum flow f_1 along this path can be 4 units only, i.e. max. flow along this route $A - D - E - G$ is $f_1 = 4$ units. Here, arc $D - E$ having smallest flow capacity, is the bottleneck arc in this route.

Step 3: Subtracting f_1 (i.e. 4) from the flow capacities in the forward direction and adding to flow capacities in the backward direction of each arc of the route $A - D - E - G$, we get the following network (residue graph) in fig. 8.34.

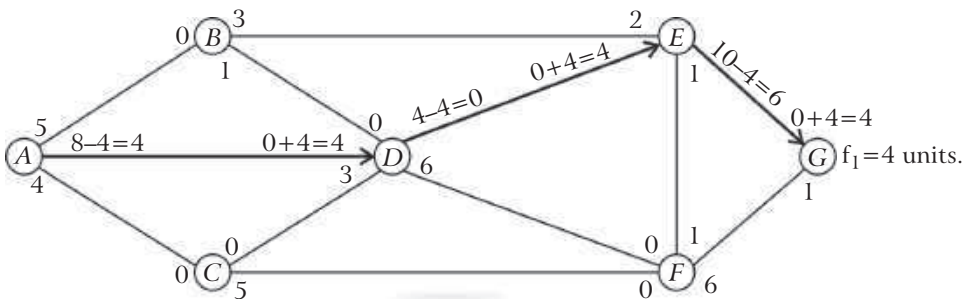


Fig. 8.34

Step 4: Now consider the route $A - B - E - G$. (see fig. 8.35).

Then proceeding as in step 2 and 3, we get the following network (residue graph) fig. 8.35 and max. flow in the route $f_2 = 3$ units. Here arc BE is the bottleneck arc.

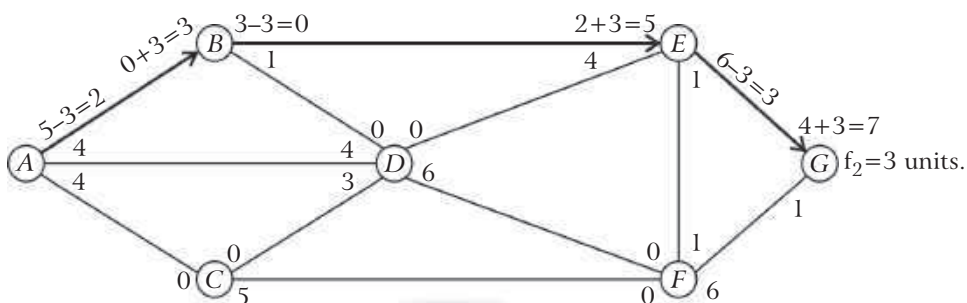


Fig. 8.35

Step 5: Now consider the route $A - C - F - G$ (see fig. 8.36). Then proceeding as in step 2 and 3, we get the following network (residue graph) fig. 8.36 and max. flow in this route $f_3 = 4$ units. Here arc AC is the bottleneck arc.

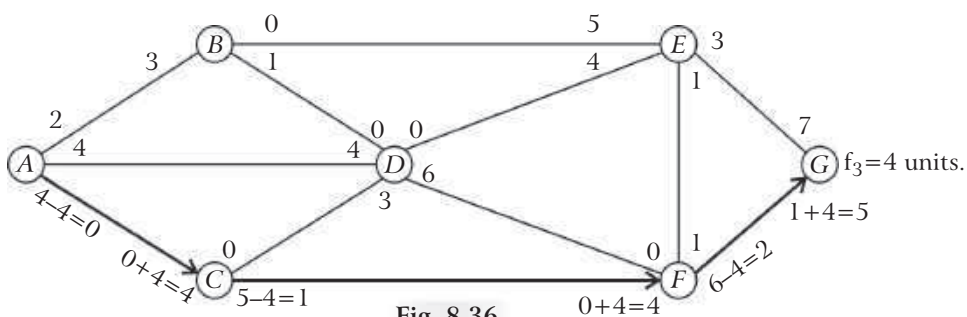


Fig. 8.36

Step 6: Consider the route $A - D - F - G$ (see fig. 8.37). Then proceeding as in steps 2 and 3, we get the following network (residue graph) fig. 8.37 and max. flow in this route $f_4 = 2$ units.

Here arc FG is the bottleneck arc.

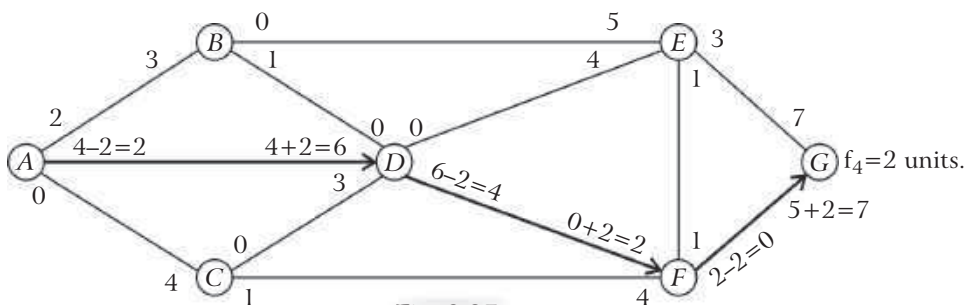


Fig. 8.37

Step 7: Now consider the route $A - D - F - E - G$ (see fig. 8.38). Then proceeding as in step 2 and 3, we get the following network (residue graph) fig. 8.38 and max. flow in this route $f_5 = 1$ unit.

Here arc FE is the bottleneck arc.

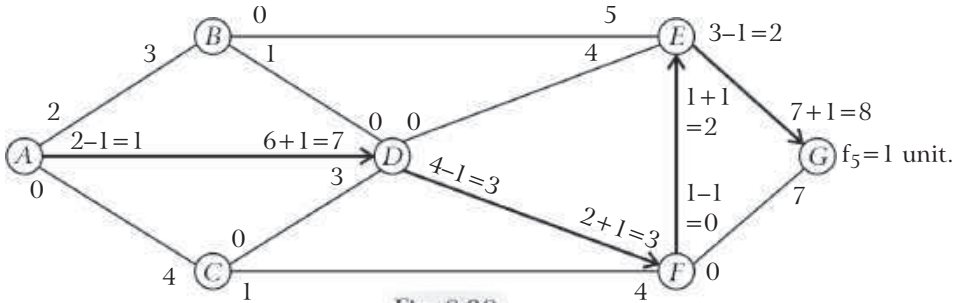


Fig. 8.38

Step 8: Now from the fig. 8.38, we note that we can not consider any other route for the flow from A to G, as the capacity of flow of at least one arc in every route is exhausted current capacity (*i.e.* reduces to zero). Thus the algorithm terminates.

Summing up all the flows from step 1 to step 7, the pattern of flows and their directions which gives the maximum simultaneous flow are shown in fig. 8.39. The arc labels in fig. 8.39 show the amount of flow in each arc. Note that the principle of flow conservation at every node is respected. For example, the flow entering the node E total $3 + 4 + 1 = 8$ which is equal to the flow leaving node E.

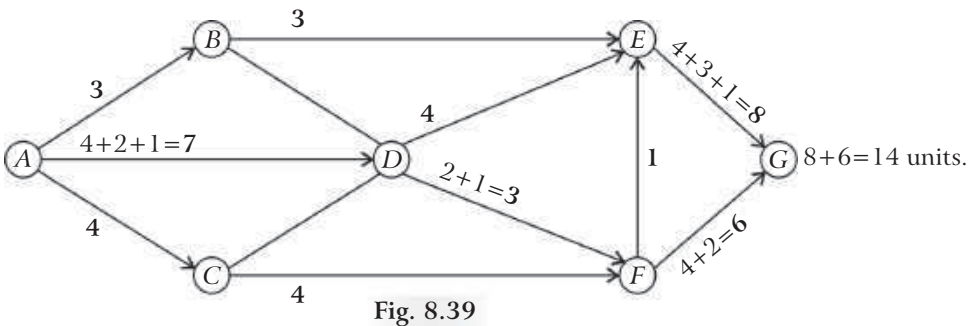


Fig. 8.39

The maximum total simultaneous flow from A to G is given by summing the flows on the 5 routes we selected and it is equal to $4 + 3 + 4 + 2 + 1 = 14$ units.

At this stage, we cannot conclude that we have reached the end of calculations *i.e.* we cannot claim that this solution is optimal. It will be optimal only if this max flow value is equal to the minimal capacity over all cuts that, when removed in the specified way from the network, causes the situation that no flow can pass from the source to the sink. Minimal cut generally involve the bottlenecks in the network.

Some such cuts are shown in figure 8.40.

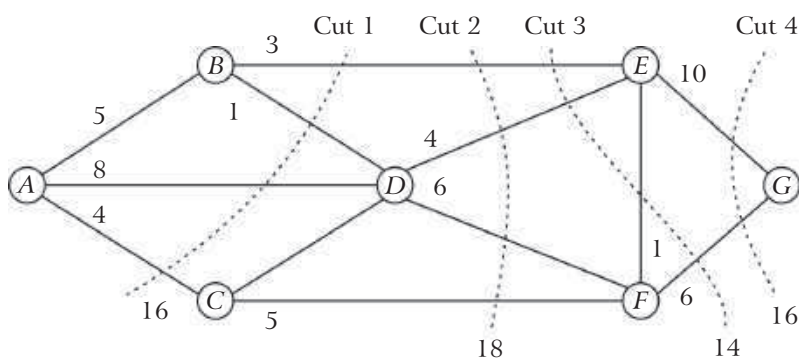


Fig. 8.40

Minimum capacity over these cuts is 14 (on cut 3) and is equal to the max. flow from A to G. Hence by max-flow min-cut theorem the maximum flow from A to G is 14 units.

Exercise

Find the minimal spanning trees of the networks in Q. 1 to Q. 4.

1.

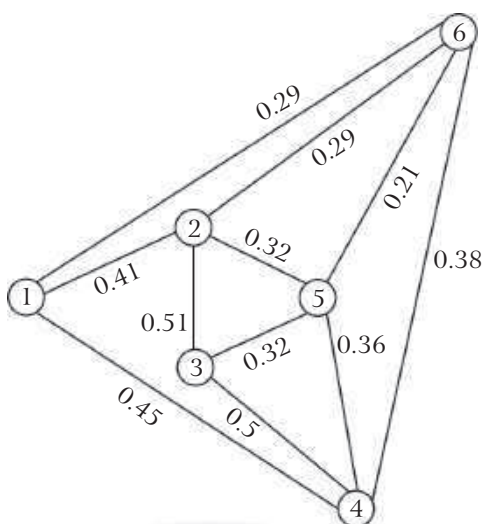


Fig. 8.41

2.

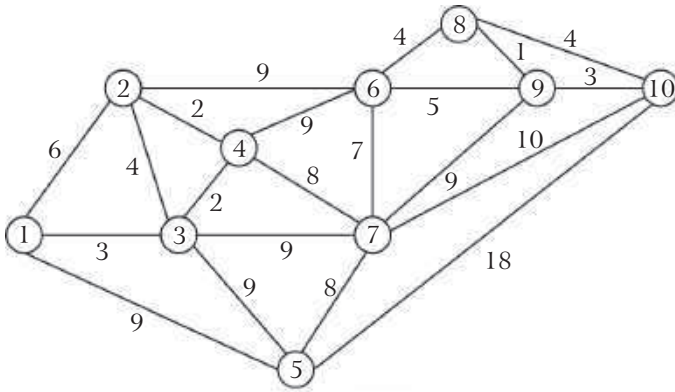


Fig. 8.42

3.

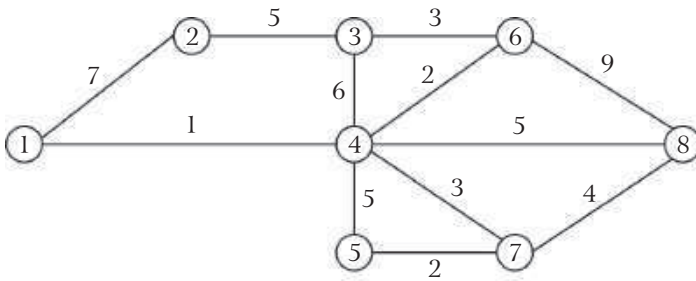


Fig. 8.43

4.

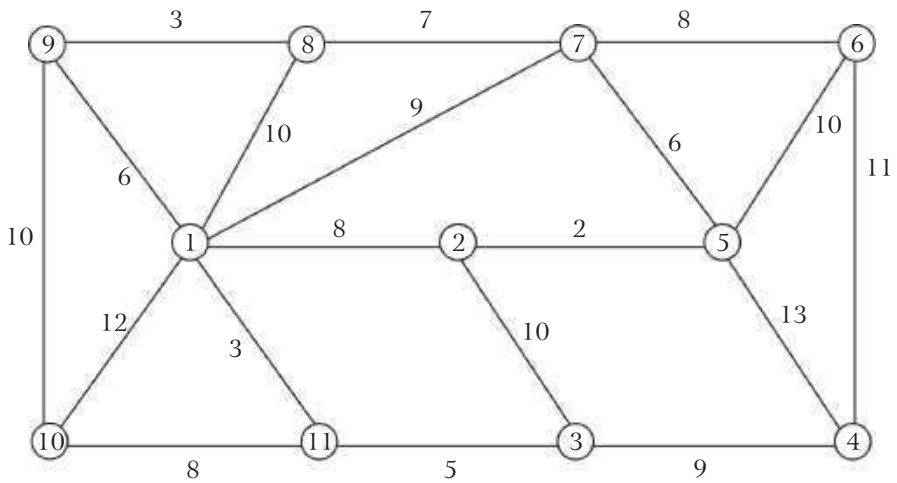


Fig. 8.44

5. Suppose that it is desired to establish a cable communication network that links the seven major cities shown in following figure 8.45. The label on each branch represents the distance between the corresponding cities in kilometers. Determine how the cities be connected such that the total used cable length is minimized.

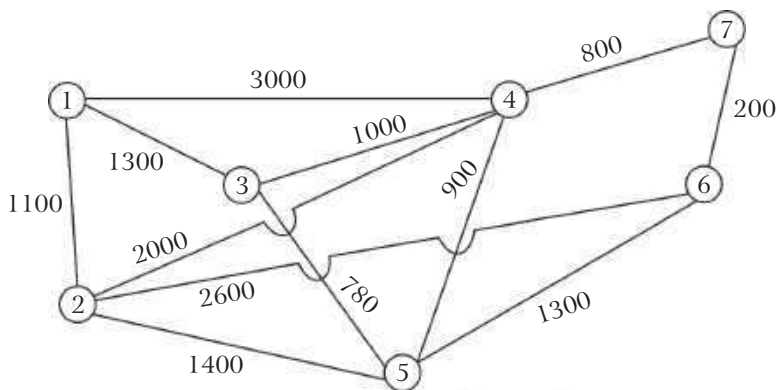


Fig. 8.45

6. The mid west TV cable company is in the process of planning a network for providing cable TV service to five new housing development areas. The cable system network is summarised in figure 8.46. The numbers associated with each branch represent the kilometers of cable needed to connect any two locations. Node 1 represents the cable TV station and the remaining nodes (2 through 6) represent the five development areas. A missing branch between two nodes implies that it is prohibitively expensive or physically impossible to connect the associated development areas. It is required to determine the links that will result in the use of minimum cable kilometers while guaranteeing that all areas are connected (directly or indirectly) to the cable TV station.

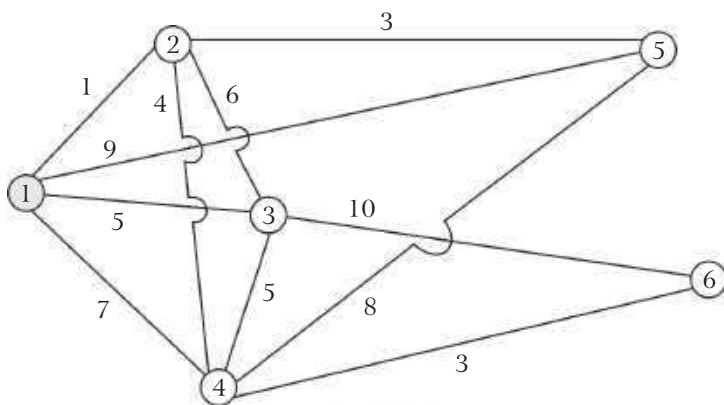


Fig. 8.46

7. Find the minimal spanning tree of the network in figure 8.46 if nodes 5 and 6 are also linked by a 2 kilometre cable.

8. Find the shortest route from node 'a' to 'i' by using graphical method. The numbers on the links represent the distance in kilometres.

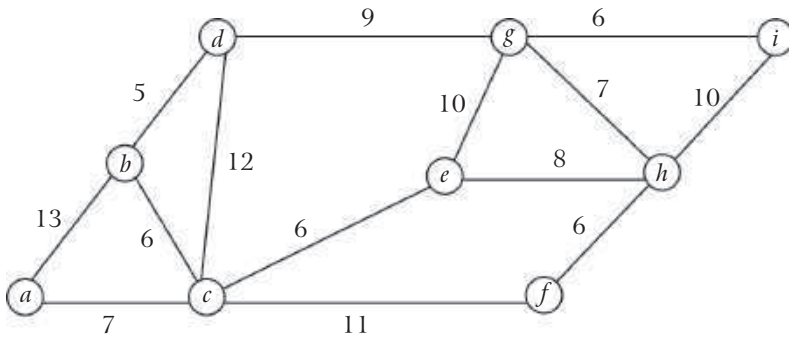


Fig. 8.47

9. Compute the shortest distance and its designated route from the starting node to the terminal node for the network given in Q.N.9 to 13

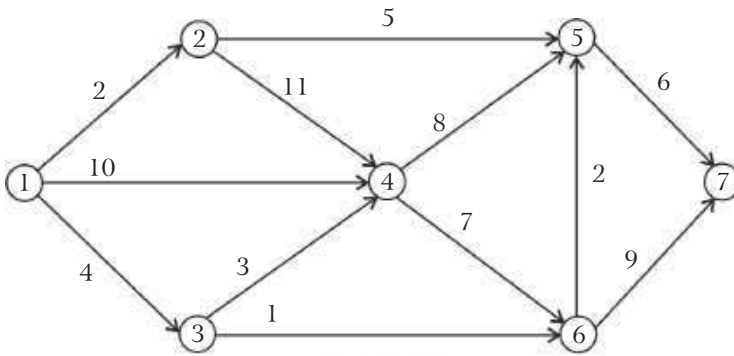


Fig. 8.48

10.

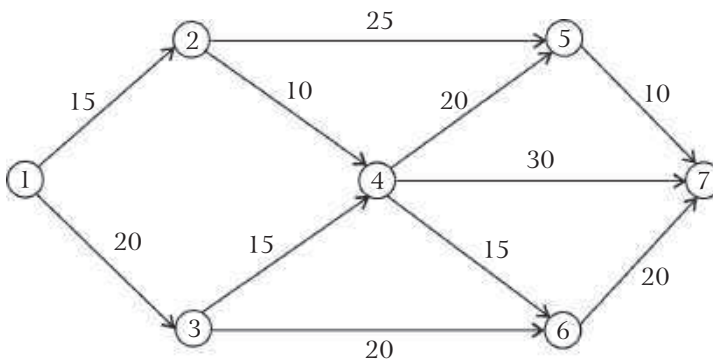


Fig. 8.49

11.

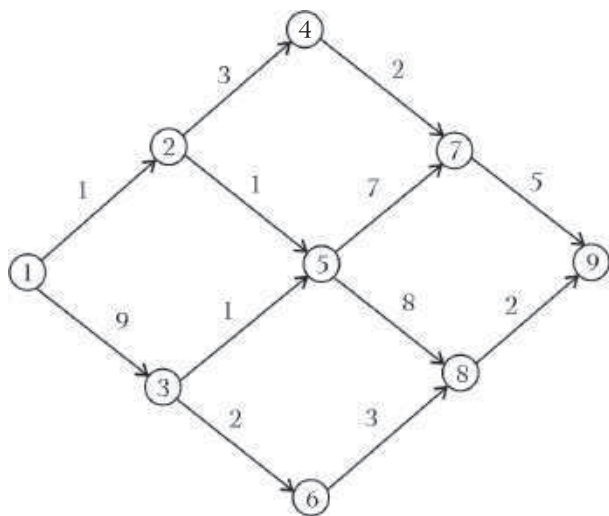


Fig. 8.50

12.

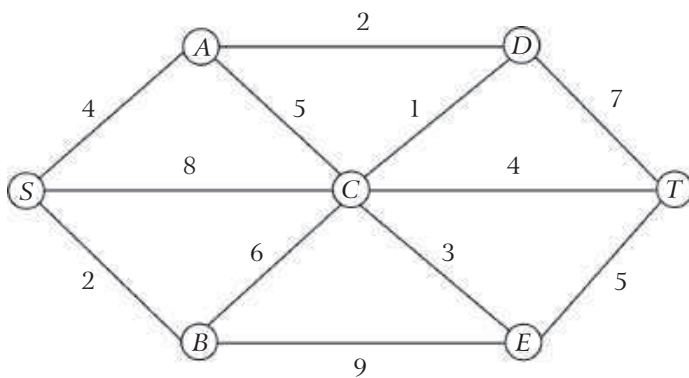


Fig. 8.51

13.

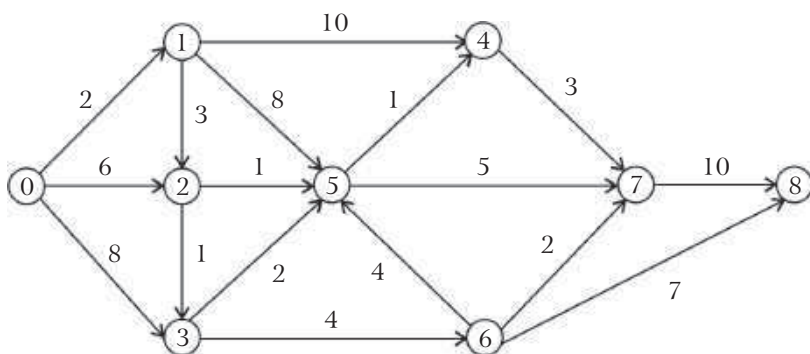


Fig. 8.52

14. Find the maximum flow from source 0 to the sink 5 in the following network. The entries on the branches indicates the flow capacity of the corresponding branch.

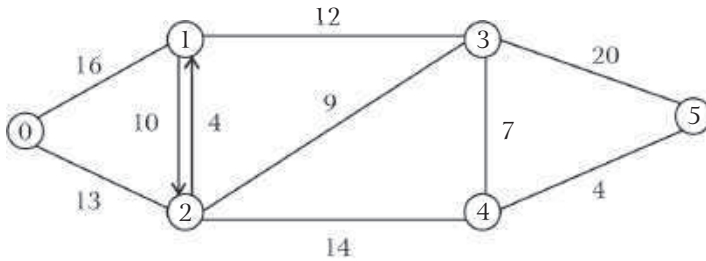


Fig. 8.53

[Hint: 12 units flow in the route $0-1-3-5$, 7 units flow in the route $0-2-4-3-5$ and 4 units flow in the route $0-2-4-5$, 1 unit flow in the route $0-2-3-5$ or $0-1-2-3-5$].

15. Find the maximum flow from the source 1 to the sink 8 in the following network. The entries on the branches indicates the flow capacity of the corresponding branch.

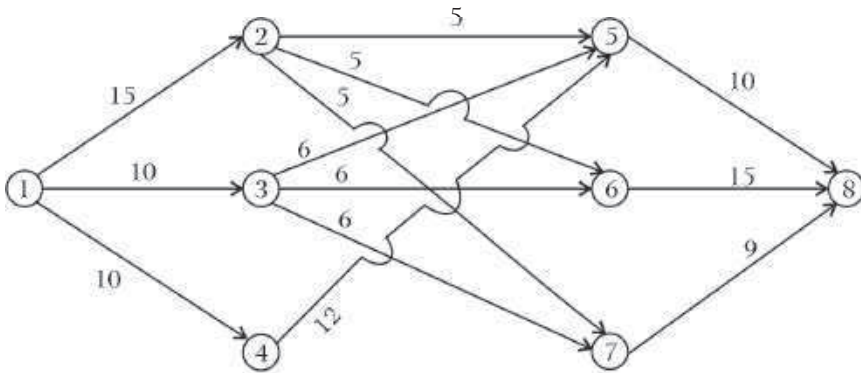


Fig. 8.54

[Hint: 10 units flow in the route $1-4-5-8$, 5 units flow in the route $1-2-6-8$, 5 units flow in the route $1-2-7-8$, 6 units flow in the route $1-3-6-8$ and 4 units flow in the route $1-3-7-8$.]

Answers

1.	All nodes and branches 1-6, 2-6, 3-5, 4-5, 5-6, minimum length = 1.47.
2.	All nodes and branches, 1-3, 3-4, 4-2, 4-7, 7-5, 7-6, 6-8, 8-9, 9-10, minimum length = 38.
3.	All nodes and branches 1-4, 4-6, 6-3, 3-2, 4-7, 7-5 and 7-8, minimum length = 20.
4.	All nodes and branches 1-9, 1-11, 2-5, 5-7, 6-7, 7-8, 8-9, 3-4, 3-11, 10-11, minimum length = 57.
5.	Cities 1-2, 1-3, 3-5, 4-5, 4-7, 6-7, minimum length of the cable = 5080 kilometres.
6.	Connect, 1-2, 2-5, 2-4, 4-6, 1-3 or 1-2, 2-5, 2-4, 4-6, 3-4, minimum cable length = 16 kilometres.
7.	Connect, 1-2, 2-5, 5-6, 4-6, 1-3 or 1-2, 2-5, 5-6, 4-6, 3-4, minimum cable length = 14 kilometres.
8.	Shortest route $a \rightarrow c \rightarrow e \rightarrow g \rightarrow i$, S.D. = 29 km.
9.	Shortest route $1 \rightarrow 2 \rightarrow 5 \rightarrow 7$ or $1 \rightarrow 3 \rightarrow 6 \rightarrow 5 \rightarrow 7$, S.D. = 13.
10.	Shortest route $1 \rightarrow 2 \rightarrow 5 \rightarrow 7$, S.D. = 50.
11.	Shortest route $1 \rightarrow 2 \rightarrow 4 \rightarrow 7 \rightarrow 9$, S.D. = 11.
12.	Shortest route $S \rightarrow A \rightarrow D \rightarrow C \rightarrow T$, S.D. = 11.
13.	Shortest route $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 8$, S.D. = 17 km.
14.	24 units.
15.	30 units.

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter (a), (b), (c) or (d).

1. A spanning tree contains:
(a) At least one vertex (b) At least two vertices.
(c) All vertices (d) None of these.
 2. The total number of spanning trees of a graph (networks) with n vertices is:
(a) 2^{n-1} (b) 2^{n-2}
(c) n^{n-2} (d) None of these.
 3. In a minimal spanning tree of a network, the sum of lengths of its branches is:
(a) Minimum (b) Maximum
(c) Any of (a) or (b) (d) None of these.
 4. The maximal flow model was first formulated by T.E. Harris and T.S. Ross in:
(a) 1950 (b) 1954
(c) 1960 (d) 1964
 5. The number of different minimal spanning trees of a network with distinct labels on its branches is:
(a) One (b) At least one
(c) At the most one (d) None of these.
 6. A connected directed graph with no cycles is called a:
(a) Tree (b) Spanning tree
(c) Minimal spanning tree (d) None of these.
 7. A tree that includes all the vertices is called a:
(a) Minimal spanning tree (b) Spanning tree
(c) Net tree (d) None of these.
-

8. The shortest route from node 1 to 7 in the following network is:

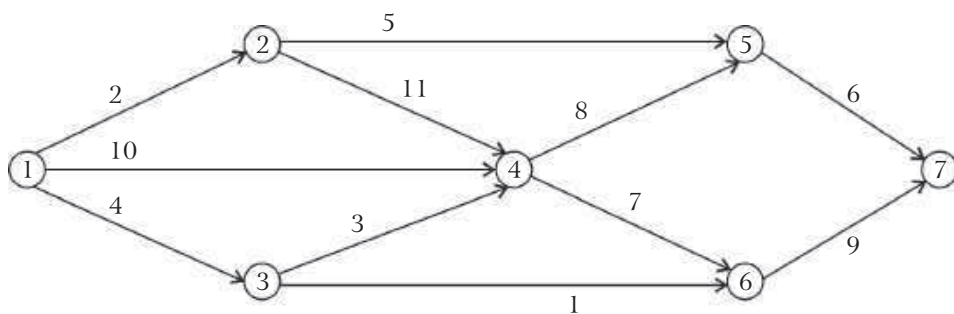


Fig. 8.55

- (a) $1 \rightarrow 2 \rightarrow 5 \rightarrow 7$ (b) $1 \rightarrow 3 \rightarrow 6 \rightarrow 7$
 (c) $1 \rightarrow 4 \rightarrow 5 \rightarrow 7$ (d) $1 \rightarrow 3 \rightarrow 4 \rightarrow 6 \rightarrow 7$
9. In the network given in fig. 8.55, the length of the shortest route from node 1 to 7 is:
- (a) 14 (b) 13
 (c) 21 (d) None of these.
10. In the network given in fig. 8.55, the node 4 is connected to node 7 by an are of length 5, then the shortest route from node 1 to node 7 is:
- (a) $1 \rightarrow 2 \rightarrow 5 \rightarrow 7$ (b) $1 \rightarrow 4 \rightarrow 7$
 (c) $1 \rightarrow 3 \rightarrow 4 \rightarrow 7$ (d) None of these.

Fill in the Blank

Fill in the blanks “.....”, so that the following statements are complete and correct.

- In a network minimization problem it is not optimum to loops (cycles).
- A tree is connected undirected graph with no
- A minimal spanning tree of a network contain all the of the network.
- The shortest route from node A to node F in the following network is

$$A \rightarrow C \rightarrow E \rightarrow D \rightarrow F.$$

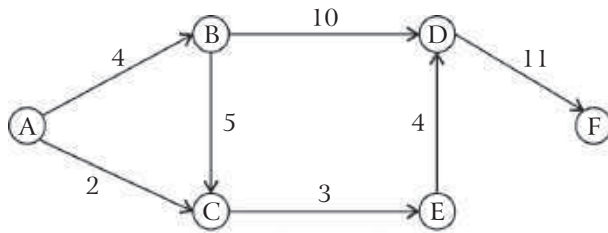


Fig. 8.56

5. In the network given in fig. 8.56, the length of shortest route from node *A* to node *F* is

True/False

Write 'T' for True and 'F' for False statement.

1. A tree may include a loop.
2. A spanning tree may include a loop.
3. A minimal spanning tree may include a loop.
4. An acyclic network does not include any loop.
5. The maximum flow between two nodes in a network may exceed the capacity of its minimal cut.

Answers

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1.	(c)	2.	(c)	3.	(a)
4.	(b)	5.	(a)	6.	(a)
7.	(b)	8.	(a)	9.	(b)
10.	(c)				

Fill in the Blank

1.	include	2.	cycle	3.	vertices
4.	C	5.	20		

True/False

1.	F	2.	F	3.	F
4.	T	5.	F		

○○○

UNIT-III

Chapter

9

Project Management (PERT/CPM)

9.1 Introduction

In the past the scheduling of a project was done with little planning. The growing complexities of today's projects have demanded more systematic and more effective planning techniques with the objective of optimizing the efficiency of executing the project. Project management has evolved a new field with the development of two analytic techniques for planning, scheduling and controlling of projects. These techniques are the **critical path method (CPM)** and **project evaluation and review technique (PERT)**. Both are basically time oriented methods in the sense that they both lead to the determination of a time schedule.

9.2 Difference between CPM and PERT

CPM and PERT are almost as a single technique and the difference between the two is only of the historical or academic interest. However the main points of difference between them are as follows:

C.P.M. :

1. **CPM is an activity oriented technique, i.e.,** CPM network emphasis more on the descriptions associated with activities.
2. **CPM is a deterministic model, i.e.** it does not take into account the uncertainties involved in the estimation of time for execution of a job or an activity.

3. **CPM** is commonly used for projects involving activities of repetitive nature.
4. **CPM** give dual emphasis to time and cost and is suitable to evaluate the trade off between project cost and project time. With additional resources, it allows the project manager to manipulate and shorten the project time within certain limits at an optimal cost.

PERT :

1. **PERT** is *event oriented*, i.e., PERT network emphasis more on the descriptions associated with events.
2. **PERT** is a probabilistic model. Here, the expected duration of each activity is probabilistic.
3. **PERT** is generally used for one time projects involving activities of non-repetitive nature.
4. **PERT** give emphasis to time only. It helps the project managers to schedule and co-ordinate various activities so that the project is completed on prescribed time.

9.3 Terminology used in PERT/CPM Network Analysis (Network Components)

PERT/CPM network consists of two major components.

1. **Activity** : An *activity* is a task (or operation or item of work) to be done that consumes time and resources. In a network, activity is represented by an arrow, the tail of which represents the start and the head represents the finish (or end) of the activity.
2. **Events** : The beginning and end points of an activity/activities are called *events* or *nodes*. Event is a particular instant in time and does not consume any time or resource. It is generally represented by a numbered circle. It may also be represented by a rectangle, square, hexagon or some other geometric shape.

For example, an activity (i, j) represented by an arrow $\rightarrow$ extends between two nodes i and j , the tail node (or event) i , called the *preceding event* represents the start of the activity and the head node j , called the *succeeding event* represents the completion of the activity. This activity is shown in the following figure.

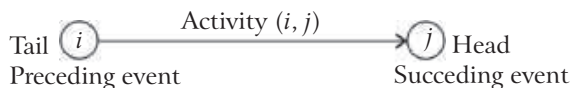


Fig. 9.1

For an activity (i, j) , the arrow is always from left to right and $i < j$ i.e., the number of head event is always higher than number of tail event.

The events can be classified by the following three categories :

- (i) **Merge Event** : An event which represents the end (or head) of more than one activity is called a *merge event* [see fig. 9.2 (a)].
- (ii) **Burst Event** : An event which represents the beginning (or tail) of more than one activity is called a *burst event* [see fig. 9.2 (b)].
- (iii) **Merge and Burst Event** : An event which represents the beginning (or tail) of some activity (or activities) and is also the end (head) of some activity (or activities) is called a *merge and burst event* [see fig. 9.2 (c)].

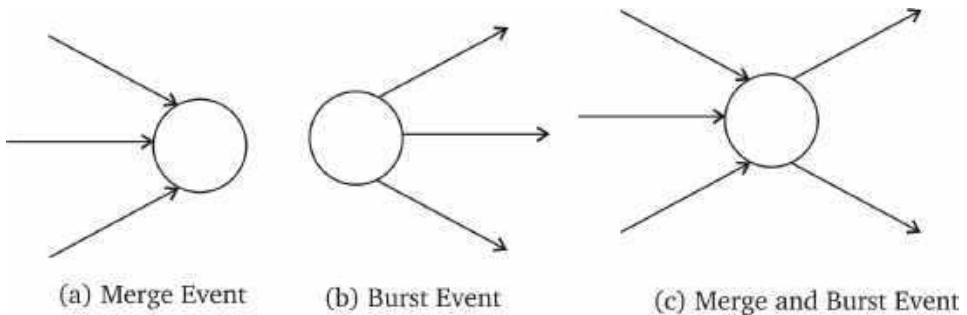


Fig. 9.2

9.4 Construction of the Network Diagram

The following questions will generate the necessary information for the construction of a network diagram.

1. Which activities (or operations) must be completed immediately before each activity (or operation) ? Such activities are called *predecessor (or preceding) activities*.
2. Which activities (or operations) can be carried out simultaneously ?
3. Which activities (or operations) follow other given activities (or operations) such activities are called *successor activities* ?

For the construction of a network diagram the following rules are followed :

Rule 1 : Each activity (or operation) is represented by one and only one arrow in the network.

This rule implies that none of the operation can be represented more than once in the network.

Rule 2 : No two activities can be identified by the same end events (nodes).

If two activities have the same end events then we introduce a **dummy activity** between end events of the activities. It should be noted that a dummy activity does not consume any time or resources.

9.5 Dummy Activity

An activity which is used simply to represent a connection between events but does not consume any time or any kind of resource is called a **Dummy Activity**. For example, consider the network, where activities B and C have activity A as their immediate preceding activity and D is the activity that is successor to both activities B and C . One can draw the above network as in fig. 9.3.

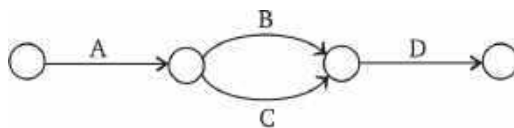


Fig. 9.3

In this figure two activities B and C have the same end events, which is against the rule 2. This drawback is removed by using a dummy activity D_1 , and the correct network may be drawn as in fig. 9.4 (a) or 9.4 (b).

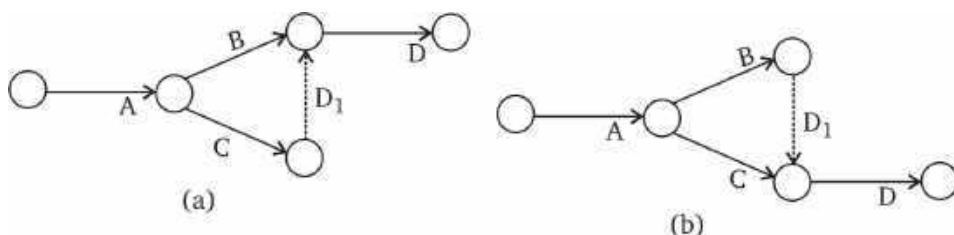


Fig. 9.4

9.6 Errors in Drawing a Network

The errors that are most commonly observed in drawing a network are as follows :

1. **Looping (or cycling) Error** : If some activities in a network form a loop (or cycle), then such error in a network is called *looping (or cycling)* error, as shown in the following figure :

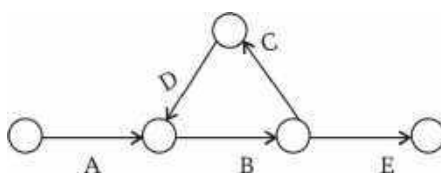


Fig. 9.5

In the above network activities *B*, *C* and *D* form a loop (cycle). Thus, the network cannot proceed. Such situations can be avoided by checking the precedence relationship of the activities.

2. **Dangling Error** : If in a network, some activity (or activities) other than the final activity (or activities) do not have any successor activity, then it is called a *dangling error* in the network. Such end-events, other than the end event of the project are called *dangling events*. Thus, in a network all events except the first and the last of the whole project must have at least one entering (predecessor) and one leaving (successor) activity.

In the network shown in figure 9.6 (a) the activity *D* leads to dangling. To avoid this dangling we introduce a dummy activity between the end event of this activity and the end event of the project. See fig. 9.6 (b) which is free of any dangling error.

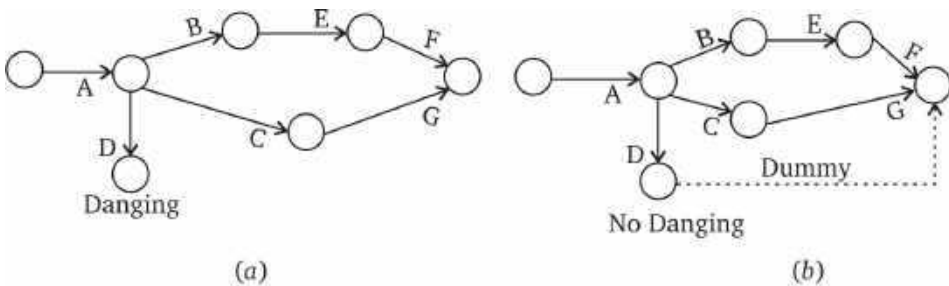


Fig. 9.6

3. **Redundancy Error** : If in a network diagram, dummy activity is the only activity emanating from an event, then it is called *redundancy error*. This redundant dummy activity can be eliminated and the network is redrawn having no redundancy error.

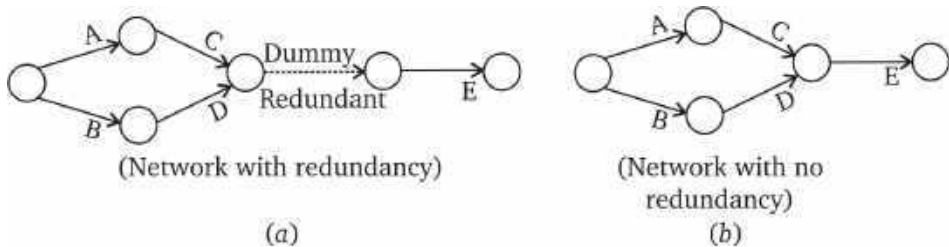


Fig. 9.7

9.7 General Procedure for the Construction of a Network Diagram

In general the following procedure is followed for the construction of a network diagram.

1. First of all prepare a list of operations (activities), their predecessor and successor operations (activities).

2. In the first column draw a circle, which represents the initial event (or node).
3. In the second column (vertical line) list (by using circles) all the operations (activities) that can be started immediately. Draw directed arrows from the initial event to these circles (events) to represent the starting activities.
4. In the third column list the operations (activities) that can be started immediately after the completion of the second column operation/operations.

Draw directed lines (arrows) between the circles (nodes) of second and third columns to indicate which operation follow other operation (activity) or operations (activities).
5. Label each circle drawn to represent the description of the operation.
6. Proceed till all operations are completed.
7. After completing the preliminary draft including all operations check for logical arrangements.
8. Try to avoid arrows (as much as possible) which cross each other.
9. Use arrows from left to right. Vertical arrows may be used if necessary.
10. After getting the final draft place consecutive number in each circle by the rule discussed in § 9.8. It may be checked that no circle will follow a circle with large number.

9.8 Numbering the Events (Fulkerson's Rule)

After the network is drawn, every event is assigned a number. The number sequence should reflect the flow of the network. These numbers are placed inside the node circle. **D.R. Fulkerson** devised a rule for the numbering of events, which consists of the following steps :

Step 1 : The initial event (node) which has all outgoing arrows with no incoming arrow in numbered '1'.

Step 2 : Delete all arrows coming out of event (node) 1. Thus, some more events (at least one) will be converted to initial events (events having no incoming arrows). Assign number 2, 3, to these events (nodes).

Step 3 : Delete all arrows coming out of the events (nodes), which are numbered in previous step 2. Thus, some more events (at least one) will be converted to initial events. Assign next numbers (in continuation) to these events.

Step 4 : Continue up to the final (end or terminal) event (node), which has no arrow (or arrows) coming out of it. Assign the next (last) number to this event.

Step 5 : Check that the numbering of events should be such that, for each activity, the number of its head event in larger than the number of its tail event.

For the clear understanding of the numbering technique, consider the following network shown in fig. 9.8.

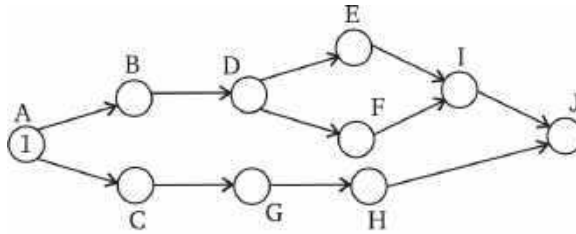


Fig. 9.8

In the above diagram A is the initial event and is numbered 1.

Deleting all arrows coming out of the event (node) 1, we get the following diagram (fig. 9.9) in which events B and C are converted to initial events, as now there is no arrow entering B and C. So we number these two events as next numbers 2 and 3.

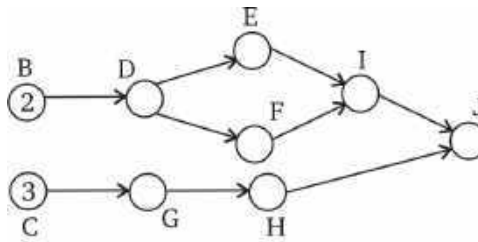


Fig. 9.9

Deleting all arrows coming out of the events (nodes) 2, 3, in the above diagram, we get the following diagram (fig. 9.10) in which events D and G are converted to initial events, as now there is no arrow entering D and G. So we number these two events as next number 4 and 5.

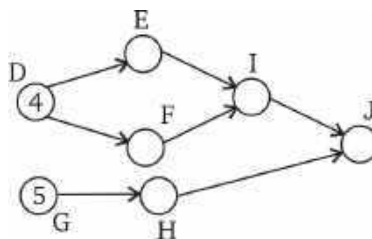


Fig. 9.10

Deleting all arrows coming out of the events (nodes) 4 and 5 in the above diagram, we get the following diagram (fig. 9.11) in which E, F and H are converted to initial events. So we number these events as next numbers 6, 7 and 8.

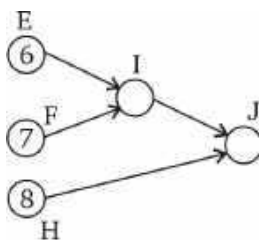


Fig. 9.11

Deleting all arrows coming out of the events (nodes) 6, 7 and 8 in the above diagram, we get the following diagram (fig. 9.12) in which event I is converted to initial event and J is the final (end) event. So we number event I as next number 9 and last event J as next number 10.



Fig. 9.12

Hence, the final diagram in which all events are numbered is as follows, (see fig. 9.13) in which for each activity (I, J), $I < J$.

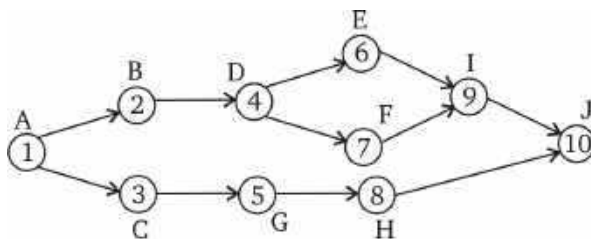


Fig. 9.13

ILLUSTRATIVE EXAMPLES

Example 1: Draw a network diagram for the following set of operations represented by separate letters.

Operations	Post-operations
A	precedes B, C
B	precedes D, E
C	precedes D
D	precedes F
E	precedes G
F	precedes G

Solution: The operation A can be started immediately as there is no operations preceding the operation. Therefore, this operation A is placed in the first column. After the completion of operation A, the operations B and C can be started immediately. Therefore, we list the operations B and C in the second column and draw the arrows shown in fig. 9.14.

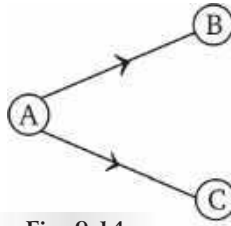


Fig. 9.14

The operation *B* precedes the operations *D*, *E* and operation *C* precedes the operation *D*. It follows that after completing the operation *B* and *C* the operation *D* and *E* can be started immediately. Therefore, these operations are listed in the third column. The arrows are drawn to indicate which operation follows the other operation as shown in fig. 9.15.

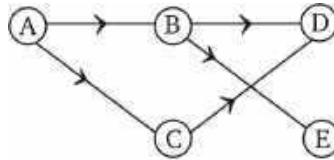


Fig. 9.15

The operation *D* precedes the operation *F*, therefore we list the operation *F* in fourth column. Again the operation *E* and *F* precedes the operation *G*, therefore we list operation *G* in fifth column. The arrows are drawn to indicate which operation follows the other operations. Thus, we get the preliminary draft as shown in fig. 9.16.

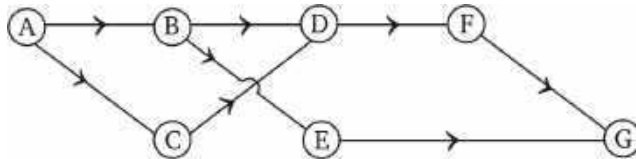


Fig. 9.16

The arrows *CD* and *BE* cross each other. To avoid this crossing we draw circle *E* above *D* in the third column. The circles are numbered according to the method described in § 9.8. Thus, we get the final network diagram as shown in fig. 9.17, which is the A.O.N. (Activity-on node) diagram of the project.

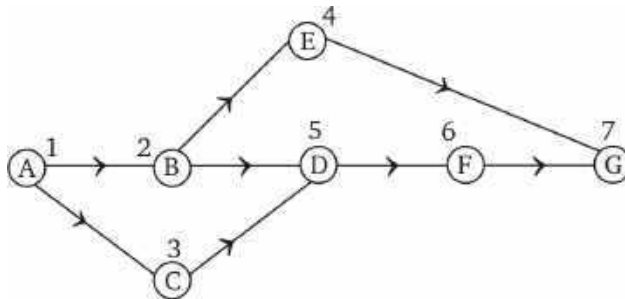


Fig. 9.17

Note : To avoid crossing of arrows CD and BE we may also draw circle C above B in second column in place of drawing circle E above D in third column.

The E.O.N. (**event-on node diagram**) of the project may be drawn as follows :

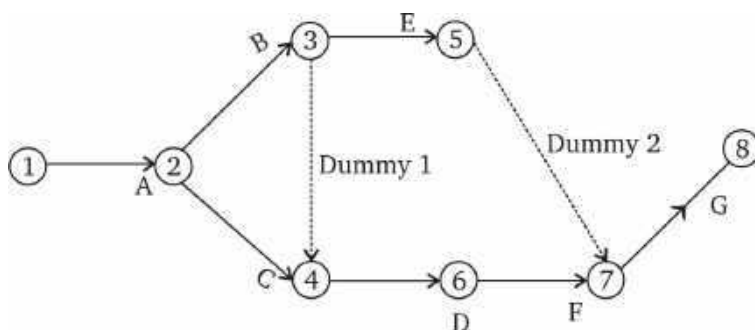


Fig. 9.18

In the above diagram two dummy activities (Dummy 1 and Dummy 2) are used to represent the predecessor and successor (post) operations (activities).

Example 2: A new type water pump is to be designed for an automobile. Prepare a network diagram from the following table showing the major activities for effective control of the project and the preceding activities.

Activity	Description	Preceding activities
A	Drawing prepared and approved	—
B	Cost analysis	A
C	Tool feasibility (economics)	A
D	Tools manufactured	C
E	Favourable cost	B, C
F	Raw materials procured	E
G	Sub-assemblies ordered	E
H	Sub-assemblies received	G
I	Parts manufactured	D, F
J	Final assembly	H, I
K	Testing and shipment	J

Solution: In the network diagram of this example the circles (called nodes) represent the points of completion of one activity and the beginning of the new ones.

- (i) The activity A has no preceding activity, so is represented by arrowed line as shown in fig. 9.19 .



Fig. 9.19

- (ii) The activities B and C can be done simultaneously and both are preceded by activity A. Therefore, they may be shown as in fig. 9.20.

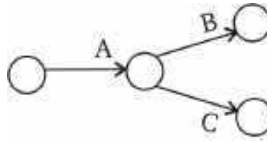


Fig. 9.20

- (iii) Activity C precedes activity D. The activity E can be completed only after the completion of activities B and C. It may easily be seen that dependence of activity E (favourable cost) of the activity C (economy of tooling) is from a technical point of view and does not consume any resource. Hence, the dummy activity is shown by a dotted line in fig. 9.21.

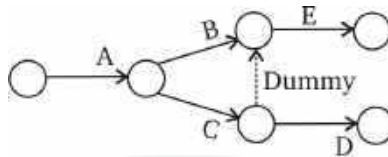


Fig. 9.21

Proceeding similarly we get the following preliminary draft as shown in fig. 9.22.

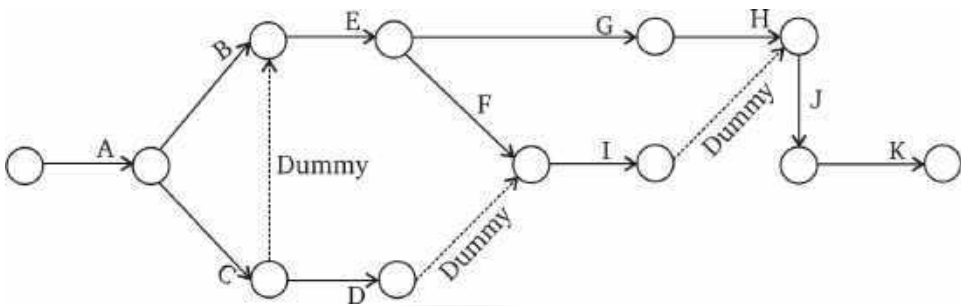


Fig. 9.22

In the above draft we have three Dummy activities. Now we proceed to reduce the Dummy activities to a minimum. Thus, we get the following final network in (**activity on arrow network**), having only one Dummy activity, as shown in fig. 9.23, in which the events (circles) are also numbered.

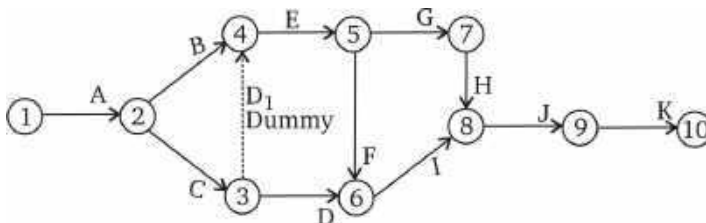


Fig. 9.23

The **activity on node diagram** of problem may be drawn as shown in figure 9.24.

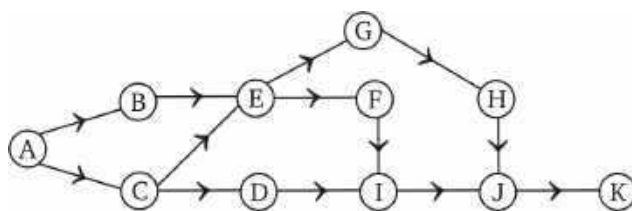


Fig. 9.24

Example 3: Following is a list and description of activities for a project XYZ building a Boat. Draw the network.

Activity	Preceding Activity	Activity Description
A	None	Design Hull
B	None	Prepare Boat Shed
C	A	Design Mast and Mast Mount
D	A	Obtain Hull
E	A	Design Sails
F	C	Obtain Mast Mount
G	C	Obtain Mast
H	C	Design Rigging
J	B, D	Prepare Hull
K	F, J	Fit Mast Mount to Hull
L	E, H, G, K	Step Mast
M	E, H	Obtain Sails and Rigging
N	L, M	Fit sails and Rigging

Solution: In the network diagram of this example the circles (called nodes) represent the points of completion of one activity and the beginning of the new ones.

- The activities A and B have no preceding activities.
- The activities C, D and E have activity A as preceding activity while activities F, G and H have activity C as preceding activity.
- The activity J, have activities B and D as preceding activities while activity K have F and J as preceding activities.

Therefore, they may be shown as in fig. 9.25.

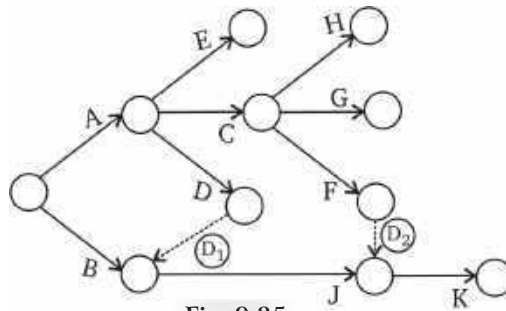


Fig. 9.25

Proceeding similarly we get the following preliminary draft as shown in fig. 9.26.

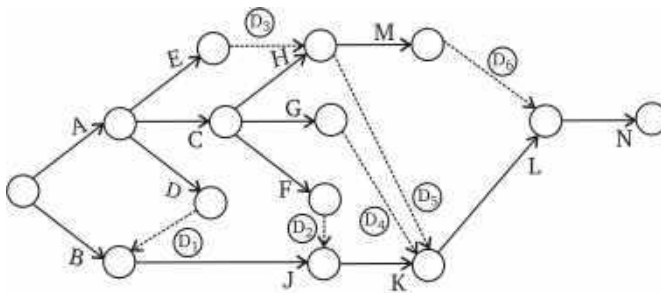


Fig. 9.26

In the above draft we have six, D_1 to D_6 Dummy activities. Now we proceed to reduce the number of Dummy activities to a minimum. Thus, we get the following final (*activity on arrow network*), as shown in fig. 9.27 in which there is only one Dummy activity D_1 . Here the events (circles) are also numbered. The *activity on node diagram* is also shown in fig. 9.28.

(A.O.A. Diagram)

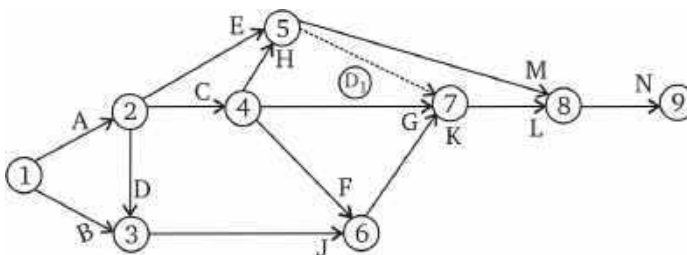


Fig. 9.27

(A.O.N. Diagram)

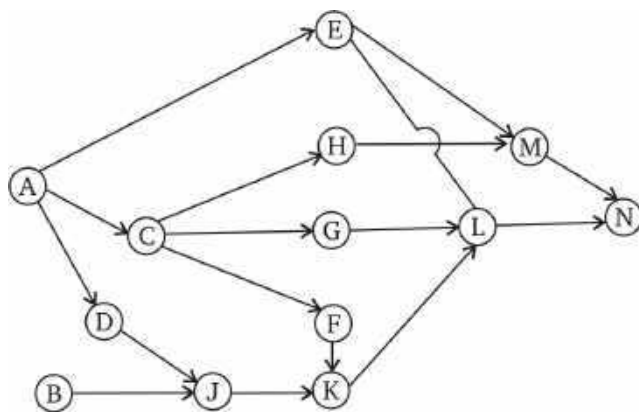


Fig. 9.28

Exercise 9.1

1. What is the dummy activity ? Why it is used ?
2. Write short note on dummy activities.
3. Distinguish between PERT and CPM.
4. Construct the network diagram for the following activity data :

Activity :	A	B	C	D	E	F	G	H	I	J	K	L
Preceded by :	-	-	A	B,C	D	E	B,C	F	F,G	H,I	B	F,G,K

5. Construct the network diagram for the following activity data :

Activity :	A	B	C	D	E	F	G	H	I	J
Preceded by :	-	A	A	A	C, D	D	E	G	F,H	B,I

6. Construct the network diagram for the following activity data :

Activity :	A	B	C	D	E	F	G	H	I	J
Preceded by :	-	A	B	B	B	C	C	F,G	D,E,H	I

7. Draw the network, given the following precedence relationship.

Event :	A	B,C	D	E	F	G
Preceded by :	-	A	B,C	C	D,E	E,F

8. A new type of water pump is to be designed for an automobile. Major specification are given in the following table. Draw the network diagram for the project.

<i>Activity</i>	<i>Description</i>	<i>Predecessor Activity</i>
<i>A</i>	<i>Drawing prepared and approved</i>	–
<i>B</i>	<i>Cost Analysis</i>	<i>A</i>
<i>C</i>	<i>Tool feasibility (economics)</i>	<i>A</i>
<i>D</i>	<i>Tool manufactured</i>	<i>C</i>
<i>E</i>	<i>Favourable cost</i>	<i>B, C</i>
<i>F</i>	<i>Raw material procured</i>	<i>D, E</i>
<i>G</i>	<i>Sub-assemblies ordered</i>	<i>E</i>
<i>H</i>	<i>Sub-assemblies received</i>	<i>G</i>
<i>I</i>	<i>Parts manufactured</i>	<i>D, F</i>
<i>J</i>	<i>Final assembly</i>	<i>I, H</i>
<i>K</i>	<i>Testing and shipment.</i>	<i>J</i>

9. The activities involved in the computer installation process are detailed below. Draw the network diagram.

<i>Activity</i>	<i>Description</i>	<i>Predecessor Activity</i>
<i>A</i>	<i>Physical Preparation</i>	–
<i>B</i>	<i>Organisational Planning</i>	–
<i>C</i>	<i>Personnel Selection</i>	<i>B</i>
<i>D</i>	<i>Equipment Installation</i>	<i>A</i>
<i>E</i>	<i>Personnel Training</i>	<i>C</i>
<i>F</i>	<i>Detailed System Design</i>	<i>C</i>
<i>G</i>	<i>File Conversion</i>	<i>F</i>
<i>H</i>	<i>Establish Standards and Controls</i>	<i>F</i>
<i>I</i>	<i>Program Preparation</i>	<i>H</i>
<i>J</i>	<i>Program Testing</i>	<i>I</i>
<i>K</i>	<i>Parallel Operations</i>	<i>D, E, G, J</i>
<i>L</i>	<i>Finalise Systems Documentation</i>	<i>I</i>
<i>M</i>	<i>Follow up</i>	<i>K, L</i>

Answers 9.1

4. Activity on Arrow Network Diagram.

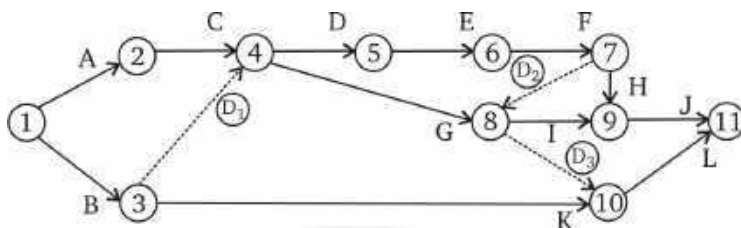


Fig. 9.29

Activity on Node Network Diagram.

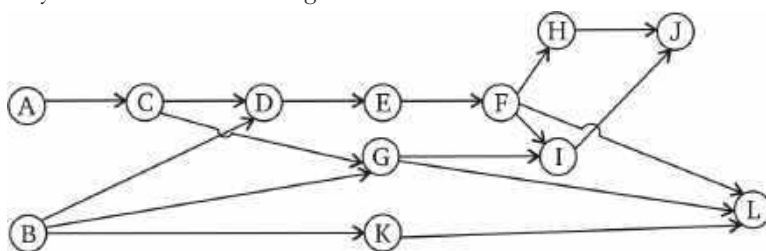


Fig. 9.30

5. Activity on Arrow Network Diagram.

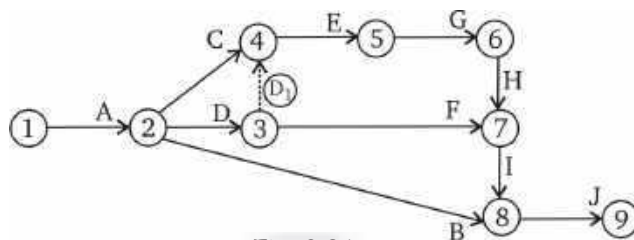


Fig. 9.31

Activity on Node Network Diagram.

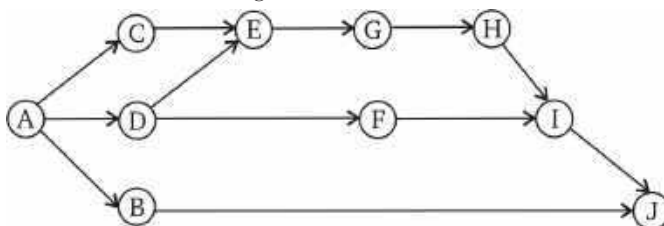


Fig. 9.32

6. Activity on Arrow Network Diagram.

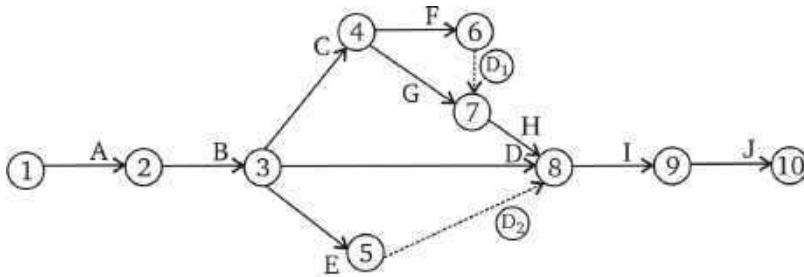


Fig. 9.33

Activity on Node Network Diagram.

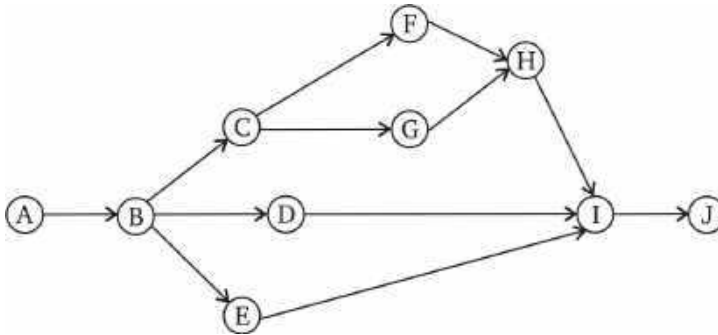


Fig. 9.34

7.

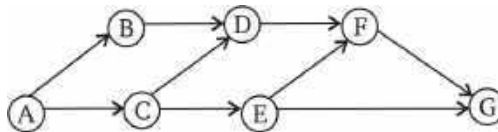


Fig. 9.35

8. Activity on Arrow Network Diagram.

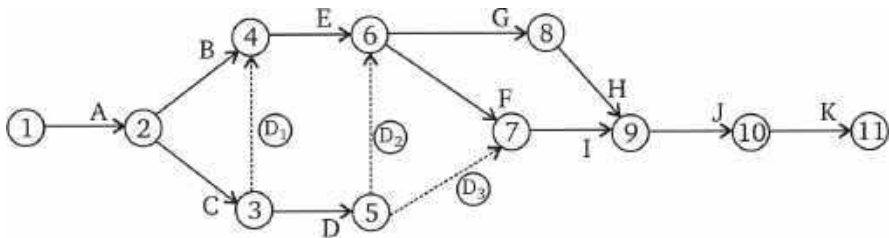


Fig. 9.36

Activity on Node Network Diagram.

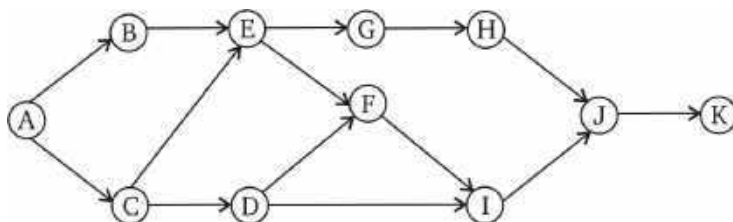


Fig. 9.37

9. Activity on Arrow Network Diagram.

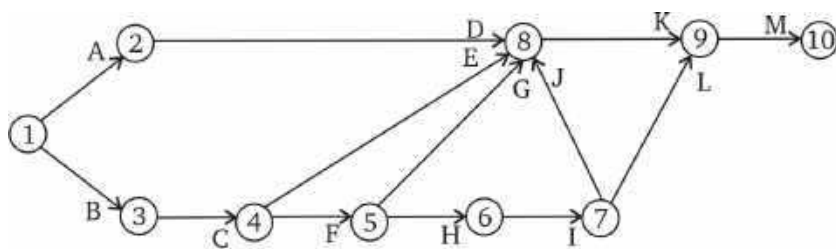


Fig. 9.38

Activity on Node Network Diagram.

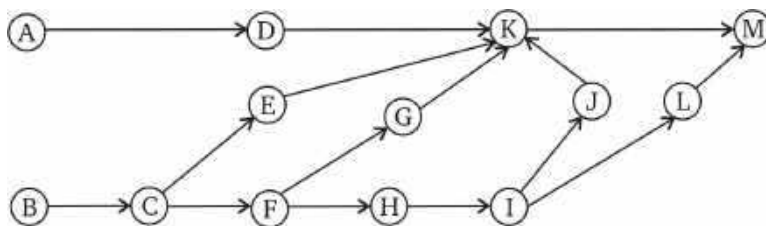


Fig. 9.39

9.9 Some CPM Terms

CPM analysis require the identification of critical activities and critical path. For this, consider following terms and notations associated to activities and events.

(i, j) = Activity (i, j) with tail event i and head event j .

E_i = Earliest occurrence time of an event i .

It is the earliest time at which an event i can occur without affecting the total project time.

L_i = Latest occurrence time of an event i .

It is the latest time at which an event i can occur without affecting the total project time.

$E S_{ij}$ or $E S_{i,j}$ = *Earliest start time of an activity (i, j).*

It is the earliest possible time at which the event i , from which the activity (i, j) emanates, can start without affecting the total project time.

$L S_{ij}$ or $L S_{i,j}$ = *Latest start time of an activity (i, j).*

It is the latest possible time by which the event i , from which the activity (i, j) emanates, must start without affecting the total project time.

$E F_{ij}$ or $E F_{i,j}$ = *Earliest finish time of an activity (i, j).*

It is the earliest possible time at which an activity (i, j) can finish without affecting the total project time.

$L F_{ij}$ or $L F_{i,j}$ = *Latest finish time of an activity (i, j).*

It is the latest possible time by which an activity (i, j) must finish without affecting the total project time.

It is the latest occurrence time of the event j at which the activity (i, j) terminates.

$t_{i,j}$ or t_{ij} = *Duration of activity (i, j).*

Obviously, for an activity (i, j)

$$E S_{ij} = E_i, \quad L F_{ij} = L_j$$

$$E F_{ij} = E S_{ij} + t_{ij} = E_i + t_{ij}, \quad L S_{ij} = L F_{ij} - t_{ij} = L_j - t_{ij}$$

Note: An activity can start only when all its preceding activities have finished (*i.e.*, completed).

These times are calculated by the following two methods.

9.10 Critical Path Calculations (Computation)

The CPM Computation consists of the following two methods.

1. Forward Pass Method (For Earliest Event Time)

Here the calculations begins from the start event 1 and passing through the events in increasing order of numbers from 1, move to the end (terminal) event of the network of the project. Here we calculate

- (i) the earliest start and earliest finish time of each activity,
- (ii) the earliest occurrence time of each event,
- (iii) the earliest finish time of the entire project.

Stepwise we proceed as follows :

Step 1 : Assume the *earliest occurrence time* of initial (start) event 1 (tail event of starting activity/activities), having no predecessor activity/activities, equal to zero.

i.e. take $E_1 = 0$

Step 2 : Proceed to the next event/events $i > 1$, for which event 1 is the preceding event *i.e.*, (1, i) is the activity/activities that emanates from event 1, and

- (i) Calculate $E S_{1,i}$ the *earliest start time* of each activity (1, i),

Obviously $E S_{1,i} = E_1 = 0$, for each activity (1, i)

- (ii) Calculate $E F_{1,i}$, the *earliest finish time* of each activity (1, i), that emanates from the tail event 1.

Obviously $E F_{1,i} = E S_{1,i} + t_{1,i} = E_1 + t_{1,i} = t_{1,i}$.

where $t_{1,i}$ is the duration of activity (1, i).

Step 3 : Proceed to the event no. 2 next to event 1 and consider all activities (2, j) emanating from event 2 and

- (i) Calculate $E S_{2,j}$, *i.e.* E_2 , the *earliest start time* of each activity (2, j), which is given by

$$E S_{2,j} = E_2 = E_1 + t_{1,2}$$

- (ii) Calculate $E F_{2,j}$, the *earliest finish time* of each activity (2, j), which is given by

$$E F_{2,j} = E S_{2,j} + t_{2,j} = E_2 + t_{2,j}$$

Step 4 : Proceed to the event no. 3 next to event 2 and consider all activities (3, k) emanating from event 3 and

- (i) Calculate $E S_{3,k}$ *i.e.*, E_3 , the *earliest start time* of each activity (3, k), which is given by

$$E S_{3,k} = E_3 = \text{Max}_s \{E_s + t_{s,3}\}$$

where s is the collection of all preceding events (nodes) of event 3 *i.e.* s is the collection of tail events of all activities that terminate at event 3.

- (ii) Calculate $E F_{3,k}$, the *earliest finish time* of each activity (3, k), which is given by

$$E F_{3,k} = E_3 + t_{3,k}$$

Step 5 : Proceed to the next event no. 4 and repeat step 4 for all activities (4, e) emanating from event 4.

Continuing in this way through events in increasing order of numbers *i.e.*, through events 1, 2, 3, in order, reach the last event say N , and

Calculate E_N , the *earliest start time* of event N , which is given by

$$E_N = \text{Max}_e \{E_e + t_{e,N}\}$$

where e is the collection of all preceding events of event N *i.e.* e is the collection of tail events of all activities that terminate at event N .

2. Backward Pass Method (For Latest Allowable Event Time)

Here, the calculations begins from the final event N and passing through the events in decreasing order of numbers from N , move to the initial (start) event 1 of the network of the project. Here we calculate

- (i) the latest finish and latest start time of each activity.
- (ii) the latest occurrence time of each event.
- (iii) the latest finish time of the entire project.

Stepwise we proceed as follows :

Step 1 : Take the *latest occurrence time* of last event number N , equal to its earliest occurrence time E_N . This is known from forward pass method. *i.e.* take

$$L_N = E_N.$$

If the target time, say T , for the completion of the project is given, then take $L_N = T$, otherwise take $L_N = E_N$.

Step 2 : Proceed backwards to the the next event/events $i < N$, for which event N is the successor event *i.e.*, (i, N) is the activity/activities that terminate at the event N , and

- (i) Calculate $L F_{i,N}$, the *latest finish time* of each activity (i, N) which is equal to the latest occurrence time L_N of last event N

$$\text{i.e. } L F_{i,N} = L_N = E_N \text{ (Earliest occurrence for each activity } (i, N), \text{ time of event } N)$$

- (ii) Calculate $L S_{i,N}$, the *latest start time* of each activity (i, N) which is given by

$$L S_{i,N} = L F_{i,N} - t_{i,N} = L_N - t_{i,N} \text{ for each activity } (i, N).$$

Where $t_{i,N}$ is the duration of activity (i, N)

Step 3 : Proceed backwards to the event no $N-1$ next to event N and consider all activities $(j, N-1)$ terminating at event $N-1$ and.

- (i) Calculate $L F_{j,N-1}$, *i.e.*, L_{N-1} the *latest finish time* of each activity $(j, N-1)$ which is given by

$$\text{i.e. } L F_{j,N-1} = L_{N-1} = L_N - t_{N-1,N}$$

- (ii) Calculate $L S_{j,N-1}$, the *latest start time* of each activity $(j, N-1)$, which is given by

$$L S_{j,N-1} = L F_{j,N-1} - t_{j,N-1} = L_{N-1} - t_{j,N-1}$$

Step 4 : Proceed backwards to the event no $N-2$, next to $N-1$ and consider all activities $(k, N-2)$ terminating at event $N-2$ and

- (i) Calculate $L F_{k,N-2}$, *i.e.* L_{N-2} the *latest finish time* of each activity $(k, N-2)$, which is given by

$$L F_{k,N-2} = L_{N-2} = \text{Mini. } \{L_s - t_{N-2,s}\}$$

where s is the collection of all successor events (nodes) of event $N-2$ *i.e.*, s is collection of head events of all activities that emanates from event $N-2$.

- (ii) Calculate $L S_{k,N-2}$, the *latest start time* of each activity $(k, N-2)$, which is given by

$$L S_{k,N-2} = L_{N-2} - t_{k,N-2}$$

Step 5: Proceed backwards to the next event no. $N-3$ and repeat step 4 for all activities $(t, N-3)$ terminating at event $N-3$.

Continuing in this way through the events in decreasing order of numbers *i.e.*, through events $N, N-1, N-2, \dots$ in order, reach the first (start) event 1 and calculate L_1 , the latest occurrence time of start event (node) 1, which is given by

$$L_1 = \text{Mini. } \{L_s - t_{1,s}\}$$

where s is the collection of all successor events (nodes) of event 1 *i.e.*, s is collection of head events of all activities emanating from event 1.

Note: If $L_N = E_N$ then always $L_1 = E_1 = 0$

9.11 The Float and Slack

The float of an activity has the same significance as the slack of events. The float corresponds to activities hence to CPM and the slack corresponds to events and hence to PERT.

There are in general three types of floats.

1. Total float
 2. Free float and
 3. Independent float
1. **Total Float** : The **total float** of an activity is defined as the amount of time by which the completion of the activity could be delayed beyond its earliest expected completion time without affecting the over all project duration time.

The **total float** of an activity can also be defined as the difference between the maximum time available to perform the activity and activity duration time. The total float of an activity. (i, j) is denoted by $T F_{i,j}$ for an activity (i, j),

Maximum time available = Latest completion time – Earliest start time of activity.

$$= L F_{i,j} - E S_{i,j} = L_j - E_i$$

Then if $t_{i,j}$ is the duration time of the activity (i, j),

$$\text{Total Float} = T F_{i,j} = (L_j - E_i) - t_{i,j}$$

We can also write,

$$\text{Total Float} = T F_{i,j} = (L_j - t_{i,j}) - E_i = L S_{i,j} - E S_{i,j}$$

$$\text{Also, Total float} = T F_{i,j} = L_j - (E_i + t_{i,j}) = L F_{i,j} - E F_{i,j}$$

Thus the **total float** of an activity is equal to (i) the difference of its latest and earliest start times or (ii) the difference of its latest and earliest finish (completion) times.

2. **Free Float** : The **Free float** of an activity is defined as the amount of time by which the completion of the activity can be delayed beyond the earliest finish time without affecting the earliest start of immediate succeeding activities.

Free float is defined by assuming that all activities start as early as possible.

The **free float** for the activity (i, j) can also be defined as the excess available time over its duration time. The **free float** of an activity (i, j) is denoted by $F F_{i,j}$.

Thus for an activity (i, j)

Time available = Earliest start time of event j – Earliest start time of event i = $E_j - E_i$.

$$\therefore \text{Free float} = F F_{i,j} = (E_j - E_i) - t_{i,j}$$

We can also write,

$$\text{Free float} = F F_{i,j} = E_j - (E_i + t_{i,j}) = E_j - E F_{i,j}$$

Thus the **free float** of an activity is equal to the difference between the earliest start time for the head event and the earliest completion time of the activity under consideration.

3. **Independent Float** : This is based on the assumption that the predecessor event occurs at its latest possible time and the successor event at its earliest possible time.
 The **Independent float** of an activity can be defined as the amount of time by which the start of the activity can be delayed without affecting the earliest start time of any immediate

preceding activities, assuming that the preceding activity has finished at its latest finish time. The independent float of an activity (i, j) is denoted by $I F_{i, j}$.

Thus for an activity (i, j)

Time available = Earliest start time of head event j – Latest finish time of tail event i
 $= E_j - L_i$

$$\therefore \text{Independent Float} = I F_{i, j} = (E_j - L_i) - t_{i, j}$$

Note:

1. The negative value of independent float is considered zero.
2. The difference between total float and free float is known as **Interference float (or interfering float)**.
3. The basic difference between the slack and float times is that slack is used for events only, where as float is applied for activities.

9.12 Inequality Relation Between Floats of an Activity

From the previous article 9.11, for activity (i, j), we have

$$\text{Total float} = T F_{i, j} = L_j - E_i - t_{i, j}$$

$$\text{Free float} = F F_{i, j} = E_j - E_i - t_{i, j}$$

$$\text{and Independent float} = I F_{i, j} = E_j - L_i - t_{i, j}$$

Since for an event, latest occurrence time cannot be less than its earliest occurrence time,

$$\therefore L_i \geq E_i \text{ and } L_j \geq E_j$$

$$\text{Thus } T F_{i, j} = L_j - E_i - t_{i, j} \geq E_j - E_i - t_{i, j}$$

$$\text{i.e., } T F_{i, j} \geq F F_{i, j} \quad \text{.....(1)}$$

$$\begin{aligned} \text{Also, } F F_{i, j} &= E_j - E_i - t_{i, j} = (E_j - t_{i, j}) - E_i \\ &\geq (E_j - t_{i, j}) - L_i \end{aligned}$$

$$\text{i.e., } F F_{i, j} \geq I F_{i, j} \quad \text{.....(2)}$$

Hence from (1) and (2) for an activity (i, j), we have

$$T F_{i, j} \geq F F_{i, j} \geq I F_{i, j}$$

i.e. Total float $\geq$ Free float $\geq$ Independent float.

9.13 Event Slacks

For any given event, the **event slack** is defined as the difference between its latest and earliest occurrence times.

Thus for an activity (i, j)

Head event slack = $L_j - E_j$ and **Tail event slack** = $L_i - E_i$

The floats of an activity (i, j) , defined in § 9.12, can be represented in terms of head and tail event slacks as follows:

For an activity (i, j) , we have

$$\text{Total Float} = L_j - E_i - t_{i,j}$$

$$\begin{aligned}\text{Free Float} &= E_j - E_i - t_{i,j} = (L_j - E_i - t_{i,j}) - (L_j - E_j) \\ &= \text{Total Float} - \text{Head event slack}\end{aligned}$$

$$\begin{aligned}\text{Independent Float} &= E_j - L_i - t_{i,j} = (E_j - E_i - t_{i,j}) - (L_i - E_i) \\ &= \text{Free Float} - \text{Tail event slack} \\ &= (\text{Total float} - \text{Head event slack}) - \text{Tail event slack} \\ &= \text{Total Float} - (\text{Head event slack} + \text{Tail event slack})\end{aligned}$$

Hence the relation between the floats of an event (i, j) are as follows.

$$\begin{aligned}\text{Independent Float} &= \text{Total Float} - (\text{Head event slack} + \text{Tail event slack}) \\ &= \text{Free Float} - \text{Tail event slack} \\ \text{Free Float} &= \text{Total Float} - \text{Head event slack}\end{aligned}$$

9.14 Critical Event, Critical Activity and Critical Path

1. **Critical Event** : An event with zero slack time is called **Critical Event**.

Thus, the event i is *critical event*, if, slack of the event $= L_i - E_i = 0$

i.e. event i is *critical event* if, $L_i = E_i$

2. **Critical Activity** : In a network diagram, an activity is called **Critical activity**, if a delay in its execution (i.e. start) will cause further delay in the project completion time.

Thus, for a critical activity, total float time is zero.

i.e., $L S_{i,j} - E S_{i,j} = 0$, $L F_{i,j} - E F_{i,j} = 0$ and $L_j - E_i - t_{i,j} = 0$

$$\Rightarrow L_i = E_i \quad \text{.....(1)}$$

$$L_j = E_j \quad \text{.....(2)}$$

$$\text{and } E_j - E_i = L_j - L_i = t_{i,j} \quad \text{.....(3)}$$

Note: (i) All relations (1), (2) and (3) are necessarily true for critical activities but not for non-critical activities : That is for non-critical activities one or two of the relations (1), (2) and (3) may be satisfied but not all.

(ii) From relations (1) and (2) it follows that the slack times of tail and head events of critical activities are zero. In other words the tail and head events of critical activities are critical events but not the converse. That is if tail and head events of an activity are critical events then it is not necessarily a critical activity. For this activity to be critical activity equation (3) should also be satisfied.

3. **Critical Path** : The unbroken (continuous) chain of critical activities from the first (start) event to the last (end) event in the network diagram is called the **Critical Path**.

It is the longest path starting from the first event to the last event and is shown by thick line or double lines in the network diagram.

The sum of individual time durations of all critical activities on the critical path is called the **total project time** and it is the minimum time required to complete the project.

9.15 Procedure of Determination of Critical Path

Procedure of determination of critical path in a network of a project is as follows.

Step 1 : Draw the network diagram of the given project. Write the individual time duration of each activity above the arrow representing the activity.

Step 2 : Calculate the *earliest start* time of each event (node) taken in increasing order of their numbers starting from the first to the last event by Forward Pass Method, by using the following method.

Take the earliest start time of start event 1, i.e., $E_1 = 0$

$$\text{Earliest start time of event } j = E_j = \text{Max. } \{E_i + t_{i,j}\}$$

where i is the collection of tail events of all activities (i, j) of duration $t_{i,j}$ terminating in head event j .

Write the earliest start time of events in the side of corresponding event.

Step 3 : Calculate the *latest start time* of each event (node) taken in decreasing order of their numbers starting from the last to the first event by Backward pass method, by using the following method :

Take the latest start time of last event (say N)

$$\text{i.e., } L_N = E_N \text{ or project completion time (if given).}$$

$$\text{Latest start time of event } i = L_i = \text{Mini. } \{L_j - t_{i,j}\}$$

Where j is the collection of head events of all activities (i, j) of duration $t_{i,j}$ which emanate from tail event i .

Write the latest start time of events in the side of corresponding event.

Step 4: Calculate the earliest finish time $E F_{i,j}$ and latest finish time $L F_{i,j}$ of all activities (i, j) , by the relations.

$$E F_{i,j} = E_i + t_{i,j} \text{ and } L F_{i,j} = L_i + t_{i,j}$$

Also determine total float, free float and independent float for each activity (i, j) by the relations.

$$\text{Total float} = (L_j - t_{i,j}) - E_i$$

$$\text{Free float} = (E_j - E_i) - t_{i,j}$$

$$\text{Independent float} = E_j - L_i - t_{i,j}$$

Step 5: Identify the critical activities (i, j) which satisfy all the following relations

(i) $E_i = L_i$ (ii) $E_j = L_j$ and (iii) $E_j - E_i = L_j - L_i = t_{i,j}$.

Step 6: Connect the critical activities, without break, from the start event to end event to get the **critical path**. Mark this path by double lines or thick line.

Step 7: Calculate the total project time which is equal to E_N , where N is last (end) event (node).

Obviously E_N = Sum of duration times of all critical activities on critical path.

Note that the slack of each node on the critical path is zero.

ILLUSTRATIVE EXAMPLES

Example I: All activities which together constitute a small engineering project are given in the following table. The table also shows the necessary immediate predecessors for each activity.

Activity :	A	B	C	D	E	F	G	H	I	J
Immediate predecessor :	–	A	A	A	B	C,D	D	B	E,F,G	G
Activity duration (days)	2	3	4	5	6	3	4	7	2	3

- Construct an activity network.
- Determine the earliest finish date for the entire project, assuming the project begins at day 0.
- The total float for each activity.
- The critical path.
- The latest start day for activity B.
- The earliest finish day for activity F.
- The effect on the project duration if activity I were to take 3 days.
- The effect on the project duration if activity F were to take 6 days.

Solution: (i) The network diagram of the given project.

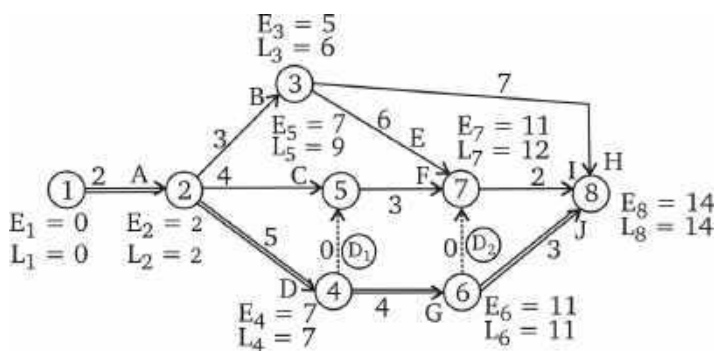


Fig. 9.40

- (ii) To find the *earliest start time* E_i of tail event i and *latest finish time* L_j of head event j of each activity (i, j) , we proceed as follows :

To find *earliest start time* E_i for tail event i of each activity (i, j) .

Taking $E_1 = 0$, since there is only one activity (1, 2) entering to the event 2, one activity (2, 3) entering to event 3, one activity (2, 4) entering to activity 4, etc,

$$\therefore E_2 = E_1 + t_{1,2} = 0 + 2 = 2 \quad , \text{ As only one activity (1, 2) is entering to event 2}$$

$$E_3 = E_2 + t_{2,3} = 2 + 3 = 5 \quad , \text{ As only one activity (2, 3) is entering to event 3}$$

$$E_4 = E_2 + t_{2,4} = 2 + 5 = 7 \quad , \text{ As only one activity (2, 4) is entering to event 4}$$

$$E_5 = \text{Max.}_{i=2,4} \{E_i + t_{i,5}\} \quad , \text{ As only two activities (2, 5) and (4, 5) are entering to event 5}$$

$$= \text{Max.} \{E_2 + t_{2,5}, E_4 + t_{4,5}\}$$

$$= \text{Max.} \{2 + 4, 7 + 0\} = 7$$

$$E_6 = E_4 + t_{4,6} = 7 + 4 = 11 \quad , \text{ As only one activity (4, 6) is entering to event 6}$$

$$E_7 = \text{Max.}_{i=3,5} \{E_i + t_{i,7}\} \quad , \text{ As only two activities (3, 7), (5, 7) are entering to event 7}$$

$$= \text{Max.} \{E_3 + t_{3,7}, E_5 + t_{5,7}\}$$

$$= \text{Max.} \{5 + 6, 7 + 3\} = 11$$

$$E_8 = \text{Max.}_{i=3,6,7} \{E_i + t_{i,8}\} \quad , \text{ As only three activities (3, 8), (6, 8), (7, 8) are entering to event 8.}$$

$$= \text{Max.} \{E_3 + t_{3,8}, E_6 + t_{6,8}, E_7 + t_{7,8}\}$$

$$= \text{Max.} \{5 + 7, 11 + 3, 11 + 2\} = 14$$

To find *latest finish time* L_j for head event j of each activity (i, j)

$$\text{Taking } L_8 = E_8 = 14,$$

$$L_7 = L_8 - t_{7,8} = 14 - 2 = 12 \quad , \text{ As only one activity (7, 8) emanate the event 7}$$

$$L_6 = \text{Min.}_{j=7,8} \{L_j - t_{6,j}\} \quad , \text{ As only two activities (6, 7) and (6, 8) emanate the event 6.}$$

$$= \text{Min.} \{L_7 - t_{6,7}, L_8 - t_{6,8}\}$$

$$= \text{Min.} \{12 - 0, 14 - 3\} = 11$$

$$L_5 = L_7 - t_{5,7} = 12 - 3 = 9 \quad , \text{ As only one activity (5, 7) emanate the event 5}$$

$$L_4 = \text{Min.}_{j=5,6} \{L_j - t_{4,j}\} \quad , \text{ As only two activities (4, 5) and (4, 6) emanate the event 4}$$

$$= \text{Min. } \{L_5 - t_{4,5}, L_6 - t_{4,6}\}$$

$$= \text{Min. } \{9 - 0, 11 - 4\} = 7$$

$$L_3 = \text{Min. } \{L_j - t_{3,j}\}, \text{ As only two activities (3, 7) and (3, 8) emanate the event 3}$$

$$= \text{Min. } \{L_7 - t_{3,7}, L_8 - t_{3,8}\}$$

$$= \text{Min. } \{12 - 6, 14 - 7\} = 6$$

$$L_2 = \text{Min. } \{L_j - t_{2,j}\}, \text{ As only three activities (2, 3), (2, 4),}$$

(2, 5) emanate the event 2

$$= \text{Min. } \{L_3 - t_{2,3}, L_4 - t_{2,4}, L_5 - t_{2,5}\} = \text{Min. } \{6 - 3, 7 - 5, 9 - 4\} = 2$$

$$L_1 = L_2 - t_{1,2} = 2 - 2 = 0 = E_1, \text{ as only one activity (1, 2) emanate the event 1.}$$

For each activity (i, j) , earliest start time E_i and earliest finish time $E F_{ij} = E_i + t_{ij}$ for tail event $i = L_j - t_{i,j}$

Latest finish time L_j and latest start time $L S_{i,j} = L_j - t_{i,j}$ of head event j and total float are tabulated in the following table.

Table 9.1

Activity (i, j)	Activ- ity durati on $t_{i,j}$ days	Earliest Time		Latest Time		Float		
		Start E_i	Finish $E F_{i,j}$ $= E_i + t_{i,j}$	Start $L S_{i,j}$ $= L_j - t_{i,j}$	Finish L_j	Total $L_j - E_i$ $- t_{i,j}$	Free T.F. - (H.Slack)	Independent T.F - (Tail Slack)
(1)	(2)	(3)	(4) = (3) + (2)	(5) = (6) - (2)	(6)	(7) = (5) - (3)	(8) = (7) - (L_j - E_j)	(8) - (L_i - E_i)
—	—	—	—	—	$L_1 = 0$	—	—	—
A (1, 2)	2	$E_1 = 0$	2	0	$L_2 = 2$	0	$0 - (L_2 - E_2) = 0$	$0 - (L_1 - E_1) = 0$
B (2, 3)	3	$E_2 = 2$	5	3	$L_3 = 6$	1	$1 - (L_3 - E_3) = 0$	$0 - (L_2 - E_2) = 0$
D (2, 4)	5	$E_2 = 2$	7	2	$L_4 = 7$	0	$0 - (L_4 - E_4) = 0$	$0 - (L_2 - E_2) = 0$
C (2, 5)	4	$E_2 = 2$	6	5	$L_5 = 9$	3	$3 - (L_5 - E_5) = 1$	$1 - (L_2 - E_2) = 1$
E (3, 7)	6	$E_3 = 5$	11	6	$L_7 = 12$	1	$1 - (L_7 - E_7) = 0$	$0 - (L_3 - E_3) = -1$

$H(3, 8)$	7	$E_3 = 5$	12	7	$L_8 = 14$	2	$2 - (L_8 - E_8) = 2$	$2 - (L_3 - E_3) = 1$
$D_1(4, 5)$	0	$E_4 = 7$	7	9	$L_5 = 9$	2	$2 - (L_5 - E_5) = 0$	$0 - (L_4 - E_4) = 0$
$G(4, 6)$	4	$E_4 = 7$	11	7	$L_6 = 11$	0	$0 - (L_6 - E_6) = 0$	$0 - (L_4 - E_4) = 0$
$F(5, 7)$	3	$E_5 = 7$	10	9	$L_7 = 12$	2	$2 - (L_7 - E_7) = 1$	$1 - (L_5 - E_5) = -1$
$D_2(6, 7)$	0	$E_6 = 11$	11	12	$L_7 = 12$	1	$1 - (L_7 - E_7) = 0$	$0 - (L_6 - E_6) = 0$
$J(6, 8)$	3	$E_6 = 11$	14	11	$L_8 = 14$	0	$0 - (L_8 - E_8) = 0$	$0 - (L_6 - E_6) = 0$
$I(7, 8)$	2	$E_7 = 11$	13	12	$L_8 = 14$	1	$1 - (L_8 - E_8) = 1$	$1 - (L_7 - E_7) = 0$
–	–	$E_8 = 14$	–	–	–	–	–	–

Hence the earliest finish date for entire project, assuring the project begins at day 0 is $E_8 = 14$ days.

(iii) From table 9.1, the total float of each activity are as follows :

<i>Activity</i>	:	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>
<i>Total Float</i>	:	0	1	3	0	1	2	0	2	1	0

(iv) From the above table 9.1 the activities with total float zero are $A(1,2)$, $D(2,4)$, $G(4,6)$ and $J(6,8)$ which are the critical activities.

Hence the critical path is $A-D-G-J$, which is shown by double lines in fig. 9.40

(v) From table 9.1 latest start day for activity B

$$= L_3 - t_{23} = 6 - 3 = 3 \text{ days.}$$

(vi) From table 9.1 earliest finish day for activity F

$$= E_5 + t_{5,7} = 7 + 3 = 10 \text{ days.}$$

(vii) If duration of activity I i.e., $t_{7,8} = 3$ days in place of 2 days, then

$$E_7 = 11 \text{ days}$$

$$E_8 = \text{Max} \{E_3 + t_{3,8}, E_6 + t_{6,8}, E_7 + t_{7,8}\} = \text{Max.} \{5 + 7, 11 + 3, 11 + 3\}$$

$$= 14 \text{ days (No change)}$$

$$L_8 = 14 \text{ days (No change)}$$

and $L_7 = L_8 - t_{7,8} = 14 - 3 = 11 \text{ days}$

Thus in this situation $E_7 = L_7 = 11 \text{ days}$.

Hence there will be no effect on the project duration which will remain the same as 14 days but $E_7 = L_7$ and $E_8 = L_8$ implies that the activity I (7, 8) will become critical activity as in this case total float = 0

(viii) If duration of activity F i.e., $t_{5,7} = 6$ days in place of 3 days, then

$$E_7 = \text{Max. } \{t_3 + t_{3,7}, E_5 + t_{5,7}\} = \text{Max. } \{5 + 6, 7 + 6\} = 13$$

and $E_8 = \text{Max. } \{E_3 + t_{3,8}, E_6 + t_{6,8}, E_7 + t_{7,8}\} = \text{Max. } \{5 + 7, 11 + 3, 13 + 2\} = 15$

Hence in this case, the project will be delayed by $15 - 14 = 1$ day.

Example 2: Draw a network diagram on the basis of the following data :

Activity	Duration (days)	Activity	Duration (days)
1 - 2	2	4 - 8	8
1 - 4	2	5 - 6	4
1 - 7	1	6 - 9	3
2 - 3	4	7 - 8	3
3 - 6	1	8 - 9	5
4 - 5	5	9 - 10	2

Find the critical path, total duration and slack times.

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Solution: The network diagram on the basis of given data is as follows :

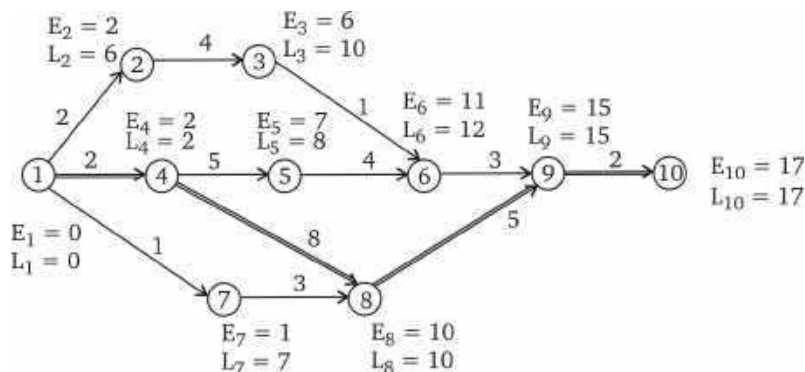


Fig. 9.41

To find earliest start time E_i for tail event i of each activity (i, j)

Taking $E_1 = 0$ we have

$$E_2 = E_1 + t_{1,2} = 0 + 2 = 2$$

$$E_3 = E_2 + t_{2,3} = 2 + 4 = 6$$

$$E_4 = E_1 + t_{1,4} = 0 + 2 = 2$$

$$E_5 = E_4 + t_{4,5} = 2 + 5 = 7$$

$$E_6 = E_5 + t_{5,6} = 7 + 4 = 11$$

$$E_7 = E_1 + t_{1,7} = 0 + 1 = 1$$

$$E_8 = \text{Max.}_{i=4,7} \{E_i + t_{i,8}\} = \text{Max.} \{E_4 + t_{4,8}, E_7 + t_{7,8}\}$$

$$E_9 = \text{Max.}_{i=6,8} \{E_i + t_{i,9}\}$$

$$= \text{Max.} \{2 + 8, 1 + 3\} = 10$$

$$= \text{Max.} \{E_6 + t_{6,9}, E_8 + t_{8,9}\},$$

$$E_{10} = E_9 + t_{9,10} = 15 + 2 = 17$$

$$= \text{Max.} \{11 + 3, 10 + 5\} = 15$$

To find latest finish time L_j for head event j of each activity (i, j)

$$\text{Taking } L_{10} = E_{10} = 17$$

$$L_9 = L_{10} - t_{9,10} = 17 - 2 = 15$$

$$L_8 = L_9 - t_{8,9} = 15 - 5 = 10$$

$$L_7 = L_8 - t_{7,8} = 10 - 3 = 7$$

$$L_6 = L_9 - t_{6,9} = 15 - 3 = 12$$

$$L_5 = L_6 - t_{5,6} = 12 - 4 = 8$$

$$L_4 = \text{Mini.}_{j=5,8} \{L_j - t_{4,j}\} = \text{Mini.} \{L_5 - t_{4,5}, L_8 - t_{4,8}\} \quad L_3 = L_6 - t_{3,6} = 11 - 1 = 10$$

$$= \text{Mini.} \{8 - 5, 10 - 8\} = 2$$

$$L_2 = L_3 - t_{2,3} = 10 - 4 = 6$$

$$L_1 = \text{Mini.}_{j=2,4,7} \{L_j - t_{1,j}\} = \text{Mini.} \{L_2 - t_{1,2},$$

$$L_4 - t_{1,4}, L_7 - t_{1,7}\} = \text{Mini.} \{6 - 2, 2 - 2, 7 - 1\} = 0$$

For each activity (i, j) earliest start time E_i and earliest finish time $E_{ij} = E_i + t_{ij}$ for tail event i , latest finish time L_j and latest start time $L_{Si,j} = L_j - t_{i,j}$ of head event j and total float (slack for head event j) are tabulated in the following table.

Table 9.2

Activity (i, j)	Activity duration $t_{i,j}$ (days)	Earliest Time		Latest Time		Total Float (slack) of activity (i, j)
		Start $ES_{i,j} = E_i$	Finish $EF_{i,j} = E_i + t_{i,j}$	Start $LS_{i,j} = L_j - t_{i,j}$	Finish $LF_{i,j} = L_j$	$LF_{i,j} - EF_{i,j}$ or $LS_{i,j} - ES_{i,j}$
(1)	(2)	(3)	(4) = (3) + (2)	(5) = (6) - 2	(6)	(6) - (4) or (5) - (3)
—	—	—	—	—	$L_1 = 0$	—
1 - 2	2	0 (E_1)	2	4	6 (L_2)	4
1 - 4	2	0 (E_1)	2	0	2 (L_4)	0

1 – 7	1	0 (E_1)	1	6	7 (L_7)	6
2 – 3	4	2 (E_2)	6	6	10 (L_3)	4
3 – 6	1	6 (E_3)	7	11	12 (L_6)	5
4 – 5	5	2 (E_4)	7	3	8 (L_5)	1
4 – 8	8	2 (E_4)	10	2	10 (L_8)	0
5 – 6	4	7 (E_5)	11	8	12 (L_6)	1
6 – 9	3	11 (E_6)	14	12	15 (L_9)	1
7 – 8	3	1 (E_7)	4	7	10 (L_8)	6
8 – 9	5	10 (E_8)	15	10	15 (L_9)	0
9 – 10	2	15 (E_9)	17	15	17 (L_{10})	0
–	–	$E_{10} = 17$	–	–	–	–

The activities with total float (slack) zero are 1 – 4, 4 – 8, 8 – 9, 9 – 10, which are the critical activities.

Hence the critical path is 1 – 4 – 8 – 9 – 10, shown by double lines in the fig. 9.41.

The *slack times* (in days) are as follows.

Activity:	1–2	1–4	1–7	2–3	3–6	4–5	4–8	5–6	6–9	7–8	8–9	9–10
Slack :	4	0	6	4	5	1	0	1	1	6	0	0

Example 3: A small maintenance project consists of the following 12 jobs :

<i>Jobs</i>	<i>Duration in days</i>	<i>Jobs</i>	<i>Duration in days</i>
1 – 2	2	5 – 8	5
2 – 3	7	6 – 7	8
2 – 4	3	6 – 10	4
3 – 4	3	7 – 9	4
3 – 5	5	8 – 9	1
4 – 6	3	9 – 10	7

Draw the arrow network of the project. Summarize CPM calculations in a tabular form calculating the three types of floats for jobs and hence determine the critical path.

Solution: The arrow network diagram of the project is as follows :

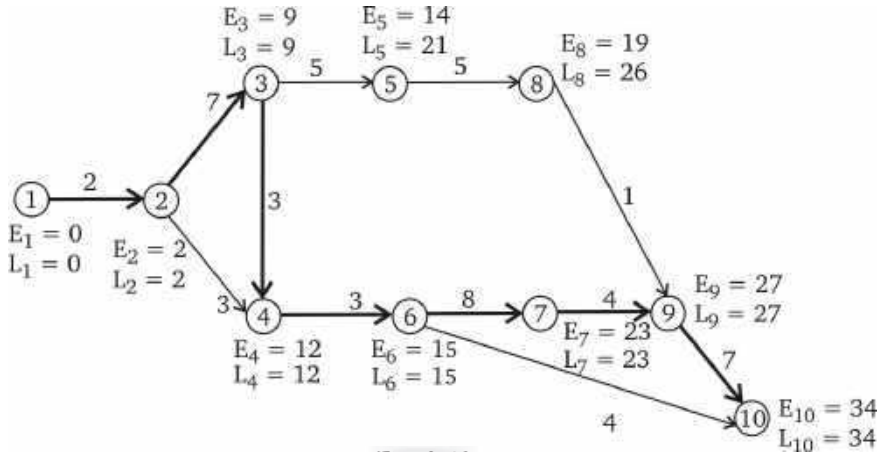


Fig. 9.42

To find earliest start time E_i for tail event i of each job (activity), $(i - j)$.

Taking $E_1 = 0$, we have $E_2 = E_1 + t_{1,2} = 0 + 2 = 2$

$$E_3 = E_2 + t_{2,3} = 2 + 7 = 9 \quad E_4 = \text{Max.}_{i=2,3} \{E_i + t_{i,4}\} = \text{Max.} \{E_2 + t_{2,4}, E_3 + t_{3,4}\}$$

$$= \text{Max.} \{2 + 3, 9 + 3\} = 12$$

$$E_5 = E_3 + t_{3,5} = 9 + 5 = 14 \quad E_6 = E_4 + t_{4,6} = 12 + 3 = 15$$

$$E_7 = E_6 + t_{6,7} = 15 + 8 = 23 \quad E_8 = E_5 + t_{5,6} = 14 + 5 = 19$$

$$E_9 = \text{Max.}_{i=7,8} \{E_i + t_{i,9}\}$$

$$= \text{Max.} \{E_7 + t_{7,9}, E_8 + t_{8,9}\} \quad E_{10} = \text{Max.}_{i=6,9} \{E_i + t_{i,10}\} = \text{Max.} \{E_6 + t_{6,10}, E_9 + t_{9,10}\}$$

$$= \text{Max.} \{23 + 4, 19 + 1\} = 27 \quad = \text{Max.} \{15 + 4, 27 + 7\} = 34$$

To find latest finish time L_j for head event j of each job (activity), $(i - j)$.

Taking $L_{10} = E_{10} = 34$ $L_9 = L_{10} - t_{9,10} = 34 - 7 = 27$

$$L_8 = L_9 - t_{8,9} = 27 - 1 = 26, \quad L_7 = L_9 - t_{7,9} = 27 - 4 = 23$$

$$L_6 = \text{Mini.}_{j=7,10} \{L_j - t_{6,j}\} \quad L_5 = L_8 - t_{5,8} = 26 - 5 = 21$$

$$= \text{Min.} \{L_7 - t_{6,7}, L_{10} - t_{6,10}\}$$

$$= \text{Min. } \{23 - 8, 34 - 4\} = 15$$

$$L_3 = \text{Mini. } \{L_j - t_{3,j}\}_{j=4,5}$$

$$L_4 = L_6 - t_{4,6} = 15 - 3 = 12$$

$$= \text{Min. } \{L_4 - t_{3,4}, L_5 - t_{3,5}\}$$

$$= \text{Mini. } \{12 - 3, 21 - 5\} = 9$$

$$L_2 = \text{Mini. } \{L_j - t_{2,j}\}_{j=3,4} = \text{Mini. } \{L_3 - t_{2,3}, L_4 - t_{2,4}\}$$

$$= \text{Min. } \{9 - 7, 12 - 3\} = 2$$

$$L_1 = L_2 - t_{1,2} = 2 - 2 = 0$$

For each job (activity) $(i-j)$ earliest start time E_i and earliest finish time $EF_{i,j} = E_i + t_{i,j}$ for tail event i , latest finish time L_j and latest start time $LS_{i,j} = L_j - t_{i,j}$ of head event j and all three types of floats are tabulated in the following table :

Table 9.3

Job (i, j)	Job duration $t_{i,j}$ days	Earliest Time		Latest Time		Float		
		Start E_i	Finish $EF_{i,j} = E_i + t_{i,j}$	Start $LS_{i,j} = L_j - t_{i,j}$	Finish L_j	Total $L_j - E_i - t_{i,j}$	Free T.F. - H. Slack	Independent F.F. - (T. Slack)
(1)	(2)	(3)	(4) = (3) + (2)	(5) = (6) - (2)	(6)	(7) = (6) - (4)	(8) = (7) - ($L_j - E_i$)	(8) - ($L_i - E_i$)
-	-	-	-	-	$L_1 = 0$	-	-	-
1 - 2	2	$E_1 = 0$	2	0	$L_2 = 2$	0	$0 - (L_2 - E_2) = 0$	$0 - (L_1 - E_1) = 0$
2 - 3	7	$E_2 = 2$	9	2	$L_3 = 9$	0	$0 - (L_3 - E_3) = 0$	$0 - (L_2 - E_2) = 0$
2 - 4	3	$E_2 = 2$	5	9	$L_4 = 12$	7	$7 - (L_4 - E_4) = 7$	$7 - (L_2 - E_2) = 7$
3 - 4	3	$E_3 = 9$	12	9	$L_4 = 12$	0	$0 - (L_4 - E_4) = 0$	$0 - (L_3 - E_3) = 0$
3 - 5	5	$E_3 = 9$	14	16	$L_5 = 21$	7	$7 - (L_5 - E_5) = 0$	$0 - (L_3 - E_3) = 0$
4 - 6	3	$E_4 = 12$	15	12	$L_6 = 15$	0	$0 - (L_6 - E_6) = 0$	$0 - (L_4 - E_4) = 0$
5 - 8	5	$E_5 = 14$	19	21	$L_8 = 26$	7	$7 - (L_8 - E_8) = 0$	$0 - (L_5 - E_5) = -7$
6 - 7	8	$E_6 = 15$	23	15	$L_7 = 23$	0	$0 - (L_7 - E_7) = 0$	$0 - (L_6 - E_6) = 0$
6 - 10	4	$E_6 = 15$	19	30	$L_{10} = 34$	15	$15 - (L_{10} - E_{10}) = 15$	$15 - (L_6 - E_6) = 15$
7 - 9	4	$E_7 = 23$	27	23	$L_9 = 27$	0	$0 - (L_9 - E_9) = 0$	$0 - (L_7 - E_7) = 0$
8 - 9	1	$E_8 = 19$	20	26	$L_9 = 27$	7	$7 - (L_9 - E_9) = 7$	$7 - (L_8 - E_8) = 0$
9 - 10	7	$E_9 = 27$	34	27	$L_{10} = 34$	0	$0 - (L_{10} - E_{10}) = 0$	$0 - (L_9 - E_9) = 0$
-	-	$E_{10} = 34$	-	-	-	-	-	-

Here the critical jobs (activities) (activities with total float zero) are

1-2, 2-3, 3-4, 4-6, 6-7, 7-9, 9-10.

Hence the *critical path* is 1-2-3-4-6-7-9-10. shown by thick lines in the figure.

Note that, the slack of events E_2 and E_4 are zero, but the activity 2-4 is not the critical job (activity) as $E_4 - E_2 = t_{2,4}$ and $L_4 - L_2 \neq t_{2,4}$.

Example 4 : A small maintenance project consists of the following 10 jobs, whose precedence relationships are identified by their node numbers :

Job	Alternative (Initial node, final node)	Estimated (duration days)
A	(1, 2)	4
B	(2, 3)	6
C	(2, 4)	10
D	(3, 5)	8
E	(3, 6)	2
F	(4, 6)	12
G	(4, 7)	4
H	(5, 8)	16
I	(6, 8)	14
J	(7, 8)	8

- Draw an arrow diagram representing the project.
- Calculate early and late start and finish time for each job.
- How much float job (3, 5), (4, 6) and (7, 8) have ?
- Which jobs are critical ? What is critical path ?
- If job (2, 3) were to take 12 days instead of 6, how would the project finish date be affected ? Is there any change in critical path ?
- Do any jobs have free and independent float ? If so, which ones and how much ?

Solution: (i) The arrow diagram of the project is as follows :

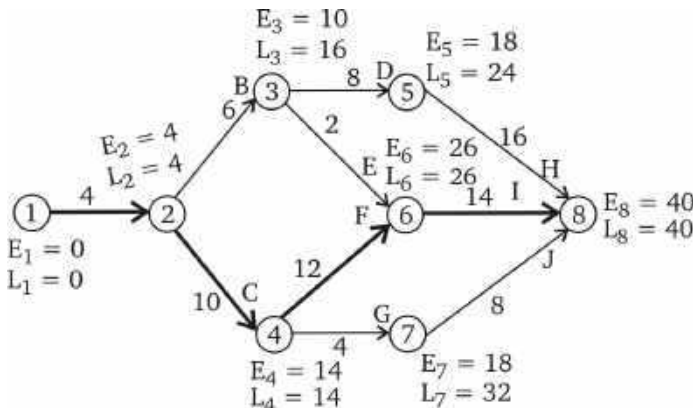


Fig. 9.43

(ii) To find earliest start time E_i for each job (i, j)

$$\begin{aligned}
 \text{Taking } E_1 = 0, \text{ we have} \quad E_2 &= E_1 + t_{1,2} = 0 + 4 = 4 \\
 E_3 &= E_2 + t_{2,3} = 4 + 6 = 10 \quad E_4 = E_2 + t_{2,4} = 4 + 10 = 14 \\
 E_5 &= E_3 + t_{3,5} = 10 + 8 = 18 \quad E_6 = \text{Max. } \{E_i + t_{i,6}\} = \text{Max. } \{E_3 + t_{3,6}, E_4 + t_{4,6}\} \\
 &= \text{Max. } \{10 + 2, 14 + 12\} = 26 \\
 E_7 &= E_4 + t_{4,7} = 14 + 4 = 18 \quad E_8 = \text{Max. } \{E_i + t_{i,8}\} \\
 &= \text{Max. } \{E_5 + t_{5,8}, E_6 + t_{6,8}, E_7 + t_{7,8}\} \\
 &= \text{Max. } \{18 + 16, 26 + 14, 18 + 8\} = 40
 \end{aligned}$$

To find latest finish time L_j for each job (i, j)

$$\begin{aligned}
 \text{Taking } L_8 = E_8 = 40, \text{ we have} \quad L_7 &= L_8 - t_{7,8} = 40 - 8 = 32 \\
 L_6 &= L_8 - t_{6,8} = 40 - 14 = 26 \quad L_5 = L_8 - t_{5,8} = 40 - 16 = 24 \\
 L_4 &= \text{Mini. } \{L_j - t_{4,j}\} \quad L_3 = \text{Mini. } \{L_j - t_{3,j}\} = \text{Mini. } \{L_5 - t_{3,5}, L_6 - t_{3,6}\} \\
 &= \text{Mini. } \{L_6 - t_{4,6}, L_7 - t_{4,7}\}, \quad = \text{Min. } \{24 - 8, 26 - 2\} = 16 \\
 &= \text{Mini. } \{26 - 12, 32 - 4\} = 14 \\
 L_2 &= \text{Mini. } \{L_j - t_{2,j}\} \quad L_1 = L_2 - t_{1,2} = 4 - 4 = 0 \\
 &= \text{Min. } \{L_3 - t_{2,3}, L_4 - t_{2,4}\}, \\
 &= \text{Min. } \{16 - 6, 14 - 10\} = 4
 \end{aligned}$$

For each job (i, j) earliest start time E_i and earliest finish time $E F_{i,j} = E_i + t_{i,j}$, latest finish time L_j and latest start time $L S_{i,j} = L_j - t_{i,j}$ and total float, free float and independent float are tabulated in the following table.

Table 9.4

Job (i, j)	Job duration t_{ij} (days)	Earliest Time		Latest Time		Float		
		Start $ES_{ij} = E_i$	Finish $EF_{ij} = E_i + t_{ij}$	Start $LS_{ij} = L_j - t_{ij}$	Finish $LF_{ij} = L_j$	Total (TF) $LF_{ij} - EF_{ij}$	Free (FF) $TF - (L_j - E_j)$	Independent (IF) $FF - (L_i - E_i)$
(1)	(2)	(3)	(4) = (3) + (2)	(5) = (6) - (2)	(6)	(7) = (6) - (4)		
—	—	—	—	—	$L_1 = 0$	—	—	—
A (1, 2)	4	$E_1 = 0$	4	0	$L_2 = 4$	0	$0 - (L_2 - E_2) = 0$	$0 - (L_1 - E_1) = 0$
B (2, 3)	6	$E_2 = 4$	10	10	$L_3 = 16$	6	$6 - (L_3 - E_3) = 0$	$0 - (L_2 - E_2) = 0$

C (2, 4)	10	$E_2 = 4$	14	4	$L_4 = 14$	0	$0 - (L_4 - E_4) = 0$	$0 - (L_2 - E_2) = 0$
D (3, 5)	8	$E_3 = 10$	18	16	$L_5 = 24$	6	$6 - (L_5 - E_5) = 0$	$0 - (L_3 - E_3) = -6$
E (3, 6)	2	$E_3 = 10$	12	24	$L_6 = 26$	14	$14 - (L_6 - E_6) = 14$	$14 - (L_3 - E_3) = 8$
F (4, 6)	12	$E_4 = 14$	26	14	$L_6 = 26$	0	$0 - (L_6 - E_6) = 0$	$0 - (L_4 - E_4) = 0$
G (4, 7)	4	$E_4 = 14$	18	28	$L_7 = 32$	32	$14 - (L_7 - E_7) = 0$	$0 - (L_4 - E_4) = 0$
H (5, 8)	16	$E_5 = 18$	34	24	$L_8 = 40$	6	$6 - (L_8 - E_8) = 6$	$6 - (L_5 - E_5) = 0$
I (6, 8)	14	$E_6 = 26$	40	26	$L_8 = 40$	0	$0 - (L_8 - E_8) = 0$	$0 - (L_6 - E_6) = 0$
J (7, 8)	8	$E_7 = 18$	26	32	$L_8 = 40$	14	$14 - (L_8 - E_8) = 14$	$14 - (L_7 - E_7) = 0$
–	–	$E_8 = 40$	–	–	–	–	–	–

(iii) The float (Total) of job (3, 5), (4, 6) and (7, 8) are 6, 0 and 14 respectively.

(iv) The total float of jobs A, C, F and I are zero, so they are the critical jobs.

∴ The critical path is $B - C - F - I$ i.e., $1 - 2 - 4 - 6 - 8$, shown by thick lines in the fig.

(v) If job (2, 3) take 12 days instead of 6 i.e., if $t_{2,3} = 12$ instead of 6. In this case increase in $t_{2,3}$ is $12 - 6 = 6$ days and the float of the job (2, 3) is also 6 days. Hence there is no effect of this change on the project. Its completion time will remain 40 days as before. In this case following earliest start time E_i changes to new values.

$$E_3 = E_2 + t_{2,3} = 4 + 12 = 16, \quad E_5 = E_3 + t_{3,5} = 16 + 8 = 24$$

The values of all other E_i and L_i remains unchanged.

$$\text{Now } E_3 = 16 = L_3, \quad E_5 = 24 = L_5 \quad L_5 - L_3 = 8 = t_{3,5} = E_5 - E_3, \quad E_8 = 40 = L_8.$$

Thus jobs (3, 5) and (5, 8) will become critical.

In this case we get one more critical path, $1 - 2 - 3 - 5 - 8$.

Hence when $t_{2,3}$ changes to 12 instead of 6. We get two critical paths

$$(a) 1 - 2 - 4 - 6 - 8 \quad \text{and} \quad (b) 1 - 2 - 3 - 5 - 8.$$

The completion time of project along both paths is 40 days.

- (vi) (a) The jobs (3, 6), (5, 8) and (7, 8) have free floats 14, 6 and 14 days respectively. (See above table)
- (b) The jobs (3, 5) and (3, 6), have independent floats, -6 and 8 days respectively.

Example 5 : Draw the network diagram from the following activities and find the critical path and total float of activities.

Job	Job Time (days)	Immediate Predecessors
A	13	–
B	8	A
C	10	B
D	9	C
E	11	B
F	10	E
G	8	D, F
H	6	E
I	7	H
J	14	G, I
K	18	J

Solution: The network diagram from the given activities is as follows :

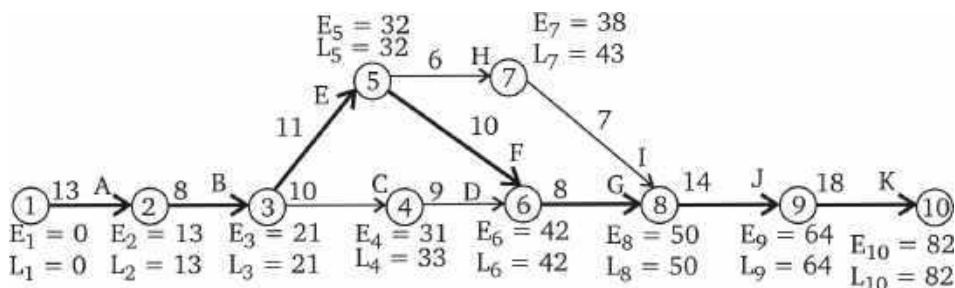


Fig. 9.44

To find earliest start time E_i for tail event i of each job (i, j)

Taking $E_1 = 0$

$$E_2 = E_1 + t_{1,2} = 0 + 13 = 13$$

$$E_3 = E_2 + t_{2,3} = 13 + 8 = 21$$

$$E_4 = E_3 + t_{3,4} = 21 + 10 = 31$$

$$E_5 = E_3 + t_{3,5} = 21 + 11 = 32,$$

$$E_6 = \text{Max.} \{E_i + t_{i,6}\} = \text{Max.} \{E_4 + t_{4,6}, E_5 + t_{5,6}\}$$

$$= \text{Max.} \{31 + 9, 32 + 10\} = 42$$

$$E_7 = E_5 + t_{5,7} = 32 + 6 = 38,$$

$$E_8 = \text{Max.} \{E_i + t_{i,8}\} = \text{Max} \{E_6 + t_{6,8}, E_7 + t_{7,8}\}$$

$$= \text{Max.} \{42 + 8, 38 + 7\} = 50$$

$$E_9 = E_8 + t_{8,9} = 50 + 14 = 64, \quad E_{10} = E_9 + t_{9,10} = 64 + 18 = 82 \text{ days.}$$

To find latest finish time L_j for head event j of each job (i, j)

$$\text{Taking } L_{10} = 82$$

$$L_9 = L_{10} - t_{9,10} = 82 - 18 = 64$$

$$L_8 = L_9 - t_{8,9} = 64 - 14 = 50,$$

$$L_7 = L_8 - t_{7,8} = 50 - 7 = 43$$

$$L_6 = L_8 - t_{6,8} = 50 - 8 = 42,$$

$$L_5 = \text{Min.}_{j=6,7} \{L_j - t_{5,j}\} = \text{Min.} \{L_6 - t_{5,6}, L_7 - t_{5,7}\} \\ = \text{Min.} \{42 - 10, 43 - 6\} = 32$$

$$L_4 = L_6 - t_{4,6} = 42 - 9 = 33,$$

$$L_3 = \text{Min.}_{j=4,5} \{L_j - t_{3,j}\} = \text{Min.} \{L_4 - t_{3,4}, L_5 - t_{3,5}\} \\ = \text{Min.} \{33 - 10, 32 - 11\} = 21$$

$$L_2 = L_3 - t_{2,3} = 21 - 8 = 13$$

$$L_1 = L_2 - t_{1,2} = 13 - 13 = 0$$

For each job (i, j) earliest start time E_i and earliest finish time $E F_{i,j} = E_i + t_{i,j}$, latest finish time L_j and latest start time $L S_{i,j} = L_j - t_{i,j}$ and the total float are tabulated in the following table :

Table 9.5

Job Activity (i, j)	Job Time $t_{i,j}$ (days)	Earliest Time		Latest Time		Total Float $L_j - E_i - t_{i,j}$
		Start E_i	Finish $E F_{i,j} = E_i + t_{i,j}$	Start $L S_{i,j} = L_j - t_{i,j}$	Finish L_j	
(1)	(2)	(3)	(4) = (3) + (2)	(5) = (6) - (2)	(6)	(7) = (6) - (4)
—	—	—	—	—	$L_1 = 0$	—
A (1, 2)	13	$E_1 = 0$	13	0	$L_2 = 13$	0
B (2, 3)	8	$E_2 = 13$	21	13	$L_3 = 21$	0
C (3, 4)	10	$E_3 = 21$	31	23	$L_4 = 33$	2
E (3, 5)	11	$E_3 = 21$	32	21	$L_5 = 32$	0
D (4, 6)	9	$E_4 = 31$	40	33	$L_6 = 42$	2
F (5, 6)	10	$E_5 = 32$	42	32	$L_6 = 42$	0
H (5, 7)	6	$E_5 = 32$	38	37	$L_7 = 43$	5
G (6, 8)	8	$E_6 = 42$	50	42	$L_8 = 50$	0
I (7, 8)	7	$E_7 = 38$	45	43	$L_8 = 50$	5
J (8, 9)	14	$E_8 = 50$	64	50	$L_9 = 64$	0
K (9, 10)	18	$E_9 = 64$	82	64	$L_{10} = 82$	0
		$E_{10} = 82$				

Since floats of jobs A, B, E, F, G, J and K are zero, so the critical activities are (1, 2), (2, 3), (3, 5), (5, 6), (6, 8), (8, 9) and (9, 10). i.e. the critical path is 1-2-3-5-6-8-9-10, shown by double lines in the fig. 9.44

Total float of activities are as follows :

Activity	:	A	B	C	D	E	F	G	H	I	J	K
Total float	:	0	0	2	2	0	0	0	5	5	0	0

Example 6: The following table gives the list of various activities and their immediate predecessors for a research project.

Activity	Preceding Activity	Duration (days)
A	–	6
B	–	5
C	B	2
D	C	2
E	A, D	2
F	D	1
G	A, D	6
H	E	5
I	G, H	6
J	I	2
K	G	4
L	J, K	3
M	L	1

- Draw a network diagram of the project.
- Find the critical path and minimum completion time of the project.
- Find the total float and the free float for each non-critical activity.

Solution: (i) Network diagram of the project.

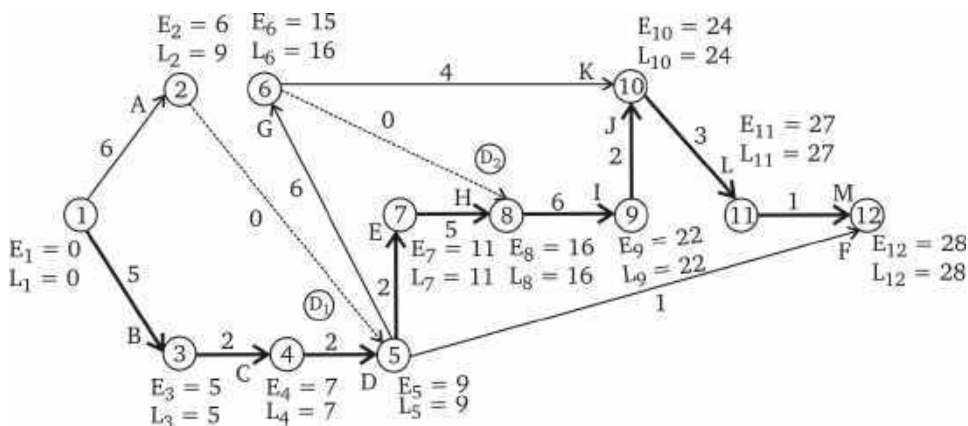


Fig. 9.45

(ii) To find earliest start time E_i for tail event i of each job (i, j)

$$\text{Taking } E_1 = 0$$

$$E_2 = E_1 + t_{1,2} = 0 + 6 = 6$$

$$E_3 = E_1 + t_{1,3} = 0 + 5 = 5$$

$$E_4 = E_3 + t_{3,4} = 5 + 2 = 7$$

$$E_5 = \text{Max.}_{i=2,4} \{E_i + t_{i,5}\}$$

$$E_6 = E_5 + t_{5,6} = 9 + 6 = 15$$

$$= \text{Max.} \{E_2 + t_{2,5}, E_4 + t_{4,5}\}$$

$$E_8 = \text{Max.}_{i=6,7} \{E_i + t_{i,8}\}$$

$$= \text{Max.} \{6 + 0, 7 + 2\} = 9$$

$$= \text{Max.} \{E_6 + t_{6,8}, E_7 + t_{7,8}\}$$

$$E_7 = E_5 + t_{5,7} = 9 + 2 = 11$$

$$= \text{Max.} \{15 + 0, 11 + 5\} = 16$$

$$E_9 = E_8 + t_{8,9} = 16 + 6 = 22$$

$$E_{10} = \text{Max.}_{i=6,9} \{E_i + t_{i,10}\}$$

$$E_{11} = E_{10} + t_{10,11} = 24 + 3 = 27$$

$$= \text{Max.} \{E_6 + t_{6,10}, E_9 + t_{9,10}\}$$

$$E_{12} = E_{11} + t_{11,12} = 27 + 1 = 28$$

$$= \text{Max.} \{15 + 4, 22 + 2\} = 24$$

To find latest finish time L_j for head event j of each activity (i, j)

$$\text{Taking } L_{12} = E_{10} = 28$$

$$L_{11} = L_{12} - t_{11,12} = 28 - 1 = 27$$

$$L_{10} = L_{11} - t_{10,11} = 27 - 3 = 24$$

$$L_9 = L_{10} - t_{9,10} = 24 - 2 = 22$$

$$L_8 = L_9 - t_{8,9} = 22 - 6 = 16$$

$$L_7 = L_8 - t_{7,8} = 16 - 5 = 11$$

$$L_6 = \text{Min.}_{j=8,10} \{L_j - t_{6,j}\}$$

$$L_5 = \text{Min.}_{k=6,7,12} \{L_k - t_{5,k}\}$$

$$= \text{Min.} \{L_8 - t_{6,8}, L_{10} - t_{6,10}\}$$

$$= \text{Min.} \{L_6 - t_{5,6}, L_7 - t_{5,7}, L_{12} - t_{5,12}\}$$

$$= \text{Min.} \{16 - 0, 24 - 4\} = 16$$

$$= \text{Min.} \{16 - 6, 11 - 2, 28 - 1\} = 9$$

$$L_4 = L_5 - t_{4,5} = 9 - 2 = 7$$

$$L_3 = L_4 - t_{3,4} = 7 - 2 = 5$$

$$L_2 = L_5 - t_{2,5} = 9 - 0 = 9$$

$$L_1 = \text{Min.}_{j=2,3} \{L_j - t_{1,j}\}$$

$$= \text{Min.} \{L_2 - t_{1,2}, L_3 - t_{1,3}\}$$

$$= \text{Min.} \{9 - 6, 5 - 5\} = 0$$

For each activity (i, j) the earliest time E_i and earliest finish time $E F_{i,j} = E_i + t_{i,j}$, latest finish time L_j and latest start time $L S_{i,j} = L_j - t_{i,j}$ are computed in the following table.

Table 9.6

Activity (i, j)	Duration $t_{i,j}$ (days)	Earliest Time		Latest Time	
		Start E_i	Finish $E F_{i,j} = E_i + t_{i,j}$	Start $L S_{i,j} = L_j - t_{i,j}$	Finish L_j
(1)	(2)	(3)	(4) = (3) + (2)	(5) = (6) - (2)	(6)
—	—	—	—	—	0 (L_1)
A (1,2)	6	0 (E_1)	6	3	9 (L_2)
B (1,3)	5	0 (E_1)	5	0	5 (L_3)
D_1 (2,5)	0	6 (E_2)	6	9	9 (L_5)
C (3,4)	2	5 (E_3)	7	5	7 (L_4)
D (4,5)	2	7 (E_4)	9	7	9 (L_5)
G (5,6)	6	9 (E_5)	15	10	16 (L_6)
E (5,7)	2	9 (E_5)	11	9	11 (L_7)
F (5,12)	1	9 (E_5)	10	27	28 (L_{12})
D_2 (6,8)	0	15 (E_6)	15	16	16 (L_8)
K (6,10)	4	15 (E_6)	19	20	24 (L_{10})
H (7,8)	5	11 (E_7)	16	11	16 (L_8)
I (8,9)	6	16 (E_8)	22	16	22 (L_9)
J (9,10)	2	22 (E_9)	24	22	24 (L_{10})
L (10,11)	3	24 (E_{10})	27	24	27 (L_{11})
M (11,12)	1	27 (E_{11})	28 (E_{12})	27 (L_{11})	28 (L_{12})
—	—	28 (E_{12})	—	—	—

Since for activities B (1,3), C (3,4), D (4,5), E (5,7)

H (7,8), I (8,9), J (9,10), L (10,11) and M (11,12)

$E_i = L_j$ and $L_j - L_i = E_j - E_i = t_{i,j}$, so these are critical activities

Thus the *critical path* is B – C – D – E – H – I – J – L – M shown by thick lines in the network diagram and project completion time = $E_{12} = 28$ days.

- (iii) **Total float and free float** for non-critical activities are tabulated in the following table.

Table 9.7

Activity (Non-Critical) (i, j)	Total Float $L_j - E_i - t_{ij}$	Free Float T. Float – Head Slack
(i)	(ii) = (6) – (4)	(ii) – ($L_j - E_j$)
A (1, 5)	$9 - 6 = 3$	$3 - (L_5 - E_5) = 3$
F (4, 12)	$28 - 10 = 18$	$18 - (L_{12} - E_{12}) = 18$
G (5, 6)	$16 - 15 = 1$	$1 - (L_6 - E_6) = 0$
K (6, 10)	$24 - 19 = 5$	$5 - (L_{10} - E_{10}) = 5$

Exercise 9.2

- What is critical path ?
- What is critical path ? State the necessary and sufficient conditions for the critical path. Can a project have multiple critical path ?
- Define total, free and independent floats.
- Write short notes on total float and free float.
- Prove that total float $\geq$ free float $\geq$ Independent float.
- A small project consists of the following jobs where precedence relationship is given below :

Job	1-2	1-3	2-3	2-5	3-4	3-6	4-5	4-6	5-6	6-7
Duration (Days)	15	15	3	5	8	12	1	14	3	14

- Draw an arrow diagram representing the project.
 - Find the critical path and the total project duration.
7. The following table gives the activities of a construction, project and duration :

Activity (i, j)	1-2	1-3	2-3	2-4	3-4	4-5
Duration (Days)	20	25	10	12	6	10

- Draw the network for the project.
- Find the critical path and the minimum project time.

8. The following table gives the activities of a project and duration.

Activity (i, j)	0-1	1-2	1-3	2-4	2-5	3-4	3-6	4-7	5-6	6-7
Duration (Days)	1	4	5	3	2	2	4	3	1	4

- (i) Draw the network for the project.
(ii) Identify the critical path and minimum completion time of the project.
(iii) Determine the total, free and independent floats of activities.
9. A certain project is composed of mini activities whose time estimates are given below :

Activity	1-2	1-3	1-4	2-5	3-5	4-6	5-6	6-7	5-7
Duration	1	3	2	1	3	2	4	6	3

Draw the project network and find out the critical path.

10. A project has the following activity and duration time schedule :

Activity	Duration (in weeks)	Activity	Duration (in weeks)
(1, 2)	4	(5, 7)	8
(1, 3)	1	(6, 8)	1
(2, 4)	1	(7, 8)	2
(3, 4)	1	(8, 9)	1
(3, 5)	6	(8, 10)	8
(4, 9)	5	(9, 10)	7
(5, 6)	4		

- (i) Draw the network diagram of the project.
(ii) Determine the float for each activity.
(iii) Find the critical path and its duration.
11. The following table gives the activities in a construction project and time duration :

Activity	Preceding Activity	Normal time (days)
1-2	—	5
1-3	—	4
2-4	1-2	6
3-4	1-3	2
4-5	2-4	1
4-6	2-4, 3-4	7
5-7	4-5	8
6-7	4-6	4
7-8	6-7, 5-7	3

- (i) Draw the activity network of the project.
- (ii) Find the total float and free float for each activity.
- (iii) Determine the critical path and the project duration.

12. The following table gives the activities in a project and time duration.

Activity	Duration in weeks	Activity	Duration in weeks
1 – 2	3	4 – 7	4
1 – 4	2	5 – 6	4
1 – 5	2	6 – 9	6
2 – 3	4	7 – 8	5
3 – 8	2	8 – 9	3
4 – 6	7		

- (i) Draw the activity network of the project.
- (ii) Compute earliest and latest event time.
- (iii) Find total, free and independent floats for each activity.
- (iv) Determine the critical path and the project duration.

13. A small project consists of seven activities for which the relevant data are given below :

Activity	Preceding Activity	Activity Duration (days)
A	–	4
B	A	7
C	–	6
D	C	5
E	B	7
F	D, E	6
G	F	5

- (i) Draw the network and find the project completion time.
- (ii) Calculate total float for each activity.

14. A project consists of a series of tasks labelled $A, B, \dots, H, I$ with the following relationships ($W < X, Y$ means X and Y cannot start until W is completed, $X, Y < W$ means W cannot start until both X and Y are completed). With this notation.

- (i) Construct the network diagram having the following constraints :

$$A < D, E; B, D < F; C < G; B, G < H; F, G < I$$

- (ii) Find also the critical path and the minimum time of completion of the project, when the time (in days) of completion of each task is as follows :

Task	A	B	C	D	E	F	G	H	I
Time (In days)	23	8	20	16	34	18	19	14	10

15. A research and development department is developing a new power supply for a console television set. It has broken the project into the following jobs with their duration times in days.

Job	Immediate Predecessors	Duration time (in days)
A	–	5
B	A	7
C	B	2
D	B	3
E	C	1
F	D	2
G	C	1
H	E, F	3
I	G, H	10

- (i) Draw a critical path scheduling diagram, identifying jobs letters and associated times with each. Indicate the critical path.
- (ii) What is the minimum time for the completion of the project ?

16. An insurance company has decided to modernise and refit one of its branch office.

The major jobs of the project have been identified as follows along with their durations and immediately preceding jobs.

Job	Immediate Predecessors	Duration (weeks)
A	E	4
B	A	2
C	B	1
D	K	12
E	–	14
F	E	2
G	F	3
H	F	2
I	F	4
J	I, L	3
K	C, G, H	4
L	D	2
M	I, L	2

- (i) Draw the network showing the inter-relations between the various element of the project.
- (ii) Determine the critical path and the minimum time that the renovation can take.
- (iii) Calculate the total, free and independent floats associated with the activities in the network diagram.

17. Consider the network shown in the following figure :

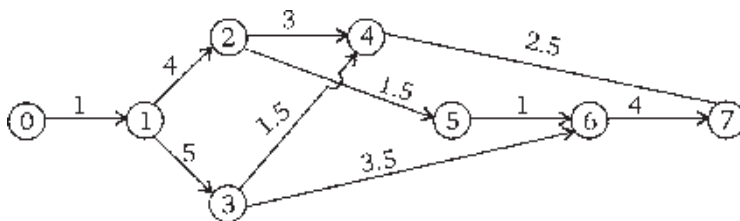


Fig. 9.46

Determine the total float, free float, independent float and identify the critical path.

Answers 9.2

6. (i)

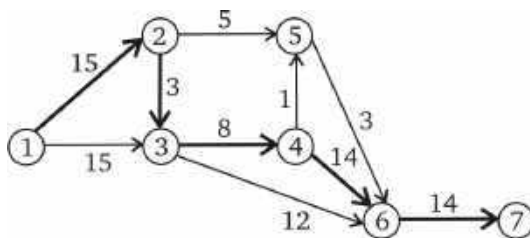


Fig.9.47

(ii) Critical path : 1-2-3-4-6-7.

Total project duration = 54 days

7. (i)

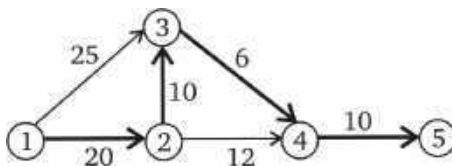


Fig.9.48

(ii) Critical path : 1-2-3-4-5

Minimum project time = 46 days

8. (i)

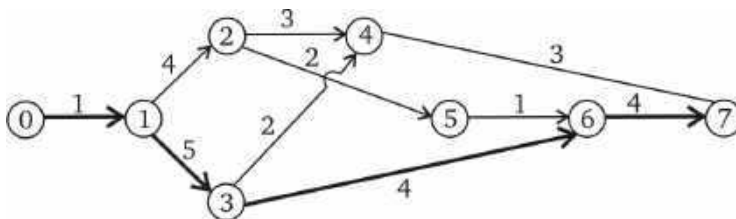


Fig.9.49

(ii) Critical path : 0-1-3-6-7

Minimum completion time = 14 days

(iii)

Floats	:	0-1	1-2	1-3	2-4	2-5	3-4	3-6	4-7	5-6	6-7
Total	:	0	2	0	3	2	3	0	3	2	0
Free	:	0	0	0	0	0	0	0	3	2	0
Independent	:	0	0	0	-2	-2	0	0	0	0	0

9. Network

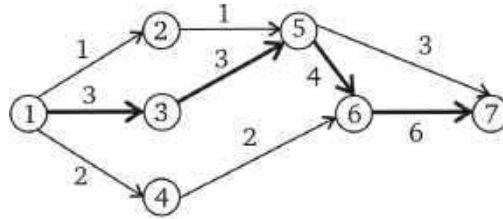


Fig. 9.50

Critical path. 1-3-5-6-7.

10. (i)

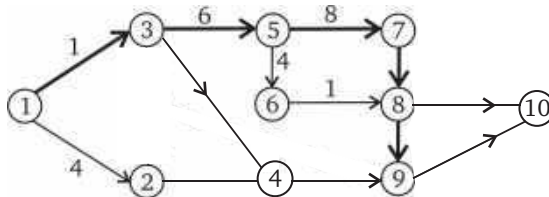


Fig. 9.51

(ii)

Activity :	(1,2)	(1,3)	(2,4)	(3,4)	(3,5)	(4,9)	(5,6)	(5-7)	(6,8)	(7,8)	(8,9)	(8,10)	(9,10)
	5	0	5	8	0	5	5	0	5	0		0	5
Total Float :	8	0	8	11	0	8	5	0	5	0	0	0	0
Free :	0	0	0	3	0	0	0	0	0	0		0	5

(iii) Critical path : (i) 1-3-5-7-8-10 (ii) 1-3-5-7-8-9-10

Duration of project = 22 weeks.

11. (i)

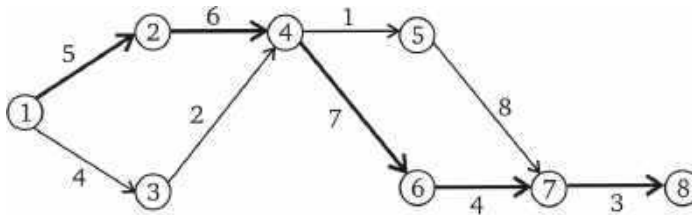


Fig. 9.52

(ii)

Activity:	1-2	1-3	2-4	3-4	4-5	4-6	5-7	6-7	7-8
Total Float:	0	5	0	5	2	0	2	0	0
Free Float:	0	0	0	5	0	0	2	0	0

(iii) Critical Path : 1-2-4-6-7-8

Project duration : 25 days.

12. (i)

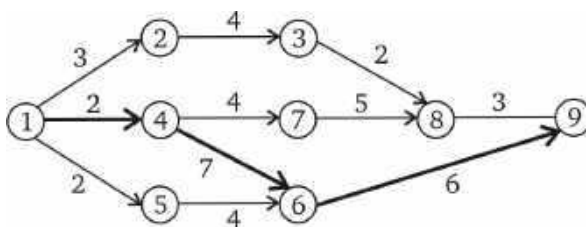


Fig. 9.53

(ii) and (iii)

Events	:	1-2	1-4	1-5	2-3	3-8	4-6	4-7	5-6	6-9	7-8	8-9
E_i or $S_{i,j}$	:	0	0	0	3	7	2	2	2	9	6	11
$E F_{i,j}$	:	3	2	2	7	9	9	6	6	15	11	14
$L S_{i,j}$	:	3	0	3	6	10	2	3	5	9	7	12
L_j or $L F_{i,j}$	:	6	2	5	10	12	9	7	9	15	12	15
T. Float	:	3	0	3	3	3	0	1	3	0	1	1
F. Float	:	0	0	0	0	2	0	0	3	0	0	1
I. Float	:	0	0	0	-3	-1	0	0	0	0	-1	-1

(iv) Critical path : 1-4-6-9

Project duration : 15 weeks.

13. (i)

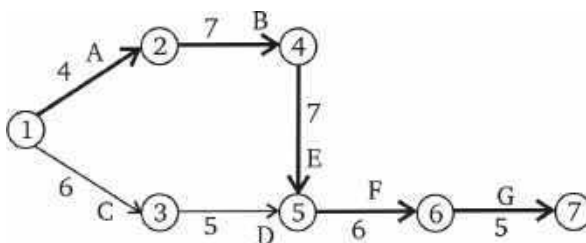


Fig. 9.54

Critical Path : A-B-E-F-G, 1-2-4-5-6-7

Project Completion Time : 29 days.

(ii)

Activity	:	A	B	C	D	E	F	G
Total Float	:	0	0	7	7	0	0	0

14. (i)

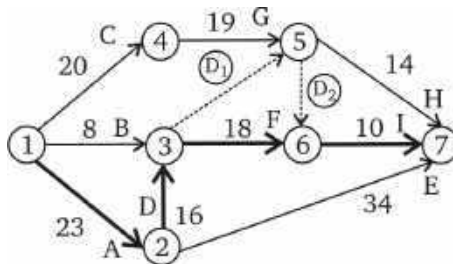


Fig. 9.55

(ii) *Critical Path* : A-D-F-I, 1-2-3-6-7

Minimum completion time of the project = 67 days.

15. (i)

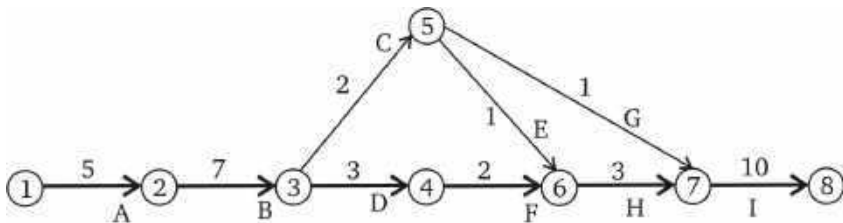


Fig. 9.56

Critical path : 1-2-3-4-6-7-8, i.e. A → B → D → F → H → I

(ii) Minimum time for the completion of the project = 30 days.

16. (i)

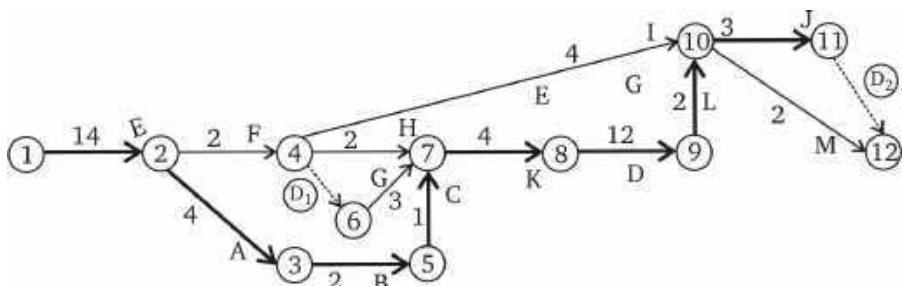


Fig. 9.57

(ii) *Critical path* : E-A-B-C-K-D-L-J

Mini time : 42 weeks.

(iii)

<i>Job</i>	:	A	B	C	D	E	F	G	H	I	J	K	L	M
T. Float	:	0	0	0	0	0	2	2	3	19	0	0	0	1
F. Float	:	0	0	0	0	0	0	2	3	19	0	0	0	0
I. Float	:	0	0	0	0	0	0	0	1	17	0	0	0	0

17.

<i>Activity</i>	:	0-1	1-2	1-3	2-4	2-5	3-4	3-6	4-7	5-6	6-7
T. float	:	0	2	0	3	2	3.5	0	3	2	0
F. float	:	0	0	0	0	0	0.5	0	3	2	0
I. float	:	0	0	0	-2	-2	0.5	0	0	0	0

9.16 PERT

The network analysis discussed so far may be termed as deterministic. In CPM analysis the time durations of activities were assumed to be certain (constant). If the activity duration time in a project are uncertain, the activity based analysis is done by using the expected values of the duration time which are based upon three different time estimates.

These *three time estimates for PERT* are as follows :

1. **Optimistic time (t_0 or a)** *Optimistic time* is the shortest possible time required to complete an activity, if every thing goes well.
2. **Pessimistic time (t_p or b)** *Pessimistic time* is the longest (maximum) possible time that an activity could take if every thing goes wrong *i.e.* under extremely bad conditions. However these conditions do not include the nature acts like floods, earth quakes etc.
3. **Most likely time (t_m or m)** *Most likely time* is the most often likely time required to complete an activity. In other words if an activity is repeated many times, then it is the duration that would occur most frequently. Statistically it is the model value of duration of an activity.

After evaluating the values of t_0 , t_p and t_m they are combined into single workable time values for each activity known as *expected times* of the activities. Also the *standard deviations* and *variance* of all activities are calculated.

Expected time : *Expected time* (t_e or μ) is the average time an activity will take if it were to be repeated on large number of times and is based on the assumption that the activity time follows Beta distribution and is given by the formula.

Expected time, t_e (or μ) = $(t_o + 4t_m + t_p) / 6$

Standard deviation and variance: The standard deviation (σ) and the variance σ^2 of an activity are given by

Standard deviation, $\sigma = (t_p - t_o) / 6$

Variance, $\sigma^2 = [(t_p - t_o) / 6]^2$

9.17 Estimate of Probability of Completing the Project by Scheduled Time

Since a variability in the activity duration time is expected, the total project may not be completed exactly in time. We can find the probability of actually completing the project as well as the activities in the schedule time.

The probability of completing the project by schedule time T_s is given by

$$\text{Prob.} \left(Z \leq \frac{T_s - T_e}{\sigma_e} \right)$$

where, T_e = Expected time of the project

= Sum of expected time of each activity lying on the critical path.

σ_e = Standard deviation of the project length

= $\sqrt{\text{Variance of critical path}}$

= $\sqrt{\text{Sum of variances of critical activities}}$

The standard normal variate value is given by $Z = \frac{T_s - T_e}{\sigma_e}$

ILLUSTRATIVE EXAMPLES

Example I : A small project consisting of eight activities has the following characteristics :

Activity	Preceding activity	Time estimates (in weeks)		
		Most optimistic t_o	Most likely t_m	Most pessimistic t_p
A	–	2	4	12
B	–	10	12	26
C	A	8	9	10
D	A	10	15	20
E	A	7	7.5	11
F	B, C	9	9	9
G	D	3	3.5	7
H	E, F, G	5	5	5

- (i) Draw the PERT network for the project.
- (ii) Find the expected duration and variance for each activity.
- (iii) Determine the critical path and the expected project length.
- (iv) Prepare the activity schedule for the project.
- (v) Calculate the variance and standard deviation of the project length.
- (vi) If a 30 weeks deadline is imposed, what is the probability that the project will be finished in time ?
- (vii) If the project manager wants to be 99% , sure that the project is completed on the schedule date, how many weeks before that data should he start the project work ?

Solution : (i) The network diagram for the project is as follows :

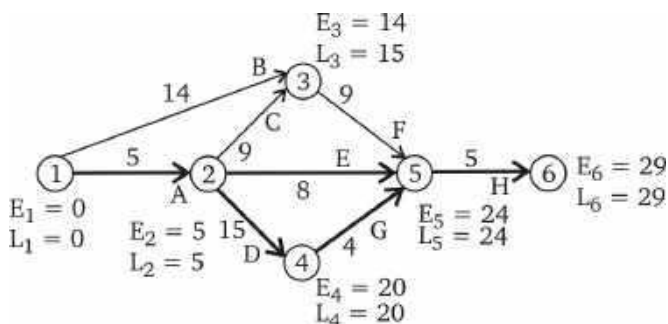


Fig. 9.58

- (ii) Expected duration time and variance of each activity are shown in the table.

Table 9.8

Activity (i, j)	Time estimates			Expected activity time $t_e = \frac{t_o + 4 t_m + t_p}{6}$	Standard deviation $\sigma = \frac{t_p - t_o}{6}$	Variance σ^2
	t_o	t_m	t_p			
A 1-2	2	4	12	5	1.67	2.79
B 1-3	10	12	26	14	2.67	7.13
C 2-3	8	9	10	9	0.33	0.11
D 2-4	10	15	20	15	1.67	2.79
E 2-5	7	7.5	11	8	0.67	0.45
F 3-5	9	9	9	9	0.00	0.00
G 4-5	3	3.5	7	4	0.67	0.45
H 5-6	5	5	5	5	0.00	0.00

(iii) To find expected earliest start time E_i of tail event of each activity (i, j) .

Taking $E_1 = 0$, We have $E_2 = E_1 + t_{e12} = 0 + 5 = 5$

$$E_3 = \text{Max.} \{E_i + t_{ei3}\} = \text{Max.} \{E_1 + t_{e13}, E_2 + t_{e23}\}$$

$$= \text{Max.} \{0 + 14, 5 + 9\} = 14$$

$$E_4 = E_2 + t_{e24} = 5 + 15 = 20$$

$$E_5 = \text{Max.} \{E_i + t_{ei5}\} \quad E_6 = E_5 + t_{e56} = 24 + 5 = 29$$

$$= \text{Max.} \{E_2 + t_{e25}, E_3 + t_{e35}, E_4 + t_{e45}\},$$

$$= \text{Max.} \{5 + 8, 14 + 9, 20 + 4\} = 24$$

To find expected latest finish time L_j for head event j of each activity $(i - j)$

Taking $L_6 = E_6 = 29$, $L_5 = L_6 - t_{e56} = 29 - 5 = 24$

$$L_4 = L_5 - t_{e45} = 24 - 4 = 20 \quad L_3 = L_5 - t_{e35} = 24 - 9 = 15$$

$$L_2 = \text{Mini.} \{L_j - t_{e2j}\} \quad L_1 = L_2 - t_{e12} = 5 - 5 = 0.$$

$$= \text{Mini.} \{L_3 - t_{e23}, L_4 - t_{e24}, L_5 - t_{e25}\}$$

$$= \text{Mini.} \{15 - 9, 20 - 15, 24 - 8\} = 5$$

$\therefore E_1 = L_1, E_2 = L_2, E_4 = L_4, E_5 = L_5$ and $E_6 = L_6$, also for these activities

$$E_j - E_i = t_{eij} = L_j - L_i.$$

So, the critical path is 1-2-4-5-6 i.e. A-D-G-H

and expected project length = $T_e = 29$ weeks.

(iv) Other activity schedule for the project are 1-3-5-6 i.e. B-F-H

and 1-2-5-6 i.e. A-E-H.

(v) Variance of the project length = Sum of variances of critical activities, A, D, G and H.

$$= 2.79 + 2.79 + 0.45 + 0.00 = 6.03$$

and standard deviation of project length = $\sigma_e = \sqrt{6.03} = 2.46$

(vi) The probability of project schedule time, $T_S = 30$ weeks, is given by

$$\text{Prob} \left\{ Z \leq \frac{T_s - T_e}{\sigma_e} \right\} = \text{Prob} \left\{ Z \leq \frac{30 - 29}{2.46} \right\} = \text{Prob} \{Z \leq 0.41\} = 0.5 + 0.1591$$

$$= 0.6591 = 0.67 \text{ (appr.)}$$

Thus, there are 67% chances of completion of the project in 30 weeks.

(vii) If T_S is the project schedule time (in weeks) for 99%. Chances of completion of the project in time, then

$$\text{Prob. } \left\{ Z \leq \frac{T_S - T_e}{\sigma_e} \right\} = 0.99$$

$$\Rightarrow \frac{T_S - 29}{2.46} = 2.33 \Rightarrow T_S = 34.7 \text{ weeks.}$$

Hence for 99% chances for the completion of project on the schedule date, the work should be started $34.7 - 30 = 4.7$ weeks before the start time.

Example 2: A project schedule has the following characteristics :

Activity	Most optimistic time (t_o)	Most likely time (t_m)	Most pessimistic time (t_p)
1 – 2	1	2	3
2 – 3	1	2	3
2 – 4	1	3	5
3 – 5	3	4	5
4 – 5	2	3	4
4 – 6	3	5	7
5 – 7	4	5	6
6 – 7	6	7	8
7 – 8	2	4	6
7 – 9	4	6	8
8 – 10	1	2	3
9 – 10	3	5	7

Construct a PERT network and find out

- (i) the earliest possible time to complete the different stages of the project.
- (ii) the latest allowable time for them.
- (iii) the slack values.
- (iv) the critical paths.
- (v) the probability factor for completing the project in 30 weeks.

Solution : Network diagram.

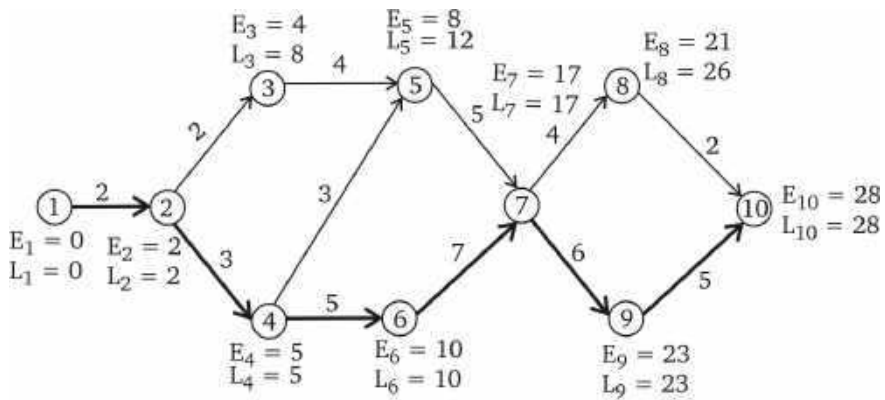


Fig. 9.59

The expected activity time and variance of each activity $i-j$ are calculated in the following table :

Table 9.9

Activity	Time estimates			Expected time	Variance
	t_o	t_m	t_p	$t_e = \frac{t_o + 4 t_m + t_p}{6}$	$\sigma^2 = \left(\frac{t_p - t_o}{6} \right)^2$
1-2	1	2	3	2	0.11
2-3	1	2	3	2	0.11
2-4	1	3	5	3	0.45
3-5	3	4	5	4	0.11
4-5	2	3	4	3	0.11
4-6	3	5	7	5	0.45
5-7	4	5	6	5	0.11
6-7	6	7	8	7	0.11
7-8	2	4	6	4	0.45
7-9	4	6	8	6	0.45
8-10	1	2	3	2	0.11
9-10	3	5	7	5	0.45

(i), (ii) and (iii)

To find the expected earliest start time E_i of tail event of each activity ($i-j$)

Taking $E_1 = 0$, we have $E_2 = E_1 + t_{e12} = 0 + 2 = 2$

$E_3 = E_2 + t_{e23} = 2 + 2 = 4$ $E_4 = E_2 + t_{e24} = 2 + 3 = 5$

$$E_5 = \text{Max.} \{E_i + t_{ei5}\} = \text{Max.} \{E_3 + t_{e35}, E_4 + t_{e45}\},$$

$$= \text{Max.} \{4 + 4, 5 + 3\} = 8 \quad E_6 = E_4 + t_{e46} = 5 + 5 = 10$$

$$E_7 = \text{Max.} \{E_i + t_{ei7}\} = \text{Max.} \{E_5 + t_{e57}, E_6 + t_{e67}\}$$

$$= \text{Max.} \{8 + 5, 10 + 7\} = 17, \quad E_8 = E_7 + t_{e78} = 17 + 4 = 21$$

$$E_9 = E_7 + t_{e79} = 17 + 6 = 23,$$

$$E_{10} = \text{Max.} \{E_i + t_{ei10}\} = \text{Max.} \{E_8 + t_{e8,10}, E_9 + t_{e910}\}$$

$$= \text{Max.} \{21 + 2, 23 + 5\} = 28$$

To find the expected latest finishing L_j of head event of each activity ($i - j$)

Taking $L_{10} = E_{10} = 28$, we have $L_9 = L_{10} - t_{e910} = 28 - 5 = 23$

$$L_8 = L_{10} - t_{e810} = 28 - 2 = 26 \quad L_7 = \text{Min.} \{L_j - t_{e7j}\}_{j=8,9}$$

$$= \text{Min.} \{L_8 - t_{e78}, L_9 - t_{e79}\}$$

$$= \text{Min.} \{26 - 4, 23 - 6\} = 17$$

$$L_6 = L_7 - t_{e67} = 17 - 7 = 10, \quad L_5 = L_7 - t_{e57} = 17 - 5 = 12$$

$$L_4 = \text{Min.} \{L_j - t_{e4j}\}_{j=5,6} = \text{Min.} \{L_5 - t_{e45}, L_6 - t_{e46}\},$$

$$= \text{Min.} \{12 - 3, 10 - 5\} = 5 \quad L_3 = L_5 - t_{e35} = 12 - 4 = 8$$

$$L_2 = \text{Min.} \{L_j - t_{e2j}\}_{j=3,4} = \text{Min.} \{L_3 - t_{e23}, L_4 - t_{e24}\}$$

$$= \text{Min.} \{8 - 2, 5 - 3\} = 2 \quad L_1 = L_2 - t_{e12} = 2 - 2 = 0$$

The earliest possible time and latest allowable time to complete the different stages of the project and the slack values of events are given tabulated here as follows.

Event	:	1	2	3	4	5	6	7	8	9	10
E_i	:	0	2	4	5	8	10	17	21	23	28
L_i	:	0	2	8	5	12	10	17	26	23	28
Slack ($L_i - E_i$)	:	0	0	4	0	4	0	0	5	0	0

(iv) The critical path connecting the events (nodes) with zero slack is

$$1-2-4-6-7-9-10$$

and expected project length $T_e = 28$ weeks.

(v) Variance of the project length = σ_e^2 = Sum of variances of the critical activities

$$1-2, 2-4, 4-6, 6-7, 7-9, 9-10.$$

$$= 0.11 + 0.45 + 0.45 + 0.11 + 0.45 + 0.45 = 2.02$$

$\therefore$ Standard deviation of project length = $\sigma_e = \sqrt{2.02} = 1.42$

The probability factor for completing the work in $T_s = 30$ weeks is given by

$$\text{Prob} \left\{ Z \leq \frac{T_s - T_e}{\sigma_e} \right\} = \text{Prob} \left\{ Z \leq \frac{30 - 28}{1.42} = 1.41 \right\} = 0.5 + 0.4207 = 0.9207$$

i.e. 92.07%

Example 3 : A certain project is composed of nine activities where time estimates are given below :

Activity	Expected duration (week)		
	Optimistic	Most likely	Pessimistic
1 – 2	1	1	7
1 – 3	3	5	7
1 – 4	2	2	8
2 – 5	1	1	1
3 – 5	3	6	9
4 – 6	2	5	8
5 – 6	4	6	14
5 – 7	3	7	11
6 – 7	6	8	10

- (i) Draw the project network and trace all the possible paths from it. What is the expected project length ?
- (ii) Calculate the variance and standard deviation of the project length.
- (iii) What is the probability that the project will be completed :
 - (a) at least 4 weeks earlier than expected time.
 - (b) no more than 4 weeks later than expected time.
- (iv) If the project date is 29 weeks, what is the probability of not meeting the due date.
- (v) What due date has about 99% chance of completing the project ?

Solution : (i) Project Network diagram.

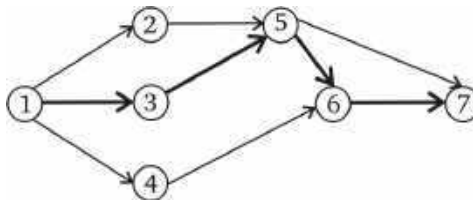


Fig. 9.60

Here all possible paths are

1-2-5-7; 1-2-5-6-7; 1-3-5-7; 1-3-5-6-7; 1-4-6-7.

The expected activity time and variance of each activity $i-j$ are calculated in the following table.

Table 9.10

Activity	Time estimates			Expected time	Variance
	t_o	t_m	t_p	$t_e = \frac{t_o + 4t_m + t_p}{6}$	$\sigma^2 = \left(\frac{t_p - t_o}{6}\right)^2$
1-2	1	1	7	2	1
1-3	3	5	7	5	0.45
1-4	2	2	8	3	1
2-5	1	1	1	1	0
3-5	3	6	9	6	1
4-6	2	5	8	5	1
5-6	4	6	14	7	2.79
5-7	3	7	11	7	1.77
6-7	6	8	10	8	0.45

To find expected earliest start time E_i for tail event of each activity $i-j$

Taking $E_1 = 0$, we have

$$E_2 = E_1 + t_{e12} = 0 + 2 = 2$$

$$E_3 = E_1 + t_{e13} = 0 + 5 = 5$$

$$E_4 = E_1 + t_{e14} = 0 + 3 = 3$$

$$E_5 = \text{Max. } \{E_i + t_{ei5}\}_{i=2,3}$$

$$E_6 = \text{Max. } \{E_i + t_{ei6}\}_{i=4,5}$$

$$= \text{Max. } \{E_2 + t_{e25}, E_3 + t_{e35}\}$$

$$= \text{Max. } \{E_4 + t_{e46}, E_5 + t_{e56}\}$$

$$= \text{Max. } \{2 + 1, 5 + 6\} = 11$$

$$= \text{Max. } \{3 + 5, 11 + 7\} = 18$$

$$E_7 = \text{Max. } \{E_i + t_{ei7}\}_{i=5,6} = \text{Max. } \{E_5 + t_{e57}, E_6 + t_{e67}\}$$

$$= \text{Max. } \{11 + 7, 18 + 8\} = 26$$

To find expected latest finish time L_j for head event of each activity $i-j$

Taking $L_7 = E_7 = 26$, We have

$$L_6 = L_7 - t_{e67} = 26 - 8 = 18$$

$$L_5 = \text{Min. } \{L_j - t_{ej5}\}_{j=6,7} = \text{Min. } \{L_6 - t_{e56}, L_7 - t_{e57}\},$$

$$= \text{Min. } \{18 - 7, 26 - 7\} = 11$$

$$L_4 = L_6 - t_{e46} = 18 - 5 = 13$$

$$L_3 = L_5 - t_{e35} = 11 - 6 = 5$$

$$L_2 = L_5 - t_{e25} = 11 - 1 = 10$$

$$L_1 = \text{Min. } \{L_j - t_{ej1}\}_{j=2,3,4} = \text{Min. } \{L_2 - t_{e12}, L_3 - t_{e13}, L_4 - t_{e14}\}$$

$$= \text{Min. } \{10 - 2, 5 - 5, 13 - 3\} = 0$$

$\therefore E_1 = L_1, E_3 = L_3, E_5 = L_5, E_6 = L_6$ and $E_7 = L_7$, also for these activities $E_j - E_i = t_{cij} = L_j - L_i$ so, the activities 1-3, 3-5, 5-6 and 6-7 are *critical activities*.

$\therefore$ The *critical path* is 1-3-5-6-7

and *expected project length*, $T_e = 26$ weeks.

(ii) For the project length,

variance $= \sigma_e^2 =$ sum of variances for critical activities, 1-3, 3-5, 5-6 and 6-7

$$= 0.45 + 1 + 2.79 + 0.45 = 4.69 \text{ weeks}$$

$\therefore$ standard deviation $= \sigma_e = \sqrt{4.69} = 2.17$ weeks.

(iii) (a) The probability of completing the project, 4 weeks earlier than expected time *i.e.* in time $T_s = 26 - 4 = 22$ weeks is given by

$$\text{Prob.} \left\{ Z \leq \frac{T_s - T_e}{\sigma_e} \right\} = \text{Prob.} \left\{ Z \leq \frac{22 - 26}{2.17} = -1.84 \right\} = 0.5 - 0.4671 = 0.0329$$

i.e. 3.29%

(b) And probability of completing the project 4 weeks later than expected time *i.e.*, in time $T_s = 26 + 4 = 30$ weeks, is given by

$$\text{Prob.} \left\{ Z \leq \frac{T_s - T_e}{\sigma_e} \right\} = \text{Prob.} \left\{ Z \leq \frac{30 - 26}{2.17} = 1.84 \right\} = 0.5 + 0.4671 = 0.9671$$

i.e., 96.71%

(iv) If $T_s = 29$ weeks, then prob of completing the work by due date is given by

$$\text{Prob.} \left\{ Z \leq \frac{T_s - T_e}{\sigma_e} \right\} = \text{Prob.} \left\{ Z \leq \frac{29 - 26}{2.17} = 1.38 \right\} = 0.5 + 0.4162 = 0.9162$$

i.e. 91.62%

Hence the probability of not meeting the project date of 29 weeks

$$= 1 - 0.9162 = 0.0838 \text{ i.e. } 8.38\%$$

(v) If T_s is the project time when the due date has about 99% chance, then

$$\text{Prob.} \left\{ Z \leq \frac{T_s - T_e}{\sigma_e} = \frac{T_s - 26}{2.17} \right\} = 0.99 \text{ (appr.)}$$

$$\Rightarrow \frac{T_s - 26}{2.17} = 2.33 \text{ (from normal distribution table)}$$

$$\Rightarrow T_s = 26 + 2.17 \times 2.33 = 26 + 5.06 = 31.06 \text{ weeks} = 31 \text{ weeks (appr.)}$$

9.18 Construction of Time Chart and Resource Labelling

The construction of the **time chart** (or **schedule**) is the end product of a network calculations, which can be converted into a calendar schedule for convenient use in the execution of a project.

Time chart construction must be made within the limitations of the available resources, since it may not be possible to execute concurrent activities because of personnel and equipment limitations. So the total floats for non-critical activity become useful. One may be able to lower the maximum resource requirements by shifting (back or forth) a non-critical activity between its maximum allowable limits (interval from earliest starting time to latest finish *i.e.*, completion time). Total floats can also be used to level resources over the duration of the entire project. This would mean a more steady work force compared to the case where the work force vary drastically from one day to the next.

For clear understanding of the procedure for the construction of **time chart** see the following illustrative example.

ILLUSTRATIVE EXAMPLES

Example 1 : For a project, the activities and their durations are given in the following table. Find total and free floats and the critical activities. Also draw the time chart of the project.

Activity	(0, 1)	(0, 2)	(1, 3)	(2, 3)	(2, 4)	(3,4)	(3, 5)	(3, 6)	(4, 5)	(4, 6)	(5, 6)
Duration	2	3	2	3	2	0	3	2	7	5	6

Solution: The arrow network diagram of the project is as follows :

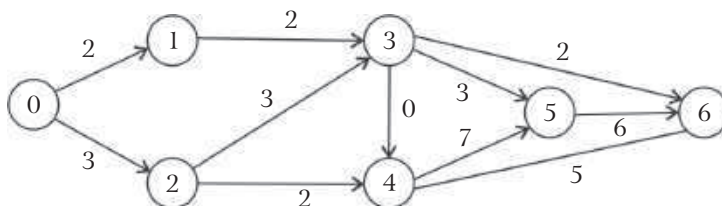


Fig. 9.61

To find the earliest start time ES_i of tail event i and latest completion time LC_j of head event j of each activity (i, j) .

Taking $ES_0 = 0$, since there is only one activity $(0, 1)$ entering to the event 1 and activity $(0, 2)$ entering to the event 2,

$$\therefore ES_1 = ES_0 + t_{0,1} = 0 + 2 = 2, \quad ES_2 = ES_0 + t_{0,2} = 0 + 3 = 3$$

$$ES_3 = \text{Max.}_{i=1,2} \{ES_i + t_{i,3}\} = \text{Max.} \{ES_1 + t_{1,3}, ES_2 + t_{2,3}\} = \text{Max.} \{2 + 2, 3 + 3\} = 6$$

$$ES_4 = \text{Max.}_{i=2,3} \{ES_i + t_{i,4}\} = \text{Max.} \{ES_2 + t_{2,4}, ES_3 + t_{3,4}\} = \text{Max.} \{2 + 2, 6 + 0\} = 6$$

$$ES_5 = \text{Max.}_{i=3,4} \{ES_i + t_{i,5}\} = \text{Max.} \{ES_3 + t_{3,5}, ES_4 + t_{4,5}\} = \text{Max.} \{6 + 3, 6 + 7\} = 13$$

$$ES_6 = \text{Max.}_{i=3,4,5} \{ES_i + t_{i,6}\} = \text{Max.} \{ES_3 + t_{3,6}, ES_4 + t_{4,6}, ES_5 + t_{5,6}\} \\ = \text{Max.} \{6 + 2, 6 + 5, 13 + 6\} = 19$$

To find latest completion time LC_j for head event j of each activity (i, j) .

$$\text{Taking } LC_6 = ES_6 = 19, LC_5 = LC_6 - t_{5,6} = 19 - 6 = 13$$

$$LC_4 = \text{Min.}_{j=5,6} \{LC_j - t_{4,j}\} = \text{Min.} \{LC_5 - t_{4,5}, LC_6 - t_{4,6}\} = \text{Min.} \{13 - 7, 19 - 5\} = 6$$

$$LC_3 = \text{Min.}_{j=4,5,6} \{LC_j - t_{3,j}\} = \text{Min.} \{LC_4 - t_{3,4}, LC_5 - t_{3,5}, LC_6 - t_{3,6}\} \\ = \text{Min.} \{6 - 0, 13 - 3, 19 - 2\} = 6$$

$$LC_2 = \text{Min.}_{j=3,4} \{LC_j - t_{2,j}\} = \text{Min.} \{LC_3 - t_{2,3}, LC_4 - t_{2,4}\} = \text{Min.} \{6 - 3, 6 - 2\} = 3$$

$$LC_1 = \text{Min.}_{j=3} \{LC_j - t_{1,j}\} = LC_3 - t_{1,3} = 6 - 2 = 4$$

$$LC_0 = \text{Min.}_{j=1,2} \{LC_j - t_{0,j}\} = \text{Min.} \{LC_1 - t_{0,1}, LC_2 - t_{0,2}\} = \text{Min.} \{4 - 2, 3 - 3\} = 0 = ES_0$$

For each activity (i, j)

earliest starting time $= ES_i$ and earliest completion time $EC_{ij} = ES_i + t_{ij}$

latest completion time $= EC_j$ and latest start time $LS_{ij} = EC_j - t_{ij}$

For the activities (i, j) , earliest and latest, start and completion (finish) times, their total and free floats are tabulated in the following table.

Activity (i, j)	Activity duration t_{ij}	Earliest Time		Latest Time		Float	
		Start ES_i	Completion EC_{ij} $= ES_i + t_{ij}$	Start LS_{ij} $= LC_j$ $- t_{ij}$	Comple- tion LC_j	Total float $= LC_j$ $- EC_{ij}$	Free float $= T.F. - (H.slack)$
(1)	(2)	(3)	(4) = (3) + (2)	(5) = (6) - (2)	(6)	(7) = (6) - (4)	$= (7) - (LC_j - ES_j)$
—	—	—	—	—	$LC_0 = 0$	—	—
(0, 1)	2	$ES_0 = 0$	2	2	$LC_1 = 4$	2	$2 - (LC_1 - ES_1) = 0$
(0, 2)	3	$ES_0 = 0$	3	0	$LC_2 = 3$	0*	$0 - (LC_2 - ES_2) = 0$
(1, 3)	2	$ES_1 = 2$	4	4	$LC_3 = 6$	2	$2 - (LC_3 - ES_3) = 2$
(2, 3)	3	$ES_2 = 3$	6	3	$LC_3 = 6$	0*	$0 - (LC_3 - ES_3) = 0$

(2, 4)	2	$ES_2 = 3$	5	4	$LC_4 = 6$	1	$1 - (LC_4 - ES_4) = 1$
(3, 4)	0	$ES_3 = 6$	6	6	$LC_4 = 6$	0*	$0 - (LC_4 - ES_4) = 0$
(3, 5)	3	$ES_5 = 6$	9	10	$LC_5 = 13$	4	$4 - (LC_5 - ES_5) = 4$
(3, 6)	2	$ES_3 = 6$	8	17	$LC_6 = 19$	11	$11 - (LC_6 - ES_6) = 11$
(4, 5)	7	$ES_4 = 6$	13	6	$LC_5 = 13$	0*	$0 - (LC_5 - ES_5) = 0$
(4, 6)	5	$ES_4 = 6$	11	14	$LC_6 = 19$	8	$8 - (LC_6 - ES_6) = 8$
(5, 6)	6	$ES_5 = 13$	19	13	$LC_6 = 19$	0*	$0 - (LC_6 - ES_6) = 0$
-	-	$ES_6 = 19$	-	-	-	-	-

The critical activities (*i.e.*, the activities with 0, T.F.) are (0, 2), (2, 3), (3, 4), (4, 5), (5, 6).

Note that the free floats of the activities whose total floats are zero must be zero. The converse is not true, *i.e.*, a non-critical activity may have zero free float.

Time Chart of the Project

For the construction of the **time chart** of the project, we proceed as follows ;

1. First consider the scheduling of the **critical activities** which are shown by **solid lines**.
2. Consider non-critical activities in order, by indicating their earliest start (*ES*) and latest completion (*LC*) time limits on the chart. The *time ranges of the non-critical activities are shown by dashed (dotted) lines*, indicating that such activities may be scheduled within those ranges provided that the precedence relationship are not disturbed.

Thus, the **time chart** of the project is given in the following fig. 9.62.

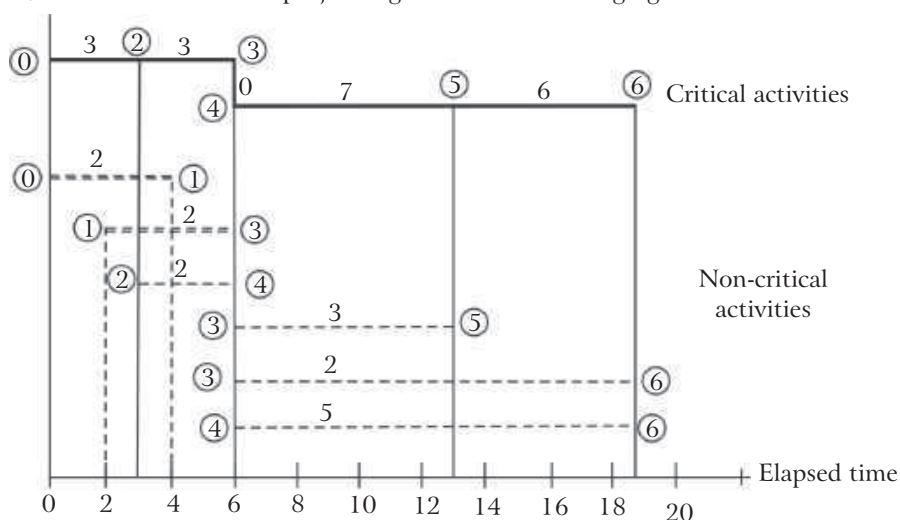


Fig. 9.62

In the above shown time chart (fig. 9.62), the dummy activity (3, 4) consumes M_0 time and so is shown by a vertical line. The numbers shown with the activities represent their durations.

9.19 The Role of Total and Free Floats in Scheduling Non-critical Activities

The roles of the **total** and **free** floats in scheduling non-critical activities is explained in terms of the following two general rules.

Rules 1: If the **total float equals the free float**, the non-critical activity can be scheduled any where between its earliest start time and latest completion time.

Rule 2 : If the **free float is less than the total float**, the starting of the non-critical activity can be delayed relative to its earliest start time by no more than the amount of its free float without effecting the scheduling of its immediately successive activities.

In the example on page 396, all non-critical activities other than activity (0, 1) can be scheduled according to rule 1, because they all have equal total and free floats. Rule 2 applies to activity (0, 1) only as here free float (0) < total float (2).

The activity (0, 1) has zero free float. So if the starting time for activity (0, 1) is not delayed beyond its earliest start time ($t = 0$), the immediately succeeding activity (1, 3) can be scheduled anywhere between its earliest start time ($t = 2$) and latest completion time ($t = 6$). If the start time of activity (0, 1) is delayed beyond its earliest start time $t = 0$, the earliest start time of succeeding activity (1, 3) must be delayed relative to its earliest start time by at least the same amount.

For example if (0, 1) start at $t = 1$, it terminates at $t = 3$, and so (1, 3) can be scheduled anywhere from $t = 3$ and $t = 6$.

This can also be seen from the time chart fig. 9.62, since (0, 1) and (1, 3) are the only two successive activities whose permissible time spans overlap.

Thus the free float less than the total float gives us a warning that the scheduling of the activity should not be finalized without first checking its effect on the start times of the immediately succeeding activities.

9.20 Resource Leveling

No technique has yet been developed to yield an optimum solution of a *resource leveling problem* i.e., minimization of the maximum required resources for the project at any point in time. Only the reasoning, past experience and advantages of the different float activities are used for the leveling of the resources in such problems of a project.

For clear understanding, see the following example :

Example 2: Consider the example on page 396 and suppose that the following worker requirements are specified for different activities :

Activity	Number of workers	Activity	Number of Workers
(0, 1)	8	(3, 5)	2
(0, 2)	5	(3, 6)	1
(1, 3)	2	(4, 5)	2
(2, 3)	7	(4, 6)	5
(2, 4)	3	(5, 6)	6

Develop a time schedule that will level the worker requirements during the project duration.

Solution: The time chart of the project is given in the fig. 9.63 (a), with activities, with their duration time and the activities that can be scheduled as early and as late as possible.

The figure 9.63 (b) shows the personnel requirements of workers over time if the non-critical activities are scheduled as early as possible. This earliest scheduling of non-critical activities requires a maximum of 13 workers for $0 \leq t \leq 2$ and minimum of 2 workers for $11 \leq t \leq 13$. This work force requirement can be improved by rescheduling activity (3, 5) at its latest possible time $t = 13$ and activity (3, 6) immediately after activity (4, 6) is completed at $t = 11$. Thus we get the new requirements for $0 \leq t \leq 19$ shown in figure 9.63. (c). Hence the requirements are as follows :

- 13 workers for $0 \leq t < 2$, 7 workers for $2 \leq t < 3$,
 12 workers for $3 \leq t < 4$, 10 workers for $4 \leq t < 5$,
 7 workers for $5 \leq t < 10$, 9 workers for $10 \leq t < 11$
 5 workers for $11 \leq t < 13$, 6 workers for $13 \leq t < 19$.

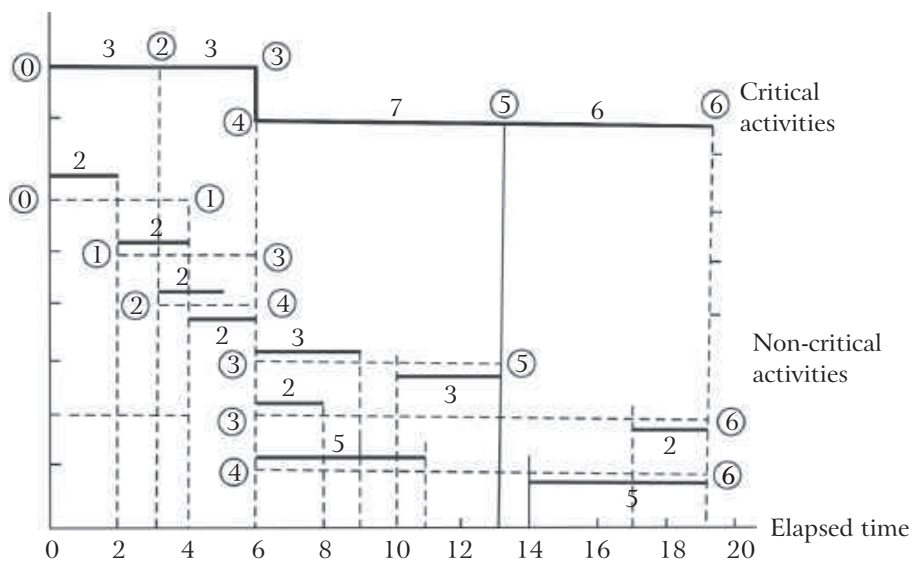


Fig. 9.63 (a)

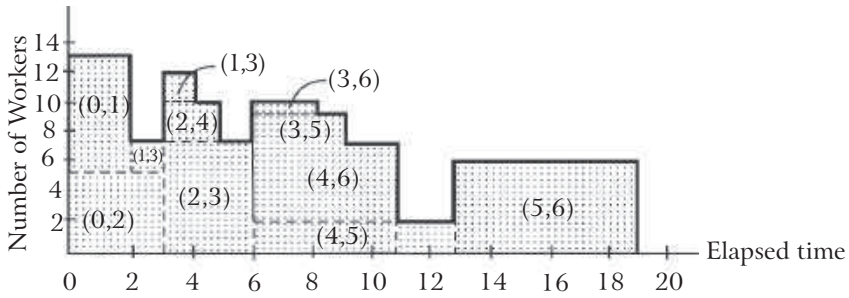


Fig. 9.63 (b)

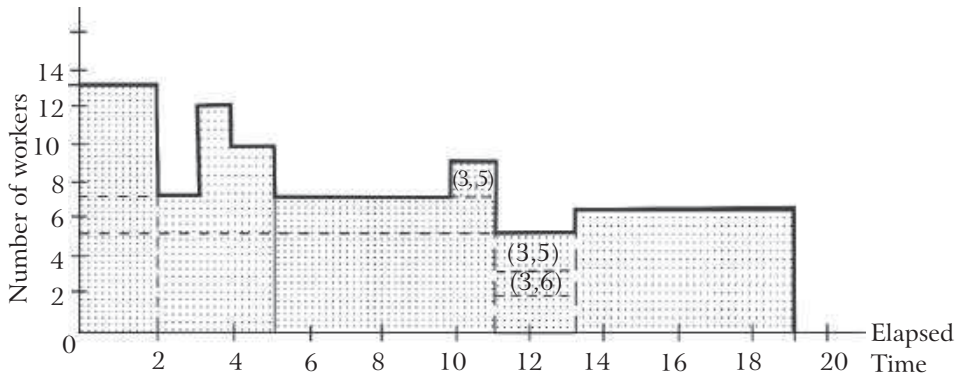


Fig.9.63(c)

Exercise 9.3

1. Write short notes on PERT.
2. Write short notes on PERT and CPM techniques.
3. A project schedule has the following characteristics.

Activity	Expected duration (Weeks)		
	Optimistic	Most likely	Pessimistic
1-2	3	3	3
2-3	3	6	9
2-4	2	4	6
3-5	4	6	8
4-6	4	6	8
5-6	0	0	0
5-7	3	4	5
6-7	2	5	8

- (i) Draw the project network and trace all the possible paths from it.
- (ii) Determine the critical path and the expected project time.
- (iii) What is the probability that the project will be completed in 20 weeks ?
- (iv) What is the probability that the project will be completed in 23 weeks ?

4. A project schedule has the following characteristics :

Activity	Expected duration (Weeks)		
	Optimistic	Most likely	Pessimistic
1-2	1	2	9
2-3	1	4	7
2-4	2	4	12
3-5	2	3	4
3-7	6	8	16
4-5	4	6	8
4-6	3	5	7
5-6	5	1	15
5-7	5	7	15
6-7	3	5	13

- (i) Draw the project network diagram and determine the critical path and the expected project time.
 - (ii) the probability of completing the project in 25 days.
5. An engineering project consists of several activities whose optimistic, most likely and pessimistic times for completion have been estimated by the project manager as shown in the following table :

Activity sequence	Estimated completion time in days		
	Optimistic	Most likely	Pessimistic
1-2	4	8	12
2-3	1	4	7
2-4	8	12	16
3-5	3	5	7
4-5	2	2	2
4-6	3	6	9
5-7	3	6	9
5-8	4	6	8
7-9	4	8	12
8-9	2	5	8
9-10	4	10	16
6-10	4	6	8

- (i) Draw a PERT diagram to represent the above project and find the time estimates.
 - (ii) Mention clearly the activities that constitute the critical path in the project network.
 - (iii) What is the minimum time required for the completion of the project ?
6. A small project is composed of 7 activities whose time estimates are listed in the table below. Activities are identified by their beginning (*i*) and ending (*j*) node numbers.

Activity	Estimated duration (Weeks)		
	Optimistic	Most likely	Pessimistic
1-2	1	1	7
1-3	1	4	7
1-4	2	2	8
2-5	1	1	1
3-5	2	5	14
4-6	2	5	8
5-6	3	6	15

- (i) Draw the project network.
- (ii) Find the expected duration and variance for each activity. What is the expected project length ?
- (iii) Calculate the variance and standard deviation of the project length. What is probability that the project will be completed ?
 - (a) At least 4 weeks earlier than expected time.
 - (b) No more than 4 weeks later than expected time.
- (iv) If the project due date is 19 weeks, what is the probability of not meeting the due date ?

Given	Z :	0.50	0.67	1.00	1.33	2.00
	Prob. :	0.3085	0.2514	0.1587	0.0918	0.0228

7. The owner of a chain of fast food restaurants, is considering a new computer system for accounting and inventory control. A computer company sent the following information :

Activity	Activity Description	Immediate Predecessor activity	Time estimates (Days)		
			Most Optimistic	Most Likely	Most Pessimistic
A	Select the computer model	–	4	6	8
B	Design input/output system	A	5	7	15
C	Design monitoring system	A	4	8	12
D	Assemble Computer Hardware	B	15	20	25
E	Develop the main programmes	B	10	18	26
F	Develop input/output routines	C	8	9	16
G	Create data base	E	4	8	12
H	Install the system	D, F	1	2	3
I	Test and implement.	G, H	6	7	8

- (i) Construct an arrow diagram for this project.
(ii) Determine the critical path and compute the expected completion time.
(iii) Determine the probability of completing the project in 55 days.

8. A project has the following activities and other characteristics :

Activity	Preceding activity	Time estimates (Days)		
		Most Optimistic	Most Likely	Most Pessimistic
A	–	4	7	16
B	–	1	5	15
C	A	6	12	30
D	A	2	5	8
E	C	5	11	17
F	D	3	6	15
G	B	3	9	27
H	E, F	1	4	7
I	G	4	19	28

- (i) Draw the PERT network diagram.
 - (ii) Identify the critical path.
 - (iii) Prepare the activity schedule for the project.
 - (iv) Determine the mean project completion time.
 - (v) Find the probability that the project is completed in 36 weeks.
 - (vi) If the project manager wishes to be 99% sure that the project is completed on June 30, 1991, when should he start the project work ?
9. Calculate the variance and expected activity time for the activities of the network shown in the following figure (Enter calculations in the tabular form).

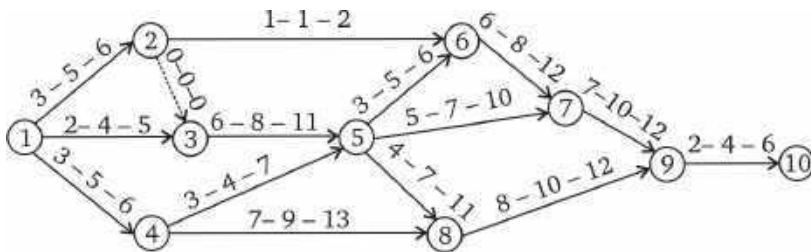


Fig. 9.64

For each activity, the three estimates t_o , t_m and t_p are given along the arrows in the $t_o - t_m - t_p$ order.

Answers 9.3

3. (i) *Pert Network*

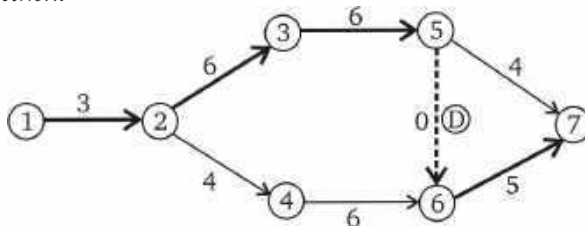


Fig. 9.65

All possible paths : 1-2-3-5-7, 1-2-3-5-6-7, 1-2-4-6-7

- (ii) Critical path : 1-2-3-5-6-7. Expected project time 20 weeks.
- (iii) 0.5 i.e. 50%
- (iv) 0.9726 i.e. 97.26%

4. (i) Network

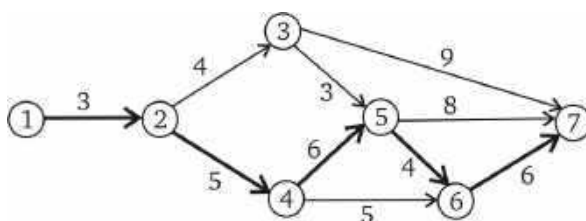


Fig. 9.66

Critical path : 1-2-4-5-6-7. Expected project time = 24 weeks.

(ii) 62.17%

5. (i) Pert diagram

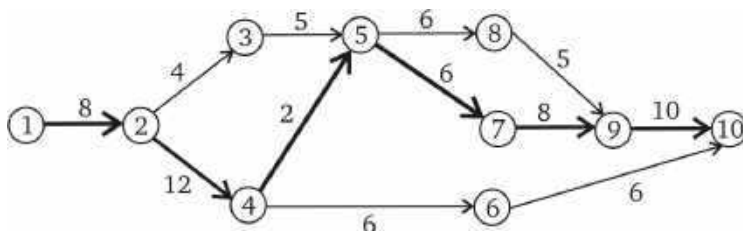


Fig. 9.67

Estimates are shown on the arrows representing activities.

(ii) 1-2, 2-4, 4-5, 5-7, 7-9, 9-10

(iii) 46 days

6. (i) Project Network

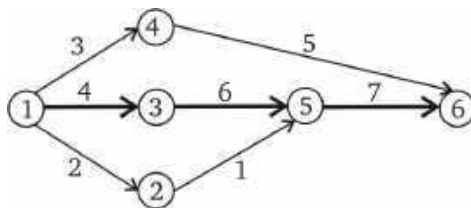


Fig. 9.68

(ii)

Activity	:	1 - 2	1 - 3	1 - 4	2 - 5	3 - 5	4 - 6	5 - 6
Expected duration	:	2	4	3	1	6	5	7
Variance	:	1	1	1	0	4	1	4

Expected project length – 17 weeks.

(iii) 9, 3 (a) 0.0918 i.e., 9.18% (b) 0.9082 i.e., 90.82%

(iv) 0.2514 i.e., 25.14%

7. (i) Arrow diagram

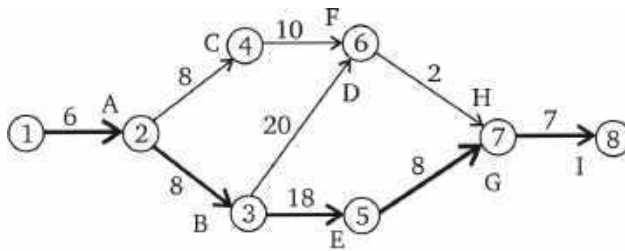


Fig. 9.69

(ii) Critical path : A-B-E-G-I i.e., 1-2-3-5-7-8

Completion time = 47 days

(iii) 0.9981 i.e., 99.81%

8. (i) Pert Network

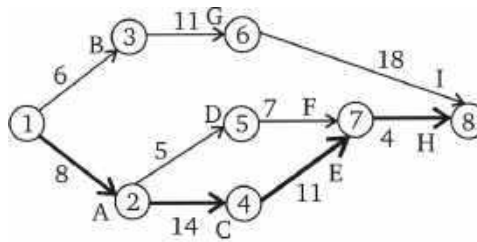


Fig. 9.70

(ii) Critical path : A → C → E → H. i.e. 1-2-4-7-8

(iii) Other activity schedule are B-G-I, A-D-F-H

(iv) 37 weeks

(v) 0.4207 i.e., 42.07%

(vi) 48.7 weeks before the due date

9.

Activity	Time estimates			Expected time	Variance
	t_o	t_m	t_p	$t_e = \frac{t_o + 4t_m + t_p}{6}$	$\sigma^2 = \left(\frac{t_p - t_o}{6}\right)^2$
1-2	3	5	6	29/6	$(1/2)^2 = 0.25$
1-3	2	4	5	23/6	$(1/2)^2 = 0.25$
1-4	3	5	6	29/6	$(1/2)^2 = 0.25$
2-3	0	0	0	0	0 = 0.00
2-6	1	1	2	7/6	$(1/6)^2 = 0.03$
3-5	6	8	11	49/6	$(5/6)^2 = 0.69$
4-5	3	4	7	13/3	$(2/3)^2 = 0.45$
4-8	7	9	13	28/3	1 = 1.00
5-6	3	5	6	29/6	$(1/2)^2 = 0.25$
5-7	5	7	10	43/6	$(5/6)^2 = 0.69$
5-8	4	7	11	43/6	$(7/6)^2 = 1.37$
6-7	6	8	12	25/3	1 = 1.00
7-9	7	10	12	59/6	$(5/6)^2 = 0.69$
8-9	8	10	12	10	$(2/3)^2 = 0.45$
9-10	2	4	6	4	$(2/3)^2 = 0.45$

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

- A graph consists of:
 - Nodes
 - Branches
 - Both (a) and (b)
 - None of these.
- An event which represents the end (or head) of more than one activity is called a:
 - Merge event
 - Node
 - Burst event
 - None of these.

3. An event which represent the beginning (or tail) of more than one activity is called a:
 - (a) Merge event
 - (b) Burst event
 - (c) Node
 - (d) None of these.
 4. A dummy activity is used in the network:
 - (a) When two parallel activities have the same tail and head events
 - (b) To satisfy the precedence requirements
 - (c) When the chain of activities have a common event yet be independent by themselves
 - (d) All (a), (b) and (c).
 5. In the network diagram, the process of numbering the events was devised by:
 - (a) Newton
 - (b) D.R Fulkeron
 - (c) JV Neumann
 - (d) Morgenstern.
 6. The objective of network analysis is to:
 - (a) Minimize total project cost
 - (b) Minimize total project duration
 - (c) Minimize production delays
 - (d) None of these.
 7. The float of an activity is/are:
 - (a) Total float
 - (b) Free float
 - (c) Independent float
 - (d) All of these.
 8. Which of the followings are true:
 - (a) Total float $\geq$ Free float
 - (b) Total float $\geq$ Independent float
 - (c) Free float $\geq$ Independent float
 - (d) All of these.
 9. An event with zero slack time is called:
 - (a) Sure event
 - (b) Critical event
 - (c) Forward event
 - (d) Backward event.
 10. The sum of individual time durations of all critical activities on the critical path in a network diagram is:
 - (a) The total project time
 - (b) Minimum time required to complete the project
 - (c) Both (a) and (b)
 - (d) None of these.
-

11. The expected time in PERT $= (t_0 + 4t_m + t_p) / l$, where $l =$
 - (a) 2
 - (b) 4
 - (c) 6
 - (d) 8.
12. If the total float of an activity is zero, it implies that the activity:
 - (a) Lies on the critical path
 - (b) Is a dummy activity
 - (c) Does not lie on critical path
 - (d) None of these.
13. For all the activities (i, j) on the critical path:
 - (a) $E_i = E_j$ and $L_i = L_j$
 - (b) $E_i = L_i$ and $E_j = L_j$
 - (c) $E_i + L_i = E_j + L_j$
 - (d) None of these.
14. The expected value of project information is:
 - (a) The utility of additional information
 - (b) Maximum expected opportunity loss
 - (c) Equal to expected regret of the optimal decision under risk
 - (d) None of these.
15. When the total project duration is a constraint taken resource scheduling is referred as:
 - (a) Resource allocation
 - (b) Resource levelling
 - (c) Resource crashing
 - (d) Resource smoothening.
16. An activity which only determines the dependency of one activity on the other, but does not consume any time is called:
 - (a) Successor activity
 - (b) Predecessor activity
 - (c) Dummy activity
 - (d) Dangling.

Fill in the Blank

1. A graph in which all the branches are oriented is called an graph.
 2. A tree always has at least two points.
 3. The beginning and end points of an activity are called
 4. If some activities in a network form a loop/or cycle, then such error in a network is called error.
 5. The total float of an activity is defined as the between the maximum time available to perform the activity and activity duration time.
 6. For an event, total float = Latest start time - start time.
 7. Independent float = Free float - Tail event
-

8. The critical path in a network diagram is the continuous chain critical activities from the first event to the
9. Optimistic time is the shortest possible time required to an activity, if every thing goes well.
10. For an activity (i, j)
Independent float = $E_j - L_i - \dots\dots\dots$.

True/False

1. CPM is a probabilistic model while PERT is a deterministic model.
2. CPM is a deterministic model while PERT is a probabilistic model.
3. CPM is event oriented technique.
4. CPM is activity oriented technique.
5. PERT is event oriented technique.
6. PERT is activity oriented technique.
7. Event slack for any event is defined as the difference between its latest and earliest occurrence times.
8. For an event, total float is equal to the sum of latest and earliest completion times.
9. Total float = Free float + Head event slack.
10. Pessimistic time is the maximum possible time that an activity could take if every thing goes well.

Answers

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1.	(c)	2.	(a)	3.	(b)
4.	(d)	5.	(b)	6.	(b)
7.	(d)	8.	(d)	9.	(b)
10.	(c)	11.	(c)	12.	(a)
13.	(b)	14.	(b)	15.	(d)
16.	(c)				

Fill in the Blank

1.	oriented	2.	extreme	3.	events (or nodes)
4.	looping (or cycling)	5.	difference	6.	earliest
7.	slack	8.	last event	9.	complete
10.	$t_{i,j}$				

True/False

1.	F	2.	T	3.	F
4.	T	5.	T	6.	F
7.	T	8.	F	9.	T
10.	T				

○○○

UNIT

IV



Chapters



10. Inventory Models

11. Dynamic Programming

12. Job Sequencing

UNIT-IV

Chapter

10



Inventory Models

10.1 Inventory

Definition : *An inventory can be defined as a stock of goods which is held for the purpose of future production or sales.* The stock of goods may be kept in the following forms.

(a) raw-materials, (b) partly finished items and (c) finished (or prepared) goods, (d) spare parts etc.

The objective of an inventory problem is to minimize total (actual or expected) cost or to maximize (actual or expected) profit.

10.2 Variables in Inventory Problems

The variables associated with the inventory problems are classified into two categories.

(A) The controlled variables and

(B) The uncontrolled variables.

(A) **The controlled variables :** The variables that may be controlled, separately or in combination are the following :

1. **The quantity acquired** (by purchase, production, or some other means) :
The decision maker may have control over the purchase or production level.
-

2. **The frequency or timing of acquisition :** The decision maker may have control over how often or when the inventory should be replenished.
3. **The stage of completion of stocked items :** The decision maker may have control over the stage at which the unfinished items be held so that there is no delay in supplying customers.

(B) **The uncontrolled variables :** The variables that may not be controlled in an inventory problem are divisible into cost variables and others.

10.3 Cost variables (or the costs) involved in inventory problems

The main cost variables involved in inventory problems are as follows :

1. Holding or storage costs.

The costs associated with the storage of the inventory until it is sold or used are known as the holding or storage costs. This cost is directly proportional to the increase in the inventory and the time for which the stocks are held. The various components of the holding costs are as follows :

- (i) *Handling cost :* Which includes the cost of labour, transportation charges etc.
- (ii) *Rent for the space or interest and the cost of depreciation on owned space.*
- (iii) *Costs of the staff required to keep records.*
- (iv) *Insurance and taxes etc.*
- (v) *Interest on the money locked for inventory.*
- (vi) *Deterioration costs etc. which arises in the case of fashion items or items that changes chemically during storage such as medicines , foods etc.*

2. **Shortage or penalty costs :** *The unsatisfied demand or shortage penalty cost is incurred when the stock proves to be inadequate to meet the demand of the customers.* The shortages may result in the cancellation of orders and loss of sales which may result in the loss of goodwill of the customers. The shortage costs are represented in two ways. One way is that the shortages can be fulfilled by back order, when the commodity next becomes available. This case is known as “**back logging**” of unsatisfied demand. These cost usually vary directly with the shortage quantity and the delay time both. Another way is that the unsatisfied demand is lost. This case is known as “**no back logging**”. This involve a fixed cost each time a shortage occurs. In this case the shortage costs are proportional to the shortage quantity only.

Obviously the shortage costs reduces with the increase in the inventory.

3. **Set up (or replenishment or ordering or procurement) costs :** *This is the cost associated with the placing of an order for purchasing goods, or it is the cost of setting a machine before it starts production.* This cost may depend on the quantity of goods purchased because of price breaks or quantity discounts.

Besides these cost variables there are other variables that may not be controlled in any inventory problem is assumed to be independent of the quantity of goods ordered or produced.

- (i) **Demand:** Demand is the number of items required per period which is not necessarily equal to the amount sold as some demand may go unfulfilled because of shortage or delays.

The demands may be of two types :

- (a) **Deterministic demands :** If the number of items required (*i.e.* demand) in a subsequent period of time is known exactly then such demands are called **deterministic demands**.
- (b) **Non deterministic or probabilistic demands :** If the demands over a subsequent period of time is not known with certainty then such demands are called **non deterministic or probabilistic demands**.
- (ii) **Lead time :** The time gap between the time of placing an order or the starting of the production and the time of arrival or delivery of goods to the inventory is called “**Lead time**”. Also the time gap between the time of demand and the time of filling the demand from the inventory is called **lead time**. If this time is known (constant) and not zero then one may order in advance by an amount of time equal to the lead time. If it is a variable *i.e.*, known only probabilistically then the question of when to order is difficult.
- (iii) **Amount delivered :** The supply of goods may be instantaneous or spread over a period of time.

10.4 Need of Inventory

The inventory in any business is maintained to decrease the set up and the shortage costs. If the demands of the customers are not fulfilled then it may result in the loss of their goodwills. If the orders are cancelled then it results in the loss of the business. Thus, there is always a need of inventory for the smooth running of any business.

10.5 Inventory Problems

An inventory problem exists if the amount of the goods in stock (*i.e.*, inventory) is subjected to control and if there is at least one cost that decreases as inventory increases.

10.6 Advantages and Disadvantages of Inventory

The advantages : Associated with increased inventory are the following :

- (i) The economy of production with large run sizes.
- (ii) The smooth and efficient running of the business.
- (iii) The economy in transportation.

- (iv) The advantage of price discounts by bulk purchasing.
- (v) Faster and adequate service to the customers, and
- (vi) Profit from speculation in the market where prices are expected to rise.

The **disadvantages** associated with increased inventory is that the holding of inventory, costs money, due to the following :

- (i) Warehouse rent.
 - (ii) Interest on invested capital.
 - (iii) Physical handling.
 - (iv) Accounting.
- and (v) Depreciation and deterioration.

10.7 Classification or Categories of Inventory Models

The inventory problems (models) may be classified into two categories.

- (i) **Deterministic models** : These are the inventory models in which demand is assumed to be fixed for a subsequent period of time.
- (ii) **Probabilistic models** : These are the inventory models in which the demand is a random variable having a known probabilistic distribution. Here the future demand is determined by collecting data from the past experience.

10.8 Some General Notations Used in Inventory Models

We shall use the following general notations in inventory models of this chapter.

I = The cost of carrying one rupee in inventory for a unit time.

C_1 = Holding cost per unit per unit time.

C_2 = Shortage cost per unit per unit time.

C_3 = Set up cost per production run.

q = Lot size per production run (*i.e.* quantity produced in one run).

R = Total demand.

r = Demand rate

K = Production rate (*i.e.* rate of replenishment of inventory)

C = Average total cost per unit time.

t = Time interval between two consecutive replenishments of inventory.

z = Order level or stock level.

L = Lead time.

q^*, t^*, z^* = Optimal values of q, t, z respectively for which the cost C is minimum.

DETERMINISTIC MODELS

First, we shall consider the deterministic models in which the demand is known and is fixed for a subsequent period of time.

10.9 Economic Lot Size Models

The most common inventory problem faced by industry concerns the situation where stock levels are depleted with time and then are replenished by the arrival of new items. The situation is given in the following **economic lot size models**. The inventory problems in which the demand is assumed to be fixed and completely predetermined are known as the **Economic Lot size problem** or **Economic Order Quantity (EOQ) problem**.

The Economic Order Quantity (EOQ) is that size of order which minimizes the total costs that include carrying costs, set up costs and the shortage costs, under the assumption that the demand is fixed and known.

10.10 Model I: Economic Lot Size Model with Uniform Rate of Demand Infinite Production Rate and having no Shortages

To derive an economic **lot size formula** and the minimum average costs under the following assumptions.

- (i) demand is uniform at a rate of r units per unit time.
- (ii) production is instantaneous (i.e. production rate is infinite)
- (iii) lead time is zero.
- (iv) C_1 = holding cost per unit per unit time,
- (v) C_3 = set up cost per production run, and
- (vi) shortages are not allowed.

Solution : Let q be the units of quantity produced (or ordered) per production run at intervals of time t .

The situation of the inventory is illustrated in fig. 10.1

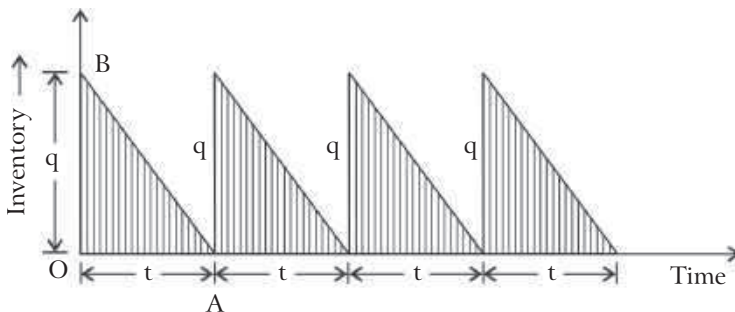


Fig. 10.1

Since the demand rate is r units per unit time.

$\therefore$ the total demand in one run of time interval t is $r t$.

$\therefore$ the quantity produced per production run (in the beginning)

$$q = r t \quad \dots(1)$$

(Since shortages are not allowed)

The cost of holding inventory (per production run)

$$= C_1 (\text{Area of } \Delta OAB) = C_1 \frac{1}{2} q t$$

and the set up cost (per production run) $= C_3$

$\therefore$ The total cost per production run of time $t = \frac{1}{2} C_1 q t + C_3$

$\therefore$ The average total cost per unit time

$$C(q) = \frac{1}{2} C_1 q + \frac{C_3}{t} = \frac{1}{2} C_1 q + \frac{C_3 r}{q} \quad \dots(2)$$

This equation is known as **cost equation**.

For minimum value of $C(q)$

$$\frac{dC}{dq} = \frac{1}{2} C_1 - \frac{C_3 r}{q^2} = 0 \quad \therefore \quad q = \sqrt{\left(\frac{2 C_3 r}{C_1} \right)}$$

Since $\frac{d^2 C}{dq^2} = \frac{2 C_3 r}{q^3}$ is positive for $q = \sqrt{\left(\frac{2 C_3 r}{C_1} \right)}$

$\therefore$ $C(q)$ given by (2) is minimum for

$$q = q^* = \sqrt{\left(\frac{2 C_3 r}{C_1} \right)} \quad \dots(3)$$

which is known as **Harris or R.H. Wilson Economic Lot-size formula** or simply **Economic lot-size formula** and q^* given by (3) is called Economic order Quantity (EOQ).

$\therefore$ From (1) the optimum value of t is given by

$$t = t^* = \sqrt{\left(\frac{2 C_3}{C_1 r} \right)} \quad \dots(4)$$

And from (2) the minimum cost per unit time is given by

$$C_{\min} = \frac{1}{2} C_1 \sqrt{\left(\frac{2 C_3 r}{C_1} \right)} + C_3 r \sqrt{\left(\frac{C_1}{2 C_3 r} \right)}$$

$$\text{or} \quad C_{\min} = \sqrt{(2 C_1 C_3 r)} \quad \dots(5)$$

If C_1 and C_3 are constants then the **minimum cost per unit time is proportional to the square root of the demand rate**.

1. If the lead time is L (not zero), *i.e.* L is the time gap between the time of placing an order and the time of receiving of the goods to the inventory, then we have to place the order in advance by the time L so that we may get the total quantity q as soon as the inventory falls to zero. Thus at the time of placing an order we have the inventory equal to rL . This point is called the **re-order point** p .

Thus, the **re-order point** (when lead time is L) is given by

$$p = rL.$$

If we have to maintain the **buffer stock** B , then the **re-order point** (when lead time is L) is given by

$$p = rL + B.$$

2. The total inventory in one cycle of time t is $\frac{1}{2}qt$.

$$\therefore \text{Average inventory at any time} = \frac{\frac{1}{2}qt}{t} = \frac{q}{2}$$

Thus, throughout the period, average level of inventory is $\frac{q}{2}$.

10.11 Another form of Model I

To derive an economic lot-size formula under the following assumptions :

- (i) λ = the demand for product in one unit of time (say one year).
- (ii) production rate is infinite.
- (iii) lead time is zero.
- (iv) P = price of one item of product in Rupees.
- (v) I = cost of carrying one rupee to the inventory for one year.
- (vi) C_3 = Set up cost per order (per cycle).
- (vii) Shortages not allowed.

Solution : Let q be the units of quantity produced (or ordered) per cycle (in time t).

From model I total inventory in one cycle (in time t) = $\frac{1}{2}qt$.

$\therefore$ the average inventory is $q/2$.

$\therefore$ the average inventory $q/2$ will continue throughout the year.

Holding cost per item for one year = IP

$$[\because C_1 = IP]$$

$\therefore$ The total holding cost per year = $\frac{q}{2}IP$

Now the no. of orders (cycles) in one year = λ/q .

$\therefore$ set up cost in one year = $\frac{\lambda}{q} \cdot C_3$

$\therefore$ The total cost in one year, $C(q) = \frac{1}{2}PqI + \frac{\lambda}{q}C_3$.

For $C(q)$ to be minimum.

$$\frac{dC}{dq} = \frac{1}{2}PI - \frac{\lambda C_3}{q^2} = 0 \quad \therefore \quad q = \sqrt{\left(\frac{2\lambda C_3}{PI}\right)}$$

$$\frac{d^2C}{dq^2} = \frac{2\lambda C_3}{q^3}, \text{ which is positive for } q = \sqrt{\left(\frac{2\lambda C_3}{PI}\right)}$$

Thus, the economic lot-size formula is

$$q^* = \sqrt{\left(\frac{2\lambda C_3}{PI}\right)}$$

This may also be obtained from model I by replacing C_1 by PI and r by λ .

10.12 Model II : Economic Lot-size Model with Different Rates of Demand in Different Production Cycles, Infinite Production Rate and having no Shortages

To derive an economic lot-size formula and the minimum average cost under the same assumptions as in model I except that the demand rates are different in different production cycles.

Solution : Let q be the units of quantity produced (ordered) per production run.

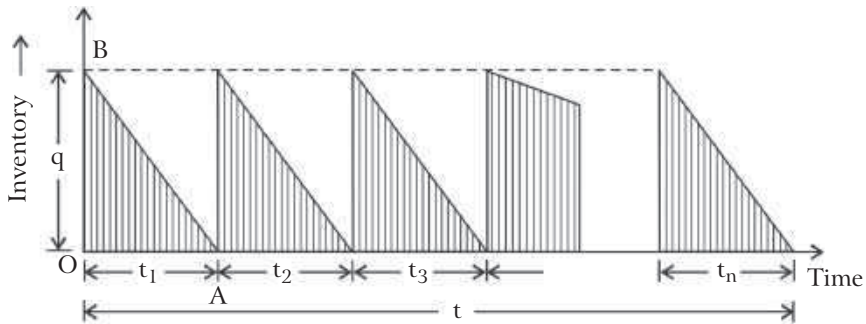


Fig. 10.2

If R is the total demand in the total period t (fixed) then the number of production cycles.

$$= R/q = n \text{ (say) (Since shortages are not allowed)}$$

Let $t_1, t_2, \dots, t_n$ be the times of the successive production cycles s.t.,

$$t = t_1 + t_2 + t_3 + \dots + t_n \quad \dots(1)$$

Thus, q , the quantity produced at the beginning of each production run, is supplied with different uniform demand rates in times $t_1, t_2, \dots, t_n$ in the successive cycles.

The situations of the inventory in different cycles is illustrated in fig. 10.2.

The cost of holding inventory for the period t

$$\begin{aligned}
 &= C_1 \text{ (sum of areas of } n \text{ triangles in the fig. 10.2)} \\
 &= C_1 \left(\frac{1}{2} q t_1 + \frac{1}{2} q t_2 + \frac{1}{2} q t_3 + \dots \frac{1}{2} q t_n \right) \\
 &= \frac{1}{2} q C_1 \cdot (t_1 + t_2 + t_3 + \dots + t_n) = \frac{1}{2} q C_1 t,
 \end{aligned}$$

and the set up cost $= n C_3 = \frac{R}{q} \cdot C_3$.

$\therefore$ The total cost for the period t (fixed) $= \frac{1}{2} q C_1 t + \frac{R}{q} \cdot C_3$.

$\therefore$ The average total cost per unit time, $C(q) = \frac{1}{2} q C_1 + \frac{R C_3}{t q}$... (2)

For minimum value of $C(q)$

$$\frac{dC}{dq} = \frac{1}{2} C_1 - \frac{R C_3}{t q^2} = 0 \quad \therefore q = q^* = \sqrt{\left(\frac{2 R C_3}{C_1 t} \right)} \quad \dots (3)$$

Since $\frac{d^2 C}{dq^2} = \frac{2 R C_3}{t q^3}$ is positive for $q^* = \sqrt{\left(\frac{2 R C_3}{C_1 t} \right)}$

$\therefore C(q)$ given by (2) is minimum for the value of q given by (3)

Hence (3) gives the required **economic lot-size formula**.

From (2) the minimum cost per unit time is given by

$$C_{\min} = \frac{1}{2} C_1 \sqrt{\left(\frac{2 R C_3}{C_1 t} \right)} + \frac{R C_3}{t} \sqrt{\left(\frac{C_1 t}{2 R C_3} \right)}$$

or $C_{\min} = \sqrt{\left(\frac{2 C_1 C_3 R}{t} \right)} \quad \dots (4)$

Note: It is clear that the relations (3) and (4) of model II that these relations may be obtained from the corresponding relations (3) and (4) of model I replacing r (demand rate) by R/t (average demand rate).

10.13 Model III: Economic Lot-size Model with Uniform Rate of Demand, Finite Rate of Replenishment having no Shortages

To derive an economic lot-size formula and minimum average cost under the same assumptions as in model I except that the replenishment rate (i.e., the production rate) is finite.

Solution : Let $K > r$ be the number of items produced per unit time.

If q is the number of items produced per production run then the production will continue for a time

$$t_1 = q/K \quad \dots(1)$$

and the time of one complete production run (*i.e.*, the interval between two runs),

$$t = q/r \quad \dots(2)$$

(Since r is the demand rate and no shortages are allowed).

The situation of the inventory in one production run is illustrated in fig. 10.3.

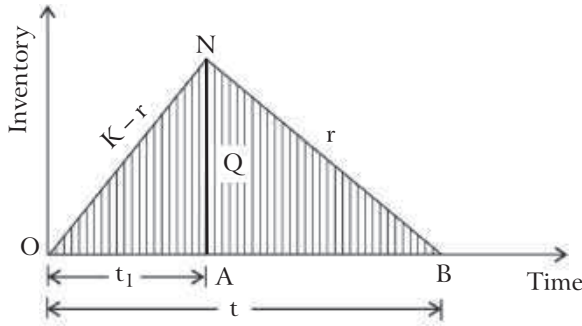


Fig. 10.3

If Q is the inventory level at the moment the production is completed (*i.e.*, at the end of time t_1)

then
$$Q = q - rt_1 = q - r \cdot \frac{q}{K} = q \left(1 - \frac{r}{K}\right) \quad \dots(3)$$

The cost of holding inventory for the period t (time interval between runs),

$$\begin{aligned} &= C_1 \cdot (\text{Areas of } \triangle ONB) \\ &= C_1 \left(\frac{tQ}{2} \right) = \frac{1}{2} q C_1 \left(1 - \frac{r}{K} \right) t \end{aligned} \quad \text{from (3)}$$

and the set up cost $= C_3$.

$\therefore$ The total cost per run of period t

$$= C_3 + \frac{1}{2} q C_1 \left(1 - \frac{r}{K} \right) t.$$

$\therefore$ The total cost per unit time

$$\begin{aligned} C(q) &= \frac{C_3}{t} + \frac{1}{2} q C_1 \left(1 - \frac{r}{K} \right) \\ C(q) &= \frac{C_3 r}{q} + \frac{1}{2} q C_1 \left(1 - \frac{r}{K} \right) \end{aligned} \quad \text{from (2)} \quad \dots(4)$$

For minimum value of $C(q)$,

$$\frac{dC}{dq} = -\frac{C_3 r}{q^2} + \frac{1}{2} C_1 \left(1 - \frac{r}{K}\right) = 0$$

$$\therefore q = q^* = \sqrt{\left\{ \frac{2C_3}{C_1} \cdot \left(\frac{rK}{K-r} \right) \right\}} \quad \dots(5)$$

Since $\frac{d^2 C}{dq^2} = \frac{2 C_3 r}{q^3}$ is positive for this value of q given by (5).

$\therefore C(q)$ given by (4) is minimum for the value of q given by (5).

Hence (5) gives the required **economic lot-size formula**.

From (4) the minimum cost per unit time is given by

$$C_{\min} = C_3 r \sqrt{\left[\frac{C_1}{2 C_3} \cdot \left(\frac{K-r}{r K} \right) \right]} + \frac{1}{2} C_1 \left(\frac{K-r}{K} \right) \cdot \sqrt{\left[\frac{2 C_3}{C_1} \left(\frac{K r}{K-r} \right) \right]}$$

or
$$C_{\min} = \sqrt{\left[2 C_1 C_3 r \left(1 - \frac{r}{K} \right) \right]}$$

Also time of one run is given by : $t^* = \frac{q^*}{r} = \frac{\sqrt{2 C_3 K}}{C_1 r (K-r)}$

DETERMINISTIC MODELS WITH SHORTAGES

Now we shall consider the deterministic models in which shortages are allowed but are to be fulfilled as soon as the items (goods) are received *i.e.*, in these models back orders are permitted.

10.14 Model IV : Fixed Time Model

Order level model* with uniform rate of demand Q to be fulfilled in constant time t infinite rate of production and having shortages which are to be fulfilled.

(*i.e.*, Back logged case)

To derive the optimal order level and the minimum average cost under the following assumptions.

- (i) demand rate is uniform at a rate of r units per unit time.
- (ii) production is instantaneous (*i.e.*, production rate of infinite),
- (iii) lead time is zero,
- (iv) C_1 = holding cost per unit per unit time,
- (v) C_2 = shortage cost per unit per unit time,
- (vi) C_3 = set up cost per production run.

* The inventory problems involving z (the level of the inventory in the beginning) are known as **order level problems**.

(vii) shortages are allowed and are back logged

(viii) Q = total demand per production run of fixed time interval t .

Let z be the order to which the inventory is raised in the beginning of a run of time interval t (constant).

This inventory is reduced to zero in time $t_1 = z/r$ (1)

Then the shortages arise increase from 0 to $Q - z$ in the remaining time $(t - t_1)$, where Q is the total demand for one run.

$\therefore Q = r t$... (2)

which is known exactly.

The situation of the inventory and the shortages are illustrated in fig. 10.4.

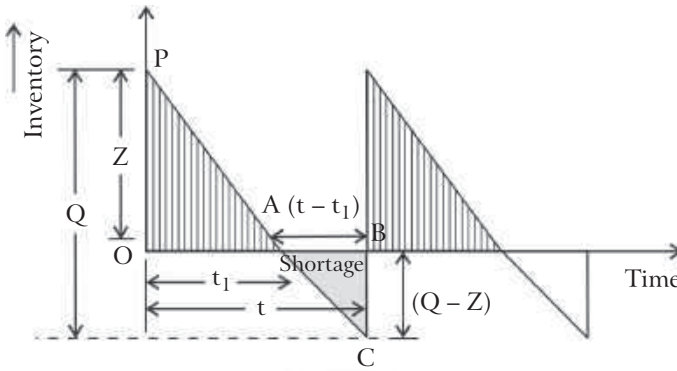


Fig. 10.4

The cost of holding inventory (per production run)

$$= C_1 \cdot (\text{Area of } \triangle OAP)$$

$$= C_1 \cdot \frac{1}{2} \cdot z \cdot t_1 = \frac{C_1 z^2}{2r}.$$

from (1)

The shortage cost for one run

$$= C_2 (\text{Area of } \triangle ABC)$$

$$= C_2 \cdot \frac{1}{2} (Q - z) \cdot (t - t_1)$$

$$= \frac{1}{2} C_2 (Q - z) \left(\frac{Q}{r} - \frac{z}{r} \right)$$

from (1) from (2)

$$= \frac{C_2}{2r} (Q - z)^2.$$

and set up cost for one run $= C_3$.

$\therefore$ The average total cost per unit time.

$$\begin{aligned}
 C(z) &= \frac{1}{t} \left[\frac{C_1}{2r} z^2 + \frac{C_2}{2r} (Q-z)^2 \right] + \frac{C_3}{t} \\
 &= \frac{r}{Q} \left[\frac{C_1}{2r} z^2 + \frac{C_2}{2r} (Q-z)^2 \right] + \frac{r C_3}{Q}
 \end{aligned}$$

$$\text{or} \quad C(z) = \frac{C_1}{2Q} z^2 + \frac{C_2}{2Q} (Q-z)^2 + \frac{r C_3}{Q} \quad \dots(3)$$

For the minimum value of $C(z)$.

$$\frac{dC}{dz} = \frac{C_1}{Q} z - \frac{C_2}{Q} (Q-z) = 0 \quad (\because r C_3/Q \text{ is constant})$$

$$\therefore \quad z = z^* = \frac{C_2 Q}{C_1 + C_2} = \frac{C_2 r t}{C_1 + C_2} \quad \dots(4)$$

which is the required **optimal order level**.

The value z given by (4) gives the minimum value of C , since $\frac{d^2 C}{dz^2} = \frac{C_1 + C_2}{Q}$ is positive.

$\therefore$ From (3) the minimum cost per unit time is given by

$$C_{\min} = \frac{C_1}{2Q} \left(\frac{C_2 Q}{C_1 + C_2} \right)^2 + \frac{C_2}{2Q} \left(Q - \frac{C_2 Q}{C_1 + C_2} \right)^2 + \frac{r C_3}{Q}$$

$$\text{or} \quad C_{\min} = \frac{C_1 C_2 Q}{2(C_1 + C_2)} + \frac{r C_3}{Q} = \frac{C_1 C_2 r t}{2(C_1 + C_2)} + \frac{r C_3}{Q}$$

Note : If $C_1 \neq 0$ then $C_1 + C_2 > C_2$

$\therefore$ from (4) $z < Q$.

i.e., the order of level to which the inventory is raised is kept less than the total demand Q to create shortages.

10.15 Model V : Economic Lot-size Model with Uniform Rate of Demand, Infinite Rate of Production and Having Shortages which are to be Fulfilled.

To derive an economic lot size formula (optimal quantity q per run) which minimises the total cost under the following conditions :

- (i) demand is uniform at a rate of r units per unit time,
- (ii) production is instantaneous,
- (iii) lead time is zero,
- (iv) C_1 = holding cost per unit per unit time,
- (v) C_2 = shortage cost per unit per unit time,

- (vi) $C_3 = \text{set up cost per run}$,
 (vii) *shortages are allowed and back logged.*

Let q be the order quantity per production run and z be the order level to which the inventory is raised at the beginning of a run of time interval t (variable).

$$\therefore q = r t \quad \dots(1)$$

Proceeding as in model IV, we have

$$\text{The cost of holding inventory (per production run)} = \frac{C_1 z^2}{2 r}$$

$$\text{The shortage cost for one run} = \frac{C_2}{2 r} (q - z)^2 = \frac{C_2}{2 r} (r t - z)^2$$

and the set up cost per run $= C_3$.

$$\therefore \text{The total cost per run} = \frac{C_1}{2 r} z^2 + \frac{C_2}{2 r} (r t - z)^2 + C_3.$$

$\therefore$ The average total cost per unit time

$$C(z, t) = \frac{C_1}{2 r} \cdot \frac{z^2}{t} + \frac{C_2}{2 r} \cdot \frac{(r t - z)^2}{t} + \frac{C_3}{t}, \quad \dots(2)$$

which is the function of two variables z and t .

For $C(z, t)$, to be minimum

$$\frac{\partial C}{\partial z} = \frac{C_1}{r} \cdot \frac{z}{t} - \frac{C_2}{t} \cdot \frac{(r t - z)}{t} = 0 \quad \dots(3)$$

$$\text{and} \quad \frac{\partial C}{\partial t} = -\frac{C_1}{2 r} \frac{z^2}{t^2} + C_2 \cdot \frac{(r t - z)}{t} - \frac{C_2}{2 r} \cdot \frac{(r t - z)^2}{t^2} - \frac{C_3}{t^2} = 0 \quad \dots(4)$$

$$\text{From (3), } \frac{z}{r t} (C_1 + C_2) = C_2$$

$$\therefore z = \frac{C_2 r t}{C_1 + C_2} \quad \dots(5)$$

Putting in (4), we have

$$-\frac{C_1}{2 r} \cdot \left(\frac{C_2 r t}{C_1 + C_2} \right)^2 + \frac{C_2}{t} \cdot \left(r t - \frac{C_2 r t}{C_1 + C_2} \right) - \frac{C_2}{2 r t^2} \left(r t - \frac{C_2 r t}{C_1 + C_2} \right)^2 - \frac{C_3}{t^2} = 0$$

$$\text{or} \quad -C_1 C_2^2 r + 2 C_1 C_2 r (C_1 + C_2) - C_1^2 C_2 r = \frac{2 C_3}{t^2} (C_1 + C_2)^2$$

$$\text{or} \quad C_1 C_2 r (-C_2 + 2 C_1 + 2 C_2 - C_1) = \frac{2 C_3}{t^2} (C_1 + C_2)^2$$

or $C_1 C_2 r (C_1 + C_2) = \frac{2 C_3}{t^2} (C_1 + C_2)^2$

or $t^2 = \frac{2 C_3 (C_1 + C_2)}{C_1 C_2 r}$

$\therefore t = t^* = \sqrt{\left(\frac{2 C_3 (C_1 + C_2)}{C_1 C_2 r} \right)}$... (6)

Now $\frac{\partial^2 C}{\partial z^2} = \frac{C_1 + C_2}{r t}$

$$\begin{aligned} \frac{\partial^2 C}{\partial t^2} &= \frac{C_1 z^2}{r t^2} + C_2 \left(\frac{z}{t^2} \right) - \frac{C_2}{2r} \left(\frac{2 r z}{t^2} - \frac{2 z^2}{t^3} \right) + \frac{2 C_3}{t^3} \\ &= \frac{C_1 z^2}{r t^3} + \frac{C_2 z^2}{r t^3} + \frac{2 C_3}{t^3} \end{aligned}$$

and $\frac{\partial^2 C}{\partial t \partial z} = -\frac{C_1 z}{r t^2} - \frac{C_2 z}{r t^2}$

$\therefore \frac{\partial^2 C}{\partial t^2} \cdot \frac{\partial^2 C}{\partial z^2} - \left(\frac{\partial^2 C}{\partial t \partial z} \right)^2 = \left(\frac{C_1 z^2}{r t^3} + \frac{C_2 z^2}{r t^3} + \frac{2 C_3}{t^3} \right) \left(\frac{C_1 + C_2}{r t} \right) - \left(-\frac{C_1 z}{r t^2} - \frac{C_2 z}{r t^2} \right)^2$

$$= \frac{2 C_3 (C_1 + C_2)}{r t^4}$$

Obviously for the value of t given by (6)

$$\frac{\partial^2 C}{\partial z^2} > 0, \frac{\partial^2 C}{\partial t^2} > 0 \text{ and } \frac{\partial^2 C}{\partial t^2} \cdot \frac{\partial^2 C}{\partial z^2} - \left(\frac{\partial^2 C}{\partial t \partial z} \right)^2 > 0$$

$\therefore C$ is minimum for the value of t given by (6)

$\therefore$ The optimum order quantity for minimum cost is given by

$$q^* = r t^* = r \sqrt{\left[\frac{2 C_3 (C_1 + C_2)}{C_1 C_2 r} \right]}$$

or $q^* = \sqrt{\left[\frac{2 r C_3 (C_1 + C_2)}{C_1 C_2} \right]}$

which is the required **economic lot size formula**.

To compute $C_{\min}$.

Substituting the value of z from (5) in (2), we have

$$C(z, t) = \frac{C_1}{2 r t} \cdot \left(\frac{C_2 r t}{C_1 + C_2} \right)^2 + \frac{C_2}{2 r t} \left(r t - \frac{C_2 r t}{C_1 + C_2} \right)^2 + \frac{C_3}{t}$$

$$\begin{aligned}
 &= \frac{C_1 C_2^2 r t}{2 (C_1 + C_2)^2} + \frac{C_2 C_1^2 r t}{2 (C_1 + C_2)^2} + \frac{C_3}{t} \\
 &= \frac{C_1 C_2 r t (C_1 + C_2)}{2 (C_1 + C_2)^2} + \frac{C_3}{t} = \frac{C_1 C_2 r t}{2 (C_1 + C_2)} + \frac{C_3}{t}
 \end{aligned}$$

∴ Substituting the value of t from (6)

$$\begin{aligned}
 C_{\min} &= \frac{C_1 C_2 r}{2 (C_1 + C_2)} \cdot \sqrt{\left[\frac{2 C_3 (C_1 + C_2)}{C_1 C_2 r} \right]} + C_3 \sqrt{\left[\frac{C_1 C_2 r}{2 C_3 (C_1 + C_2)} \right]} \\
 &= \sqrt{\left[\frac{C_1 C_2 C_3 r}{2 (C_1 + C_2)} \right]} + \sqrt{\left[\frac{C_1 C_2 C_3 r}{2 (C_1 + C_2)} \right]} \\
 &= 2 \sqrt{\left[\frac{C_1 C_2 C_3 r}{2 (C_1 + C_2)} \right]} = \sqrt{\left[\frac{2 C_1 C_2 C_3 r}{C_1 + C_2} \right]} \quad \dots(7)
 \end{aligned}$$

Deductions : If shortages are not permitted *i.e.*, $C_2 = \infty$ then

$$\frac{C_1 + C_2}{C_2} = \frac{C_1}{C_2} + 1 = 1$$

$$\therefore \quad q^* = \sqrt{\left(\frac{2rC_3}{C_1} \right)}$$

which is **economic lot size formula**, for model I.

Remark :

1. Comparing with model I, we see that, here allowing shortages the holding cost per unit time is reduced in the ratio $\frac{C_2}{C_1 + C_2}$ from that of model I.

2. Note that

$$\begin{aligned}
 &C_{\min} \text{ in Model V} - C_{\min} \text{ in Model I} \\
 &= \sqrt{\left(\frac{2 C_1 C_2 C_3 r}{C_1 + C_2} \right)} - \sqrt{2 C_1 C_3 r} \\
 &= \sqrt{\left(\frac{2 C_1 C_3 r}{C_1 + C_2} \right)} \left[\sqrt{C_2} - \sqrt{C_1 + C_2} \right] \\
 &< 0 \quad \because \quad C_2 < C_1 + C_2
 \end{aligned}$$

Thus, $C_{\min}$ in Model V $<$ $C_{\min}$ in Model I

i.e., the minimum cost per unit time is reduced by allowing shortages which are to be back logged.

10.16 Model VI: Economic Lot-size Model with Uniform Rate of Demand, Finite Rate of Production and having Shortages which are to be Fulfilled

To derive an economic lot-size formula and the average minimum cost per unit time under the following conditions.

- (i) demand is uniform at a rate of r units per unit time,
- (ii) production rate is finite say K ($>r$) units per unit time,
- (iii) lead time is zero,
- (iv) C_1 = holding cost per unit per unit time,
- (v) C_2 = shortage cost per unit per unit time,
- (vi) C_3 = set up cost per run, and
- (vii) shortages are allowed and backlogged.

Initially the stock is zero and the production starts with a finite rate K ($>r$) units per unit time while the demand is r units per unit time. Thus, the inventory (stock) increases with a rate $(K - r)$ units per unit time. Let the production continue for a period t_1 .

$\therefore$ The inventory level at the end of time t_1 is

$$Q = (K - r) t_1 \quad \dots(1)$$

Now the production is stopped, and so this inventory Q falls to zero with a rate r units per unit time in time t_2 (say) and then shortage arises with a rate r unit per unit time and continue for a period t_3 (say). Let S be the maximum shortage at the end of time t_3 .

$$\therefore Q = r t_2 \quad \dots(2)$$

$$\text{and} \quad S = r t_3 \quad \dots(3)$$

At the end of time t_3 , production starts again and the shortages are fulfilled in time t_4 (say) i.e., at the end of time t_4 shortages reaches to zero.

The cycle then repeats itself, after the time $t_1 + t_2 + t_3 + t_4$.

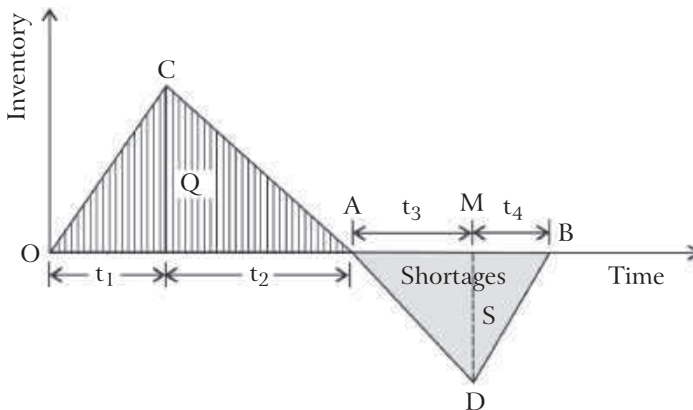


Fig. 10.5

Since the production rate is K units per unit time and the demand rate is r units per unit time, therefore the rate of reductions of shortages is $(K - r)$ units per unit time during the time t_4 .

$$\therefore S = (K - r) t_4 \quad \dots(4)$$

The situation of inventory and shortage etc. are illustrated in fig. 10.5.

From (1) and (2), we have

$$Q = (K - r) t_1 = r t_2 \quad \dots(5)$$

and from (3) and (4), we have

$$S = r t_3 = (K - r) t_4 \quad \dots(6)$$

If q units is the total demand to be fulfilled in time $t_1 + t_2 + t_3 + t_4$ of one cycle, then

$$q = r (t_1 + t_2 + t_3 + t_4) \quad \dots(7)$$

Eliminating t_1 and t_4 from (7) with the help of (5) and (6), we have

$$q = r \left(\frac{r t_2}{K - r} + t_2 + t_3 + \frac{r t_3}{K - r} \right)$$

$$\text{or} \quad q = \frac{K(t_2 + t_3)r}{K - r} \quad \dots(8)$$

The holding cost for one cycle = $C_1 \cdot (\text{Area of } \Delta OCA) = C_1 \cdot \frac{1}{2} Q (t_1 + t_2)$

$$\begin{aligned} &= \frac{C_1}{2} \cdot r t_2 \left\{ \frac{r t_2}{K - r} + t_2 \right\} \quad \text{from (5)} \\ &= \frac{C_1 r K t_2^2}{2 (K - r)} \end{aligned}$$

Shortage cost for one cycle = $C_2 \cdot (\text{Area of } \Delta ABD) = C_2 \cdot \frac{1}{2} S (t_3 + t_4)$

$$\begin{aligned} &= \frac{C_2}{2} \cdot r t_3 \cdot \left\{ t_3 + \frac{r t_3}{K - r} \right\} \quad \text{from (6)} \\ &= \frac{C_2 r K t_3^2}{2 (K - r)}. \end{aligned}$$

Set up cost for one cycle = C_3

$$\begin{aligned} \therefore \text{The total cost per cycle} &= \frac{C_1 r K t_2^2}{2 (K - r)} + \frac{C_2 r K t_3^2}{2 (K - r)} + C_3 \\ &= \frac{r K (C_1 t_2^2 + C_2 t_3^2) + 2 C_3 \cdot (K - r)}{2 (K - r)} \end{aligned}$$

∴ The total average cost per per cycle

$$C = \frac{\frac{r K (C_1 t_2^2 + C_2 t_3^2) + 2 C_3 \cdot (K - r)}{2 (K - r)}}{t_1 + t_2 + t_3 + t_4}$$

Eliminating t_1 and t_4 with the help of (5) and (6), we have

$$C(t_2, t_3) = \frac{r K (C_1 t_2^2 + C_2 t_3^2) + 2 C_3 \cdot (K - r)}{2 (K - r) \left[\frac{r t_2}{K - r} + t_2 + t_3 + \frac{r t_3}{K - r} \right]}$$

$$\text{or} \quad C(t_2, t_3) = \frac{r K (C_1 t_2^2 + C_2 t_3^2) + 2 C_3 \cdot (K - r)}{2 K (t_2 + t_3)} \quad \dots(9)$$

which is a function of two variables t_2 and t_3 .

For $C(t_2, t_3)$ to be minimum

$$\frac{\partial C}{\partial t_2} = \frac{r K \cdot 2 C_1 t_2 \cdot 2 K (t_2 + t_3) - \{r K (C_1 t_2^2 + C_2 t_3^2) + 2 C_3 (K - r)\} 2 K}{(2 K)^2 (t_2 + t_3)} = 0$$

$$\text{or} \quad C_1 t_2^2 + 2 C_1 t_2 t_3 - C_2 t_3^2 - \frac{2 C_3}{r K} (K - r) = 0 \quad \dots(10)$$

$$\text{and} \quad \frac{\partial C}{\partial t_3} = \frac{r K \cdot 2 C_2 t_3 \cdot 2 K (t_2 + t_3) - \{r K (C_1 t_2^2 + C_2 t_3^2) + 2 C_3 (K - r)\} 2 K}{(2 K)^2 (t_2 + t_3)^2} = 0$$

$$\text{or} \quad C_1 t_2^2 - 2 C_2 t_2 t_3 - C_2 t_3^2 + \frac{2 C_3}{r K} (K - r) = 0 \quad \dots(11)$$

Subtracting (10) and (11), we have

$$t_2 t_3 = \frac{2 (K - r) C_3}{r K (C_1 + C_2)} \quad \dots(12)$$

Adding (10) and (11), we have

$$2 C_1 t_2^2 + 2 C_1 t_2 t_3 - 2 C_2 t_2 t_3 - 2 C_2 t_3^2 = 0$$

$$\text{or} \quad (C_1 t_2 - C_2 t_3) (t_2 + t_3) = 0$$

$$\therefore \text{ either } t_3 = -t_2 \quad \text{or} \quad t_3 = \frac{C_1}{C_2} t_2$$

$$\text{If } t_3 = -t_2, \text{ then from (12) we have, } t_2^2 = -\frac{2 (K - r) C_2}{r K (C_1 + C_2)} < 0 \quad (\because K > r)$$

∴ t_2 is imaginary.

When $t_3 = \frac{C_1}{C_2} t_2$, then from (12), we have

$$t_2^2 = \frac{2 C_2 C_3 (K - r)}{C_1 r K (C_1 + C_2)}$$

$$\therefore t_2^* = t_2 = \sqrt{\frac{2 C_2 C_3 \left(1 - \frac{r}{K}\right)}{r C_1 (C_1 + C_2)}}$$

$$\text{and } t_3^* = \frac{C_1}{C_2} t_2^* = \sqrt{\frac{2 C_1 C_3 \left(1 - \frac{r}{K}\right)}{r C_2 (C_1 + C_2)}}$$
...(13)

It can be verified that

$$\frac{\partial^2 C}{\partial t_2^2} > 0, \frac{\partial^2 C}{\partial t_3^2} > 0 \text{ and } \frac{\partial^2 C}{\partial t_2^2} \cdot \frac{\partial^2 C}{\partial t_3^2} - \left(\frac{\partial^2 C}{\partial t_2 \partial t_3} \right)^2 > 0$$

for the values of t_2 and t_3 given by (13).

Hence for the values of t_2 and t_3 given by (13), the average cost per unit time given by (9) is minimum.

$\therefore$ From (9), the **minimum cost** per unit time is given by

$$C_{\min} = \sqrt{\frac{2 r C_1 C_2 C_3 \left(1 - \frac{r}{K}\right)}{(C_1 + C_2)}}$$

and from (8) the **optimum order quantity (EOQ)** is given by

$$q^* = \sqrt{\left(\frac{2 r C_3}{C_1}\right) \cdot \left(\frac{C_1 + C_2}{C_2}\right) \cdot \left(\frac{1}{1 - (r/K)}\right)}$$

ILLUSTRATIVE EXAMPLES

Example 1 : If in model I, the set up cost instead of being fixed is equal to $C_3 + b q$, where b is the set up cost per item produced, then show that there is no change in the optimum order quantity produced due to change in the set up cost.

Solution : Replacing C_3 by $C_3 + b q$ in Model I eqn. (2).

Average total cost per unit time.

$$C(t) = \frac{1}{2} C_1 r t + \frac{C_3 + b q}{t}$$

$$\text{or } C(t) = \frac{1}{2} C_1 r t + \frac{C_3}{t} + b r \quad (\because q = rt)$$

For minimum of $C(t)$

$$\frac{d}{dt} C(t) = \frac{C_1 r}{2} - \frac{C_3}{t^2} = 0 \quad \therefore t = t^* = \sqrt{\frac{2 C_3}{C_1 r}}$$

$$\frac{d^2 C(t)}{dt^2} = \frac{2 C_3}{t^3} > 0$$

$$\therefore C(t) \text{ is minimum for } t = t^* = \sqrt{\frac{2 C_3}{C_1 r}} \text{ and } q^* = r t^* = \sqrt{\left(\frac{2 C_3 r}{C_1} \right)}$$

which is the same as in Model I.

Example 2 : A manufacturer has to supply his customer with 600 units of his product per year. Shortages are not allowed and the storage cost amount to ₹ 0.60 per unit per year. The set up cost per run is ₹ 80.00. Find the optimum run size and the minimum average yearly cost.

Solution : Considering one year as one unit of time, we have

$$C_1 = ₹ 0.60, C_3 = ₹ 80.00, r = 600$$

$$q^* = \sqrt{\left(\frac{2 C_3 r}{C_1} \right)}$$

From Article 10.9

$$= \sqrt{\left(\frac{2 \times 80 \times 600}{0.60} \right)} = 400$$

and
$$t^* = \frac{q^*}{r} = \frac{400}{600} = \frac{2}{3} \text{ year} = \frac{2}{3} \times 12 = 8 \text{ months}$$

Thus, the manufacturer should produce 400 units of his product at an interval of 8 months.

$$\text{Minimum average cost} = \sqrt[3]{(2 C_1 C_3 r)}$$

$$= \sqrt[3]{(2 \times 0.60 \times 80 \times 600)} = ₹ 240 \text{ units / year}$$

Example 3 : The storage cost of one item is ₹1 per month and the set up cost is ₹ 25 per run. If the production is instantaneous and the demand is 200 units per month, find the optimal size of the batch and the best time for the replenishment of inventory.

Solution : Considering one month as one unit of time, we have

$$C_1 = ₹ 1.00, C_3 = ₹ 25, r = 200$$

$$\therefore q^* = \sqrt{\left(\frac{2 C_3 r}{C_1} \right)} = \sqrt{\left(\frac{2 \times 25 \times 200}{1} \right)} = 100$$

and $t^* = \frac{q^*}{r} = \frac{100}{200} = \frac{1}{2}$ month = 15 days.

Thus, he should produce 100 units of his product at interval of 15 days.

Minimum average cost = $\sqrt{(2 C_1 C_3 r)} = \sqrt{(2 \times 1 \times 25 \times 200)} = ₹ 100/\text{month}$.

Note : Average inventory throughout the month = $q/2 = 100$.

∴ If he produces all 200 items in the beginning of a month then the cost = $25 \times 1 + 100 = ₹ 125$, which is more than the cost, if the items that are produced twice in a month.

Example 4 : Consider the inventory system with the following data in usual notation:

$r = 1000$ units/year, $I = 0.30$, $P = ₹ 0.50$ per unit.

$C_3 = ₹ 10.00$, $L = 2$ yrs. (lead time).

Determine the following

- (i) Optimal order quantity,
- (ii) Re-order point,
- (iii) Minimum average cost.

Solution : Here $C_1 = IP = (0.30) \times (0.50) = 0.15$ per unit/year

$C_3 = ₹ 10.00$, $r = 1000$ units/year and $L = 2$ yrs.

$$\begin{aligned} \text{(i) Optimal order quantity (EOQ), } q^* &= \sqrt{\left(\frac{2 C_3 r}{C_1}\right)} \\ &= \sqrt{\frac{2 \times 10 \times 1000}{0.15}} = 365 \text{ units} \end{aligned}$$

(ii) Re-order point, $p = rL = 1000 \times 2 = 2000$ units

(iii) Minimum average cost = $\sqrt{(2 C_1 C_3 r)}$
 $= \sqrt{\{2 \times (0.15) \times 10 \times 1000\}} = ₹ 54.80$

Example 5 : In a central grain store, it takes about 15 days to get the stock after placing the order and daily 500 tons are despatched to neighbouring markets. On an adhoc basis safety stock is assumed to be 10 days stock. Calculate the recorder point p .

Solution : Here safety stock $B = 10$ days stock

$$= 10 \times 500 = 5000 \text{ tons}$$

$L = 15$ days and $r = 500$ tons.

Reorder point $p = B + Lr = 5000 + 15 \times 500 = 12500$ tons

Example 6 : You have to supply your customers with 100 units of a certain product every Monday and only then, you obtain the product from a local supplier at ₹ 60 per unit. The costs of ordering and transportation from the supplier are ₹ 150 per order. The cost of carrying inventory is estimated at 15% per year of the cost of the product carried.

- (i) Find the total size which will minimize the cost of the system.
(ii) Determine the optimal cost.

Solution : Here $C_1 = 15\%$ per year of the cost of the product carried

$$= ₹ \frac{15}{100} \times 60 = ₹ 2 \text{ per unit/year.}$$

$$= ₹ \frac{9}{52} \text{ per unit/week}$$

$$C_3 = ₹ 150 \text{ and } r = 100 \text{ units/week}$$

$$(i) \quad \text{Optimal lot size, } q^* = \sqrt{\left(\frac{2 C_3 r}{C_1}\right)} = \sqrt{\left(\frac{2 \times 150 \times 100 \times 52}{9}\right)} = 416 \text{ units}$$

$$(ii) \quad \text{Optimal cost} = \sqrt{(2 C_1 C_3 r)} + 60 r$$

$$= \sqrt{\left(2 \times \frac{9}{52} \times 150 \times 100\right)} + 60 \times 100 = ₹ 6072$$

Example 7 : ABC manufacturing company purchases 9000 parts of a machine for its annual requirements, ordering one month usage at a time. Each part costs ₹ 20. The ordering cost per order is ₹ 15, and the carrying charges are 15% of the average inventory per year.

You have been asked to suggest a more economical purchasing policy for the company. What advice would you offer, and how much would it save per year ?

Solution : Here $r = 9000$ parts/year

$$C_1 = 15\% \text{ of the average inventory per year}$$

$$= ₹ 20 \times \frac{15}{100} = ₹ 3 \text{ each part / year.}$$

$$C_3 = ₹ 15$$

$$\therefore \text{Economic lot size, } q^* = \sqrt{\frac{2 C_3 r}{C_1}} = \sqrt{\frac{2 \times 15 \times 9000}{3}} = 300 \text{ units.}$$

$$\text{and} \quad t^* = \frac{q^*}{r} = \frac{300}{9000} = \frac{1}{30} \text{ year}$$

minimum cost per year = $\sqrt{(2 C_1 C_3 r)} = \sqrt{(2 \times 3 \times 15 \times 9000)} = ₹ 900$

If the company follows the policy of ordering each month.

Then, annual ordering cost = $₹ 12 \times 15 = ₹ 180$

and lot size each month = $\frac{9000}{12} = 750 = q$

$\therefore$ average inventory at any time = $\frac{q}{2} = \frac{750}{2} = 375$

$\therefore$ storage (or carrying) cost at any time = $375 C_1$
 $= 375 \times 3 = ₹ 1125.$

$\therefore$ Total annual cost = $₹ 1125 + ₹ 180 = ₹ 1305$

Hence if the company purchase 300 parts at time intervals of $\frac{1}{30}$ yrs. in place of order 750 parts each month then the net saving of the company is

$₹ 1305 - ₹ 900 = ₹ 405$ per year

Example 8 : A company uses annually 24000 units of a raw material which costs ₹ 1.25/unit. Placing each order costs ₹ 22.5 and the carrying cost is 5.4% per year of the average inventory. Find the economic lot size and the total inventory cost (including the cost of material).

Solution : Here $r = 24000$ units/year.

$C_1 = 5.4\%$ of the average inventory per year

$= ₹ 1.25 \times \frac{5.4}{100} = ₹ \frac{27}{400}$ per unit/year

$C_3 = ₹ 22.5$

$\therefore$ Economic lot size, $q^* = \sqrt{\frac{2 C_3 r}{C_1}} = \sqrt{\frac{2 \times 22.5 \times 24000 \times 400}{27}} = 4000$ units

and $t^* = \frac{q^*}{r} = \frac{4000}{24000} = \frac{1}{6}$ year = 2 months.

The optimal lot size is 4,000 units, after every two months.

Total annual inventory cost (including cost of material)

$= \sqrt{(2 C_1 C_3 r)} + \text{purchasing cost per year}$

$$\begin{aligned}
 &= \sqrt{2 \times \frac{27}{400} \times 22.5 \times 24000 + (1.25) \times 24000} \\
 &= ₹ 270 + ₹ 30000 = ₹ 30270
 \end{aligned}$$

Example 9 : Neon lights are replaced at the rate of 100 units per day. The physical plant orders the neon lights periodically. It costs ₹100 to initiate a purchase order. A neon light kept in storage is estimated to cost ₹ 0.02 per day. The lead time between placing and receiving an order is 12 days. Determine the optimal inventory policy for ordering the neon lights.

Solution : Given that, $C_3 = ₹ 100$, $C_1 = ₹ 0.02$, $r = 100$ units per day

$$\therefore \text{Optimal order quantity } q^* = \sqrt{\frac{2 C_3 r}{C_1}} = \sqrt{\frac{2 \times 100 \times 100}{0.02}} = 1000 \text{ lights}$$

$$\therefore \text{Length of the cycle, } t^* = \frac{q^*}{r} = \frac{1000}{100} = 10 \text{ days}$$

Here, lead time $L = 12$ days and the length of the cycle = 10 days

i.e., the lead time is greater than the length of the cycle.

$\therefore$ the re-ordering will take place when the level of inventory is sufficient to satisfy the demands for $12 - 10 = 2$ days.

Hence the re-order point (ROP) = $2 \times 100 = 200$ lights.

i.e., the quantity 100 lights is to be ordered when the level of inventory reaches **200 lights**.

Note : When the lead time L is greater than the length of the cycle t^* , then the effective lead time is taken equal to $L - t^*$. In this case effective lead time is taken as $L - t^* = 12 - 10 = 2$ days instead of 12 days.

Example 10 : An item is produced at the rate of 50 items per day. The demand occurs at the rate of 25 items per day. If the set up cost is ₹ 100 per set up and holding cost is 0.01 per unit of item per day, find the economic lot size for one run, assuming that the shortages are not permitted.

Solution : It is the problem of Model III.

Here $K = 50$ items/day, $r = 25$ items/day,

$C_1 = ₹ 0.01/\text{day}$, $C_3 = ₹ 100/\text{run}$.

$\therefore$ lot size formula is

$$q^* = \sqrt{\frac{C_3}{C_1} \cdot \left(\frac{rK}{K-r} \right)} = \sqrt{\frac{2 \times 100}{0.01} \cdot \left(\frac{25 \times 50}{50 - 25} \right)} = 1000 \text{ items}$$

and

$$t^* = \frac{q^*}{r} = \frac{1000}{25} = 40 \text{ days}$$

$$\begin{aligned} \text{Minimum cost} &= \sqrt{2 C_1 C_3 r \left(1 - \frac{r}{K}\right)} \\ &= \sqrt{2 \times 0.01 \times 100 \times 25 \left(1 - \frac{25}{50}\right)} = ₹ 5 \text{ per day} \end{aligned}$$

Hence, total cost per run = ₹ 5 × 40 = ₹ 200.

Example 11 : A contractor has to supply 10000 bearing per day to an automobile manufacturer. He finds that when he starts a production run, he can produce 25000 bearing per day. The cost of holding a bearing in stock for one year is 2 paise, and the set up cost of a production run is ₹ 18. How frequently should production run be made ?

Solution : It is the problem of model III.

Here $K = 25,000$ bearing/day, $r = 10,000$ bearing/day

$C_1 = 2$ paise per bearing per year.

$$= ₹ \frac{2}{100 \times 365} = ₹ \frac{1}{50 \times 365} \text{ per bearing per day.}$$

$$C_3 = ₹ 18$$

$$\begin{aligned} \therefore \text{lot size formula is } q^* &= \sqrt{\frac{2 C_3}{C_1} \left(\frac{r K}{K - r} \right)} \\ &= \sqrt{2 \times 18 \times 50 \times 365 \left(\frac{10000 \times 25000}{25000 - 10000} \right)} \\ &= 104642 \text{ bearings.} \end{aligned}$$

and

$$q^* = \frac{104642}{10000} = 10.46 \text{ days.}$$

Example 12 : A contractor has to supply diesel engines to a truck manufacturer at the rate of 25 per day. There is a clause in the contract penalizing him ₹10 per engine per day late for missing the scheduled delivery date. He finds that the cost of holding a completed engine in stock is ₹ 16 per month. The production process is such that each month (30 days) he starts a batch of engines through the shops, and all these engines are available for delivery any time after the end of the month. What should his inventory level be at the beginning of each month ?

Solution : It is the problem of Model IV.

Here

$$r = 25 \text{ engines/day}$$

$$C_1 = ₹ \frac{16}{30} \text{ per engine per day} = ₹ \frac{8}{15} \text{ per engine per day.}$$

$$C_2 = ₹ 10 \text{ per engine per day}$$

∴ Optimal inventory level

$$z^* = \frac{C_2 r t}{C_1 + C_2} = \frac{10 \times 25 \times 30}{(8/15) + 10} = 712 \text{ engines.}$$

Example 13 : The demand of an item is uniform at a rate of 25 units per month. The fixed cost is ₹ 15 each time a production run is made. The production cost is ₹ 1 per item, and the inventory carrying cost is ₹ 0.30 per item per month. If the shortage cost is ₹ 1.50 per item per month, determine how often to make a production run, and of what size it should be.

Solution : It is a problem of model V.

In this problem set up cost is a variable and is equal to $C + Pq$, where P is production cost of an item and q is the order quantity per production run.

Proceeding as in model V, average total cost per unit time is given by

$$\begin{aligned} C(z, t) &= \frac{C_1}{2r} \cdot \frac{z^2}{t} + \frac{C_2}{2r} \cdot \frac{(rt - z)^2}{t} + \frac{C_3 + Pq}{t} \\ &= \frac{C_1}{2r} \cdot \frac{z^2}{t} + \frac{C_2}{2r} \cdot \frac{(rt - z)^2}{t} + \frac{C_3}{t} + Pr \end{aligned} \quad (\because q = rt)$$

Since Pr is constant ∴ as in model V, $C(z, t)$ is minimum, when,

$$t^* = \sqrt{\frac{2 C_3 (C_1 + C_2)}{C_1 C_2 r}}$$

$$\therefore q^* = r t^* = \sqrt{\frac{2 r (C_1 + C_2) C_3}{C_1 C_2}}$$

Here $r = 25$ units/month.

$$C_1 = ₹ 30/\text{item/month},$$

$$C_2 = ₹ 1.50/\text{item/month},$$

$$C_3 = ₹ 15 \text{ per production run.}$$

$$\therefore q^* = \sqrt{\frac{2 \times 15 \times (0.30 + 1.50) \times 25}{(0.30) \times (1.50)}} = 55 \text{ items}$$

and

$$t^* = \frac{q^*}{r} = \frac{55}{25} = 2.2 \text{ months}$$

Example 14 : A company, for one of the A-class item, placed 6 orders each of size 200 in a year. Giving ordering cost = ₹ 600, holding cost = 40%, cost per unit = ₹ 40, find out the loss to the company in not operating scientific inventory policy? What are your recommendations for the future ?

Solution : Considering one year as one unit of time, we have

$$C_1 = IP = ₹ \frac{40}{100} \times 40 = ₹ 16 \text{ per unit/year.}$$

$$C_3 = ₹ 600.$$

Total demand in one year, $R = 6 \times 200 = ₹ 1200$ items

$$\therefore \quad (\text{EOQ}) \quad q^* = \sqrt{\frac{2 C_3 R}{C_1}} = \sqrt{\frac{2 \times 600 \times 1200}{16}} = 300 \text{ units}$$

$\therefore$ Minimum cost per year

= The cost of holding inventory + Ordering cost

$$= \frac{1}{2} C_1 q^* + (R C_3 / q^*) = ₹ \left(\frac{1}{2} \times 16 \times 300 + \frac{1200}{300} \times 600 \right) = ₹ 4800$$

Total cost per year under the existing policy

= The cost of holding inventory + Ordering cost

$$= \frac{1}{2} C_1 q + \frac{R}{q} C_3 = ₹ \left(\frac{1}{2} \times 16 \times 200 + \frac{1200}{200} \times 600 \right) = ₹ 5200$$

$\therefore$ Loss to the company in not operating scientific inventory policy

$$= ₹ 5200 - ₹ 4800 = ₹ 400$$

Recommendation for the future : The company should place four orders each of size 300 units in a year.

Example 15: A company uses annually 50000 units of an item each costing ₹ 1.20. Each order costs ₹ 45 and inventory carrying costs 15% of the annual average inventory value.

(i) Find EOQ

(ii) If the company operates 250 days a years, the procurement time is 10 days and safety stock is 500 units, find reorder level, maximum, minimum and average inventory.

Solution : We have

$$C_1 = IP = ₹ \frac{15}{100} \times 1.20 = ₹ \frac{9}{50} \text{ per unit/year.}$$

$$C_3 = ₹ 45$$

Total demand in one year $R = 50000$ units

$$(i) \quad EOQ = \sqrt{\frac{2 C_3 R}{C_1}} = \sqrt{\frac{2 \times 45 \times 50000 \times 50}{9}}$$

$$= 5000 \text{ units}$$

(ii) Number of days the company operates = 250

$$\text{Lead time } L = 10 \text{ day}$$

$$\text{and safety stock } B = 500 \text{ units}$$

$\therefore$ Recorder level = Lead time demand + Safety stock

$$= L \times \frac{50000}{250} + B$$

$$= 10 \times 200 + 500 = 2500 \text{ units}$$

$$\text{Maximum inventory} = EOQ + \text{Safety stock}$$

$$= 5000 + 500 = 5500 \text{ units.}$$

$$\text{Minimum inventory} = \text{Safety stock} = 500 \text{ units}$$

$$\text{Average inventory} = \left(\frac{O + EOQ}{2} \right) + \text{Safety stock}$$

$$= \frac{1}{2} \times 5000 + 500$$

$$= 3000 \text{ units.}$$

Exercise 10.1

1. A shop keeper has a uniform demand of an item at the rate of 50 items per month, he buys from a supplier at a cost of ₹ 6 per item and the cost of ordering is ₹ 10 each time. If the stock holding costs are 20% per year of stock value, how frequently should he replenish his stocks.
2. A contractor has to supply diesel engines to a truck manufacturer at a rate of 20 per day. The penalty in the contract is ₹ 10 per engine per day late for missing the scheduled delivery data. The cost of holding an engine in stock for one month is ₹ 15. His production process is such that each month (30 days) he starts a batch of engines through the agencies and all are available for supply after the end of the month, what should be the inventory level in the beginning of each month.
3. A precision Engineering Factory consumes 50000 units of a component per year. The ordering receiving and handling costs are ₹ 3 per order while the trucking costs are ₹ 12 per order. Further details are as follows :

Interest cost ₹ 0.06 per unit per year.

Deterioration and observation cost ₹ 0.004 per unit per year. Shortage cost ₹ 1000 per year for 50000 units; calculate the economic order quantity.

[Hint : It is the problem on model I

Here $C_3 = ₹ 3 + ₹ 12 = ₹ 15/\text{order}$, $r = 50000$,

$$C_1 = ₹ 0.06 + ₹ 0.004 + ₹ (1000/50000) \\ = ₹ 0.084/\text{unit}.$$

4. A company uses annually 12000 units of a raw material costing ₹ 1.25 per unit. Placing each order costs ₹ 0.45 and the carrying costs are 15% per year per unit of the average inventory. Find the Economic Order Quantity.
5. A certain item costs ₹ 235 per ton. The monthly requirements are 5 tons and each time the stock is replenished there is a set up cost of ₹ 1000. The cost of carrying inventory has been estimated as 10% of the value of the stock per year. What is the optimal order quantity ?
6. An electric appliance manufacturer wishes to know what the economic quantity should be for a plastic impeller when the following information is available.

The average daily requirement is 120 units and the company has 250 working days a year, so that the total yearly requirement is approximately 30000 units a year. The manufacturing cost is 50 paise per part. The sum of the annual rate for interest, insurance, taxes and so forth is 20% of the unit cost, and the cost of preparation is ₹ 50 per lot.

[Hint : Model I, $r = 3000$ units, $C_1 = ₹ 0.50 \times 0.20 = ₹ 0.1$]

7. A manufacturing company uses a certain part at a constant rate of 4000 units per year. Each unit costs ₹ 2 and the company personnel estimate that it cost ₹ 50 to place an order, and that the carrying cost of inventory is 20% per year. Find the optimal size of each order and minimum yearly cost.
8. A product is produced at the rate of 50 items per day. The demand occurs at the rate of 30 items per day. Given that, $C_3 = ₹ 100$, $C_1 = ₹ 0.05$, find the economic lot size and the associated total cost per cycle assuming that no shortage is allowed.
9. In question 7, suppose that shortage is allowed. If it is decided to use a lot-size of 600 items, what is the simplified shortage cost per unit under optimal conditions ?
10. An oil engine manufacturer purchases lubricants at the rate of ₹ 42 per piece from a vendor. The requirement of these lubricants is 1800 per year. What should be the order quantity per order, if the cost per placement of an order is ₹ 16 and inventory carrying charges per rupee per year is only 20 paise.

[Hint : From Model I, $r = 42 \times 1800$, $q^* = \sqrt{\frac{2 C_3 r}{C_1}} = 3478$ appr.]

∴ Optimum inventory quantity = $3478/42 = 83$ lubricants]

11. In Ex. 10, how frequently should production run be made if $C_1 = 20$ paise/ bearing / year and $C_3 = ₹ 180$.
 12. A contractor has a requirement for a cement that amounts to 300 bags per day. No shortages are allowed Cement cost ₹ 12 per bag. Inventory carrying cost is 10% of the average inventory valuation per day and it costs ₹ 20 to purchase order. Find the minimum cost purchase quantity.
- [Hint : Mode I. Here $r = 300$, $C_1 = ₹ 0.10 \times 12 = ₹ 1.20$, $C_3 = ₹ 20$.]
13. An aircraft company uses rivets at an approximate customer rate of 2500 kg per year. The rivets cost ₹ 30 per kg and the company personnel estimate that it costs ₹ 130 to place an order and the inventory carrying cost is 10% per year. How frequently should orders for rivets be placed and what quantities should be ordered?

Answers 10.1

1.	Lot-size 100 items after every two months	2.	571 engines
3.	$q^* = 4226$ units	4.	240 units
5.	71.5 tons	6.	5477 units
7.	1000 units, ₹ 400	8.	$130\sqrt{3}$, ₹ 11.00
9.	$C_2 = 1/220$	10.	83 lubricants
11.	10.46 days	12.	$q^* = 100$ bags, $C_{\min} = ₹ 120$
13.	466 kg. appr., 5.3 orders per year		

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1. What aims at optimizing inventory levels ?
(a) Inventory control (b) Inventory capacity
(c) Inventory planning (d) None of these
 2. What can be defined as a useful idle resource which has economic value e.g. raw materials, spare parts, finished items, etc. ?
(a) Inventory control (b) Inventory
(c) Inventory Planning (d) None of these
 3. The stock of materials kept in the stores in anticipation of future demand is known as :
(a) Raw materials (b) Storage of materials
(c) Inventory (d) Stock of materials
 4. If small orders are placed frequently (rather than placing large orders infrequently), then the average level of inventory hold in stock will be :
(a) Reduced (b) Increased
(c) Either increased or reduced (d) Minimized
 5. If small orders are placed frequently (rather than placing large orders infrequently), then the total inventory cost will :
(a) Increase (b) Decrease
(c) Either increase or decrease (d) Minimized
 6. Which cost can vary with order quantity ?
(a) Unit cost only (b) Holding cost only
(c) Re-order cost only (d) All of the above
 7. If C_1 holding cost per unit, C_3 -set up cost per production run, r -demand rate for the product, then the optimal period for placing an order is given by :
(a) $2 C_1 C_3 r$ (b) $(2 C_1 C_3 r)^{1/2}$
(c) $(2 C_3 / r C_1)^{1/2}$ (d) None of these
 8. In the *EOQ* model :
(a) Orders arrive in a batch
(b) Demand is known as occurs at a constant rate
(c) All demand must be satisfied
(d) All of these
-

9. If orders are placed with size determined by the *EOQ*, then the re-order costs component is :
 - (a) Equal to the holding cost component
 - (b) Greater than the holding cost component
 - (c) Less than the holding cost component
 - (d) Either greater or less than the holding cost component
10. If the total investment in stock is limited will the best order quantity for each item be :
 - (a) Greater than *EOQ*
 - (b) Equal to *EOQ*
 - (c) Less than *EOQ*
 - (d) Either greater or less than *EOQ*

Fill in the blank

1. The purpose of the *EOQ* inventory model is to determine the order quantity which will total annual costs for an inventory item.
2. The level of inventory at reorder point (*i.e.*, at the time of placing order) is called
3. The lead time is the time between the order of inventory and its arrival, lead time demand is the amount of used during a lead time period.
4. If $C_1 = ₹ 0.30$, $C_3 = ₹ 80.00$ and $r = 300$ units, then the optimal order quantity is units.
5. If the procurement (or replenishment) cost used in the formula to compute *EOQ* is 1/4 of the actual procurement cost, then the *EOQ* so obtained will be of the *EOQ*.

True/False

1. If demand is constant and known, and stockouts are not permitted, then reorder point is the same as the lead time demand.
2. Economic lot size is the optimal amount of an order which should be placed at a time and the reordering point gives the idea when the next order should be placed.
3. The average inventory is equal to half the order quantity under the *EOQ* model because we assume that all inventory arrives at one time when the inventory level reaches zero and demand is constant.
4. The optimal number of orders per year increases when carry cost decreases.
5. Wilson *EOQ* formula is given by

$$q^* = \sqrt{(C_3 r / C_1)}$$

Answers

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1.	(c)	2.	(b)	3.	(c)
4.	(a)	5.	(b)	6.	(d)
7.	(c)	8.	(d)	9.	(a)
10.	(c)				

Fill in the Blank

1.	minimize	2.	reorder point	3.	inventory
4.	400 units	5.	1/2		

True/False

1.	T	2.	T	3.	T
4.	F	5.	F		

○○○

UNIT-IV

Chapter

11



Dynamic Programming

11.1 Introduction

Dynamic programming is a newly developed mathematical technique which is often useful for making a sequence of inter-related decisions. It provides a dynamic procedure for determining the combination of decisions which maximize over all effectiveness. Classical mathematics has proved inadequate in handling many optimization problems that involve large number of decision variables and/or large number of inequality constraints. Such problems are handled by the technique of dynamic programming. The technique decomposes the original problem in n -variables into n -subproblems (stages) each in one variable. The solution is obtained in an orderly manner by starting from one stage to the next, and is completed after the final stage is reached. Dynamic programming method is applicable in solving a wide variety of problems, including allocation, inventory control and replacement etc.

11.2 Bellman's Principle of Optimality in Dynamic Programming

The technique of dynamic programming was developed by **Richard Bellman** in the early 1950. **Bellman's principle of optimality** states that

"An optimal policy (set of decisions) has the property that whatever be the initial state and initial decisions, the remaining decisions must constitute an optimal policy for the state resulting from the first decision".

We shall consider the implication of this principle (Bellman's principle of optimality) as a multistage decision problem. It is important to note that a problem which does not satisfy the principle of optimality cannot be solved by dynamic programming.

11.3 Multistage Decision Problem

A problem in which the decisions have to be made at successive stages is called a multistage decision problem. Multistage problems can be classified on the basis of the following properties:

1. *The outcome of a decision may be deterministic or stochastic (probabilistic). In case of deterministic, if the stage of a process is given, the outcome of the decision at any stage is uniquely determined and known. In stochastic case, there is a set of possible outcomes given by a known probability distribution.*
2. *The no. of possible decisions at any stage, from which we have to choose one may be finite or infinite.*
3. *The total number of stages in the process may be finite or infinite and may be known and unknown.*

11.4 Characteristics of Dynamic Programming Problems

The basic features which characterize dynamic programming problem may be presented as follows:

1. *The problem can be divided up into stages, with a policy decision required at each stage.*
2. *Each stage has a number of **states** associated with it.*
The states are various possible conditions in which the system might find itself at that stage of the problem. The number of states may be either finite or infinite.
3. *The effect of the policy decision at each stage is to transform the current state into a state associated with the next stage.*
This suggests that dynamic programming problems can be interpreted in terms of the networks.
4. *Given the current stage, an optimal policy for the remaining stages is independent of the policy adopted in previous stages.*
For dynamic problem, in general, knowledge of the current state of the system conveys all of the informations about its previous behaviour necessary for determining the optimal policy henceforth. This is the **Markovian property**.
5. *The solution procedure begins by finding the optimal policy for each state of the last stage.*
6. *A recursive relationship (**functional equation or transition function or state transformation**) is available which identifies the optimal policy for each stage with r stages remaining, given the optimal policy for each stage with $(n - 1)$ stages remaining. This formula allows to go from one stage to another.*

7. Using the recursive relationship (functional equation), the solution procedure moves backward, stage by stage each time finding the optimal policy for each state of that stage – until it finds the optimal policy when starting at the initial stage.

11.5 Solution of a Multistage Problem by Dynamic Programming with Finite Number of Stages

The solution of a problem (multistage problem) by dynamic programming involves the following steps:

1. The mathematical formulation of the problem and the development of the functional equations i.e., to develop a recurrence relation connecting the optimal decision function for the n stage problem with the optimal decision function for the $(n-1)$ stage sub-problem $n = 1, 2, 3, \dots, n$.
2. To solve the functional equations for determining the optimal policy.

For this, first we write the optimal decision function for one stages sub-problem and solve it. Then we solve the optimal decision function for 2-stage, 3-stage, $(n-1)$ stage and n -stage problem.

Generally, the solution of a recursive equation involve two types of computations according as the system is discrete or continuous. In case of discrete system a tabulator computational scheme is followed at each stage. In each table, the number of row and columns are equal to the no. of corresponding feasible state values and the no. of possible decisions respectively. In case of continuous system, the optimal decisions at each stage are obtained by using the usual classical techniques such as differentiation etc.

ILLUSTRATIVE EXAMPLES

Example 1: A positive quantity c is to be divided into n parts in such a way that the product of the n parts is to be a maximum. Obtain the optimal subdivision.

Solution: **Classical formulation:** If the number c is divided into n parts $y_1, y_2, y_3, \dots, y_n$ then the problem (in classical form) is to find $y_1, y_2, \dots, y_n$ which

maximize $Z = y_1 \cdot y_2 \cdot \dots \cdot y_n$

such that $y_1 + y_2 + \dots + y_n = c$.

To solve the problem by dynamic programming we proceed as follows:

Step 1: To develop the functional equations (recursive relation)

First we shall develop a recursive relation connecting the optimal decision function for the n -stage problem with the optimal decision function for the $(n-1)$ stage sub-problem, $n = 1, 2, \dots, n$.

Let y_i ($i = 1, 2, \dots, n$) be the i -th part of c . Here each part i may be regarded as a stage. Since y_i may assume any non-negative values such that $y_1 + y_2 + y_3 + \dots + y_n = c$. Hence, the alternatives at each stage are infinite. Thus, it is the problem of a continuous system and hence the optimal decisions at each stage are obtained by using the usual classical technique (differentiation).

Let $f_n(c)$ denote the maximum attainable product, when the quantity (number) c is divided into n parts. Clearly $f_n(c)$ depends on n because the quantity c is fixed. It is clear that $f_n(c)$ is a function of the discrete variable n ($n = 1, 2, \dots, n$).

For $n = 1$, i.e. if c is divided into one part only then $y_1 = c$.

$$\therefore \text{clearly } f_1(c) = c \quad \dots(1)$$

For $n = 2$, i.e., if c is divided into two parts y_1 and y_2 i.e., $y_1 + y_2 = c$,

$$\text{let } y_1 = z, \quad \therefore y_2 = c - z$$

$$\therefore f_2(c) = \text{Max. } y_1 \cdot y_2 = \text{Max. } \{z(c - z)\}$$

$$\text{or } f_2(c) = \text{Max.}_{0 \leq z \leq c} \{z \cdot f_1(c - z)\} \quad \dots(2)$$

[since $f_1(c - z) = c - z$ from (1)]

For $n = 3$, i.e. if c is divided into three parts y_1, y_2, y_3 i.e., $y_1 + y_2 + y_3 = c$,

$$\text{let } y_1 = z, \quad \text{then } y_2 + y_3 = c - z.$$

$\therefore$ the part $c - z$ is further divided into two parts y_2, y_3 whose maximum attainable product is $f_2(c - z)$ by the def. of $f_n(c)$

$$\therefore f_3(c) = \text{Max. } y_1 \cdot y_2 \cdot y_3 = \text{Max.}_{0 \leq z \leq c} \{z \cdot f_2(c - z)\} \quad \dots(3)$$

Proceeding in this way, we get the recursive relation (functional equation) for $n = m$.

$$f_m(c) = \text{Max.}_{0 \leq z \leq c} \{z \cdot f_{m-1}(c - z)\} \quad \dots(4)$$

Step 2: To solve the functional equations for determining optimal policy

From (1), we have $f_1(c) = c$.

$$\begin{aligned} \text{From (2), } f_2(c) &= \text{Max.}_{0 \leq z \leq c} \{z \cdot f_1(c - z)\} \\ &= \text{Max.}_{0 \leq z \leq c} \{z \cdot (c - z)\} \end{aligned}$$

Now $z(c - z)$ is max. or mini. for that value of z for which

$$\frac{d}{dz} \{z(c - z)\} = c - 2z = 0$$

$$\therefore z = \frac{c}{2}, c - z = c - \frac{c}{2} = \frac{c}{2}$$

$z(c - z)$ is max. at $z = \frac{c}{2}$. Since $\frac{d^2}{dz^2} \{z(c - z)\} = -2$. (Negative)

$$\therefore f_2(c) = \frac{c}{2} \cdot \frac{c}{2} = \left(\frac{c}{2}\right)^2 \quad \dots(5)$$

and the optimal policy for two parts is $\left(\frac{c}{2}, \frac{c}{2}\right)$, i.e., division of c in two equal parts.

$$\begin{aligned} \text{From (3)} \quad f_3(c) &= \text{Max.}_{0 \leq z \leq c} \{z f_2(c-z)\} \\ &= \text{Max.}_{0 \leq z \leq c} \left\{ z \left(\frac{c-z}{2} \right)^2 \right\} \\ &\quad \left[\text{since } f_2(c-z) = \left(\frac{c-z}{2} \right)^2, \text{ from (5)} \right] \end{aligned}$$

Now $z \cdot \left(\frac{c-z}{2} \right)^2$ is max. for that value of z , for which

$$\frac{d}{dz} \left\{ z \left(\frac{c-z}{2} \right)^2 \right\} = 1 \cdot \left(\frac{c-z}{2} \right)^2 - 2z \left(\frac{c-z}{2} \right) \cdot \frac{1}{2} = 0.$$

$$\therefore z = \frac{c}{3}, \quad c-z = \frac{2c}{3}.$$

i.e., $c-z = \frac{2c}{3}$ is to be divided into two parts whose product is max.

By the policy for two parts, $f_2(c-z)$ i.e., $f_2\left(\frac{2c}{3}\right)$ is attained when the two parts are $\frac{1}{2}\left(\frac{2c}{3}\right)$

and $\frac{1}{2}\left(\frac{2c}{3}\right)$ i.e., $\frac{c}{3}, \frac{c}{3}$.

$$\therefore f_3(c) = \frac{c}{3} \left\{ \frac{c - (c/3)}{2} \right\}^2 = \left(\frac{c}{3}\right)^3$$

and the optimal-policy for three parts is $\left(\frac{c}{3}, \frac{c}{3}, \frac{c}{3}\right)$ i.e., division of c in three equal parts.

The calculation for stages 1, 2, 3 suggests that in general for n parts (stages), the optimal policy is

$$\left(\frac{c}{n}, \frac{c}{n}, \dots, \frac{c}{n}\right)$$

$$\text{and} \quad f_n(c) = \left(\frac{c}{n}\right)^n \quad \dots(6)$$

This result can be proved by induction.

Assuming that the result (6) hold for $n=2, 3, \dots, m$

$$\text{i.e.,} \quad f_m(c) = \left(\frac{c}{m}\right)^m$$

Now we shall prove that (6) also hold for $n = m + 1$.

From (4), we have

$$\begin{aligned} f_{m+1}(c) &= \text{Max.}_{0 \leq z \leq c} \{z \cdot f_m(c-z)\} \\ &= \text{Max.}_{0 \leq z \leq c} \left\{ z \cdot \left(\frac{c-z}{m} \right)^m \right\} \end{aligned}$$

Now $z \cdot \left(\frac{c-z}{m} \right)^m$ is max. for the value of z for which

$$\frac{d}{dz} \left\{ z \cdot \left(\frac{c-z}{m} \right)^m \right\} = 1 \cdot \left(\frac{c-z}{m} \right)^m - z \cdot m \left(\frac{c-z}{m} \right)^{m-1} \left(\frac{1}{m} \right) = 0$$

$$\therefore z = \frac{c}{m+1}$$

$$\therefore f_{m+1}(c) = \frac{c}{m+1} \cdot \left\{ \frac{c - \frac{c}{m+1}}{m} \right\}^m = \left(\frac{c}{m+1} \right)^{m+1}$$

and the optimal policy in this case is

$$\left(\frac{c}{m+1}, \frac{c}{m+1}, \dots, \frac{c}{m+1} \right)$$

Hence, the required optimal policy is $\left(\frac{c}{n}, \frac{c}{n}, \dots, \frac{c}{n} \right)$ and maximum product is $\left(\frac{c}{n} \right)^n$.

Example 2: Use dynamic programming to show that

$$- \sum_{i=1}^n p_i \log p_i,$$

subject to $\sum_{i=1}^n p_i = 1$ is maximum

when $p_1 = p_2 = \dots = p_n = 1/n$.

Solution: Step 1: To develop the functional equations

Here we may consider the problem as follows:

Divide 1 in n parts $p_1, p_2, \dots, p_n$ i.e., let $p_1 + p_2 + \dots + p_n = 1$, such that

$$- \sum_{i=1}^n p_i \log p_i = -(p_1 \log p_1 + p_2 \log p_2 + \dots + p_n \log p_n) \text{ is maximum.}$$

Let $f_n(l)$ denote the maximum attainable value of $-\sum_{i=1}^n p_i \log p_i$ when l is divided in n parts

$$p_1, p_2, \dots, p_n$$

$$\text{s.t.} \quad p_1 + p_2 + \dots + p_n = l$$

Clearly $f_n(l)$ is a function of discrete variable and it is a continuous system problem since p_i may assume any non-negative value.

For $n=1$, i.e., if l is divided into one part only then $p_1 = l$

$$\therefore f_1(l) = \text{Max.} (-p_1 \log p_1) = -l \log l. \quad \dots(1)$$

For $n=2$, i.e., if l is divided into parts p_1 and p_2 i.e., $p_1 + p_2 = l$,

$$\begin{aligned} \text{let} \quad p_1 &= z \quad \therefore \quad p_2 = l - z \\ \therefore f_2(l) &= \text{Max.} (-p_1 \log p_1 - p_2 \log p_2) \\ &= \text{Max.} [-z \log z - (l - z) \log (l - z)] \\ &= \text{Max.}_{0 \leq z \leq l} [-z \log z + f_1(l - z)] \quad \dots(2) \end{aligned}$$

For $n=3$, i.e., if l is divided into three parts p_1, p_2, p_3 i.e., $p_1 + p_2 + p_3 = l$

$$\text{let} \quad p_1 = z, \text{ then } p_2 + p_3 = l - z.$$

$\therefore$ The part $(l - z)$ is further divided into two parts p_2, p_3 whose maximum attainable product is $f_2(l - z)$

$$\begin{aligned} f_3(l) &= \text{Max.} \{-p_1 \log p_1 - p_2 \log p_2 - p_3 \log p_3\} \\ &= \text{Max.}_{0 \leq z \leq l} \{-z \log z + f_2(l - z)\} \quad \dots(3) \end{aligned}$$

Proceeding in this way, we get the recursive relation (functional equation) for $n = m$.

$$f_m(l) = \text{Max.}_{0 \leq z \leq l} \{-z \log z + f_{m-1}(l - z)\} \quad \dots(4)$$

Step 2: To solve the functional equations for determining the optimal policy

$$\text{From (1), we have} \quad f_1(l) = -l \log l.$$

$$\begin{aligned} \text{From (2),} \quad f_2(l) &= \text{Max.}_{0 \leq z \leq l} [-z \log z + f_1(l - z)] \\ &= \text{Max.}_{0 \leq z \leq l} [-z \log z - (l - z) \log (l - z)] \\ -z \log z - (l - z) \log (l - z) &\text{ is max. or mini. for that value of } z \text{ for which} \end{aligned}$$

$$\frac{d}{dz} [-z \log z - (l - z) \log (l - z)] = -\log z + \log (l - z) - 1 + 1 = 0$$

$$\text{or} \quad \log z = \log (l - z) \text{ or } z = l - z$$

$$\therefore z = \frac{1}{2}, 1 - z = 1 - \frac{1}{2} = \frac{1}{2}. \text{ When } z = \frac{1}{2},$$

$$\frac{d^2}{dz^2} [-z \log z - (1 - z) \log (1 - z)] = -\frac{1}{z} - \frac{1}{1 - z} = -4. \text{ (Negative), when } z = \frac{1}{2}$$

$$\therefore -z \log z - (1 - z) \log (1 - z) \text{ is max. when } z = \frac{1}{2}$$

$$\text{i.e., } f_2(1) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} = 2 \left(-\frac{1}{2} \log \frac{1}{2} \right) \quad \dots(5)$$

and the optimal policy for two parts is $p_1 = p_2 = \frac{1}{2}$.

Again from (3), we have

$$\begin{aligned} f_3(1) &= \text{Max.}_{0 \leq z \leq 1} \{-z \log z + f_2(1 - z)\} \\ &= \text{Max.}_{0 \leq z \leq 1} \left[-z \log z + 2 \left\{ -\left(\frac{1 - z}{2}\right) \log \left(\frac{1 - z}{2}\right) \right\} \right] \end{aligned}$$

Now $-z \log z - (1 - z) \log \left(\frac{1 - z}{2}\right)$ is max. when

$$\frac{d}{dz} \left\{ -z \log z - (1 - z) \log \left(\frac{1 - z}{2}\right) \right\} = -\log z + \log \left(\frac{1 - z}{2}\right) - 1 + 1 = 0$$

$$\text{or } \log z = \log \frac{(1 - z)}{2}$$

$$\text{or } z = \frac{1 - z}{2}$$

$$\therefore z = \frac{1}{3}, 1 - z = 1 - \frac{1}{3} = \frac{2}{3}$$

i.e., $\frac{2}{3}$ is to be divided into two parts p_2 and p_3

s.t. $-p_2 \log p_2 - p_3 \log p_3$ is max.

By the policy for two parts, $f_2(1 - z)$ i.e., $f_2\left(\frac{2}{3}\right)$ is attained when the two part are

$$p_2 = \frac{1}{2} \left(\frac{2}{3}\right) = \frac{1}{3}, p_3 = \frac{1}{2} \left(\frac{2}{3}\right) = \frac{1}{3}.$$

$$f_2(1 - z) = f_2\left(\frac{2}{3}\right) = 2 \cdot \left(-\frac{2/3}{2} \cdot \log \frac{2/3}{2} \right) = 2 \left(-\frac{1}{3} \log \frac{1}{3} \right)$$

$$\therefore f_3(1) = -\frac{1}{3} \log \frac{1}{3} + 2 \left(-\frac{1}{3} \log \frac{1}{3} \right) = 3 \left(-\frac{1}{3} \log \frac{1}{3} \right) \quad \dots(6)$$

and the optimal policy for three parts is $p_1 = p_2 = p_3 = \frac{1}{3}$.

Now calculations for stages 1, 2, 3 suggest that in general for n parts (stages) the optimal policy is $p_1 = p_2 = p_3 = \dots = \frac{1}{n}$

and
$$f_n(1) = n \left(-\frac{1}{n} \log \frac{1}{n} \right) \quad \dots(7)$$

This result can be proved by induction.

Assuming that the result (7) holds for $n = 2, 3, \dots, m$.

i.e.,
$$f_m(1) = m \left(-\frac{1}{m} \log \frac{1}{m} \right)$$

Now we shall prove that (7) also hold for $n = m + 1$.

From (4), we have

$$\begin{aligned} f_{m+1}(1) &= \text{Max.}_{0 \leq z \leq 1} \{-z \log z + f_m(1-z)\}. \\ &= \text{Max.}_{0 \leq z \leq 1} \left\{ -z \log z + (m) \left(-\frac{1-z}{m} \log \frac{1-z}{m} \right) \right\}. \end{aligned}$$

Now $u = -z \log z + (m) \left(-\frac{1-z}{m} \log \frac{1-z}{m} \right)$ is max. for that value of z for which

$$\frac{du}{dz} = -\log z + \log \left(\frac{1-z}{m} \right) - 1 + 1 = 0, \text{ or } \log z = \log \left(\frac{1-z}{m} \right), \text{ or } z = \log \frac{1-z}{m}$$

$$\therefore z = \frac{1}{1+m}.$$

Since
$$f_m(1-z) = f_m \left(1 - \frac{1}{1+m} \right)$$

$$= f_m \left(\frac{m}{1+m} \right) = m \left[-\frac{\left(\frac{m}{1+m} \right)}{m} \cdot \log \frac{m}{1+m} \right]$$

$$= m \left(-\frac{1}{1+m} \log \frac{1}{1+m} \right)$$

$$\therefore f_{m+1}(1) = -\frac{1}{1+m} \cdot \log \frac{1}{1+m} + m \left(-\frac{1}{1+m} \log \frac{1}{1+m} \right)$$

$$= (1+m) \cdot \left(-\frac{1}{1+m} \log \frac{1}{1+m} \right)$$

and the optimal policy is $p_1 = p_2 = \dots = p_{m+1} = \frac{1}{1+m}$

Hence, the required optimal policy is

$$p_1 = p_2 = \dots = p_n = \frac{1}{n}$$

Example 3: Find min. $Z = x_1 + x_2 + \dots + x_n$, when

$$x_1 \cdot x_2 \dots x_n = d, x_1, x_2, \dots, x_n \geq 0$$

Solution: Step 1: To develop the functional equations: Let $f_n(d)$ denote the minimum attainable sum $Z = x_1 + x_2 + \dots + x_n$ when d is factorized in n factors $x_1, x_2, \dots, x_n$ i.e., for n stage problem.

For $n=1$, i.e., if d is factorized into one factor only then $x_1 = d$. i.e., $x_1 \cdot x_2 = d$

$$\therefore f_1(d) = \text{Min. } z = \text{Min. } x_1 = d \quad \dots(1)$$

For $n=2$ i.e., if d is factorized into two factors x_1, x_2 i.e., $x_1 \cdot x_2 = d$

let $x_1 = y, x_2 = d/y$

$$\begin{aligned} f_2(d) &= \text{Min. } Z = \text{Min.}_{0 \leq y \leq d} \left(y + \frac{d}{y} \right) \\ &= \text{Min.}_{0 \leq y \leq d} \{ y + f_1(d/y) \} \end{aligned} \quad \dots(2)$$

For $n=3$, i.e., if d is factorized into three parts x_1, x_2, x_3 , i.e., $x_1 \cdot x_2 \cdot x_3 = d$

let $x_1 = y, x_2 \cdot x_3 = d/y$

i.e., the part d/y is further divided into two part whose minimum attainable product is $f_2(d/y)$.

$$\begin{aligned} \therefore f_3(d) &= \text{Min. } z = \text{Min. } \{ x_1 + x_2 + x_3 \} \\ &= \text{Min.}_{0 \leq y \leq d} \{ y + f_2(d/y) \} \end{aligned} \quad \dots(3)$$

Proceeding in this way, we get the following recursive relation (functional equation) for $n=m$.

$$f_m(d) = \text{Min.}_{0 \leq y \leq d} \{ y + f_{m-1}(d/y) \} \quad \dots(4)$$

Step 2: To solve the functional equations for determining optimal policy.

From (1), we have $f_1(d) = d$

From (2), we have $f_2(d) = \text{Min.}_{0 \leq y \leq d} \{ y + f_1(d/y) \}$

$$= \text{Min.}_{0 \leq y \leq d} \{ y + (d/y) \}$$

$y + d/y$ is max. or mini. for that value of y , for which

$$\frac{d}{dy} (y + d/y) = 1 - d/y^2 = 0 \quad \therefore \quad y = \pm d^{1/2}$$

$$\frac{d^2}{dy^2} (y + d/y) = 2d/y^3 = \text{positive for } y = d^{1/2}$$

$$\therefore \quad f_2(d) = d^{1/2} + \frac{d}{d^{1/2}} = 2d^{1/2} \quad \dots(5)$$

and optimal policy is $\left(d^{1/2}, \frac{d}{d^{1/2}}\right)$ i.e., $(d^{1/2}, d^{1/2})$.

$$\begin{aligned} \text{From (3), we have} \quad f_3(d) &= \text{Min.}_{0 \leq y \leq d} \{y + f_2(d/y)\} \\ &= \text{Min.}_{0 \leq y \leq d} \{y + 2(d/y)^{1/2}\} \quad [\text{from (5)}] \end{aligned}$$

Now $y + 2(d/y)^{1/2}$ is max. or mini. for that value of y , for which

$$\frac{d}{dy} \{y + 2(d/y)^{1/2}\} = 1 - \frac{d^{1/2}}{y^{3/2}} = 0$$

$$\therefore \quad y = d^{1/3} \quad \text{so that } d/y = d^{2/3}$$

It can be verified that $y = d^{1/3}$ is a point of minima.

$$\therefore \quad f_3(d) = d^{1/3} + 2(d/d^{1/3})^{1/2} = 3d^{1/3} \quad \dots(6)$$

and optimal policy is $(d^{1/3}, (d^{2/3})^{1/2}, (d^{2/3})^{1/2})$ i.e., $(d^{1/3}, d^{1/3}, d^{1/3})$

Proceeding in this way, we get in general

$$f_n(d) = nd^{1/n}$$

$$\text{and optimal policy is } (d^{1/n}, d^{1/n}, \dots, d^{1/n}). \quad \dots(7)$$

The result can be proved by induction.

Assuming that result (7) hold for $n = 2, 3, \dots, m$.

$$\text{i.e.,} \quad f_m(d) = md^{1/m}$$

It may be verified for $n = m + 1$

$$\text{i.e.,} \quad f_{m+1}(d) = (m+1)d^{1/(m+1)}$$

and optimal policy is $(d^{1/(m+1)}, d^{1/(m+1)}, \dots, d^{1/(m+1)})$.

Hence, the optimal solution is

$$\text{Min. } Z = nd^{1/n},$$

and

$$x_1 = x_2 = \dots = x_n = d^{1/n}.$$

Example 4: Use the principle of optimality to find the maximum value of $b_1x_1 + b_2x_2 + \dots + b_nx_n$ when

$$x_1 + x_2 + \dots + x_n = c,$$

$$x_i \geq 0, \quad i = 1, 2, \dots, n.$$

Solution: **Step 1: To develop the functional equations:** Let $f_n(c)$ denote the maximum attainable sum $b_1x_1 + b_2x_2 + \dots + b_nx_n$ when the positive quantity c is divided into n parts $x_1, x_2, \dots, x_n$.

For $n = 1$, i.e., if c is divided into one part, then $x_1 = c$.

$$\therefore f_1(c) = \text{Max. } (b_1x_1) = b_1c \quad \dots(1)$$

For $n = 2$, i.e., if c is divided into two parts x_1, x_2

$$\text{s.t.} \quad x_1 + x_2 = c, \quad \text{let } x_2 = y \quad \therefore x_1 = c - y$$

$$\begin{aligned} f_2(c) &= \text{Max. } (b_1x_1 + b_2x_2) \\ &= \text{Max.}_{0 \leq y \leq c} \{b_2y + b_1(c - y)\} \\ &= \text{Max.}_{0 \leq y \leq c} \{b_2y + f_1(c - y)\} \quad \dots(2) \end{aligned}$$

Proceeding in this way if y is the m -th part of c , then

$$\begin{aligned} f_m(c) &= \text{Max.}_{0 \leq y \leq c} \{b_my + f_{m-1}(c - y)\} \quad \dots(3) \\ m &= 1, 2, \dots, n. \end{aligned}$$

Step 2: To solve the functional equations for determining the optimal policy

$$\begin{aligned} \text{From (2),} \quad f_2(c) &= \text{Max.}_{0 \leq y \leq c} \{b_2y + f_1(c - y)\} \\ &= \text{Max.}_{0 \leq y \leq c} \{b_2y + b_1(c - y)\} \\ &= \text{Max.}_{0 \leq y \leq c} \{(b_2 - b_1)y + b_1c\} \end{aligned}$$

If $b_2 - b_1$ is positive, then this is maximum for $y = c$, otherwise will be minimum.

$$\therefore f_2(c) = (b_2 - b_1)c + b_1c = b_2c$$

and optimal policy is

$$x_1 = c - c = 0, x_2 = c.$$

For $m = 3$,

$$\begin{aligned}
 b_3(c) &= \text{Max.}_{0 \leq y \leq c} \{b_3 y + f_2(c - y)\} \\
 &= \text{Max.}_{0 \leq y \leq c} \{b_3 y + b_2(c - y)\} \\
 &= \text{Max.}_{0 \leq y \leq c} \{(b_3 - b_2)y + b_2 c\}
 \end{aligned}$$

If $b_3 - b_2$ is positive, then this is maximum for $y = c$, otherwise it will be minimum

$$\therefore f_3(c) = (b_3 - b_2)c + b_2 c = b_3 c$$

and optimal policy is $x_1 = 0 = x_2, x_3 = c$

Proceeding in this way, we get

$b_n(c) = b_n c$ and the optimal policy for n parts is

$$x_1 = 0 = x_2 = \dots = x_{n-1}, x_n = c.$$

Example 5: By dynamic programming technique, solve the problem.

$$\text{Minimize } z = x_1^2 + x_2^2 + x_3^2$$

subject to the constraint

$$x_1 + x_2 + x_3 \geq 15, x_1, x_2, x_3 \geq 0.$$

Solution: Step 1: To develop the functional equations: It is a three stage problem which can be defined as follows:

$$s_3 = x_1 + x_2 + x_3 \geq 15$$

$$s_2 = x_1 + x_2 = s_3 - x_3$$

and

$$s_1 = x_1 = s_2 - x_2$$

Let $f_i(s_i)$ be the minimum value of z at the i -th stage where

$$s_i = x_1 + x_2 + \dots + x_i, (i = 1, 2, 3)$$

Thus, the functional equations are

$$f_1(s_1) = \text{Min.}_{0 \leq x_1 \leq s_1} x_1^2 = (s_2 - x_2)^2 \quad \dots(1)$$

$$\begin{aligned}
 f_2(s_2) &= \text{Min.}_{0 \leq x_2 \leq s_2} (x_1^2 + x_2^2) \\
 &= \text{Min.}_{0 \leq x_2 \leq s_2} \{x_2^2 + f_1(s_1)\} \quad \dots(2)
 \end{aligned}$$

and

$$\begin{aligned}
 f_3(s_3) &= \text{Min.}_{0 \leq x_3 \leq s_3} \{x_1^2 + x_2^2 + x_3^2\} \\
 &= \text{Min.}_{0 \leq x_3 \leq s_3} \{x_3^2 + f_2(s_2)\} \quad \dots(3)
 \end{aligned}$$

Step 2: To solve the functional equations

From (1) $f_1(s_1) = (s_2 - x_2)^2$

From (2)
$$\begin{aligned} f_2(s_2) &= \text{Min.}_{0 \leq x_2 \leq s_2} \{x_2^2 + f_1(s_1)\} \\ &= \text{Min.}_{0 \leq x_2 \leq s_2} \{x_2^2 + (s_2 - x_2)^2\} \\ &= \text{Min.}_{0 \leq x_2 \leq s_2} \{2x_2^2 - 2s_2x_2 + s_2^2\} \end{aligned}$$

Now $u = 2x_2^2 - 2s_2x_2 + s_2^2$ is max. or mini, when

$$\frac{du}{dx_2} = \frac{d}{dx_2} (2x_2^2 - 2s_2x_2 + s_2^2) = 4x_2 - 2s_2 = 0 \quad \therefore x_2 = \frac{1}{2}s_2$$

$\therefore \frac{d^2u}{dx_2^2} = 4$ is positive when $x_2 = \frac{1}{2}s_2$, $\therefore u$ is minimum when $x_2 = \frac{1}{2}s_2$,

$\therefore f_2(s_2) = \left(\frac{1}{2}s_2\right)^2 + \left(s_2 - \frac{1}{2}s_2\right)^2 = \frac{1}{2}s_2^2$

Now $f_3(s_3) = \text{Min.}_{0 \leq x_3 \leq s_3} \{x_3^2 + f_2(s_2)\}$

$$= \text{Min.}_{0 \leq x_3 \leq s_3} \{x_3^2 + \frac{1}{2}(s_3 - x_3)^2\} \quad \therefore f_2(s_2) = \frac{1}{2}s_2^2 = \frac{1}{2}(s_3 - x_3)^2.$$

Now $v = x_3^2 + \frac{1}{2}(s_3 - x_3)^2$ is max. or mini., when

$$\frac{dv}{dx_3} = 2x_3 - 2 \cdot \frac{1}{2} \cdot (s_3 - x_3) = 3x_3 - s_3 = 0 \quad \therefore x_3 = \frac{1}{3}s_3.$$

$\therefore \frac{d^2v}{dx_3^2} = 3$, which is positive.

$\therefore v$ is minimum when $x_3 = \frac{1}{3}s_3$.

$\therefore f_3(s_3) = \left(\frac{1}{3}s_3\right)^2 + \frac{1}{2}\left(s_3 - \frac{1}{3}s_3\right)^2 = \frac{1}{3}s_3^2.$

But $s_3 \geq 15$, i.e., minimum $s_3 = 15$.

$\therefore z$ is minimum when $x_3 = \frac{1}{3}s_3 = 5$,

$$x_2 = \frac{1}{2}s_2 = \frac{1}{2}(s_3 - x_3) = 5, \quad x_1 = 5.$$

Hence, the optimal policy is

$$x_1 = x_2 = x_3 = 5 \quad \text{and} \quad \text{mini. } z = 75.$$

Exercise 11.1

1. State Bellman's principle of optimality in dynamic programming.

2. Solve by dynamic programming method:

$$\begin{aligned} \text{maximize} \quad & Z = x_1 x_2 x_3 \\ \text{subject to} \quad & x_1 + x_2 + x_3 = 5 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

3. Use dynamic programming to show that

$$\sum_{i=1}^n p_i \log p_i \text{ subject to } \sum_{i=1}^n p_i = 1 \text{ is minimum.}$$

$$\text{when } p_1 = p_2 = \dots = p_n = 1/n.$$

4. Solve the following problem by dynamic programming

$$\text{Min.} \quad Z = \sum_{i=1}^n y_i^2$$

$$\text{s.t.} \quad y_1 \cdot y_2 \cdot \dots \cdot y_n = c, \quad y_1, y_2, \dots, y_n \geq 0$$

5. Find the maximum value of

$$Z = x_1^2 + 2x_2^2 + 4x_3$$

$$\text{s.t.} \quad x_1 + 2x_2 + x_3 \leq 8, \quad x_1, x_2, x_3 \geq 0$$

Answers 11.1

2.	$x_1 = 5/3 = x_2 = x_3$, max. $Z = 125/27$
4.	$(c^{1/n}, c^{1/n}, \dots, c^{1/n})$, min. $Z = nc^{2/n}$
5.	$x_1 = 8, x_2 = 0, x_3 = 0$, max. $Z = 64$

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

- The technique of dynamic programming was developed by:
(a) Rechard Bellman (b) George Dantzig
(c) J.V. Neumann (d) None of these.
 - The technique of dynamic programming was developed in:
(a) 1900 (b) 1920
(c) 1950 (d) 1970.
 - A dynamic programming problem can be decomposed into a sequence of smaller subproblems, called:
(a) States (b) Stages
(c) Both (a) and (b) (d) None of (a) or (b).
 - If $x_1 + x_2 + x_3 = 9$, $x_1, x_2, x_3 \geq 0$, then $Z = x_1 x_2 x_3$ is maximum when:
(a) $x_1 = 1, x_2 = 4, x_3 = 4$ (b) $x_1 = 4 = x_2, x_3 = 1$
(c) $x_1 = 3 = x_2 = x_3$ (d) None of these.
 - If $x_1 \cdot x_2 \cdot x_3 = 5$, $x_1, x_2, x_3 \geq 0$, then $Z = x_1 + x_2 + x_3$ is minimum when:
(a) $x_1 = x_2 = x_3 = 5^{1/3}$ (b) $x_1 = 1, x_2 = x_3 = \sqrt{5}$
(c) $x_1 = x_3 = \sqrt{5}, x_2 = 1$ (d) None of these.
 - Using dynamic programming, the maximum value of $Z = y_1 \cdot y_2 \cdot y_3$,
s.t. $y_1 + y_2 + y_3 = 5, y_1, y_2, y_3 \geq 0$ is:
(a) 125/27 (b) 125
(c) 125/9 (d) None of these.
 - The return function in a dynamic programming depends upto:
(a) The state of the problem (b) The stage of the problem
(c) The optimal decision (d) None of these.
 - The relationship between stages of a dynamic programming problem is called:
(a) Random variables (b) Note
(c) State (d) Transformation.
 - When classical method is used in solving a dynamic programming problem, the objective may be linear or non-linear but the constraints must be:
(a) Linear (b) Non-linear
(c) Linear or non-linear (d) None of these.
-

10. Dynamic programming deals with:
 - (a) Single stage decision-making problems
 - (b) Multistage decision-making problems
 - (c) Problems with fix the levels of different variables so as to max. profit or min. cost.
 - (d) Time-independent decision making problem.
11. Dynamic programming problem:
 - (a) Can be solved by simplex method
 - (b) Can be solved by Monte Carlo method
 - (c) Can not be dealt with non-linear constraints
 - (d) Is solved starting from the initial stage to the next till the final stage is finished.

Fill in the Blank

1. Bellman's principle of optimality states that, an optimal policy has the property that whatever be the initial state and initial decisions, the remaining decisions must constitute an for the state resulting from the first decision.
2. A problem which does not satisfy the principle of optimality be solved by dynamic programming.
3. A problem in which the decisions have to be made at successive stages is called a decision problem.
4. Bellman's principle of optimality can be used to solve a decision problem.
5. Each stage in a dynamic programming problem has a certain number of associated with it.

True/False

1. Dynamic programming is not itself a method of optimization, but rather a method for decomposing one problem into a set of smaller problems.
 2. A dynamic programming problem can be divided into a number of stages where decision must be made at each stage.
 3. A L.P.P. in n decision variables and m constraints can be converted into n stage dynamic programming problem with m states.
 4. A L.P.P. can not be solved by dynamic programming technique.
 5. In dynamic programming, there is no set procedure as in linear programming that can be used to solve all problems.
-

Answers

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1.	(a)	2.	(c)	3.	(b)
4.	(c)	5.	(a)	6.	(a)
7.	(c)	8.	(d)	9.	(a)
10.	(b)	11.	(d)		

Fill in the Blank

1.	optimal policy	2.	cannot	3.	multistage
4.	multistage	5.	states		

True/False

1.	T	2.	T	3.	T
4.	F	5.	T		

○○○



Job Sequencing

12.1 Introduction

The selection of an appropriate order in which to service waiting customers (or jobs) is called sequencing. Here, the effectiveness measure which may be time, cost, mileage etc. is a function of the order or sequence in which a series of tasks are performed (or processed). There are the problems in which we have n tasks to be processed on some or all of m different machines in which the total effectiveness depends on the order of processing. There are $(n!)^m$ possible sequences for the problem. It is very difficult to find the effectiveness measure of all the sequences which are $(n!)^m$ (a large number) and select one most suitable sequence, which optimizes the effectiveness measure. So, here in this chapter, we shall discuss methods of dealing with such sequencing problems.

12.2 Sequencing Problem

A general sequencing problem may be defined as follows:

Let there be n -jobs, which have to be processed one at a time at each of m machines $A, B, C, \dots$. The order of the machines for each job, in which it should go to the machines is given. The time required by the jobs on each of the machines is also given. Then, the problem is to find the sequence, out of $(n!)^m$ sequences, which optimizes (minimizes) the total time elapsed from the start of the first job to the completion of the last job.

Mathematically: If $A_i, B_i, \dots$, etc. are the times for job i on machines $A, B, C, \dots$ and T is the total time elapsed from the start of the first job to the completion of the last job, then the problem of sequencing is to find for each machine a sequence, $(i_1, i_2, \dots, i_n)$, of jobs $i (i=1, 2, \dots, n)$, where $(i_1, i_2, \dots, i_n)$ is a permutation of the integers $(1, 2, \dots, n)$, which minimize T .

So far, a very limited work is done on this topic of sequencing. All types of sequencing problems cannot be solved satisfactorily. Here, we shall discuss the following three special cases of sequencing problems:

1. n jobs to be processed on two machines A and B , all jobs to be processed in the order AB .
2. n jobs to be processed on three machine A, B and C . All jobs to be processed in the order ABC .
3. n jobs to be processed on m machines.
4. Two jobs to be processed on m machines. Each job to be processed through the machines in a prescribed order which is not necessarily the same for both jobs.

12.3 General Assumptions

We shall solve the sequencing problems under the following assumptions:

1. The processing times of jobs on machines are independent of their processing order.
2. Each job, once started on a machine, must be performed upto the completion on that machine.
3. No machine may process more than one job at a time.
4. The time taken by the jobs in going from one machine to another is negligible.
5. There is only one machine of each type.
6. A job is processed as soon as possible subject only to ordering requirements.
7. All jobs are known and ready to start processing.

12.4 Sequencing Decision Problem for n -jobs on two Machines

Johnson's Method: Here, we consider the problem of processing n -jobs $1, 2, \dots, n$ on two machines A and B , under the following assumptions:

1. Each job is processed in the order AB .
2. A_i = Processing time of i -th job on machine A . ($i = 1, 2, \dots, n$)
3. B_i = Processing time of i -th job on machine B . ($i = 1, 2, \dots, n$)

The problem is to find the sequence of jobs to be performed on two machines so that, the total time (T) elapsed from the start of the first job to the completion of the last job is minimized.

The procedure for the solution of the above problem was developed by **Johnson and Bellman**. The method is based on minimizing the idle time for second machine. The **Johnson's procedure for determining an optimal sequence** is as follows:

1. Select the smallest processing time in the lists $A_1, A_2, \dots, A_n$ (processing times of jobs on machine A) and $B_1, B_2, \dots, B_n$ (processing times of jobs on machine B). If there is a tie then either of these smallest processing times may be chosen.
2. If the smallest processing time is A_r (in the list $A_1, A_2, \dots, A_n$) then do the r -th job first. If it is B_s (in the list $B_1, B_2, \dots, B_n$) then do the s -th job last. If the smallest processing times A_i and B_i for the two machines are equal and are for the same job i , then do this i -th job in the last.
3. Delete the times of the job already assigned from the two lists of A_i 's and B_i 's i.e., if r -th job is assigned previously, then delete A_r and B_r both and if s -th job is assigned previously then delete A_s and B_s both.
4. After assigning one job, we are left with $(n - 1)$ jobs. Repeat step 1 to 3 on the processing times of the remaining $(n - 1)$ jobs.
5. Continue in this way until all jobs have been ordered. In this way, we get an optimal sequence of jobs. Finally, the minimum time T elapsed can be found by forming a chart.

For clear understanding of the procedure see Example 1 to 3.

ILLUSTRATIVE EXAMPLES

Example 1: Six jobs go first over machine I and then over II. The order of the completion of jobs has no significance. The following table gives the machine times in hours for six jobs on the two machines:

Job No. i	1	2	3	4	5	6
Time on machine I (A_i)	5	9	4	7	8	6
Time on machine II (B_i)	7	4	8	3	9	5

Find the sequence of the jobs that minimizes the total elapsed time to complete the jobs. Find the minimum time by using Gantt's chart or by other method.

Solution: Here, the ordering of the jobs on machine is I, II. The minimum time in the given table is 3 which is B_4 [i.e., the time for job 4 on machine II]. Therefore, we shall do the job 4 in the last and list the element as follows:

					4
--	--	--	--	--	---

After assigning 4th job, we are left with 5 jobs with their processing times as follows:

Job No. i	1	2	3	5	6
Time on machine I (A_i)	5	9	4*	8	6
Time on machine II (B_i)	7	4	8	9	5

Now, the minimum time in this table is 4 which is A_3 . [i.e., the time for job 3 on machine I]. Therefore, we shall do job 3 first and list the elements as follows:

3					4
---	--	--	--	--	---

After assigning 3rd job also, we are left with 4 jobs with their processing times as follows:

Job No. i	1	2	5	6
Time on machine I [A_i]	5	9	8	6
Time on machine II [B_i]	7	4*	9	5

Again, the minimum time in this table is 4 which in B_2 (i.e., time for job 2 on machine II). Therefore, we shall do job 2 just before the last job 4 and list the elements as follows:

3				2	4
---	--	--	--	---	---

Proceeding in this way, we get the following optimal sequence of jobs.

3	1	5	6	2	4
---	---	---	---	---	---

To find the minimum elapsed time: The minimum elapsed time from the start of the first job to the completion of the last job in the above optimal sequencing is calculated as shown in the following table:

Jobs	Machine I		Machine II		Idle time of machine II
	Time in	Time out	Time in	Time out	
3	0	4	4	12	4
1	4	9	12	19	
5	9	17	19	28	
6	17	23	28	33	
2	23	32	33	37	
4	32	39	39	42*	2

From the above table, it is clear that the total elapsed time is 42 hours and the idle time of machine I is $42 - 39 = 3$ hours and that for machine II is 6 hours.

Example 2: We have five jobs, each of which must go through two machines A and B in the order AB. Processing times in hours are given in the table below:

Jobs	1	2	3	4	5
Machine A (A_i)	5	1	9	3	10
Machine B (B_i)	2	6	7	8	4

Determine the sequence for the five jobs that will minimize the elapsed time T .

Solution: The minimum time in the table is 1 which is A_2 . Therefore, we must do the job 2 first. We list the sequencing in table. After assigning jobs 2, we are left with four jobs 1, 3, 4, 5. The minimum time for these four jobs in 2 which is B_1 . Therefore, we must do the job 1 last. Proceeding in this way, we get the following optimal sequencing.

2	4	3	5	1
---	---	---	---	---

To find minimum elapsed time: To find minimum elapsed time we construct the following table:

Jobs	Machine A		Machine B		Idle time of machine B
	Time in	Time out	Time in	Time out	
2	0	1	1	7	1
4	1	4	7	15	
3	4	13	15	22	
5	13	28	23	27	1
1	23	28	28	30*	1

Thus, the minimum elapsed time is 30 hours and the idle time for machine A is $30 - 28 = 2$ hours and that for machine B is 3 hours.

Example 3: Find the sequence that minimizes the total elapsed time required to complete the following tasks on two machines:

Tasks	A	B	C	D	E	F	G	H	I
Machine I	2	5	4	9	6	8	7	5	4
Machine II	6	8	7	4	3	9	3	8	11

Solution: The minimum time in the table is 2 which is for job A on machine I.

$\therefore$ We must do the job A first. After assignment job A, we are left with the remaining 8 jobs with their processing times. Now the minimum of the processing time for the remaining jobs is 3 which is for job E and G both on machine II.

Therefore, we must do the job E or G (either of them) in the last. Proceeding in this way, we get the following optimal sequencing:

1.

A	C	I	B	H	F	D	G	E
-----	-----	-----	-----	-----	-----	-----	-----	-----
2.

A	I	C	H	B	F	D	G	E
-----	-----	-----	-----	-----	-----	-----	-----	-----
3.

A	I	C	B	H	F	D	G	E
-----	-----	-----	-----	-----	-----	-----	-----	-----
4.

A	C	I	H	B	F	D	G	E
-----	-----	-----	-----	-----	-----	-----	-----	-----
5.

A	C	I	B	H	F	D	E	G
-----	-----	-----	-----	-----	-----	-----	-----	-----
6.

A	I	C	H	B	F	D	E	G
-----	-----	-----	-----	-----	-----	-----	-----	-----
7.

A	C	I	H	B	F	D	E	G
-----	-----	-----	-----	-----	-----	-----	-----	-----
8.

A	I	C	B	H	F	D	E	G
-----	-----	-----	-----	-----	-----	-----	-----	-----

It may be seen that the total time elapsed for all sequencing are equal.

Exercise 12.1

Find the sequence that minimizes the total elapsed time required to complete the following jobs in Q.N. 1 and 2:

1.	Jobs No.	1	2	3	4	5	6
	Processing time $\begin{cases} A_i \\ B_i \end{cases}$	$\begin{matrix} 2 \\ 6 \end{matrix}$	$\begin{matrix} 5 \\ 8 \end{matrix}$	$\begin{matrix} 4 \\ 1 \end{matrix}$	$\begin{matrix} 3 \\ 2 \end{matrix}$	$\begin{matrix} 2 \\ 3 \end{matrix}$	$\begin{matrix} 1 \\ 5 \end{matrix}$

2.	Jobs No.	1	2	3	4	5
	Machine A	10	2	18	6	20
	Machine B	4	12	14	16	8

3. Seven jobs, each of which has to go through two machines M_1 and M_2 , in order M_1 M_2 take time on the machines as follows. Find, in which order the jobs should be performed to minimise the time.

Jobs	1	2	3	4	5	6	7
M_1	3	12	15	6	10	11	9
M_2	8	10	10	6	12	1	3

4. Determine the optimal sequence of jobs that minimizes the total elapsed time to complete the following jobs in the order *AB*:

Jobs	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
Machine <i>A</i>	6	2	10	4	11
Machine <i>B</i>	3	7	8	9	5

5. Six jobs go over two machines I and II in that order. The order of the completion of the jobs has no significance. From the data given below, find the sequence that minimizes the total elapsed time and also find that minimum time.

Jobs		1	2	3	4	5	6
Time in hours	Machine <i>I</i>	4	8	3	6	7	5
	Machine <i>II</i>	6	3	7	2	8	4

6. Six jobs are performed first over Machine I and then over Machine II. The order of the completion of the jobs has no significance. Find the sequence of the jobs that minimizes the total elapsed time. The time for each job on each machine is given below:

Hours

Jobs No.	1	2	3	4	5	6
Machine <i>I</i>	8	12	7	10	11	9
Machine <i>II</i>	10	7	11	6	12	8

7. The following table given the machine times (in hours) for 5 jobs to be processed on two-different machines.

Jobs	1	2	3	4	5
Machine <i>A</i>	3	7	4	5	7
Machine <i>B</i>	6	2	7	3	4

Passing time is not allowed. Find the optimal sequence in which jobs should be processed.

8. A book binder has one printing press, one binding machine and the manuscripts of a number of different books. The times required to perform the printing and binding operations for each book are known. We wish to determine the order in which books should be processed on the machines, in order to minimize the total time required.

Books	1	2	3	4	5	6
Printing time (hrs.)	30	120	50	20	90	110
Binding time (hrs.)	80	100	90	60	30	10

9. A ready-made garment manufacture has to process 6 items through two stages of production viz, cutting and sewing. The time taken for each of these items at the different stages is given below in appropriate units:

Items	1	2	3	4	5	6
Cutting time	30	120	50	20	90	110
Sewing time	80	100	90	60	30	10

Find an order in which, these items are to be processed through these stages so as to minimise the total processing time. Also calculate total elapsed time and idle times.

10. A company plant to fill six positions. Since, the position are known to very considerably with respect to skill and responsibility, different types of aptitude tests and interviews are required for each. While the aptitude test are conducted by people from the clerical positions, the job interviews are hold by the personnel from the management cadre. The job interviews immediately follows the aptitude test. The time required (in minutes) by each of the positions is given here.

Positions	P_1	P_2	P_3	P_4	P_5	P_6
Aptitude test	140	180	150	200	170	100
Job interview	70	120	110	80	100	90

It is desired to minimize the waiting time of the management personnel. In what order is the position filling to be handled? Also, calculate the idle time of jobs interview personnels.

11. A ready-made garment manufacturer has to process seven items through two stages of production, *i.e.*, cutting and sewing. The time taken for each of these item at the different stages are given below in appropriate units:

Items No.	Processing Time Cutting	Processing Time Sewing
1	5	2
2	7	6
3	3	7
4	4	5
5	6	9
6	7	5
7	12	8

Find an order in which these are to be processed through these stages, so as to minimize the total processing time.

Answers 12.1

1.	6 - 1 - 5 - 2 - 4 - 3, Min. Time = 26 hours.
2.	2 - 4 - 3 - 5 - 1, Min. Time = 60 hours.
3.	1 - 4 - 5 - 3 - 2 - 7 - 6, 1 - 5 - 3 - 2 - 4 - 7 - 6, 1 - 4 - 5 - 2 - 3 - 7 - 6 and 1 - 5 - 2 - 3 - 4 - 7 - 6, Min. Time = 67 hours.
4.	B - D - C - E - A, Min. Time = 36 hours. Idle time, for A - 3 hours, for B - 4 hours.
5.	3 - 1 - 5 - 6 - 2 - 4, Min. Time = 35 hours.
6.	3 - 1 - 5 - 6 - 2 - 4, Min. Time = 57 hours.
7.	1 - 3 - 5 - 4 - 2, Min. Time = 28 hours.
8.	4 - 1 - 3 - 2 - 5 - 6, Min. Time = 430 hours.
9.	4 - 1 - 3 - 2 - 5 - 6, Min. Time = 430 hours. Idle time, for A - 10 units, for B - 60 units.
10.	$P_2 - P_3 - P_5 - P_6 - P_4 - P_1$, Min. Time = 440 minutes.
11.	3 - 4 - 5 - 7 - 2 - 6 - 1, Min. Time = 46 hours.

125 Sequencing Decision Problem for n -Jobs on Three Machines

Modified Johnson's Method: Now we consider the problem of processing n jobs 1, 2, ..., n , on three machines A, B, C, under the following assumptions:

- Each job is performed in the order ABC.
- A_i = Processing time of i -th job on machine A, $i = 1, 2, \dots, n$.
- B_i = Processing time of i -th job on machine B, $i = 1, 2, \dots, n$.
- C_i = Processing time of i -th job on machine C, $i = 1, 2, \dots, n$.

The problem is, to find the sequence of jobs to be performed on three machines, so that the total time T elapsed from the start of the first job to the completion of the last job is minimized.

No method is yet available for the solution of the above problem. The method of the previous article 12.4 can be extended to the cases in which either or both of the following conditions hold:

- (i) The smallest processing time for machine $A \geq$ the largest processing time for machine B .
- (ii) The smallest processing time for machine $C \geq$ the largest processing time for machine B .

The method is to replace the problem with an equivalent problem involving n jobs on two machines. If G and H denote the two machines (fictitious) then the processing times G_i and H_i of i -th [$i = 1, 2, \dots, n$] job on these two machines are given by

$$G_i = A_i + B_i$$

and

$$H_i = B_i + C_i$$

Now find the optimal sequence of jobs in the order GH on these two machines G and H by the method of article 12.4. These resulting optimal sequence on two machines G and H will also be the optimal sequence on three machines A, B, C .

For clear understanding of the procedure see Examples 1 and 5. on pages 477 and 478.

12.6 Sequencing Decision Problem for n Jobs on m Machines

Now we consider the problem of processing n jobs 1, 2, ..., n on m machines $M_1, \dots, M_m$ under the following assumptions.

1. Each job is performed in the order $M_1, M_2, \dots, M_m$.
2. M_{ij} = Processing of i -th job on j -th machine M_j .

$$i = 1, 2, \dots, n, \quad j = 1, 2, \dots, m$$

The problem is to find sequence of jobs to be performed on the machines in the order $M_1, M_2, \dots, M_m$, so that the total time T elapsed from the start of the first job to the completion of the last job is minimized. There is no general method available for the solution of the above problem. The method of article 12.4 can be extended to the cases in which either or both of the following conditions hold:

- (i) $\min_i M_{i1} \geq \max_i M_{ij}$, for $j = 2, 3, \dots, (m-1)$
- (ii) $\min_i M_{im} \geq \max_i M_{ij}$, for $j = 2, 3, \dots, (m-1)$

The method is to replace the problem with an equivalent problem involving n jobs on two machines. If G and H denote the two machines (fictitious) then the processing times M_{iG} and M_{iH} of i -th [$i = 1, 2, \dots, n$] job on these two machines are given by:

$$M_{iG} = M_{i1} + M_{i2} + \dots + M_{i(m-1)}$$

$$M_{iH} = M_{i2} + M_{i3} + \dots + M_{im}$$

Now, find the optimal sequence of n jobs in the order GH on the two fictitious machines G and H by the method of article 12.4. The resulting optimal sequence on two machines G and H will also be the optimal sequence on m machines $M_1, M_2, \dots, M_m$.

For clear understanding of the procedure see Example 3 on pages 479.

Note: If in addition to the conditions [i] and [ii], if

$$M_{i2} + M_{i3} + \dots + M_{i(m-1)} = c \text{ (fixed positive constant)}$$

for all $i = 1, 2, \dots, n$ then determine the optimal sequence of n jobs on two machines M_1 and M_m in the order $M_1 M_m$ by the method of article 1.

ILLUSTRATIVE EXAMPLES

Example 1: We have five jobs, each of which must go through the machines A, B and C in the order ABC .

Processing times in hours

Job no. i	1	2	3	4	5
Machine A (A_i)	5	7	6	9	5
Machine B (B_i)	2	1	4	5	3
Machine C (C_i)	3	7	5	6	7

Determine a sequence for the jobs, that will minimize the total elapsed time.

Solution: Here, $\text{Mini. } A_i = 5$, $\text{Max. } B_i = 5$, $\text{Mini. } C_i = 3$

Since, $\text{Mini. } A_i \geq \text{Max. } B_i$.

Therefore, we may consider two fictitious machines G and H .

The processing times on these fictitious machines are given in table:

Job i	Processing times	
	$G_i = A_i + B_i$	$H_i = B_i + C_i$
1	7	5
2	8	8
3	10	9
4	14	11
5	8	10

The minimum time in this table is 5 which is H_1 (time for job 1 on machine H). Therefore, the job 1 will be done last. Proceeding in usual manner, we get the following optimal sequencing:

2	5	4	3	1
5	4	3	2	1
5	2	4	3	1

To find minimum elapsed time: We shall calculate minimum elapsed time for the first optimal sequencing in the following table:

Jobs	Machine A		Machine B		Machine C		Idle time of Machine B	Idle time of Machine C
	Time in	Time out	Time in	Time out	Time in	Time out		
2	0	7	7	8	8	15	7	8
5	7	12	12	15	15	22	4	–
4	12	21	21	26	26	32	6	4
3	21	27	27	31	32	37	1	
1	27	32	32	34	37	40*	1 + 6	

Thus, the minimum elapsed time is 40 hours. Students may verify the time for other alternative sequencing also. Idle times for the machines A, B, C and 8, 25 and 12 hours respectively.

Example 2: We have five jobs, each of which must go through the machines A, B, C in the order ABC. Processing times are:

Jobs	A	B	C
1	4	5	8
2	9	6	10
3	8	2	6
4	6	3	7
5	5	4	11

Determine a sequence for the five jobs that will minimize the elapsed time T .

Solution: Here, Mini. $A_i = 4$, Max. $B_i = 6$, Mini. $C_i = 6$

Since, Max. $B_i \leq$ Mini. C_i

∴ We may consider two fictitious machines G and H whose processing times are given in the following table:

Jobs i	$G_i = A_i + B_i$	$H_i = B_i + C_i$
1	9	13
2	15	16
3	10	8
4	9	10
5	9	15

Proceeding in usual manner, we get the following optimal sequencings:

1	4	5	2	3
4	1	5	2	3
5	1	4	2	3

1	5	4	2	3
4	5	1	2	3
5	4	1	2	3

Example 3: Solve the following sequencing problem when passing time is not allowed. Processing time (in hours) are given below:

Machines					
Items	A	B	C	D	E
I	9	7	4	5	11
II	8	8	6	7	12
III	7	6	7	8	10
IV	10	5	5	4	8

Solution: Here, $\min_i M_{i1} = 7$, $\min_i M_{i5} = 8$

$\max_i M_{ij} = 8$, for $j = 2, 3, 4$

The condition

$\min_i M_{i5} \geq \max_i M_{ij}$ for $j = 2, 3, 4$

is satisfied. Therefore, we can consider two fictitious machines G and H . The processing time on these two fictitious machines G and H are given in table.

<i>Jobs</i>	$M_{iG} = \sum_{j=1}^4 M_{ij}$	$M_{iH} = \sum_{j=2}^5 M_{ij}$
<i>I</i>	25	27
<i>II</i>	29	33
<i>III</i>	28	31
<i>IV</i>	24	22

Proceeding in usual manner, we get the following optimal sequence,

<i>I</i>	<i>III</i>	<i>II</i>	<i>IV</i>
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Exercise 12.2

- There are five jobs, each of which must go through the machines A , B and C in the order ABC . Processing times are:

Jobs	<i>A</i>	<i>B</i>	<i>C</i>
1	4	5	8
2	9	6	10
3	8	2	6
4	6	3	7
5	5	4	11

Determine a sequence for the five jobs that will minimize the elapsed time T .

- We have five jobs, each of which must go through machines A , B and C in the order ABC . Processing times are given in the following table:

Jobs	1	2	3	4	5
Machine A (A_i)	8	10	6	7	11
Machine B (B_i)	5	6	2	3	4
Machine C (C_i)	4	9	8	6	5

Determine a sequence for the five jobs that will minimize the elapsed time T .

3. Determine the optimal sequence of jobs which minimizes the total elapsed time based on the following information:

Jobs	1	2	3	4	5
Machine $A (A_i)$	3	8	7	5	2
Machine $B (B_i)$	3	4	2	1	5
Machine $C (C_i)$	5	8	10	7	6

4. Find the sequence that minimizes the total time required in performing the following jobs on three machines in the order ABC :

Jobs	1	2	3	4	5	6
Machine A	8	3	7	2	5	1
Machine B	3	4	5	2	1	6
Machine C	8	7	6	9	10	9

5. Find the time that minimizes the total time required to complete the following tasks:

Tasks	A	B	C	D	E	F	G
Machine I	3	8	7	4	9	8	7
Machine II	4	3	2	5	1	4	3
Machine III	6	7	5	11	5	6	12

6. Find the optimal sequence of jobs which minimizes the total elapsed time based on the following information. The mode of operation being ABC for each job.

Jobs	A_i	B_i	C_i
1	3	3	5
2	8	4	8
3	7	2	10
4	5	1	7
5	2	5	6

7. Find the sequence and idle time of each machine that minimizes the total elapsed time required to complete all the jobs on machines A, B, C in the order ABC .

Machines	Jobs				
	1	2	3	4	5
A	7	12	11	9	8
B	8	9	5	6	7
C	11	13	9	10	14

8. Determine the optimal sequence of jobs which minimizes the total elapsed time based on the following information. Also find the idle time of each machine:

Processing times on machines A , B , C			
Jobs	A_i	B_i	C_i
1	3	8	10
2	8	9	13
3	7	7	15
4	5	6	12
5	2	10	11
6	15	10	10

9. A company has six jobs which go through three machines X , Y and Z in the order XYZ . The processing time in minute for each job on each machine is as follows:

		Jobs					
		1	2	3	4	5	6
Machine	X	18	12	29	36	43	37
Machine	Y	7	12	11	2	6	12
Machine	Z	19	12	23	47	28	36

What would be the sequence of jobs that will minimize the elapsed time. Find this minimum time and idle time for each machine.

10. Find the sequence, the total elapsed time and idle times of A , B , C which minimizes the total elapsed time for the following sequencing problem:

Jobs	Processing Times		
	Machine A	Machine B	Machine C
1	7	2	7
2	2	3	6
3	6	4	5
4	1	1	8
5	4	0	9
6	0	5	8

11. Find the sequence that minimizes the total elapsed time required to complete the following jobs. Each job is processed in the order *ACB*.

Jobs	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>	<i>V</i>	<i>VI</i>	<i>VII</i>
Machine <i>A</i>	12	6	5	11	5	7	6
Machine <i>B</i>	7	8	9	4	7	8	3
Machine <i>C</i>	3	4	1	5	2	3	4

Time on machines is given in hours.

12. Solve the following sequencing problem giving an optimal solution when passing time is not allowed.

Machines	Jobs				
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
M_1	10	12	8	15	16
M_2	3	2	4	1	5
M_3	5	6	4	7	3
M_4	14	7	12	8	10

[Hint: Here (i) $\min_i M_{i1} \geq \max_i M_{ij}$ (ii) $\min_i M_{i4} \geq \max_i M_{ij}$ for $j = 2, 3$

and $M_{i2} + M_{i3} = 8$ (fixed positive constant) for $i = 1, 2, \dots, 5$.

$\therefore$ The problem reduces to the problem of 5 jobs on two machines *G* and *H* in the order *GH*.

13. Find an optimal sequence for processing four jobs on five machines when passing is not allowed, if each processing time (in hours) is given as:

Machines \ Jobs	1	2	3	4
M_1	6	5	4	7
M_2	4	5	3	2
M_3	1	3	4	2
M_4	2	4	5	1
M_5	8	9	7	5

Also find the total time.

14. Solve the following sequencing problem of jobs on six machines $M_j, j = 1, 2, \dots, 6$, in the order $M_1, M_2, \dots, M_6$. Processing times (in hours) are given below:

Jobs	Machines					
	M_1	M_2	M_3	M_4	M_5	M_6
A	18	8	7	2	10	25
B	17	6	9	6	8	19
C	11	5	8	5	7	15
D	20	4	3	4	8	12

15. Find the optimal sequence for processing 4 jobs A, B, C, D on four machines A_1, A_2, A_3, A_4 in the order A_1, A_2, A_3, A_4 . Processing times are given as follows:

Jobs/Machines	Processing times (a_{ij}) in hours			
	$A_1 (a_{11})$	$A_2 (a_{12})$	$A_3 (a_{13})$	$A_4 (a_{14})$
A	15	5	4	14
B	12	2	10	12
C	13	3	6	15
D	16	0	3	19

Answers 12.2

1.	4-1-5-2-3, 4-5-1-2-3, 1-4-5-2-3, 1-5-4-2-3, 5-1-4-2-3, 5-4-1-2-3. Total elapsed time $T = 51$ hours.
2.	3-2-1-4-5, 3-2-4-1-5, 3-2-4-5-1, 3-2-5-4-1, 3-2-1-5-4 and 3-2-5-1-4.
3.	1-4-5-3-2 and 4-1-5-3-2, Min. Time = 42 hours.
4.	4-5-2-6-1-3, Min. Time = 53 hours.
5.	$A-D-G-F-B-C-E$, $A-D-G-B-F-C-E$ Min. Time = 59 hours.
6.	1-4-5-3-2, 4-1-5-3-2, Total elapsed time = 42 hours.

7.	1 - 5 - 4 - 2 - 3. Total elapsed time = 72 minutes Idle times for A - 25 m, for B - 37 m and for C - 15 m.
8.	1 - 4 - 5 - 3 - 2 - 6, 4 - 1 - 5 - 3 - 2 - 6. Total elapsed time = 82 hours, Idle times for A - 42 hours, for B - 32 hours and for C - 11 hours.
9.	2 - 1 - 4 - 6 - 5 - 3, Total elapsed time = 209 minutes. Idle times for X - 34 m, for Y - 159 m and for Z - 44 m.
10.	4 - 5 - 2 - 6 - 1 - 3, 4 - 5 - 6 - 2 - 1 - 3, 4 - 5 - 2 - 6 - 3 - 1, 4 - 5 - 6 - 2 - 3 - 1. Total elapsed time = 45 hours. Idle times are; for A - 25h, for B - 30h and fore C - 2 hours.
11.	III - V - II - VI - I - IV - VII, III - V - VI - II - I - IV - VII Min. Time = 59 hours. Idle times, for A - 7h, for B - 13h and for C - 37 hours.
12.	C - A - E - D - B.
13.	1 - 3 - 2 - 4 Total elapsed time = 43 hours.
14.	C - A - B - D.
15.	D - C - B - A.

12.7 Processing Two Jobs Through m Machines

Now, we shall consider the following problem:

1. There are two jobs 1 and 2 to be performed.
2. There are m machines denoted by $A, B, C, \dots, M$.
3. The technological ordering of each of the two jobs 1 and 2 through m machines is given in advance. This ordering need not be the same for jobs.
4. The exact or expected processing times $A_1, B_1, C_1, \dots, M_1$, for job 1 and $A_2, B_2, C_2, \dots, M_2$ for job 2 on the given m machines are given.

The problem is to minimize the total elapsed time T from the start of the first job on first machine to the completion of last job on the last machine.

It is obvious that for each machine, there are two sequences of jobs *i.e.*, (1, 2) or (2, 1). Thus, we have to select one sequence out of these two for each machine. Graphical method (discussed in article 12.8) is a simple method for the solution of this problem.

12.8 Graphical Method

For the solution of a sequencing problem involving 2 jobs and m machines, we draw two lines horizontal and vertical representing Job 1 and Job 2 respectively. On these two lines (axes) we mark the processing times of Jobs on different machines in the given order. A horizontal line in the graph will represent the work on job 1 only while job 2 remains idle and the vertical line will represent the work on job 2 only while job 1 remains idle. A line making an angle of 45° to the horizontal will represent the work on the two jobs simultaneously. A job will be processed on a machine if the machine is idle. Both the jobs cannot be processed on the same machine. These possibilities will be shown in the graph by shaded rectangular blocks. Now, we shall start with zero time (starting point at the origin) and move on doing jobs avoiding the shaded rectangular blocks until we reach at the finish point. It will be seen that the path is limited to a series segment which are horizontal, vertical or diagonals inclined at 45° to the horizontal. Movement to the right (along a horizontal line) which mean that the job 1 is proceeding while job 2 is idle, movement upwards will mean that job 2 is proceeding while job 1 is idle, and movement diagonally will mean that both jobs are proceeding simultaneously. A best path is that which minimizes the idle time for job 1 and job 2 *i.e.*, minimizes the horizontal and vertical line in the path. Hence we shall try to move as much as we can along the line (diagonal) inclined at 45° to the horizontal.

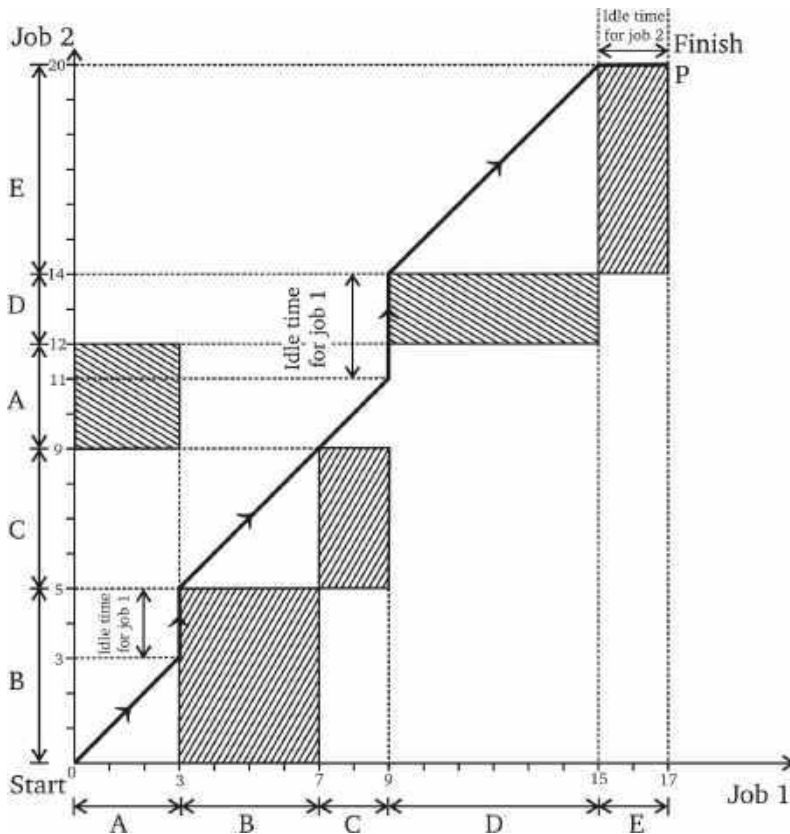
For the clear understanding of the procedure see the following Example 1.

Example 1: Use graphical method to minimize the time needed to process the following jobs on the machine shown, *i.e.*, for each machine, find the job which should be done first. Also calculate the total time needed to complete both the jobs.

Job 1		Job 2	
Sequence of Machines	Time	Sequence of Machines	Time
A	3	B	5
B	4	C	4
C	2	A	3
D	6	D	2
E	2	E	6

Solution: We draw the graph of the problem in the manner described in article 12.8 as follows:

1. We draw lines (horizontal and vertical) OX and OY representing the processing time of Job 1 and Job 2 respectively.
2. Then, we mark the processing times of the jobs on the machines in the given order, as shown in figure.
3. Then, we draw shaded rectangular blocks corresponding to each machines as shown in figure.



4. Now, starting from the starting point O we move on doing jobs avoiding the shaded rectangular blocks until the finish point is reached. Here it is important to note that we shall try to move as much as we can along a line inclined at angle 45° to the horizontal. Wherever movement along this line is not possible we shall move only horizontally or vertically (as needed).

The best path is shown in figure by arrows.

It is clear from the graph that we have processed.

Job 1 before 2 on machine *A*

Job 2 before 1 on machine *B*

Job 2 before 1 on machine *C*

Job 2 before 1 on machine *D*

and Job 2 before 1 on machine *E*

The elapsed time

$$= \text{Processing time of Job 1} + \text{idle time of Job 1}$$

$$= 17 + (2 + 3) = 22 \text{ hours}$$

or
$$= \text{Processing time of Job 2} + \text{idle time of Job 2}$$

$$= 20 + 2 = 22 \text{ hours}$$

Exercise 12.3

1. What is sequencing problem?
2. State clearly the assumptions used in the study of sequencing problems.
3. Give the Johnson's method for determining an optimal sequence for processing of n jobs on two machines.
4. Modify Johnson's method to solve the problem of processing n jobs through three machines.
5. Use the graphical method to minimize the time needed to process the following two jobs on four machines *A*, *B*, *C* and *D* with the technological orderings:

Job 1	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
Job 2	<i>D</i>	<i>B</i>	<i>A</i>	<i>C</i>

and the processing times:

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
Job 1	2	4	5	1
Job 2	2	5	3	6

6. Use graphical method to minimize the time required to process the following jobs on the machines *i.e.*, for each machine specify the job which should be done first. Also calculate the table elapsed time to complete both jobs.

Job 1	Sequence	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
	Time (in hrs.)	6	8	4	12	4
Job 2	Sequence	<i>B</i>	<i>C</i>	<i>A</i>	<i>D</i>	<i>E</i>
	Time (in hrs.)	10	8	6	4	12

7. Use graphical method to minimize the time needed to process the following jobs on two machines shown below *i.e.*, for each machine find the job which should be done first. Also calculate the total time needed to complete both jobs:

Job 1		Job 2	
Sequence of Machines	Time	Sequence of Machines	Time
<i>A</i>	2	<i>C</i>	4
<i>B</i>	3	<i>A</i>	5
<i>C</i>	4	<i>D</i>	3
<i>D</i>	6	<i>E</i>	2
<i>E</i>	2	<i>B</i>	6

8. The time spent in processing of two jobs on six machines A, B, C, D, E and F and the necessary technological orderings of machines are as follows:

Job I	$A - 20, C - 10, D - 10, B - 30, E - 25, F - 15$
Job II	$A - 10, C - 30, B - 15, D - 10, F - 15, E - 20$

Use graphical method to determine an optimal sequence of jobs, which minimizes the elapsed time.

9. Two jobs are to be processed on four machines a, b, c and d . The technological order for these jobs on machines are as follows:

Job 1 : a, b, c, d .

Job 2 : d, b, a, c .

Processing times are given in the following table:

	Machines			
	a	b	c	d
Job 1	4	6	7	3
Job 2	4	7	5	8

Find the optimal sequence of jobs on each of the machines.

Answers 12.3

5.	Both Jobs can be worked simultaneously until Job 1 is completed and then Job 2 is completed. Total elapsed time = 16 hours. Idle time for job 1 = 4 hours.
6.	Job 1 before 2 on machine <i>A</i> Job 2 before 1 on machine <i>B</i> Job 2 before 1 on machine <i>C</i> Job 2 before 1 on machine <i>D</i> Job 2 before 1 on machine <i>E</i> Total elapsed time = 44 hours. Idle time for Job 1 = 10 hours. and for Job 2 = 4 hours.
7.	Job 1 before 2 on machine <i>A</i> Job 1 before 2 on machine <i>B</i> Job 1 before 2 on machine <i>C</i> Job 2 before 1 on machine <i>D</i> Job 1 before 2 on machine <i>E</i> Total elapsed time = 20 hours. Idle time for Job 1 = 3 hours.
8.	Job I before II on machine <i>A</i> Job I before II on machine <i>C</i> Job I before II on machine <i>D</i> Job I before II on machine <i>B</i> Job I before II on machine <i>E</i> Job II before I on machine <i>F</i> Total elapsed time = 130 hours. Idle time for Job I = 20 hours. Idle time for Job II = 30 hours.

9. Both Jobs can be worked simultaneous until Job 1 is computed and then Job 2 is computed.

Total elapsed time = 26 hours.

Idle time for Job 1 = 6 hours

Idle time for Job 2 = 2 hours.

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

- The procedure for determining an optimal sequence of n -jobs on two machines was developed by:
 - Newton
 - Johnson
 - Legendre
 - J.V. Neumann.
- The sequencing order of n -jobs on three machines A, B & C in the order ABC can be obtained if:
 - The smallest processing time for machine $A \geq$ the largest processing time for machine B .
 - The smallest processing time for machine $C \geq$ the largest processing time for machine B .
 - (a) or (b)
 - None of these.
- If A_i, B_i, C_i are the processing times of i -th job on machines A, B and C respectively, then to find the sequencing of n -jobs on these machines in the order ABC , these machines are replaced by two machines G & H where the processing times of i -th job on then these two machines i.e. G_i and H_i are given by:
 - $G_i = A_i + B_i, H_i = B_i + C_i$
 - $G_i = A_i - B_i, H_i = B_i + C_i$
 - $G_i = A_i - B_i, H_i = B_i + C_i$
 - $G_i = A_i - B_i, H_i = B_i - C_i$.
- If $M_{ij}, i = 1, 2, \dots, n; j = 1, 2, \dots, m$ are the processing times of n -jobs on m -machines, and $M_{i_2} + M_{i_3} + \dots + M_{i_{(m-1)}} = \text{fixed positive constant}, \forall i = 1, 2, \dots, n$, then the sequencing of n -jobs on m -machines in the order $M_1 M_2 \dots M_m$ is the same as the sequencing of n -jobs on two machines:
 - M_1, M_m in the order $M_1 M_m$
 - M_1, M_m in the order $M_m M_1$
 - M_1, M_m in the order $M_m M_1$
 - None of these.

5. Johnson's method for sequencing of n -jobs on three machines is known as his:
- (a) Modified method (b) Extended method
(c) Normal method (d) None of these.
6. A sequencing problem involving 5 jobs and three machines requires evolution of:
- (a) $\lfloor 5 \rfloor + \lfloor 5 \rfloor + \lfloor 5 \rfloor$ sequences (b) $(\lfloor 5 \rfloor)^3$ sequences
(c) $5 \times 5 \times 5$ sequences (d) $5+5+5$ sequences.

Fill in the Blank

- The selection of an appropriate order in which to service waiting customers is called
- Time taken by the jobs in going from one machine to another is
- No machine can process more than job at a time.
- The processing times of jobs on machines are of their processing times.
- In graphical solution of 2 jobs and m -machines sequencing problem, an optimal path is one that minimizes the idle time for job.
- The minimum elapsed time for the sequencing problem :

		Job				
		J_1	J_2	J_3	J_4	J_5
A		5	1	9	3	10
B		2	6	7	8	4

is hrs.

True/False

- Each job, once started on a machine, must be performed upto the completion on that machine.
- In sequencing problems of jobs on machines, there is only one machine of each type.
- A sequencing problem involving six jobs and two machines requires evolution of $(\lfloor 6 \rfloor)^2$ sequences.
- A sequencing problem involving processing of two jobs on n -machines cannot be solved graphically.

Answers

OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1.	(b)	2.	(c)	3.	(a)
4.	(a)	5.	(a)	6.	(b)

Fill in the Blank

1.	sequencing	2.	negligible	3.	one
4.	independent	5.	second	6.	30

True/False

1.	T	2.	T	3.	T
4.	F				

○○○

Table 1: Normal Distribution

An entry in the table are the areas under the entire normal curve between normal variate $Z = 0$ and a positive value of Z . The curve is symmetrical about $Z = 0$.

<i>Z</i> to First Decimal	<i>Second Decimal</i>									
	·00	·01	·02	·03	·04	·05	·06	·07	·08	·09
0·0	·0000	·0040	·0080	·0120	·0160	·0199	·0239	·0279	·0319	·0359
0·1	·0398	·0438	·0478	·0517	·0557	·0596	·0636	·0675	·0714	·0753
0·2	·0793	·0832	·0871	·0910	·0948	·0987	·1026	·1064	·1103	·1141
0·3	·1179	·1217	·1255	·1293	·1331	·1368	·1406	·1443	·1480	·1517
0·4	·1554	·1591	·1628	·1664	·1700	·1736	·1772	·1808	·1844	·1879
0·5	·1915	·1950	·1985	·2019	·2054	·2088	·2123	·2157	·2190	·2224
0·6	·2257	·2291	·2324	·2357	·2389	·2422	·2454	·2486	·2518	·2549
0·7	·2580	·2611	·2642	·2674	·2704	·2734	·2764	·2794	·2823	·2852
0·8	·2881	·2910	·2939	·2967	·2995	·3023	·3051	·3078	·3106	·3133
0·9	·3159	·3186	·3212	·3238	·3264	·3289	·3315	·3340	·3365	·3389
1·0	·3413	·3438	·3461	·3485	·3508	·3531	·3554	·3577	·3599	·3621
1·1	·3643	·3665	·3686	·3708	·3729	·3749	·3770	·3790	·3810	·3830
1·2	·3849	·3869	·3888	·3907	·3925	·3944	·3962	·3980	·3997	·4015
1·3	·4032	·4049	·4066	·4082	·4099	·4115	·4131	·4147	·4162	·4177
1·4	·4192	·4207	·4222	·4236	·4251	·4265	·4279	·4292	·4306	·4319
1·5	·4332	·4345	·4357	·4370	·4382	·4394	·4406	·4418	·4429	·4441
1·6	·4452	·4463	·4474	·4484	·4495	·4505	·4515	·4525	·4535	·4545
1·7	·4554	·4564	·4573	·4582	·4591	·4599	·4608	·4616	·4625	·4633
1·8	·4641	·4649	·4656	·4664	·4671	·4678	·4686	·4693	·4699	·4706
1·9	·4713	·4719	·4726	·4732	·4738	·4744	·4750	·4756	·4761	·4767
2·0	·4772	·4778	·4783	·4788	·4793	·4798	·4803	·4808	·4812	·4817
2·1	·4821	·4826	·4830	·4834	·4838	·4842	·4846	·4850	·4854	·4857
2·2	·4861	·4865	·4868	·4871	·4874	·4878	·4881	·4884	·4887	·4890
2·3	·4893	·4896	·4898	·4901	·4904	·4906	·4909	·4911	·4913	·4916
2·4	·4918	·4920	·4922	·4925	·4927	·4929	·4931	·4932	·4934	·4936
2·5	·4938	·4940	·4941	·4943	·4945	·4946	·4948	·4949	·4951	·4952
2·6	·4953	·4955	·4956	·4957	·4959	·4960	·4961	·4962	·4963	·4964
2·7	·4965	·4966	·4967	·4968	·4969	·4970	·4971	·4972	·4973	·4974
2·8	·4974	·4975	·4976	·4977	·4977	·4978	·4979	·4979	·4980	·4981
2·9	·4981	·4982	·4982	·4983	·4984	·4984	·4985	·4985	·4986	·4986
3·0	·4987	·4987	·4987	·4988	·4988	·4989	·4989	·4989	·4990	·4990

Each of the Areas under the normal curve between $Z = -\infty$ to $Z = 0$ and that between $Z = 0$ to $Z = \infty$ is equal to 0.5.

Method to read Prob. from the table : Let $Z = 1.21$, then

$P(Z \leq 1.21) = \text{Area under the curve from } Z = -\infty \text{ to } Z = 1.21$

$$= 0.5 + 0.3869 = 0.8869 \quad \text{i.e. } 88.69\%$$

Where, 0.3869 = value opposite to 1.21 in the table

and $P(Z \leq -1.21) = \text{Area under the curve from } Z = -\infty \text{ to } Z = -1.21$

$$= 0.5 - 0.3869 = 0.1131. \quad \text{i.e. } 11.31\%$$

Table 2: Present Values

Year	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
1	.990	.980	.971	.962	.952	.943	.935	.926	.917	.909
2	.980	.961	.943	.925	.907	.890	.873	.857	.842	.826
3	.971	.942	.915	.889	.864	.840	.816	.794	.772	.751
4	.961	.924	.888	.855	.823	.792	.763	.735	.708	.683
5	.951	.906	.863	.822	.784	.747	.713	.681	.650	.621
6	.942	.888	.837	.790	.746	.705	.666	.630	.596	.564
7	.933	.871	.813	.760	.711	.665	.623	.583	.547	.513
8	.923	.853	.789	.731	.677	.627	.582	.540	.502	.467
9	.914	.837	.766	.703	.645	.592	.544	.500	.460	.424
10	.865	.820	.744	.676	.614	.558	.508	.463	.422	.386
11	.896	.804	.722	.650	.585	.527	.475	.429	.388	.350
12	.887	.789	.701	.625	.557	.497	.444	.397	.356	.319
13	.879	.773	.681	.601	.530	.469	.415	.368	.326	.290
14	.870	.758	.661	.577	.505	.442	.388	.340	.299	.263
15	.861	.743	.642	.555	.481	.417	.362	.315	.275	.239
16	.853	.728	.623	.534	.458	.394	.339	.292	.252	.218
17	.844	.714	.605	.513	.436	.371	.317	.270	.231	.198
18	.836	.700	.587	.494	.416	.350	.296	.250	.212	.180
19	.828	.686	.570	.475	.396	.331	.277	.232	.194	.164
20	.820	.673	.554	.456	.377	.312	.258	.215	.178	.149
21	.811	.660	.538	.439	.359	.294	.242	.199	.164	.135
22	.803	.647	.522	.422	.342	.278	.226	.184	.150	.123
23	.795	.634	.507	.406	.326	.262	.211	.170	.138	.112
24	.788	.622	.492	.390	.310	.247	.197	.158	.126	.102
25	.780	.610	.478	.375	.295	.233	.184	.146	.116	.092
30	.742	.552	.412	.308	.231	.174	.131	.099	.075	.057
35	.706	.500	.355	.253	.181	.130	.094	.068	.049	.036
40	.672	.453	.307	.208	.142	.097	.067	.046	.032	.022
45	.639	.410	.264	.171	.111	.073	.048	.031	.021	.014
50	.806	.372	.228	.141	.087	.054	.034	.021	.013	.009

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